(d) Install the four mounting screws [6] and washers [7].

NOTE: A safety cable or lockwire is not necessary.

(e) Tighten the mounting screws [6] to 20 ±1 in-lb (2.3 ±0.2 N·m).

(f) Remove excess sealant with a dry cotton wiper, G00034.

NOTE: Do not remove sealant from under the mounting screw [6].

(g) Make sure that the sealant is continuously squeezed out along the edges of the contact surfaces. If gaps, bubbles or voids show in the sealant, disassemble and apply more sealant.

(h) Make a fillet seal.

(i) Make sure that the sealant is flushed with the mating part edges.

(j) Remove all unnecessary sealant.

SUBTASK 28-22-11-765-001

28-AWL-21: CDCCL

(2) Measure the electrical bonding resistance between the upper housing of the actuator [4] and the front spar (SWPM 20-20-10).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

28-AWL-21: CDCCL

(a) Measure the electrical bonding resistance with the bonding jumper [3] and the electrical connector [5] disconnected.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

NOTE: Make sure that the bonding jumper [3] does not touch the actuator [4].

(b) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-21: CDCCL

(c) If the actuator [4], index plate [8] and valve adapter plate [9] were removed, make sure that the electrical bonding resistance is 0.0030 ohm (3.0 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) If the electrical bonding resistance is more than 0.0030 ohm (3.0 milliohms), do the applicable procedure in this task: Engine Fuel Spar Valve Electrical Faying Surface Bonds Repair, TASK 28-22-11-300-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

28-AWL-21: CDCCL

(d) If the actuator [4] and index plate [8] were removed, make sure that the electrical bonding resistance is 0.0040 ohm (4.0 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) If the electrical bonding resistance is more than 0.0040 ohm (4.0 milliohms), do the applicable procedure in this task: Engine Fuel Spar Valve Electrical Faying Surface Bonds Repair, TASK 28-22-11-300-801.

28-AWL-21: CDCCL

(e) If only the actuator [4] was removed, make sure the electrical bonding resistance is 0.0040 ohm (4.0 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) If the electrical bonding resistance is more than 0.0040 ohm (4.0 milliohms), do the applicable procedure in this task: Engine Fuel Spar Valve Electrical Faying Surface Bonds Repair, TASK 28-22-11-300-801.

SUBTASK 28-22-11-100-002

28-AWL-21: CDCCL

(3) If the terminal of the bonding jumper [3] is disconnected from the stiffener, install the bonding jumper [3] with a fay surface bond between the stiffener and the terminal of bonding jumper [3] (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

SUBTASK 28-22-11-420-006

(4) If the terminal of the bonding jumper [3] is disconnected from the stiffener, install the bonding jumper [3]:

(a) Install the bonding jumper [3], washer [10], and nut [11] on the ground stud.

(b) Tighten the nut [11] to 31 ±3 in-lb (3.5 ±0.3 N·m).

SUBTASK 28-22-11-765-002

28-AWL-21: CDCCL

(5) Measure the electrical bonding resistance between the loose end of the bonding jumper [3] and the front spar (SWPM 20-20-10).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

28-AWL-21: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0015 ohm (1.5 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(c) If the electrical bonding resistance is more than 0.0015 ohm (1.5 milliohms), do the applicable procedure in this task: Engine Fuel Spar Valve Electrical Faying Surface Bonds Repair, TASK 28-22-11-300-801.

SUBTASK 28-22-11-390-002

(6) If the bonding jumper [3] was disconnected from the stiffener, apply a fillet seal to the ground stud, terminal on the bonding jumper [3], and nut [11] (SWPM 20-20-10):

28-AWL-21: CDCCL

(a) Apply a fillet seal of sealant, A50337 to the terminal of the bonding jumper [3] at the stiffener (SWPM 20-20-10).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

SUBTASK 28-22-11-100-003

(7) Do these steps to prepare the bonding jumper [3] and the actuator [4] for installation:

(a) Apply a fay surface bond between the surfaces of the actuator [4], and bonding jumper [3] (SWPM 20-20-00):

SUBTASK 28-22-11-420-025

28-AWL-21: CDCCL

(8) If the bonding jumper is attached to the stiffener, measure the resistance between the loose end of the bonding jumper [3] and the spar (SWPM 20-20-10).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-21: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0020 ohm (2.0 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(c) If the electrical bonding resistance is more than 0.0020 ohm (2.0 milliohms), do the applicable procedure in this task: Engine Fuel Spar Valve Electrical Faying Surface Bonds Repair, TASK 28-22-11-300-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

28-AWL-21: CDCCL

(9) Apply a fay surface bond between the mating surfaces of the actuator [4] and the terminal of the bonding jumper [3] (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

SUBTASK 28-22-11-420-009

(10) Tighten the screw [1] to 23 ±2 in-lb (2.6 ±0.2 N·m).

SUBTASK 28-22-11-765-003

28-AWL-21: CDCCL

(11) Measure the electrical bonding resistance between the upper housing of the actuator [4] and the attached terminal of the bonding jumper [3] (SWPM 20-20-10).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

NOTE: Do not measure the bonding resistance from the screw [1].

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-21: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0010 ohm (1.0 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) If the electrical bonding resistance is more than 0.0010 ohm (1.0 milliohm), do the applicable procedure in this task: Engine Fuel Spar Valve Electrical Faying Surface Bonds Repair, TASK 28-22-11-300-801.

SUBTASK 28-22-11-390-003

28-AWL-21: CDCCL

(12) Apply a fillet seal between the mating surfaces of the actuator [4] and the terminal of the bonding jumper [3] (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

28-AWL-21: CDCCL

(13) Apply a cap seal on the screw [1], and the terminal lug of the bonding jumper [3] attached to the actuator [4] (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Apply sealant, A50337 to the fay surfaces to the area around the bonding jumper [3].

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SUBTASK 28-22-11-916-001

(14) Apply protective finishes with coating, C00064 or coating, C00307 to the bare metal areas on the front spar (TASK 28-11-00-600-801).

NOTE: Apply protective coating to the area where the finish was removed from the electrical bonding probe touching the surface.

SUBTASK 28-22-11-916-002

(15) Apply Bonderite M-CR 1200S Aero coating, C50033 to areas that have bare metal on the actuator [4] (TASK 51-21-41-370-802).

NOTE: Apply the protective coating to the actuator [4] where the electrical bonding probe removed the finish and bare metal areas around the bonding jumper [3] that is not covered with sealant.

SUBTASK 28-22-11-430-001

(16) Connect the electrical connector [5] to the actuator [4].

J. Spar Valve Actuator Operational Test

SUBTASK 28-22-11-865-001

(1) For the left spar valve actuator, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1
B	5	C00540	FUEL SPAR VALVE IND

SUBTASK 28-22-11-865-002

(2) For the right spar valve actuator, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	5	C00540	FUEL SPAR VALVE IND

SUBTASK 28-22-11-710-001

(3) For the applicable actuator [4], do this task: Engine Fuel Spar Valve - Electrical Control and Indication Test, TASK 28-22-00-710-801.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-11-410-004

(1) Close these access panels:

Number	Name/Location
521BB	Outboard Lower Fixed Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel

SUBTASK 28-22-11-860-002

(2) Activate the leading edge flaps and slats (TASK 27-81-00-440-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-11-000-802

4. Engine Fuel Spar Valve - Valve Adapter Removal

(Figure 402)

A. General

(1) The engine fuel spar valve - valve adapter is referred to as the valve adapter in this task.

(2) The valve adapter and shaft assembly connect the actuator to the valve body. The valve adapter assembly consists of these components:

(a) Index plate

(b) Adapter plate (adapter shaft attached)

(c) Adapter shaft (with U-joint connection).

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-26-00-650-802	Tank to Tank Fuel Transfer (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-4746	Handle - Sealant Removal Tool Part #: 310/1 (for SkyScraper) Supplier: F6892 Part #: SHR0272-T (for PnJ scraper) Supplier: 7RKH2 Part #: ST982LH-F-3 (for ST982LF) Supplier: 81205 Opt Part #: SHW0215 (for PnJ scraper) Supplier: 7RKH2

D. Location Zones

Zone	Area
521	Left Wing - Leading Edge to Front Spar
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
621	Right Wing - Leading Edge to Front Spar
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

E. Access Panels

Number	Name/Location
521BB	Outboard Lower Fixed Leading Edge Access Panel
532BB	Main Tank Access Door - Wing Station 265
621BB	Outboard Lower Fixed Leading Edge Access Panel
632BB	Main Tank Access Door - Wing Station 265

F. Prepare for the Removal

SUBTASK 28-22-11-650-001

(1) Defuel or transfer fuel from the applicable main tank (TASK 28-26-00-650-801, TASK 28-26-00-650-802).

SUBTASK 28-22-11-010-012

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(2) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove the applicable access door(s):

(TASK 28-11-11-000-801)

Number	Name/Location
532BB	Main Tank Access Door - Wing Station 265
632BB	Main Tank Access Door - Wing Station 265

SUBTASK 28-22-11-860-003

(3) Open these access panels:

Number	Name/Location
521BB	Outboard Lower Fixed Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel

SUBTASK 28-22-11-010-013

(4) Remove the applicable actuator [4] (TASK 28-22-11-000-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-11-010-014

(5) Remove the sealant from the adapter plate edge with sealant removal tool, COM-2481 and sealant removal tool handle, COM-4746 or equivalent.

G. Engine Fuel Spar Valve - Valve Adapter Removal

SUBTASK 28-22-11-010-015

(1) Remove the valve adapter plate [9] from the front spar:

(a) Remove the sealant from the four bolts [26].

(b) Remove the four bolts [26] and washers [27].

(c) Disengage the adapter shaft from the valve body [21].

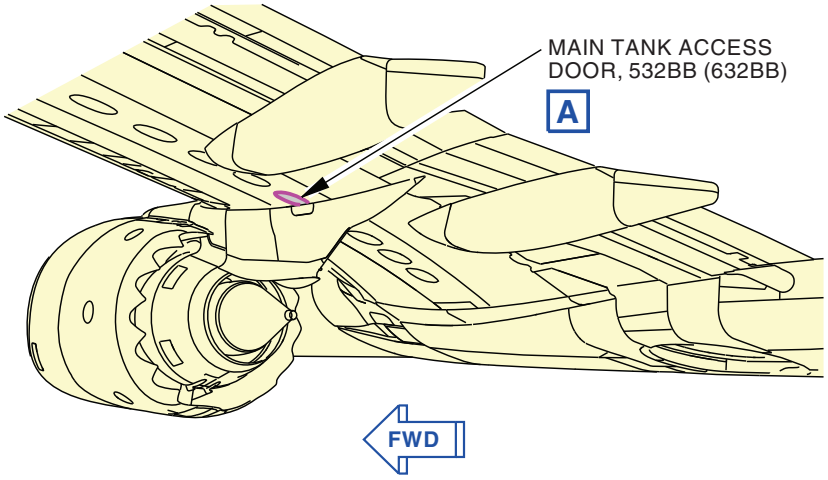
SUBTASK 28-22-11-010-016

(2) Remove the valve adapter plate [9].

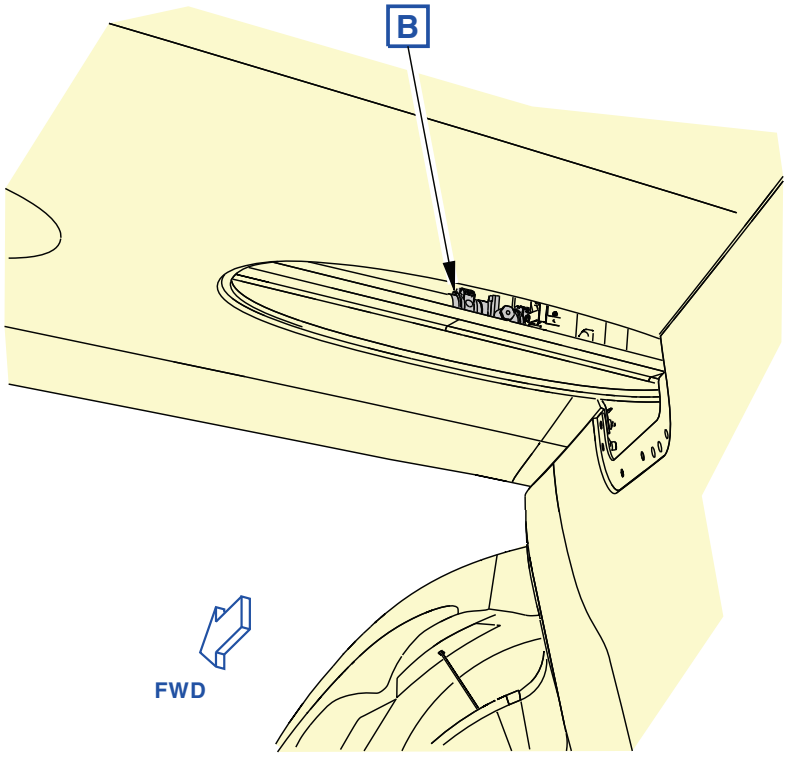
NOTE: An additional person is necessary to help remove the valve adapter plate [9] to keep the valve adapter plate [9] from dropping.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS EQUIVALENT)



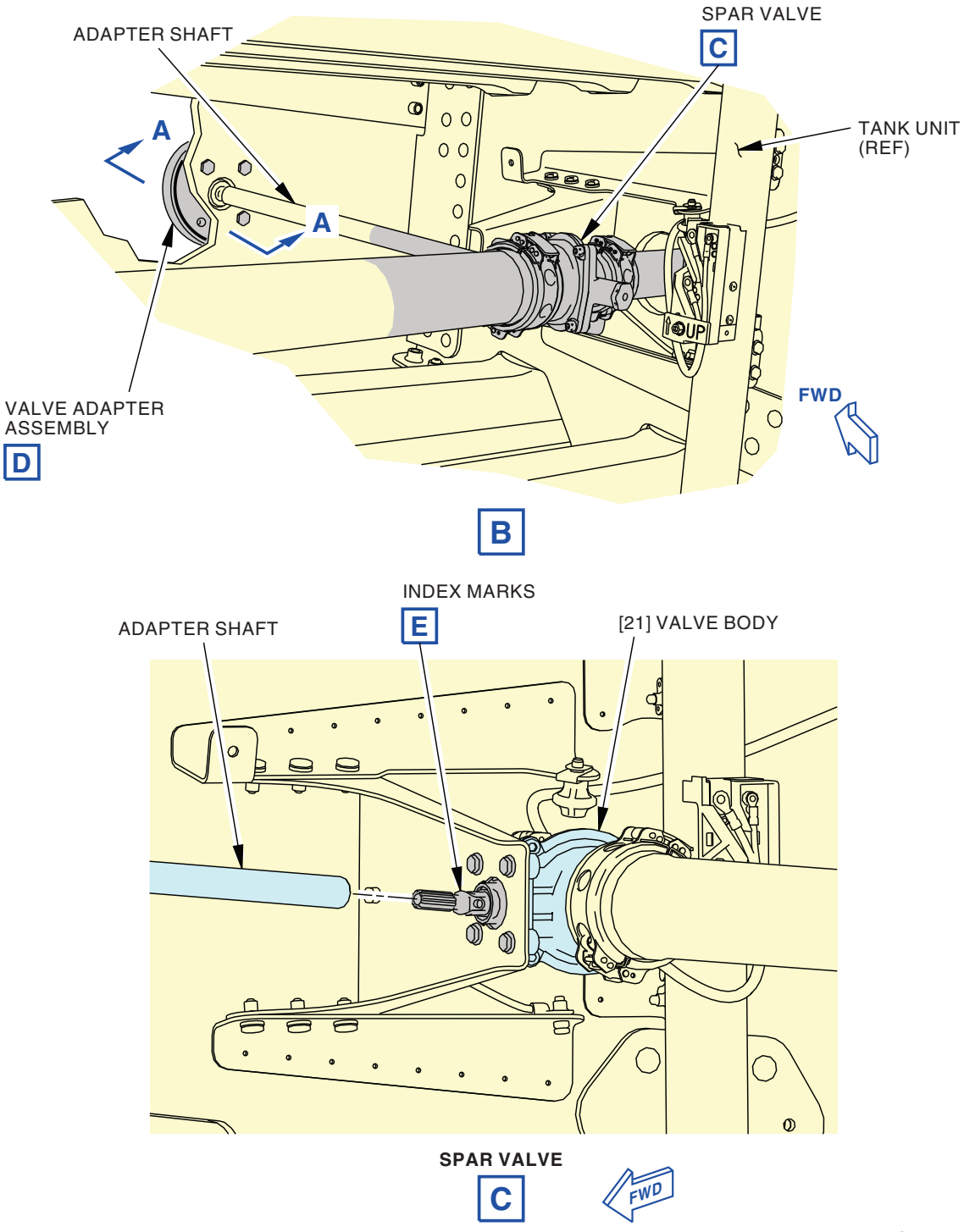
MAIN TANK ACCESS DOOR, 532BB (632BB)

A

2413587 S00061532880_V2

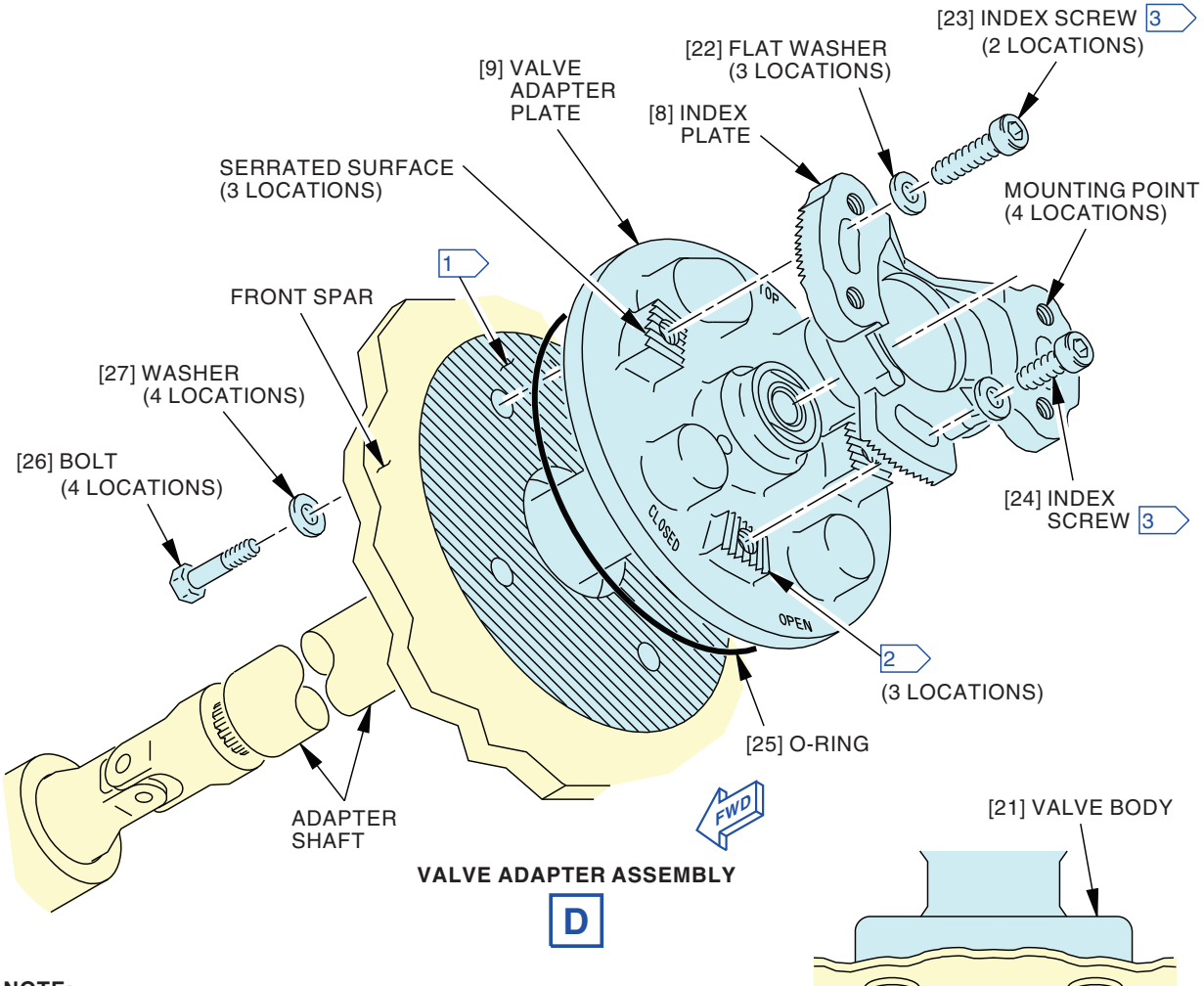
Spar Valve Adapter Installation
Figure 402/28-22-11-990-802 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Spar Valve Adapter Installation
Figure 402/28-22-11-990-802 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

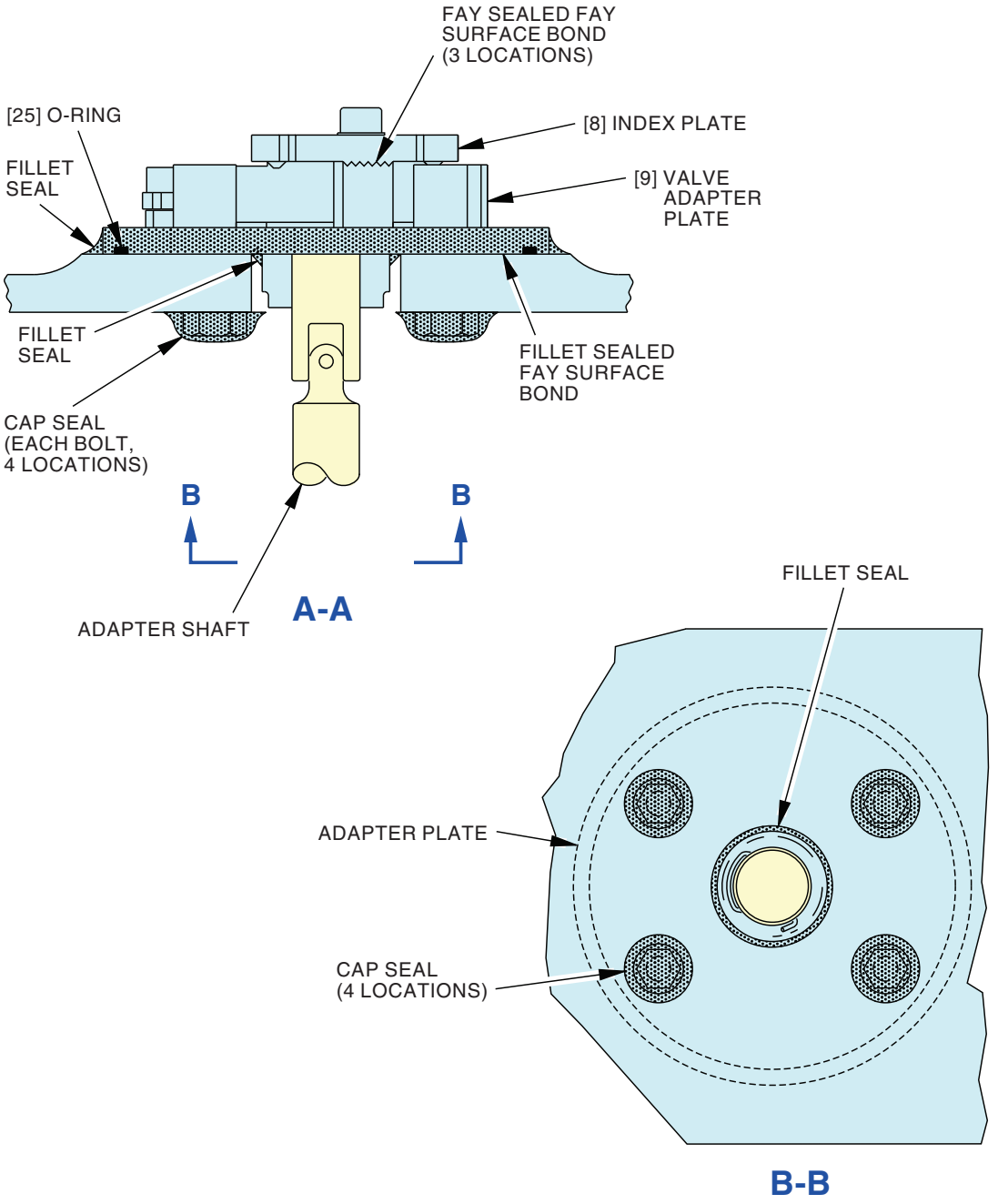
LEFT VALVE ADAPTER ASSEMBLY IS SHOWN. RIGHT VALVE ADAPTER ASSEMBLY HAS INSTALLED 180° UPSIDE DOWN, COMPARED TO THE LEFT SIDE. THE "TOP" MARKING ON VALVE ADAPTER PLATE IS POINTED TO THE BOTTOM.

1. PREPARE AND INSTALL THE VALVE ADAPTER TO THE FRONT SPAR WITH A FILLET SEALED FAY SURFACE BOND (SWPM 20-20-00).
2. PREPARE AND INSTALL THE SERRATED CONTACT SURFACES OF THE VALVE ADAPTER AND INDEX PLATE WITH A FAY SEAL FAY SURFACE BOND (3 LOCATIONS) (SWPM 20-20-00).
3. APPLY SEALANT TO THE SHANK AND THREADS OF THE FASTENERS DURING THE INSTALLATION.

2413589 S00061532882_V3

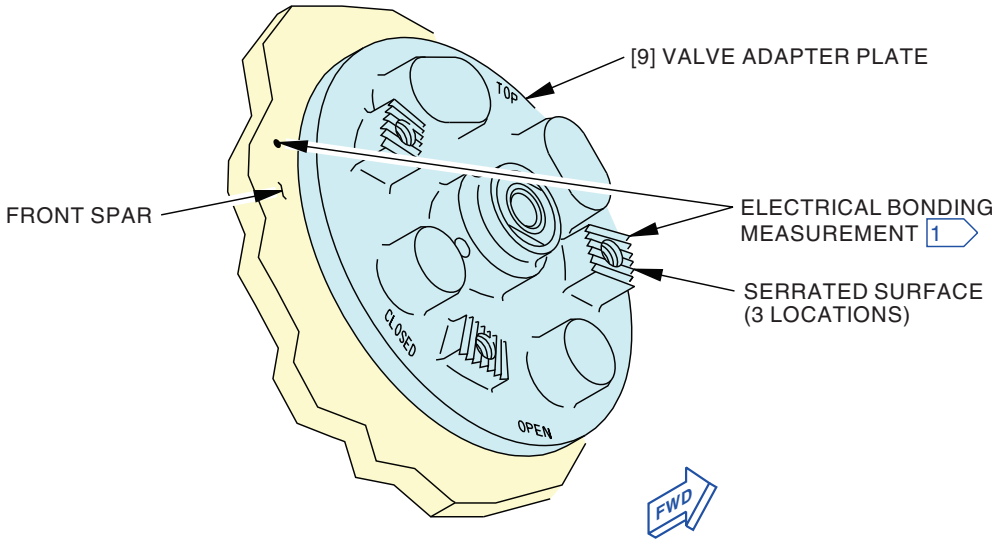
Spar Valve Adapter Installation
Figure 402/28-22-11-990-802 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

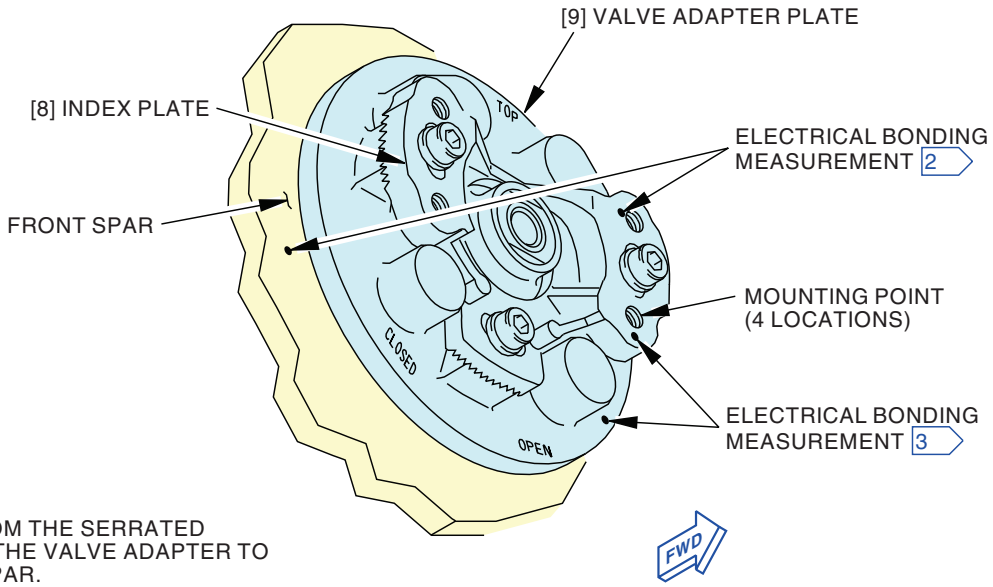


Spar Valve Adapter Installation
Figure 402/28-22-11-990-802 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



VALVE ADAPTER ELECTRICAL BONDING MEASUREMENT



INDEX PLATE ELECTRICAL BONDING MEASUREMENT

NOTE:

1. MEASURE FROM THE SERRATED SURFACE ON THE VALVE ADAPTER TO THE FRONT SPAR.
2. MEASURE FROM ONE OF THE MOUNTING POINTS ON THE INDEX PLATE TO THE FRONT SPAR.
3. MEASURE FROM ONE OF THE MOUNTING POINTS ON THE INDEX PLATE TO THE VALVE ADAPTER.

2413591 S00061532884_V2

Engine Fuel Spar Valve Resistance Measurement - Probe Locations
Figure 403/28-22-11-990-807

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-11-400-802

5. Engine Fuel Spar Valve - Valve Adapter Installation

(Figure 402, Figure 403)

A. General

(1) The engine fuel spar valve - valve adapter is referred to as the valve adapter in this task.

(2) The valve adapter and shaft assembly connect the actuator to the valve body. The valve adapter assembly consists of these components:

(a) Index plate

(b) Adapter plate (adapter shaft attached)

(c) Adapter shaft (with U-joint connection).

(3) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-00-600-801	Apply the Corrosion Resistant Finish (Topcoat) (P/B 701)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
28-22-00-710-801	Engine Fuel Spar Valve - Electrical Control and Indication Test (P/B 501)
28-22-00-720-802	Engine Fuel Spar Valve Installation - Operational Test (P/B 501)
28-22-11-300-801	Engine Fuel Spar Valve Electrical Faying Surface Bonds Repair (P/B 301)
51-22-00-160-801	Prepare For Sealing (P/B 201)
51-22-00-390-806	Fillet Seal Application (P/B 201)
51-22-00-390-808	Fastener Seal Application (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50153	Sealant - Fuel Tank - Class A-2	BMS5-45 Class A-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50337	Sealant - Fuel Tank	BMS5-45 Class B
C50315	Coating - Chemical Conversion - Bonderite M-CR 600 Aero (Formerly Alodine 600)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
D00504	Grease - Petrolatum	VV-P-236
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51372	Kit - Safety Cable, 321 CRES - 0.035 +/-0.003 Inch (1.0 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	Valve adapter plate	28-22-11-01-070	PTF ALL
		28-22-11-02-070	PTF ALL
25	O-ring	28-22-11-01-100	PTF ALL
		28-22-11-02-100	PTF ALL

F. Location Zones

Zone	Area
521	Left Wing - Leading Edge to Front Spar
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
621	Right Wing - Leading Edge to Front Spar
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

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G. Access Panels

Number	Name/Location
521BB	Outboard Lower Fixed Leading Edge Access Panel
532BB	Main Tank Access Door - Wing Station 265
621BB	Outboard Lower Fixed Leading Edge Access Panel
632BB	Main Tank Access Door - Wing Station 265

H. Prepare for Installation

SUBTASK 28-22-11-110-001

(1) Do these steps to prepare the valve adapter plate [9] and the front spar for installation:

(a) Remove the sealant from the front spar, do this task: Clean Bare, Clad, or Plated Metal with Solvent, TASK 20-10-34-110-802.

SUBTASK 28-22-11-760-005

28-AWL-21: CDCCL

(2) Apply a fay sealed fay surface bond between the valve adapter plate [9] and the front spar, outside the fuel tank (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

I. Valve Adapter Installation

SUBTASK 28-22-11-640-001

(1) Apply a thin layer of grease, D00504 on the new O-ring [25].

SUBTASK 28-22-11-420-010

(2) Put the O-ring [25] in the groove of the valve adapter plate [9].

SUBTASK 28-22-11-430-002

(3) Put the adapter shaft through the hole in the front spar.

SUBTASK 28-22-11-420-011

(4) Install the valve adapter plate [9]:

(a) Make sure that the index marks on the valve body [21] align with the adapter shaft.

NOTE: A small misalignment of the index marks is permitted.

(b) Engage the adapter shaft spline with the spline on the valve body [21].

1) Make sure that the shoulder on the valve adapter plate [9] does not bind in the front spar opening.

(c) Install the four bolts [26] and washers [27] to secure the valve adapter plate [9].

(d) Tighten the bolts [26] to 65 in-lb (7.3 N·m).

SUBTASK 28-22-11-420-012

(5) Apply sealant to the four bolts [26], as follows:

(a) Clean the surface areas to be sealed (TASK 51-22-00-160-801).

(b) Apply sealant, A50153 brushcoat to the surfaces and area around the four bolts [26].

28-AWL-21: CDCCL

(c) In the fuel tank, apply a cap seal with sealant, A50337 to the four bolts [26] (TASK 51-22-00-390-808).

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SUBTASK 28-22-11-420-013

28-AWL-21: CDCCL

(6) On the inside of the tank, apply a fillet seal around the edge of the hole and the stationary part of the valve adapter plate [9] (Fillet Seal Application, TASK 51-22-00-390-806).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Clean the surface areas to be sealed (TASK 51-22-00-160-801).

(b) Apply sealant, A50153 brushcoat to the around the edge of the hole and the stationary part of the valve adapter plate [9].

(c) Make sure that the sealant does not touch the adapter shaft or other parts of the valve adapter plate [9] that turn.

SUBTASK 28-22-11-420-014

(7) On the outside of the tank, apply a fillet seal around the periphery of the valve adapter plate [9] and the front spar (TASK 51-22-00-390-806) (View A-A).

(a) Use sealant, A50110 or sealant, A50155 to apply the fillet seal.

SUBTASK 28-22-11-765-004

28-AWL-21: CDCCL

(8) Measure the electrical bonding resistance between one of the raised serrated surfaces of the valve adapter plate [9] (bare metal surface) and the front spar (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-21: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(c) If the electrical bonding resistance is more than 0.0005 ohm (0.5 milliohm), then do the applicable steps in this task: Engine Fuel Spar Valve Electrical Faying Surface Bonds Repair, TASK 28-22-11-300-801.

SUBTASK 28-22-11-760-006

28-AWL-21: CDCCL

(9) Measure the electrical bonding resistance between the index plate [8] and the valve adapter plate [9] (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

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28-AWL-21: CDCCL

(b) Make sure the electrical bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(c) If the electrical bonding resistance is more than 0.0005 ohm (0.5 milliohm), then do the applicable procedure in this task: Engine Fuel Spar Valve Electrical Faying Surface Bonds Repair, TASK 28-22-11-300-801.

SUBTASK 28-22-11-820-001

(10) Do an alignment of the valve body [21] and adapter shaft (TASK 28-22-11-820-801).

SUBTASK 28-22-11-020-003

(11) Remove the index plate [8] from the valve adapter plate [9].

(a) Remove the lockwires from the two index screws [23] and index screw [24].

(b) Remove the two index screws [23] and flat washers [22].

(c) Remove the index screw [24] and flat washer [22].

(d) Carefully disassemble the index plate [8] from the valve adapter plate [9].

NOTE: The two plates are bonded with faying surface sealant.

SUBTASK 28-22-11-390-004

28-AWL-21: CDCCL

(12) Apply a fay sealed fay surface bond with sealant, A50110 or sealant, A50155 to the serrated mating surfaces on the index plate [8] (three locations) and the valve adapter plate [9] (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

NOTE: Protect the alignment reference mark when you clean the valve adapter plate [9].

SUBTASK 28-22-11-420-015

(13) Do a wet installation of the index screws [23], index screw [24], and flat washers [22].

(a) Apply sealant, A50231 or sealant, A50337 to each side of the three flat washers [22].

(b) Apply sealant to the shank and threads for the two index screws [23] and index screw [24].

1) Use sealant, A50231 or sealant, A50337 for the wet installation of the index bolts.

(c) Apply sealant, A50231 or sealant, A50337 in the holes on the index plate [8] as necessary to create a continuous squeeze-out around the fastener heads.

SUBTASK 28-22-11-420-016

(14) Install the index plate [8]:

(a) Align the reference marks between the index plate [8] and valve adapter plate [9].

(b) Install the two index screws [23] and flat washers [22].

(c) Install the index screw [24] and flat washer [22].

(d) Tighten the two index screws [23] and index screw [24] to 22 ±1 in-lb (2.5 ±0.1 N·m).

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(e) Make sure that the sealant has a continuous squeeze-out along the edges of the contact surfaces.

1) If there are gaps, bubbles or voids in the sealant squeeze out, disassemble and apply more sealant.

(f) Shape the squeezed out sealant into a fillet seal (TASK 51-22-00-390-806).

(g) As an option, remove excess sealant with a dry cotton wiper, G00034.

NOTE: Do not remove the sealant below the index screws.

SUBTASK 28-22-11-765-005

(15) Measure the electrical bonding resistance between one of the mounting points on the index plate [8] and the valve adapter plate [9] (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) If you installed the valve adapter plate [9], make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

(c) If the electrical bonding resistance is more than 0.001 ohm (1.0 milliohm), do the applicable procedure in this task: Engine Fuel Spar Valve Electrical Faying Surface Bonds Repair, TASK 28-22-11-300-801.

(d) If you did not install the valve adapter plate [9], make sure the electrical bonding resistance is 0.002 ohm (2.0 milliohms) or less.

(e) If the electrical bonding resistance is more than 0.002 ohm (2.0 milliohms), do the applicable procedure in this task: Engine Fuel Spar Valve Electrical Faying Surface Bonds Repair, TASK 28-22-11-300-801.

SUBTASK 28-22-11-420-017

(16) Install MS20995NC32 lockwire, G01912 or safety cable kit, G51372 from the two index screws [23] to the index plate [8].

SUBTASK 28-22-11-420-018

(17) Install MS20995NC32 lockwire, G01912 or safety cable kit, G51372 from the index screw [24] to the index plate [8].

SUBTASK 28-22-11-790-001

(18) Install the actuator [4] (TASK 28-22-11-400-801).

SUBTASK 28-22-11-370-001

(19) Apply protective finish Bonderite M-CR 600 Aero coating, C50315 to the bare areas on the front spar (Apply the Corrosion Resistant Finish (Topcoat), TASK 28-11-00-600-801).

NOTE: Apply protective finish to the front spar bare areas not covered by the fillet seal and where the electrical bonding probe removed the finish.

J. Spar Valve Operational Tests

NOTE: The tests that follow will make sure that the engine fuel spar valve electrical operation and index plate [8] alignment are correct, and that no fuel leaks from the front spar.

SUBTASK 28-22-11-860-004

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Install the applicable access doors:

(TASK 28-11-11-400-801).

Number	Name/Location
532BB	Main Tank Access Door - Wing Station 265

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
632BB	Main Tank Access Door - Wing Station 265

SUBTASK 28-22-11-760-007

(2) Do these tests for the applicable engine fuel spar valve:

(a) Engine Fuel Spar Valve - Electrical Control and Indication Test, TASK 28-22-00-710-801.

NOTE: This test is done when you install the actuator [4].

(b) Engine Fuel Spar Valve Installation - Operational Test, TASK 28-22-00-720-802.

SUBTASK 28-22-11-790-002

(3) Do this leak check if you have replaced the valve adapter plate [9]:

NOTE: The leak check is only necessary if you replaced the valve adapter plate [9] or entered the fuel tank. It is not necessary if you have replaced or aligned the index plate [8] for adjustment purposes only.

(a) Refuel the applicable fuel tank to a minimum of 4400 lb (1996 kg) to check for leaks at the front spar (TASK 12-11-00-650-803).

(b) Make sure there are no fuel leaks at the valve adapter plate [9] on the front spar.

K. Put the Airplane Back to the Usual Condition

SUBTASK 28-22-11-410-005

(1) Close the applicable access panel(s):

Number	Name/Location
521BB	Outboard Lower Fixed Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel

——— END OF TASK ———

TASK 28-22-11-820-801

6. Engine Fuel Spar Valve Alignment

(Figure 402, Figure 404)

A. General

(1) This task has two options to align the adapter shaft:

(a) Engine Fuel Spar Valve Alignment (Alignment Tool with Handle)

(b) Engine Fuel Spar Valve Alignment (Alignment Tool with Wrench).

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1587	Wrench - Torque, 30 in-lbs, +/-5% accuracy Part #: 5718A42 Supplier: 3A054 Opt Part #: TE3FUA Supplier: 55719
SPL-1771	Alignment Equipment - Fuel Shutoff Valve Part #: B28009-1 Supplier: 81205

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C. Location Zones

Zone	Area
521	Left Wing - Leading Edge to Front Spar
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
621	Right Wing - Leading Edge to Front Spar
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Prepare for the Procedure

SUBTASK 28-22-11-010-017

(1) Remove the actuator [4] (TASK 28-22-11-000-801).

SUBTASK 28-22-11-020-004

(2) Remove the index plate [8] from the valve adapter plate [9].

(a) Remove the lockwires from the two index screws [23] and index screw [24].

(b) Remove the two index screws [23] and flat washers [22].

(c) Remove the index screw [24] and flat washer [22].

(d) Carefully disassemble the index plate [8] from the adapter plate.

NOTE: The two plates are bonded with faying surface sealant.

SUBTASK 28-22-11-480-001

(3) Install the alignment tool [43] (valve alignment equipment, SPL-1771) in the valve adapter plate [9], as follows:

(a) Position and align the serrated surfaces on the index plate [8] with the valve adapter plate [9].

(b) Loosely install the index screw [24] and flat washer [22] in the index plate [8].

(c) Install the alignment tool [43].

(d) Align the alignment tool [43] guide pins with the index screw holes and attach to the index plate [8].

(e) Make sure that the spline on the alignment tool [43] engages with the shaft spline.

E. Engine Fuel Spar Valve Alignment (Alignment Tool with Handle)

SUBTASK 28-22-11-820-002

(1) To align the adapter shaft on the valve adapter plate [9], do these steps:

NOTE: This procedure finds the point where the internal valve butterfly engages the valve seal. The torque necessary to turn the shaft increases when the butterfly engages the seal and decreases when the butterfly continues to turn. When you find the seal edge, turn the valve shaft to the opposite direction to center the butterfly on the seal.

(a) Align the handle in the counterclockwise position.

(b) Carefully turn the alignment tool handle clockwise until the torque necessary to turn the handle increases.

(c) If you cannot find where the torque increases, move the adapter shaft counterclockwise to its initial position.

(d) Turn the alignment tool [43] clockwise until the toque necessary to turn the handle increases, then immediately stop turning the handle.

(e) If you cannot find where the torque increases, go to the next procedure.

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(f) Read the scale indicator on the alignment tool [43].

NOTE: This is the point where the valve butterfly starts to engage the edge of the seal.

(g) Turn the handle 13 degrees clockwise to center the valve seal.

NOTE: If you turn the handle more than 13 degrees, do not turn the handle counterclockwise. Start the procedure again with the handle in the counterclockwise position.

SUBTASK 28-22-11-480-002

(2) Tighten the lockscrew [42] to keep the alignment tool [43] in its position on the valve adapter plate [9].

NOTE: The lockscrew [42] keeps the adapter shaft calibrated while the index plate [8] is aligned.

SUBTASK 28-22-11-480-003

(3) Turn the index plate [8] until the actuator mounting holes in the index plate [8] align with the alignment screws [41].

SUBTASK 28-22-11-480-004

(4) Install the two alignment screws [41] to secure the index plate [8] to the valve adapter plate [9].

SUBTASK 28-22-11-820-003

(5) Tighten the index screw [24].

SUBTASK 28-22-11-080-001

(6) Disengage the two alignment screws [41].

SUBTASK 28-22-11-080-002

(7) Loosen the lockscrew [42].

SUBTASK 28-22-11-080-003

(8) Carefully remove the alignment tool [43].

SUBTASK 28-22-11-930-001

(9) Use a fine point felt pen to apply a reference mark between the index plate [8] and the valve adapter plate [9] at one of the serrated attachment points.

F. Engine Fuel Spar Valve Alignment (Alignment Tool with Wrench)

SUBTASK 28-22-11-820-004

(1) To align the adapter shaft on the valve adapter plate [9], do these steps:

NOTE: A torque wrench (0-30 lbs), COM-1587 is necessary to do this procedure.

(a) Align the handle in the counterclockwise position.

(b) Put the torque wrench (0-30 lbs), COM-1587 in the base of the handle.

(c) Carefully turn the torque wrench (0-30 lbs), COM-1587, on the alignment tool handle, clockwise until the torque necessary to turn the handle increases.

(d) Write down the maximum torque that you applied, while you turned the torque wrench (0-30 lbs), COM-1587 in the clockwise position.

(e) Align the adapter shaft back to the initial counterclockwise position.

(f) Turn the torque wrench (0-30 lbs), COM-1587 clockwise at approximately the same rate.

(g) Stop at the position where the torque increases to approximately two-thirds of the maximum torque that you wrote.

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(h) Read the scale indicator on the alignment tool [43].

NOTE: This is the point where the valve butterfly starts to engage the edge of the seal.

(i) Turn the torque wrench (0-30 lbs), COM-1587 13 degrees clockwise to center the valve seal.

NOTE: If you turn the handle more than 13 degrees, do not turn the handle counterclockwise. Start the procedure again with the handle in the counterclockwise position.

SUBTASK 28-22-11-480-005

(2) Tighten the lockscrew [42] to keep the alignment tool [43] in its position on the valve adapter plate [9].

NOTE: The lockscrew [42] keeps the adapter shaft calibrated while the index plate [8] is aligned.

SUBTASK 28-22-11-820-005

(3) Turn the index plate [8] until the actuator mounting holes in the index plate [8] align with the alignment screws [41].

SUBTASK 28-22-11-820-006

(4) Install the two alignment screws [41] to secure the index plate [8] to the valve adapter plate [9].

SUBTASK 28-22-11-480-006

(5) Tighten the index screw [24].

SUBTASK 28-22-11-080-004

(6) Disengage the two alignment screws [41].

SUBTASK 28-22-11-080-005

(7) Loosen the lockscrew [42].

SUBTASK 28-22-11-080-006

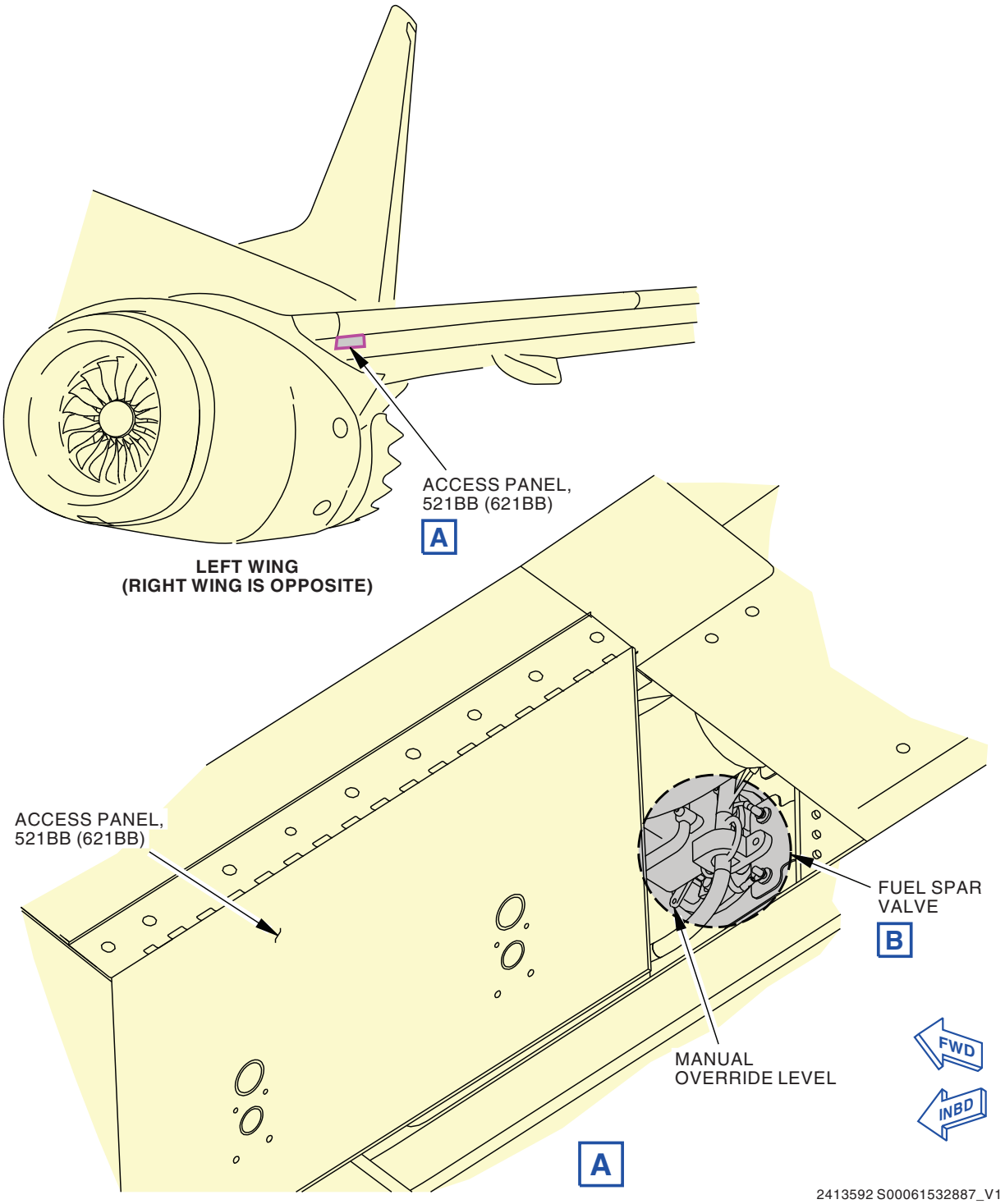
(8) Carefully remove the alignment tool [43].

SUBTASK 28-22-11-930-002

(9) Use a fine point felt pen to apply a reference mark between the index plate [8] and the valve adapter plate [9] at one of the serrated attachment points.

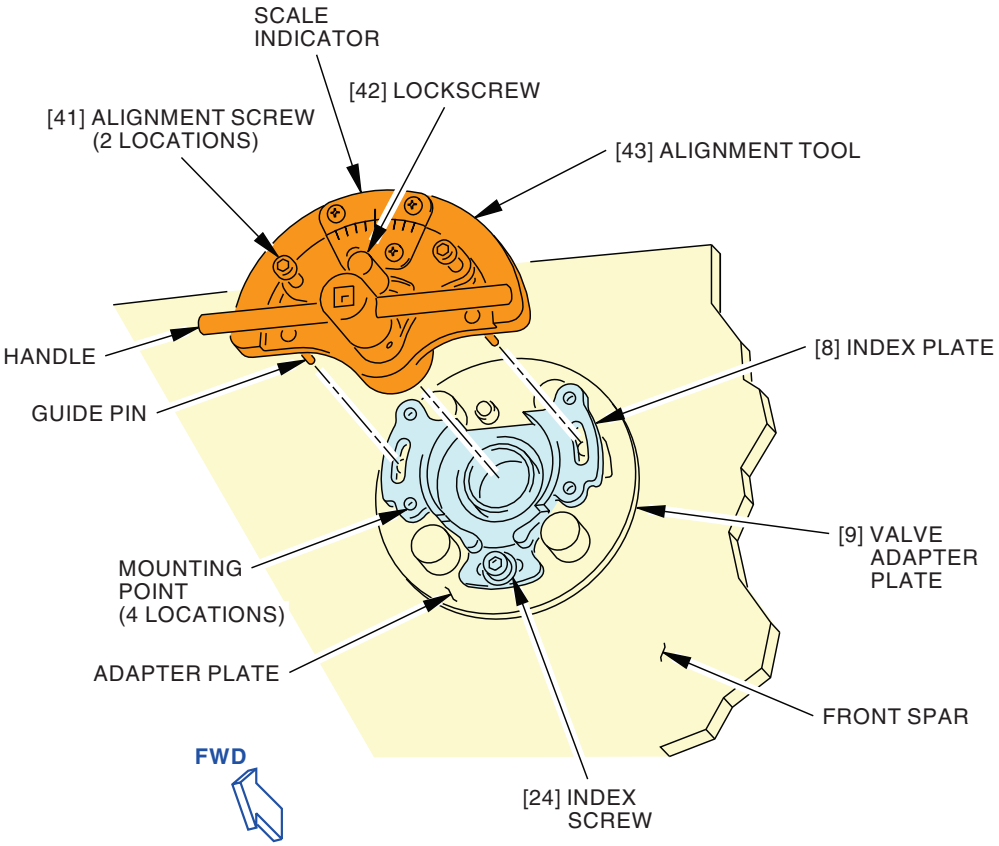
——— END OF TASK ———

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Engine Fuel Spar Valve Alignment
Figure 404/28-22-11-990-803 (Sheet 1 of 2)

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Engine Fuel Spar Valve Alignment
Figure 404/28-22-11-990-803 (Sheet 2 of 2)

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TASK 28-22-11-000-803

7. Engine Fuel Spar Valve - Valve Body Removal

(Figure 405)

A. General

(1) The objective of the task is to provide instructions to remove the valve body.

B. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-22-15-000-803	Couplings Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-26-00-650-802	Tank to Tank Fuel Transfer (P/B 201)
73-11-02-000-801-G00	Fuel Filter Cartridge Removal (P/B 401)

C. Location Zones

Zone	Area
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Access Panels

Number	Name/Location
532BB	Main Tank Access Door - Wing Station 265
632BB	Main Tank Access Door - Wing Station 265

E. Prepare for the Removal

SUBTASK 28-22-11-860-005

(1) On the P8 Aisle Stand, set the Engine Start Levers to the CUTOFF position.

SUBTASK 28-22-11-650-002

(2) Defuel or transfer fuel from the applicable main tank (TASK 28-26-00-650-801 or TASK 28-26-00-650-802).

SUBTASK 28-22-11-860-006

(3) Make sure that the APU master switch, on the P5 Overhead Panel, is in the OFF position.

SUBTASK 28-22-11-860-007

(4) Do these steps to make sure that the engine fuel valves are in the correct positions:

NOTE: There are two sets of engine fuel valves. The first set are the high pressure shutoff valves (HPSOV) which are installed on the engines. The second set are the spar valves which are installed in the main fuel tanks. For this test, the applicable HPSOV is closed when the applicable engine fuel spar valve is open.

(a) Make sure that Engine Start Levers are in the CUTOFF position.

(b) Make sure that ENGINE START 1 and 2 switches are in the OFF position on the P5 Overhead Panel.

(c) Put a DO-NOT-OPERATE placard on the ENGINE START 1 and 2 switches.

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(d) Open the applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

(e) Make sure that the fuel crossfeed VALVE OPEN light is off, on the P5-2 Fuel System Module.

(f) Set the applicable Engine Start Lever to the IDLE position.

NOTE: This opens the applicable engine fuel spar valve.

(g) Make sure that the applicable SPAR VALVE CLOSED light goes off, on the P5-2 Fuel System Module.

SUBTASK 28-22-11-860-008

(5) Make sure that the defueling valve handle is in the CLOSED position.

SUBTASK 28-22-11-860-009

WARNING

REMOVE ELECTRICAL POWER BEFORE YOU DISCONNECT THE FUEL, HYDRAULIC, AND ELECTRICAL LINES. INJURIES TO PERSONNEL CAN OCCUR.

(6) Make sure that electrical power has been removed (TASK 24-22-00-860-802).

SUBTASK 28-22-11-650-003

(7) Drain fuel from the fuel supply line (TASK 73-11-02-000-801-G00).

SUBTASK 28-22-11-860-010

(8) Manually turn the manual override lever on the actuator [4] to the CLOSED position.

(a) On the P8 Aisle Stand, set the Engine Start Levers to the CUTOFF position.

SUBTASK 28-22-11-650-004

WARNING

OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(9) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove the applicable access doors:

(TASK 28-11-11-000-801)

Number	Name/Location
532BB	Main Tank Access Door - Wing Station 265
632BB	Main Tank Access Door - Wing Station 265

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SUBTASK 28-22-11-860-011

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(10) Go into the fuel tank (TASK 28-11-00-910-802).

F. Valve Body Removal

SUBTASK 28-22-11-020-005

(1) Disconnect the clamshell coupling [52] (TASK 28-22-15-000-803).

(a) Discard the two O-rings [53].

SUBTASK 28-22-11-020-006

(2) Disconnect the clamshell coupling [54] (TASK 28-22-15-000-803).

(a) Discard the two O-rings [56].

SUBTASK 28-22-11-020-007

(3) Hold the valve body [21] and remove the four screws [58] and washers [57].

SUBTASK 28-22-11-020-008

(4) Disengage the valve body [21] from the adapter shaft.

(a) Do not let the adapter shaft fall and cause damage.

NOTE: Temporarily attach the adapter shaft to the structure or put soft material on or below the adapter shaft.

SUBTASK 28-22-11-020-009

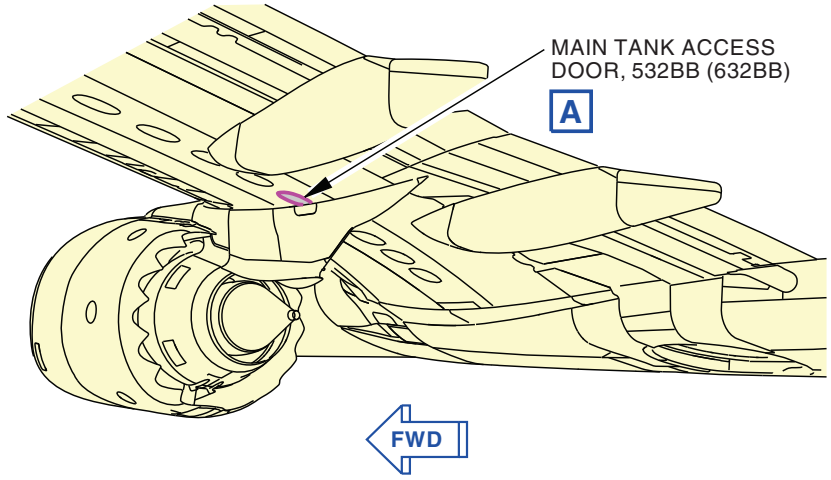
(5) Remove the valve body [21].

SUBTASK 28-22-11-480-007

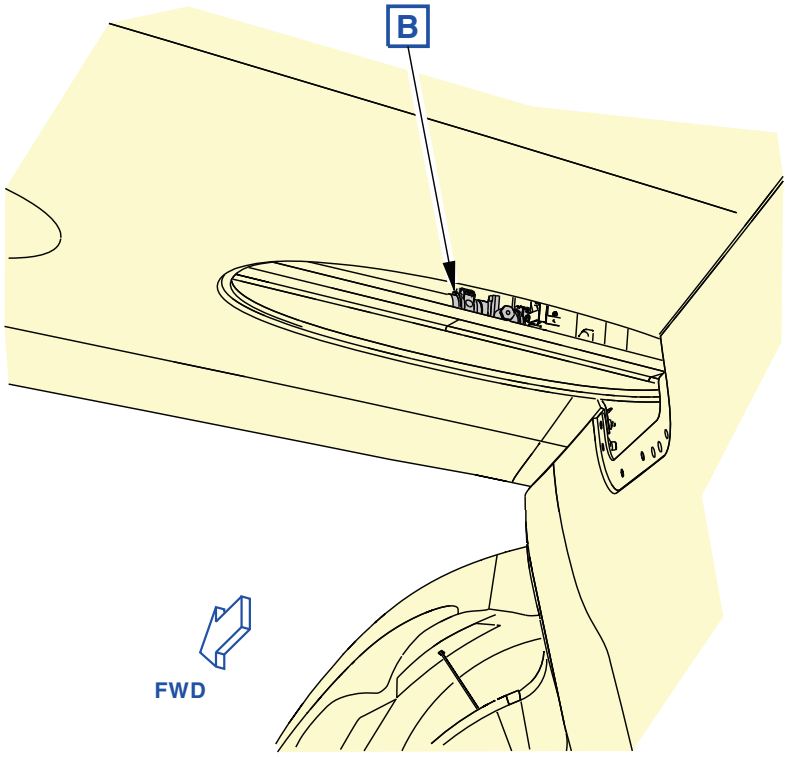
(6) Put covers on the open ends of the engine fuel line [51] and engine fuel line [55] to keep unwanted materials out.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS EQUIVALENT)



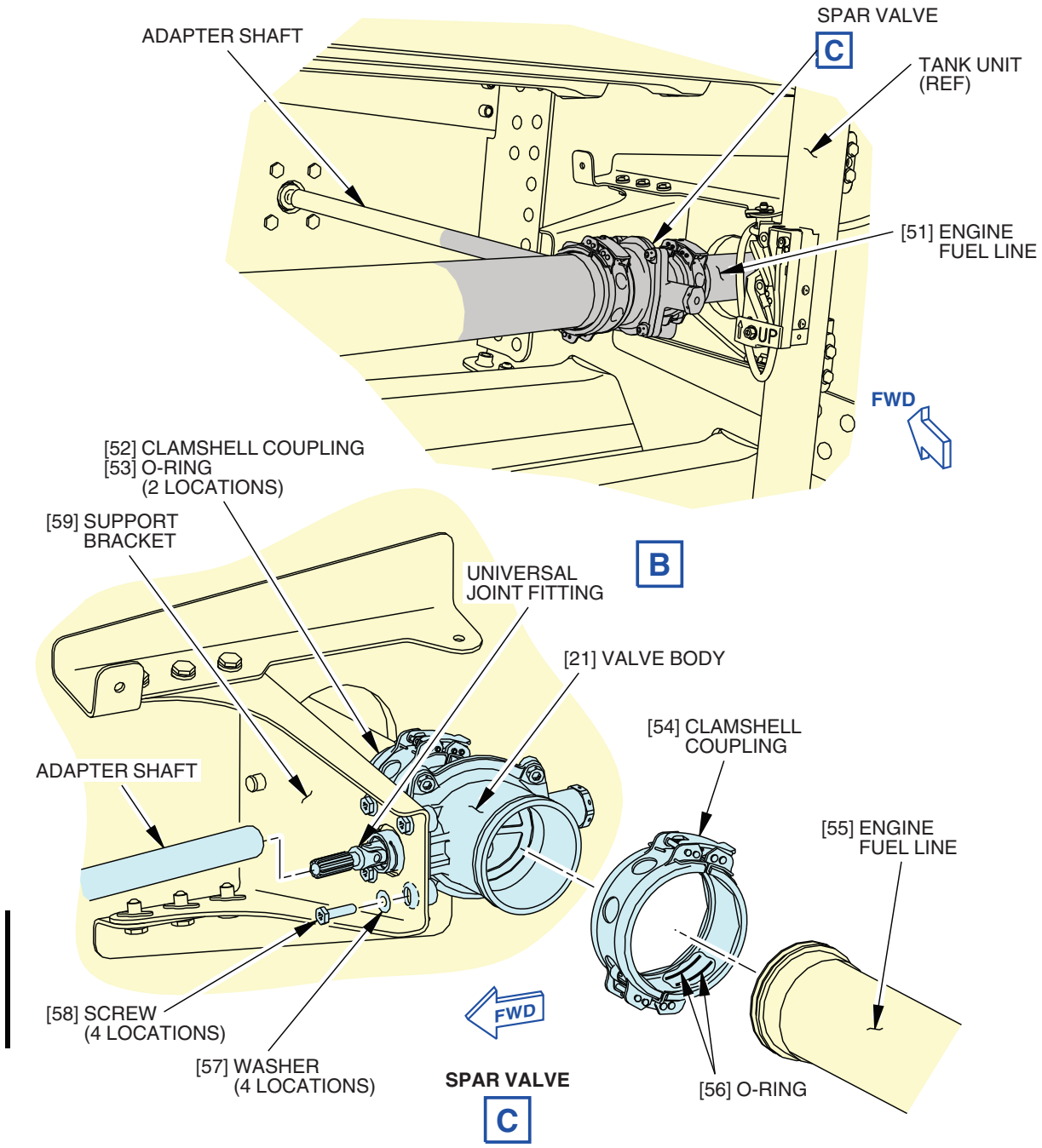
MAIN TANK ACCESS DOOR, 532BB (632BB)

A

2413587 S00061532880_V2

Valve Body Installation
Figure 405/28-22-11-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413594 S00061532890_V3

Valve Body Installation
Figure 405/28-22-11-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-11-400-803

8. Engine Fuel Spar Valve - Valve Body Installation

(Figure 405)

A. General

(1) The objective of the task is to provide instructions to install the valve body.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
28-22-00-710-802	Engine Fuel Suction Feed - Operational Test (P/B 501)
28-22-00-720-802	Engine Fuel Spar Valve Installation - Operational Test (P/B 501)
28-22-15-400-803	Couplings Installation (P/B 401)
28-22-15-710-801	Engine Fuel Feed Manifold - Leak Test (P/B 601)
71-00-00-910-808-G00	Dry Motor Procedure (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C50475	Coating - Chemical Conversion - Bonderite M-CR 600 Aero	BAC5719 Type II Class D

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G51293	Cotton Wiper - Knit, Process Cleaning Absorbent Wiper	BMS15-5 Class B

E. Location Zones

Zone	Area
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

F. Access Panels

Number	Name/Location
532BB	Main Tank Access Door - Wing Station 265
632BB	Main Tank Access Door - Wing Station 265

G. Prepare for the Installation

SUBTASK 28-22-11-860-012

(1) Make sure that the manual override lever on the actuator [4] is in the CLOSED position.

SUBTASK 28-22-11-080-007

(2) Remove the covers from the engine fuel lines [51] and engine fuel line [55].

SUBTASK 28-22-11-110-002

(3) Clean the fay surfaces of the aluminum valve body [21] and the aluminum support bracket [59] (SWPM 20-20-00):

(a) Use Cleaning Procedure 5 to clean the fay surfaces of the valve body [21] and the support bracket [59].

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

1) Use a new cotton wiper, G00034 or knit cotton wiper, G51293 soaked with solvent, B00062 or solvent, B00184.

2) Use a new cotton wiper to dry the surfaces.

3) Continue to clean and dry the surface until the cotton wiper shows clean.

H. Valve Body Installation

SUBTASK 28-22-11-910-002

(1) Apply a layer of sealant, A50110, or sealant, A50155, to the fay surface of the valve body [21] (SWPM 20-20-00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-11-420-019

(2) Turn the universal joint fitting, on the valve body [21], until the index mark aligns with the adapter shaft.

NOTE: A small misalignment of the index marks is satisfactory.

SUBTASK 28-22-11-420-020

(3) Align the valve body [21] with the adapter shaft.

SUBTASK 28-22-11-210-001

(4) Make sure that TOP shows on the top of the valve body [21].

SUBTASK 28-22-11-080-008

(5) Remove items used for protection of the adapter shaft.

(a) Do not let the adapter shaft fall and cause damage.

SUBTASK 28-22-11-420-021

(6) Engage the universal joint fitting on the valve body [21] into the adapter shaft.

SUBTASK 28-22-11-820-007

(7) Align the valve body [21] with the support bracket [59].

SUBTASK 28-22-11-420-022

(8) Install the four screws [58] and washers [57].

SUBTASK 28-22-11-220-001

(9) Measure the bonding resistance between the valve body [21] and the support bracket [59] with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) Make sure that the measured resistance is below the maximum bonding resistance.

SUBTASK 28-22-11-910-003

(10) Apply coating, C00064 or Bonderite M-CR 600 Aero coating, C50475 to the abraded surfaces of the valve body [21] and the support bracket [59].

SUBTASK 28-22-11-420-023

(11) Connect the clamshell coupling [52] between the valve body [21] and the engine fuel lines [51] (TASK 28-22-15-400-803).

(a) Install the two O-rings [53].

SUBTASK 28-22-11-420-024

(12) Connect the clamshell coupling [54] between the valve body [21] and the engine fuel line [55] (TASK 28-22-15-400-803).

(a) Install the two O-rings [56].

SUBTASK 28-22-11-860-013

(13) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Close these access doors:

(TASK 28-11-11-400-801)

Number	Name/Location
532BB	Main Tank Access Door - Wing Station 265
632BB	Main Tank Access Door - Wing Station 265

I. Valve Body Installation Test

SUBTASK 28-22-11-860-014

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-11-860-015

(2) Remove the applicable safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 28-22-11-710-002

(3) Do one of these tests:

(a) Do a leak test of the engine fuel feed manifold.

1) Do this task: Engine Fuel Feed Manifold - Leak Test, TASK 28-22-15-710-801.

2) On the P5 Overhead Panel, remove the DO-NOT-OPERATE placards from the ENGINE START 1 and 2 switches.

3) Do this task: Engine Fuel Spar Valve Installation - Operational Test, TASK 28-22-00-720-802.

CAUTION

DO NOT MOTOR THE ENGINE BEFORE VERIFYING THAT THE FUEL SPAR VALVE IS IN THE OPEN POSITION AND FUEL BOOST PUMP PRESSURE IS APPLIED TO THE FUEL PUMP INLET. THE FUEL PUMP AND THE HYDRO MECHANICAL UNIT ARE FUEL LUBRICATED, ZERO FUEL PRESSURE CAN CAUSE DAMAGE TO THE FUEL PUMP AND THE HYDRO MECHANICAL UNIT.

4) If it is necessary to dry motor the engine, apply the boost pump pressure to the fuel pump inlet (TASK 71-00-00-910-808-G00).

(b) Do an operational test for the engine fuel suction feed system.

NOTE: When you do this test, you must first do a leak check of the engine fuel spar valve that was replaced.

1) On the P5 Overhead Panel, remove the DO-NOT-OPERATE placards from the ENGINE START 1 and 2 switches.

2) Do this task: Engine Fuel Spar Valve Installation - Operational Test, TASK 28-22-00-720-802.

3) Do this task: Engine Fuel Suction Feed - Operational Test, TASK 28-22-00-710-802.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-11-650-005

(1) Refuel the applicable main tank (TASK 12-11-00-650-803).

SUBTASK 28-22-11-860-016

(2) Remove electrical power when necessary (TASK 24-22-00-860-802).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FUEL SPAR VALVE - INSPECTION/CHECK

1. General

A. This procedure has this task for the engine fuel spar valve system:

(1) Engine Fuel Spar Valve Actuator - Bonding Resistance Check.

TASK 28-22-11-200-801

2. Engine Fuel Spar Valve Actuator - Bonding Resistance Check

(Figure 601)

A. General

(1) The engine fuel spar valve actuator is referred to as the actuator in this task.

B. References

Reference	Title
28-22-00-710-801	Engine Fuel Spar Valve - Electrical Control and Indication Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-10	Replacement of Ground Studs and Bonding Jumper Installation

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Location Zones

Zone	Area
521	Left Wing - Leading Edge to Front Spar
621	Right Wing - Leading Edge to Front Spar

E. Access Panels

Number	Name/Location
521BB	Outboard Lower Fixed Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Bonding Resistance Check

SUBTASK 28-22-11-865-003

(1) For the left spar valve, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1
B	5	C00540	FUEL SPAR VALVE IND

SUBTASK 28-22-11-865-004

(2) For the right spar valve, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	5	C00540	FUEL SPAR VALVE IND

SUBTASK 28-22-11-010-018

(3) Open the applicable access panels:

Number	Name/Location
521BB	Outboard Lower Fixed Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel

G. Electrical Bonding Measurement

SUBTASK 28-22-11-710-003

(1) On the actuator, move the manual override lever to the CLOSED position.

SUBTASK 28-22-11-862-003

(2) Disconnect the electrical connector from the actuator.

SUBTASK 28-22-11-765-006

(3) Measure the electrical bonding resistance between the actuator (at the electrical connector flange) and the front spar with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00 and SWPM 20-20-10).

(a) Make sure that the electrical bonding resistance is 0.005 ohm (5.0 milliohms) or less.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-11-861-002

(1) Connect the electrical connector to the actuator.

SUBTASK 28-22-11-410-006

(2) Close the applicable access panels:

Number	Name/Location
521BB	Outboard Lower Fixed Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel

SUBTASK 28-22-11-865-005

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-3

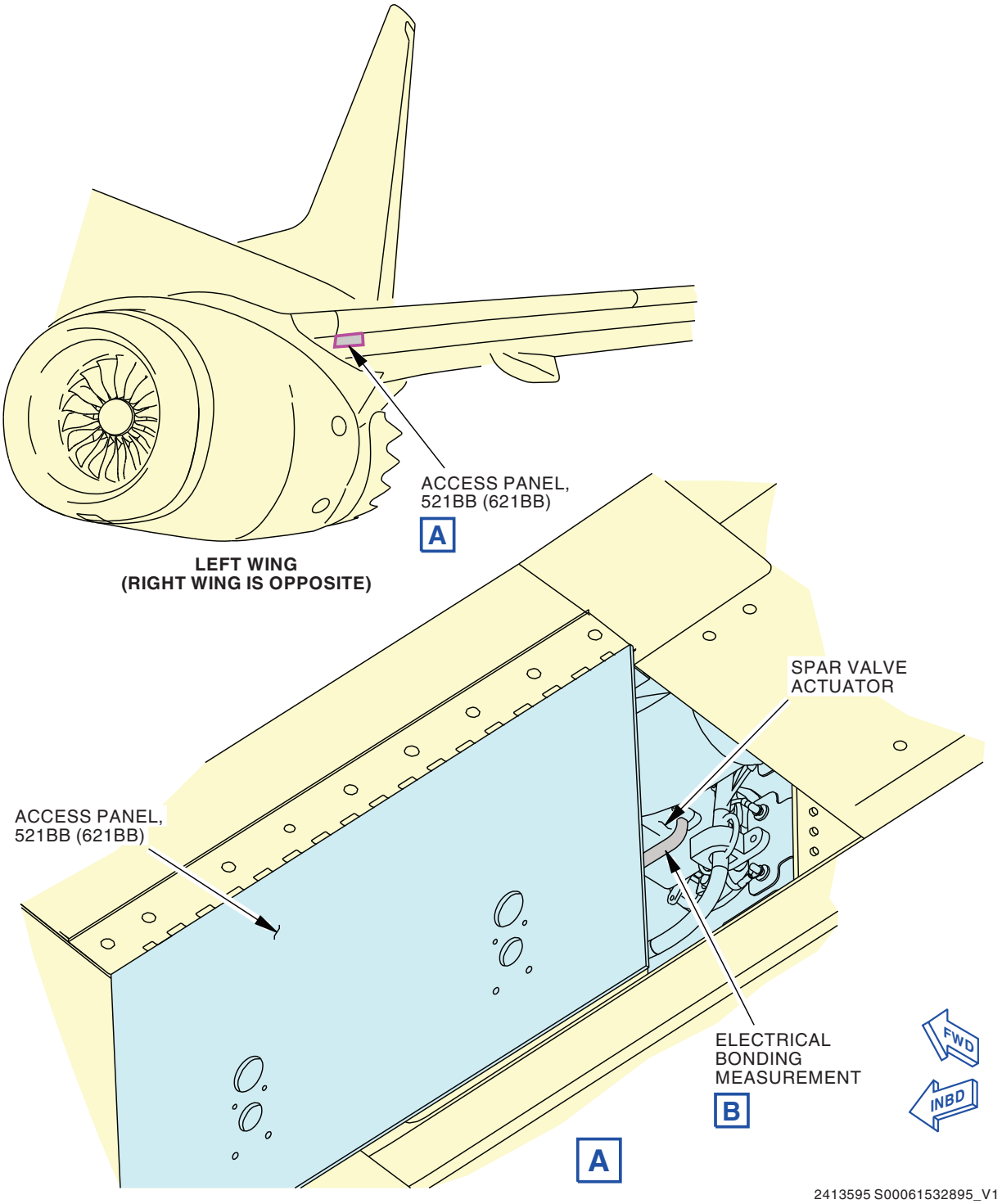
Row	Col	Number	Name
B	5	C00540	FUEL SPAR VALVE IND

SUBTASK 28-22-11-760-008

(4) Do this task: Engine Fuel Spar Valve - Electrical Control and Indication Test, TASK 28-22-00-710-801.

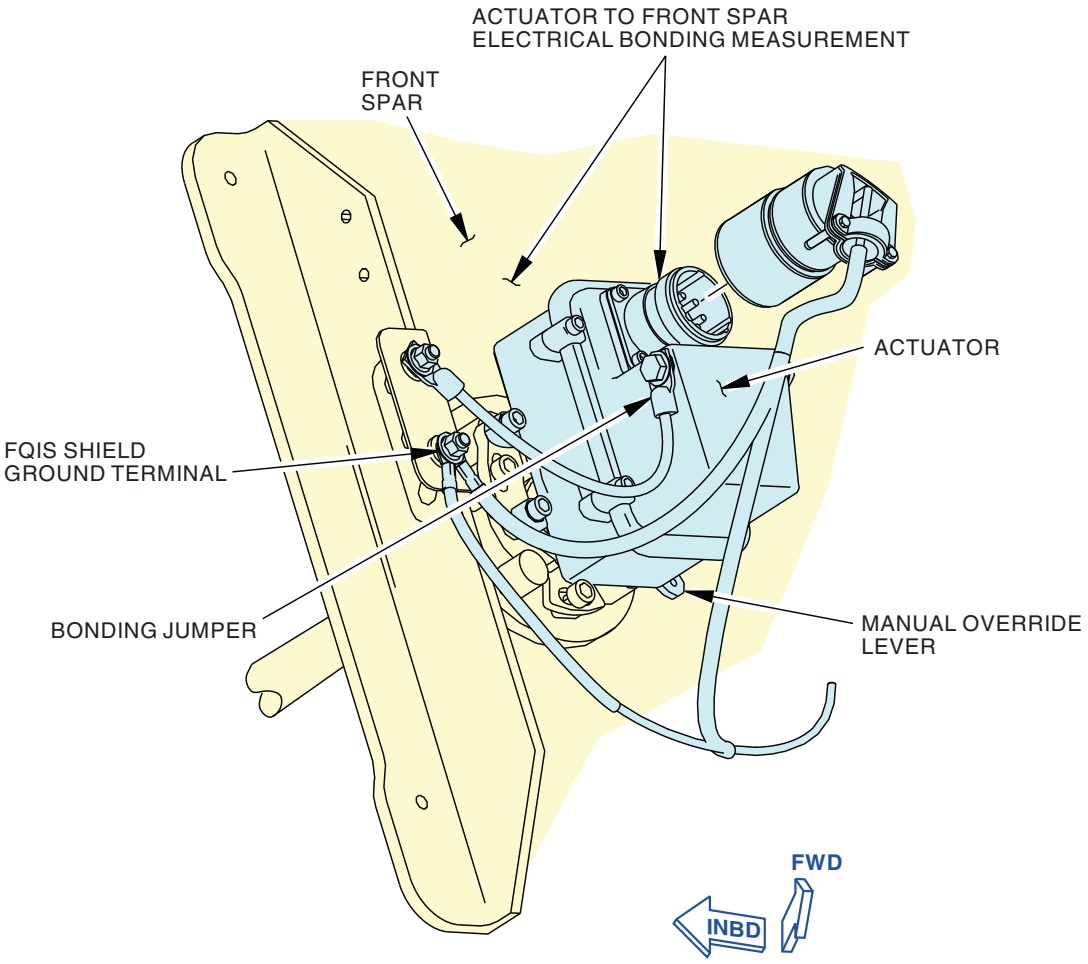
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fuel Spar Valve Actuator - Bonding Resistance Check
Figure 601/28-22-11-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELECTRICAL BONDING MEASUREMENT

B

2413596 S00061532896_V1

Engine Fuel Spar Valve Actuator - Bonding Resistance Check
Figure 601/28-22-11-990-806 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER SCAVENGE JET PUMP - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the Center Tank Water Scavenge Jet Pump Nozzle Assembly

(2) An installation of the Center Tank Water Scavenge Jet Pump Nozzle Assembly

(3) A removal of the Main Tank Water Scavenge Jet Pump Nozzle Assembly

(4) An installation of the Main Tank Water Scavenge Jet Pump Nozzle Assembly

(5) A removal of the Center Tank Water Scavenge Jet Pump

(6) An installation of the Center Tank Water Scavenge Jet Pump

(7) A removal of the Main Tank Water Scavenge Jet Pump

(8) An installation of the Main Tank Water Scavenge Jet Pump.

B. There is one water scavenge jet pump in each main tank.

C. There are two water scavenge jet pumps in the center tank.

TASK 28-22-13-000-801

2. Center Tank Water Scavenge Jet Pump Nozzle Assembly Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the center tank water scavenge jet pump nozzle assembly.

(2) The center tank water scavenge jet pump nozzle assembly is referred to as the nozzle assembly in this task.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-6646	Container - Fuel Resistant

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Removal

SUBTASK 28-22-13-860-001

(1) On the P5-2 Fuel System Module, set the FUEL PUMPS CTR L and R switches to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-13-860-022

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-22-13-010-014

(3) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-13-860-002

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-13-480-001

(5) Put a fuel resistant container, STD-6646, below the nozzle assembly [4].

G. Center Tank Water Scavenge Jet Pump Nozzle Assembly Removal

SUBTASK 28-22-13-020-002

(1) Remove the safety cable [1] (or lockwire) from the nozzle assembly [4].

SUBTASK 28-22-13-010-001

(2) Remove the nozzle assembly [4]:

NOTE: The nozzle assembly consists of a pump nozzle and nozzle plug that are removed as one unit.

(a) Remove the nozzle assembly [4] and O-ring [3] from the pump housing [2].

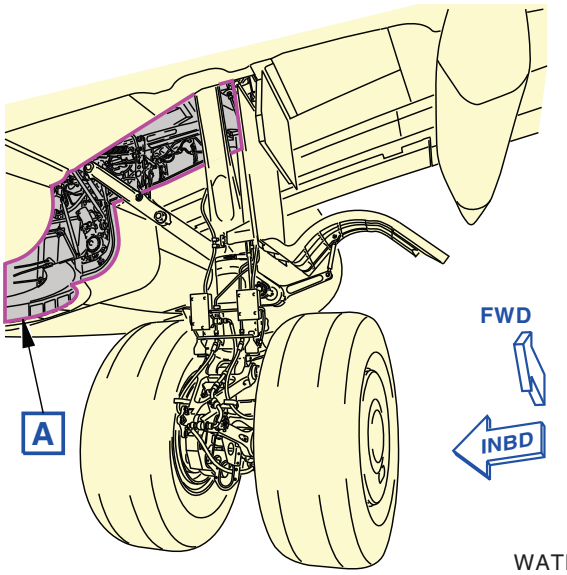
NOTE: The internal check valve in the pump housing closes to prevent fuel spills.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

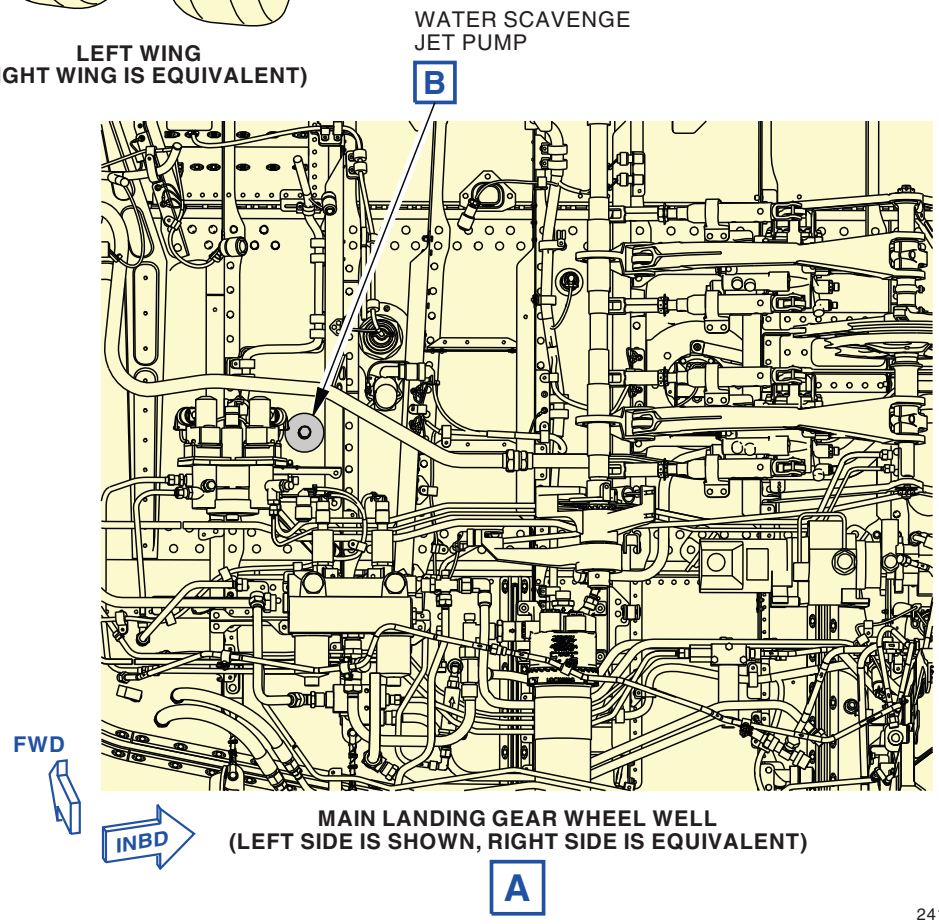
(b) Discard the O-ring [3].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS EQUIVALENT)

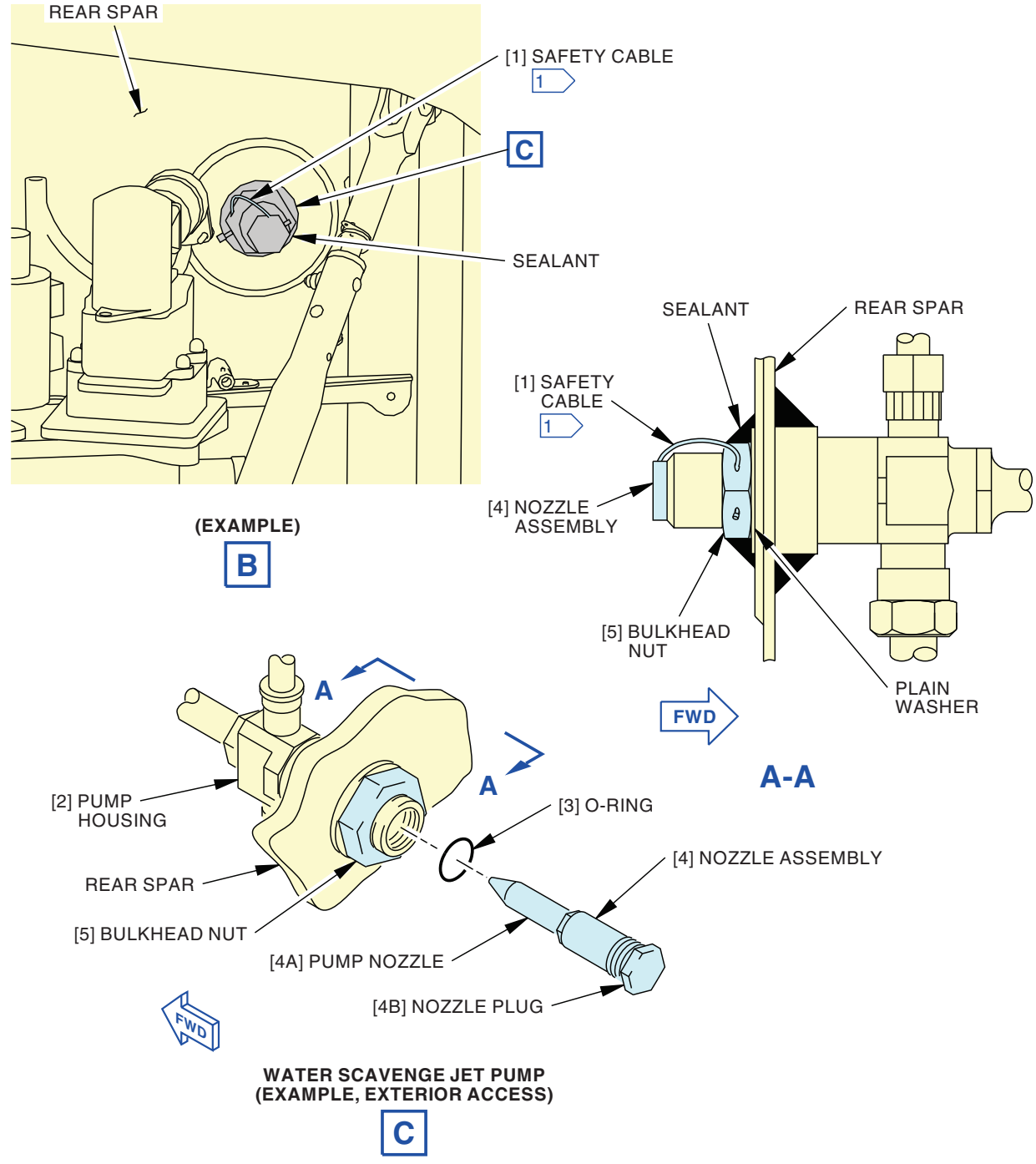


MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

2413597 S00061532900_V1

Center Tank Water Scavenge Jet Pump Nozzle Assembly Installation
Figure 401/28-22-13-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 SAFETY CABLE PREFERRED, LOCKWIRE ALTERNATE

2413598 S00061532901_V2

Center Tank Water Scavenge Jet Pump Nozzle Assembly Installation
Figure 401/28-22-13-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-13-400-801

3. Center Tank Water Scavenge Jet Pump Nozzle Assembly Installation

(Figure 401)

A. General

(1) This task gives instructions to install the center tank water scavenge jet pump nozzle assembly.

(2) The center tank water scavenge jet pump nozzle assembly is referred to as the nozzle assembly in this task.

B. References

Reference	Title
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
51-22-00-390-806	Fillet Seal Application (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B50226	Solvent - Citra-Safe	BAC5750
D00504	Grease - Petrolatum	VV-P-236
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51372	Kit - Safety Cable, 321 CRES - 0.035 +/-0.003 Inch (1.0 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	O-ring	28-22-13-01-097	PTF ALL
4A	Pump nozzle	28-22-13-01-098	PTF ALL
		28-22-13-05-158	PTF ALL
		28-22-13-06-158	PTF ALL
4B	Nozzle plug	28-22-13-01-096	PTF ALL
		28-22-13-05-156	PTF ALL
		28-22-13-06-156	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

G. Prepare for the Installation

SUBTASK 28-22-13-100-001

(1) Do these steps to prepare the nozzle assembly [4] and the rear spar for installation:

(a) Remove the sealant from the rear spar, do this task: Clean Bare, Clad, or Plated Metal with Solvent, TASK 20-10-34-110-802.

(b) Clean the bulkhead nut and the fay surfaces on the aluminum pump housing [2].

1) Clean the surface beyond the area to be sealed to prevent contamination.

2) Use a new cotton wiper, G00034, made wet a citra-safe solvent, B50226 or solvent, B00184.

3) Dry the area with a clean cotton wiper, G00034.

4) Continue to clean and dry the surface until the cotton wiper shows clean.

SUBTASK 28-22-13-211-005

(2) Examine the pump housing [2] for these conditions:

(a) pump nozzle [4A]

1) Nicks or scratches on the conical taper

2) Nicks or scratches on the orifice tip

3) Worn surface areas through the anodize finish.

(b) nozzle plug [4B]

1) Crossed, stripped or damaged threads

2) Nicks or scratches on the packing groove

3) Wear on the hex head

4) Worn surface areas through the anodize finish.

SUBTASK 28-22-13-900-001

(3) If the nozzle plug [4B] is damaged or worn, replace it.

(a) Remove the pump nozzle [4A] from the nozzle plug [4B].

(b) Screw the new pump nozzle [4A] in the nozzle plug [4B].

(c) Tighten the nozzle plug [4B] to 71 ±4 in-lb (8.0 ±0.5 N·m).

SUBTASK 28-22-13-900-002

(4) If the pump nozzle [4A] is damaged or worn, replace it.

(a) Remove the pump nozzle [4A] from the nozzle plug [4B].

(b) Screw the new pump nozzle [4A] in the nozzle plug [4B].

(c) Tighten the nozzle plug [4B] to 71 ±4 in-lb (8.0 ±0.5 N·m).

SUBTASK 28-22-13-640-001

(5) Apply a thin layer of grease, D00504 to the new O-ring [3].

SUBTASK 28-22-13-420-005

(6) Install the O-ring [3] on the nozzle assembly [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Center Tank Water Scavenge Jet Pump Nozzle Assembly Installation

SUBTASK 28-22-13-420-006

(1) Install the nozzle assembly [4] in the pump housing [2].

SUBTASK 28-22-13-420-007

(2) Tighten the nozzle assembly [4] to 160 ±5 in-lb (18.1 ±0.6 N·m).

SUBTASK 28-22-13-420-008

(3) Install safety cable [1] between the nozzle assembly [4] and bulkhead nut [5] (TASK 20-10-44-400-801).

NOTE: The safety cable kit, G51372 is preferred to MS20995NC32 lockwire, G50347 for the safety cable.

SUBTASK 28-22-13-390-001

(4) Apply sealant, A02315 between the rear spar and nozzle assembly [4] (TASK 51-22-00-390-806).

I. Center Tank Water Scavenge Jet Pump Nozzle Assembly Installation Test

SUBTASK 28-22-13-860-015

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-13-410-015

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-13-860-016

(3) On the P5-2 Fuel System Module, set the applicable FUEL PUMPS CTR L or R switch to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-13-210-003

(4) Look for leakage at the pump housing [2].

(a) If leakage occurs, repeat this procedure.

SUBTASK 28-22-13-860-017

(5) Set the FUEL PUMPS CTR L or R switch to the OFF position.

——— END OF TASK ———

TASK 28-22-13-000-802

4. Main Tank Water Scavenge Jet Pump Nozzle Assembly Removal

(Figure 402)

A. General

(1) This task gives instructions to remove the main tank water scavenge jet pump nozzle assembly.

(2) The main tank water scavenge jet pump nozzle assembly is referred to as the nozzle assembly in this task.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

E. Prepare for the Removal

SUBTASK 28-22-13-860-003

(1) On the P5-2 Fuel System Module, set these switches to the OFF position:

(a) FUEL PUMPS 1 AFT

(b) FUEL PUMPS 1 FWD

(c) FUEL PUMPS 2 AFT

(d) FUEL PUMPS 2 FWD.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-13-480-006

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-22-13-010-015

(3) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-13-860-004

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Open the applicable circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT

SUBTASK 28-22-13-010-009

(5) Open the applicable access panels:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

F. Main Tank Water Scavenge Jet Pump Nozzle Assembly Removal

SUBTASK 28-22-13-020-003

(1) Remove the safety cable [1] (or lockwire) from the nozzle assembly [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-13-010-002

(2) Remove the nozzle assembly [4]:

NOTE: The nozzle assembly consists of a pump nozzle and nozzle plug that are removed as one unit.

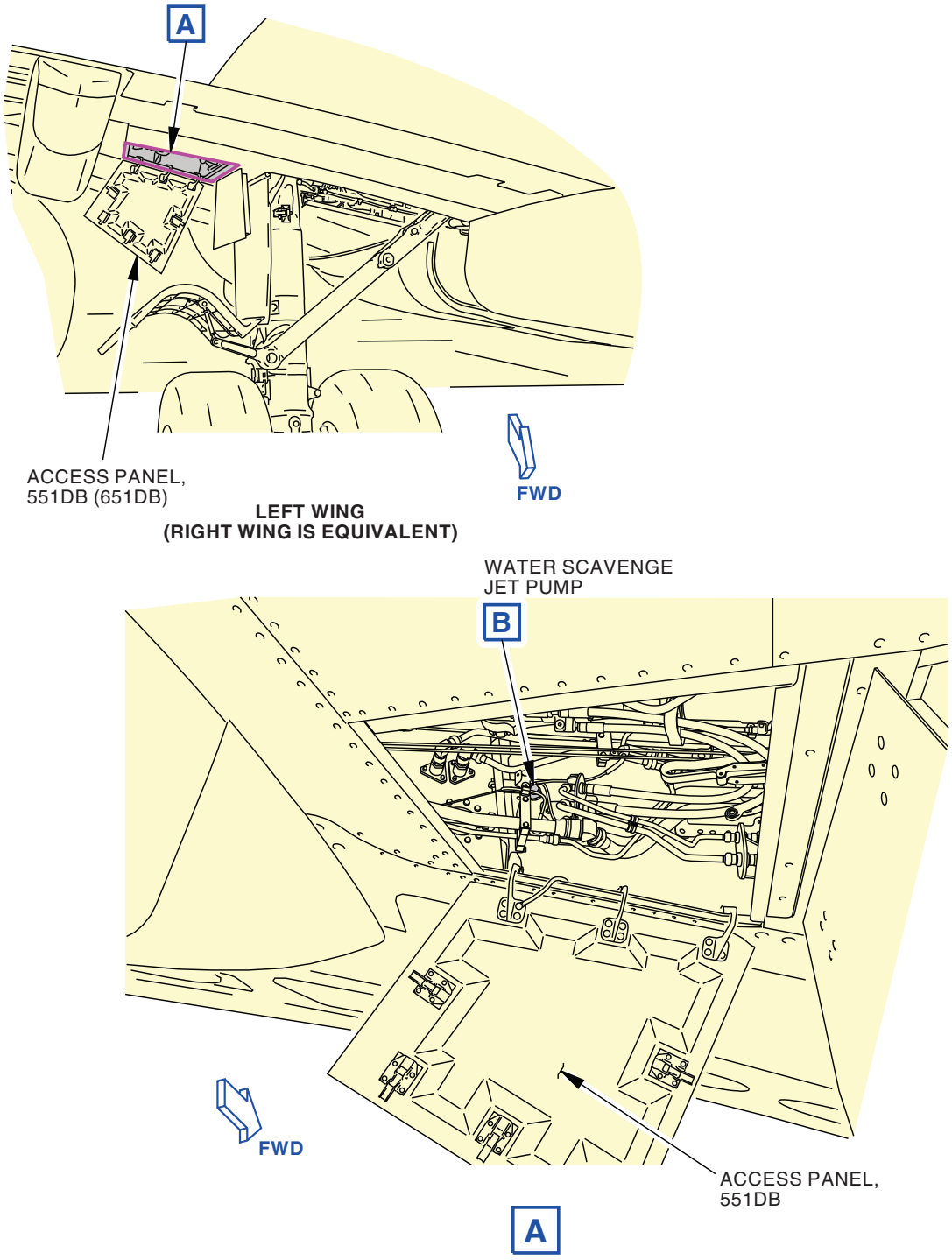
(a) Remove the nozzle assembly [4] and O-ring [5] from the pump housing [2].

NOTE: An internal check valve in the pump housing closes to prevent fuel spills.

(b) Discard the O-ring [5].

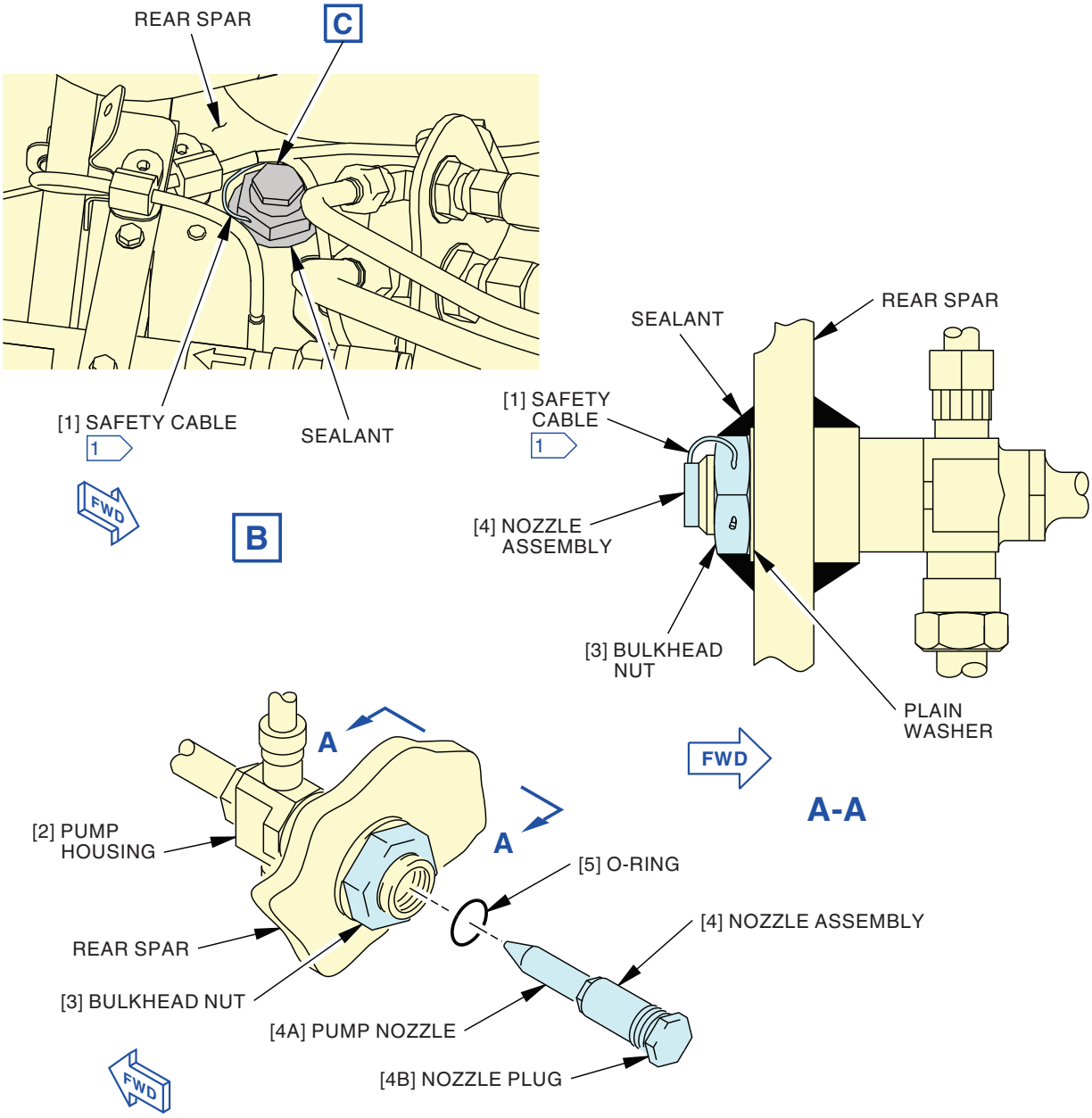
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Tank Water Scavenge Jet Pump Nozzle Assembly Installation
Figure 402/28-22-13-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



WATER SCAVENGE JET PUMP (EXAMPLE, EXTERIOR ACCESS)

C

1 SAFETY CABLE PREFERRED, LOCKWIRE ALTERNATE

2413600 S00061532905_V2

Main Tank Water Scavenge Jet Pump Nozzle Assembly Installation
Figure 402/28-22-13-990-802 (Sheet 2 of 2)

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TASK 28-22-13-400-802

5. Main Tank Water Scavenge Jet Pump Nozzle Assembly Installation

(Figure 402)

A. General

(1) This task gives instructions to install the main tank water scavenge jet pump nozzle assembly.

(2) The main tank water scavenge jet pump nozzle assembly is referred to as the nozzle assembly in this task.

B. References

Reference	Title
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
20-10-44-400-802	Safety Cable Installation (P/B 401)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
51-22-00-390-806	Fillet Seal Application (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B50226	Solvent - Citra-Safe	BAC5750
D00504	Grease - Petrolatum	VV-P-236
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51372	Kit - Safety Cable, 321 CRES - 0.035 +/-0.003 Inch (1.0 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4A	Pump nozzle	28-22-13-01-098	PTF ALL
		28-22-13-05-158	PTF ALL
		28-22-13-06-158	PTF ALL
4B	Nozzle plug	28-22-13-01-096	PTF ALL
		28-22-13-05-156	PTF ALL
		28-22-13-06-156	PTF ALL
5	O-ring	28-22-13-05-157	PTF ALL
		28-22-13-06-157	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
551	Left Wing - Rear Spar To Landing Gear Support Beam

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
651	Right Wing - Rear Spar to Landing Gear Support Beam

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

G. Prepare for the Installation

SUBTASK 28-22-13-100-002

(1) Do these steps to prepare the nozzle assembly [4] and rear spar for installation:

(a) Remove the sealant from the rear spar, do this task: Hand Clean Metal Surfaces with Abrasives, TASK 20-10-34-120-801.

(b) Clean the bulkhead nut [3] and the fay surfaces on the aluminum pump housing [2].

1) Clean the surface beyond the area to be sealed to prevent contamination.

2) Use a new cotton wiper, G00034, made wet with citra-safe solvent, B50226 or solvent, B00184.

3) Dry the area with a clean cotton wiper, G00034.

4) Continue to clean and dry the surface until the cotton wiper shows clean.

SUBTASK 28-22-13-211-006

(2) Examine the nozzle assembly [4] for these conditions:

(a) pump nozzle [4A]

1) Nicks or scratches on the conical taper

2) Nicks or scratches on the orifice tip

3) Worn surface areas through the anodize finish.

(b) nozzle plug [4B]

1) Crossed, stripped or damaged threads

2) Nicks or scratches on the packing groove

3) Wear on the hex head

4) Worn surface areas through the anodize finish.

SUBTASK 28-22-13-900-003

(3) If the nozzle plug [4B] is damaged or worn, replace it.

(a) Remove the pump nozzle [4A] from the nozzle plug [4B].

(b) Screw the new pump nozzle [4A] in the nozzle plug [4B].

(c) Tighten the nozzle plug [4B] to 71 ±4 in-lb (8.0 ±0.5 N·m).

SUBTASK 28-22-13-900-004

(4) If the pump nozzle [4A] is damaged or worn, replace it.

(a) Remove the pump nozzle [4A] from the nozzle plug [4B].

(b) Screw the new pump nozzle [4A] in the nozzle plug [4B].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Tighten the nozzle plug [4B] to 71 ±4 in-lb (8.0 ±0.5 N·m).

SUBTASK 28-22-13-640-002

(5) Apply a thin layer of grease, D00504 to the new O-ring [5].

SUBTASK 28-22-13-420-009

(6) Install the O-ring [5] on the nozzle assembly [4].

H. Main Tank Water Scavenge Jet Pump Nozzle Assembly Installation

SUBTASK 28-22-13-420-010

(1) Install the nozzle assembly [4] in the pump housing [2].

SUBTASK 28-22-13-420-011

(2) Tighten the nozzle assembly [4] to 160 ±5 in-lb (18.1 ±0.6 N·m).

SUBTASK 28-22-13-420-012

(3) Install a safety cable [1] between the nozzle assembly [4] and the bulkhead nut [3] (TASK 20-10-44-400-802).

NOTE: The safety cable kit, G51372 is preferred to MS20995NC32 lockwire, G50347 for the safety cable.

SUBTASK 28-22-13-390-002

(4) Apply sealant, A02315 between the rear spar and nozzle assembly [4] (TASK 51-22-00-390-806).

I. Main Tank Water Scavenge Jet Pump Nozzle Assembly Installation Test

SUBTASK 28-22-13-860-018

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-13-410-016

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-13-860-019

(3) On the P5-2 Fuel System Module, set the FUEL PUMPS 1 AFT or FUEL PUMPS 2 AFT switch to the ON position.

SUBTASK 28-22-13-210-004

(4) Look for leakage at the pump housing [2].

(a) If leakage occurs, repeat this procedure.

SUBTASK 28-22-13-860-020

(5) Set the FUEL PUMPS 1 AFT or FUEL PUMPS 2 AFT switch to the OFF position.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-13-410-003

(1) Close the applicable access panels:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

SUBTASK 28-22-13-080-004

WARNING DO NOT REMOVE THE DOWNLOCK PINS UNLESS ALL MAINTENANCE AND SERVICING OF THE AIRPLANE IS COMPLETE. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) If all maintenance and servicing of the airplane is complete, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

END OF TASK

TASK 28-22-13-000-803

6. Center Tank Water Scavenge Jet Pump Removal

(Figure 401, Figure 403)

A. General

(1) This task gives instructions to remove the center tank water scavenge jet pump.

(2) The center tank water scavenge jet pump nozzle assembly is referred to as the nozzle assembly in this task.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-26-00-650-802	Tank to Tank Fuel Transfer (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

Reference	Description
STD-7423	Cover - Protective Tube

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
131AB	Center Tank Access

F. Prepare for the Removal

SUBTASK 28-22-13-480-007

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-22-13-840-001

(2) Defuel or transfer fuel from the center tank and No. 1 main tank (TASK 28-26-00-650-801 or TASK 28-26-00-650-802).

SUBTASK 28-22-13-860-005

(3) On the P5-2 Fuel System Module, set the FUEL PUMPS CTR L and R switches to the OFF position.

SUBTASK 28-22-13-010-017

(4) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-13-860-006

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-13-010-010

(6) Remove the nozzle assembly [4] (TASK 28-22-13-000-801).

SUBTASK 28-22-13-840-002

WARNING

OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(7) Do this task for the center tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access door:

(TASK 28-11-31-000-801)

Number	Name/Location
131AB	Center Tank Access

G. Center Tank Water Scavenge Jet Pump Removal

SUBTASK 28-22-13-020-004

(1) Remove the bulkhead nut [5] and plain washer [18] from the pump housing [2], from out of the fuel tank.

SUBTASK 28-22-13-010-011

WARNING

OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-22-13-020-005

(3) Remove the sealant from the rear spar and pump housing [2], in the fuel tank.

SUBTASK 28-22-13-020-006

(4) Disconnect the B-nut [12] between the motive flow line [11] and the pump housing [2].

SUBTASK 28-22-13-020-007

(5) Disconnect the B-nut [16] between the scavenge inlet line [15] and the pump housing [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-13-020-008

(6) Disconnect the B-nut [13] between the outlet line [14] and the pump housing [2].

SUBTASK 28-22-13-020-009

(7) Remove the pump housing [2].

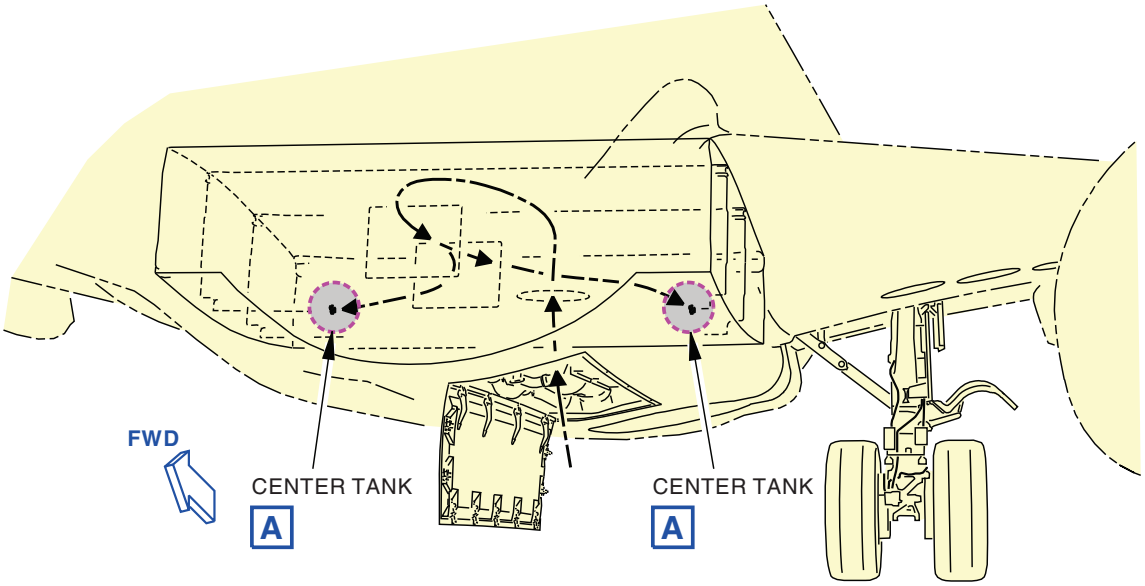
(a) Discard the O-ring [17].

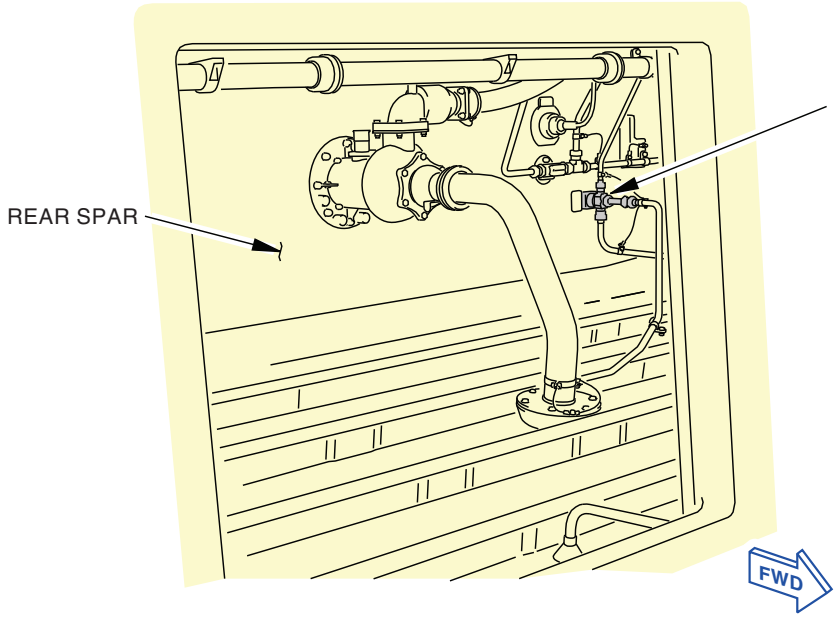
SUBTASK 28-22-13-480-002

(8) Install protective covers, STD-7423, on the open fuel lines to keep unwanted materials out.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





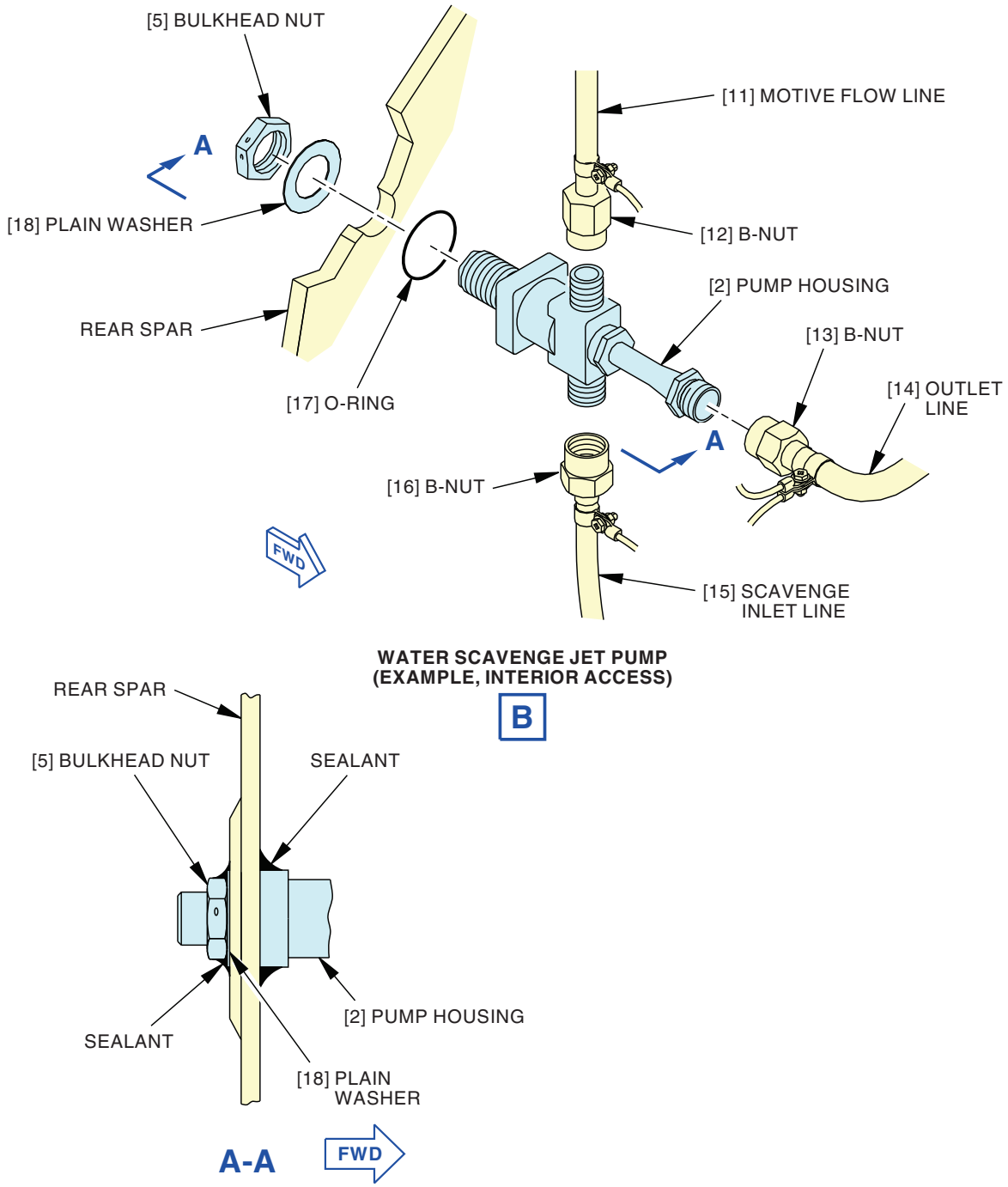
CENTER TANK
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

A

2413601 S00061532908_V1

Center Tank Water Scavenge Jet Pump Installation
Figure 403/28-22-13-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413602 S00061532909_V2

Center Tank Water Scavenge Jet Pump Installation
Figure 403/28-22-13-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-13-400-803

7. Center Tank Water Scavenge Jet Pump Installation

(Figure 403)

A. General

(1) This task gives instructions to install the center tank water scavenge jet pump.

B. References

Reference	Title
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
20-10-51-400-801	Flareless Tubing Assembly Installation (P/B 401)
28-11-00-300-803	Repair of Sealant Leaks in the Fuel Tank Structure (P/B 801)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
28-22-41-710-801	Fuel Boost Pump - Operational Test (P/B 501)
51-22-00-390-806	Fillet Seal Application (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A00767	Sealant - Fuel Tank	BMS5-45
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C50315	Coating - Chemical Conversion - Bonderite M-CR 600 Aero (Formerly Alodine 600)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
D00504	Grease - Petrolatum	VV-P-236
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50398	Pad - Adbrasive, Scotch-Brite Type S, Abrasive Pad	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
G50492	Pad - 3M Scotch Brite 7448 Ultra Fine Pad	MIL-A-9962A Type III Grade AAA
G51293	Cotton Wiper - Knit, Process Cleaning Absorbent Wiper	BMS15-5 Class B

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Pump housing	28-22-13-01-095	PTF ALL
		28-22-13-05-155	PTF ALL
		28-22-13-06-155	PTF ALL
17	O-ring	28-22-13-01-100	PTF ALL
		28-22-13-05-160	PTF ALL
		28-22-13-06-160	PTF ALL

F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
131AB	Center Tank Access

H. Prepare for the Installation

SUBTASK 28-22-13-100-003

(1) Do these steps to prepare the pump housing [2] and rear spar for installation:

(a) Remove the sealant from the rear spar, do this task: Clean Bare, Clad, or Plated Metal with Solvent, TASK 20-10-34-110-802.

NOTE: Sealant is on the inner and outer side of the rear spar.

(b) Clean the fay surfaces of the aluminum pump housing [2] and aluminum rear spar (SWPM 20-20-00):

1) Use Cleaning Procedure 1 to manually clean the rear spar (dry side) fay surface.

a) Use a very fine Scotch-Brite Type S pad, G50398, or an ultra fine Scotch-Brite 7448 pad, G50492.

b) Move the pad in a circular movement while you press with your thumb.

2) Use Cleaning Procedure 1 to manually clean the rear spar (wet side) fay surface.

a) Use an abrasive pad to clean the fay surface on the pump housing [2].

b) Move the pad in a circular movement while you press with your thumb.

3) Use Cleaning Procedure 5 to clean the pump housing [2] fay surface.

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WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034, or knit cotton wiper, G51293, soaked with solvent, B00062, or solvent, B00184.

b) Use the new cotton wiper to dry the surface.

c) Continue to clean and dry the surface until the cotton wiper shows cleans.

4) Final clean the rear spar fay surfaces.

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a) Use a new cotton wiper, G00034, or knit cotton wiper, G51293, soaked with solvent, B00062, or solvent, B00184.

b) Rub dry with a clean dry cotton wiper.

c) Continue to clean and dry the surface until the cotton wiper shows clean.

(c) Manually apply Bonderite M-CR 600 Aero coating, C50315, to the fay surface on the rear spar.

SUBTASK 28-22-13-080-001

(2) Remove the protective covers from the open fuel lines.

I. Center Tank Water Scavenge Jet Pump Installation

SUBTASK 28-22-13-640-003

(1) Lubricate a new O-ring [17] with a thin layer of grease, D00504.

SUBTASK 28-22-13-420-013

(2) Install the O-ring [17] on the pump housing [2].

SUBTASK 28-22-13-820-001

(3) Align the pump housing [2] in position on the rear spar.

SUBTASK 28-22-13-420-014

(4) Install the plain washer [18] and bulkhead nut [5] to attach the pump housing [2] to the rear spar.

(a) Make sure that the motive flow line [11] and scavenge inlet line [15] align with the pump housing [2].

SUBTASK 28-22-13-420-015

(5) Tighten the bulkhead nut [5] to 360 ±20 in-lb (40.7 ±2.3 N·m).

SUBTASK 28-22-13-760-001

(6) Measure the electrical bonding resistance between the pump housing [2] and rear spar with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) Make sure that the electrical bonding resistance is 0.00050 ohm (0.50 milliohms) or less.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-13-410-004

(7) Connect the B-nut [12] between the motive flow line [11] and the pump housing [2] (TASK 20-10-51-400-801).

SUBTASK 28-22-13-410-005

(8) Connect the B-nut [16] between the scavenge inlet line [15] and the pump housing [2] (TASK 20-10-51-400-801).

SUBTASK 28-22-13-410-006

(9) Connect the B-nut [13] between the outlet line [14] and the pump housing [2] (TASK 20-10-51-400-801).

SUBTASK 28-22-13-390-003

(10) Apply a fillet seal of sealant, A00767, between the pump housing [2] and the rear spar (wet side) (TASK 28-11-00-300-803).

NOTE: Use sufficient sealant to cover un-primed areas of the structure for corrosion protection.

SUBTASK 28-22-13-410-007

(11) Install the nozzle assembly [4] (TASK 28-22-13-400-801).

SUBTASK 28-22-13-410-008

(12) Apply a fillet seal of sealant, A02315, between the bulkhead nut [5] and the rear spar (dry side) (TASK 51-22-00-390-806).

J. Center Tank Water Scavenge Jet Pump Installation Test

SUBTASK 28-22-13-390-004

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Install this access panel:

(TASK 28-11-31-400-801)

Number	Name/Location
131AB	Center Tank Access

SUBTASK 28-22-13-410-009

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-13-410-017

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-13-710-001

(4) Do the fuel boost pump operational test for the center tank (TASK 28-22-41-710-801).

(a) Look for leakage at the pump housing [2].

(b) If leakage occurs around the pump housing [2], repeat this procedure.

——— END OF TASK ———

TASK 28-22-13-000-804

8. Main Tank Water Scavenge Jet Pump Removal

(Figure 402, Figure 404)

A. General

(1) This task gives instructions to remove the main tank water scavenge jet pump.

(2) The main tank water scavenge jet pump nozzle assembly is referred to as the nozzle assembly in this task.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-26-00-650-802	Tank to Tank Fuel Transfer (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-7423	Cover - Protective Tube

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
532AB	Main Tank Access Door - Wing Station 216
632AB	Main Tank Access Door - Wing Station 216

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

SUBTASK 28-22-13-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-22-13-840-003

(2) Defuel or transfer fuel from the applicable main tank (TASK 28-26-00-650-801 or TASK 28-26-00-650-802).

SUBTASK 28-22-13-860-007

(3) On the P5-2 Fuel System Module, set the FUEL PUMPS 1 AFT or FUEL PUMPS 2 AFT switches to the OFF position.

SUBTASK 28-22-13-010-016

(4) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-13-860-008

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Open the applicable circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT

SUBTASK 28-22-13-010-013

(6) Remove the nozzle assembly [4] (TASK 28-22-13-000-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-13-840-004

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(7) Do this task for the center tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove the applicable access door:

(TASK 28-11-11-000-801)

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216
632AB	Main Tank Access Door - Wing Station 216

G. Main Tank Water Scavenge Jet Pump Removal

SUBTASK 28-22-13-020-010

(1) Remove the bulkhead nut [5] and plain washer [18] from the pump housing [2] (dry side).

SUBTASK 28-22-13-480-004

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-22-13-020-011

(3) Remove the sealant from the rear spar and pump housing [2] (wet side).

SUBTASK 28-22-13-020-012

(4) Disconnect the B-nut [12] between the motive flow line [11] and the pump housing [2].

SUBTASK 28-22-13-020-013

(5) Disconnect the B-nut [16] between the scavenge inlet line [15] and the pump housing [2].

SUBTASK 28-22-13-020-014

(6) Disconnect the B-nut [13] between the outlet line [14] and the pump housing [2].

SUBTASK 28-22-13-020-015

(7) Remove the pump housing [2].

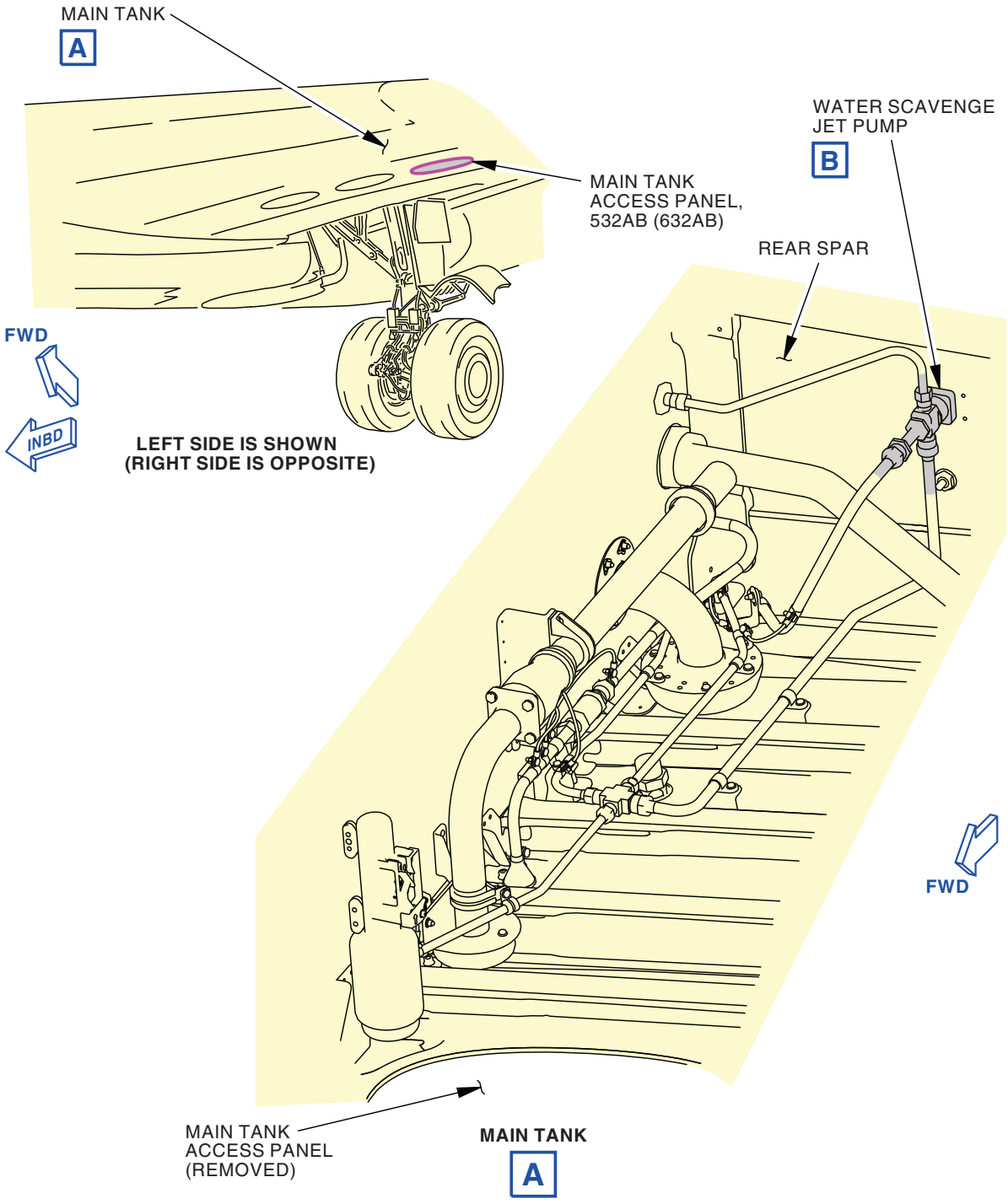
(a) Discard the O-ring [17].

SUBTASK 28-22-13-480-005

(8) Install protective covers, STD-7423, on the open fuel lines to keep unwanted materials out.

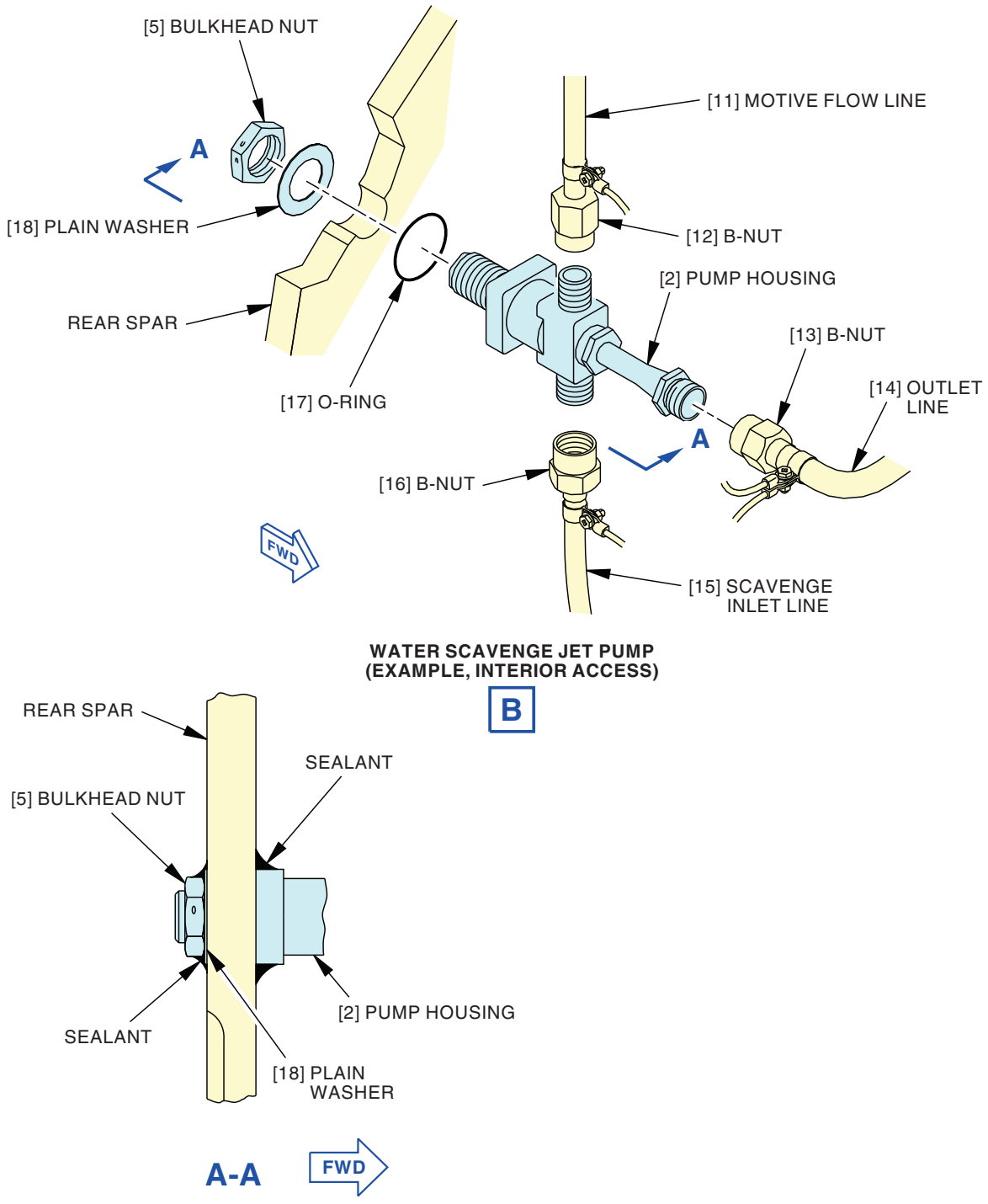
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Tank Water Scavenge Jet Pump Installation
Figure 404/28-22-13-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Main Tank Water Scavenge Jet Pump Installation
Figure 404/28-22-13-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-13-400-804

9. Main Tank Water Scavenge Jet Pump Installation

(Figure 404)

A. General

(1) This task gives instructions to install the main tank water scavenge jet pump.

B. References

Reference	Title
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
20-10-51-400-801	Flareless Tubing Assembly Installation (P/B 401)
28-11-00-300-803	Repair of Sealant Leaks in the Fuel Tank Structure (P/B 801)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
28-22-41-710-801	Fuel Boost Pump - Operational Test (P/B 501)
51-22-00-390-806	Fillet Seal Application (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A00767	Sealant - Fuel Tank	BMS5-45
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C50315	Coating - Chemical Conversion - Bonderite M-CR 600 Aero (Formerly Alodine 600)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
D00504	Grease - Petrolatum	VV-P-236
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50398	Pad - Adbrasive, Scotch-Brite Type S, Abrasive Pad	

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(Continued)

Reference	Description	Specification
G50492	Pad - 3M Scotch Brite 7448 Ultra Fine Pad	MIL-A-9962A Type III Grade AAA
G51293	Cotton Wiper - Knit, Process Cleaning Absorbent Wiper	BMS15-5 Class B

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Pump housing	28-22-13-01-095	PTF ALL
		28-22-13-05-155	PTF ALL
		28-22-13-06-155	PTF ALL
17	O-ring	28-22-13-01-100	PTF ALL
		28-22-13-05-160	PTF ALL
		28-22-13-06-160	PTF ALL

F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
532AB	Main Tank Access Door - Wing Station 216
632AB	Main Tank Access Door - Wing Station 216

H. Prepare for the installation

SUBTASK 28-22-13-100-004

(1) Do these steps to prepare the applicable pump housing [2] and rear spar for installation:

(a) Remove the sealant from the rear spar, do this task: Clean Bare, Clad, or Plated Metal with Solvent, TASK 20-10-34-110-802.

NOTE: Sealant is on the inner and outer side of the rear spar.

(b) Clean the fay surfaces of the aluminum pump housing [2] and aluminum rear spar (SWPM 20-20-00):

1) Use Cleaning Procedure 1 to manually clean the rear spar (dry side) fay surface.

a) Use a very fine Scotch-Brite Type S pad, G50398, or an ultra fine Scotch-Brite 7448 pad, G50492.

b) Move the pad in a circular movement while you press with your thumb.

2) Use Cleaning Procedure 1 to manually clean the rear spar (wet side) fay surface.

a) Use an abrasive pad to clean the fay surface on the pump housing [2].

b) Move the pad in a circular movement while you press with your thumb.

3) Use Cleaning Procedure 5 to clean the pump housing [2] fay surface.

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a) Use a new cotton wiper, G00034, or knit cotton wiper, G51293, soaked with solvent, B00062 or solvent, B00184.

b) Use the new cotton wiper to dry the surface.

c) Continue to clean and dry the surface until the cotton wiper shows cleans.

4) Final clean the rear spar fay surfaces.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034, or knit cotton wiper, G51293, soaked with solvent, B00062 or solvent, B00184.

b) Rub dry with a clean dry cotton wiper.

c) Continue to clean and dry the surface until the cotton wiper shows clean.

(c) Manually apply Bonderite M-CR 600 Aero coating, C50315, to the fay surface on the rear spar.

SUBTASK 28-22-13-080-002

(2) Remove the protective covers from the open fuel lines.

I. Main Tank Water Scavenge Jet Pump Installation

SUBTASK 28-22-13-640-004

(1) Lubricate the new O-ring [17] with a thin layer of grease, D00504.

SUBTASK 28-22-13-420-016

(2) Install the O-ring [17] on the pump housing [2].

SUBTASK 28-22-13-820-002

(3) Align the pump housing [2] in position on the rear spar.

SUBTASK 28-22-13-420-017

(4) Install the plain washer [18] and bulkhead nut [5] to attach the pump housing [2] to the rear spar.

(a) Make sure that the motive flow line [11] and scavenge inlet line [15] align with the pump housing [2].

SUBTASK 28-22-13-420-018

(5) Tighten the bulkhead nut [5] to 360 ±20 in-lb (40.7 ±2.3 N·m).

SUBTASK 28-22-13-760-002

(6) Measure the electrical bonding resistance between the pump housing [2] and rear spar with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) Make sure that the electrical bonding resistance is 0.00050 ohm (0.50 milliohms) or less.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-13-410-010

(7) Connect the B-nut [12] between the motive flow line [11] and the pump housing [2] (TASK 20-10-51-400-801).

SUBTASK 28-22-13-410-011

(8) Connect the B-nut [16] between the scavenge inlet line [15] and the pump housing [2] (TASK 20-10-51-400-801).

SUBTASK 28-22-13-410-012

(9) Connect the B-nut [13] between the outlet line [14] and the pump housing [2] (TASK 20-10-51-400-801).

SUBTASK 28-22-13-390-005

(10) Apply a fillet seal of sealant, A00767, between the pump housing [2] and the rear spar (wet side) (TASK 28-11-00-300-803).

NOTE: Use sufficient sealant to cover un-primed areas of the structure for corrosion protection.

SUBTASK 28-22-13-410-013

(11) Install the nozzle assembly [4] (TASK 28-22-13-400-802).

SUBTASK 28-22-13-390-006

(12) Apply a fillet seal of sealant, A02315, between the bulkhead nut [5] and the rear spar (dry side) (TASK 51-22-00-390-806).

J. Main Tank Water Scavenge Jet Pump Installation Test

SUBTASK 28-22-13-410-014

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Install the applicable access panels:

(TASK 28-11-11-400-801)

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216
632AB	Main Tank Access Door - Wing Station 216

SUBTASK 28-22-13-860-021

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT

SUBTASK 28-22-13-410-018

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-13-710-002

(4) Do the fuel boost pump operational test for the applicable main tank (TASK 28-22-41-710-801).

NOTE: The aft fuel boost pump operates the water scavenge jet pump.

(a) Look for leakage at the pump housing [2].

(b) If leakage occurs around the pump housing [2], repeat this procedure.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER SCAVENGE JET PUMP - CLEANING/PAINTING

1. General

A. This procedure has these tasks:

(1) Center Tank Water Scavenge Jet Pump Cleaning

(2) Main Tank Water Scavenge Jet Pump Cleaning

B. The water scavenge jet pump operates continuously while the related fuel boost pump operates. These are the four fuel boost pumps that operate the water scavenge jet pumps: left and right center fuel boost pump, and aft fuel boost pump in the No. 1 and No. 2 main tank. Regular inspections are necessary to make sure that there is no pump blockage.

TASK 28-22-13-100-801

2. Center Tank Water Scavenge Jet Pump Cleaning

(Figure 701)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
28-22-13-000-801	Center Tank Water Scavenge Jet Pump Nozzle Assembly Removal (P/B 401)
28-22-13-400-801	Center Tank Water Scavenge Jet Pump Nozzle Assembly Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-123	Brush - Soft Bristle

C. Consumable Materials

Reference	Description	Specification
B00074	Solvent - Degreasing	MIL-PRF-680 (Supersedes P-D-680)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Prepare for the Cleaning

SUBTASK 28-22-13-010-006

(1) Do this task: Center Tank Water Scavenge Jet Pump Nozzle Assembly Removal, TASK 28-22-13-000-801.

F. Center Tank Water Scavenge Jet Pump Nozzle Assembly Cleaning

SUBTASK 28-22-13-010-007

(1) Remove the jet nozzle [1] from the nozzle plug [2].

SUBTASK 28-22-13-210-001

(2) Make sure that the holes on the jet nozzle [1] are clean and have no blockage.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-13-110-001

WARNING KEEP THE CLEANING SOLUTIONS AWAY FROM SPARKS, FLAMES, AND HEAT. CLEANING SOLUTIONS ARE FLAMMABLE, AND CAN CAUSE INJURIES TO PERSONNEL.

CAUTION DO NOT USE ABRASIVE MATERIALS. ABRASIVE MATERIALS WILL CAUSE DAMAGE TO THIS COMPONENT.

(3) If necessary, clean the nozzle plug [2] and jet nozzle [1], as follows:

(a) Clean the nozzle plug [2] and jet nozzle [1] with solvent, B00074, or equivalent.

(b) For harder grit, use a soft bristle brush, STD-123.

(c) Dry the nozzle plug [2] and jet nozzle [1] with moisture-free compressed air or nitrogen.

SUBTASK 28-22-13-420-001

(4) Install the jet nozzle [1] into the nozzle plug [2].

SUBTASK 28-22-13-420-002

(5) Tighten the nozzle plug [2] to 71 ±4 in-lb (8.0 ±0.5 N·m).

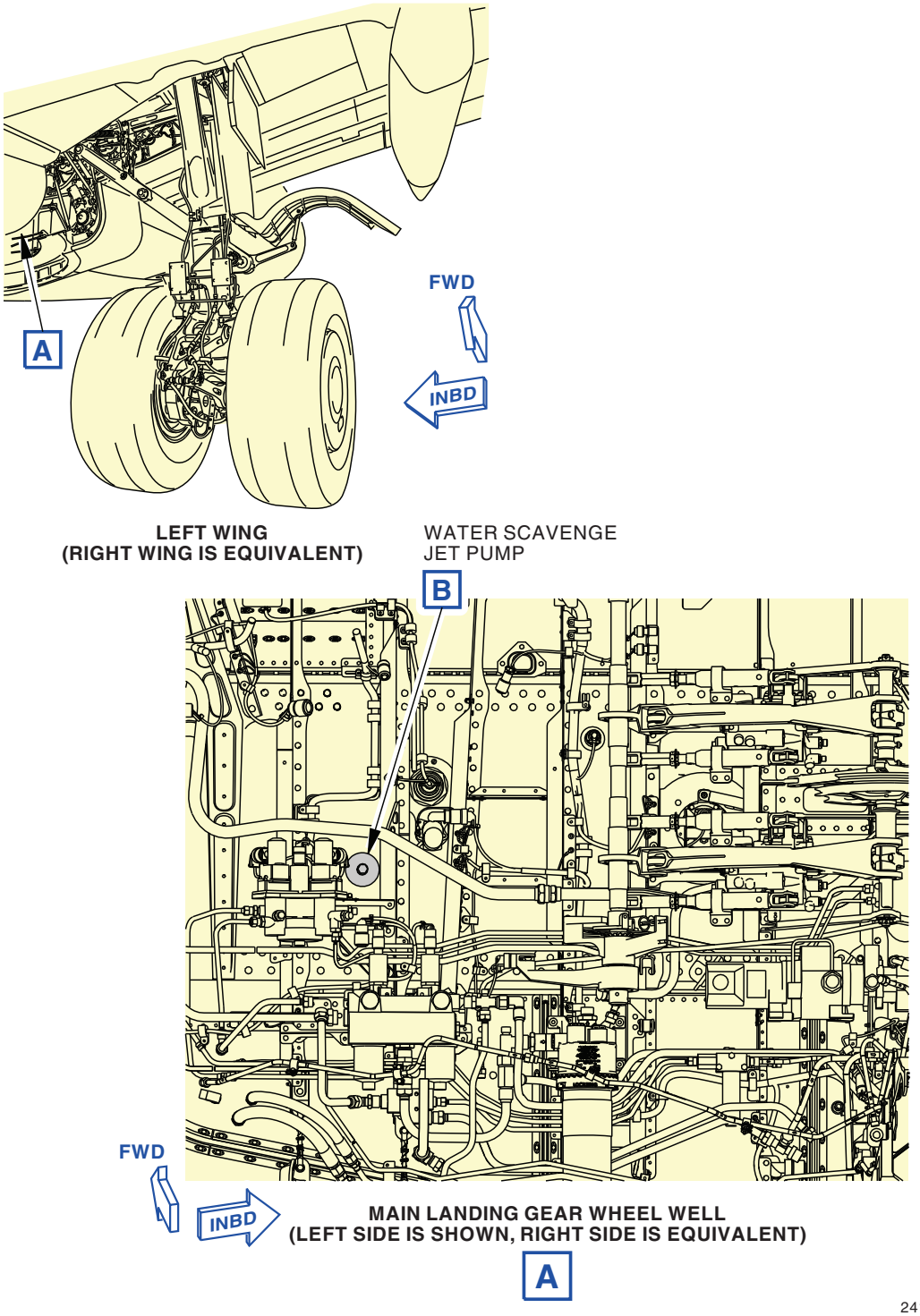
G. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-13-410-001

(1) Do this task: Center Tank Water Scavenge Jet Pump Nozzle Assembly Installation, TASK 28-22-13-400-801.

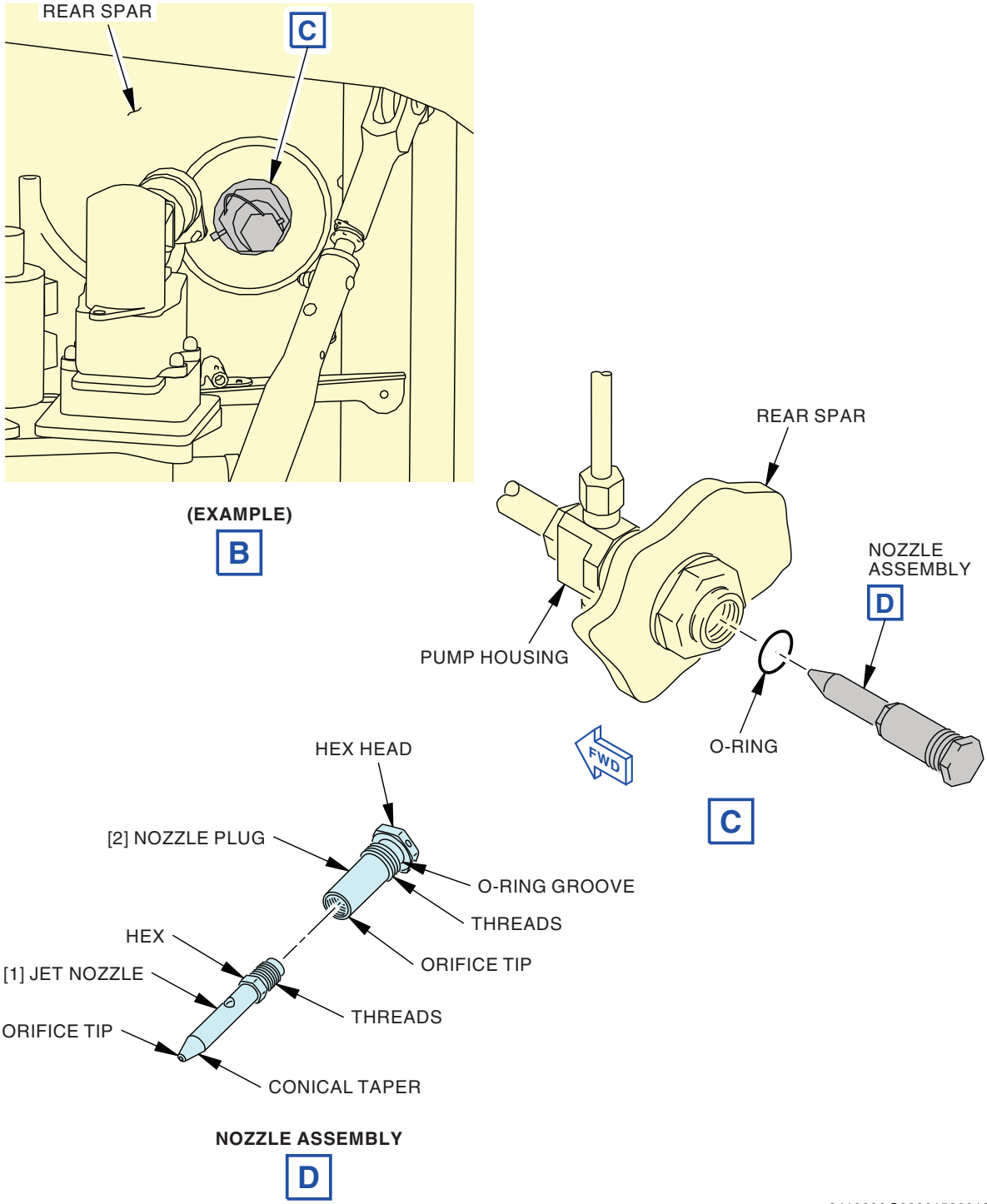
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Center Tank Water Scavenge Jet Pump Cleaning
Figure 701/28-22-13-990-805 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Center Tank Water Scavenge Jet Pump Cleaning
Figure 701/28-22-13-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-13-100-802

3. Main Tank Water Scavenge Jet Pump Cleaning

(Figure 702)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
28-22-13-000-802	Main Tank Water Scavenge Jet Pump Nozzle Assembly Removal (P/B 401)
28-22-13-400-802	Main Tank Water Scavenge Jet Pump Nozzle Assembly Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-123	Brush - Soft Bristle

C. Consumable Materials

Reference	Description	Specification
B00074	Solvent - Degreasing	MIL-PRF-680 (Supersedes P-D-680)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam

E. Prepare for the Cleaning

SUBTASK 28-22-13-010-008

(1) Do this task: Main Tank Water Scavenge Jet Pump Nozzle Assembly Removal, TASK 28-22-13-000-802.

F. Main Tank Water Scavenge Jet Pump Nozzle Assembly Cleaning

SUBTASK 28-22-13-020-001

(1) Remove the jet nozzle [1] from the nozzle plug [2].

SUBTASK 28-22-13-210-002

(2) Make sure that the holes on the jet nozzle [1] are clean and have no blockage.

SUBTASK 28-22-13-110-002

WARNING KEEP THE CLEANING SOLUTIONS AWAY FROM SPARKS, FLAMES, AND HEAT. CLEANING SOLUTIONS ARE FLAMMABLE, AND CAN CAUSE INJURIES TO PERSONNEL.

CAUTION DO NOT USE ABRASIVE MATERIALS. ABRASIVE MATERIALS WILL CAUSE DAMAGE TO THIS COMPONENT.

(3) If necessary, clean the nozzle plug [2] and jet nozzle [1], as follows:

(a) Clean the nozzle plug [2] and jet nozzle [1] with solvent, B00074, or equivalent.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) For harder grit, use a soft bristle brush, STD-123.

(c) Dry the nozzle plug [2] and jet nozzle [1] with moisture-free compressed air or nitrogen.

SUBTASK 28-22-13-420-003

(4) Install the jet nozzle [1] into the nozzle plug [2].

SUBTASK 28-22-13-420-004

(5) Tighten the nozzle plug [2] to 71 ±4 in-lb (8.0 ±0.5 N·m).

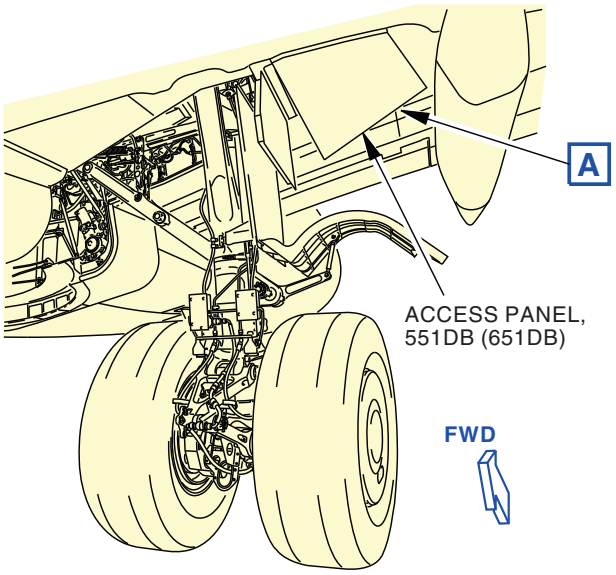
G. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-13-410-002

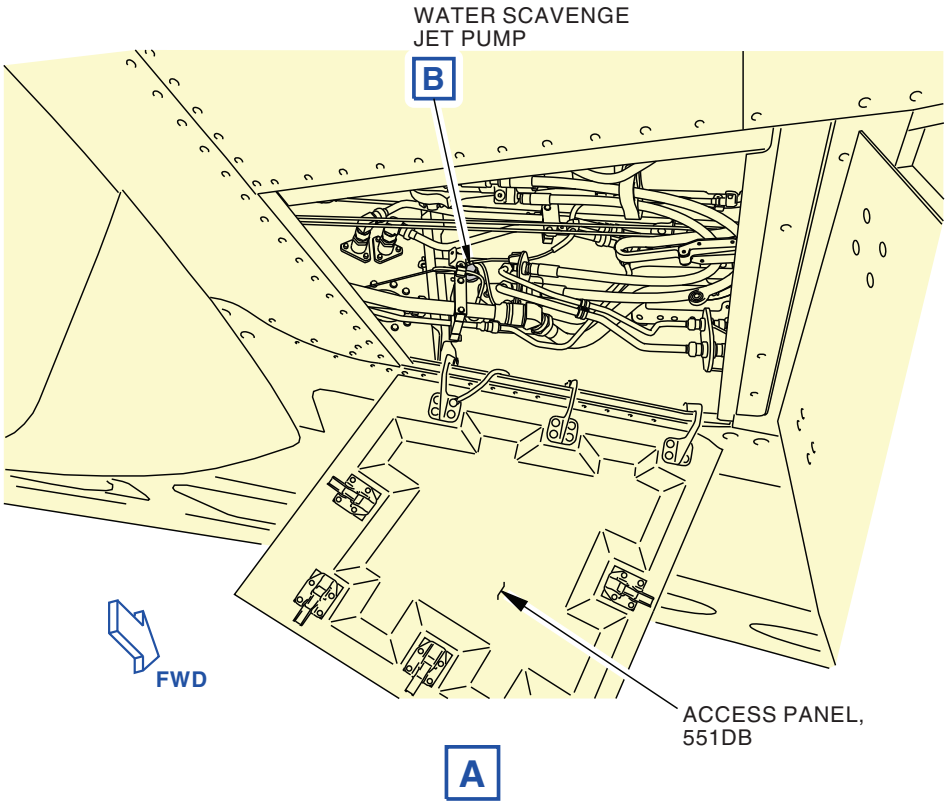
(1) Do this task: Main Tank Water Scavenge Jet Pump Nozzle Assembly Installation, TASK 28-22-13-400-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



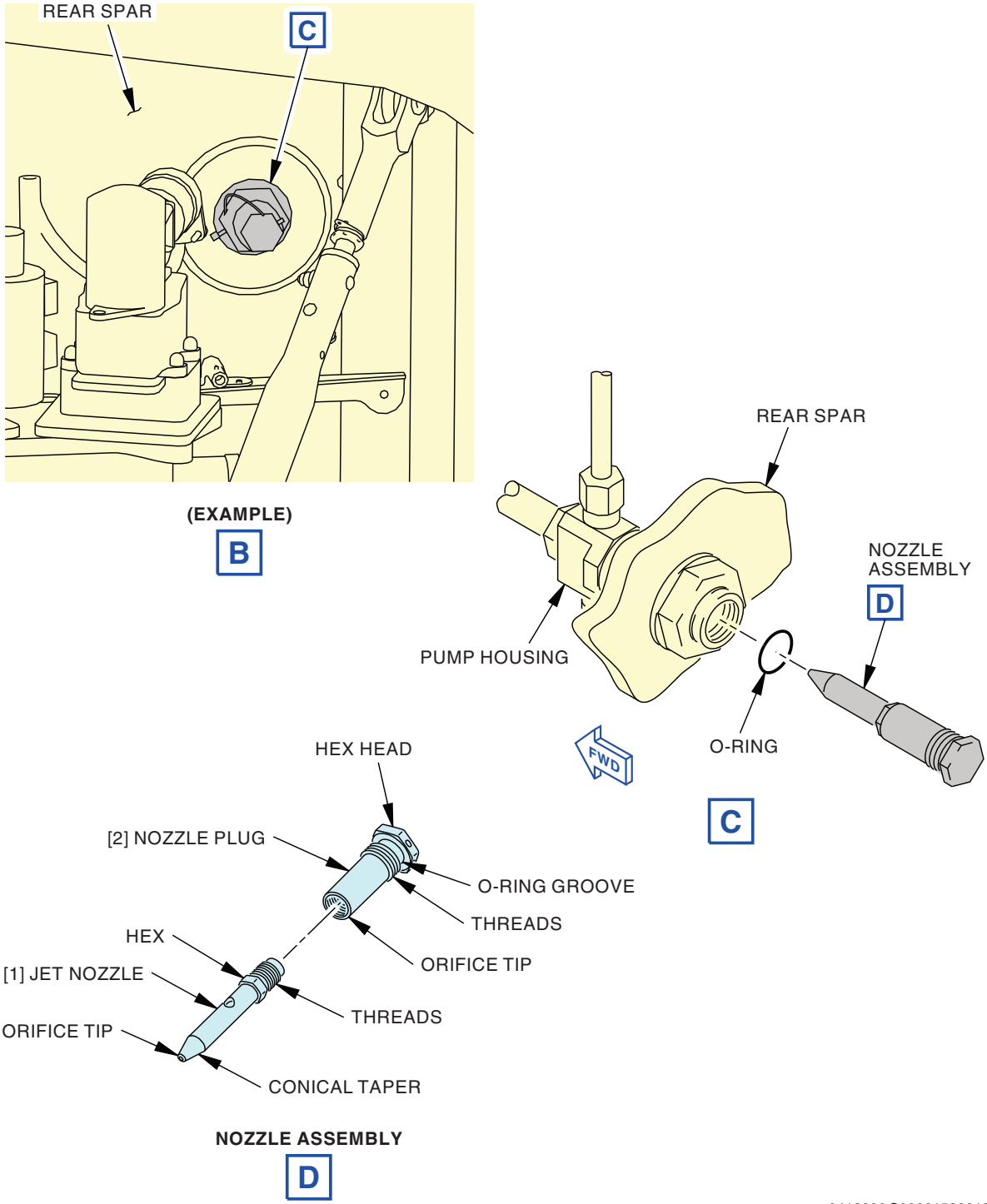
LEFT WING
(RIGHT WING IS EQUIVALENT)



2413607 S00061532920_V1

Main Tank Water Scavenge Jet Pump Cleaning
Figure 702/28-22-13-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413606 S00061532918_V1

Main Tank Water Scavenge Jet Pump Cleaning
Figure 702/28-22-13-990-806 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EMERGENCY FUEL SHUTOFF BATTERY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the emergency fuel shutoff battery

(2) An installation of the emergency fuel shutoff battery.

TASK 28-22-14-000-801

2. Emergency Fuel Shutoff Battery Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives instructions to remove the emergency fuel shutoff battery.

(2) The battery and charger system is not a Boeing supplied part. Refer to the Component Maintenance Manual (CMM) for the battery and charger system instructions for disassembly.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
25-11-01-000-801	Captain and First Officer Seat Removal (P/B 401)
25-11-03-000-801	Log Book Holder and Stowage Box Removal (P/B 401)

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 28-22-14-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	4	C01471	FUEL SHUTOFF VALVES PWR PACK
C	6	C01472	FUEL SHUTOFF VALVES BUS

SUBTASK 28-22-14-020-005

(2) If it is necessary, do this task: Log Book Holder and Stowage Box Removal, TASK 25-11-03-000-801.

SUBTASK 28-22-14-020-006

(3) If it is necessary, remove the first officer's seat, do this task: Captain and First Officer Seat Removal, TASK 25-11-01-000-801.

SUBTASK 28-22-14-020-001

(4) Open the P6-5 F/O electrical system panel, on the right side of the flight compartment, behind the first officer's seat.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Emergency Fuel Shutoff Battery Removal

SUBTASK 28-22-14-020-007

CAUTION

DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(1) Before the battery and charger system [1] is touched, do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 28-22-14-020-002

(2) Disconnect the electrical connector [3].

SUBTASK 28-22-14-020-003

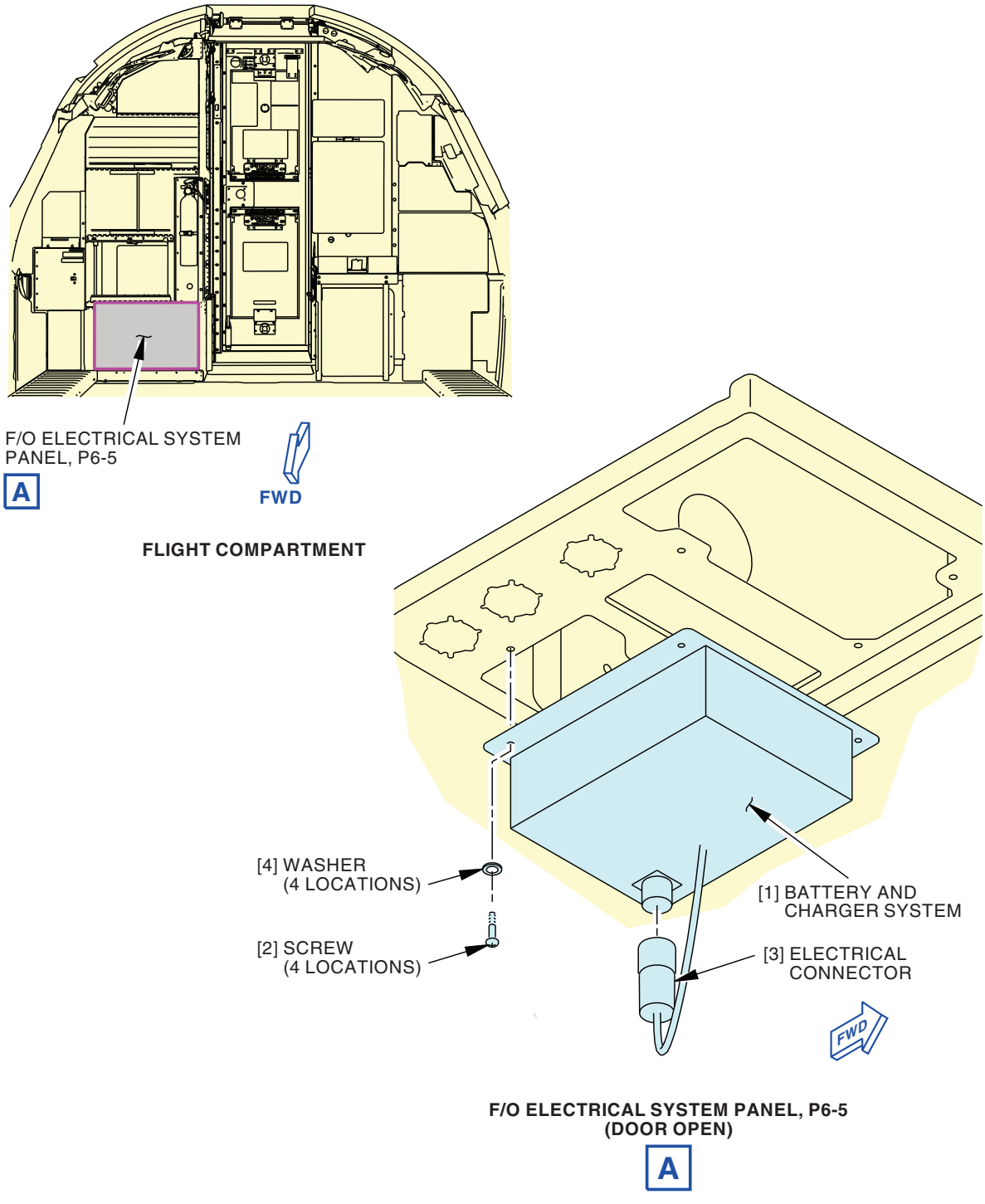
(3) Remove the screws [2] and washers [4] that attach the battery and charger system [1] to the top of the stowage bin.

SUBTASK 28-22-14-020-004

(4) Remove the battery and charger system [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Emergency Fuel Shutoff Battery Installation
Figure 401/28-22-14-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-14-400-801

3. Emergency Fuel Shutoff Battery Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives instructions to install the emergency fuel shutoff battery.

(2) The battery and charger system is not a Boeing supplied part. Refer to the Component Maintenance Manual (CMM) for the battery and charger system instructions for assembly.

B. References

Reference	Title
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
25-11-01-400-801	Captain and First Officer Seat Installation (P/B 401)
25-11-03-400-801	Log Book Holder and Stowage Box Installation (P/B 401)
28-22-00-720-801	Emergency Fuel Shutoff Battery - Operational Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Battery and charger system	28-22-14-01A-005	PTF ALL

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Emergency Fuel Shutoff Battery Installation

SUBTASK 28-22-14-420-006

CAUTION

DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(1) Before the battery and charger system [1] is touched, do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 28-22-14-420-001

(2) Align the battery and charger system [1] in its position.

SUBTASK 28-22-14-420-002

(3) Install the screws [2] and washers [4].

SUBTASK 28-22-14-420-003

(4) Connect the electrical connector [3] to the battery and charger system [1].

SUBTASK 28-22-14-410-001

(5) Close the P6-5 F/O electrical system panel.

F. Emergency Fuel Shutoff Battery Installation Test

SUBTASK 28-22-14-860-002

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	4	C01471	FUEL SHUTOFF VALVES PWR PACK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	6	C01472	FUEL SHUTOFF VALVES BUS

SUBTASK 28-22-14-860-003

(2) Charge the battery and charger system [1] for one hour.

SUBTASK 28-22-14-710-001

(3) Test the engine fuel spar valve (TASK 28-22-00-720-801).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-14-420-004

(1) If the stowage box was removed, do this task: Log Book Holder and Stowage Box Installation, TASK 25-11-03-400-801.

SUBTASK 28-22-14-420-005

(2) If it is removed, install the first officer's seat, do this task: Captain and First Officer Seat Installation, TASK 25-11-01-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL FEED MANIFOLD AND COUPLINGS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Fuel Line, Fitting and Coupling Removal

(2) Fuel Line, Fitting and Coupling Installation

(3) Engine Fuel Feed Tube Bulkhead Fitting Removal

(4) Engine Fuel Feed Tube Bulkhead Fitting Installation

(5) Couplings Removal

(6) Couplings Installation.

B. This procedure has example procedures to remove and install the fuel feed lines and fittings.

C. There are five types of coupling fittings: a flexible full coupling, flexible half coupling, a rigid coupling, self-bonding flexible coupling (with a tie strap), and self-bonding flexible coupling (without a tie strap).

D. It is not necessary to replace the aluminum tubing for the fuel system (Types 2024, 5052, and 6061) if the damage is not more than the damage limits in (TASK 28-22-15-700-801).

TASK 28-22-15-000-801

2. Fuel Line, Fitting and Coupling Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the fuel lines, fittings, and couplings.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)

C. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
533	Left Wing - Surge Tank, Rib 22 to Rib 25, Wing Station 643.50 to Wing BL 616.75
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50
633	Right Wing - Surge Tank, Rib 22 to Rib 25, Wing Station 643.50 to Wing Buttock Line 616.75

D. Prepare for the Removal

SUBTASK 28-22-15-650-001

(1) Defuel all fuel tanks (TASK 28-26-00-650-801).

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SUBTASK 28-22-15-010-003

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802

(a) Remove the applicable main tank access doors (TASK 28-11-11-000-801).

(b) Remove the applicable center tank access doors (TASK 28-11-31-000-801).

SUBTASK 28-22-15-860-007

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Go into the fuel tank (TASK 28-11-00-910-802).

E. Fuel Line, Fitting and Coupling Removal

SUBTASK 28-22-15-020-001

(1) To remove the single bonding jumper clamp (Figure 401).

(a) Remove the screw [1], washer [2], bonding jumper [3], bonding jumper clamp [6], washer [4], and nut [5] to disconnect the bonding jumper [3] from the bonding jumper clamp [6].

(b) Remove the bonding jumper clamp [6] from the fuel line.

SUBTASK 28-22-15-020-007

(2) To remove the double bonding jumper clamp (Figure 401).

(a) Remove the screw [1], washer [2], bonding jumper [3], bonding jumper clamp [6], bonding jumper [7], washer [4], nut [5] to disconnect the bonding jumper [3] and bonding jumper [7] from the bonding jumper clamp [6].

(b) Remove the bonding jumper clamp [6] from the fuel line.

SUBTASK 28-22-15-020-002

(3) To remove a loop clamp (Figure 402).

(a) Remove the screw [11], washer [13], and nut [14] to disconnect the loop clamp [12] from the structure.

SUBTASK 28-22-15-020-008

(4) To remove a double loop clamp (Figure 402).

(a) Remove the screw [11], washer [13], and nut [14] to disconnect the small loop clamp [15] from the large loop clamp [16].

SUBTASK 28-22-15-020-009

(5) To remove a self-bonding saddle clamp assembly (Figure 402).

(a) Remove the two screws [11] and washers [13].

(b) Remove the self-bonding saddle clamp top [19] from the fuel line and self-bonding saddle clamp base [18].

(c) Remove the self-bonding saddle clamp base [18] from the bracket.

(d) Remove the clip-on nut [17].

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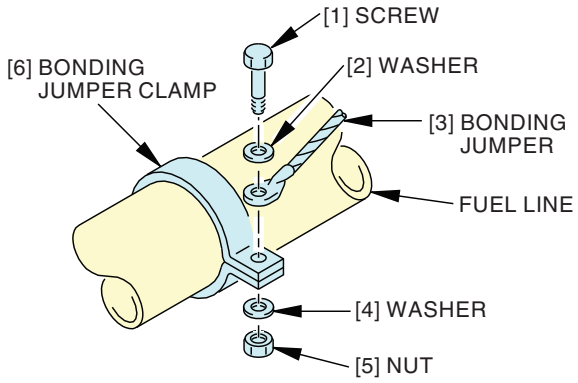
SUBTASK 28-22-15-020-003

(6) To remove the applicable coupling, do this task: Couplings Removal, TASK 28-22-15-000-803.

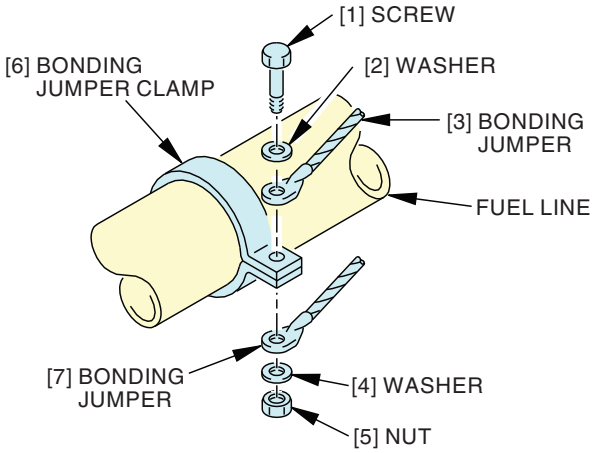
NOTE: These are different couplings used in the fuel system: flexible half coupling, flexible full coupling, rigid coupling, and self-bonding flexible coupling.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SINGLE BONDING JUMPER CLAMP (EXAMPLE)

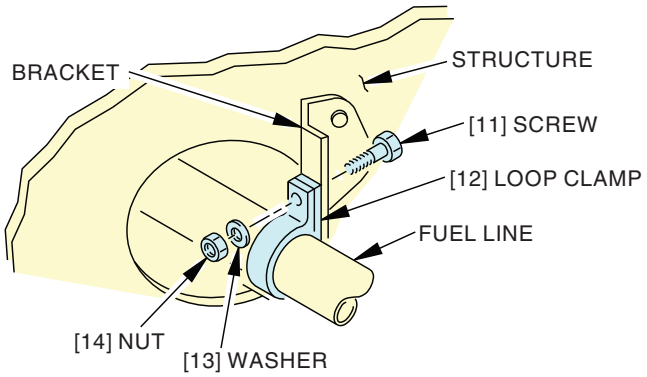


DOUBLE BONDING JUMPER CLAMP (EXAMPLE)

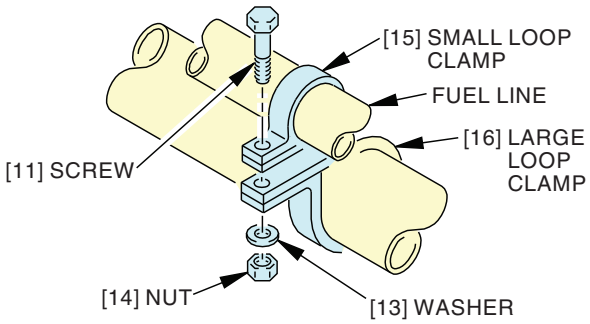
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Bonding Jumper Clamp Installation
Figure 401/28-22-15-990-801

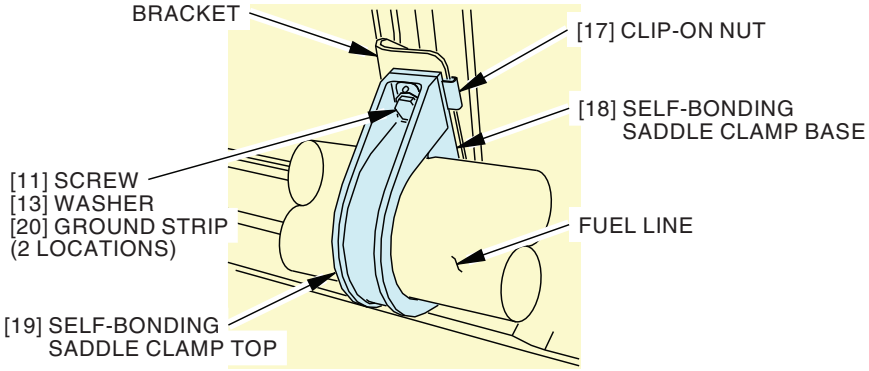
ECCN 9E991 BOEING PROPRIETARY - See title page for details



LOOP CLAMP (EXAMPLE)



DOUBLE LOOP CLAMP (EXAMPLE)



SELF-BONDING SADDLE CLAMP ASSEMBLY (EXAMPLE)

2413611 S00061532931_V1

Fuel Feed Manifold and Couplings Installation
Figure 402/28-22-15-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-15-400-801

3. Fuel Line, Fitting and Coupling Installation

(Figure 401)

A. General

(1) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
28-22-00-710-802	Engine Fuel Suction Feed - Operational Test (P/B 501)
28-22-15-710-801	Engine Fuel Feed Manifold - Leak Test (P/B 601)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
533	Left Wing - Surge Tank, Rib 22 to Rib 25, Wing Station 643.50 to Wing BL 616.75
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

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(Continued)

Zone	Area
633	Right Wing - Surge Tank, Rib 22 to Rib 25, Wing Station 643.50 to Wing Buttock Line 616.75

E. Fuel Line, Fitting and Coupling Installation

SUBTASK 28-22-15-420-010

(1) To install the applicable coupling, do this task: Couplings Installation, TASK 28-22-15-400-803.

NOTE: These are different couplings used in the fuel system: flexible half coupling, flexible full coupling, rigid coupling, and self-bonding flexible coupling.

SUBTASK 28-22-15-400-017

(2) Tighten the clamps on the fuel line to the structure or support brackets as follows (Figure 402):

(a) To install a loop clamp.

1) Put the loop clamp [12] on the fuel line.
2) Align the loop clamp [12] with the bracket.
3) Install the screw [11], washer [13], and nut [14] to attach the loop clamp [12] to the bracket.

(b) To install a double loop clamp.

1) Put the large loop clamp [16] on the tubing.
2) Put the small loop clamp [15] on the fuel line.
3) Align the large loop clamp [16] with the small loop clamp [15].
4) Install the screw [11], washer [13], and nut [14] to attach the large loop clamp [16] to the small loop clamp [15].

(c) To install the self-bonding saddle clamp assembly.

1) Install and then align the holes on the clip-on nuts [17] and bracket.
2) Make sure that the ground strips [20] are installed on the self-bonding saddle clamp top [19].
3) Align the fuel line, self-bonding saddle clamp base [18] and self-bonding saddle clamp top [19] with the bracket.
4) Install the two screws [11] and washers [13].

SUBTASK 28-22-15-420-005

(3) Tighten the support brackets to the structure with bolts, washers, and nuts (if the support brackets are not riveted).

SUBTASK 28-22-15-210-001

(4) Make sure that the fuel line does not touch the structure.

SUBTASK 28-22-15-210-002

(5) Make sure that the distance between the fuel line and the structure is a minimum of 0.25 in. (6.35 mm).

NOTE: This does not include the area near the support bracket.

SUBTASK 28-22-15-420-006

(6) To install a single bonding jumper clamp (Figure 401).

(a) Clean the fay surfaces of the aluminum fuel line, aluminum bonding jumper clamp [6], and the terminal of the bonding jumper [3] (SWPM 20-20-00, Paragraph 8).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Install the bonding jumper clamp [6] on the fuel line.

(c) Install the screw [1], washer [2], bonding jumper [3], washer [4], and nut [5] on the bonding jumper clamp [6].

(d) Tighten the nut [5] to 31.5 ±3.5 in-lb (3.6 ±0.4 N·m).

(e) Measure the electrical bonding resistance between the terminal on the bonding jumper [3] and the fuel line with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00, Paragraph 21).

1) Make sure that the measured resistance is less than the maximum bonding resistance as specified in SWPM 20-20-00, Paragraph 21.

SUBTASK 28-22-15-400-018

(7) To install a double bonding jumper clamp (Figure 401).

(a) Clean the fay surfaces of the aluminum fuel line, aluminum bonding jumper clamp [6], the terminal of the bonding jumper [3], and the terminal of the bonding jumper [7] (SWPM 20-20-00, Paragraph 8).

(b) Install the bonding jumper clamp [6] on the fuel line.

(c) Install the screw [1], washer [2], bonding jumper [3], bonding jumper [7], washer [4], and nut [5] on the bonding jumper clamp [6].

(d) Tighten the nut [5] (SWPM 20-20-00, Paragraph 8).

(e) Measure the electrical bonding resistance between the terminal on the bonding jumper [3] and the fuel line with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00, Paragraph 21).

1) Make sure that the measured resistance is less than the maximum bonding resistance as specified in SWPM 20-20-00, Paragraph 21.

(f) Measure the electrical bonding resistance between the terminal on the bonding jumper [7] and the fuel line with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00, Paragraph 21).

1) Make sure that the measured resistance is less than the maximum bonding resistance as specified in SWPM 20-20-00, Paragraph 21.

SUBTASK 28-22-15-420-007

28-AWL-11: CDCCL

(8) If the self-bonding saddle clamp assembly in the fuel tank was removed from the first fuel line that mates with the bulkhead fitting on the front spar, install the self-bonding saddle clamp (Figure 403).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-15-220-001

28-AWL-11: CDCCL

(9) For the first fuel tube in the fuel tank that mates with the bulkhead fitting, if the self-bonding saddle clamp was removed or replaced, measure the electrical bonding resistance between the structure (wing rib) in the fuel tank and the fuel tube (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-11: CDCCL

(b) Make sure that the electrical bonding resistance is 0.100 ohm (100.0 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

SUBTASK 28-22-15-200-003

(10) For all other fuel lines that are replaced, measure the electrical bonding resistance between the fuel lines and the airplane structure.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure the electrical bonding resistance between the fuel lines and the airplane structure are 0.100 ohm (100 milliohms) or less.

F. Fuel Line, Fitting and Coupling Installation Test

SUBTASK 28-22-15-710-005

(1) Do this task: Engine Fuel Feed Manifold - Leak Test, TASK 28-22-15-710-801.

SUBTASK 28-22-15-080-001

(2) Remove all tools and equipment from the fuel tank.

SUBTASK 28-22-15-410-001

(3) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Install the applicable access doors (TASK 28-11-11-400-801, TASK 28-11-31-400-801).

SUBTASK 28-22-15-710-001

(4) Do the Engine Fuel Suction Feed Operational Test (TASK 28-22-00-710-802).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-15-650-002

(1) Refuel the fuel tanks (TASK 12-11-00-650-803).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-15-000-802

4. Engine Fuel Feed Tube Bulkhead Fitting Removal

(Figure 403)

A. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-11-11-000-803	Main Tank Inner Access Door Removal (P/B 401)
28-22-15-700-801	Engine and APU Fuel Feed, Shroud, Fuel Vent Line and Couplings, and NEADS Lines Dent Criteria - Inspection/Check (P/B 601)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)

B. Location Zones

Zone	Area
430	Subzone - Engine 1, Nacelle Strut
440	Subzone - Engine 2, Nacelle Strut
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

C. Access Panels

Number	Name/Location
521AT	Outbd Leading Edge - Gap Cover Access
532AB	Main Tank Access Door - Wing Station 216
532AZ	Main Tank Inner Access at Rib 6
621AT	Outbd Leading Edge - Gap Cover Access
632AB	Main Tank Access Door - Wing Station 216
632AZ	Main Tank Inner Access at Rib 6

D. Prepare for the Removal

SUBTASK 28-22-15-860-008

(1) Defuel the center tank and main tanks (TASK 28-26-00-650-801).

SUBTASK 28-22-15-010-004

(2) Remove the applicable access panels:

Number	Name/Location
521AT	Outbd Leading Edge - Gap Cover Access
621AT	Outbd Leading Edge - Gap Cover Access

SUBTASK 28-22-15-010-005

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

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(a) Remove the applicable access doors:

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216
632AB	Main Tank Access Door - Wing Station 216

1) Do this task: Main Tank Access Door Removal, TASK 28-11-11-000-801.

(b) Remove the applicable access doors:

Number	Name/Location
532AZ	Main Tank Inner Access at Rib 6
632AZ	Main Tank Inner Access at Rib 6

1) Do this task: Main Tank Inner Access Door Removal, TASK 28-11-11-000-803.

SUBTASK 28-22-15-860-009

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Go into the fuel tank (TASK 28-11-00-910-802).

E. Bulkhead Fitting Removal

SUBTASK 28-22-15-150-001

(1) Remove the sealant.

SUBTASK 28-22-15-020-010

(2) Remove the safety cable (lockwire) from the coupling nut [61].

SUBTASK 28-22-15-020-011

(3) Remove the coupling nut [61], two retaining rings [70], and O-ring [62] from the body [69].

(a) Discard the O-ring [62].

SUBTASK 28-22-15-210-006

(4) Inspect the retaining rings [70] and coupling nut [61] for wear (TASK 28-22-15-700-801).

(a) Replace these components as necessary.

SUBTASK 28-22-15-020-012

(5) Remove the nut [65] from the body [69].

SUBTASK 28-22-15-020-013

(6) Remove the bulkhead nut [67] and plain washer [68] from the body [69].

SUBTASK 28-22-15-020-014

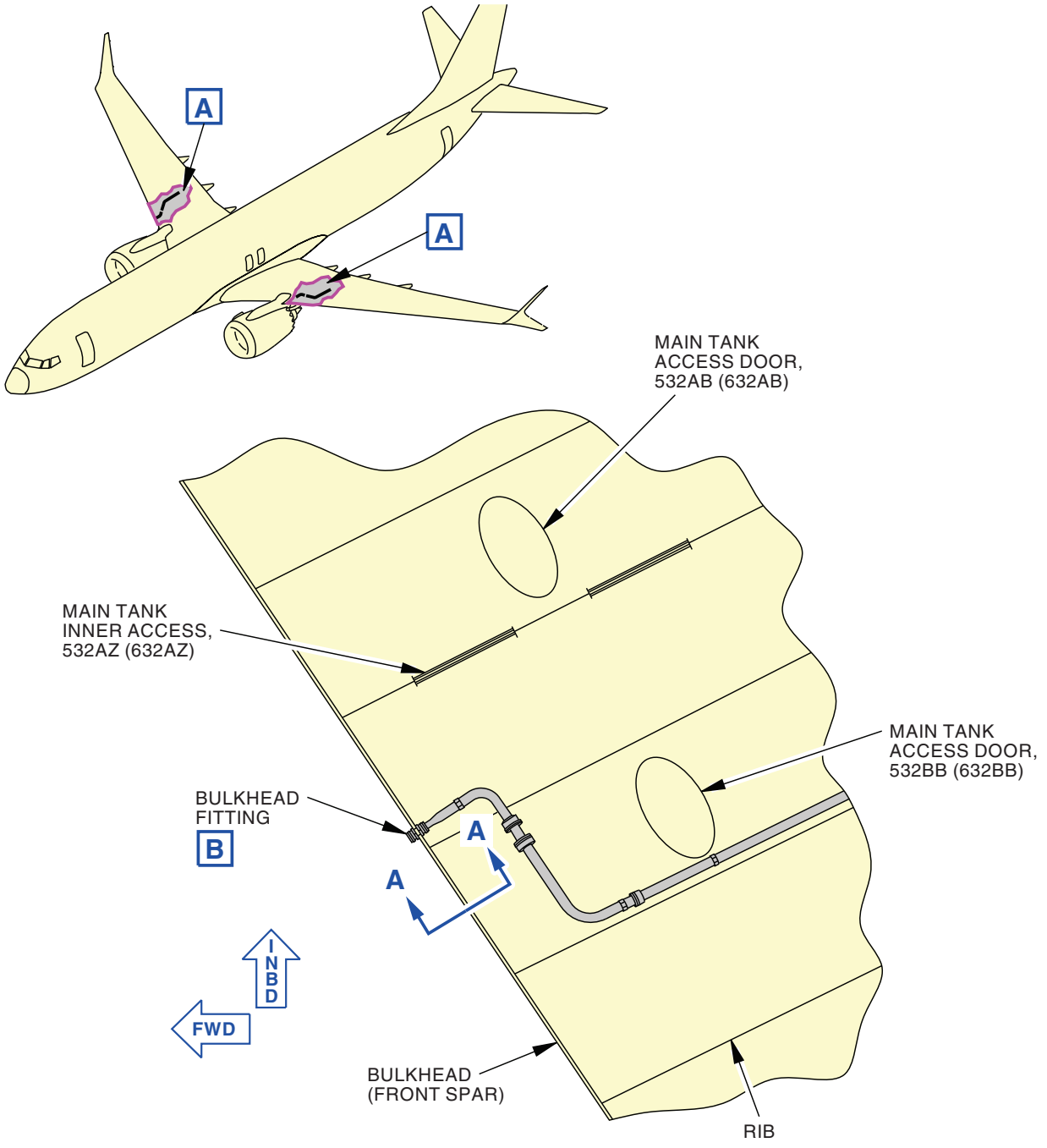
(7) Remove the body [69] from the bulkhead.

SUBTASK 28-22-15-020-015

(8) Discard the O-ring [64].

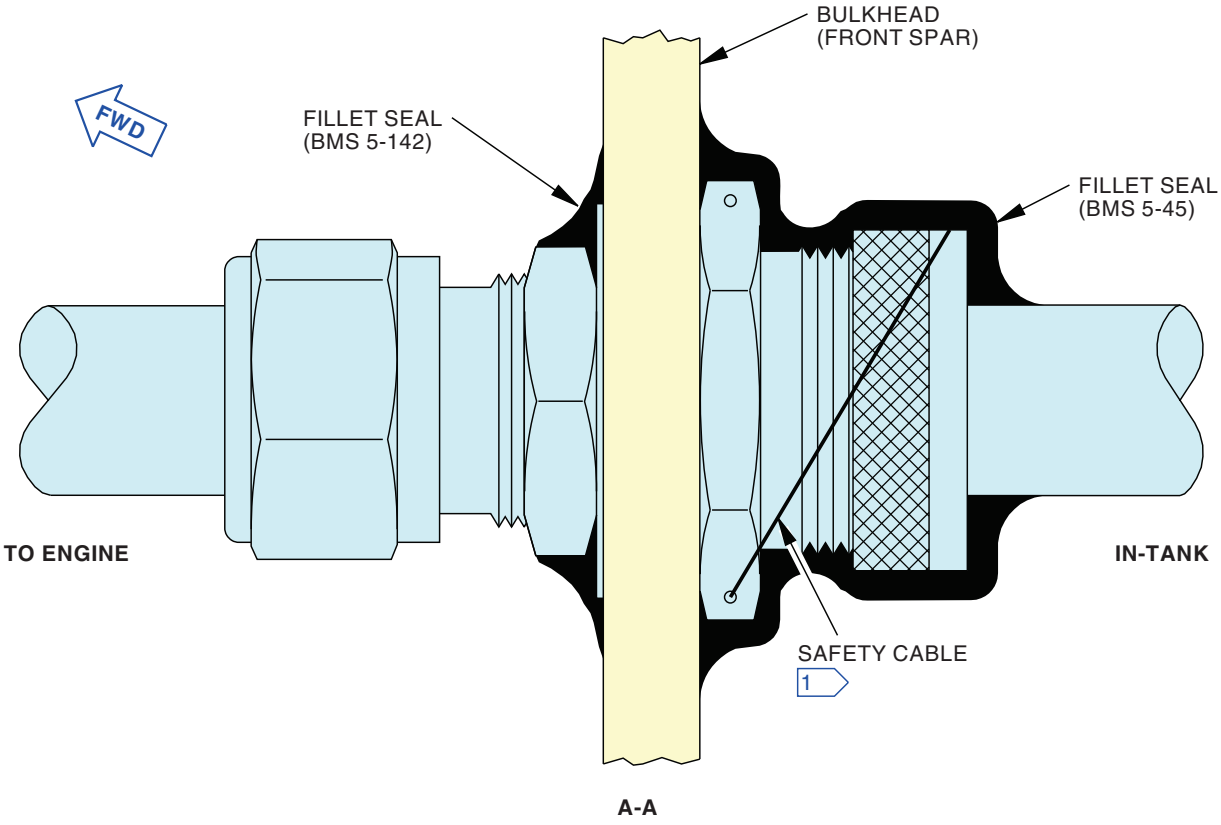
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fuel Feed Tube Bulkhead Fitting Installation
Figure 403/28-22-15-990-811 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

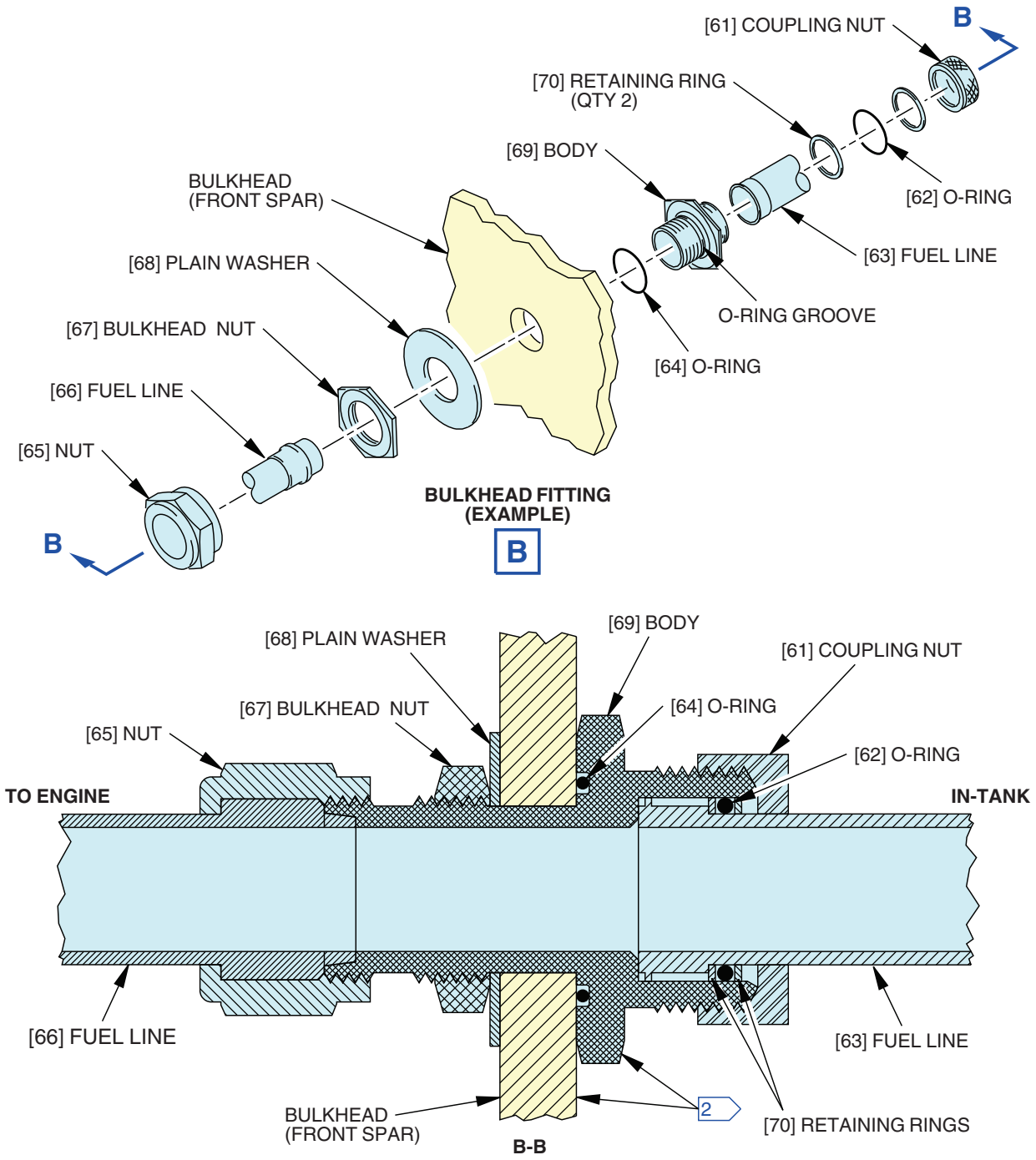


1 SAFETY CABLE PREFERRED, LOCKWIRE ALTERNATE.

2413617 S00061532937_V1

Engine Fuel Feed Tube Bulkhead Fitting Installation
Figure 403/28-22-15-990-811 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2 MEASURE THE BONDING RESISTANCE BETWEEN THE BODY THREADS AND BULKHEAD.

2413618 S00061532938_V2

Engine Fuel Feed Tube Bulkhead Fitting Installation
Figure 403/28-22-15-990-811 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-15-400-802

5. Engine Fuel Feed Tube Bulkhead Fitting Installation

(Figure 403)

A. General

(1) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

B. References

Reference	Title
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
28-11-11-400-804	Main Tank Inner Access Door Installation (P/B 401)
28-22-00-710-802	Engine Fuel Suction Feed - Operational Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-00 Paragraph 9	Electrical Bonding - Tube Bulkhead Fitting Installations
SWPM 20-20-00, Paragraph 2	CLEANING OF FAY SURFACES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A00767	Sealant - Fuel Tank	BMS5-45
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
C00862	Coating - Chemical Conversion - Bonderite M-CR 600 Aero (Formerly Alodine 600)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
D00504	Grease - Petrolatum	VV-P-236
G50078	Abrasive - Aluminum Oxide Paper, 320 grit or finer	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51372	Kit - Safety Cable, 321 CRES - 0.035 +/-0.003 Inch (1.0 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
62	O-ring	28-22-61-01-395	PTF ALL
		28-22-61-10-400	PTF ALL
64	O-ring	28-22-61-01-400	PTF ALL
		28-22-61-10-405	PTF ALL

F. Location Zones

Zone	Area
430	Subzone - Engine 1, Nacelle Strut
440	Subzone - Engine 2, Nacelle Strut
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

G. Access Panels

Number	Name/Location
521AT	Outbd Leading Edge - Gap Cover Access
532AB	Main Tank Access Door - Wing Station 216
532AZ	Main Tank Inner Access at Rib 6
621AT	Outbd Leading Edge - Gap Cover Access
632AB	Main Tank Access Door - Wing Station 216
632AZ	Main Tank Inner Access at Rib 6

H. Prepare for the Installation

SUBTASK 28-22-15-400-001

(1) Loosely connect the self-bonding saddle clamp assembly for the fuel lines to the support brackets (Figure 402).

I. Bulkhead Fitting Installation

SUBTASK 28-22-15-120-001

28-AWL-11: CDCCL

(1) Apply a fay surface bond between the body [69] (bulkhead fitting) and the front spar inside the tank.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

(a) Clean and remove the finish from the threads and surfaces on the body [69] (SWPM 20-20-00, SWPM 20-20-00 Paragraph 9).

(b) Polish the flange of the body [69].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Remove the anodic surface finish with 320 grit or finer abrasive paper, G50078, to make a flat bare metal surface all the way around the mating surface (SWPM 20-20-00).

2) Remove a minimum quantity of material.

(c) Clean the wing front spar web area, wet and dry sides (SWPM 20-20-00, Paragraph 2).

1) Remove contamination and dirt from the contact areas of the front spar.

2) Clean the surrounding surface to prepare for applying alodine and sealant.

(d) Polish the fay surface area of the front spar web at the mating surface of the flange of the body [69] with 320 grit or finer abrasive paper, G50078.

1) Remove a minimum quantity of material.

NOTE: The anodic surface finish is less than 0.0001 in. (0.0025 mm).

2) Polish and clean to a minimum of 0.0625 in. (1.59 mm) diameter more than the spar fitting on the dry side.

3) Polish and clean to a minimum of 0.0625 in. (1.59 mm) diameter more than the nut on the wet side.

CAUTION

KEEP THE TIME THAT THE METAL SURFACES ARE BARE TO A MINIMUM. IF THE SURFACES STAY BARE FOR MORE THAN FOUR HOURS, RUB THEM LIGHTLY AGAIN BEFORE YOU APPLY THE CONVERSION COATING.

(e) Apply Bonderite M-CR 600 Aero coating, C00862, to the abraded surfaces on the front spar (wet and dry side).

SUBTASK 28-22-15-640-001

(2) Apply a thin layer of grease, D00504, to the new O-ring [64].

SUBTASK 28-22-15-400-002

(3) Put the O-ring [64] in the O-ring groove on the body [69].

SUBTASK 28-22-15-400-003

(4) Put the body [69] in its position in the bulkhead.

SUBTASK 28-22-15-400-004

(5) Install the bulkhead nut [67] (jam nut) and plain washer [68] on the body [69] (dry side).

NOTE: A second person is necessary to hold the body and fuel line [63] in the fuel tank.

(a) Tighten the bulkhead nut [67] to 750 ±75 in-lb (85 ±9 N·m).

SUBTASK 28-22-15-200-001

28-AWL-11: CDCCL

(6) Measure the electrical bonding resistance between the bulkhead fitting and the front spar in the fuel tank (View B-B, Figure 403) (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

28-AWL-11: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

SUBTASK 28-22-15-400-005

(7) Put the coupling nut [61] on the ferrule of the fuel line [63].

SUBTASK 28-22-15-400-006

CAUTION

DO NOT TWIST THE RETAINING RING OUT OF SHAPE. IF YOU TWIST THE RETAINING RING, FUEL LEAKS AT THE FITTING WILL OCCUR.

(8) Put a retaining ring [70] on the ferrule to hold the coupling nut [61] on the fuel line [63].

SUBTASK 28-22-15-600-001

(9) Apply a thin layer of petrolatum grease, D00504, to the new O-ring [62].

SUBTASK 28-22-15-400-007

CAUTION

DO NOT USE THE SAME O-RING FOR THE FUEL LINE FITTINGS AFTER YOU TIGHTEN THE COUPLING ONE TIME. YOU MUST INSTALL NEW O-RINGS TO PREVENT FUEL LEAKS.

(10) Install the O-ring [62] on the ferrule on the fuel line [63].

SUBTASK 28-22-15-400-008

CAUTION

DO NOT TWIST THE RETAINING RING OUT OF SHAPE. IF YOU TWIST THE RETAINING RING, FUEL LEAKS AT THE FITTING WILL OCCUR.

(11) Put the second retaining ring [70] on the ferrule of the fuel line [63].

SUBTASK 28-22-15-400-009

(12) Align the fuel line [63] with the body [69].

SUBTASK 28-22-15-400-010

CAUTION

DO NOT USE TOOLS TO TIGHTEN THE FITTINGS. DAMAGE TO THE FITTING COMPONENTS COULD OCCUR. THE ADJACENT FITTINGS COULD BECOME LOOSE.

(13) Manually tighten the coupling nut [61] to connect the fuel line [63] to the body [69].

SUBTASK 28-22-15-400-011

(14) Loosen the coupling nut [61] one turn, and then manually tighten the coupling nut [61].

SUBTASK 28-22-15-400-012

(15) Install the safety cable kit, G51372, or MS20995NC32 lockwire, G50347, between the coupling nut [61] and body [69].

SUBTASK 28-22-15-400-013

(16) Install the nut [65], as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Align the fuel line [66] with the body [69].

(b) Connect the nut [65] on the body [69].

(c) Tighten the nut [65] to 750 ±38 in-lb (85 ±5 N·m).

(d) Loosen the nut [65] to relieve tension in the fuel line [66].

(e) Tighten the nut [65] to 750 ±38 in-lb (85 ±5 N·m) to connect the fuel line [66].

SUBTASK 28-22-15-390-001

28-AWL-11: CDCCL

(17) Apply sealant, encapsulating the first fuel tube coupling to the fitting interface in the fuel tank (View A-A, Figure 403).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

(a) Use sealant, A00767.

(b) Seal to a minimum thickness of 0.10 in. (2.54 mm).

(c) Seal to a minimum of 0.1875 in. (4.7625 mm) diameter more than the spar fitting on the front spar web.

SUBTASK 28-22-15-390-002

28-AWL-11: CDCCL

(18) Apply sealant, encapsulating the fitting to the tank wall interface in the tank (View A-A, Figure 403).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

(a) Use sealant, A00767.

(b) Make sure that the sealant extends to the flat dimension of the coupling nut [61] and at least two threads of the bulkhead fitting.

SUBTASK 28-22-15-390-003

28-AWL-11: CDCCL

(19) Apply a fillet seal between the bulkhead nut [67] (jam nut) and the front spar outside the tank (View A-A, Figure 403).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

(a) Use sealant, A02315, to apply the fillet seal.

(b) Seal to a minimum of 0.1875 in. (4.7625 mm) diameter more than the bulkhead nut [67] (jam nut) on the other side of the front spar web.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-15-400-016

28-AWL-11: CDCCL

(20) Install a self-bonding saddle clamp [71] between the first fuel tube mating with the bulkhead fitting and structure in the fuel tank (TASK 28-22-15-400-801).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

SUBTASK 28-22-15-200-002

28-AWL-11: CDCCL

(21) If the tube self-bonding saddle clamp [71] was removed and reinstalled or replaced, measure the electrical bonding resistance from the structure to the first fuel tube mating with the bulkhead fitting in the fuel tank (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-11: CDCCL

(b) Make sure that the electrical bonding resistance is 0.100 ohm (100 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-11.

J. Engine Fuel Feed Tube Bulkhead Fitting Installation Test

SUBTASK 28-22-15-410-006

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Close the applicable access panels:

Number	Name/Location
532AZ	Main Tank Inner Access at Rib 6
632AZ	Main Tank Inner Access at Rib 6

1) Do this task: Main Tank Inner Access Door Installation, TASK 28-11-11-400-804.

(b) Close the applicable access panels:

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216
632AB	Main Tank Access Door - Wing Station 216

1) Do this task: Main Tank Access Door Installation, TASK 28-11-11-400-801.

SUBTASK 28-22-15-710-004

(2) Do the Engine Fuel Suction Feed Operational Test (TASK 28-22-00-710-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

K. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-15-410-007

(1) Close the applicable access panels:

Number	Name/Location
521AT	Outbd Leading Edge - Gap Cover Access
621AT	Outbd Leading Edge - Gap Cover Access

END OF TASK

TASK 28-22-15-000-803

6. Couplings Removal

(Figure 404, Figure 405, Figure 406, Figure 407, and Figure 408)

A. General

(1) This task includes the steps to remove these couplings from fluid transfer tubes and ducts:

(a) Flexible half coupling

(b) Flexible full coupling

(c) Rigid coupling

(d) Self-Bonding Flexible Coupling (Without Tie Strap)

(e) Self-Bonding Flexible Coupling (With Tie Strap).

B. References

Reference	Title
28-22-15-700-801	Engine and APU Fuel Feed, Shroud, Fuel Vent Line and Couplings, and NEADS Lines Dent Criteria - Inspection/Check (P/B 601)

C. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Prepare for the Removal

SUBTASK 28-22-15-010-006

(1) Get access to the applicable coupling.

E. Flexible Half Coupling Removal

(Figure 404)

SUBTASK 28-22-15-020-016

CAUTION

HOLD THE ADJACENT TUBE WHEN YOU REMOVE THE COUPLING. IF YOU DO NOT HOLD THE TUBE, YOU CAN CAUSE DAMAGE TO EQUIPMENT.

(1) To remove a flexible half coupling.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the safety cable or lockwire from the flexible half coupling.

(b) Loosen the coupling nut [31].

(c) Remove the retaining rings [32] and O-ring [33].

1) Discard the O-ring [33].

(d) Inspect the retaining rings [32] and coupling nut [31] for wear (TASK 28-22-15-700-801).

F. Flexible Full Coupling Removal

(Figure 405)

SUBTASK 28-22-15-020-017

CAUTION

HOLD THE ADJACENT TUBE WHEN YOU REMOVE THE COUPLING. IF YOU DO NOT HOLD THE TUBE, YOU CAN CAUSE DAMAGE TO EQUIPMENT.

(1) To remove a flexible full coupling.

(a) Remove the safety cable or lockwire from the flexible full coupling.

(b) Loosen the coupling nut [42].

(c) Move the coupling nut [42] away from the ferrule.

(d) Move the coupling [41] away from the ferrule.

(e) Remove the retainer halves [47], retaining rings [46], and O-rings [45].

1) Discard the O-rings [45].

(f) Inspect the retaining rings [46], retainer halves [47], coupling [41], and coupling nut [42] for wear (TASK 28-22-15-700-801).

G. Rigid Coupling Removal

(Figure 406)

SUBTASK 28-22-15-020-018

CAUTION

HOLD THE ADJACENT TUBE WHEN YOU REMOVE THE COUPLING. IF YOU DO NOT HOLD THE TUBE, YOU CAN CAUSE DAMAGE TO EQUIPMENT.

(1) To remove the rigid coupling.

(a) Remove the safety cable or lockwire from the rigid coupling.

(b) Loosen the coupling nut [51].

(c) Remove the retaining ring [54] and O-ring [55].

1) Discard the O-ring [55].

(d) Inspect the retaining ring [54] and coupling nut [51] for wear (TASK 28-22-15-700-801).

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H. Self-Bonding Flexible Coupling (Without Tie Strap) Removal

(Figure 407)

SUBTASK 28-22-15-020-019

CAUTION

HOLD THE ADJACENT TUBE WHEN YOU REMOVE THE COUPLING. IF YOU DO NOT HOLD THE TUBE, YOU CAN CAUSE DAMAGE TO EQUIPMENT.

(1) For a self-bonding flexible coupling installed without a tie strap:

(a) Lift the two latches at the same time to disengage the coupling body [94].

(b) Remove the coupling body [94].

(c) Remove the sleeve [92] from the ferrules.

1) Discard the O-rings [91].

(d) Examine the coupling body [94] and the sleeve [92] for worn areas or damage.

1) Replace these components if it is necessary.

I. Self-Bonding Flexible Coupling (With Tie Strap) Removal

(Figure 408)

SUBTASK 28-22-15-020-020

CAUTION

HOLD THE ADJACENT TUBE WHEN YOU REMOVE THE COUPLING. IF YOU DO NOT HOLD THE TUBE, YOU CAN CAUSE DAMAGE TO EQUIPMENT.

(1) For a self-bonding flexible coupling installed with a tie strap:

(a) Cut the tie strap and remove it from the coupling body [83].

(b) Lift the two latches at the same time to disengage the coupling body [83].

(c) Remove the coupling body [83].

(d) Remove the sleeve [84] from the ferrules.

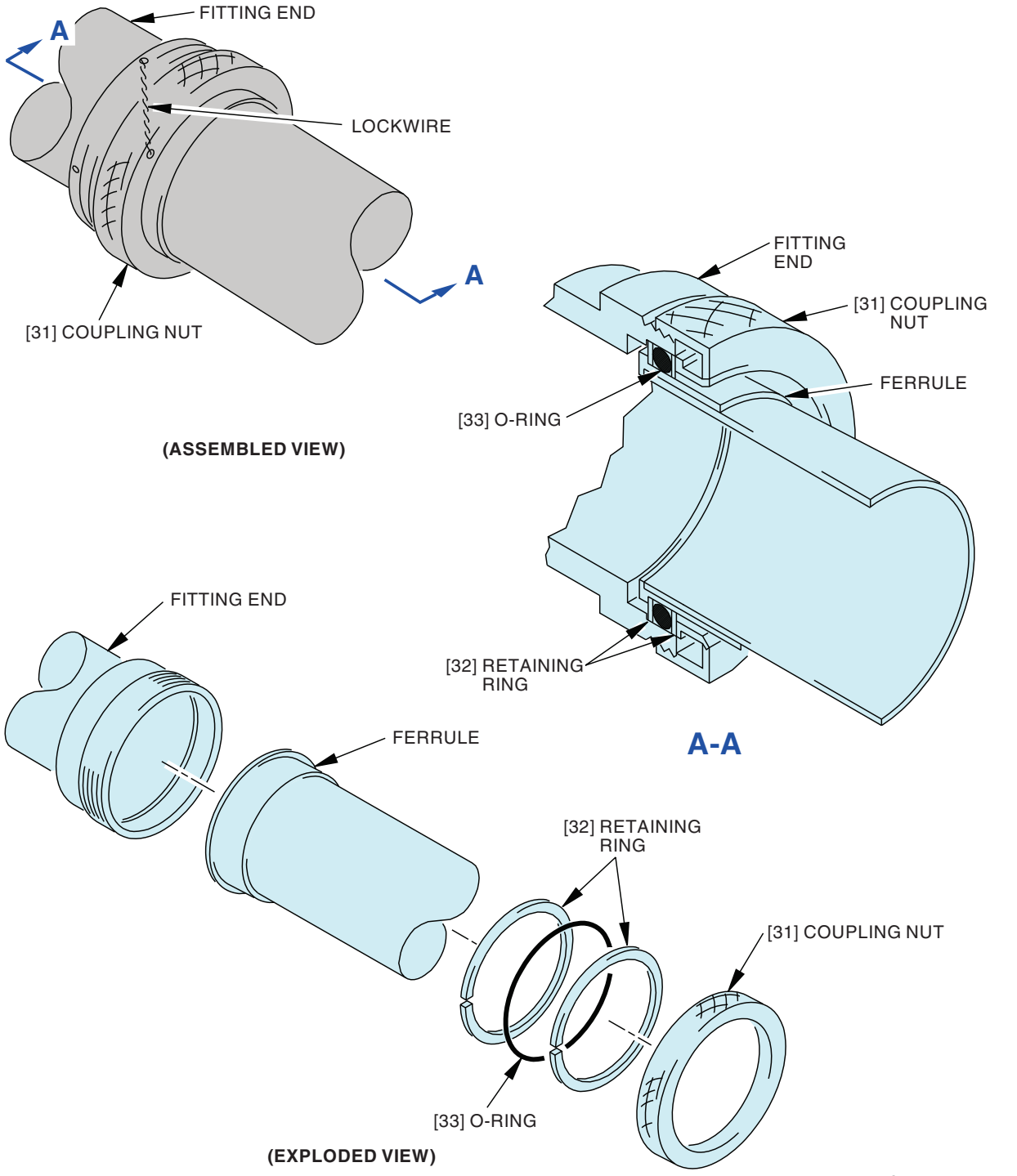
1) Discard the O-rings [81].

(e) Examine the coupling body [83] and the sleeve [84] for worn areas or damage.

1) Replace these components if it is necessary.

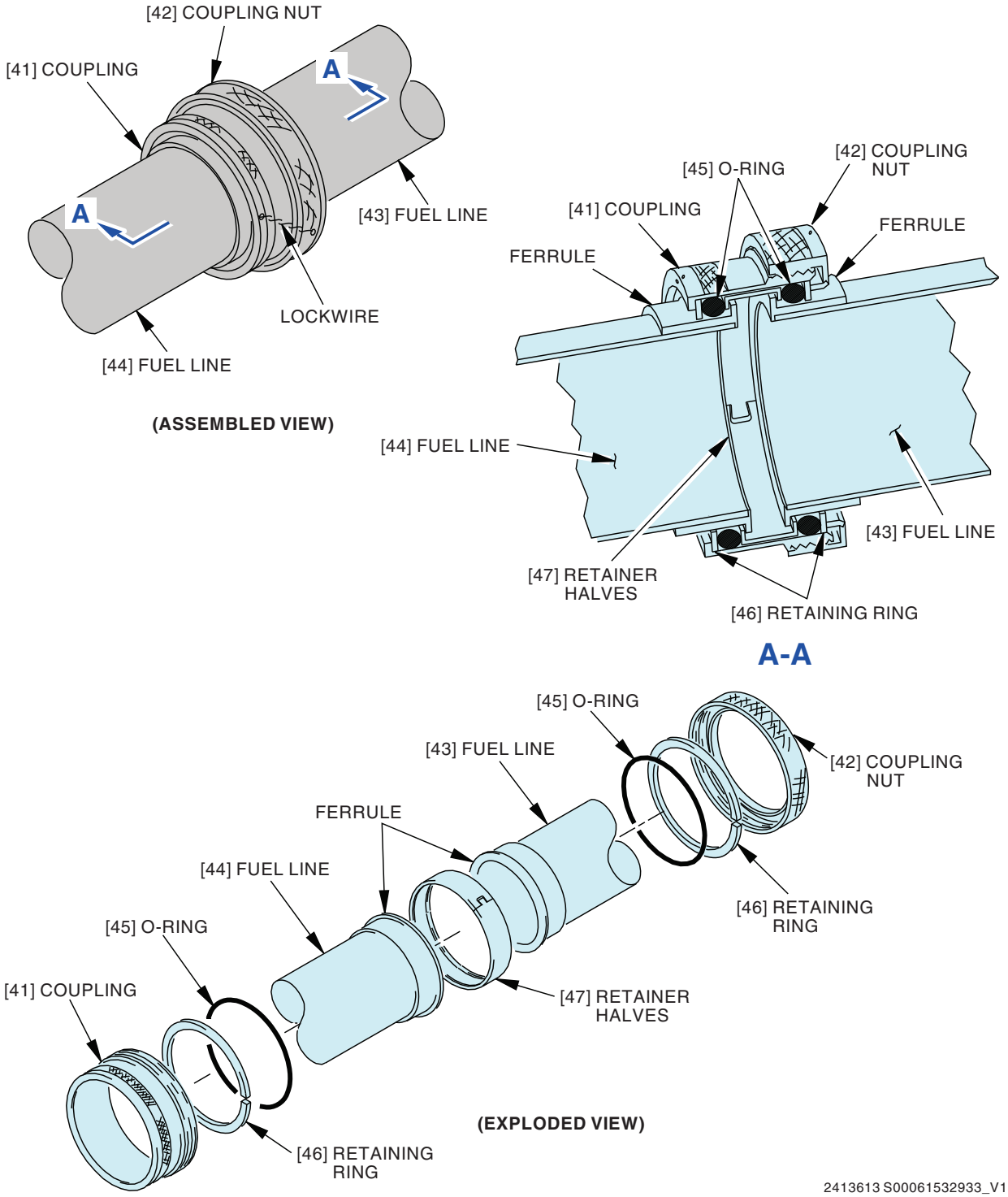
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



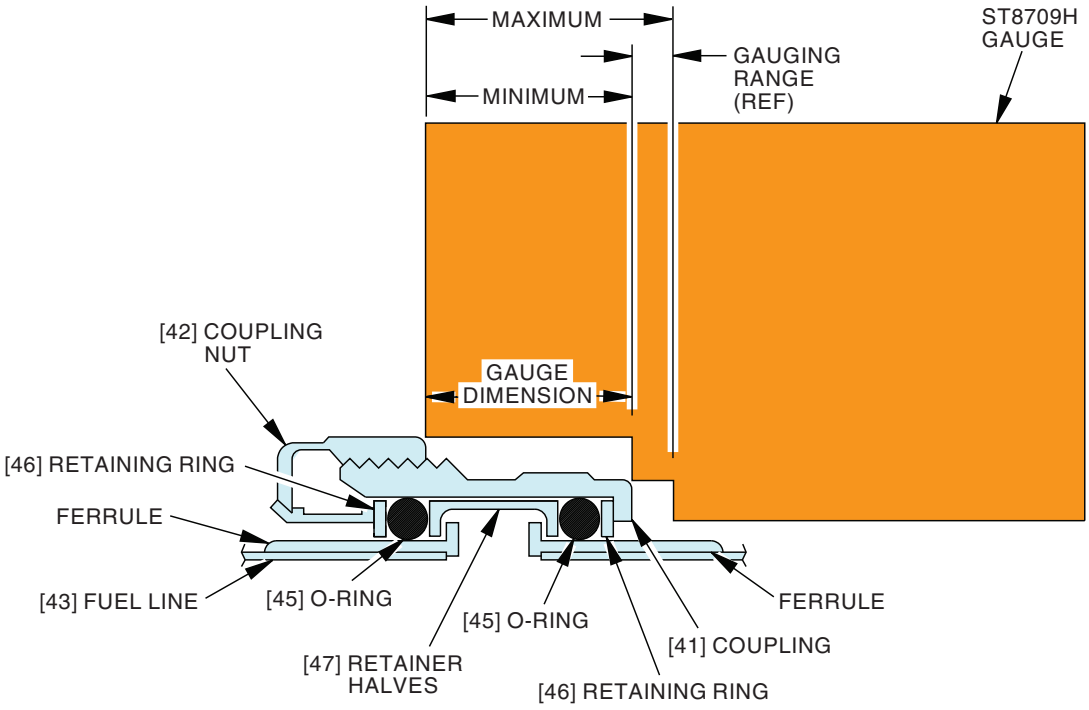
Flexible Half Coupling Installation
Figure 404/28-22-15-990-812

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flexible Full Coupling Installation
Figure 405/28-22-15-990-813 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLEXIBLE FULL COUPLING WITH GAUGE

NOTE:

GO/NO-GO GAUGE IS FOR WIGGINS BACC42R COUPLINGS.

ST8709H-X USAGE

TUBE DIA.	BACC42R SIZE	MAXIMUM GAUGE (IN.) 1	MINIMUM GAUGE (IN.) 1	ST8709H GAUGE NO.
1/2	08	0.590	0.470	-1
5/8	10	0.590	0.470	-1
3/4	12	0.690	0.550	-2
1	16	0.750	0.600	-3
1 1/4	20	0.750	0.600	-3
1 1/2	24	0.930	0.760	-4
1 3/4	28	0.930	0.760	-4
2	32	0.930	0.760	-4

ST8709H-X USAGE (CONT.)

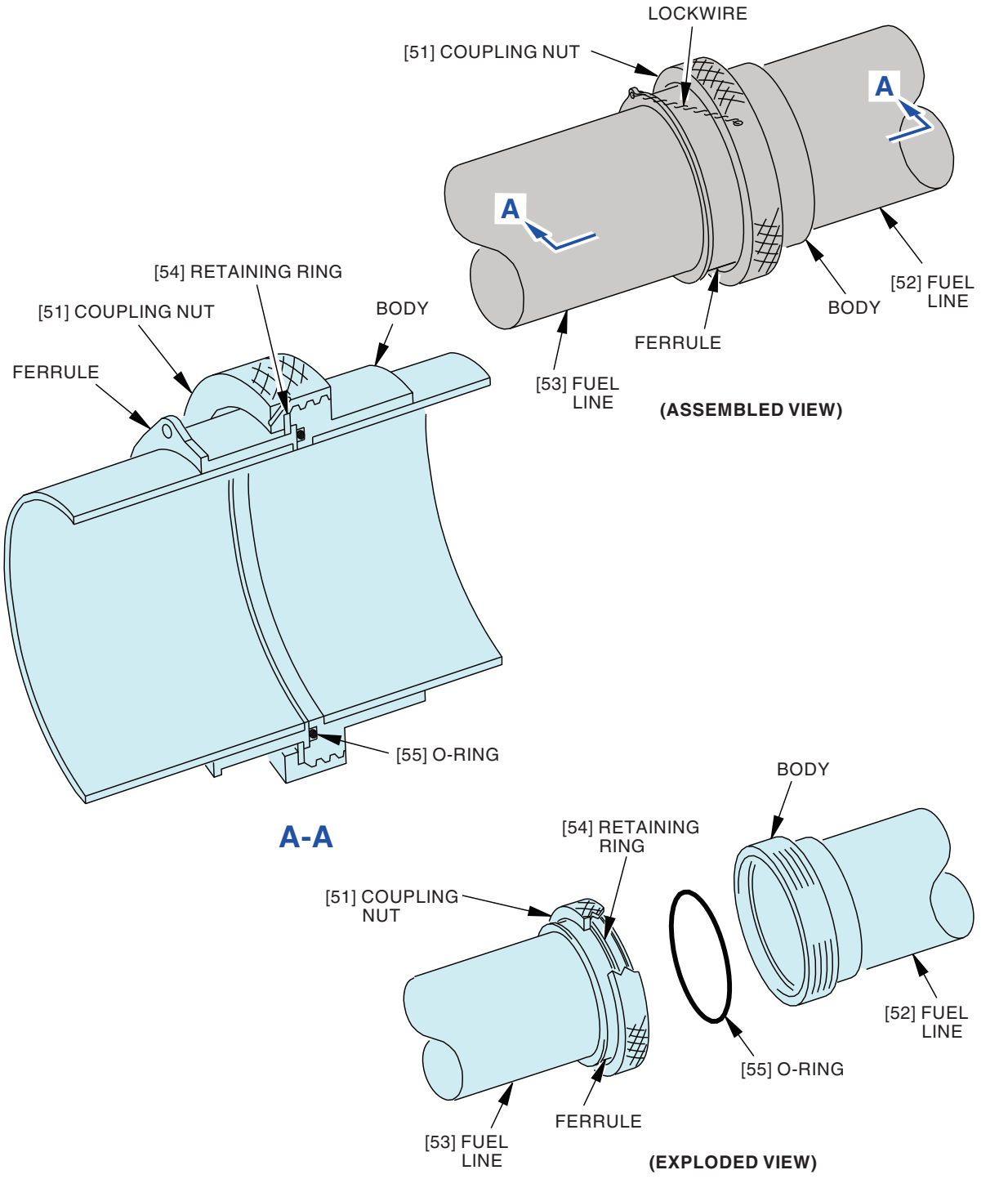
TUBE DIA.	BACC42R SIZE	MAXIMUM GAUGE (IN.) 1	MINIMUM GAUGE (IN.) 1	ST8709H GAUGE NO.
2 1/4	36	0.930	0.760	-4
2 1/2	40	0.930	0.760	-4
3	48	0.930	0.760	-4
3 1/2	56	1.030	0.880	-5
4	64	1.030	0.880	-5
4 1/2	72	1.160	0.990	-6
5	80	1.390	1.220	-7
5 1/2	88	1.390	1.220	-7

1 DIMENSIONAL REQUIREMENTS TO ASSEMBLE THE FLEXIBLE FULL COUPLING

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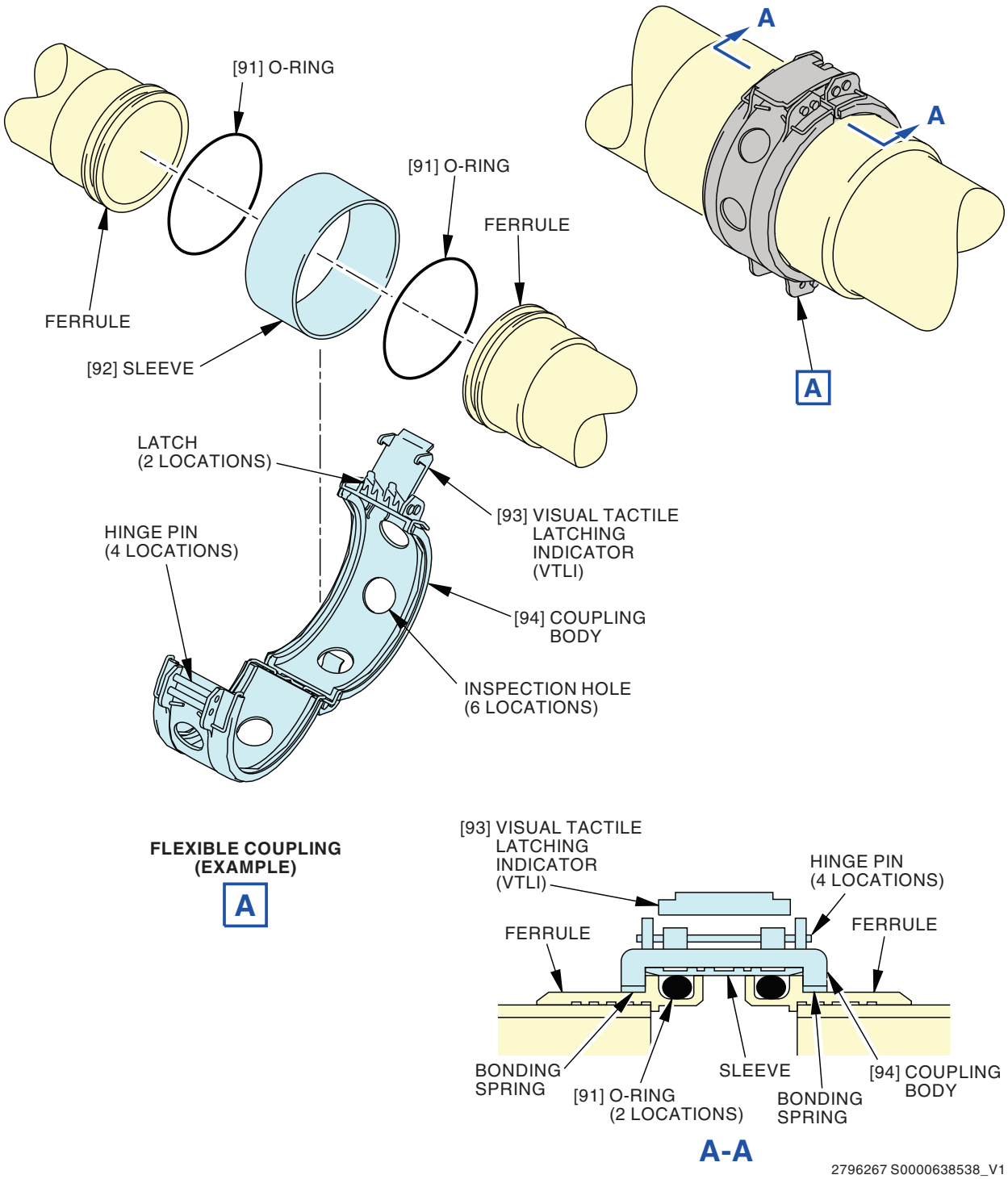
Flexible Full Coupling Installation
Figure 405/28-22-15-990-813 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



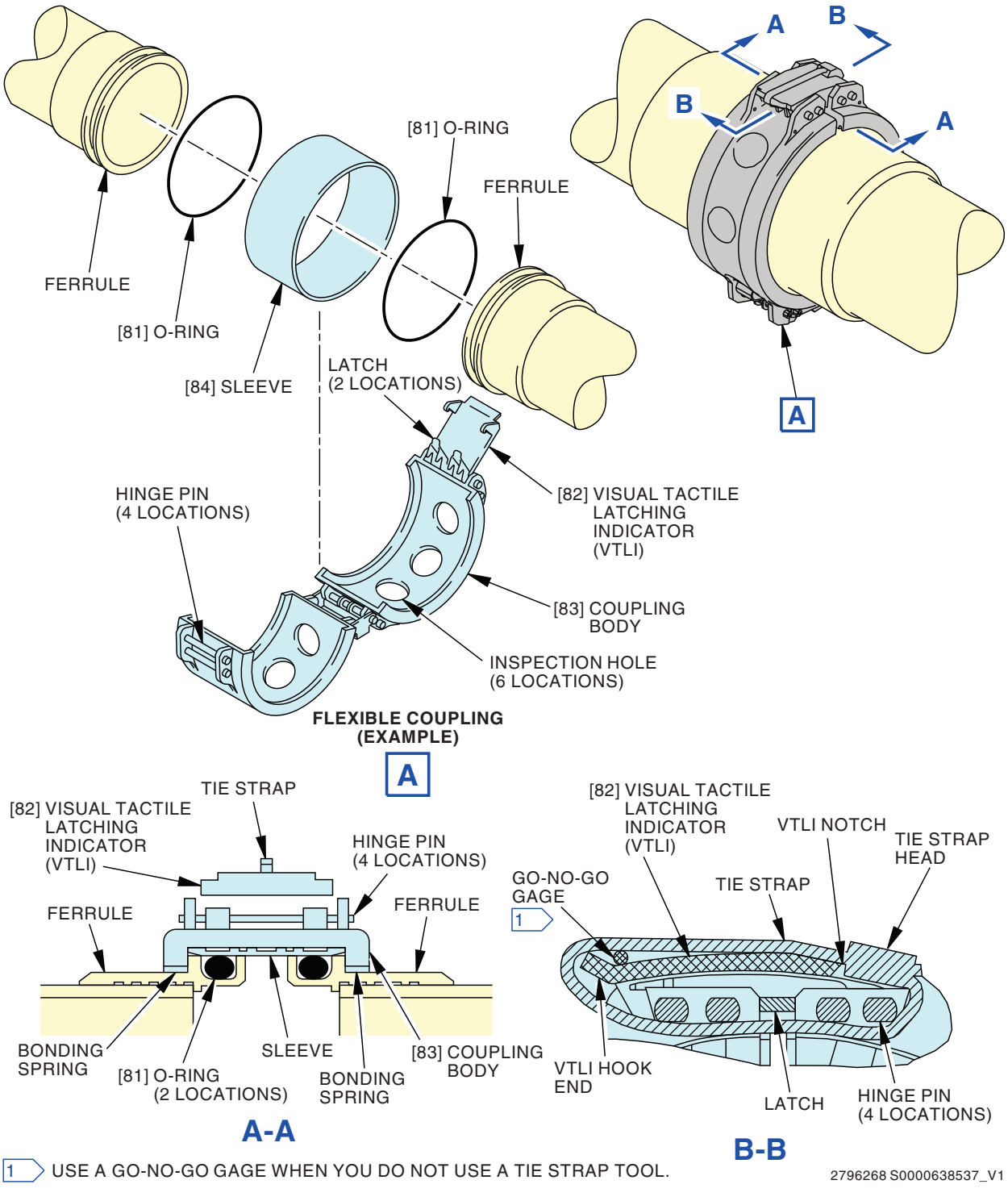
Rigid Coupling Installation
Figure 406/28-22-15-990-814

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Self-Bonding Flexible Coupling (Without Tie Strap) Installation
Figure 407/28-22-15-990-815

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Self-Bonding Flexible Coupling (With Tie Strap) Installation
Figure 408/28-22-15-990-816

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-15-400-803

7. Couplings Installation

(Figure 404, Figure 405, Figure 406, Figure 407, and Figure 408)

A. General

(1) This task includes the steps to install these couplings to fluid transfer tubes and ducts:

(a) Flexible half coupling

(b) Flexible full coupling

(c) Rigid coupling

(d) Self-Bonding Flexible Coupling (Without Tie Strap)

(e) Self-Bonding Flexible Coupling (With Tie Strap).

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
COM-3923	Tool - Installation, Tie Strap Part #: GS2B-E Supplier: 06383 Part #: GS4H-E Supplier: 06383 Part #: GTS-E Supplier: 06383 Part #: PPTS Supplier: 06383 Opt Part #: GS2B Supplier: 06383 Opt Part #: GS2BL Supplier: 06383 Opt Part #: GS4H Supplier: 06383 Opt Part #: GTS Supplier: 06383
SPL-7464	Go/No-Go Gage for Assembly of Wiggins BACC42R Coupling Part #: ST8709H-KIT (-1 through -7) Supplier: 62176
STD-13259	Gage - Wire, 0.098 in. dia. nominal to 0.101 in. dia. max.

D. Consumable Materials

Reference	Description	Specification
D00504	Grease - Petrolatum	VV-P-236
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51074	Strap - Plastic, Wire Harness Tie, Self-locking, 7.25 Inches (184.15 mm) Long	BACS38W1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
G51372	Kit - Safety Cable, 321 CRES - 0.035 +/-0.003 Inch (1.0 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K

E. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

F. Flexible Half Coupling Installation

(Figure 404)

SUBTASK 28-22-15-420-011

(1) Install the flexible half coupling on the fuel line, as follows:

(a) Put the coupling nut [31] on the fuel line with a ferrule.

CAUTION

DO NOT TWIST THE RETAINING RING OUT OF SHAPE. IF YOU TWIST THE RETAINING RING, FUEL LEAKS AT THE FITTING WILL OCCUR.

(b) Put a retaining ring [32] on the ferrule.

(c) Apply a thin layer of grease, D00504, on the new O-ring [33].

(d) Put the O-ring [33] on the ferrule.

CAUTION

DO NOT TWIST THE RETAINING RING OUT OF SHAPE. IF YOU TWIST THE RETAINING RING, FUEL LEAKS AT THE FITTING WILL OCCUR.

(e) Put the second retaining ring [32] on the ferrule.

(f) Align the ferrule with the fitting end.

CAUTION

DO NOT USE TOOLS TO TIGHTEN THE FITTINGS. DAMAGE TO THE FITTING COMPONENTS COULD OCCUR. THE ADJACENT FITTINGS COULD BECOME LOOSE.

(g) Manually tighten the coupling nut [31] to the fitting end.

SUBTASK 28-22-15-420-012

(2) Install a safety cable kit, G51372 or MS20995NC32 lockwire, G50347 between the coupling nut [31] and the ferrule (TASK 20-10-44-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Flexible Full Coupling Installation

(Figure 405)

SUBTASK 28-22-15-420-013

(1) Install a flexible full coupling on the fuel line, as follows:

(a) Put the coupling nut [42] on the fuel line [43].

(b) Put the coupling [41] on the fuel line [44].

CAUTION

DO NOT TWIST THE RETAINING RING OUT OF SHAPE. IF YOU TWIST THE RETAINING RING, FUEL LEAKS AT THE FITTING WILL OCCUR.

(c) Put one retaining ring [46] on the ferrule of fuel line [44] and fuel line [43].

(d) Apply a thin layer of grease, D00504, to the new O-rings [45].

CAUTION

DO NOT TWIST THE RETAINING RING OUT OF SHAPE. IF YOU TWIST THE RETAINING RING, FUEL LEAKS AT THE FITTING WILL OCCUR.

(e) Put one O-ring [45] on each ferrule.

(f) Align the ferrules of fuel line [44] and fuel line [43].

1) Make sure that the ferrules have a clearance of 0.12 ±0.03 in. (3.05 ±0.76 mm).

(g) Install the retainer halves [47] together on the ferrules to connect fuel line [44] and fuel line [43].

CAUTION

DO NOT USE TOOLS TO TIGHTEN THE FITTINGS. DAMAGE TO THE FITTING COMPONENTS COULD OCCUR. THE ADJACENT FITTINGS COULD BECOME LOOSE.

(h) Manually tighten the coupling nut [42] to the coupling [41].

1) Measure the gauge dimension on the Go/No-Go Gage, SPL-7464 tool to make sure that the flexible full coupling is in the gauging range.

NOTE: The flexible full coupling can catch and feel tight when the coupling nut [42] is not fully installed. The Go/No-Go Gage, SPL-7464, shows a coupling nut [42] that is not installed correctly.

SUBTASK 28-22-15-420-014

(2) Install a safety cable kit, G51372 or MS20995NC32 lockwire, G50347 between the coupling nut [42] and coupling [41] (TASK 20-10-44-400-801).

H. Rigid Coupling Installation

(Figure 406)

SUBTASK 28-22-15-020-021

(1) Loosen the adjacent clamps and couplings before you install the rigid coupling.

NOTE: This prevents a preload on the rigid coupling.

SUBTASK 28-22-15-420-015

(2) Install the rigid coupling as follows:

(a) Put the coupling nut [51] on the ferrule of the fuel line [53].

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CAUTION

DO NOT TWIST THE RETAINING RING OUT OF SHAPE. IF YOU TWIST THE RETAINING RING, FUEL LEAKS AT THE FITTING WILL OCCUR.

(b) Put the retaining ring [54] on the ferrule to hold the coupling nut [51] on the fuel line [53].

(c) Align the fuel line [52] and fuel line [53].

NOTE: The fuel lines must align correctly to prevent fuel leaks.

(d) Apply a thin layer of grease, D00504, to the new O-ring [55].

CAUTION

DO NOT USE THE SAME O-RING FOR THE FUEL LINE FITTINGS AFTER YOU TIGHTEN THE COUPLING ONE TIME. YOU MUST INSTALL NEW O-RINGS TO PREVENT FUEL LEAKS.

1) Install the O-ring [55] in the groove of the body on the fuel line [52].

CAUTION

DO NOT APPLY FORCE TO THE TUBE TO TRY TO ALIGN THE TUBE ENDS. TOO MUCH FORCE CAN CAUSE DAMAGE TO THE TUBES.

(e) Make sure that the fuel line [52] and fuel line [53] align correctly.

1) Make sure that the ferrule is parallel to the coupling body.

2) Adjust and tighten the adjacent clamps and couplings.

CAUTION

DO NOT USE TOOLS TO TIGHTEN THE FITTINGS. DAMAGE TO THE FITTING COMPONENTS COULD OCCUR. THE ADJACENT FITTINGS COULD BECOME LOOSE.

(f) Manually tighten the coupling nut [51] on the body until the coupling nut [51] is tight against the body.

SUBTASK 28-22-15-420-016

(3) Install a safety cable kit, G51372 or MS20995NC32 lockwire, G50347 between the coupling nut [51] and ferrule (TASK 20-10-44-400-801).

I. Self-Bonding Flexible Coupling (Without Tie Strap) Installation

(Figure 407)

SUBTASK 28-22-15-100-001

(1) Prepare a coupling body [94] for the installation of a self-bonding flexible coupling (SWPM 20-20-00):

(a) Clean the mating surfaces of these components:

1) The bonding springs on the coupling body [94].

2) The external mating surfaces of the ferrules.

3) The bonding surfaces of the sleeve [92].

SUBTASK 28-22-15-420-017

(2) Install the coupling body [94] as follows:

(a) Apply a thin layer of grease, D00504, to the two O-rings [91].

1) Install the two O-rings [91] in the grooves of the ferrules.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Apply a thin layer of grease, D00504, to the inner side of the sleeve [92].

1) Install the sleeve [92] on one of the O-rings [91].

(c) Make sure that the tubes align correctly.

1) If the two ferrules are male, the angle between the center lines of the tubes cannot be more than 4°arc (0 rad).

2) If one ferrule is female, the angle between the center lines of the tubes cannot be more than 2°arc (0 rad).

(d) Move the sleeve [92] on top of the two O-rings [91].

NOTE: A small movement from side to side can help the entry of the second O-ring [91].

(e) Install the coupling body [94] on the sleeve [92].

1) Put the open coupling body [94] on the sleeve [92].

2) Close the coupling body [94].

3) Make sure that the two latches engage correctly on the hinge pins.

4) Make sure that the coupling body [94] engages correctly on the ferrules.

5) Look through the inspection holes in the coupling body [94] to make sure that the sleeve [92] is installed correctly.

6) Manually turn the coupling body [94] to make sure that you installed it correctly.

SUBTASK 28-22-15-760-003

(3) Measure the electrical bonding resistance across the flexible coupling for the tube-to-tube, tube-to-component, or tube-to-isolator assembly with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) For a coupling body [94] that connects to an isolator, put the probe on the ferrule of the isolator.

(b) Make sure that the bonding resistance is 1.0 ohm or less.

NOTE: The probe distance between the coupling and tube is approximately 0.25 in. (6.35 mm).

J. Self-Bonding Flexible Coupling (With Tie Strap) Installation

(Figure 408)

SUBTASK 28-22-15-100-002

(1) Prepare a coupling body [83] for the installation of a self-bonding flexible coupling (SWPM 20-20-00):

(a) Clean the mating surfaces of these components:

1) The bonding springs on the coupling body [83].

2) The external mating surfaces of the ferrules.

3) The bonding surfaces of the sleeve [84].

SUBTASK 28-22-15-420-018

(2) Install the coupling body [83] as follows:

(a) Apply a thin layer of grease, D00504, to the two O-rings [81].

1) Install the two new O-rings [81] in the grooves of the ferrules.

(b) Apply a thin layer of grease, D00504, to the inner side of the sleeve [84].

1) Install the sleeve [84] on one of the O-rings [81].

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(c) Make sure that the tubes align correctly.

1) If the two ferrules are male, the angle between the center lines of the tubes cannot be more than 4°arc (0 rad).

2) If one ferrule is female, the angle between the center lines of the tubes cannot be more than 2°arc (0 rad).

(d) Move the sleeve [84] on top of the two O-rings [81].

NOTE: A small movement from side to side can help the entry of the second O-ring [81].

(e) Install the coupling body [83] on the sleeve [84].

1) Put the open coupling body [83] in its position on the sleeve [84].

a) If possible, put the coupling body [83] on the sleeve [84] with the notch of the visual tactile latching indicator (VTLI) [82] near you.

NOTE: This will help the installation of the tie strap.

2) Close the coupling body [83].

3) Make sure that the two latches engage correctly on the hinge pin.

4) Put the visual tactile latching indicator (VTLI) [82] to the closed position.

5) Make sure that the visual tactile latching indicator (VTLI) [82] engages correctly on the hinge pin.

6) Make sure that the coupling body [83] engages correctly on the ferrules.

7) Look through the inspection holes in the coupling body [83] to make sure that the sleeve [84] is installed correctly.

SUBTASK 28-22-15-420-019

CAUTION

MAKE SURE THAT YOU PUT THE STRAP IN THE CENTER LUGS BELOW ALL FOUR HINGE PINS. THE STRAP CAN MOVE ABOVE THE SECOND SET OF PINS. THESE WILL CAUSE THE VTLI TO BECOME UNSTABLE AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Install the tie strap, G51074on the coupling body [83], as follows:

(a) Put the end of the new tie strap, G51074, around the visual tactile latching indicator (VTLI) [82] and below the four hinge pins of the coupling body [83].

(b) Put the end of the tie strap, G51074, over the visual tactile latching indicator (VTLI) [82] and into the head of the tie strap, G51074.

1) Make sure that the head of the tie strap, G51074, is on the same side with the notch in the visual tactile latching indicator (VTLI) [82].

SUBTASK 28-22-15-420-020

CAUTION

DO NOT USE MORE THAN THE NECESSARY TENSION TO TIGHTEN THE STRAP ON THE COUPLING. TOO MUCH TENSION CAN CUT THE STRAP TOO SHORT AND CAUSE IT TO COME OFF DURING FLEX COUPLING MOTION OR WHEN EXPOSED TO AIRPLANE NORMAL VIBRATION. DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Tighten the tie strap with the tie strap installation tool, COM-3923 as follows:

NOTE: This is the preferred procedure.

(a) Adjust the tension setting of the tie strap.

1) Adjust the dial to MIN.

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2) Adjust the fine adjust to 1.

(b) Put the end of the tie strap, G51074, in the tie strap installation tool, COM-3923.

(c) Tighten the tie strap, G51074, with the tie strap installation tool, COM-3923.

1) Make sure that the head of the tie strap, G51074, is in the notch of the visual tactile latching indicator (VTLI) [82].

2) Make sure that the head of the tie strap, G51074, is not the tallest part of the coupling installation.

SUBTASK 28-22-15-420-021

CAUTION

DO NOT USE MORE THAN THE NECESSARY TENSION TO TIGHTEN THE STRAP ON THE COUPLING. TOO MUCH TENSION CAN CUT THE STRAP TOO SHORT AND CAUSE IT TO COME OFF DURING FLEX COUPLING MOTION OR WHEN EXPOSED TO AIRPLANE NORMAL VIBRATION. DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Tighten the tie strap without the tie strap installation tool, COM-3923, as follows:

NOTE: The installed components or structure can prevent the operation of the tool. This is an alternate procedure.

(a) Put the go-no-go gage, STD-13259, between the tie strap, G51074, and near the VTLI hook end.

(b) Tighten the tie strap, G51074, with your hand until the tie strap, G51074, touches the go-no-go gage, STD-13259.

1) Make sure that the head of the tie strap, G51074, is in the notch of the visual tactile latching indicator (VTLI) [82].

2) Make sure that the head of the tie strap, G51074, is not the tallest part of the coupling installation.

(c) Make sure that the go-no-go gage, STD-13259, is tight between the tie strap, G51074, and the visual tactile latching indicator (VTLI) [82].

(d) Remove the go-no-go gage, STD-13259.

SUBTASK 28-22-15-420-022

(6) Cut the loose end of the tie strap, G51074.

(a) Make sure that the cut end of the tie strap, G51074, is not more than 0.05 in. (1.27 mm) from the head of the tie strap, G51074.

(b) Make sure that the end of the tie strap, G51074, does not have sharp edges.

(c) Make sure that a minimum of 2 teeth show after the tooth engaged in the tie strap, G51074.

SUBTASK 28-22-15-420-023

(7) Manually turn the coupling body [83] to make sure that you installed it correctly.

SUBTASK 28-22-15-760-004

(8) Measure the electrical bonding resistance across the flexible coupling for the tube-to-tube, tube-to-component, or tube-to-isolator assembly with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) For a coupling body that connects to an isolator, put the probe on the ferrule of the isolator.

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(b) Make sure that the bonding resistance is 1.0 ohm or less.

NOTE: The probe distance between the coupling and tube is approximately 0.25 in. (6.35 mm).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL FEED MANIFOLD COUPLINGS - INSPECTION/CHECK

1. General

A. This procedure contains these tasks:

(1) Engine Fuel Feed Manifold - Leakage Test

(2) Engine and APU Fuel Feed, Shroud, Fuel Vent Line and Couplings, and NEADS Lines Dent Criteria - Inspection/Check

TASK 28-22-15-710-801

2. Engine Fuel Feed Manifold - Leak Test

(Figure 601, Figure 602, Figure 603, Figure 604)

A. General

(1) This leak test is a positive pressure test of the engine fuel feed manifold. This task isolates leakage in the engine fuel feed manifold.

(2) It is necessary to defuel the applicable fuel tank when there is a leak.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-22-11-000-801	Engine Fuel Spar Valve Actuator Removal (P/B 401)
28-22-11-000-803	Engine Fuel Spar Valve - Valve Body Removal (P/B 401)
28-22-11-400-801	Engine Fuel Spar Valve Actuator Installation (P/B 401)
28-22-11-400-803	Engine Fuel Spar Valve - Valve Body Installation (P/B 401)
28-22-11-820-801	Engine Fuel Spar Valve Alignment (P/B 401)
28-22-15-000-801	Fuel Line, Fitting and Coupling Removal (P/B 401)
28-22-15-400-801	Fuel Line, Fitting and Coupling Installation (P/B 401)
28-22-21-000-801	Engine Fuel Crossfeed Valve Actuator Removal (P/B 401)
28-22-21-000-803	Engine Fuel Crossfeed Valve - Valve Body Removal (P/B 401)
28-22-21-400-801	Engine Fuel Crossfeed Valve Actuator Installation (P/B 401)
28-22-21-400-803	Engine Fuel Crossfeed Valve - Valve Body Installation (P/B 401)
28-22-21-820-801	Engine Fuel Crossfeed Valve Alignment (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-26-00-650-802	Tank to Tank Fuel Transfer (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

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C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1776	Test Equipment - Engine Fuel Feed Manifold Part #: C28014-1 Supplier: 81205
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00504	Grease - Petrolatum	VV-P-236

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Preformed packing	73-11-03-01-030	PTF ALL

F. Location Zones

Zone	Area
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Prepare for the Test

SUBTASK 28-22-15-650-003

(1) Defuel each main tank until the fuel is below the spar valves (TASK 28-26-00-650-801 or TASK 28-26-00-650-802).

SUBTASK 28-22-15-650-004

(2) On the P5 Overhead Panel, make sure that the APU master switch is in the OFF position.

(a) Get access to the APU fuel shutoff valve through the left main landing gear wheel well.

(b) Make sure that the red manual override lever shows that the valve is closed.

NOTE: If the APU fuel shutoff valve is not closed, the valve can be inoperative.

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SUBTASK 28-22-15-650-005

(3) Make sure that the engine fuel valves are in the correct positions, as follows:

NOTE: There are two sets of engine fuel valves. The first set are the high pressure shutoff valves (HPSOV) which are installed on the engines. The second set are the spar valves which are installed in the main tanks. For this test, the two HPSOV are closed when the two engine fuel spar valves are open.

(a) On the P10 Control Stand, make sure that the Engine Start Levers are in the CUTOFF position.

(b) On the P5 Overhead Panel, make sure that the ENGINE START 1 and 2 switches are in the OFF position.

(c) Put a DO NOT OPERATE tag, STD-858, on the ENGINE START 1 and 2 switches.

(d) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

(e) On the P5-2 Fuel System Module, turn the CROSSFEED switch to the open position.

1) On the P5-2 Fuel System Module, make sure that the fuel crossfeed VALVE OPEN light comes on bright while the valve goes from the closed position to the open position.

2) Make sure that the fuel crossfeed VALVE OPEN light dims in 6 seconds.

(f) Put the Engine Start Levers to the IDLE position.

NOTE: This opens the engine fuel spar valves.

(g) On the P5-2 Fuel System Module, make sure that the left and right SPAR VALVE CLOSED lights goes off.

SUBTASK 28-22-15-860-001

(4) Make sure that the defueling valve handle is in the CLOSED position.

SUBTASK 28-22-15-010-002

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1

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(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 28-22-15-680-002

(6) Drain the fuel supply line:

(a) Put a fuel resistant container (5 gal), STD-201, below the fuel filter bowl.

NOTE: Have an extra fuel resistant container (5 gal), STD-201, in case it is necessary.

(b) Remove the drain plug [3] from the fuel filter bowl [1].

1) Cut and remove the safety wire from the drain plug [3].

2) Remove the drain plug [3] from the fuel filter bowl [1].

3) Let the fuel drain.

4) Remove and discard the preformed packing [2] from the groove of the drain plug [3].

SUBTASK 28-22-15-420-009

(7) Install the drain plug [3].

(a) Make sure that the drain plug [3] and fuel filter bowl [1] mating faces are clean and clear of unwanted materials.

(b) Lubricate a new preformed packing [2] with grease, D00504.

(c) Install the new preformed packing [2] onto the drain plug [3].

(d) Install the drain plug [3] in the fuel filter bowl [1].

(e) Tighten the drain plug [3] to 140 in-lb (15.8 N·m) - 150 in-lb (16.9 N·m).

(f) Install the safety wire on the drain plug [3].

SUBTASK 28-22-15-020-006

(8) Disconnect the fuel supply hose for the left and right engines at the disconnect panel (Figure 601).

SUBTASK 28-22-15-480-001

(9) Connect the manifold test equipment, SPL-1776, to each engine fuel supply line (Figure 603).

I. Engine Fuel Feed Manifold - Leak Test

SUBTASK 28-22-15-480-002

(1) Close the bleed valve on the manifold test equipment, SPL-1776, at the right engine.

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SUBTASK 28-22-15-780-001

(2) Use the regulator to apply 40 ±2 psig (276 ±14 kPa) air pressure to the fuel supply line at the left engine.

(a) Make sure that the air pressure is stable.

(b) Close the shutoff valve on the manifold test equipment, SPL-1776, at the left engine to hold a pressure of 40 ±2 psig (276 ±14 kPa).

SUBTASK 28-22-15-970-001

(3) Record the initial air pressure.

SUBTASK 28-22-15-860-004

(4) Wait 10 minutes.

SUBTASK 28-22-15-970-002

(5) Record the final air pressure.

SUBTASK 28-22-15-860-003

(6) Make sure that the air pressure did not decrease 1.5 psi (10.3 kPa) or more.

(a) If the air pressure did not decrease more than 1.5 psi (10.3 kPa) after 10 minutes, the fuel feed manifold has no leaks.

1) Do this procedure: Put the Airplane Back to Its Usual Condition.

(b) If the air pressure decreases more than 1.5 psi (10.3 kPa) in 10 minutes, there is a leak in the fuel feed manifold.

1) Do this procedure: Engine Fuel Feed Manifold - Leak Location Test.

J. Engine Fuel Feed Manifold - Leak Location Test

SUBTASK 28-22-15-790-001

(1) To find the general location of the fuel manifold leak, isolate the fuel feed manifold at the right spar valve, as follows (Figure 604):

(a) Close the right spar valve.

1) On the P10 Control Stand, put the Engine 2 Start Lever to the CUTOFF position.

(b) Do this procedure: Engine Fuel Feed Manifold - Leak Test.

1) If the air pressure does not decrease by more than 1.5 psi (10.3 kPa) in 10 minutes, the leak is between the right spar valve and the right engine fuel valve.

2) If the air pressure decreases more than 1.5 psi (10.3 kPa) in 10 minutes, continue.

SUBTASK 28-22-15-790-003

(2) Isolate the fuel feed manifold at the fuel crossfeed valve, as follows:

(a) On the P5-2 Fuel System Module, turn the CROSSFEED switch to the closed position.

(b) Do this procedure: Engine Fuel Feed Manifold - Leak Test.

(c) If the air pressure does not decrease by more than 1.0 psi (6.9 kPa) in 10 minutes, the leak is between the fuel crossfeed valve and the right spar valve.

(d) If the air pressure decreases more than 1.0 psi (6.9 kPa) in 10 minutes, continue.

SUBTASK 28-22-15-860-005

(3) Isolate the fuel feed manifold at the left spar valve, as follows:

(a) Close the left spar valve.

1) On the P10 Control Stand, put the Engine 1 Start Lever to the CUTOFF position.

(b) Open the right spar valve.

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1) On the P10 Control Stand, put the Engine 2 Start Lever to the IDLE or RUN position.

(c) On the P5-2 Fuel System Module, turn the CROSSFEED switch to the open position.

(d) Do this procedure: Engine Fuel Feed Manifold - Leak Test with the following conditions.

NOTE: Because of the thermal relief feature on the fuel spar valves when in the closed position, the spar valve is to be pressurized from the airplane side of the valve, rather than the engine side, for this procedure.

1) Use the regulator to apply 40 ±2 psig (276 ±14 kPa) air pressure to the fuel supply line at the right engine

2) Open the bleed valve on the manifold test equipment, SPL-1776, at the left engine supply line.

(e) If the air pressure does not decrease by more than 1.0 psi (6.9 kPa) in 10 minutes, the leak is between the left spar valve and the left engine fuel valve.

(f) If the air pressure decreases more than 1.0 psi (6.9 kPa) in 10 minutes, the leak is between the left spar valve and the test equipment on the left engine.

SUBTASK 28-22-15-360-001

(4) If there is a leak, do these steps:

(a) Turn the CROSSFEED valve to the open position.

(b) Open the left and right fuel spar valves.

(c) Open the bleed valve on the manifold test equipment, SPL-1776, to drain fuel and release the air pressure from the fuel supply line.

(d) Remove the two manifold test equipment, SPL-1776, from the fuel supply line.

(e) Connect the fuel supply hose to the engine fuel supply line (Figure 601).

(f) Defuel the applicable fuel tank (TASK 28-26-00-650-801).

(g) At the disconnect panel, disconnect the fuel supply hose for the No. 1 and No. 2 engine (Figure 601).

(h) Connect the manifold test equipment, SPL-1776, to each engine fuel supply line (Figure 603).

SUBTASK 28-22-15-360-002

(5) Find the correct location of the leak:

NOTE: The leak test found the general location of the leak.

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(a) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

1) Remove the applicable fuel tank access doors (TASK 28-11-11-000-801, TASK 28-11-31-000-801).

(b) Pressurize the fuel feed manifold at the inlet line for the left engine.

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WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Go in the applicable fuel tank to isolate the leak (TASK 28-11-00-910-802).

(d) Do a leak check of the fuel feed lines and couplings in the area where the pressure test failed.

CAUTION DO NOT USE THE SAME O-RING FOR THE FUEL LINE FITTINGS AFTER YOU TIGHTEN THE COUPLING ONE TIME. YOU MUST INSTALL NEW O-RINGS TO PREVENT FUEL LEAKS.

(e) If a coupling has a leak, remove and install the coupling.

1) Do this task: Fuel Line, Fitting and Coupling Removal, TASK 28-22-15-000-801.

2) Do this task: Fuel Line, Fitting and Coupling Installation, TASK 28-22-15-400-801.

(f) If a fuel line has a leak or is damaged, remove and then install a new fuel line.

1) Do this task: Fuel Line, Fitting and Coupling Removal, TASK 28-22-15-000-801.

2) Do this task: Fuel Line, Fitting and Coupling Installation, TASK 28-22-15-400-801.

SUBTASK 28-22-15-790-002

(6) After you repair a leak, do these steps:

(a) Pressurize the complete fuel feed manifold to 40 psi (276 kPa) from the inlet fuel line for the left engine to the right engine.

(b) If the pressure does not decrease by more than 1.5 psi (10.3 kPa) in 10 minutes, the system passed.

K. Right Fuel Spar Valve - Leak Test

SUBTASK 28-22-15-790-004

(1) To do a test of the right spar valve, put the valves to these positions:

NOTE: The valve must be electrically commanded to the open or closed position.

(a) On the P5 Overhead Panel, put the APU master switch to the OFF position.

1) Get access to the APU fuel shutoff valve through the left main landing gear wheel well.

2) Make sure that the red manual override lever shows that the valve is closed.

NOTE: If the APU fuel shutoff valve is not closed, the valve can be inoperative.

(b) On the P5-2 Fuel System Module, turn the CROSSFEED switch to the open position.

(c) On the P10 Control Stand, put the Engine 1 Start Lever to the IDLE or RUN position.

(d) On the P10 Control Stand, put the Engine 2 Start Lever to the CUTOFF position.

(e) Make sure that the defuel valve is in the closed position.

SUBTASK 28-22-15-780-002

(2) Use the regulator to apply 40 ±2 psig (276 ±14 kPa) air pressure to the fuel supply line at the left engine.

(a) Make sure that the air pressure is stable.

(b) Close the shutoff valve on the manifold test equipment, SPL-1776, at the left engine to hold a pressure of 40 ±2 psig (276 ±14 kPa).

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SUBTASK 28-22-15-970-003

(3) Record the initial air pressure.

SUBTASK 28-22-15-860-006

(4) Wait 10 minutes.

SUBTASK 28-22-15-970-004

(5) Record the final air pressure.

SUBTASK 28-22-15-790-005

(6) Make sure that the pressure is 40 psig (276 kPa) on the pressure gauge for the left engine pressure check equipment.

(a) If the pressure decreases, find the section of the leak: Engine Fuel Feed Manifold - Leak Test.

SUBTASK 28-22-15-790-006

(7) Make sure that the outlet pressure at the right engine disconnect is 0 psig (0 kPa).

SUBTASK 28-22-15-790-007

(8) If the outlet pressure is more than 0 psig (0 kPa), there is a leak in the right spar valve.

NOTE: The leak can be caused by the valve body, the actuator, or the adapter shaft is not aligned.

(a) Align the right fuel spar valve (Engine Fuel Spar Valve Alignment, TASK 28-22-11-820-801).

1) Do the right fuel spar valve - leak test procedure.

(b) If the leak test is not correct, replace the right fuel spar actuator:

1) Do this task: Engine Fuel Spar Valve Actuator Removal, TASK 28-22-11-000-801.

2) Do this task: Engine Fuel Spar Valve Actuator Installation, TASK 28-22-11-400-801.

3) Do the right fuel spar valve - leak test procedure.

(c) If the leak test is not correct, replace the right valve body:

1) Do this task: Engine Fuel Spar Valve - Valve Body Removal, TASK 28-22-11-000-803.

2) Do this task: Engine Fuel Spar Valve - Valve Body Installation, TASK 28-22-11-400-803.

3) Do the right fuel spar valve - leak test procedure.

L. Fuel Crossfeed Valve - Leak Test

SUBTASK 28-22-15-710-002

(1) To do a test of the fuel crossfeed valve, put the valves to these positions:

NOTE: The valves must be electrically commanded to the open or closed position.

(a) On the P5 Overhead Panel, put the APU master switch to the OFF position.

1) Get access to the APU fuel shutoff valve through the left main landing gear wheel well.

2) Make sure that the red manual override lever shows that the valve is closed.

NOTE: If the APU fuel shutoff valve is not closed, the valve can be inoperative.

(b) On the P5-2 Fuel System Module, turn the CROSSFEED switch to the closed position.

(c) On the P10 Control Stand, put the Engine 1 Start Lever to the IDLE or RUN position.

(d) On the P10 Control Stand, put the Engine 2 Start Lever to the IDLE or RUN position.

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(e) Make sure that the defuel valve is in the closed position.

SUBTASK 28-22-15-790-008

(2) Use the regulator to apply 40 ±2 psig (276 ±14 kPa) air pressure to the fuel supply line at the left engine.

(a) Make sure that the air pressure is stable.

(b) Close the shutoff valve on the manifold test equipment, SPL-1776, at the left engine to hold a pressure of 40 ±2 psig (276 ±14 kPa).

SUBTASK 28-22-15-970-005

(3) Record the initial air pressure.

SUBTASK 28-22-15-970-006

(4) Wait 10 minutes.

SUBTASK 28-22-15-970-007

(5) Record the final air pressure.

SUBTASK 28-22-15-790-009

(6) Make sure that the pressure is 40 psig (276 kPa) on the pressure gauge for the left engine pressure check equipment.

(a) If the pressure decreases, find the section of the leak: Engine Fuel Feed Manifold - Leak Test procedure.

SUBTASK 28-22-15-790-010

(7) Make sure that the outlet pressure at the right engine disconnect is 0 psig (0 kPa).

SUBTASK 28-22-15-790-011

(8) If the outlet pressure is more than 0 psig (0 kPa), there is a leak in the fuel crossfeed valve.

NOTE: The leak can be caused by the valve body, the actuator, or the adapter shaft is not aligned.

(a) Align the fuel crossfeed valve (Engine Fuel Crossfeed Valve Alignment, TASK 28-22-21-820-801).

1) Do the Fuel Crossfeed Valve - Leak Test procedure.

(b) If the leak test is not correct, replace the actuator:

1) Do this task: Engine Fuel Crossfeed Valve Actuator Removal, TASK 28-22-21-000-801.

2) Do this task: Engine Fuel Crossfeed Valve Actuator Installation, TASK 28-22-21-400-801.

3) Do the Fuel Crossfeed Valve - Leak Test procedure.

(c) If the leak test is not correct, replace the valve body:

1) Do this task: Engine Fuel Crossfeed Valve - Valve Body Removal, TASK 28-22-21-000-803.

2) Do this task: Engine Fuel Crossfeed Valve - Valve Body Installation, TASK 28-22-21-400-803.

3) Do the Fuel Crossfeed Valve - Leak Test procedure.

M. Left Fuel Spar Valve - Leak Test

SUBTASK 28-22-15-710-003

(1) To do a test of the left spar valve, put the valves to these positions:

NOTE: The valves must be electrically commanded to the open or closed position.

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(a) On the P5 Overhead Panel, put the APU master switch to the OFF position.

1) Get access to the APU fuel shutoff valve through the left main landing gear wheel well.

2) Make sure that the red manual override lever shows that the valve is closed.

NOTE: If the APU fuel shutoff valve is not closed, the valve can be inoperative.

(b) On the P5-2 Fuel System Module, turn the CROSSFEED switch to the open position.

(c) On the P10 Control Stand, put the Engine 1 Start Lever to the CUTOFF position.

(d) On the P10 Control Stand, put the Engine 2 Start Lever to the IDLE or RUN position.

(e) Make sure that the defuel valve is in the closed position.

SUBTASK 28-22-15-790-012

(2) Use the regulator to apply 40 ±2 psig (276 ±14 kPa) air pressure to the fuel supply line at the right engine.

(a) Make sure that the air pressure is stable.

(b) Close the shutoff valve on the manifold test equipment, SPL-1776, at the right engine to hold a pressure of 40 ±2 psig (276 ±14 kPa).

SUBTASK 28-22-15-970-008

(3) Record the initial air pressure.

SUBTASK 28-22-15-970-009

(4) Wait 10 minutes.

SUBTASK 28-22-15-970-010

(5) Record the final air pressure.

SUBTASK 28-22-15-790-013

(6) Make sure that the pressure is 40 psig (276 kPa) on the pressure gauge for the right engine pressure check equipment.

(a) If the pressure decreases, find the section of the leak: Engine Fuel Feed Manifold - Leak Test.

SUBTASK 28-22-15-790-014

(7) Make sure that the outlet pressure at the left engine disconnect is 0 psig (0 kPa).

SUBTASK 28-22-15-790-015

(8) If the outlet pressure is more than 0 psig (0 kPa), there is a leak in the left spar valve.

NOTE: The leak can be caused by the valve body, the actuator, or the adapter shaft is not aligned.

(a) Align the left fuel spar valve (TASK 28-22-11-820-801).

1) Do the left fuel spar valve leak test procedure.

(b) If the leak test is not correct, replace the left fuel spar valve actuator:

1) Do this task: Engine Fuel Spar Valve Actuator Removal, TASK 28-22-11-000-801.

2) Do this task: Engine Fuel Spar Valve Actuator Installation, TASK 28-22-11-400-801.

3) Do the left fuel spar valve leak test procedure.

(c) If the leak test is not correct, replace the left valve body:

1) Do this task: Engine Fuel Spar Valve - Valve Body Removal, TASK 28-22-11-000-803.

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2) Do this task: Engine Fuel Spar Valve - Valve Body Installation, TASK 28-22-11-400-803.

3) Do the left fuel spar valve leak test procedure.

N. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-15-410-003

(1) Remove the manifold test equipment, SPL-1776, from the engine fuel supply line, as follows:

(a) Open the bleed valve on the manifold test equipment, SPL-1776, to drain fuel and release the air pressure from the fuel supply line.

NOTE: Fuel can spill from the fuel supply line when the fuel has not been drained.

(b) Remove the two manifold test equipment, SPL-1776, from the fuel supply line.

(c) Connect the left and right engine fuel supply hose to the fuel supply line (Figure 601).

(d) On the P5 Overhead Panel, turn the CROSSFEED switch to the closed position.

(e) On the P10 Control Stand, put the Engine Start Levers to the CUTOFF position.

NOTE: This closes the fuel spar valves.

(f) On the P5 Overhead Panel, make sure that the two SPAR VALVE CLOSED lights are dim.

SUBTASK 28-22-15-760-002

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 28-22-15-410-004

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1

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(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 28-22-15-650-006

(4) On the P5 Overhead Panel, remove the DO-NOT-OPERATE placards from the ENGINE START 1 and 2 switches.

SUBTASK 28-22-15-650-007

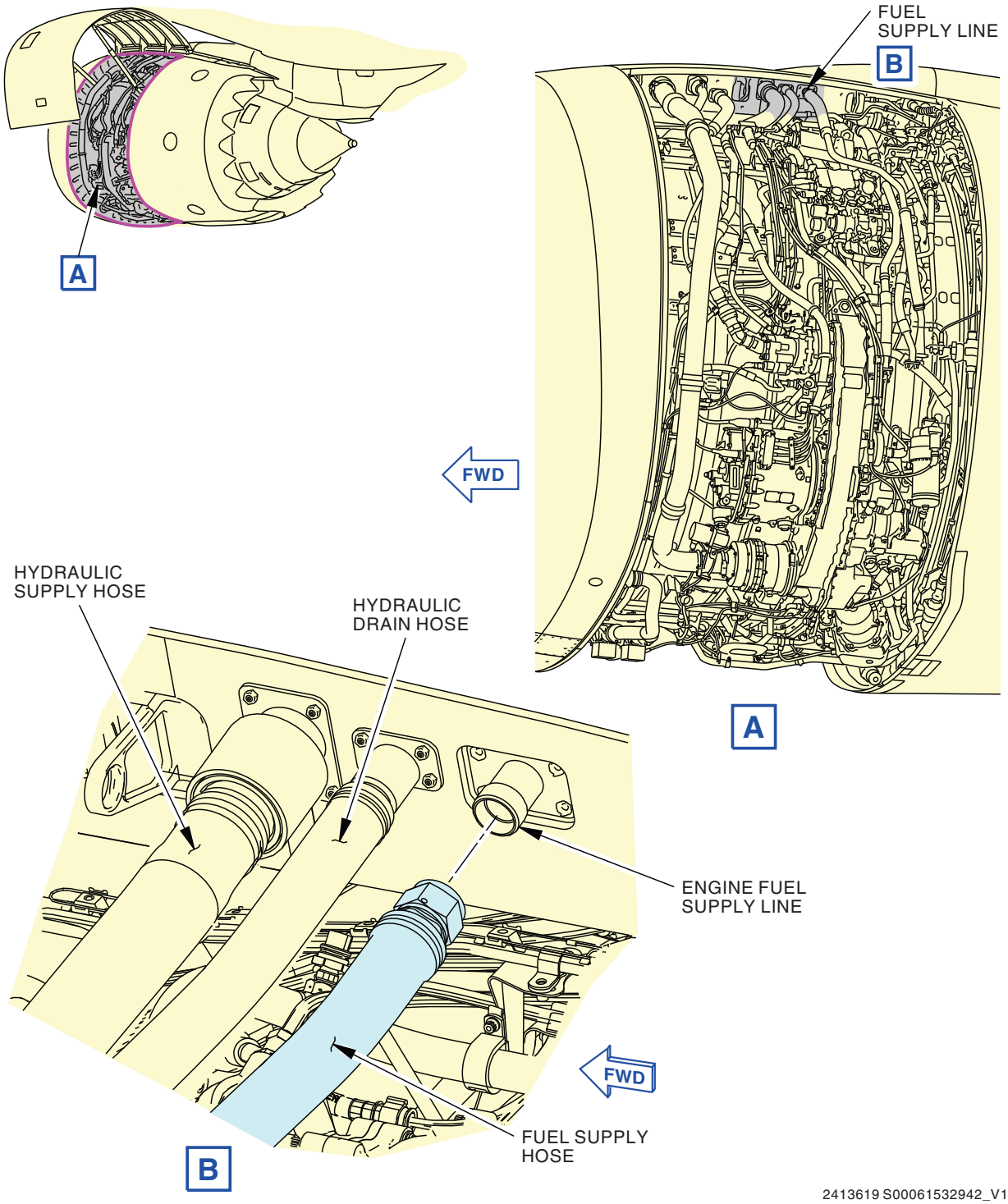
(5) Fuel the fuel tanks (TASK 12-11-00-650-803).

SUBTASK 28-22-15-700-001

(6) Do a leak check of the main fuel supply line (TASK 71-00-00-800-804-G00).

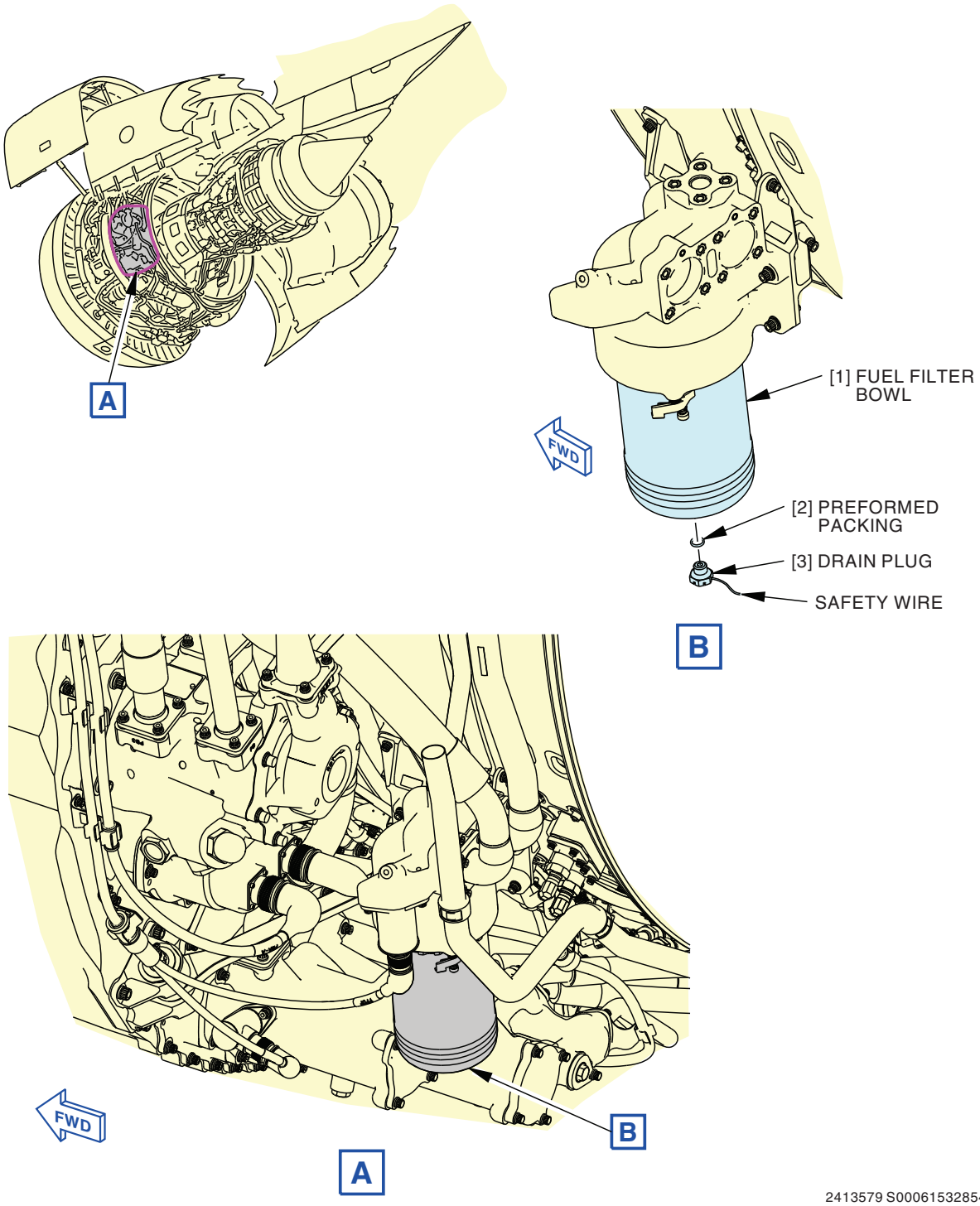
——— END OF TASK ———

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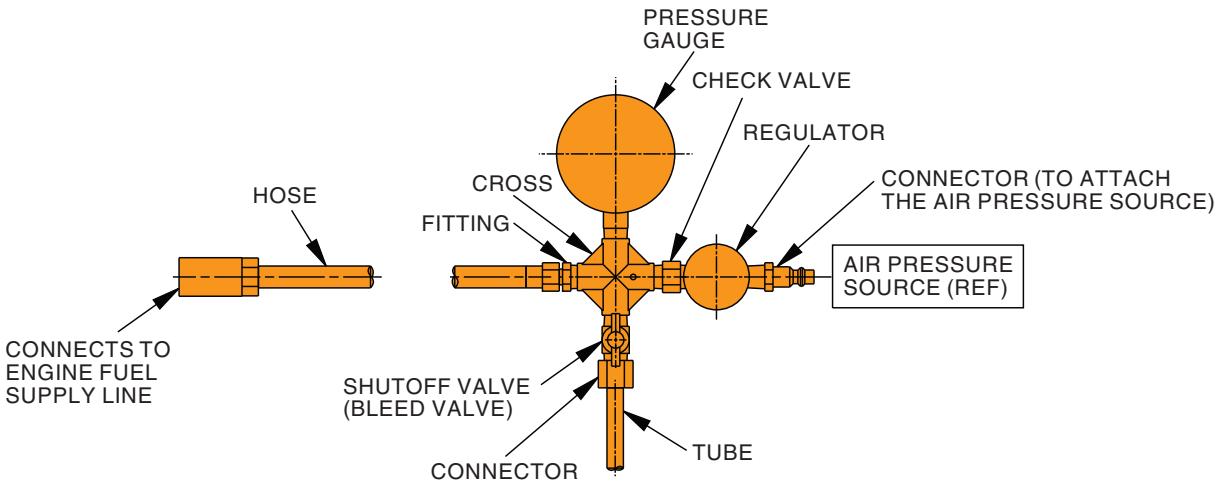
Fuel Supply Line Disconnect for Fuel-Feed Manifold Pressure Check
Figure 601/28-22-15-990-807

ECCN 9E991 BOEING PROPRIETARY - See title page for details

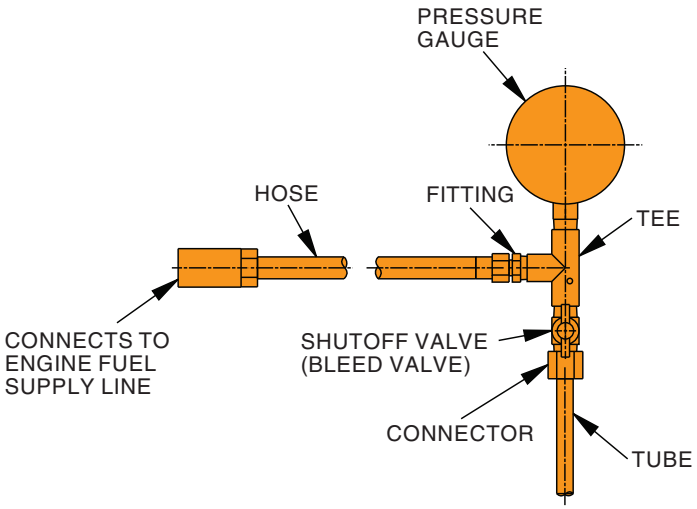


Fuel Filter Drain Location
Figure 602/28-22-15-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PRESSURE CHECK EQUIPMENT - LEFT FUEL SUPPLY LINE

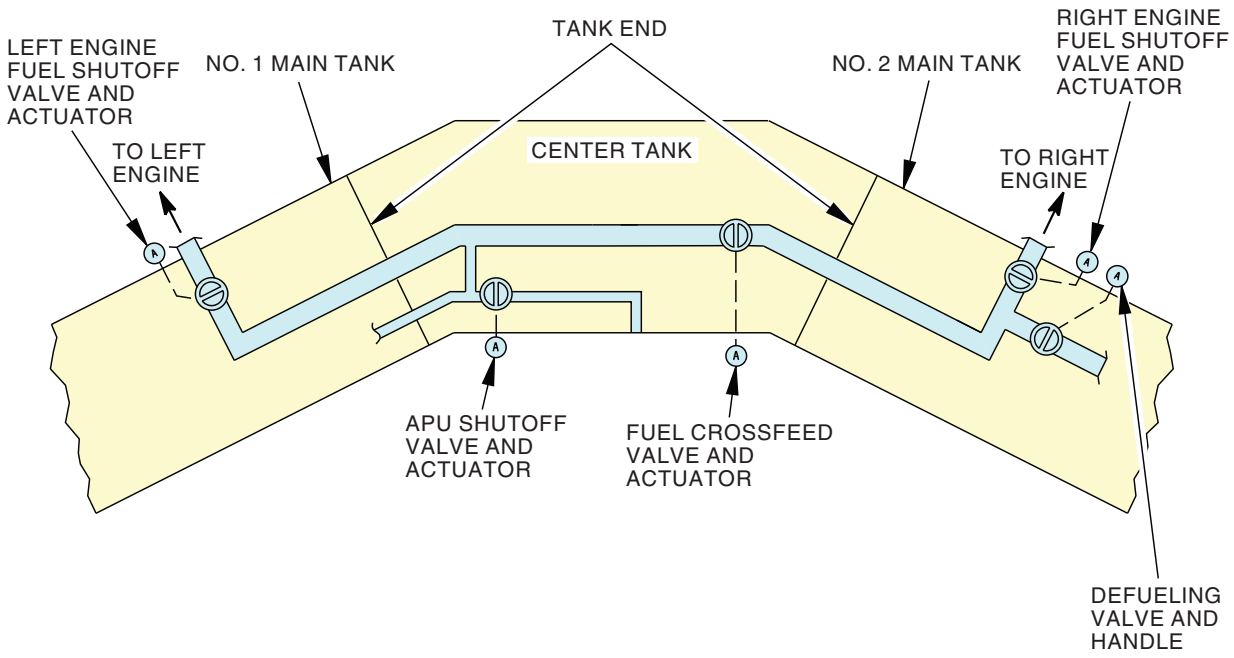


PRESSURE CHECK EQUIPMENT - RIGHT FUEL SUPPLY LINE

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Pressure Check Equipment
Figure 603/28-22-15-990-809

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413622 S00061532945_V1

Fuel Feed Manifold - Simplified Schematic
Figure 604/28-22-15-990-810

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-15-700-801

3. Engine and APU Fuel Feed, Shroud, Fuel Vent Line and Couplings, and NEADS Lines Dent Criteria - Inspection/Check

A. General

(1) The inspection criteria apply to:

(a) Engine fuel feed lines

(b) Pressure refueling lines

(c) APU shroud

(d) Fuel vent lines and related couplings

(e) NEADS lines.

(2) The engine fuel feed lines, pressure refueling lines, APU shrouds, and fuel vent lines are referred to as fuel lines in this task.

B. References

Reference	Title
20-10-51-000-801	Flareless Tubing Assembly Removal (P/B 401)
28-22-15-000-801	Fuel Line, Fitting and Coupling Removal (P/B 401)

C. Fuel Lines Inspection

SUBTASK 28-22-15-210-003

(1) Visually examine all fuel lines to make sure that they are tightly and correctly attached.

SUBTASK 28-22-15-420-008

(2) Adjust incorrectly attached fuel lines.

SUBTASK 28-22-15-210-004

(3) Visually examine all fuel lines for damage and then make sure that the damage is permitted.

(a) Do an inspection for cracks.

1) Make sure that there are no cracks.

(b) Do an inspection for dents.

NOTE: Dents are areas that are pushed in the fuel line.

1) All dents must have an area that is large compared to the depth of the dent.

2) A dent cannot decrease the diameter of the fuel line by more than 2.5 percent.

3) For several dents, the total dent depth (some dents depths added together) in a section, five times the fuel line diameter long, must not be more than 10 percent of the fuel line diameter.

4) Dents are not permitted near the fuel line flanges. The minimum dent distance is the diameter of the fuel line.

5) The dents in the fuel line must not have an area of more than 0.125 in^2 (80.6 mm^2).

6) For dents in the outer part of a fuel line bend, the dent depth must be 0.015 in. (0.38 mm) or less.

(c) Do an inspection for nicks.

1) The nicks in the fuel line must have a rounded bottom.

2) For nicks in the fuel line, the nick depth must be 0.004 in. (0.102 mm) or less.

3) If the nick has burrs, remove all burrs from the nicks.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Do an inspection for fretting.

1) Fretting must have a maximum depth of 0.004 in. (0.102 mm) or less.

(e) Do an inspection for scoring.

1) If some small scoring are found, you can make the scoring smooth, for these conditions:

a) The scoring must not be on the outer part of a bend.

b) You must not remove more than 0.004 in. (0.102 mm) of metal.

(f) Do an inspection for fuel lines with a cross section that is not fully circular (oval fuel lines).

1) Calculate the percent ovality with this formula:

percent ovality = [(OD Max - OD Min) ÷ (OD Nominal)] X 100%

a) OD Max is the maximum outer diameter of the fuel line

b) OD Min is the minimum outer diameter of the fuel line

c) OD Nominal is the outer diameter of the fuel line at a position where the fuel line is circular or almost circular.

2) The calculated percent ovality must not be more than 10 percent.

3) The calculated percent ovality must be more than 5 percent for fuel lines with a diameter of more than 2 in. (51 mm).

(g) Do an inspection for wrinkles.

NOTE: The height of a wrinkle is the distance between a straight line that touches the high point of a wrinkle and a straight line that touches the low point of a wrinkle.

1) Wrinkles must not have a height larger than the applicable maximum wrinkle height given in Table 601.

Table 601/28-22-15-993-801 Allowable Wrinkle Height

Fuel Line Diameter (inch (mm))	Maximum Wrinkle Height (inch (mm))
less than 1.00 in. (25.4 mm)	0.020 in. (0.508 mm)
1.00 in. (25.40 mm) to 1.99 in. (50.55 mm)	0.030 in. (0.762 mm)
2.00 in. (50.8 mm) to 2.99 in. (75.95 mm)	0.040 in. (1.016 mm)
3.00 in. (76.2 mm) or more	0.050 in. (1.270 mm)

SUBTASK 28-22-15-350-001

(4) Repair or replace all fuel lines that have damage that is not permitted (TASK 20-10-51-000-801, TASK 28-22-15-000-801).

D. Coupling Inspection

SUBTASK 28-22-15-210-005

(1) Visually examine all couplings for external indications of damage and make sure that the damage is permitted, as follows:

(a) Do an inspection for cracks.

1) No cracks are permitted.

(b) Do an inspection for scratches.

1) The scratches in the couplings must not have a depth of more than 0.004 in. (0.102 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Do an inspection for nicks.

1) The nicks in the couplings must have rounded bottoms.

2) The nicks in the couplings must not have a depth of more than 0.004 in. (0.102 mm).

3) Remove all the burrs around the nick.

(d) Do an inspection for worn areas (galling).

1) No worn areas in the surfaces that have an O-ring seal are permitted.

SUBTASK 28-22-15-350-002

(2) Replace couplings that have damage that is not permitted (TASK 28-22-15-000-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL SCAVENGE FLOAT OPERATED SHUTOFF VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Fuel Scavenge Float Operated Shutoff Valve Removal

(2) Fuel Scavenge Float Operated Shutoff Valve Installation.

TASK 28-22-16-000-801

2. Fuel Scavenge Float Operated Shutoff Valve Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the fuel scavenge float operated shutoff valve.

(2) The fuel scavenge float operated shutoff valve is referred to as shutoff valve in this task.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-26-00-650-802	Tank to Tank Fuel Transfer (P/B 201)

C. Location Zones

Zone	Area
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50

D. Access Panels

Number	Name/Location
532BB	Main Tank Access Door - Wing Station 265

E. Prepare for the Removal

SUBTASK 28-22-16-650-001

(1) Defuel or transfer fuel from the No. 1 main tank (TASK 28-26-00-650-802 or TASK 28-26-00-650-801).

SUBTASK 28-22-16-940-001

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task for the No. 1 main tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Open this access door:

(TASK 28-11-11-000-801)

Number	Name/Location
532BB	Main Tank Access Door - Wing Station 265

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-16-480-001

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Go into the fuel tank (TASK 28-11-00-910-802).

F. Fuel Scavenge Float Operated Shutoff Valve Removal

SUBTASK 28-22-16-020-001

(1) Remove the screw [1], washer [2], washer [4], and nut [3] that attaches the shutoff valve [7] and bonding jumper [6] to the support bracket.

SUBTASK 28-22-16-020-002

(2) Remove the three screws [1], six washers [2], and three nuts [3] that attaches the shutoff valve [7] to the support bracket.

SUBTASK 28-22-16-020-003

(3) Loosen the coupling nut [5] to disconnect the shutoff valve [7] from the fuel scavenge tube.

SUBTASK 28-22-16-480-002

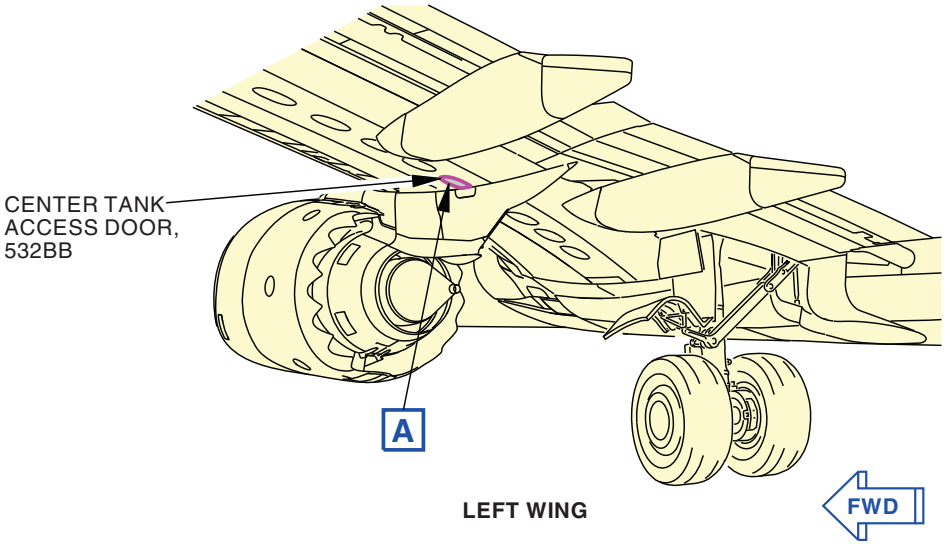
(4) Remove the shutoff valve [7].

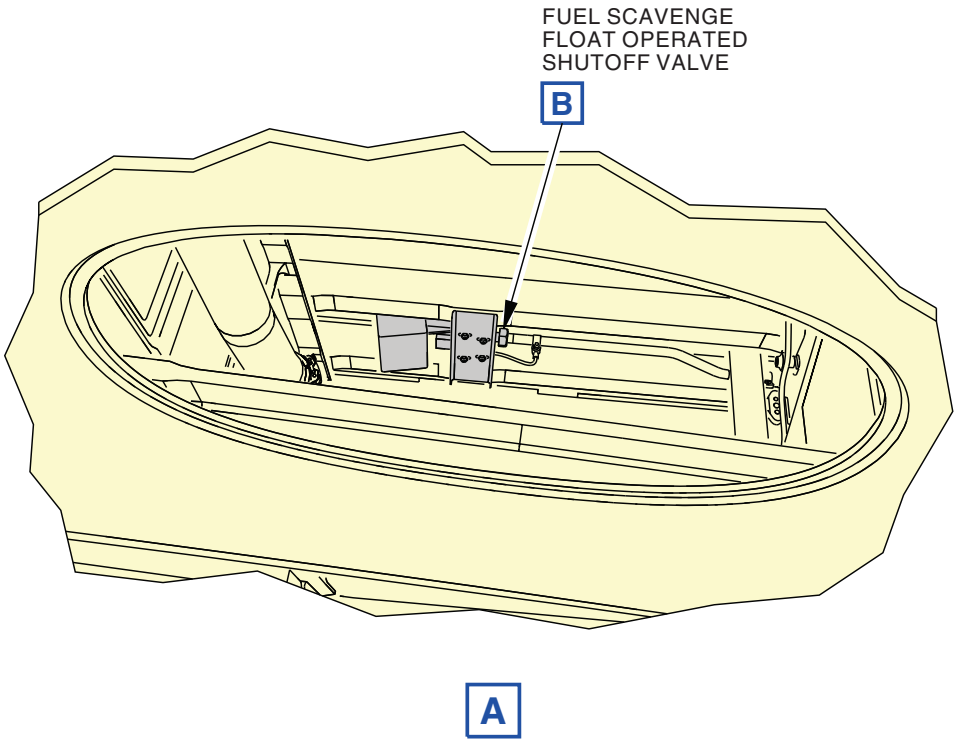
SUBTASK 28-22-16-020-004

(5) Put a cap on the open end of the fuel scavenge tube.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

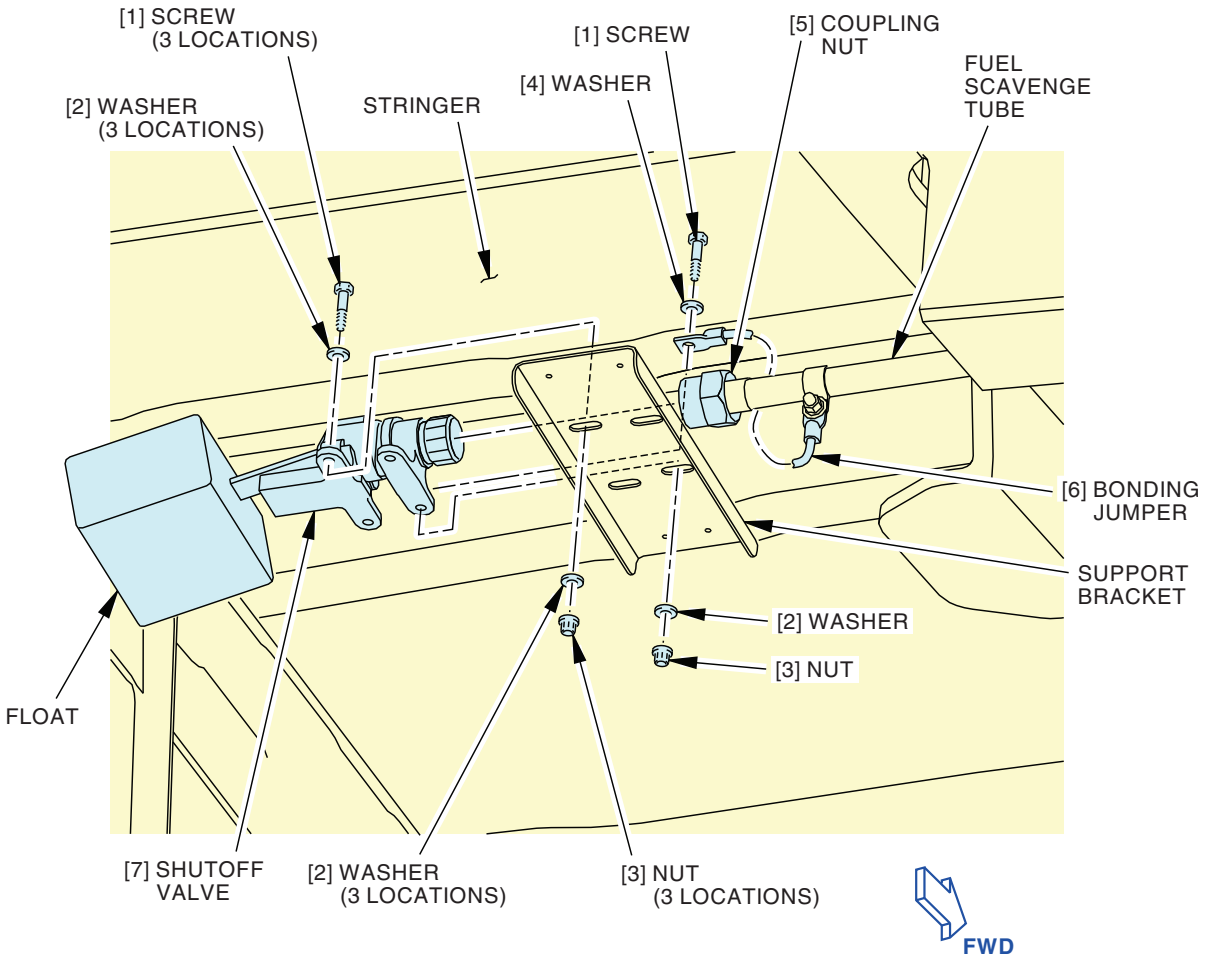




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Fuel Scavenge Float Operated Shutoff Valve Installation
Figure 401/28-22-16-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FUEL SCAVENGE FLOAT OPERATED SHUTOFF VALVE

B

2413624 S00061532951_V1

Fuel Scavenge Float Operated Shutoff Valve Installation
Figure 401/28-22-16-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-16-400-801

3. Fuel Scavenge Float Operated Shutoff Valve Installation

(Figure 401)

A. General

(1) This task gives instructions to install the fuel scavenge float operated shutoff valve.

(2) The fuel scavenge float operated shutoff valve is referred to as shutoff valve in this task.

B. References

Reference	Title
20-10-51-000-801	Flareless Tubing Assembly Removal (P/B 401)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
28-22-00-720-803	Fuel Scavenge System - Operational Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G51293	Cotton Wiper - Knit, Process Cleaning Absorbent Wiper	BMS15-5 Class B

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Shutoff valve	28-22-16-01-035	PTF ALL

F. Location Zones

Zone	Area
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Access Panels

Number	Name/Location
532BB	Main Tank Access Door - Wing Station 265

H. Fuel Scavenge Float Operated Shutoff Valve Installation

SUBTASK 28-22-16-100-001

(1) Do these steps to prepare the shutoff valve [7] and support plate for installation:

(a) Clean the fay surfaces of the aluminum shutoff valve [7] and aluminum support plate (SWPM 20-20-00):

1) Use Cleaning Procedure 5 to clean the fay surfaces on the shutoff valve [7].

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034 or knit cotton wiper, G51293 soaked with solvent, B00062 or solvent, B00184.

b) Use a new cotton wiper to dry the surfaces.

c) Continue to clean and dry the surfaces until the cotton wiper shows clean.

2) Use Cleaning Procedure 5 to clean the terminal fay surface on the bonding jumper [6].

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034 (alternate: knit cotton wiper, G51293) soaked with solvent, B00062 or solvent, B00184.

b) Use a new cotton wiper to dry the surfaces.

c) Continue to clean and dry the surfaces until the cotton wiper shows clean.

3) Use Cleaning Procedure 5 to clean the support plate fay surface.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034 (alternate: knit cotton wiper, G51293) soaked with solvent, B00062 or solvent, B00184.

b) Use a new cotton wiper to dry the surfaces.

c) Continue to clean and dry the surfaces until the cotton wiper shows clean.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-16-080-001

(2) Remove the cap on the open end of the fuel scavenge tube.

SUBTASK 28-22-16-420-001

(3) Loosely install the three screws [1], six washers [2], three nuts [3] and shutoff valve [7] to the support bracket.

SUBTASK 28-22-16-820-001

(4) Align the shutoff valve [7] with the fuel scavenge tube.

SUBTASK 28-22-16-420-002

(5) Tighten the coupling nut [5] to secure the shutoff valve [7] to the fuel scavenge tube (TASK 20-10-51-000-801).

SUBTASK 28-22-16-420-003

(6) Tighten the three screws [1].

SUBTASK 28-22-16-420-004

(7) Install the screw [1], washer [2], washer [4], bonding jumper [6], and nut [3] to secure the shutoff valve [7] to the support bracket.

SUBTASK 28-22-16-760-001

(8) Measure the electrical bonding resistance between the shutoff valve [7] and bonding jumper [6] with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohms) or less.

I. Fuel Scavenge Float Operated Shutoff Valve Installation Test

SUBTASK 28-22-16-410-001

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Install this access door:

(TASK 28-11-11-400-801)

Number	Name/Location
532BB	Main Tank Access Door - Wing Station 265

SUBTASK 28-22-16-700-001

(2) Do the operational test for the fuel scavenge system (TASK 28-22-00-720-803).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL SCAVENGE JET PUMP - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Fuel Scavenge Jet Pump Nozzle Assembly Removal

(2) Fuel Scavenge Jet Pump Nozzle Assembly Installation

(3) Fuel Scavenge Jet Pump Removal

(4) Fuel Scavenge Jet Pump Installation.

TASK 28-22-17-020-801

2. Fuel Scavenge Jet Pump Nozzle Assembly Removal

(Figure 401)

A. General

(1) The fuel scavenge jet pump nozzle assembly is referred to as the nozzle assembly in this task.

(2) The objective of the task is to provide instructions to remove the nozzle assembly.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
551	Left Wing - Rear Spar To Landing Gear Support Beam

D. Prepare for the Removal

SUBTASK 28-22-17-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-22-17-010-001

(2) Get access to the fuel scavenge jet pump on the aft side of the rear spar.

E. Nozzle Assembly Removal

SUBTASK 28-22-17-010-002

(1) Remove the safety cable (or lockwire) from the nozzle assembly [2].

SUBTASK 28-22-17-020-001

(2) Remove the nozzle assembly [2] as follows:

NOTE: The nozzle assembly [2] consists of a pump plug and pump nozzle that are removed as one unit.

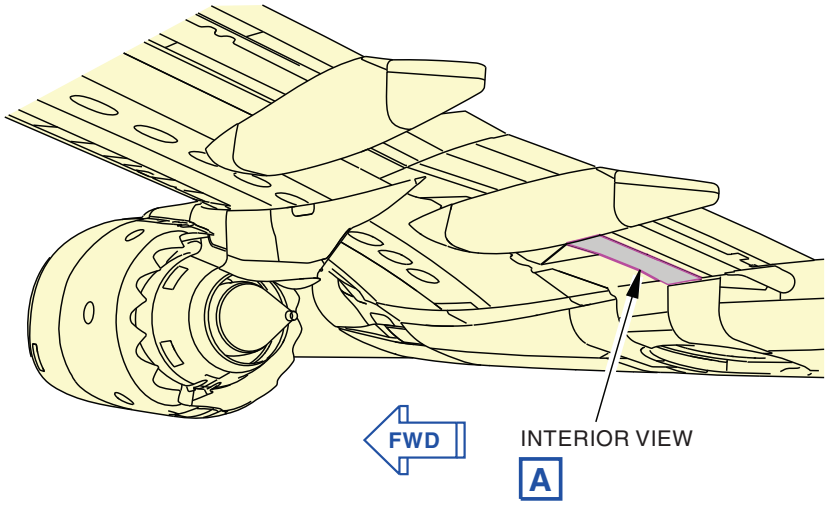
(a) Remove the nozzle assembly [2] and O-ring [3] from the pump housing [1].

NOTE: An internal check valve in the pump housing [1] prevents fuel spills.

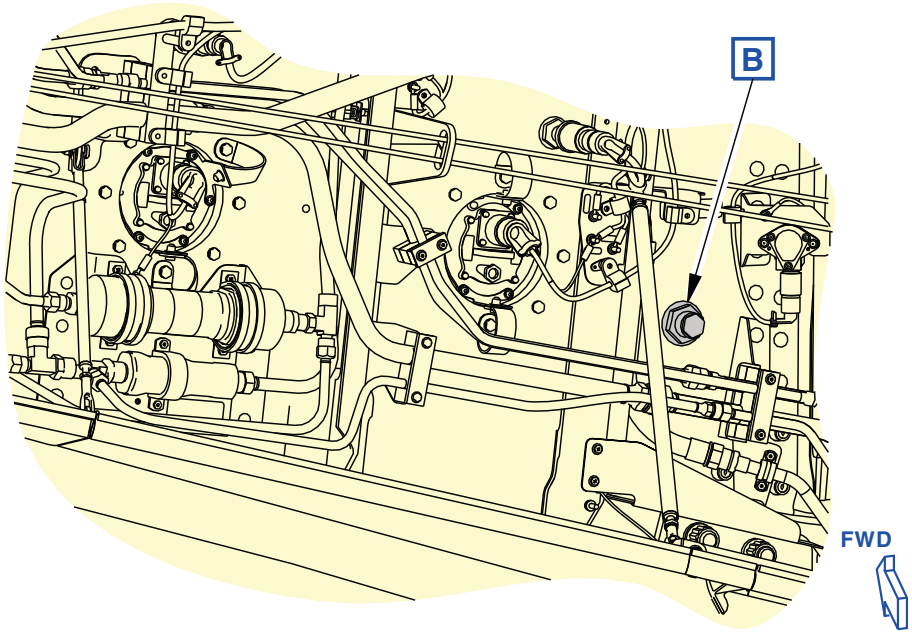
(b) Discard the O-ring [3].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS EQUIVALENT)

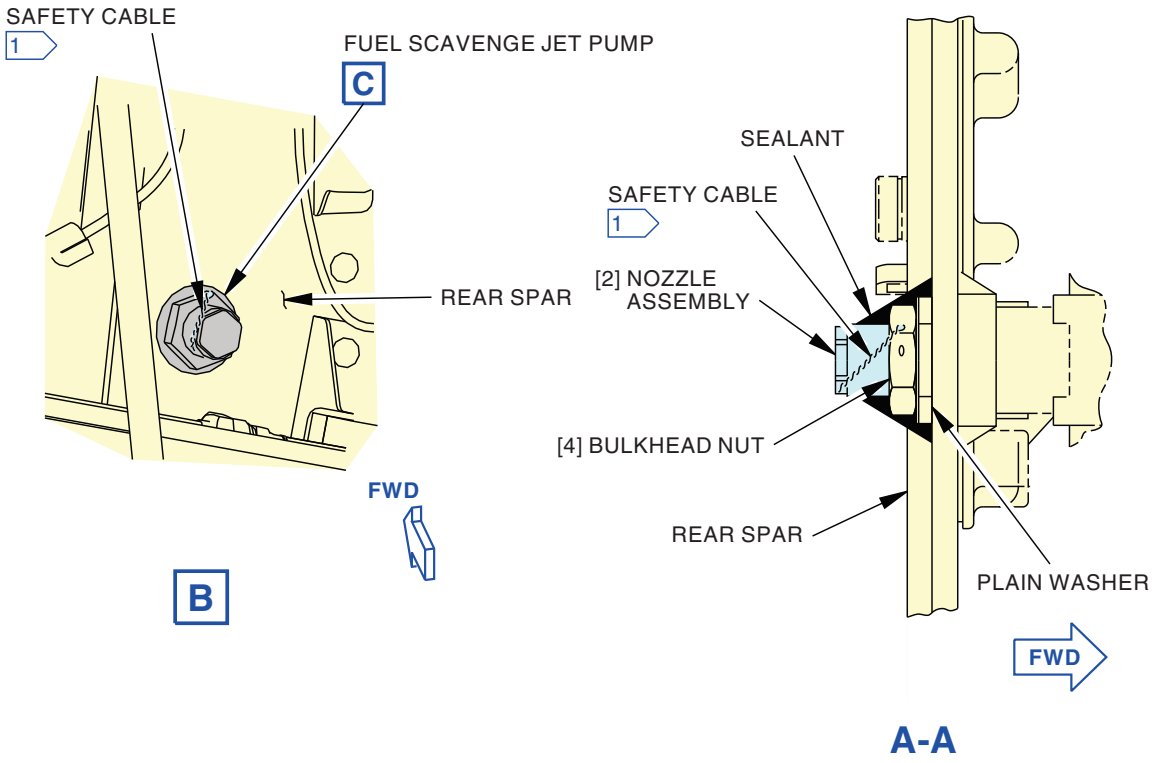


INTERIOR VIEW
A

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Fuel Scavenge Jet Pump Nozzle Assembly Installation
Figure 401/28-22-17-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Scavenge Jet Pump Nozzle Assembly Installation
Figure 401/28-22-17-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-17-420-801

3. Fuel Scavenge Jet Pump Nozzle Assembly Installation

(Figure 401)

A. General

(1) The fuel scavenge jet pump nozzle assembly is referred to as the nozzle assembly in this task.

(2) The objective of the task is to provide instructions to install the nozzle assembly.

B. References

Reference	Title
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
51-22-00-390-806	Fillet Seal Application (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B50226	Solvent - Citra-Safe	BAC5750
D00504	Grease - Petrolatum	VV-P-236
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50375	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule)	BACC13AT3K, AMS 5689

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	O-ring	28-22-17-01-147	PTF ALL

E. Location Zones

Zone	Area
551	Left Wing - Rear Spar To Landing Gear Support Beam

F. Prepare for the Installation

SUBTASK 28-22-17-100-001

(1) Do these step to prepare the nozzle assembly [2] and the rear spar for installation:

(a) Remove the sealant from the rear spar, do this task: Clean Bare, Clad, or Plated Metal with Solvent, TASK 20-10-34-110-802.

(b) Clean the fasteners and the fay surfaces of the aluminum pump housing [1].

NOTE: Clean the surface beyond the area to be sealed to prevent contamination.

1) Use a new cotton wiper, G00034, made wet with a citra-safe solvent, B50226 (alternate: solvent, B00184).

2) Dry the area with a clean cotton wiper, G00034.

SUBTASK 28-22-17-210-001

(2) Examine the nozzle assembly [2] for these conditions:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) pump nozzle [2A]

1) Nicks or scratches on the conical taper
2) Nicks or scratches on the orifice tip
3) Worn surface areas through the anodize finish.

(b) pump plug [2B]

1) Crossed, stripped or damaged threads
2) Nicks or scratches on the packing groove
3) Wear on the hex head
4) Worn surface areas through the anodize finish.

SUBTASK 28-22-17-960-001

(3) If the pump plug [2B] is damaged or worn, replace it.

(a) Remove the pump nozzle [2A] from the pump plug [2B].
(b) Screw the new pump nozzle [2A] in the pump plug [2B].
(c) Tighten the pump plug [2B] to 90 ±5 in-lb (10.2 ±0.6 N·m).

SUBTASK 28-22-17-960-002

(4) If the pump nozzle [2A] is damaged or worn, replace it.

(a) Remove the pump nozzle [2A] from the pump plug [2B].
(b) Screw the new pump plug [2B] in the pump nozzle [2A].
(c) Tighten the pump plug [2B] to 90 ±5 in-lb (10.2 ±0.6 N·m).

SUBTASK 28-22-17-640-001

(5) Apply a thin layer of petrolatum grease, D00504, to the new O-ring [3].

SUBTASK 28-22-17-420-001

(6) Install the O-ring [3] on the nozzle assembly [2].

G. Fuel Scavenge Jet Pump Nozzle Assembly Installation

SUBTASK 28-22-17-420-002

(1) Install the nozzle assembly [2] and O-ring [3] in the pump housing [1].

NOTE: The nozzle assembly [2] consists of a pump plug [2B] and pump nozzle [2A] that are remove as one unit.

SUBTASK 28-22-17-420-003

(2) Tighten the nozzle assembly [2] to 200 ±10 in-lb (22.6 ±1.1 N·m).

SUBTASK 28-22-17-420-004

(3) Install safety cable kit, G50375 (alternate: lockwire) between the nozzle assembly [2] and the bulkhead nut (TASK 20-10-44-400-801).

SUBTASK 28-22-17-420-005

(4) Apply sealant, A02315, between the rear spar and pump housing [1] (TASK 51-22-00-390-806).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-17-000-801

4. Fuel Scavenge Jet Pump Removal

(Figure 401)

A. General

(1) The objective of the task is to provide instructions to remove the fuel scavenge jet pump.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-22-15-000-803	Couplings Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-26-00-650-802	Tank to Tank Fuel Transfer (P/B 201)

C. Tools/Equipment

Reference	Description
STD-7423	Cover - Protective Tube

D. Location Zones

Zone	Area
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
551	Left Wing - Rear Spar To Landing Gear Support Beam

E. Access Panels

Number	Name/Location
531AB	Center Tank Access Door - Wing Station 168

F. Prepare for the Removal

SUBTASK 28-22-17-020-002

(1) Remove the nozzle assembly [2] (TASK 28-22-17-020-801).

SUBTASK 28-22-17-650-001

(2) Defuel or transfer fuel from the center tank and No. 1 main tank (TASK 28-26-00-650-801 or TASK 28-26-00-650-802).

SUBTASK 28-22-17-480-002

WARNING DO THE PURGING AND FUEL TANK ENTRY PROCEDURE. FAILURE TO FOLLOW THE PURGING AND FUEL TANK ENTRY PROCEDURE COULD CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(3) Prepare the center tank for entry, do this task: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

SUBTASK 28-22-17-010-003

(4) Open this access panel:

(TASK 28-11-31-000-801)

Number	Name/Location
531AB	Center Tank Access Door - Wing Station 168

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Fuel Scavenge Jet Pump Removal

SUBTASK 28-22-17-020-003

(1) Remove the bulkhead nut [4] and plain washer [20] from the pump housing [1], from out of the fuel tank.

SUBTASK 28-22-17-940-001

(2) Go into the center tank.

SUBTASK 28-22-17-120-001

(3) Remove the sealant from the rear spar and pump housing [1] in the fuel tank.

SUBTASK 28-22-17-020-004

(4) Disconnect the rigid coupling [13] from the pump housing [1] (TASK 28-22-15-000-803).

(a) Discard the O-ring [14].

SUBTASK 28-22-17-020-005

(5) Disconnect the coupling nut [18] from the pump housing [1].

SUBTASK 28-22-17-020-006

(6) Disconnect the coupling nut [15] from the pump housing [1].

SUBTASK 28-22-17-020-007

(7) Remove the pump housing [1].

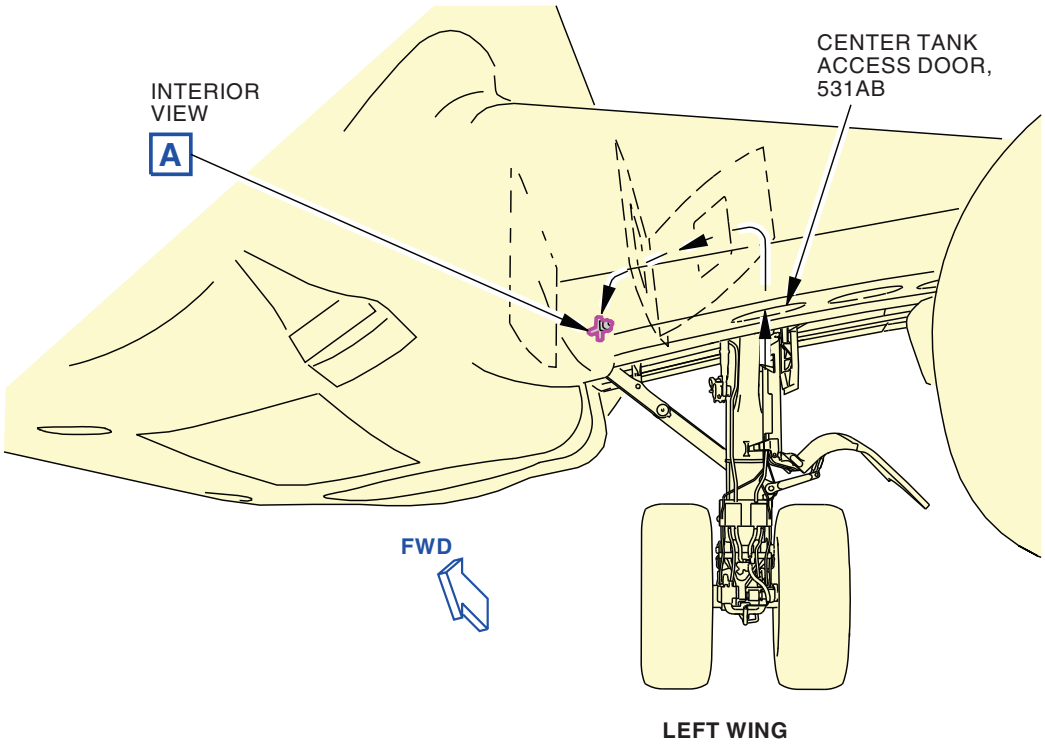
(a) Discard the O-ring [19].

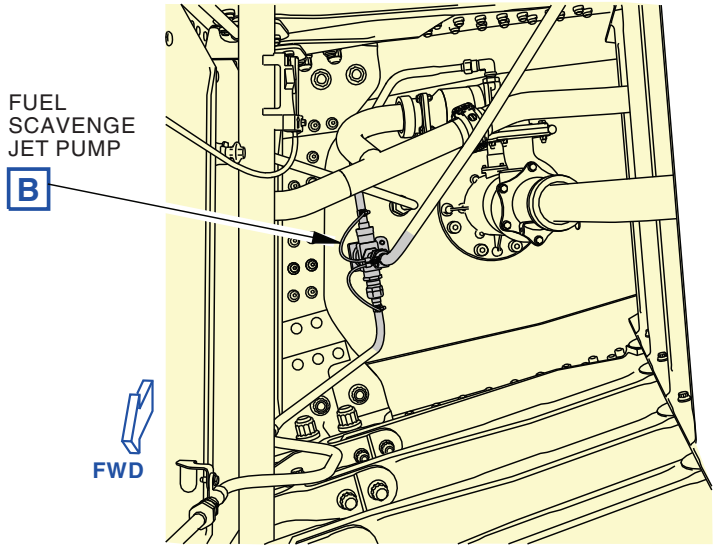
SUBTASK 28-22-17-480-003

(8) Install protective covers, STD-7423, on the open fuel lines to keep unwanted materials out.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

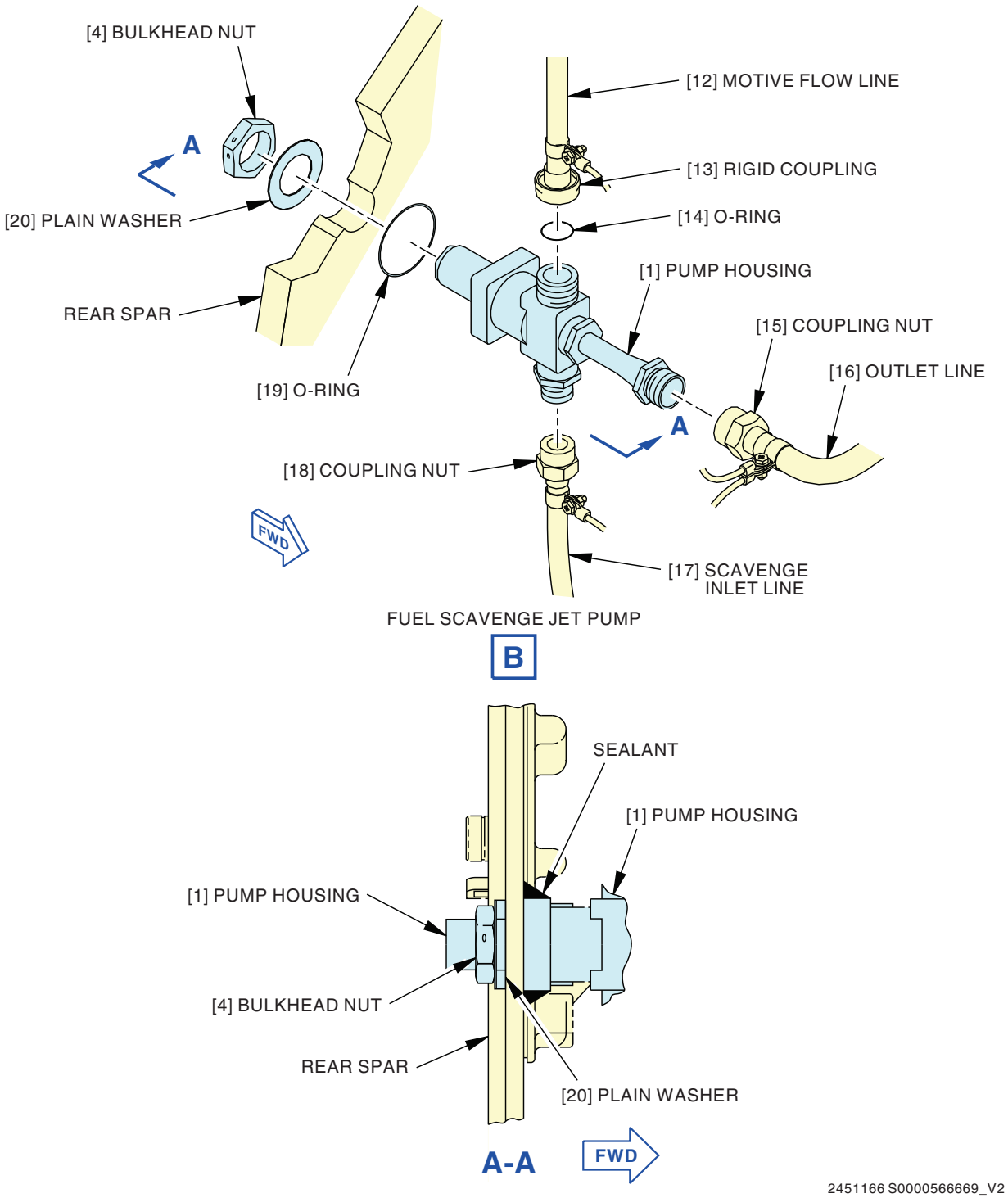




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Fuel Scavenge Jet Pump Installation
Figure 402/28-22-17-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Scavenge Jet Pump Installation
Figure 402/28-22-17-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-17-400-801

5. Fuel Scavenge Jet Pump Installation

(Figure 401)

A. General

(1) The objective of the task is to provide instructions to install the pump housing.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
28-22-00-720-803	Fuel Scavenge System - Operational Test (P/B 501)
28-22-15-400-803	Couplings Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A00767	Sealant - Fuel Tank	BMS5-45
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C50315	Coating - Chemical Conversion - Bonderite M-CR 600 Aero (Formerly Alodine 600)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
D00504	Grease - Petrolatum	VV-P-236
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G51293	Cotton Wiper - Knit, Process Cleaning Absorbent Wiper	BMS15-5 Class B

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
14	O-ring	28-22-17-01-140	PTF ALL

F. Location Zones

Zone	Area
511	Left Wing - Leading Edge To Front Spar
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5

G. Access Panels

Number	Name/Location
531AB	Center Tank Access Door - Wing Station 168

H. Prepare for the Installation

SUBTASK 28-22-17-100-002

(1) Do these steps to prepare the pump housing [1] and rear spar for installation:

(a) Remove the sealant from the rear spar, do this task: Clean Bare, Clad, or Plated Metal with Solvent, TASK 20-10-34-110-802.

NOTE: Sealant is on the inner and outer side of the rear spar.

(b) Clean the fay surfaces of the aluminum pump housing [1] and the aluminum rear spar (SWPM 20-20-00):

1) Use Cleaning Procedure 1 to manually clean the rear spar.

a) Use an abrasive pad to clean the fay surface on the rear spar.

b) Move the pad in a circular movement while you press with your thumb.

2) Use Cleaning Procedure 5 to clean the pump assembly fay surface and the fasteners.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034 (alternate: knit cotton wiper, G51293) made moist with alcohol, B00130 or solvent, B00184.

b) Use a new cotton wiper to dry the surfaces.

c) Continue to clean and dry the surface until the cotton wiper stays clean.

3) Final clean the fay surfaces of the rear spar.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034 (alternate: knit cotton wiper, G51293) made moist with solvent, B00184 or alcohol, B00130.

b) Rub dry with a clean dry cotton wiper.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) Continue to clean and dry the surface until the cotton wiper stays clean.

(c) Manually apply Bonderite M-CR 600 Aero coating, C50315, to the fay surface on the rear spar.

SUBTASK 28-22-17-080-001

(2) Remove the protective covers from the opened fuel lines.

I. Fuel Scavenge Jet Pump Installation

SUBTASK 28-22-17-640-002

(1) Apply a thin layer of grease, D00504, to the new O-ring [19].

SUBTASK 28-22-17-420-006

(2) Install the O-ring [19] onto the pump housing [1].

SUBTASK 28-22-17-820-001

(3) Align the pump housing [1] in position on the rear spar.

SUBTASK 28-22-17-420-007

(4) Install the plain washer [20] and bulkhead nut [4] to attach the pump housing [1] to the rear spar.

(a) Make sure that the motive flow line [12] and scavenge inlet line [17] align correctly.

(b) Tighten the bulkhead nut [4] to 425 in-lb (48 N·m) - 480 in-lb (54 N·m).

SUBTASK 28-22-17-860-001

(5) Measure the bonding resistance between the base on the pump housing [1] and the rear spar with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) Make sure that the measured resistance is not more than the maximum bonding resistance value (SWPM 20-20-00).

SUBTASK 28-22-17-640-003

(6) Apply a thin layer of grease, D00504, to the new O-ring [14].

SUBTASK 28-22-17-420-008

(7) Connect the rigid coupling [13] and O-ring [14] to the pump housing [1] (TASK 28-22-15-400-803).

SUBTASK 28-22-17-420-009

(8) Connect the coupling nut [18] to the pump housing [1].

(a) Tighten the coupling nut [18] to 280 ±14 in-lb (32 ±2 N·m).

SUBTASK 28-22-17-420-010

(9) Connect the coupling nut [15] to the pump housing [1].

(a) Tighten the coupling nut [15] to 360 ±18 in-lb (41 ±2 N·m).

SUBTASK 28-22-17-390-001

(10) Apply a fillet seal of sealant, A00767, between the pump housing [1] and the rear spar (in the fuel tank).

NOTE: Use sufficient sealant to cover un-primed areas of the structure for corrosion protection.

SUBTASK 28-22-17-020-008

(11) Install the nozzle assembly [2] (TASK 28-22-17-420-801).

SUBTASK 28-22-17-010-004

(12) Apply a fillet seal of sealant, A02315, between the pump housing [1] and the rear spar (out of the fuel tank).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

J. Fuel Scavenge Jet Pump Installation Test

SUBTASK 28-22-17-410-001

(1) Close this access panel:

(TASK 28-11-31-400-801)

Number	Name/Location
531AB	Center Tank Access Door - Wing Station 168

SUBTASK 28-22-17-710-001

(2) Do this task: Fuel Scavenge System - Operational Test, TASK 28-22-00-720-803.

SUBTASK 28-22-17-650-002

(3) Do this task: Pressure Refuel Procedure, TASK 12-11-00-650-803.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GROUND FAULT INTERRUPTER (GFI) RELAY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the Ground Fault Interrupter (GFI) relay

(2) An installation of the GFI relay.

B. There are six GFI relays for the fuel boost pumps. The R19, R20, and R54 GFI relays are located in the P91 power distribution panel. The R18, R21, and R55 GFI relays are located in the P92 power distribution panel.

C. Mounting hardware is included with a new GFI relay.

TASK 28-22-18-000-801

2. Ground Fault Interrupter (GFI) Relay Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the applicable Ground Fault Interrupter (GFI) relay.

B. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
123	Forward Cargo Compartment - Left
124	Forward Cargo Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Removal

SUBTASK 28-22-18-860-001

(1) On the P5-2 fuel system module, set these switches to the OFF position:

(a) FUEL PUMPS 1 AFT

(b) FUEL PUMPS 1 FWD

(c) FUEL PUMPS 2 AFT

(d) FUEL PUMPS 2 FWD

(e) FUEL PUMPS CTR L

(f) FUEL PUMPS CTR R.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-18-860-002

WARNING REMOVE ELECTRICAL POWER BEFORE YOU REMOVE OR INSTALL COMPONENTS IN THE POWER DISTRIBUTION PANELS. THERE ARE HIGH VOLTAGES THAT CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Remove electrical power (TASK 24-22-00-860-802).

SUBTASK 28-22-18-860-003

(3) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

F. GFI Relay Removal

SUBTASK 28-22-18-860-004

(1) Make sure that all power warning lights on the P91 or P92 power distribution panels are off.

SUBTASK 28-22-18-010-001

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open the applicable power distribution panel, P91 or P92.

NOTE: The GFI relays are located on the upper left corner of the open panel.

(a) The GFI relays in the P91 panel:

1) R19, Forward Fuel Boost Pump - No. 1 Main Tank
2) R20, Aft Fuel Boost Pump - No. 2 Main Tank
3) R54, Left Center Fuel Boost Pump - Center Tank.

(b) The GFI relays in the P92 panel:

1) R18, Aft Fuel Boost Pump - No. 1 Main Tank
2) R21, Forward Fuel Boost Pump - No. 2 Main Tank
3) R55, Right Center Fuel Boost Pump - Center Tank.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-18-860-005

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open the applicable circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-18-020-001

(4) Remove the lock nuts [1] and flat washers [2] from the GFI relay [3].

(a) Discard the lock nuts [1] and flat washers [2].

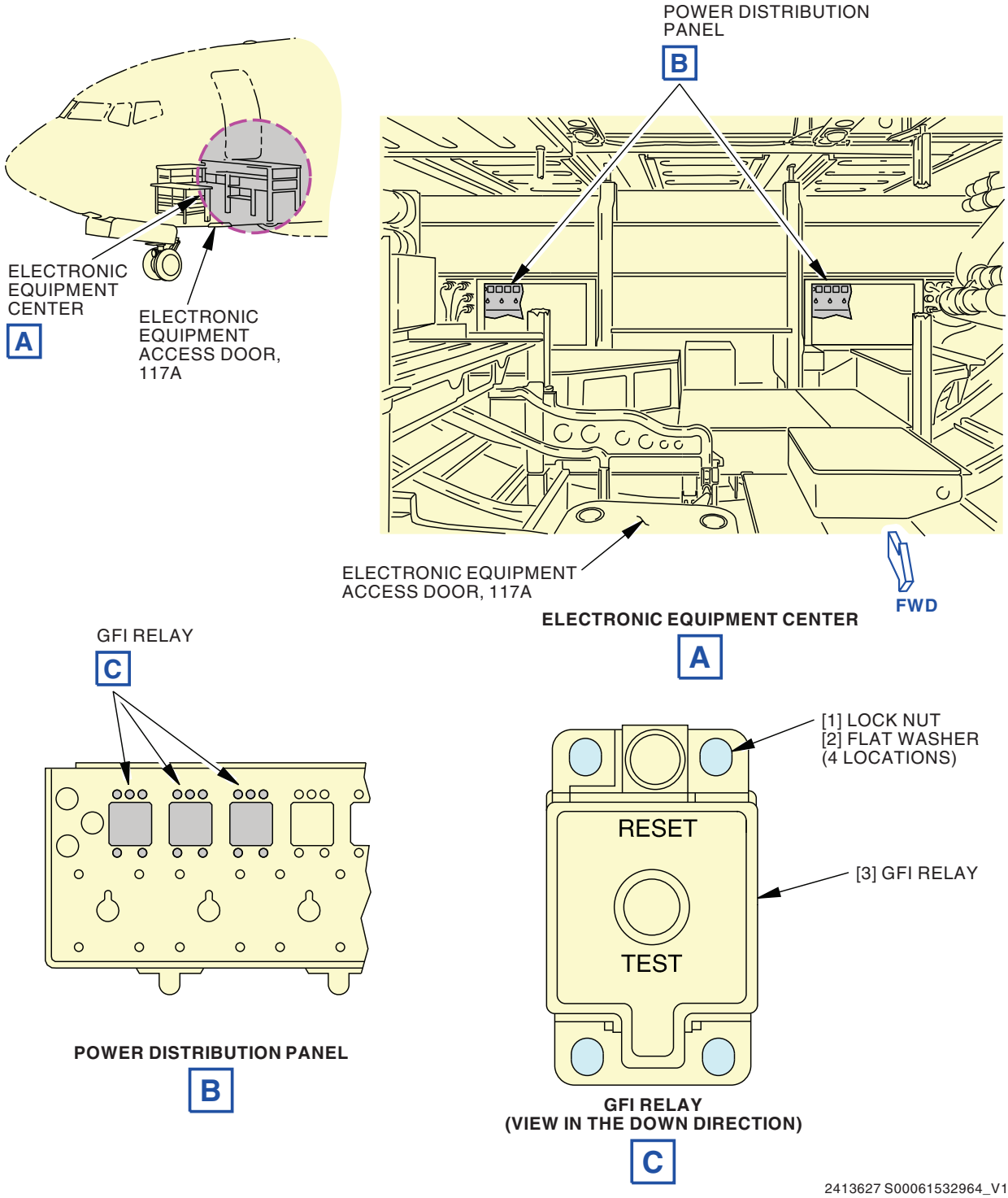
NOTE: If the GFI relay is removed for troubleshooting or testing reasons, it is not necessary to discard the lock nuts and washers.

SUBTASK 28-22-18-020-002

(5) Remove the GFI relay [3].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ground Fault Interrupter (GFI) Relay Installation
Figure 401/28-22-18-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-18-400-801

3. Ground Fault Interrupter (GFI) Relay Installation

(Figure 401)

A. General

(1) This task gives instructions to install the Ground Fault Interrupter (GFI) relay.

B. References

Reference	Title
28-22-41-720-801	Ground Fault Interrupter (GFI) - Operational Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426

D. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
123	Forward Cargo Compartment - Left
124	Forward Cargo Compartment - Right

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

G. GFI Relay Installation

SUBTASK 28-22-18-160-001

(1) Use a clean cotton wiper, G00034, moist with alcohol, B00130, to clean the contact surface of the mounting hardware and bonding surface of the GFI relay [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-18-420-001

(2) Align the GFI relay [3] in the mounting socket on the power distribution panel.

SUBTASK 28-22-18-420-002

(3) Install the supplied flat washers [2] and lock nuts [1] to secure the GFI relay [3] to the power distribution panel.

NOTE: If the GFI relay was removed for troubleshooting or testing reasons, it is possible to use the removed lock nuts and washers.

SUBTASK 28-22-18-420-003

(4) Tighten the lock nuts [1] to 6 in-lb (0.7 N·m) - 7 in-lb (0.8 N·m).

SUBTASK 28-22-18-760-001

(5) Measure the electrical bonding resistance between the GFI relay [3] mounting flange to a point on the panel cross member that is a distance of 1 in. (2.54 cm) or more from the point on the GFI relay [3] mounting flange (SWPM 20-20-00).

(a) Use a non-intrinsically safe bonding meter, COM-614.

(b) Make sure that the resistance is 0.0025 ohm (2.5 milliohms) or less.

H. GFI Relay Installation Test

SUBTASK 28-22-18-860-006

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-18-720-001

(2) Do this task: Ground Fault Interrupter (GFI) - Operational Test, TASK 28-22-41-720-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-18-860-007

WARNING	WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING	DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Close the applicable power distribution panel.

SUBTASK 28-22-18-860-008

(2) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FUEL CROSSFEED VALVE - REMOVAL/INSTALLATION

1. General

A. The engine fuel crossfeed valve has an actuator installed out of the fuel tank. An adapter shaft and valve body are installed in the fuel tank.

B. The engine fuel crossfeed valve actuator is installed on the rear spar of the center fuel tank, to the right of the body centerline. It is not necessary to defuel the center fuel tank to remove the actuator. It is necessary to defuel the center fuel tank to remove the adapter shaft. It is necessary to defuel all the fuel tanks to remove the valve body.

C. This procedure contains these tasks:

(1) Engine Fuel Crossfeed Valve Actuator Removal

(2) Engine Fuel Crossfeed Valve Actuator Installation

(3) Engine Fuel Crossfeed Valve - Valve Adapter Removal

(4) Engine Fuel Crossfeed Valve - Valve Adapter Installation

(5) Engine Fuel Crossfeed Valve Alignment

(6) Engine Fuel Crossfeed Valve - Valve Body Removal

(7) Engine Fuel Crossfeed Valve - Valve Body Installation

TASK 28-22-21-000-801

2. Engine Fuel Crossfeed Valve Actuator Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the engine fuel crossfeed valve actuator [15].

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975

D. Location Zones

Zone	Area
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right

E. Prepare for the Removal

SUBTASK 28-22-21-862-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	7	C00361	FUEL CROSS FEED VALVE CONT

SUBTASK 28-22-21-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-22-21-010-001

(3) Go to the crossfeed valve location on the rear spar in the main landing gear wheel well (right side) (View A, Figure 401).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-21-420-001

(4) On the crossfeed valve actuator [15], move the manual override lever [9] to the CLOSED position.

F. Engine Fuel Crossfeed Valve Actuator Removal

SUBTASK 28-22-21-420-002

(1) Disconnect the electrical connector [1] from the crossfeed valve actuator [15].

SUBTASK 28-22-21-140-001

(2) Remove the sealant from the bonding jumper [13] and fasteners at the crossfeed valve actuator [15] attachment location.

(a) Use sealant removal tool, COM-2481, or equivalent.

SUBTASK 28-22-21-420-003

(3) Remove the screw [14], two washers [12], and bonding jumper [13] from the crossfeed valve actuator [15].

SUBTASK 28-22-21-420-004

(4) Remove the four mounting screws [3] and washers [4].

SUBTASK 28-22-21-420-005

(5) Carefully disassemble the crossfeed valve actuator [15] from the index plate [5].

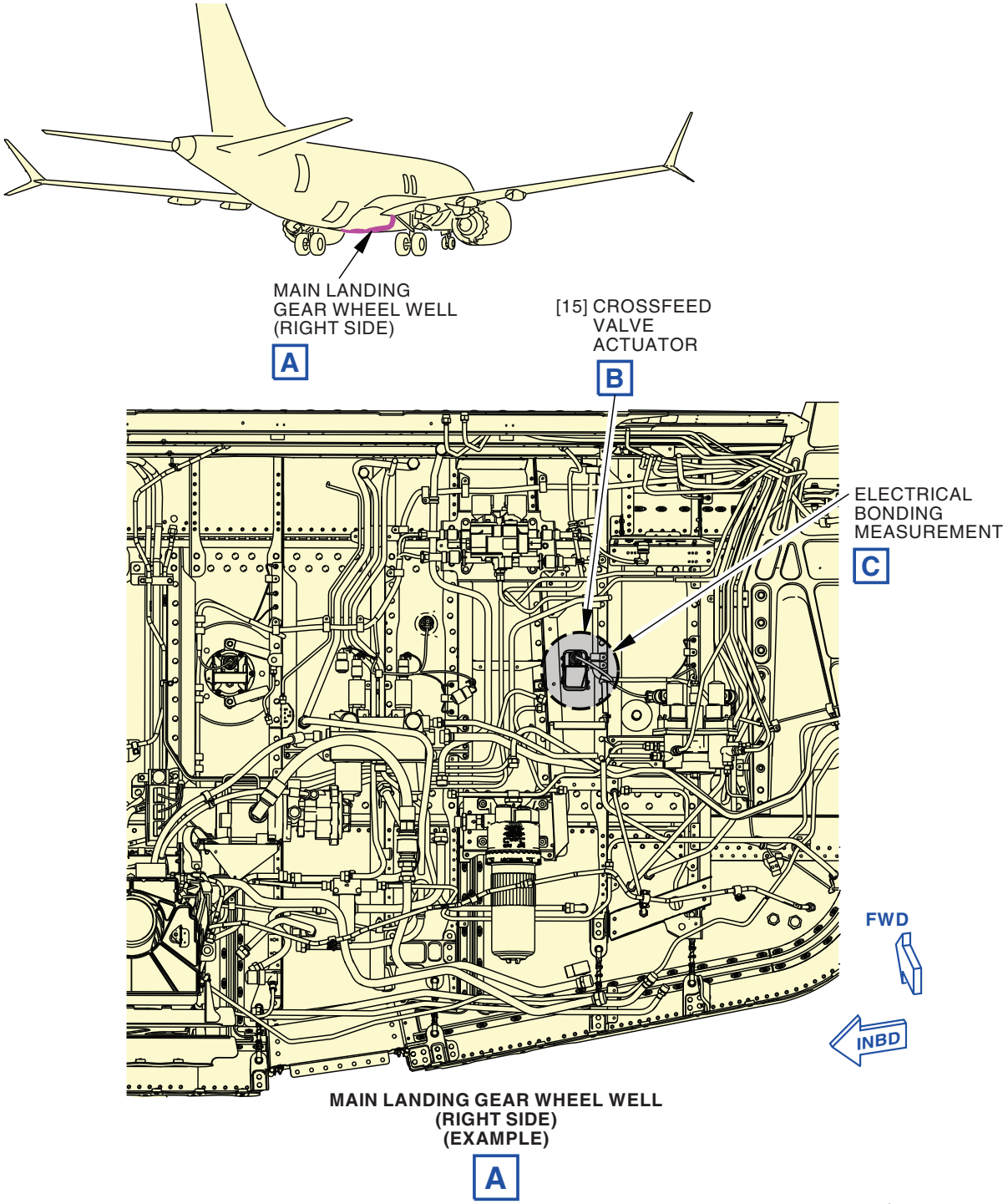
NOTE: The crossfeed valve actuator [15] and index plate [5] are bonded together with sealant.

SUBTASK 28-22-21-420-006

(6) Remove the crossfeed valve actuator [15].

——— END OF TASK ———

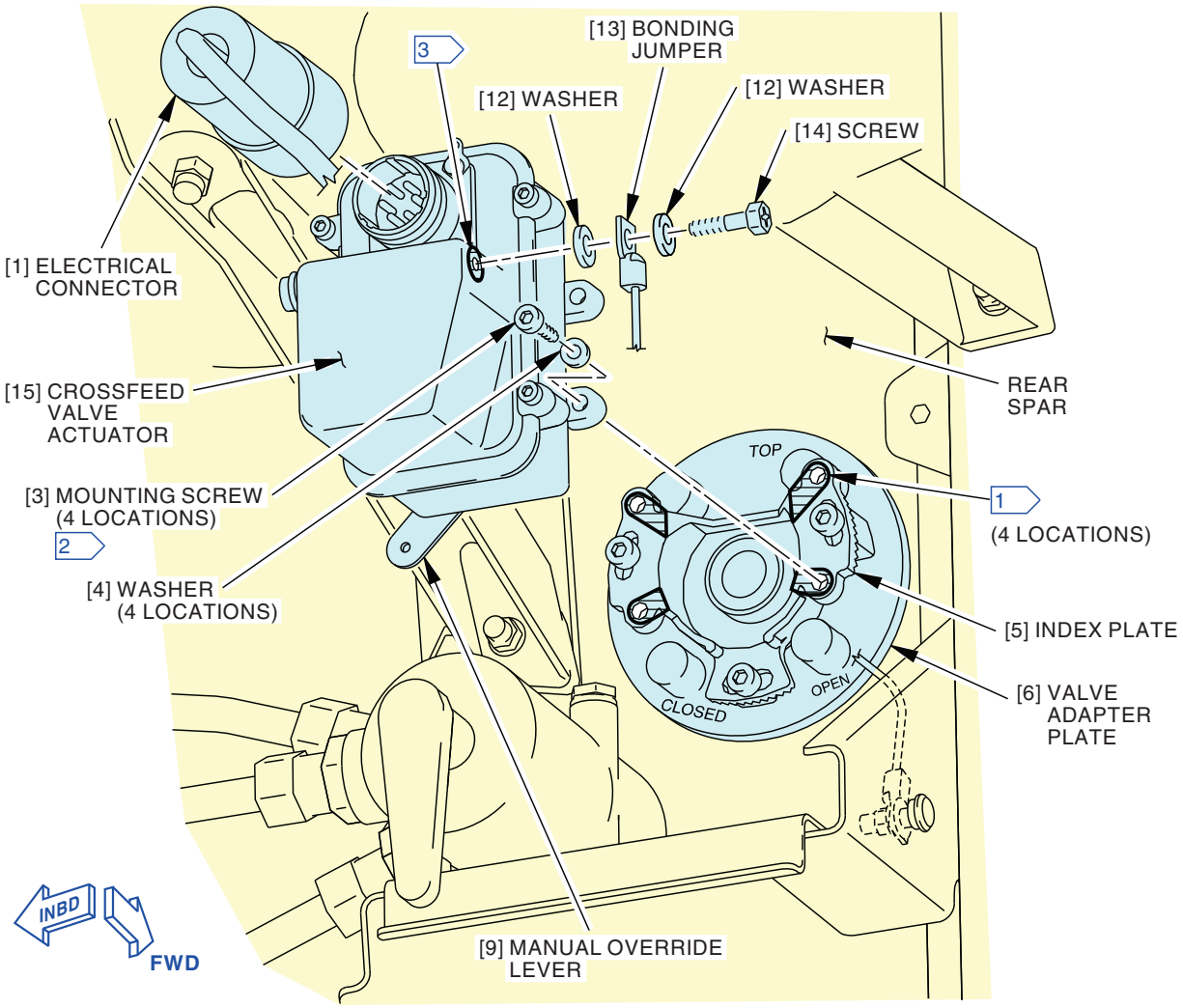
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413628 S00061532969_V2

Crossfeed Valve Actuator Installation
Figure 401/28-22-21-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CROSSFEED VALVE ACTUATOR INSTALLATION

B

1 PREPARE AND INSTALL THE CONTACT SURFACES OF THE INDEX PLATE AND ACTUATOR FOOTPADS (4 LOCATIONS) WITH A FAY SEALED FAY SURFACE BOND (SWPM 20-20-00).

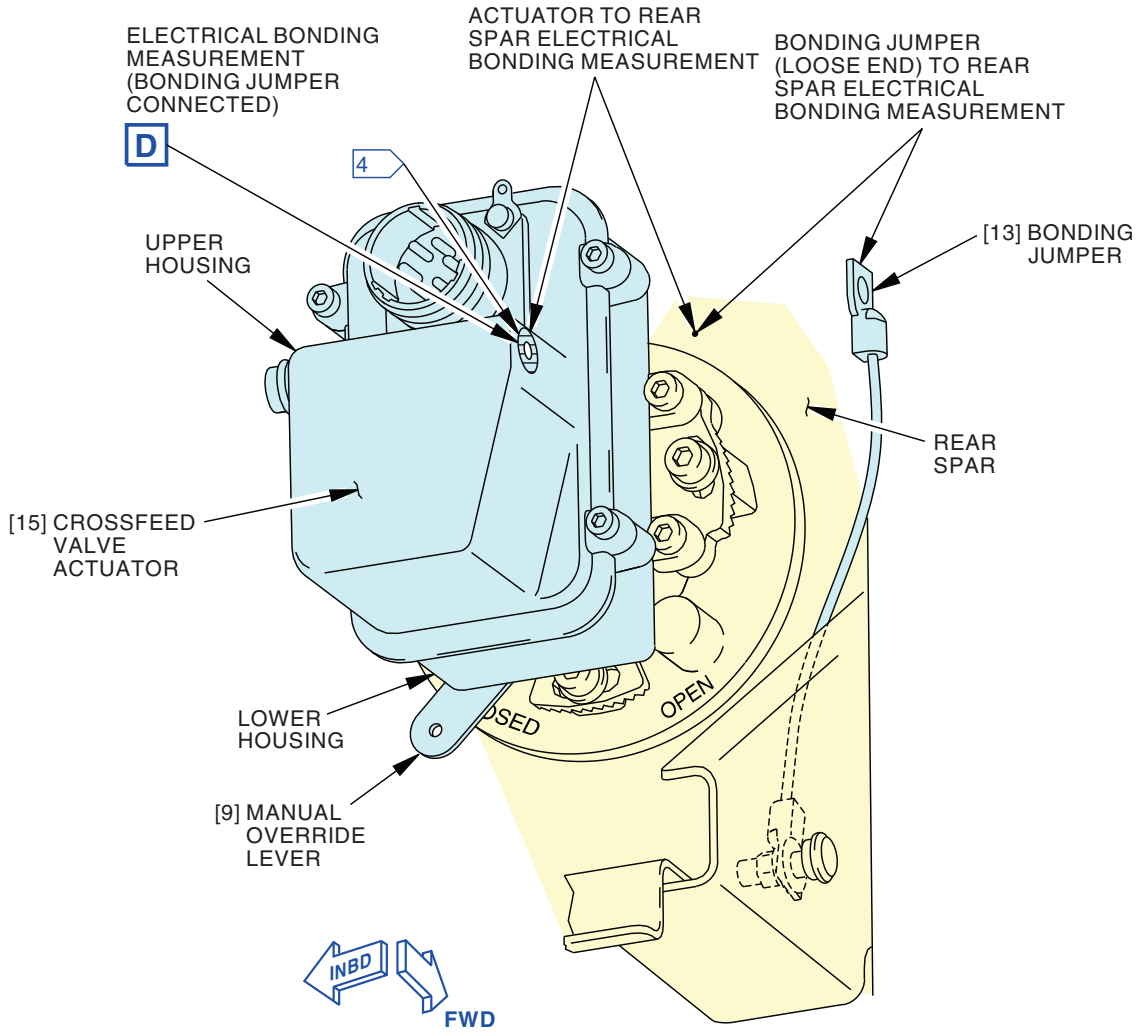
2 APPLY SEALANT TO THE SHANK AND THREADS OF THE MOUNTING SCREWS.

3 CLEAN THE CONTACT SURFACES OF THE ACTUATOR AND BONDING JUMPER. AFTER ELECTRICAL BONDING MEASUREMENT, APPLY A CAP SEAL TO THE BONDING JUMPER SCREW.

2413629 S00061532970_V1

Crossfeed Valve Actuator Installation
Figure 401/28-22-21-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELECTRICAL BONDING MEASUREMENT (BONDING JUMPER DISCONNECTED)

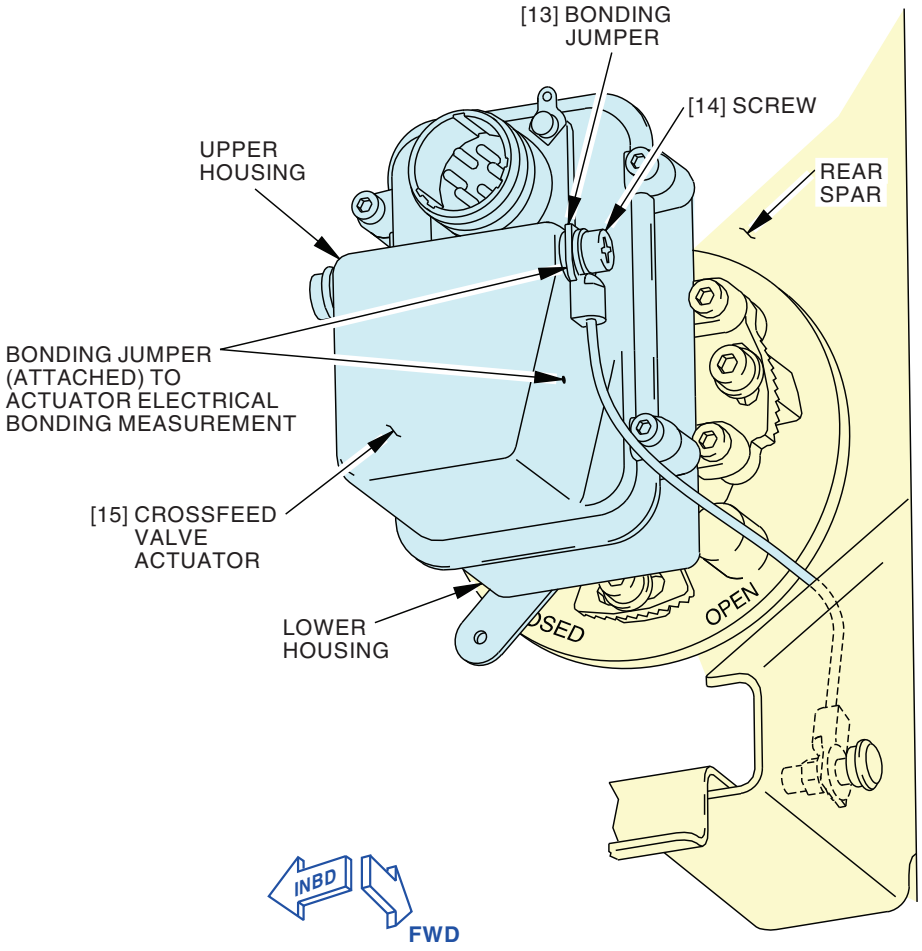
C

4 BARE METAL CONTACT SURFACE OF THE ACTUATOR AT THE BONDING JUMPER INSTALLATION.

2413630 S00061532971_V1

Crossfeed Valve Actuator Installation
Figure 401/28-22-21-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELECTRICAL BONDING MEASUREMENT (BONDING JUMPER CONNECTED)

D

2413631 S00061532972_V1

Crossfeed Valve Actuator Installation
Figure 401/28-22-21-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-21-400-801

3. Engine Fuel Crossfeed Valve Actuator Installation

(Figure 401)

A. General

(1) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, do this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

B. References

Reference	Title
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-11-00-600-801	Apply the Corrosion Resistant Finish (Topcoat) (P/B 701)
28-22-21-710-802	Crossfeed Valve Operational Test (P/B 501)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
51-21-41-370-802	Apply Bonderite M-CR 600, Bonderite M-CR 1200, Bonderite M-CR 600RTU, or Bonderite M-CR 1200S Solution (P/B 701)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
COM-4746	Handle - Sealant Removal Tool Part #: 310/1 (for SkyScraper) Supplier: F6892 Part #: SHR0272-T (for PnJ scraper) Supplier: 7RKH2 Part #: ST982LH-F-3 (for ST982LF) Supplier: 81205 Opt Part #: SHW0215 (for PnJ scraper) Supplier: 7RKH2
STD-123	Brush - Soft Bristle

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00307	Coating - Corrosion Resistant Finish For Integral Fuel Tanks	BMS10-20 Type II

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
C50033	Coating - Chemical Conversion - Bonderite M-CR 1200S Aero (Formerly Alodine 1200)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
15	Crossfeed valve actuator	28-22-21-01-050	PTF ALL

F. Location Zones

Zone	Area
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right

G. Engine Fuel Crossfeed Valve Actuator Installation

SUBTASK 28-22-21-490-001

WARNING MAKE SURE THAT YOU INSTALL THE GROUND LOCK ASSEMBLIES ON THE LANDING GEAR. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR IF YOU DO NOT INSTALL THE GROUND LOCK ASSEMBLIES.

WARNING CHECK THAT GROUND LOCK ASSEMBLIES ARE INSTALLED IN MAIN LANDING GEAR TO PREVENT INADVERTENT OPERATION OF GEAR. INJURY TO PERSONNEL AND/OR EQUIPMENT COULD RESULT IF GEAR RETRACTS.

(1) Make sure that the ground locks are installed on the landing gear (TASK 32-00-01-480-801).

SUBTASK 28-22-21-110-001

(2) Remove the old sealant and clean the new crossfeed valve actuator [15], index plate [5], and the fasteners (if reused):

(a) Use sealant removal tool, COM-2481, and sealant removal tool handle, COM-4746 (or equivalent), to remove the old sealant.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS) AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(b) To remove the remaining sealant, grease, oil, dirt, and other contamination from the components, use one of these brushes/pads soaked with solvent, B00184 or solvent, B00148:

1) soft bristle brush, STD-123

2) cotton wiper, G00034.

SUBTASK 28-22-21-110-002

(3) Do these steps to prepare the crossfeed valve actuator [15] and index plate [5] for a fay sealed fay surface bond (SWPM 20-20-00, SWPM 20-20-10):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE. OBEY THE MATERIAL SAFETY DATA SHEETS (MSDS) FOR SOLVENTS. OBEY LOCAL REGULATIONS FOR THE CORRECT PROCEDURES TO USE OR DISCARD SOLVENTS. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Final clean the contact surfaces with a cotton wiper, G00034, soaked with solvent, B00062 or solvent, B00184.

This, in turn, causes and places stress at the CV joints and differential output shaft bearings. The result is a prominent “clicking” and, worse, rear-end vibrations.

(b) Rub dry with a clean, dry cotton wiper, G00034.

(c) Continue to clean and dry the surface until the dry cotton wiper, G00034, stays clean.

28-AWL-21: CDCCL

(d) Apply a fay sealed fay surface bond to the mating surfaces between the crossfeed valve actuator [15] and the index plate [5] (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) Use BMS 5-45 Class B-2 sealant, A50110, BMS 5-45 Class C sealant, A50155, BMS 5-95 Class B sealant, A50231, or BMS 5-95 Class C sealant, A50296, for the faying surface seal that is outside the tank.

2) Make sure that the sealant layer is approximately 0.005 in. (0.127 mm) thick.

(e) Apply sealant to the shank and the threads of the four mounting screws [3].

1) Use BMS 5-45 Class B-2 sealant, A50110, BMS 5-45 Class C sealant, A50155, BMS 5-95 Class B sealant, A50231, or BMS 5-95 Class C sealant, A50296 for the wet installation of the mounting screws [3].

2) Make sure that the sealant applied to the fasteners is approximately 0.060 in. (1.524 mm) thick.

SUBTASK 28-22-21-420-007

(4) Do these steps to install the crossfeed valve actuator [15] to the index plate [5]:

(a) Make sure that the manual override lever [9] on the crossfeed valve actuator [15] is in the CLOSED position.

(b) Align the actuator output shaft with the valve adapter plate [6].

(c) Make sure that the two spaces for teeth on the valve adapter plate [6] align with the crossfeed valve actuator [15] output shaft.

(d) Put the crossfeed valve actuator [15] output shaft into the valve adapter plate [6].

NOTE: The mounting feet on the crossfeed valve actuator [15] automatically align with the mounting points on the index plate [5].

(e) Install the four mounting screws [3] and the washers [4].

NOTE: Lockwire is not necessary for the mounting screws [3].

(f) Tighten the mounting screws [3] to 20 in-lb (2.3 N·m).

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(g) Make sure that the sealant is continuously squeezed out along the edges of the contact surfaces.

(h) If there are gaps, bubbles or voids in the sealant squeeze out, then disassemble and apply more sealant.

(i) Shape the squeezed out sealant into a fillet seal.

(j) As an option, remove the extra squeezed out sealant.

NOTE: Make sure that the sealant that remains is flush with the mating part edges.

SUBTASK 28-22-21-765-001

28-AWL-21: CDCCL

(5) Before the bonding jumper [13] and the electrical connector [1] is attached, measure the electrical bonding resistance between the upper housing of the crossfeed valve actuator [15] and the rear spar (SWPM 20-20-10) (View C).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Make sure that the bonding jumper [13] does not touch the crossfeed valve actuator [15] during the bonding measurement.

(b) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-21: CDCCL

(c) If the crossfeed valve actuator [15], index plate [5], and valve adapter plate [6] were removed, make sure that the electrical bonding resistance is 0.0030 ohm (3.0 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) If the electrical bonding resistance is more than 0.0030 ohm (3.0 milliohms), then do the applicable procedure in this task: Rework the Electrical Faying Surface Bonds for the Engine Crossfeed Valve, TASK 28-22-21-300-801.

28-AWL-21: CDCCL

(d) If the crossfeed valve actuator [15] and index plate [5] were removed, make sure the electrical bonding resistance is 0.0040 ohm (4.0 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) If the electrical bonding resistance is more than 0.0040 ohm (4.0 milliohms), then do the applicable procedure in this task: Rework the Electrical Faying Surface Bonds for the Engine Crossfeed Valve, TASK 28-22-21-300-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

28-AWL-21: CDCCL

(e) If only the crossfeed valve actuator [15] was removed, make sure the electrical bonding resistance is 0.0040 ohm (4.0 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) If the electrical bonding resistance is more than 0.0040 ohm (4.0 milliohms), then do the applicable procedure in this task: Rework the Electrical Faying Surface Bonds for the Engine Crossfeed Valve, TASK 28-22-21-300-801.

SUBTASK 28-22-21-390-001

28-AWL-21: CDCCL

(6) If the terminal of the bonding jumper [13] is disconnected from the stiffener, install the bonding jumper [13] with a fay surface bond between the stiffener and the terminal of the bonding jumper [13] (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

SUBTASK 28-22-21-765-002

28-AWL-21: CDCCL

(7) Before the bonding jumper [13] is attached to the crossfeed valve actuator [15], measure the electrical bonding resistance between the loose end of the bonding jumper [13] and the rear spar (SWPM 20-20-10) (View C).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-21: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0015 ohm (1.5 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) If the electrical bonding resistance is more than 0.0015 ohm (1.5 milliohms), then do the applicable procedure in this task: Rework the Electrical Faying Surface Bonds for the Engine Crossfeed Valve, TASK 28-22-21-300-801.

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28-AWL-21: CDCCL

(8) If the terminal of the bonding jumper [13] is disconnected from the stiffener, apply a fillet seal to the terminal of the bonding jumper [13] at the stiffener (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

SUBTASK 28-22-21-420-008

28-AWL-21: CDCCL

(9) If the terminal of the bonding jumper [13] was not disconnected from the stiffener, measure the resistance between the loose end of the bonding jumper [13] and the rear spar (SWPM 20-20-10).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-21: CDCCL

(b) Make sure the electrical bonding resistance is 0.0020 ohm (2.0 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) If the electrical bonding resistance is more than 0.0020 ohm (2.0 milliohms), then do the applicable procedure in this task: Rework the Electrical Faying Surface Bonds for the Engine Crossfeed Valve, TASK 28-22-21-300-801.

SUBTASK 28-22-21-420-032

28-AWL-21: CDCCL

(10) Apply a fay surface bond between the crossfeed valve actuator [15] and the terminal of the bonding jumper [13] (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

SUBTASK 28-22-21-400-001

(11) Install the bonding jumper [13] to the crossfeed valve actuator [15] with the screw [14] and two washers [12].

(a) Tighten the screw [14] to 35 in-lb (4 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-21-765-003

28-AWL-21: CDCCL

(12) Measure the electrical bonding resistance between the upper housing of the crossfeed valve actuator [15] and the attached terminal of the bonding jumper [13] (SWPM 20-20-10) (View D).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Do not touch the screw [14] when you make the bonding measurement.

(b) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-21: CDCCL

(c) Make sure the electrical bonding resistance is 0.0010 ohm (1.0 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) If the electrical bonding resistance is more than 0.0010 ohm (1.0 milliohm), then do the applicable procedure in this task: Rework the Electrical Faying Surface Bonds for the Engine Crossfeed Valve, TASK 28-22-21-300-801.

SUBTASK 28-22-21-390-004

28-AWL-21: CDCCL

(13) Apply a fillet seal between the terminal of the bonding jumper [13] and the crossfeed valve actuator [15].

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

SUBTASK 28-22-21-390-002

28-AWL-21: CDCCL

(14) Apply a cap seal over the screw [14] and the terminal lug of the bonding jumper [13] attached to the crossfeed valve actuator [15].

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Use BMS 5-45 Class B-2 sealant, A50110, BMS 5-45 Class C sealant, A50155, BMS 5-95 Class B sealant, A50231, or BMS 5-95 Class C sealant, A50296 to apply the cap seal.

SUBTASK 28-22-21-916-001

(15) Do this task to apply protective finishes to the bare metal areas of the rear spar: Apply the Corrosion Resistant Finish (Topcoat), TASK 28-11-00-600-801.

NOTE: Re-apply the protective coatings to the rear spar where the electrical bonding probe removed the finish.

(a) These are the protective finishes:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Alodine 600 coating, C00064

2) BMS 10-20 coating, C00307.

SUBTASK 28-22-21-916-002

(16) Do this task to apply Bonderite M-CR 1200S Aero coating, C50033, to any bare metal areas of the crossfeed valve actuator [15]: Apply Bonderite M-CR 600, Bonderite M-CR 1200, Bonderite M-CR 600RTU, or Bonderite M-CR 1200S Solution, TASK 51-21-41-370-802.

NOTE: Re-apply the protective coating to the crossfeed valve actuator [15] where the electrical bonding probe removed the finish and any bare metal areas around the bonding jumper installation that are not covered with sealant.

SUBTASK 28-22-21-410-001

(17) Connect the electrical connector [1] to the crossfeed valve actuator [15].

H. Crossfeed Valve Actuator Operational Test

SUBTASK 28-22-21-861-001

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	7	C00361	FUEL CROSS FEED VALVE CONT

SUBTASK 28-22-21-760-001

(2) Do this task: Crossfeed Valve Operational Test, TASK 28-22-21-710-802.

I. Put the Airplane Back to the Usual Condition

SUBTASK 28-22-21-090-001

WARNING DO THE APPLICABLE PROCEDURE TO REMOVE THE LANDING-GEAR DOWNLOCK PINS INSTALLED ON THE LANDING GEAR. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR IF THE LANDING GEAR OPERATES WITHOUT WARNING.

(1) Remove the ground locks installed on the landing gear if they are not needed for other tasks (TASK 32-00-01-080-801).

——— END OF TASK ———

TASK 28-22-21-000-802

4. Engine Fuel Crossfeed Valve - Valve Adapter Removal

(Figure 402)

A. General

(1) This task gives instructions to remove the engine fuel crossfeed valve adapter plate [6].

(2) The valve adapter plate [6] and shaft assembly (valve adapter) is used to connect the actuator to the crossfeed valve body [21]. The valve adapter plate [6] consists of these components:

(a) index plate [5]

(b) Adapter plate

(c) Adapter shaft (with U-joint connection).

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)

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(Continued)

Reference	Title
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
COM-4746	Handle - Sealant Removal Tool Part #: 310/1 (for SkyScraper) Supplier: F6892 Part #: SHR0272-T (for PnJ scraper) Supplier: 7RKH2 Part #: ST982LH-F-3 (for ST982LF) Supplier: 81205 Opt Part #: SHW0215 (for PnJ scraper) Supplier: 7RKH2

D. Location Zones

Zone	Area
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right

E. Access Panels

Number	Name/Location
131AB	Center Tank Access

F. Prepare for the Removal

SUBTASK 28-22-21-650-001

(1) Defuel the center fuel tank (TASK 28-26-00-650-801).

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SUBTASK 28-22-21-860-001

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Obey the fuel tank entry precautions (TASK 28-11-00-910-802).

SUBTASK 28-22-21-650-002

(3) Drain and purge the center fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-22-21-010-002

(4) Remove this access panel:

Number	Name/Location
131AB	Center Tank Access

(a) Do this task: Center Tank Access Door Removal, TASK 28-11-31-000-801.

G. Engine Fuel Crossfeed Valve Adapter Removal

SUBTASK 28-22-21-020-001

(1) Do these steps on the outside of the fuel tank:

(a) Do this task: Engine Fuel Crossfeed Valve Actuator Removal, TASK 28-22-21-000-801.

(b) Use a sealant removal tool to remove the old sealant from the periphery of the adapter plate (View A-A, Figure 402).

1) Use sealant removal tool, COM-2481, and sealant removal tool handle, COM-4746, or equivalent.

SUBTASK 28-22-21-020-002

WARNING DO THE PURGING AND FUEL-TANK ENTRY PROCEDURE. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the center tank and do these steps:

(a) Go to the valve adapter plate [6] location (View B, Figure 402).

(b) Remove the old sealant from the four bolts [26] and the valve adapter plate [6] rear spar penetration (View A-A, Figure 402).

(c) Remove the four bolts [26] and the washers [27] to disconnect the valve adapter plate [6].

(d) Disengage the adapter shaft that attached to the valve adapter plate [6] and crossfeed valve body [21] (View C, Figure 402).

SUBTASK 28-22-21-020-003

(3) Go out of the tank and remove the valve adapter plate [6] from the rear spar.

H. Remove the Index Plate

NOTE: Do these steps if you will reuse the valve adapter plate [6] and the attached index plate [5] or if you need to remove the index plate [5] to align the crossfeed valve body [21]. If the task is to discard the old valve adapter plate [6] and install a new valve adapter plate [6], it is not necessary to disassemble the index plate [5].

SUBTASK 28-22-21-020-004

(1) Do these steps to remove the index plate [5] from the valve adapter plate [6] (View D, Figure 402):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the lockwire from the two index screws [23] and index screw [24].

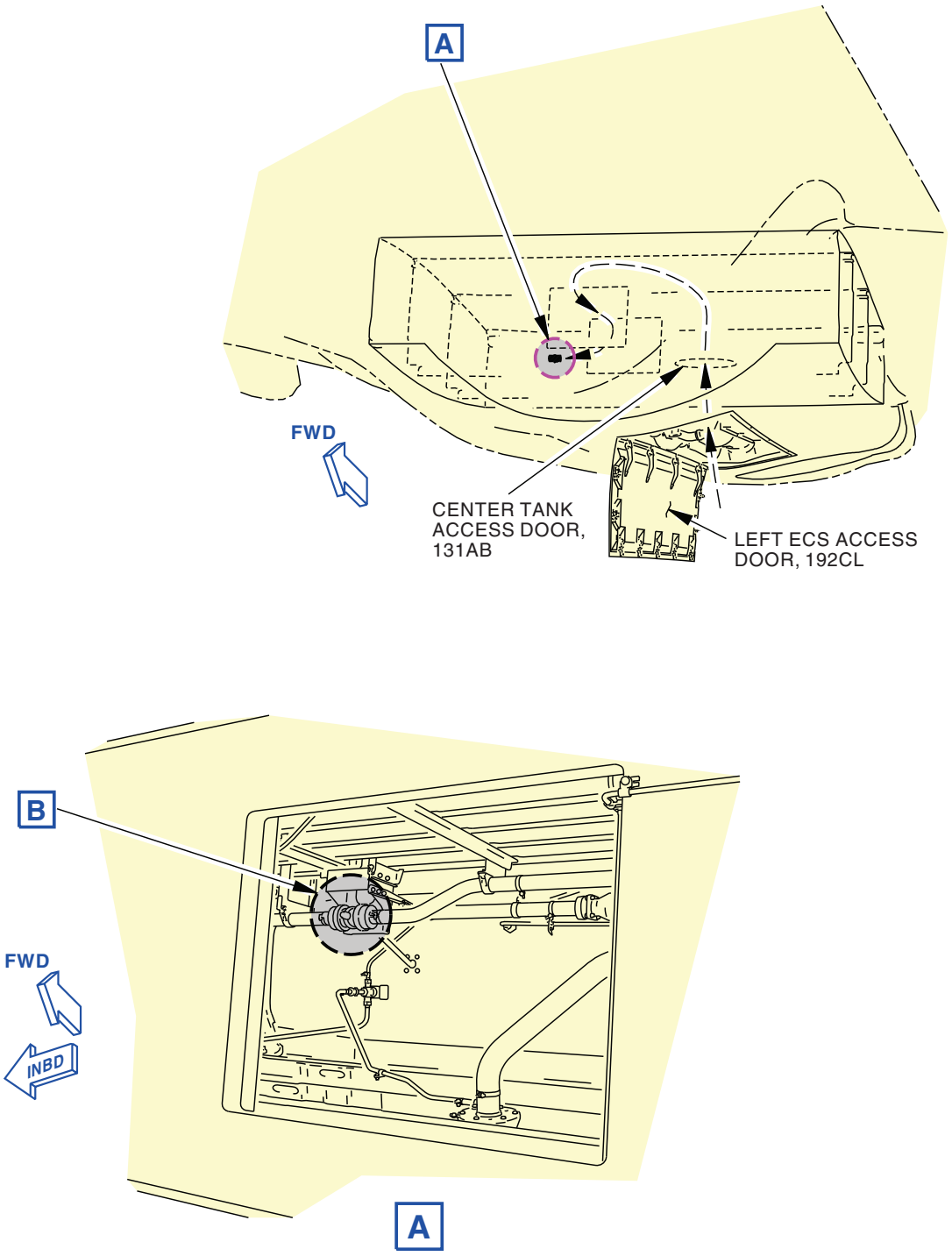
(b) Remove index screws [23], index screw [24], and the washers [22].

(c) Carefully disassemble the index plate [5] from the valve adapter plate [6].

NOTE: The two plates are bonded with faying surface sealant.

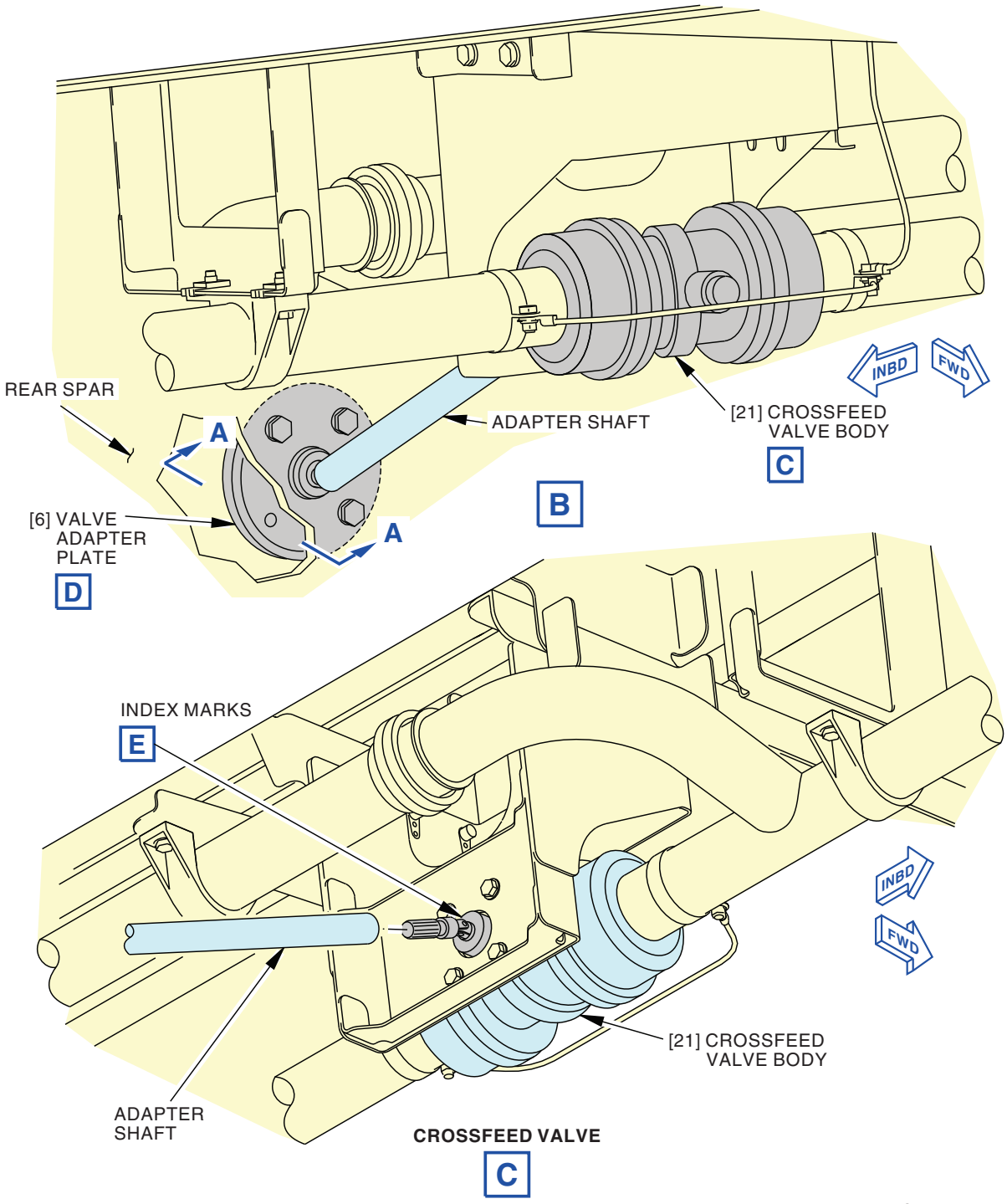
——— END OF TASK ———

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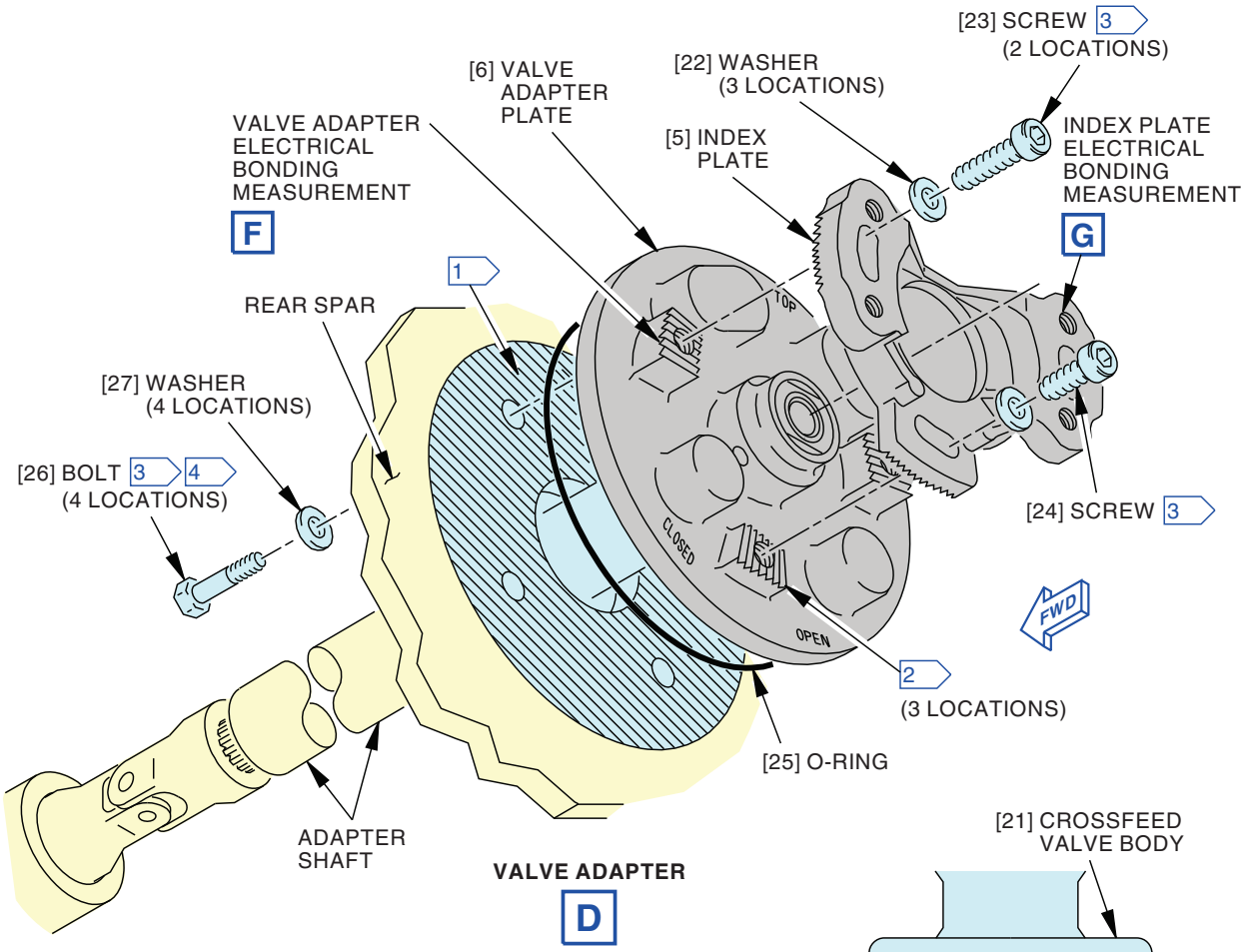
Crossfeed Valve Adapter Installation
Figure 402/28-22-21-990-802 (Sheet 1 of 5)

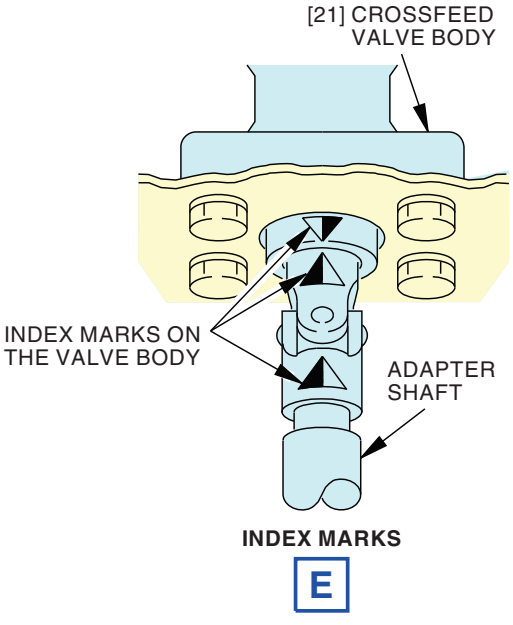
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Crossfeed Valve Adapter Installation
Figure 402/28-22-21-990-802 (Sheet 2 of 5)

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1 PREPARE AND INSTALL THE VALVE ADAPTER TO THE REAR SPAR WITH A FILLET SEAL FAY SURFACE BOND (SWPM 20-20-00).

2 PREPARE AND INSTALL THE SERRATED CONTACT SURFACES OF THE VALVE ADAPTER AND INDEX PLATE WITH A FAY SEALED FAY SURFACE BOND (3 LOCATIONS) (SWPM 20-20-00).

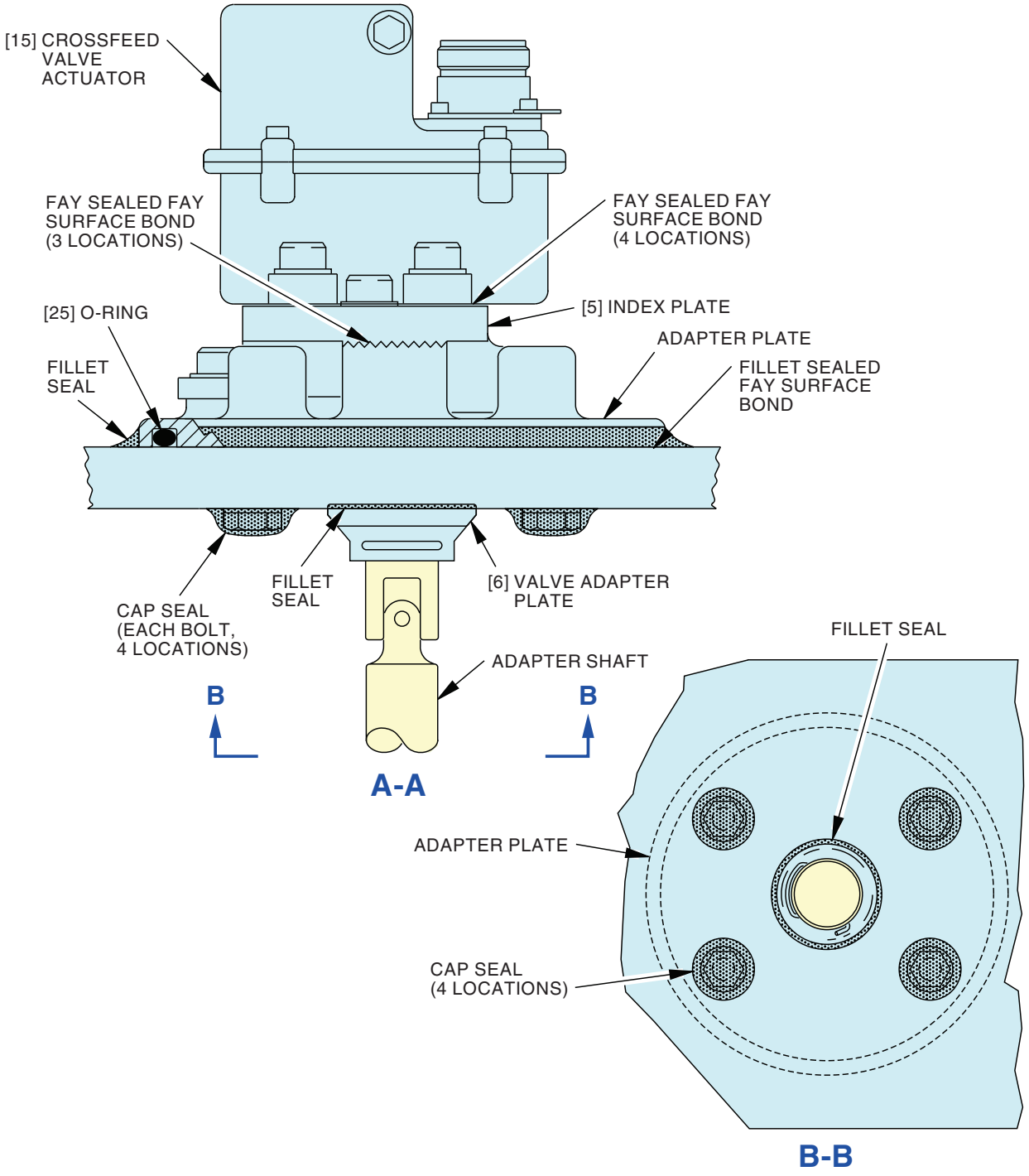
3 APPLY SEALANT TO THE SHANK AND THREADS OF THE FASTENERS DURING THE INSTALLATION.

4 AFTER INSTALLATION, APPLY A CAP SEAL TO THE HEAD OF THE FASTENERS.

2413634 S00061532977_V1

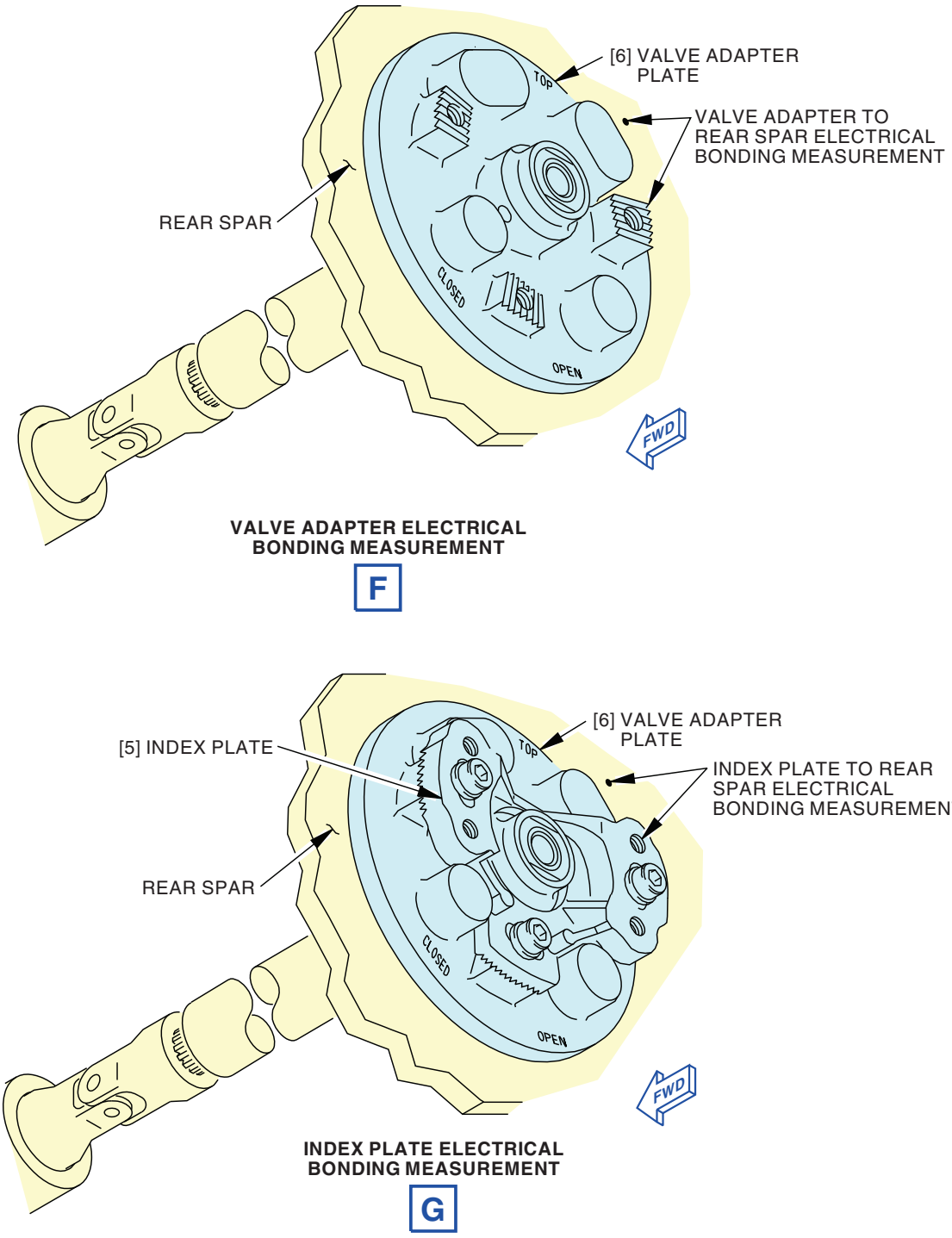
Crossfeed Valve Adapter Installation
Figure 402/28-22-21-990-802 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Crossfeed Valve Adapter Installation
Figure 402/28-22-21-990-802 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Crossfeed Valve Adapter Installation
Figure 402/28-22-21-990-802 (Sheet 5 of 5)

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TASK 28-22-21-400-802

5. Engine Fuel Crossfeed Valve - Valve Adapter Installation

(Figure 402)

A. General

(1) This task has these procedures:

(a) Install the valve adapter plate [6]

(b) Install the index plate [5]

(c) Crossfeed Valve Operational Tests

(d) Put the Airplane Back to the Usual Condition.

(2) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-00-600-801	Apply the Corrosion Resistant Finish (Topcoat) (P/B 701)
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
28-22-00-730-802	Crossfeed Valve - Functional Test (P/B 501)
28-22-21-710-802	Crossfeed Valve Operational Test (P/B 501)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
51-22-00-390-806	Fillet Seal Application (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
COM-4746	Handle - Sealant Removal Tool Part #: 310/1 (for SkyScraper) Supplier: F6892 Part #: SHR0272-T (for PnJ scraper) Supplier: 7RKH2 Part #: ST982LH-F-3 (for ST982LF) Supplier: 81205 Opt Part #: SHW0215 (for PnJ scraper) Supplier: 7RKH2
STD-123	Brush - Soft Bristle

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
A50337	Sealant - Fuel Tank	BMS5-45 Class B
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00307	Coating - Corrosion Resistant Finish For Integral Fuel Tanks	BMS10-20 Type II

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(Continued)

Reference	Description	Specification
D00504	Grease - Petrolatum	VV-P-236
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
6	Valve adapter plate	28-22-21-01-060	PTF ALL
25	O-ring	28-22-21-01-095	PTF ALL

F. Location Zones

Zone	Area
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right

G. Access Panels

Number	Name/Location
131AB	Center Tank Access

H. Valve Adapter Installation

SUBTASK 28-22-21-140-002

(1) Remove the old sealant and clean the rear spar, valve adapter plate [6], index plate [5], and the fasteners (if re-used).

(a) Use sealant removal tool, COM-2481, and sealant removal tool handle, COM-4746 (or equivalent), to remove the old sealant from the rear spar.

(b) If new, use an abrasive pad to remove the anodized finish from the faying surfaces of the valve adapter plate [6] and index plate [5] (SWPM 20-20-00).

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(c) To remove the remaining sealant, grease, oil, dirt, and other contamination from the components, use one of these brushes/pads soaked with solvent, B00184, or solvent, B00666:

1) soft bristle brush, STD-123

2) cotton wiper, G00034.

SUBTASK 28-22-21-110-003

(2) Do these steps to prepare the valve adapter plate [6] and the rear spar for a fay sealed fay surface bond (View D, Figure 402):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(a) Final clean the contact surfaces of the valve adapter plate [6] and the rear spar with a cotton wiper, G00034, soaked with alcohol, B00130, or solvent, B00184.

(b) Rub dry with a clean, dry cotton wiper, G00034.

(c) Continue to clean and dry the surface until the dry cotton wiper, G00034, stays clean.

SUBTASK 28-22-21-390-003

28-AWL-21: CDCCL

(3) Apply a fay sealed fay surface bond between the valve adapter plate [6] and the rear spar outside the tank (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

SUBTASK 28-22-21-420-009

(4) On the outside of the airplane, do these steps:

(a) Put a thin layer of petrolatum grease, D00504, on the new O-ring [25].

(b) Put the new O-ring [25] in the O-ring groove of the valve adapter plate [6].

(c) Put the adapter shaft attached to the valve adapter plate [6] through the hole in the rear spar.

(d) Make sure that the shoulder of the valve adapter plate [6] does not bind in the opening of the rear spar.

SUBTASK 28-22-21-410-002

WARNING DO THE PURGING AND FUEL-TANK ENTRY PROCEDURE. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Go into the center tank to the valve adapter plate [6] location (TASK 28-11-00-910-802) (View A, Figure 402).

SUBTASK 28-22-21-420-010

(6) Do these steps to install the valve adapter plate [6]:

(a) Make sure that the index marks on the body of the crossfeed valve body [21] are aligned (View E, Figure 402).

NOTE: A small misalignment of the index marks is satisfactory.

(b) Engage the adapter shaft of the valve adapter plate [6] with the crossfeed valve body [21] (View C, Figure 402).

(c) Apply sealant to the shank and the threads of the four bolts [26].

1) Use sealant, A50110, for the wet installation of the bolts [26].

2) Make sure that the sealant applied to the fasteners is approximately 0.060 in. (1.524 mm) thick.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Install the four bolts [26] and washers [27] to attach the valve adapter plate [6].

SUBTASK 28-22-21-420-011

28-AWL-21: CDCCL

(7) On the inside of the tank, apply a cap seal to each of the four bolts [26] (TASK 51-22-00-390-806) (View A-A).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Use sealant, A50110, to apply the cap seal.

SUBTASK 28-22-21-420-012

28-AWL-21: CDCCL

(8) On the inside of the tank, apply a fillet seal around the edge of the hole and the stationary part of the valve adapter plate [6] (TASK 51-22-00-390-806) (View A-A).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Make sure that the sealant does not touch the adapter shaft or other parts of the valve adapter plate [6] that turn.

(b) Use sealant, A50110, to apply the fillet seal.

SUBTASK 28-22-21-420-013

(9) On the outside of the tank, apply a fillet seal around the periphery of the valve adapter plate [6] and the spar (TASK 51-22-00-390-806) (View A-A, Figure 402).

(a) Use sealant, A50110, to apply the fillet seal.

SUBTASK 28-22-21-765-004

28-AWL-21: CDCCL

(10) On the outside of the fuel tank, measure the electrical bonding resistance between one of the raised serrated surfaces of the valve adapter plate [6] and the rear spar (View F) (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-21: CDCCL

(b) Make sure the electrical bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) If the electrical bonding resistance is more than 0.0005 ohm (0.5 milliohm), then do the applicable procedure in this task: Rework the Electrical Faying Surface Bonds for the Engine Crossfeed Valve, TASK 28-22-21-300-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-21-410-003

(11) Do this task to close the fuel tank: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Install this access panel:

Number	Name/Location
131AB	Center Tank Access

Do this task: Center Tank Access Door - Installation, TASK 28-11-31-400-801.

SUBTASK 28-22-21-820-001

(12) Do this task: Engine Fuel Crossfeed Valve Alignment, TASK 28-22-21-820-801.

I. Install the Index Plate

SUBTASK 28-22-21-820-002

(1) Do this procedure after you have correctly aligned and marked the alignment position of the index plate [5] and adapter plate per this task: Engine Fuel Crossfeed Valve Alignment, TASK 28-22-21-820-801.

SUBTASK 28-22-21-020-005

(2) Remove the index plate [5] from the valve adapter plate [6].

SUBTASK 28-22-21-110-004

(3) Do these steps to prepare the index plate [5] and the valve adapter plate [6] for a sealed electrical faying surface bond (SWPM 20-20-00):

(a) Protect the alignment reference mark when you clean the index plate [5] and the valve adapter plate [6] for the electrical bond.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(b) Final clean the contact surfaces with a cotton wiper, G00034, soaked with alcohol, B00130, or solvent, B00184.

(c) Rub dry with a clean, dry cotton wiper, G00034.

(d) Continue to clean and dry the surface until the dry cotton wiper, G00034, stays clean.

28-AWL-21: CDCCL

(e) Apply a fay sealed fay surface bond between the serrated mating surfaces of the index plate [5] (three locations) and the valve adapter plate [6].

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) Use BMS 5-45 Class B-2 sealant, A50110, BMS 5-45 Class C sealant, A50155, BMS 5-95 Class B sealant, A50231, or BMS 5-95 Class C sealant, A50296, for the faying surface seal that is outside the tank.

2) Make sure that the sealant layer is approximately 0.005 in. (0.127 mm) thick.

(f) Apply sealant to the shank and the threads of the two top index screws [23] and the bottom index screw [24].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Use sealant, A50337 (BMS 5-45 Class B), for the wet installation of the index screws.

2) Make sure that the sealant applied to the fasteners is approximately 0.060 in. (1.524 mm) thick.

SUBTASK 28-22-21-420-014

(4) Do these steps to install the index plate [5]:

(a) Align the index plate [5] and the valve adapter plate [6] with the alignment reference mark that you made during the crossfeed valve alignment procedure.

(b) Install the two longer index screws [23] and washers [22] in the top index mounting hole positions.

(c) Install the shorter index screw [24] and washer [22] in the bottom index mounting hole location (6 o'clock position).

(d) Tighten the index screws to 22 ±1 in-lb (2.5 ±0.1 N·m).

(e) Make sure that the sealant is continuously squeezed out along the edges of the contact surfaces.

(f) If there are gaps, bubbles or voids in the sealant squeeze out, then disassemble and apply more sealant.

(g) Shape the squeezed out sealant into a fillet seal (TASK 51-22-00-390-806).

(h) As an option, remove the extra squeezed out sealant.

NOTE: Make sure the sealant that remains is flush with the mating part edges.

SUBTASK 28-22-21-765-005

(5) Measure the electrical bonding resistance between one of the crossfeed valve actuator [15] feet contact surfaces on the index plate [5] (bare metal surface) and the rear spar (View G, Figure 402).

(a) Use an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(b) Make sure that the electrical bonding resistance is 0.002 ohm (2.0 milliohms) or less.

1) If the electrical bonding resistance is more than 0.002 ohm (2.0 milliohms), then do the applicable procedure in this task: Rework the Electrical Faying Surface Bonds for the Engine Crossfeed Valve, TASK 28-22-21-300-801.

SUBTASK 28-22-21-760-002

28-AWL-21: CDCCL

(6) Measure the electrical bonding resistance between the index plate [5] and the valve adapter plate [6] (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

(a) Use an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

28-AWL-21: CDCCL

(b) Make sure the electrical bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-21.

1) If the electrical bonding resistance is more than 0.0005 ohm (0.5 milliohm), then do the applicable procedure in this task: Rework the Electrical Faying Surface Bonds for the Engine Crossfeed Valve, TASK 28-22-21-300-801.

SUBTASK 28-22-21-420-015

(7) Lockwire the two top two screws [23] and the bottom screw [24] to the index plate [5].

SUBTASK 28-22-21-420-016

(8) Do this task: Engine Fuel Crossfeed Valve Actuator Installation, TASK 28-22-21-400-801.

SUBTASK 28-22-21-370-001

(9) Do this task to apply protective finishes to the bare metal areas of the rear spar: Apply the Corrosion Resistant Finish (Topcoat), TASK 28-11-00-600-801.

NOTE: Re-apply the protective coating to the rear spar at all bare metal areas not covered by the fillet seal, and where the electrical bonding probe removed the finishes.

(a) These are the protective finishes:

1) Alodine 600 coating, C00064

2) BMS 10-20 coating, C00307.

J. Crossfeed Valve Operational Tests

NOTE: These tests will make sure that the valve adapter plate electrical operation is correct, the valve adapter plate alignment is correct, and that there are no fuels leaks through the rear spar.

SUBTASK 28-22-21-760-003

(1) Do these tasks:

(a) Crossfeed Valve Operational Test, TASK 28-22-21-710-802.

NOTE: This test is done when you install the crossfeed valve actuator.

(b) Crossfeed Valve - Functional Test, TASK 28-22-00-730-802.

SUBTASK 28-22-21-790-001

(2) Do this leak check if you have replaced the valve adapter plate [6]:

NOTE: The leak check is only necessary if you have done a fuel tank entry and replaced the valve adapter plate. It is not necessary if you have replaced or aligned the index plate for adjustment purposes only.

(a) Refuel the center fuel tank to a minimum of 13,000 lb (5897 kg) (TASK 12-11-00-650-803).

NOTE: This is the fuel quantity necessary to make sure that there are no leaks at the rear spar.

(b) Make sure that there are no fuel leaks at the valve adapter plate [6] installation on the rear spar.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Make sure that there are no fuel leaks at this access door:

Number	Name/Location
131AB	Center Tank Access

K. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-21-090-002

WARNING MAKE SURE THE LANDING GEAR DOWNLOCK PINS ARE REMOVED, BEFORE THE LANDING GEAR CONTROL LEVER CAN BE MOVED TO THE UP POSITION. DAMAGE TO EQUIPMENT AND INJURIES TO PERSONNEL CAN OCCUR.

WARNING DO THE APPLICABLE PROCEDURE TO REMOVE THE LANDING-GEAR-DOWNLOCK PINS INSTALLED ON THE LANDING GEAR. PERSONNEL INJURY AND DAMAGE TO EQUIPMENT CAN OCCUR IF YOU OPERATE THE LANDING GEAR BEFORE YOU REMOVE THE DOWNLOCK PINS.

(1) Remove the ground locks installed on the landing gear if they are not needed for other tasks (TASK 32-00-01-080-801).

——— END OF TASK ———

TASK 28-22-21-820-801

6. Engine Fuel Crossfeed Valve Alignment

(Figure 403)

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1587	Wrench - Torque, 30 in-lbs, +/-5% accuracy Part #: 5718A42 Supplier: 3A054 Opt Part #: TE3FUA Supplier: 55719
SPL-1771	Alignment Equipment - Fuel Shutoff Valve Part #: B28009-1 Supplier: 81205

C. Location Zones

Zone	Area
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right

D. Prepare for the Procedure

SUBTASK 28-22-21-490-002

WARNING MAKE SURE THAT YOU INSTALL THE GROUND LOCK ASSEMBLIES ON THE LANDING GEAR. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR IF YOU DO NOT INSTALL THE GROUND LOCK ASSEMBLIES.

(1) Do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-21-010-003

(2) Go to the crossfeed valve location on the rear spar in the main landing gear wheel well (right side) (View A, Figure 403).

SUBTASK 28-22-21-010-004

(3) If not done previously, do this task: Engine Fuel Crossfeed Valve Actuator Removal, TASK 28-22-21-000-801.

SUBTASK 28-22-21-010-005

(4) If not done previously, remove the index plate [5] from the valve adapter plate [6].

E. Crossfeed Valve Alignment

SUBTASK 28-22-21-820-003

(1) Do these steps to install the alignment tool [64] (valve alignment equipment, SPL-1771) in the valve adapter plate [6] (View B, Figure 403):

(a) Position the index plate [5] and align the serrated surfaces of the index plate [5] and valve adapter plate [6].

(b) Loosely install the bottom index screw [66] to the index plate [5].

(c) Install the alignment tool [64].

(d) Put the alignment tool [64] in its position with the two guide pins in the holes left by the two alignment screws [61] that you removed (the scale [62] should be on the top of the tool).

(e) Make sure that the spline on the alignment tool [64] engages with the spline on the shaft.

SUBTASK 28-22-21-820-004

(2) Do these steps to align the valve adapter plate [6] (Procedure 1, optional to Procedure 2):

NOTE: This procedure finds the point where the internal valve butterfly engages the valve seal. The torque necessary to turn the shaft increases as the butterfly engages the seal, then decreases again as the butterfly leaves the other side of the seal. Once the edge of the seal is found, the valve shaft will then be turned to center the butterfly on the seal.

(a) Put the tool handle in the extreme counterclockwise position.

(b) Slowly and smoothly turn the handle to the extreme clockwise position while you note the changes in the torque necessary to turn the handle.

(c) Do the previous steps as many times as necessary to find the point in the rotation where the torque necessary to turn the handle increases significantly.

(d) If you cannot find the position in the rotation where the torque increases significantly, then use Procedure 2 below.

(e) Put the shaft in the extreme counterclockwise position again.

(f) Slowly and smoothly turn the handle of the tool clockwise until a distinct increase in torque is found and immediately stop turning the handle.

(g) Read the scale [62] on the tool.

NOTE: This is the point where the valve butterfly starts to engage the edge of the seal.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Turn the handle 13 degrees more in the clockwise direction (the scale is calibrated in degrees).

NOTE: The valve butterfly is now centered on the valve seal.

NOTE: If you turn the handle past the target value, do not turn the handle back in the counterclockwise direction again. You must start the procedure again with the handle in the extreme counterclockwise position.

SUBTASK 28-22-21-820-005

(3) Do these steps to align the valve adapter plate [6] (Procedure 2, optional to Procedure 1):

NOTE: A torque wrench (0-30 lbs), COM-1587, is necessary to do this procedure.

(a) Put the tool handle in the extreme counterclockwise position.

(b) Slowly and smoothly turn the tool handle to the extreme clockwise position while you monitor the torque that you apply to the handle.

(c) Write down the maximum torque that you apply to the tool while you turn it from the extreme counterclockwise position to the extreme clockwise position.

(d) Turn the tool back to the extreme counterclockwise position.

(e) Turn the shaft clockwise again at approximately the same rate that you used before.

(f) Stop at the position where the torque increases to approximately two-thirds of the maximum torque that you wrote down before.

(g) Read the scale [62] on the tool at this position.

NOTE: This is the position where valve butterfly starts to engage the edge of the seal.

(h) Turn the handle 13 more degrees in the clockwise direction past the reading from the previous step.

NOTE: The butterfly valve is now centered on the valve seal.

NOTE: If you turn the valve past the target value, do not turn it back in the counterclockwise direction. You must start again with the tool handle in the extreme counterclockwise position.

SUBTASK 28-22-21-820-006

(4) Tighten the lockscrew [63] on the alignment tool [64] to keep the tool in its position on the valve adapter plate [6].

SUBTASK 28-22-21-820-007

(5) Turn the index plate [5] until the actuator mounting holes in the index plate are aligned with the two upper holes for the alignment screws [61].

SUBTASK 28-22-21-820-008

(6) Insert the two alignment screws [61] into the holes for the actuator mounting screws to hold the index plate [5] in alignment with the adapter plate (this is the final adjustment of the position of the index plate).

SUBTASK 28-22-21-820-009

(7) Tighten the bottom index screw [66].

SUBTASK 28-22-21-820-010

(8) Carefully remove the two alignment screws [61].

SUBTASK 28-22-21-820-011

(9) Loosen the lockscrew [63] and carefully remove the alignment tool [64].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-21-820-012

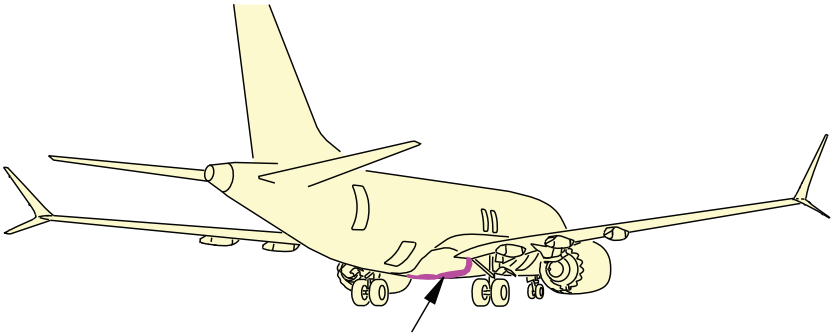
(10) Use a fine point felt pen to apply a reference mark between the index plate [5] and the valve adapter plate [6] at one of the serrated attachment points.

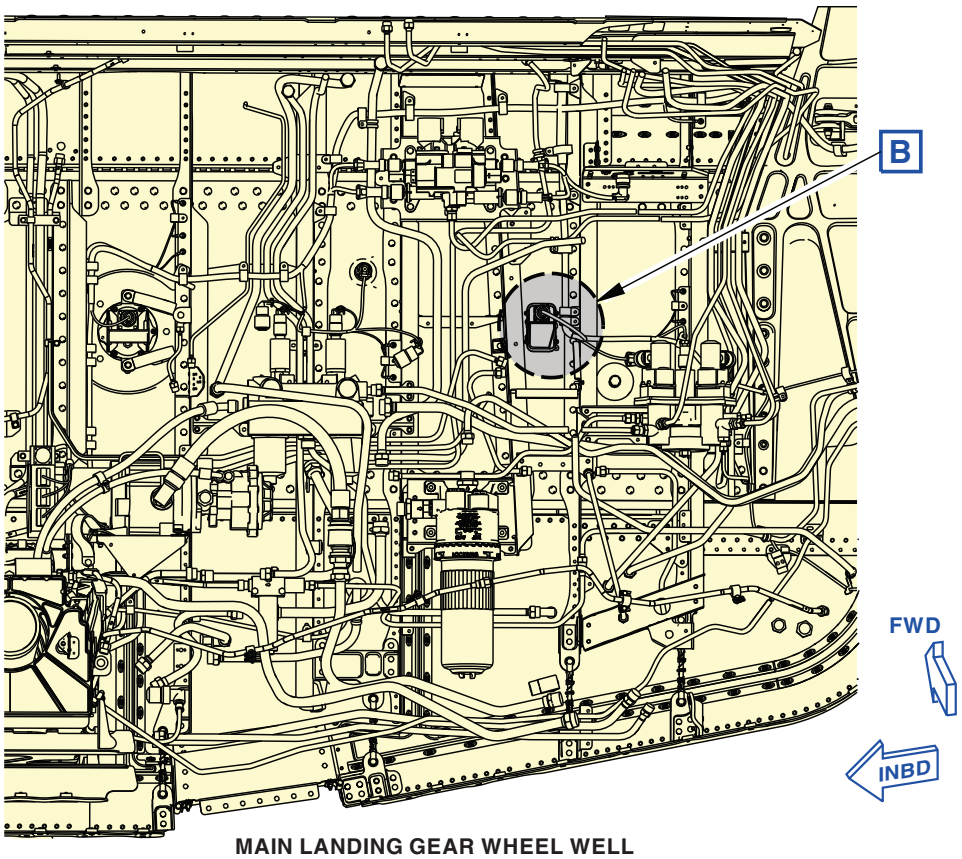
SUBTASK 28-22-21-420-017

(11) Continue with the index plate installation procedure in this task: Engine Fuel Crossfeed Valve - Valve Adapter Installation, TASK 28-22-21-400-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



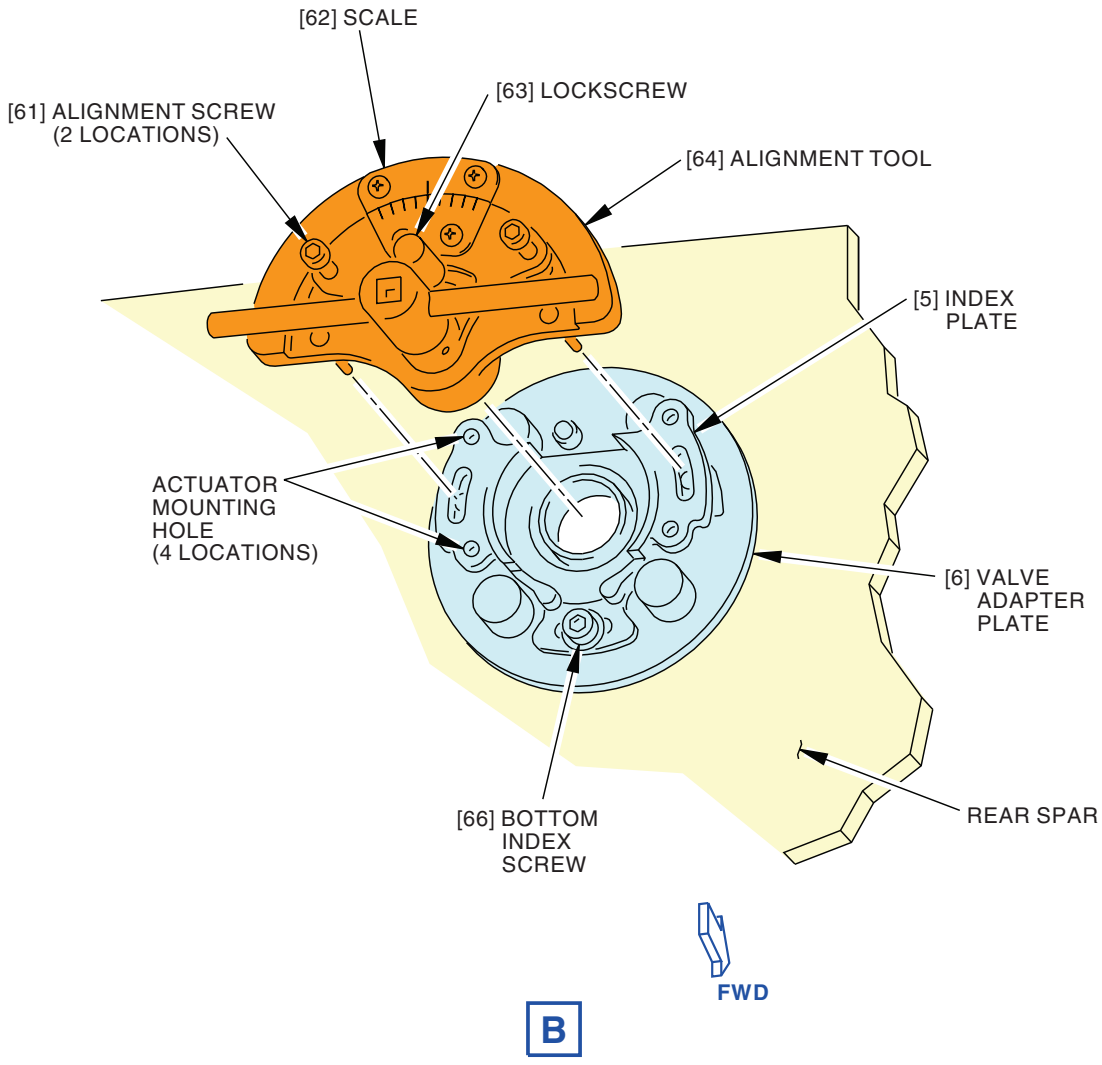


MAIN LANDING GEAR WHEEL WELL
(RIGHT SIDE)
(EXAMPLE)
A

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Crossfeed Valve Alignment
Figure 403/28-22-21-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Crossfeed Valve Alignment
Figure 403/28-22-21-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-21-000-803

7. Engine Fuel Crossfeed Valve - Valve Body Removal

(Figure 404)

A. General

(1) This task gives instructions to remove the engine fuel crossfeed valve body [21].

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-22-15-000-803	Couplings Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-77	Air Source, Regulated - Dry Filtered, 0-50 psig
STD-200	Container - Fuel Resistant, 10-gallon (38-Liter)

D. Location Zones

Zone	Area
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
413	Engine 1 - Fan Cowl, Left
423	Engine 2 - Fan Cowl, Left

E. Access Panels

Number	Name/Location
131AB	Center Tank Access

F. Prepare for the Removal

SUBTASK 28-22-21-650-003

(1) Defuel the center, No. 1 and No. 2 fuel tanks (TASK 28-26-00-650-801).

SUBTASK 28-22-21-650-004

(2) To reduce the quantity of fuel that drains from fuel lines while in the center tank, you can do these steps to drain the fuel supply line for the left and right engines (optional) (Figure 405):

(a) Do these steps to make sure that the engine fuel valves (spar and crossfeed) are in the correct positions:

1) Make sure that the engine start levers on the control stand are in the CUTOFF position.

2) Make sure that the engine START switches are in the OFF position on the overhead panel, P5.

3) Put a DO-NOT-OPERATE placard on the left and right engine START switches on the overhead panel, P5.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

5) On the fuel system panel, P5, set the CROSSFEED switch to the open position.

6) Make sure that the VALVE OPEN light for the crossfeed valve comes on bright during valve transit and then stays on dim to show that the crossfeed valve is open.

7) Move the engine start levers on the control stand, P10, to the IDLE position.

NOTE: This opens the fuel spar valves.

8) On the forward overhead panel, P5, make sure that the two SPAR VALVE CLOSED lights are off.

WARNING

BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING

DO NOT LET THE ENGINE FUEL STAY ON YOUR SKIN. FUEL IS POISONOUS AND CAN CAUSE INJURY.

(b) Do this task: TASK 71-11-04-010-801-G00.

(c) Put a fuel resistant container (10 gal), STD-200, below each fuel filter drain plug.

(d) Do the applicable steps in this task to remove the drain plug from the fuel filter:

(e) Let the fuel drain into the fuel resistant container (10 gal), STD-200.

(f) Apply air pressure at the left engine to remove the trapped fuel in the fuel line [31].

1) Supply air pressure with a regulated air source, STD-77, to a maximum of 40 psi (276 kPa), to the left engine fuel filter drain hole.

2) Let the remaining fuel drain completely into the fuel resistant container (10 gal), STD-200, at the right engine fuel filter drain hole.

(g) Do the applicable steps in this task to install the drain plug on the fuel filter:

(h) Do this task: TASK 71-11-04-410-801-G00.

(i) Put the fuel crossfeed switch, on the overhead panel, P5, to the CLOSED position.

(j) Put the engine start levers on the control stand, P10, to the CUTOFF position.

NOTE: This closes the fuel spar valves.

(k) On the forward overhead panel, P5, make sure that the two SPAR VALVE CLOSED lights are dim.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(l) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

(m) Remove the DO-NOT-OPERATE placards from the left and right engine START switches, on the overhead panel, P5.

SUBTASK 28-22-21-910-001

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do the fuel tank purging and entry procedure for the center tank (TASK 28-11-00-910-802).

(a) Remove the center tank access door:

Number	Name/Location
131AB	Center Tank Access

(TASK 28-11-31-000-801).

G. Engine Fuel Crossfeed Valve Body Removal

SUBTASK 28-22-21-010-006

WARNING DO THE PURGING AND FUEL-TANK ENTRY PROCEDURE. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Go into the center fuel tank to the crossfeed valve location (TASK 28-11-00-910-802) (View A, Figure 404).

SUBTASK 28-22-21-480-001

(2) Put a fuel resistant container (10 gal), STD-200, below the crossfeed valve body [21] to collect fuel.

SUBTASK 28-22-21-020-006

(3) Disconnect the two flexible full couplings [32] from each side of the crossfeed valve body [21] (TASK 28-22-15-000-803) (View B, Figure 404).

SUBTASK 28-22-21-020-007

(4) Hold the crossfeed valve body [21] and remove the four screws [34] and washers [35] that attach the crossfeed valve body [21] to the support bracket [33] (View C, Figure 404).

SUBTASK 28-22-21-020-008

(5) Disengage the crossfeed valve body [21] from the adapter shaft attached to the valve adapter plate [6] (View C, Figure 404).

SUBTASK 28-22-21-020-009

(6) Remove the crossfeed valve body [21].

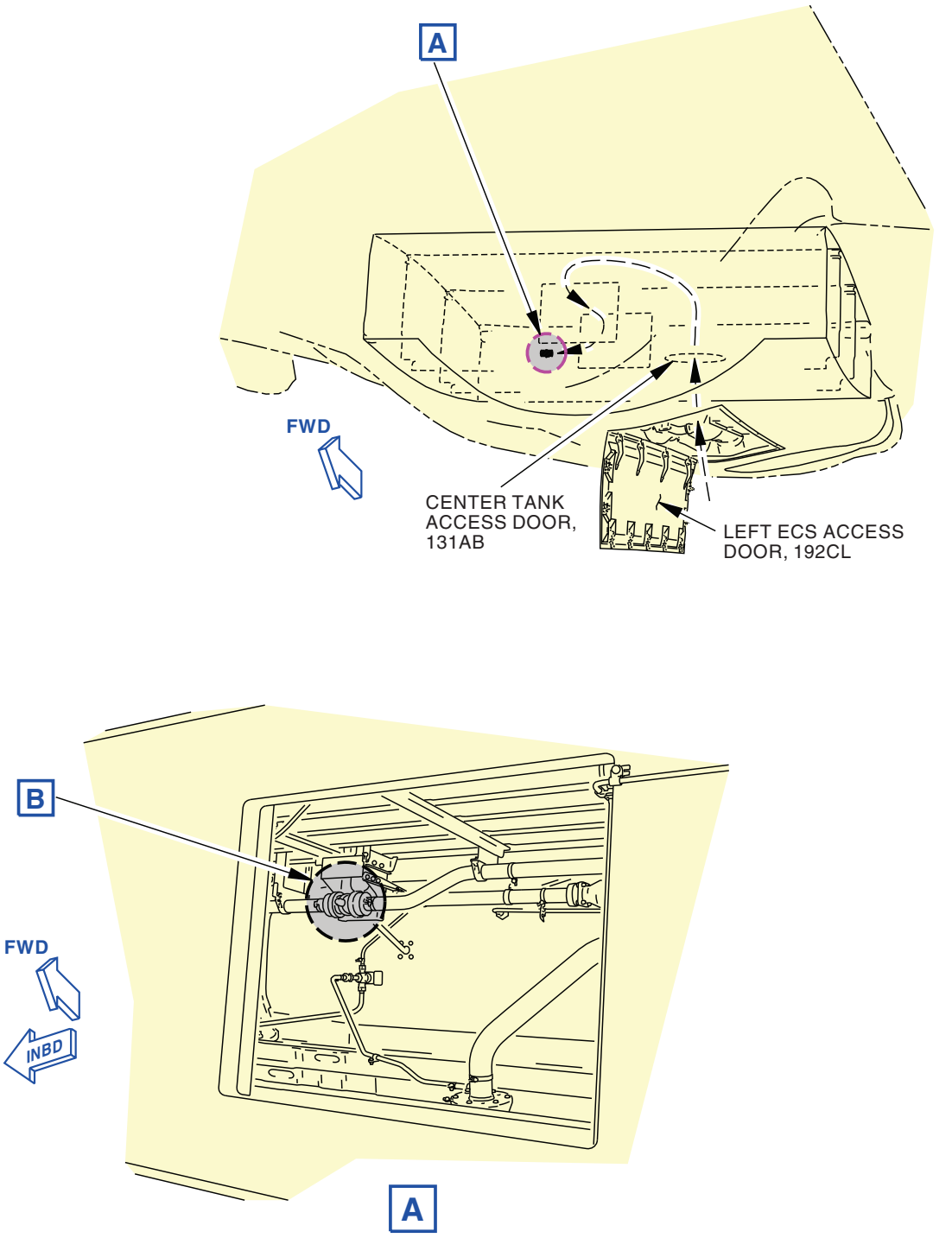
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-21-480-002

(7) Put the covers on the openings in the fuel line [31] to keep unwanted materials out.

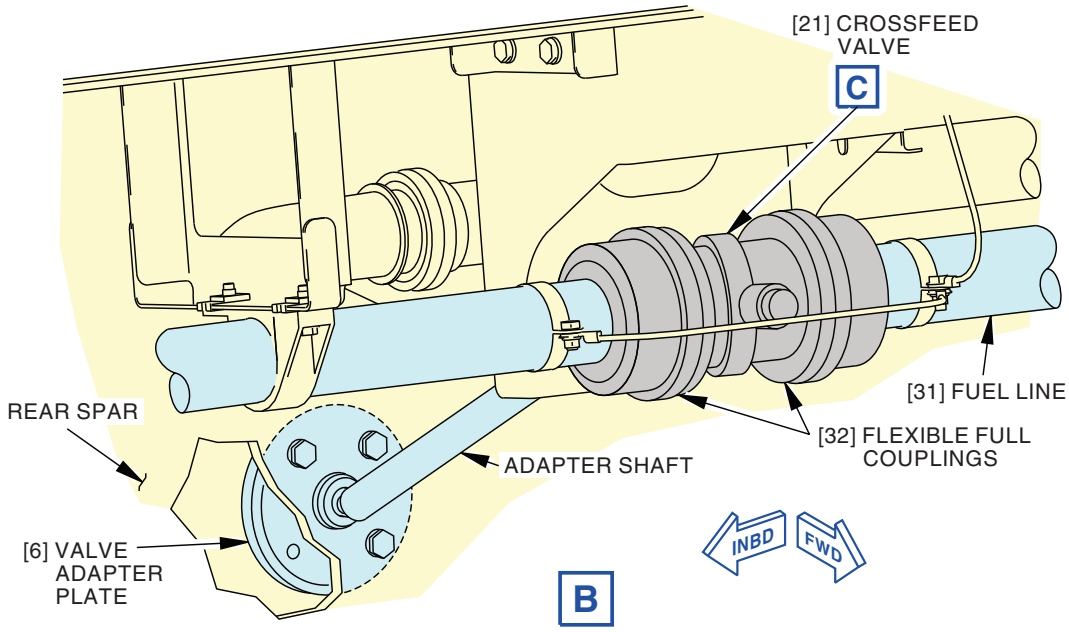
——— END OF TASK ———

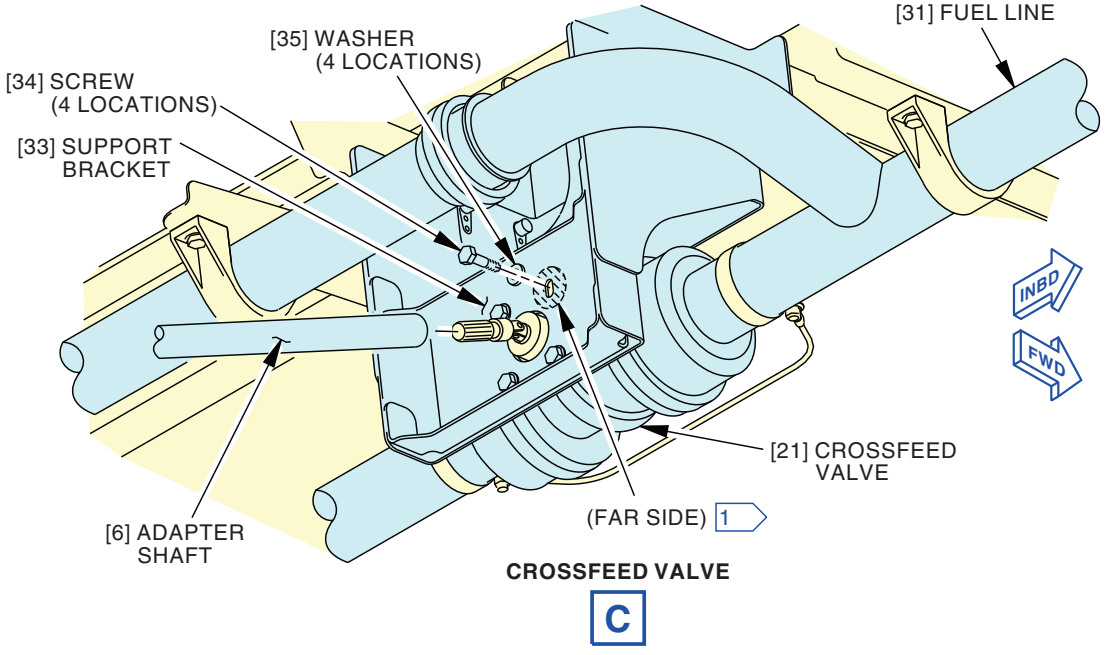
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Crossfeed Valve Body Installation
Figure 404/28-22-21-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



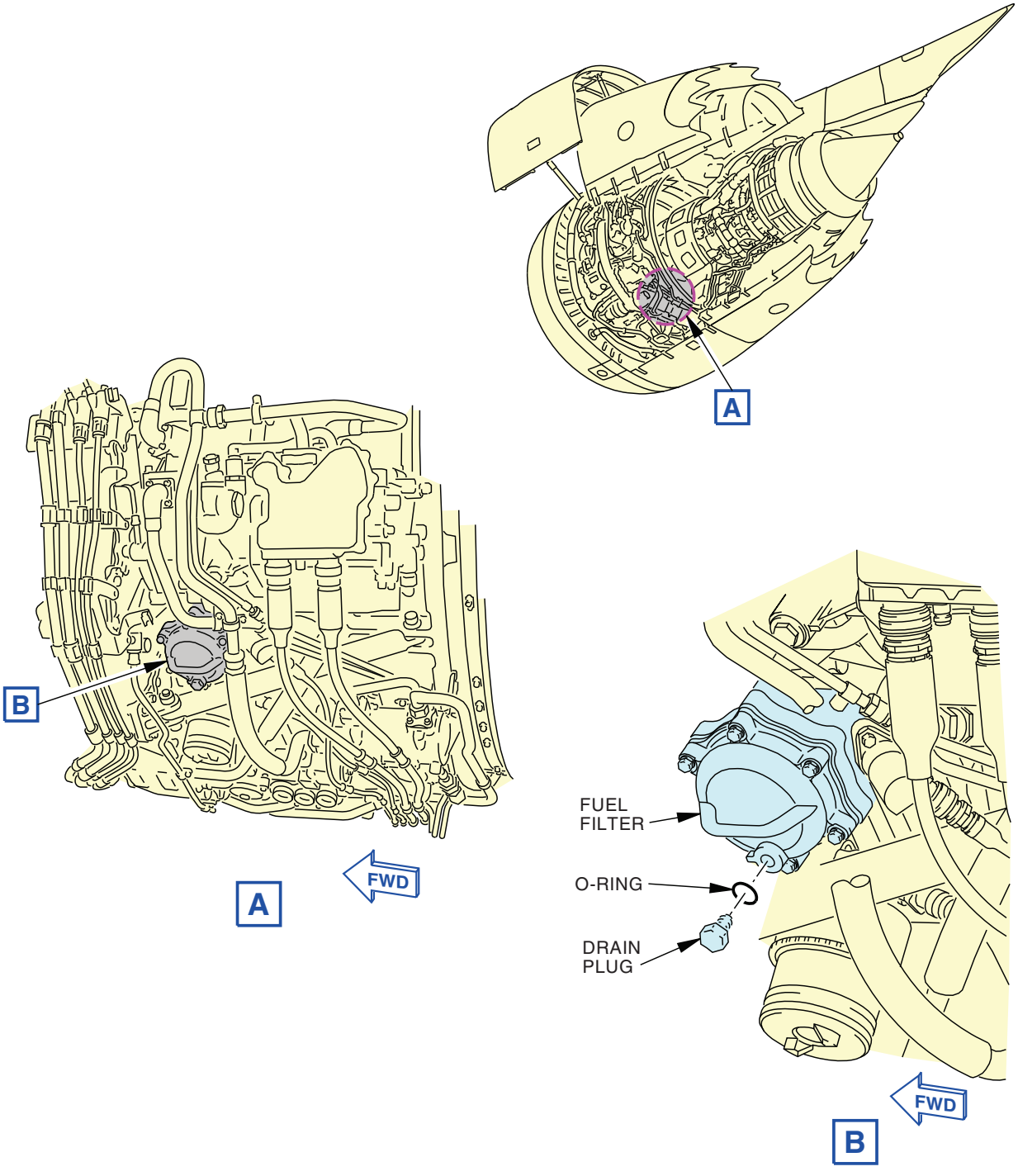


1 PREPARE THE AREA AROUND THE INBOARD TOP FASTENER HOLE AND THE CONTACT SURFACE OF THE VALVE BODY FOR AN UNSEALED ELECTRICAL FAYING SURFACE BOND (SWPM 20-20-00).

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Crossfeed Valve Body Installation
Figure 404/28-22-21-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Filter Drain Location
Figure 405/28-22-21-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-21-400-803

8. Engine Fuel Crossfeed Valve - Valve Body Installation

(Figure 404)

A. References

Reference	Title
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
28-22-00-710-802	Engine Fuel Suction Feed - Operational Test (P/B 501)
28-22-00-730-802	Crossfeed Valve - Functional Test (P/B 501)
28-22-15-400-803	Couplings Installation (P/B 401)
28-22-15-710-801	Engine Fuel Feed Manifold - Leak Test (P/B 601)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
21	Crossfeed valve body	28-22-21-01-100	PTF ALL

E. Location Zones

Zone	Area
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right

F. Access Panels

Number	Name/Location
131AB	Center Tank Access

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Engine Fuel Crossfeed Valve Body Installation

SUBTASK 28-22-21-010-007

WARNING DO THE PURGING AND FUEL TANK ENTRY PROCEDURE. FAILURE TO FOLLOW THE PURGING AND FUEL TANK ENTRY PROCEDURE COULD CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Go into the center wing tank to the crossfeed valve location (TASK 28-11-00-910-802).

SUBTASK 28-22-21-020-010

(2) Remove the covers from the fuel lines [31].

SUBTASK 28-22-21-110-005

(3) Do these steps to prepare the inboard top fastener hole of the support bracket [33] and the contact surface of the crossfeed valve body [21] for an unsealed electrical faying surface bond (View C, Figure 404):

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE. OBEY THE MATERIAL SAFETY DATA SHEETS (MSDS) FOR SOLVENTS. OBEY LOCAL REGULATIONS FOR THE CORRECT PROCEDURES TO USE OR DISCARD SOLVENTS. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Final clean the contact surfaces of the support bracket [33] and the crossfeed valve body [21] with a cotton wiper, G00034, soaked with alcohol, B00130 or solvent, B00184.

(b) Rub dry with a clean, dry cotton wiper, G00034.

(c) Continue to clean and dry the surface until the dry cotton wiper, G00034, stays clean.

SUBTASK 28-22-21-420-018

(4) Turn the universal joint fitting on the crossfeed valve body [21] until the index marks are aligned.

NOTE: A small misalignment of the index marks is satisfactory.

SUBTASK 28-22-21-420-019

(5) Put the crossfeed valve body [21] into position.

SUBTASK 28-22-21-210-001

WARNING INSTALL THE VALVE WITH THE TOP SIDE OF THE VALVE POINTED UP. THERE IS A STENCIL OF THE WORD TOP ON THE TOP SIDE. IF YOU INSTALL THE VALVE BODY INCORRECTLY, IT CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(6) Make sure that "TOP" shows on the top of the crossfeed valve body [21].

SUBTASK 28-22-21-420-020

(7) Engage the universal joint fitting on the crossfeed valve body [21] into the valve adapter plate [6].

SUBTASK 28-22-21-420-021

(8) Align the crossfeed valve body [21] with the support bracket [33].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-21-420-022

(9) Engage the crossfeed valve body [21] with the adapter shaft attached to the valve adapter plate [6].

SUBTASK 28-22-21-420-023

(10) Install the four screws [34] and the washers [35] to attach the mounting flange on the support bracket [33] to the crossfeed valve body [21].

SUBTASK 28-22-21-765-006

(11) Measure the bonding resistance between the crossfeed valve body [21] and the airplane structure with an electrical intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) Make sure that the bonding resistance is 0.010 ohm (10 milliohms) or less.

SUBTASK 28-22-21-420-024

(12) Connect the two flexible full couplings [32] between the crossfeed valve body [21] and the fuel line [31] (TASK 28-22-15-400-803).

SUBTASK 28-22-21-410-004

(13) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

SUBTASK 28-22-21-410-005

(14) Install this access panel:

Number	Name/Location
131AB	Center Tank Access

(TASK 28-11-31-400-801).

SUBTASK 28-22-21-710-001

(15) Do one of these tasks:

(a) Do a test of the engine fuel feed manifold for leakage (TASK 28-22-15-710-801).

(b) Do an operational test for the engine fuel suction feed system (TASK 28-22-00-710-802).

SUBTASK 28-22-21-710-002

(16) Do this task: Crossfeed Valve - Functional Test, TASK 28-22-00-730-802.

——— END OF TASK ———

TASK 28-22-21-300-801

9. Rework the Electrical Faying Surface Bonds for the Engine Crossfeed Valve

A. General

(1) This task contains these procedures:

(a) Rework the Faying Surface Bond - crossfeed valve actuator [15] to index plate [5]

(b) Rework the Faying Surface Bond - index plate [5] to valve adapter plate [6]

(c) Rework the Faying Surface Bond - valve adapter plate [6] to rear spar

(d) Rework the Bonding Jumper Bond.

(2) Do this task if the measurement for the electrical bonding resistance is more than the allowed resistance value.

(3) The motor operated valve uses two electrical bonding paths:

(a) One bonding path is from the upper housing crossfeed valve actuator [15] through the bonding jumper [13] to the airplane structure.

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(b) The second bonding path is from the upper housing crossfeed valve actuator [15] to the bare metal surface of the rear spar. The bond path is through the bare metal faying surfaces of the crossfeed valve actuator [15] (at the mounting feet), index plate [5], and valve adapter plate [6] to the bare metal surface of the rear spar.

NOTE: The crossfeed valve actuator [15] is divided into two sections, the upper housing section and the lower housing section. The two housing sections are separated by an electrical faying surface. You cannot repair or rework the faying surface between the housing sections at the airplane. When you do an electrical bonding measurement, make sure that the measurement is done on the upper housing section of the crossfeed valve actuator [15].

(4) If the electrical bonding resistance is more than the maximum allowed resistance, you must disassemble the components, rework the faying surface, assemble the components, and then re-check the electrical bonding resistance. Continue to rework the faying surfaces until the electrical resistance between the components is within allowable resistance values.

(5) Because the build-up of the crossfeed valve includes several faying surface bonds, you may not know which faying surface bond is the cause of the problem. This task will help you isolate the location of the faying surface bond that needs to be reworked. The procedure does a check of the faying surface bond downstream of the bond that is in question.

B. Consumable Materials

Reference	Description	Specification
A50337	Sealant - Fuel Tank	BMS5-45 Class B
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

C. Location Zones

Zone	Area
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right

D. Rework the Faying Surface Bond - Actuator to Index Plate

(Figure 401),

(Figure 402)

NOTE: Do the steps in this procedure if the measurement for the electrical bonding resistance between the upper housing crossfeed valve actuator [15] and the rear spar is more than 0.004 ohm (4.0 milliohms). The measurement is done with the bonding jumper [13] and the electrical connector disconnected from the crossfeed valve actuator [15].

NOTE: This is the repair bonding resistance value. The valve adapter installation bonding resistance value is 0.003 ohm (3.0 milliohm).

SUBTASK 28-22-21-010-008

(1) Do this task: Engine Fuel Crossfeed Valve Actuator Removal, TASK 28-22-21-000-801.

SUBTASK 28-22-21-765-007

(2) Do these steps to check the index plate [5] to rear spar bonding path:

NOTE: These steps will tell you if the crossfeed valve actuator [15] to index plate [5] faying surface bond is the cause of the problem, or if the index plate [5] to valve adapter plate [6] faying surface bond is the cause of the problem.

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(a) Remove the sealant and clean one of the contact surfaces on the index plate [5].

(b) Measure the bonding resistance between the contact surface on the index plate [5] (bare metal surface) and the rear spar (SWPM 20-20-00) (View G, Figure 402).

(c) Make sure that the bonding resistance is 0.002 ohm (2.0 milliohms) or less.

(d) If the bonding resistance is not satisfactory, then do the steps in this procedure: Rework the Faying Surface Bond - Index Plate to Adapter Plate.

(e) If the bonding resistance is satisfactory, then continue with the subsequent steps.

SUBTASK 28-22-21-420-025

(3) Do this task: Engine Fuel Crossfeed Valve Actuator Installation, TASK 28-22-21-400-801.

NOTE: This task will rework the faying surface bond and re-install the crossfeed valve actuator [15].

SUBTASK 28-22-21-765-008

(4) During the installation, do a check of the electrical bonding resistance between the upper housing crossfeed valve actuator [15] and rear spar (SWPM 20-20-00, SWPM 20-20-10) (View C, Figure 401):

(a) Do this measurement with the bonding jumper [13] and the electrical connector [1] disconnected.

(b) Make sure that the bonding jumper [13] does not touch the crossfeed valve actuator [15] during the bonding measurement.

(c) Make sure that the bonding resistance is 0.004 ohm (4.0 milliohms) or less.

NOTE: This is the repair bonding resistance value. The valve adapter plate [6] installation bonding resistance value is 0.003 ohm (3.0 milliohms).

(d) If the bonding resistance is not satisfactory, then repeat the steps in this procedure.

(e) If the bonding resistance is satisfactory, then continue with the steps to install the crossfeed valve actuator [15] (TASK 28-22-21-400-801).

SUBTASK 28-22-21-710-003

(5) If you have replaced the valve adapter plate [6] or adjusted the index plate [5], then continue with the steps in this task: Crossfeed Valve Operational Tests (TASK 28-22-21-400-802).

E. Rework the Faying Surface Bond - Index Plate to Adapter Plate

(Figure 402)

NOTE: Do the steps in this procedure if the measurement for the electrical bonding resistance between the index plate [5] and the rear spar is more than 0.002 ohm (2.0 milliohms). The measurement is done with the crossfeed valve actuator [15] removed.

SUBTASK 28-22-21-020-011

(1) Do this task: Remove the index plate [5] (TASK 28-22-21-000-801).

SUBTASK 28-22-21-765-009

(2) Do these steps to check the valve adapter plate [6] to rear spar bonding path:

NOTE: These steps will tell you if the index plate [5] to valve adapter plate [6] faying surface bond is the cause of the problem, or if the valve adapter plate [6] rear spar faying surface bond is the cause of the problem.

(a) Remove the sealant and clean one of the serrated contact surfaces on the valve adapter plate [6].

(b) Measure the bonding resistance between the serrated contact surface on the valve adapter plate [6] and the rear spar (SWPM 20-20-00) (View F, Figure 402).

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(c) Make sure that the bonding resistance is 0.0015 ohm (1.5 milliohms) or less.

NOTE: This is the repair bonding resistance value. The valve adapter plate [6] installation bonding resistance value is 0.0005 ohm (0.5 milliohm).

(d) If the bonding resistance is not satisfactory, then do the steps in this procedure: Rework the Faying Surface Bond - valve adapter plate [6] to rear spar.

(e) If the bonding resistance is satisfactory, then continue with the subsequent steps.

SUBTASK 28-22-21-420-026

(3) Do these steps to install the index plate [5]:

NOTE: These steps will re-align the crossfeed valve, rework the faying surface, and re-install the index plate [5].

(a) Do this task: Engine Fuel Crossfeed Valve Alignment, TASK 28-22-21-820-801.

(b) Do this procedure: Install the index plate [5] (Engine Fuel Crossfeed Valve - Valve Adapter Installation, TASK 28-22-21-400-802).

SUBTASK 28-22-21-765-010

(4) During the installation, check the electrical bonding resistance between the index plate [5] and the rear spar (SWPM 20-20-00) (View G, Figure 402):

(a) Make sure that the bonding resistance is 0.002 ohm (2.0 milliohms) or less.

(b) If the bonding resistance is not satisfactory, then repeat the steps in this procedure.

(c) If the bonding resistance is satisfactory, then continue with the procedure to install the index plate [5] (TASK 28-22-21-400-802)

SUBTASK 28-22-21-710-004

(5) After you install the index plate [5], do this procedure: Crossfeed Valve Operational Tests (TASK 28-22-21-400-802).

F. Rework the Faying Surface Bond - Adapter Plate to Rear Spar

(Figure 402)

NOTE: Do the steps in this procedure if the measurement for the electrical bonding resistance between the adapter plate and the rear spar is more than 0.0015 ohm (1.5 milliohms).

SUBTASK 28-22-21-020-012

(1) Do this procedure: Remove the Valve Adapter (TASK 28-22-21-000-802).

NOTE: A fuel tank entry is necessary to rework the faying surface of the valve adapter.

SUBTASK 28-22-21-420-027

(2) Do this procedure: Install the Valve Adapter (TASK 28-22-21-400-802).

NOTE: These steps will rework the faying surface, and re-install the valve adapter.

SUBTASK 28-22-21-765-011

(3) During the installation, check the electrical bonding resistance between the valve adapter plate [6] and the rear spar (SWPM 20-20-00) (View F, Figure 402).

(a) Make sure the bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

(b) If the bonding resistance is not satisfactory, then repeat the steps in this procedure.

(c) If the bonding resistance is satisfactory, then continue with the steps to install the valve adapter plate [6] (TASK 28-22-21-400-802).

SUBTASK 28-22-21-420-028

(4) After you install the valve adapter plate [6], do this procedure: Install the index plate [5] (TASK 28-22-21-400-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Rework the Bonding Jumper Bond

(Figure 401)

NOTE: Do the steps in this procedure if the measurement for the electrical bonding resistance between the bonding jumper [13] terminal and the rear spar is more than 0.0015 ohm (1.5 milliohms). The measurement is done with the bonding jumper [13] disconnected from the crossfeed valve actuator [15].

SUBTASK 28-22-21-020-013

(1) Remove the sealant from the bonding jumper [13] terminal attached to the structure.

SUBTASK 28-22-21-420-029

(2) Remove the nut, two washers, and the bonding jumper [13].

SUBTASK 28-22-21-420-030

(3) Do these steps to install the bonding jumper [13] to the structure:

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS) AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE. OBEY THE MATERIAL SAFETY DATA SHEETS (MSDS) FOR SOLVENTS. OBEY LOCAL REGULATIONS FOR THE CORRECT PROCEDURES TO USE OR DISCARD SOLVENTS. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Final clean the contact surfaces with a cotton wiper, G00034, soaked with alcohol, B00130 or solvent, B00184.

1) Rub dry with a clean, dry cotton wiper, G00034.

2) Continue to clean and dry the surface until the dry cotton wiper, G00034, stays clean.

(b) Install the nut, washers, and the bonding jumper [13] to the structure.

(c) Tighten the nut to 31 ±3 in-lb (4 ±1 N·m).

(d) Measure the electrical bonding resistance between the loose end of bonding jumper [13] and the rear spar (SWPM 20-20-10) (View C, Figure 401).

1) Make sure that the bonding resistance is 0.0015 ohm (1.5 milliohms) or less.

(e) Apply a cap seal of sealant, A50337, (BMS 5-45 Class B) over the nut and the terminal of the bonding jumper [13].

SUBTASK 28-22-21-420-031

(4) Continue with this task: Engine Fuel Crossfeed Valve Actuator Installation, TASK 28-22-21-400-801.

SUBTASK 28-22-21-710-005

(5) If you have replaced the valve adapter plate [6] or adjusted the index plate [5], then continue with the steps in this procedure: Crossfeed Valve Operational Tests (TASK 28-22-21-400-802).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FUEL CROSSFEED VALVE - INSPECTION/CHECK

1. General

A. This procedure contains a functional check of the bonding resistance of the engine fuel crossfeed actuator.

TASK 28-22-21-200-801

2. Engine Fuel Crossfeed Valve Actuator - Bonding Resistance Check

(Figure 601)

A. References

Reference	Title
28-22-21-710-802	Crossfeed Valve Operational Test (P/B 501)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-10	Replacement of Ground Studs and Bonding Jumper Installation

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Location Zones

Zone	Area
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right

D. Prepare for the Procedure

SUBTASK 28-22-21-490-003

WARNING MAKE SURE YOU INSTALL THE GROUND LOCK ASSEMBLIES IN ALL LANDING GEAR. ACCIDENTAL RETRACTION OF THE LANDING GEAR CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 28-22-21-865-001

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	7	C00361	FUEL CROSS FEED VALVE CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-21-560-001

(3) Go to the crossfeed valve location on the rear spar in the main landing gear wheel well (right side).

E. Electrical Bonding Measurement

SUBTASK 28-22-21-710-006

(1) Move the manual override lever to the fully CLOSED position.

SUBTASK 28-22-21-862-002

(2) Disconnect the electrical connector from the actuator.

SUBTASK 28-22-21-765-012

(3) Measure the electrical bonding resistance between the crossfeed valve actuator (at the connector flange) and the spar with a intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00, SWPM 20-20-10) (View B).

(a) Make sure the bonding resistance is 0.005 ohm (5.0 milliohms) or less.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-21-861-003

(1) Connect the electrical connector to the actuator.

SUBTASK 28-22-21-865-002

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	7	C00361	FUEL CROSS FEED VALVE CONT

SUBTASK 28-22-21-710-007

(3) Do this task: Crossfeed Valve Operational Test, TASK 28-22-21-710-802

SUBTASK 28-22-21-090-003

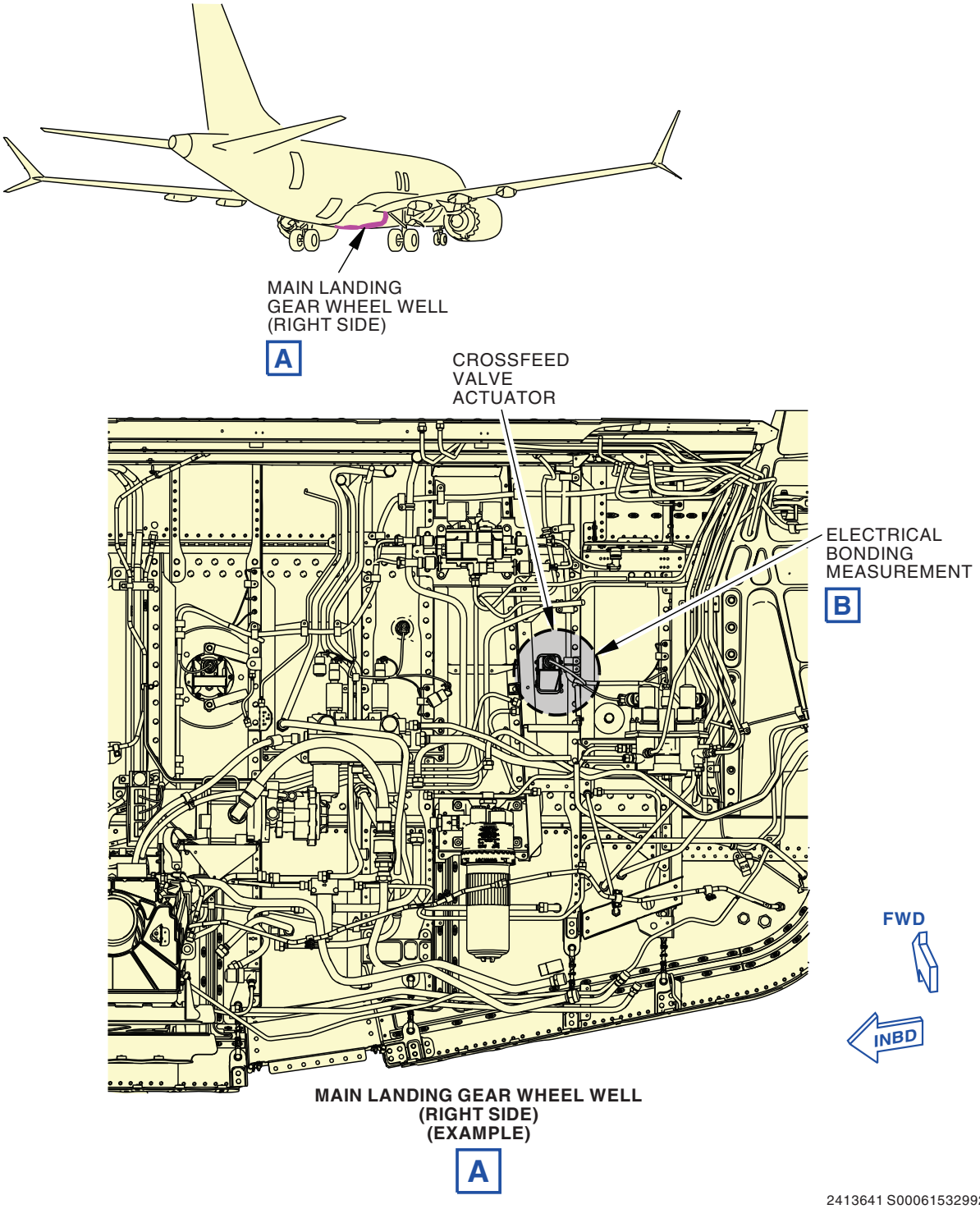
WARNING

OBEY THE PROCEDURE FOR THE REMOVAL OF THE DOWNLOCK PINS. IF YOU MOVE THE CONTROL LEVER FOR THE LANDING GEAR TO THE UP POSITION, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(4) Remove the ground locks installed on the landing gear if they are not needed for other tasks (TASK 32-00-01-080-801).

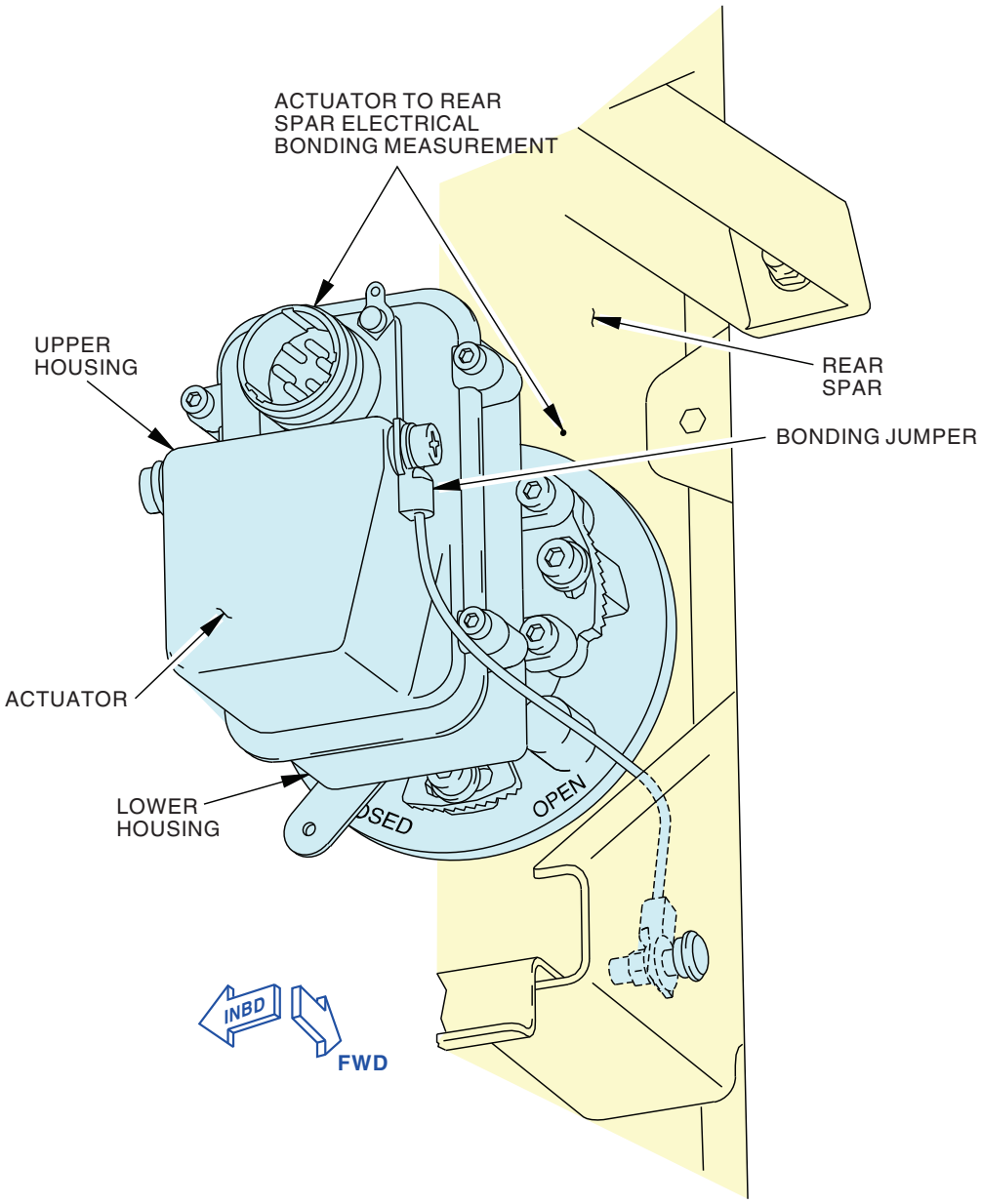
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fuel Crossfeed Valve Actuator - Bonding Resistance Check
Figure 601/28-22-21-990-807 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELECTRICAL BONDING MEASUREMENT

B

2413642 S00061532993_V1

Engine Fuel Crossfeed Valve Actuator - Bonding Resistance Check
Figure 601/28-22-21-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FUEL CROSSFEED VALVE - ADJUSTMENT/TEST

1. General

A. This procedure contains this task:

(1) Crossfeed Valve Operational Test.

TASK 28-22-21-710-802

2. Crossfeed Valve Operational Test

(Figure 501)

A. Location Zones

Zone	Area
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right

B. Crossfeed Valve Operational Test

SUBTASK 28-22-21-860-007

(1) Set the CROSSFEED VALVE switch on the forward overhead panel, P5, to the OPEN position.

SUBTASK 28-22-21-210-006

(2) Make sure that the position indication light shows these effects:

(a) The position indication light comes on brightly to show a change in the crossfeed valve position.

(b) The position indication light stays on dimly to show that the crossfeed valve is open.

SUBTASK 28-22-21-210-007

(3) Make sure that the manual override lever [9] for the crossfeed valve is in the OPEN position.

SUBTASK 28-22-21-860-008

(4) Set the CROSSFEED VALVE switch on the forward overhead panel, P5, to the CLOSED position.

SUBTASK 28-22-21-210-008

(5) Make sure that the position indication light for the crossfeed valve shows these effects:

(a) The position indication light comes on brightly to show a change in the crossfeed valve position.

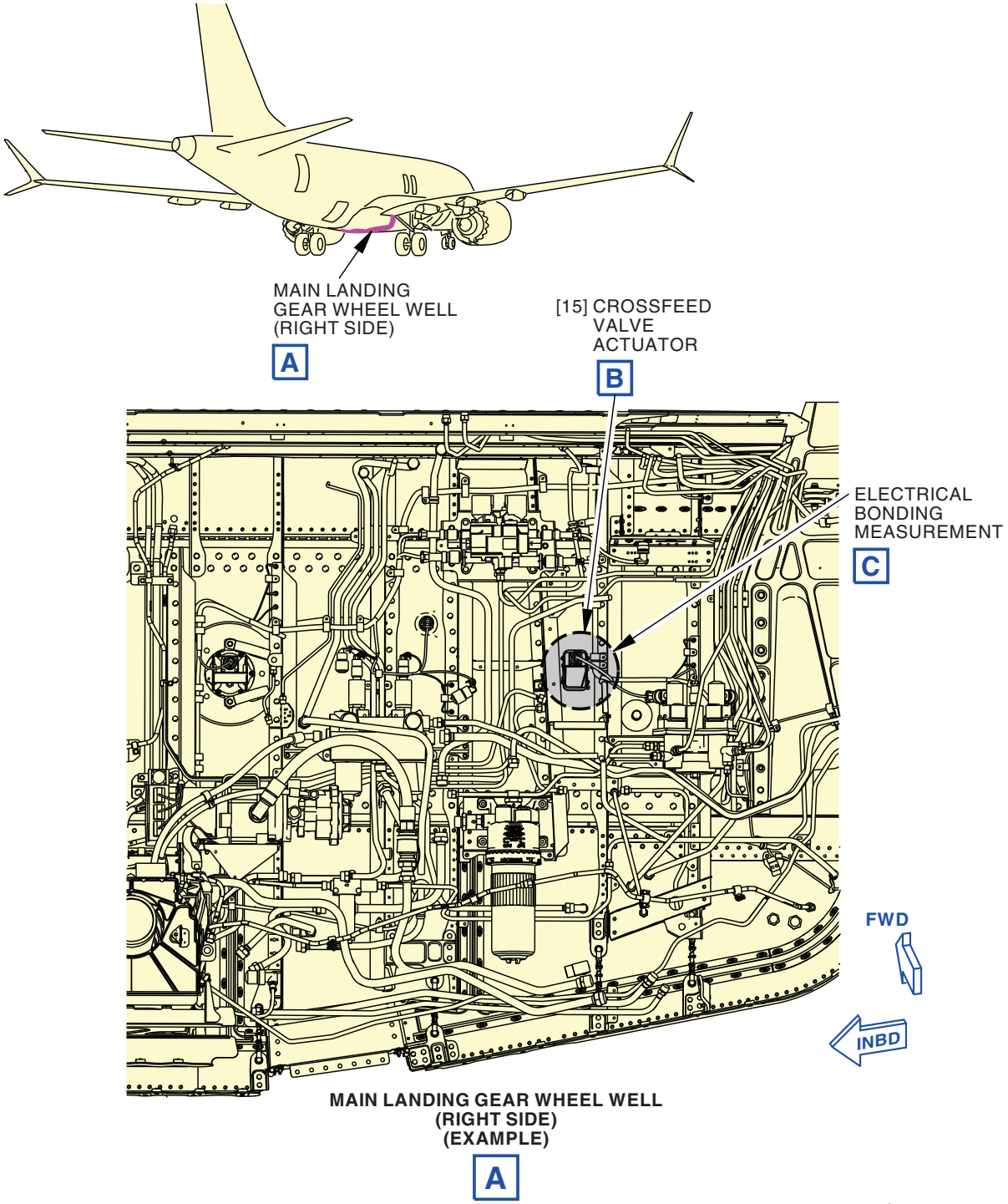
(b) The position indication light goes off to show the crossfeed valve is closed.

SUBTASK 28-22-21-210-009

(6) Make sure that the manual override lever [9] for the crossfeed valve is in the CLOSED position.

——— END OF TASK ———

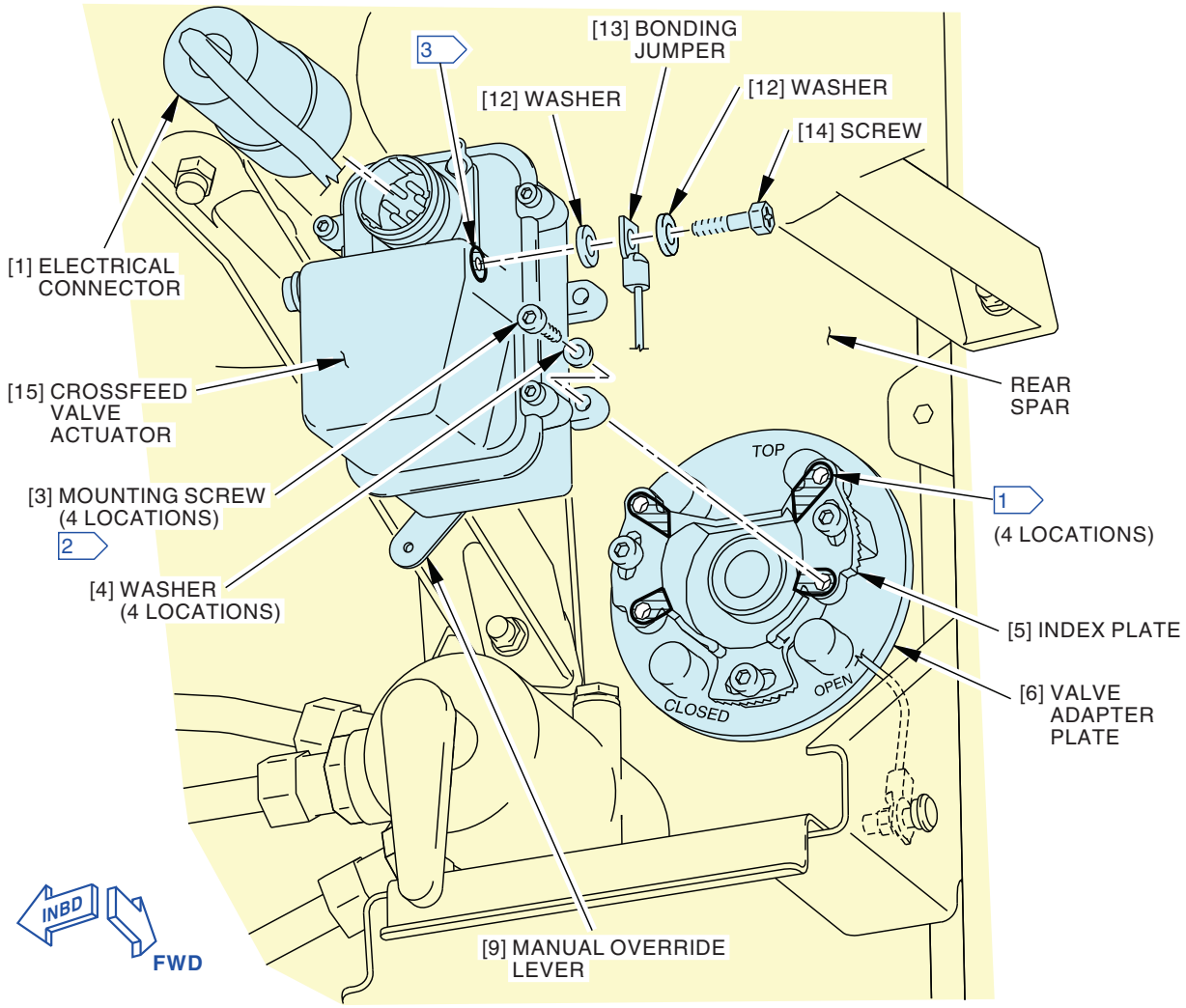
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413628 S00061532969_V2

Crossfeed Valve Actuator Installation
Figure 501/28-22-21-990-808 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CROSSFEED VALVE ACTUATOR INSTALLATION

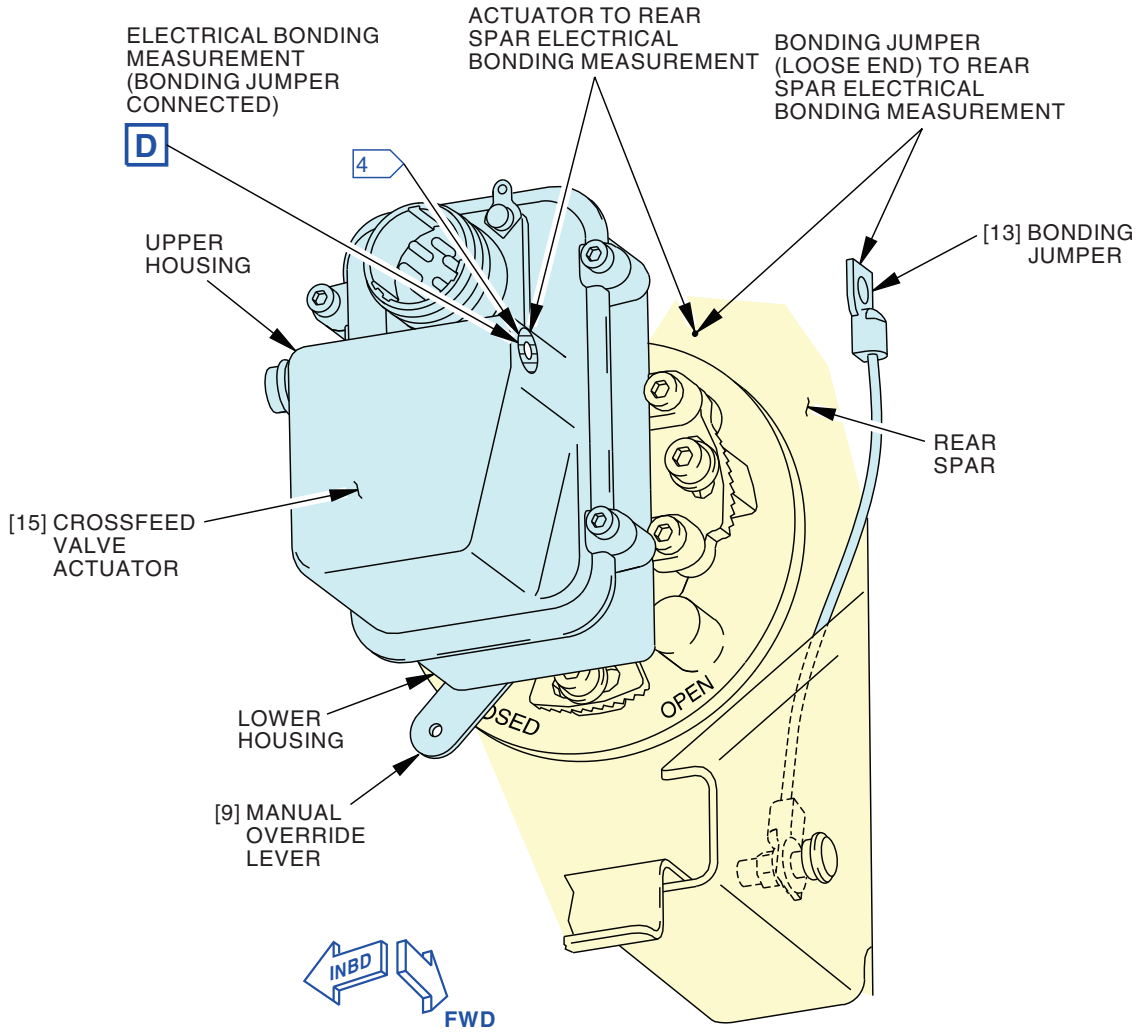
B

1. PREPARE AND INSTALL THE CONTACT SURFACES OF THE INDEX PLATE AND ACTUATOR FOOTPADS (4 LOCATIONS) WITH A FAY SEALED FAY SURFACE BOND (SWPM 20-20-00).
2. APPLY SEALANT TO THE SHANK AND THREADS OF THE MOUNTING SCREWS.
3. CLEAN THE CONTACT SURFACES OF THE ACTUATOR AND BONDING JUMPER. AFTER ELECTRICAL BONDING MEASUREMENT, APPLY A CAP SEAL TO THE BONDING JUMPER SCREW.

2413629 S00061532970_V1

Crossfeed Valve Actuator Installation
Figure 501/28-22-21-990-808 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELECTRICAL BONDING MEASUREMENT (BONDING JUMPER DISCONNECTED)

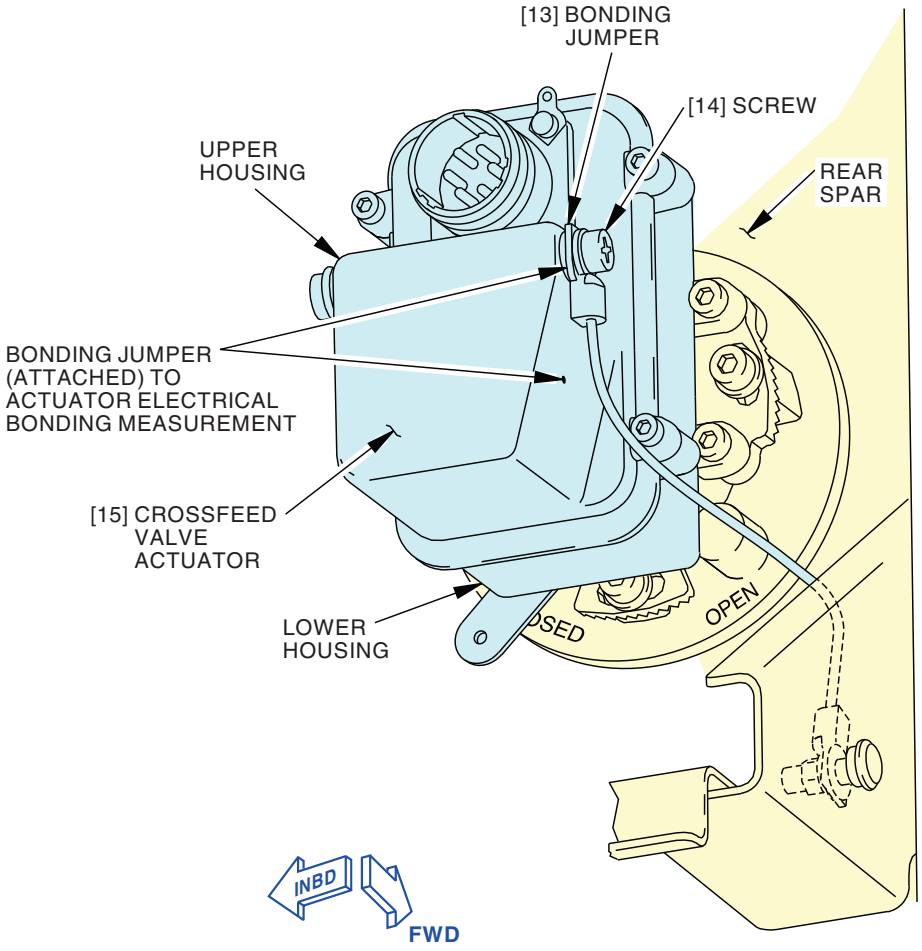
C

4 BARE METAL CONTACT SURFACE OF THE ACTUATOR AT THE BONDING JUMPER INSTALLATION.

2413630 S00061532971_V1

Crossfeed Valve Actuator Installation
Figure 501/28-22-21-990-808 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELECTRICAL BONDING MEASUREMENT (BONDING JUMPER CONNECTED)

D

2413631 S00061532972_V1

Crossfeed Valve Actuator Installation
Figure 501/28-22-21-990-808 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL BOOST PUMP - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Fuel Boost Pump Motor-Impeller Removal

(2) Fuel Boost Pump Motor-Impeller Installation

(3) Fuel Boost Pump and Override Pump Priming

(4) Fuel Boost Pump Housing Removal

(5) Fuel Boost Pump Housing Installation.

B. The fuel system consists of these fuel boost pumps:

(1) Left, Forward and Aft Fuel Boost Pumps

(2) Right, Forward and Aft Fuel Boost Pumps

(3) Left and Right Center Fuel Boost Pumps.

C. The left and right forward fuel boost pumps are installed on the inboard rear spar forward of each main landing gear door. The left and right aft fuel boost pumps are installed outboard of the left and right forward fuel boost pumps. The left and right center fuel boost pumps are installed on the rear spar of the center section of the center tank. To get access to the left center fuel boost pump, remove the pogo which is part of the aileron control unit linkage in the left wheel well.

D. Each fuel boost pump has four parts: the motor-impeller, housing, discharge check valve and a removal check valve.

E. The fuel tank must be defueled to remove the housing, discharge check valve, and the removal check valve. To remove the motor-impeller, the fuel tanks do not have to be defueled.

F. To replace the removal check valve, do these tasks:

(1) Removal Check Valve Removal, TASK 28-22-51-000-801

(2) Removal Check Valve Installation, TASK 28-22-51-400-801.

G. To replace the discharge check valve, do these tasks:

(1) Fuel Boost Pump Discharge Check Valve Removal, TASK 28-22-71-000-801

(2) Fuel Boost Pump Discharge Check Valve Installation, TASK 28-22-71-400-801.

TASK 28-22-41-000-801

2. Fuel Boost Pump Motor-Impeller Removal

(Figure 401, Figure 402, Figure 403, Figure 404, Figure 405, Figure 406)

A. General

(1) The fuel boost pump motor-impeller is referred to as the motor-impeller in this task.

(2) This task gives instructions to remove the motor-impeller.

B. References

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1769	Puller - Fuel Pump Part #: B28003-1 Supplier: 81205
STD-4049	Container - Fuel Resistant, 1 Gallon (4 Liter)
STD-8881	Screw - Retention (MS35275-250)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
551	Left Wing - Rear Spar To Landing Gear Support Beam
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
651	Right Wing - Rear Spar to Landing Gear Support Beam

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Removal

SUBTASK 28-22-41-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-22-41-010-001

(2) To access the applicable motor-impeller [8] for the forward or aft fuel boost pumps, go through the left or right main shock strut door.

SUBTASK 28-22-41-010-002

(3) To access the applicable motor-impeller [8] for the center tank fuel boost pump, do these steps:

(a) Access the right center fuel boost pump in the right wheel well.

(b) Access the left center fuel boost pump, do these additional steps (Figure 404):

1) Deactivate the aileron hydraulic system (TASK 27-11-00-860-801).

2) Remove the cotter pin [28], castellated nut [27], plain washer [26], plain bushing [24], countersunk washer [22], and hex bolt [23] to disconnect the pogo assembly [21] from the power control unit [25].

NOTE: Be careful not to change the length of the pogo assembly [21] or it will need to be adjusted after installation.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Remove the three bolts [29] and washers [30] to disconnect the bracket assembly from the power control unit [25].

4) Pull the pogo assembly [21] away from the rear spar.

NOTE: The bolt that attaches the pogo assembly [21] to the aileron feel and centering unit does not need to be removed.

SUBTASK 28-22-41-010-006

(4) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-41-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Do these steps for the applicable tanks:

(a) For the No. 1 main tank, do these steps:

1) For the aft fuel boost pump, open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT

2) For the forward fuel boost pump, open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD

(b) For the No. 2 main tank, do these steps:

1) For the aft fuel boost pump, open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT

2) For the forward fuel boost pump, open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD

(c) For the center tank, do these steps:

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1) For the left center fuel boost pump, open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

2) For the right center fuel boost pump, open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

G. Fuel Boost Pump Motor-Impeller Removal

SUBTASK 28-22-41-020-001

(1) Disconnect the electrical connector [3] from the motor-impeller [8].

SUBTASK 28-22-41-020-002

(2) Remove the lockwires from the cap screws [10] and drain plug [7].

SUBTASK 28-22-41-480-002

(3) Put a 1 gallon (4 l) fuel resistant container, STD-4049 below the motor-impeller [8] to collect fuel.

SUBTASK 28-22-41-020-003

(4) Loosen, but do not remove the drain plug [7].

SUBTASK 28-22-41-020-004

(5) Remove the two bonding jumpers [2]:

(a) Remove the sealant around the hex screw [4] and nut [6] at each bonding clip [1].

(b) Remove the hex screw [4], three washers [5], nut [6], and bonding jumper [2] from each of the bonding clips [1].

SUBTASK 28-22-41-020-005

(6) Remove the two cap screws [10] and washers [9], at the 1 O'clock and 7 O'clock positions, from the housing [14].

SUBTASK 28-22-41-480-003

(7) Install the retention screws, STD-8881, at the 1 O'clock and 7 O'clock position on the housing [14].

SUBTASK 28-22-41-020-006

(8) Remove the three cap screws [10], washers [9], and two bonding clips [1] from the housing [14].

SUBTASK 28-22-41-020-007

(9) Pull the motor-impeller [8] out 0.6 in. (15.2 mm) (or to the retention screws, STD-8881) from the housing [14].

NOTE: This allows the removal check valve to close.

(a) For fuel boost pumps in the No. 1 and No. 2 main tanks, pull the motor-impeller [8] with the puller, SPL-1769.

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SUBTASK 28-22-41-020-008

WARNING

DO NOT LET FUEL SPILL. WHEN YOU REMOVE THE DRAIN PLUG, A SMALL QUANTITY OF FUEL WILL COME OUT OF THE MOTOR/IMPELLER UNIT. IF THERE IS A CONTINUOUS FLOW OF FUEL, THE VALVE IS NOT CLOSED. FUEL CAN CAUSE A FIRE OR AN EXPLOSION. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(10) Remove the drain plug [7].

SUBTASK 28-22-41-650-001

(11) If fuel continuously drains:

NOTE: The removal check valve did not close.

(a) Install the drain plug [7].

(b) Install the motor-impeller [8].

(c) Defuel the applicable fuel tanks (TASK 28-26-00-650-801).

NOTE: Defuel the No. 1 main tank for the forward or aft fuel boost pump. Defuel the center tank and the No. 1 main tank for the left center fuel boost pump. Defuel the center tank and the No. 2 main tank for the right center fuel boost pump. Defuel the No. 2 main tank for the forward or aft fuel boost pump.

SUBTASK 28-22-41-020-009

(12) Remove and discard the O-ring [15].

SUBTASK 28-22-41-080-001

(13) After the fuel drains from the motor-impeller [8], remove the retention screws, STD-8881.

SUBTASK 28-22-41-020-010

CAUTION

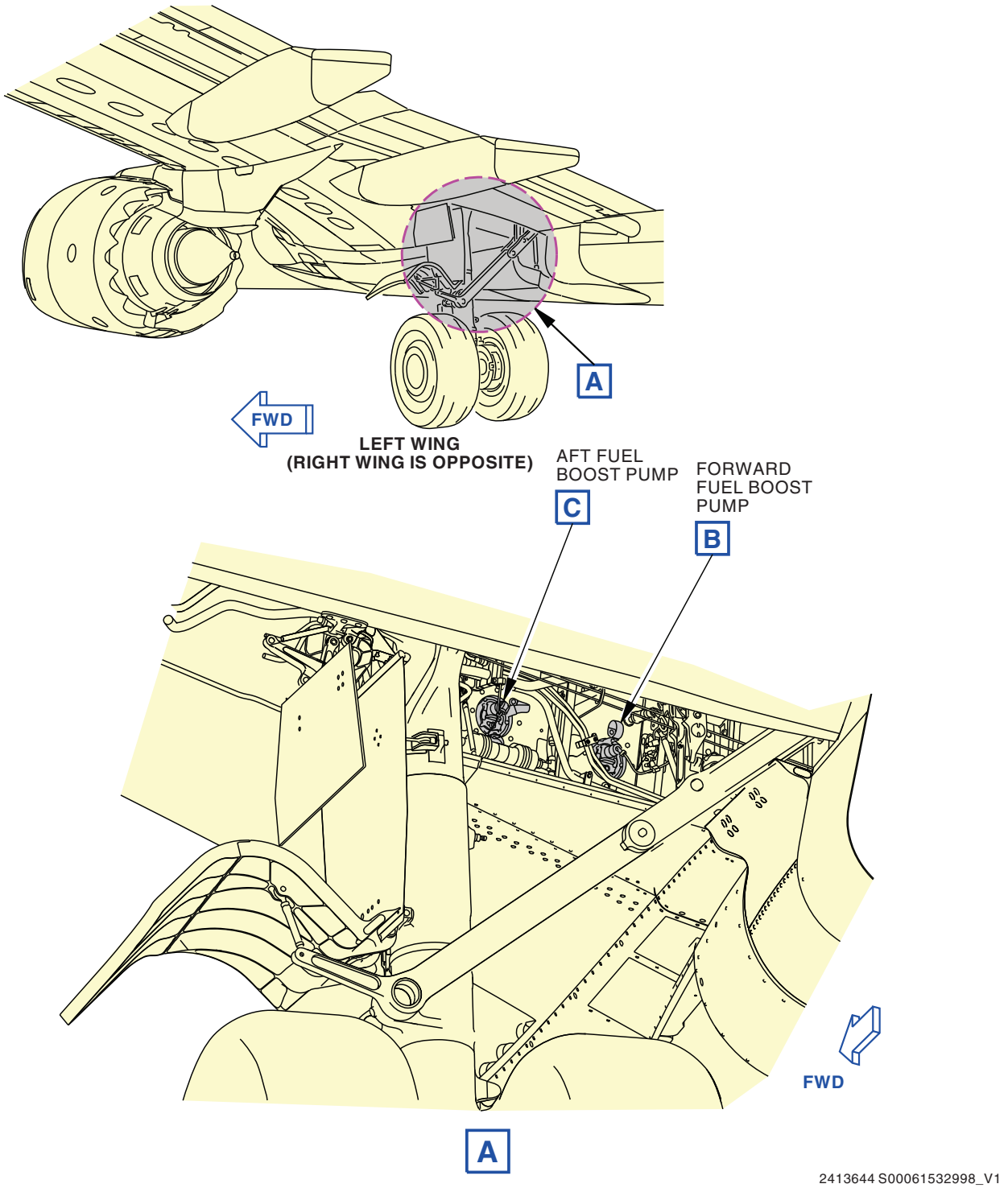
QUICKLY REPLACE THE MOTOR IMPELLER TO PREVENT FUEL DROPS INTO THE HOUSING. DAMAGE TO EQUIPMENT CAN OCCUR.

(14) Remove the motor-impeller [8] from the housing [14].

(a) Discard the O-ring [11], O-ring [12], and O-ring [13].

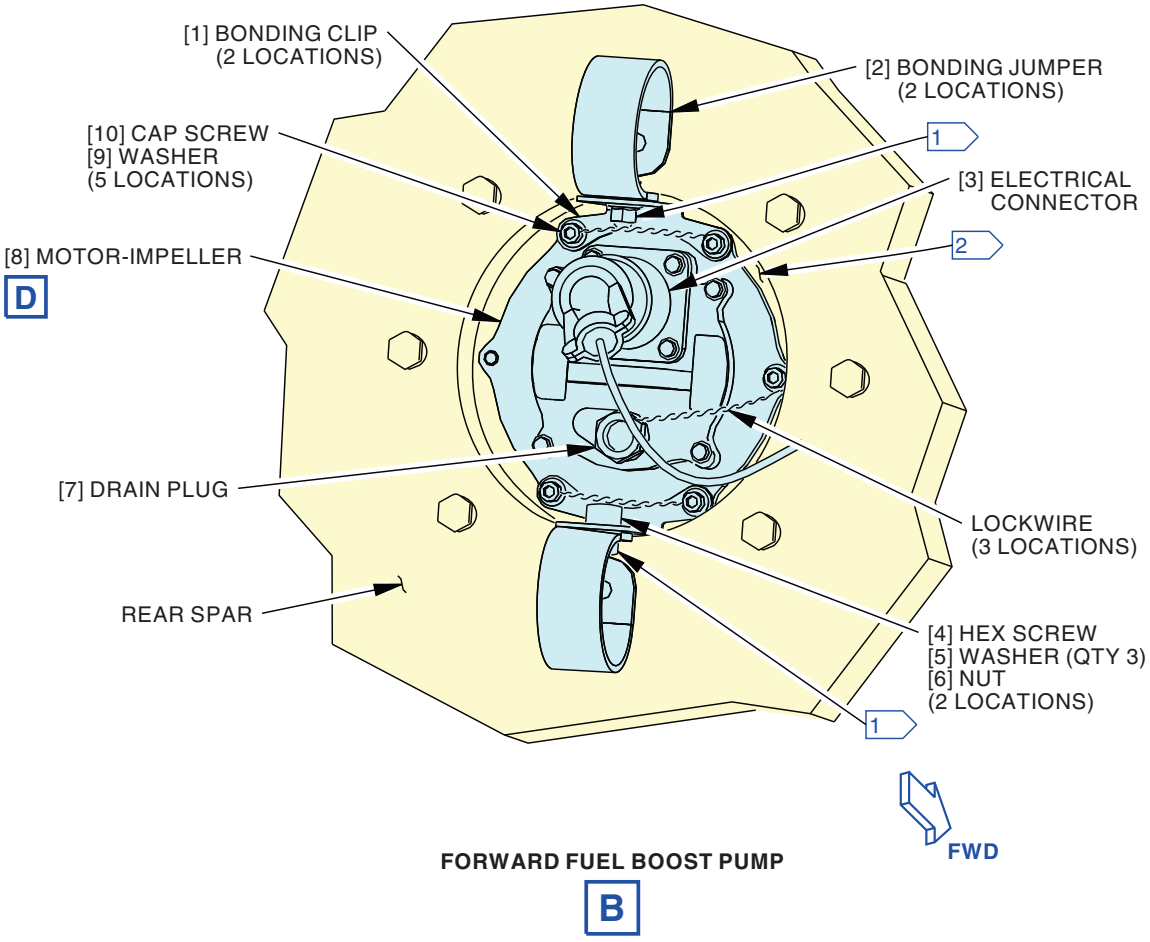
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Tanks - Forward or Aft Fuel Boost Pump Installation
Figure 401/28-22-41-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



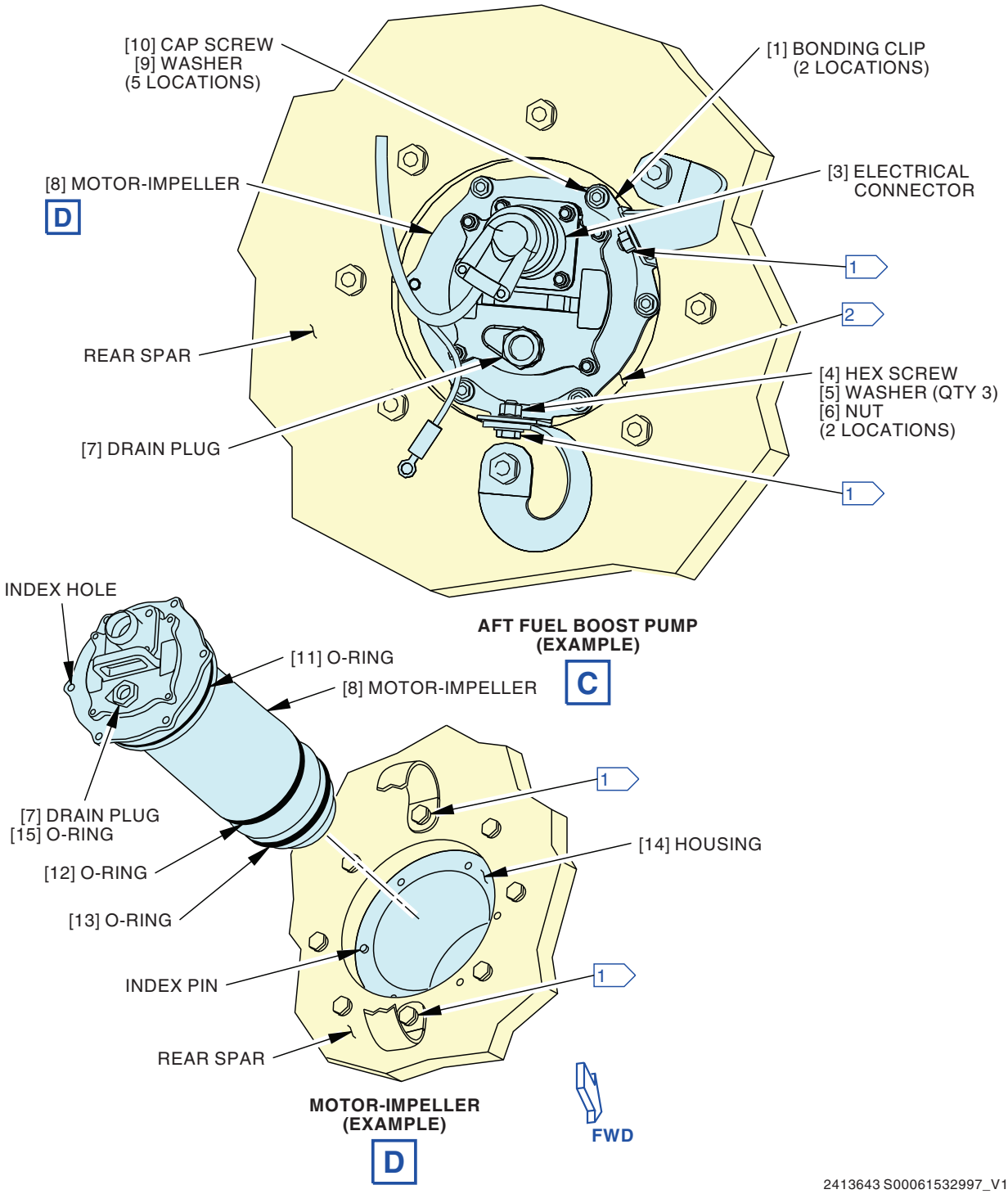
1 FAY SEAL WITH BMS5-45

2 CORROSION PREVENTIVE FILLET SEAL (AROUND COMPLETE PERIMETER OF HOUSING)

2413645 S00061532999_V1

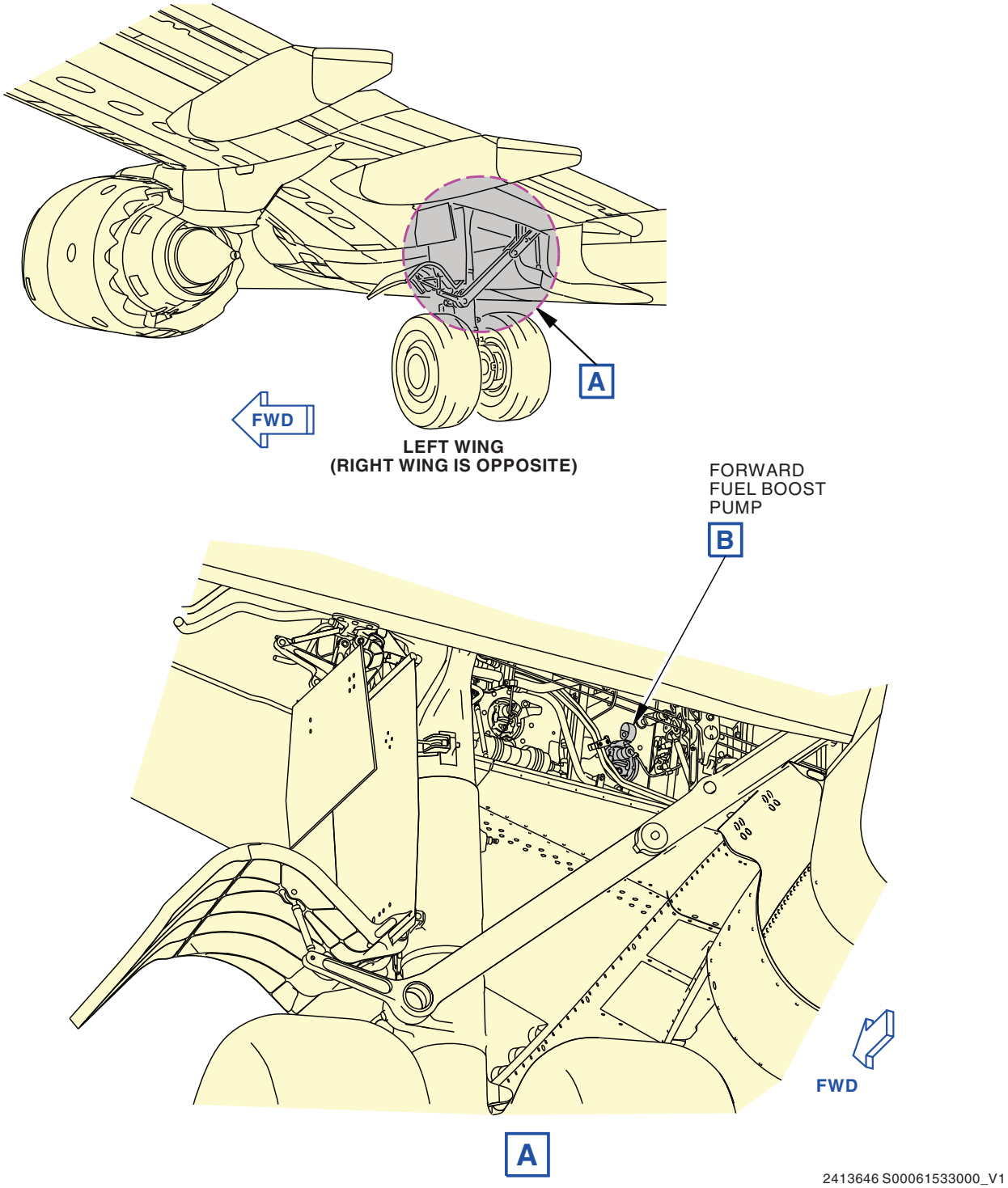
Main Tanks - Forward or Aft Fuel Boost Pump Installation
Figure 401/28-22-41-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



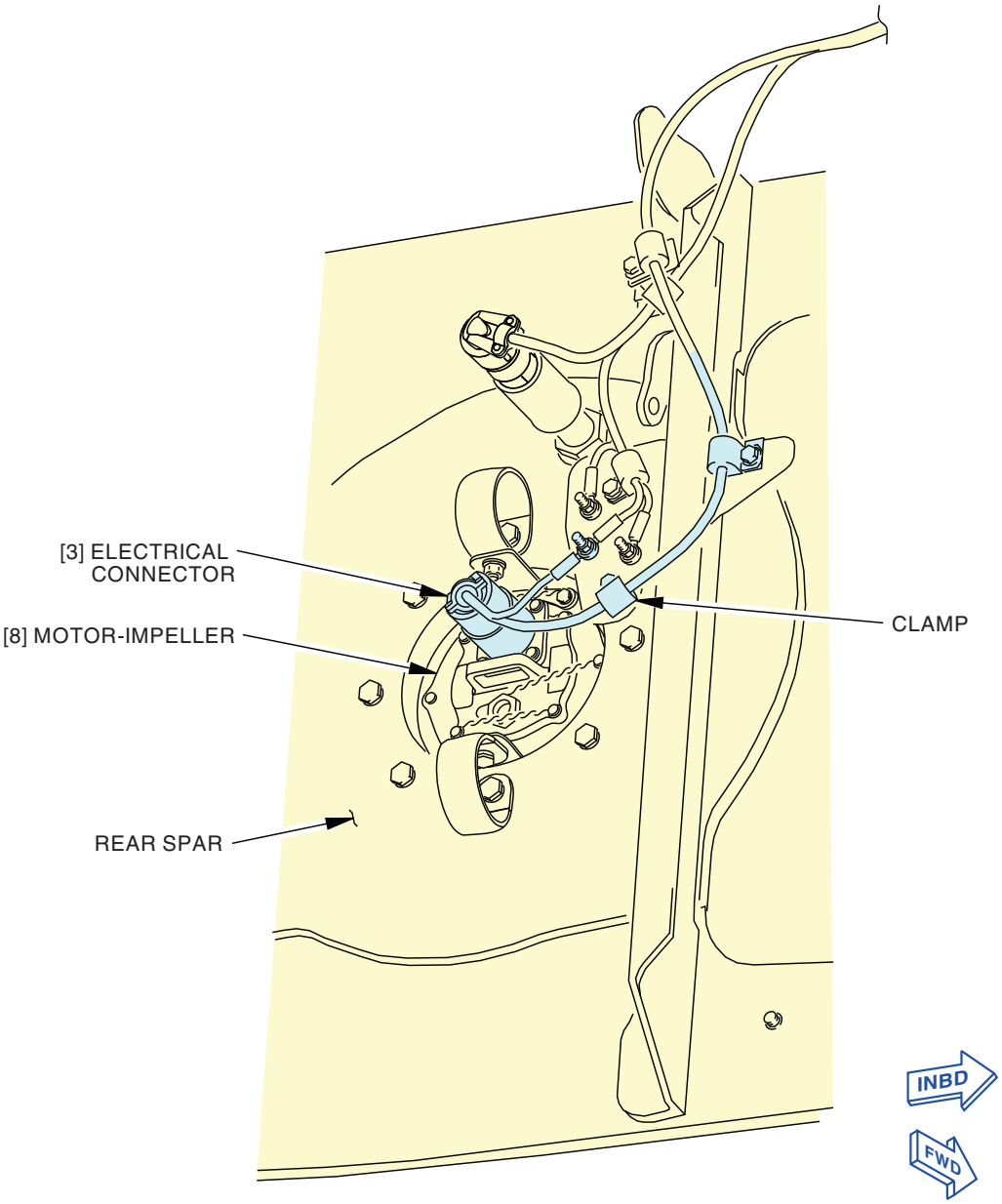
Main Tanks - Forward or Aft Fuel Boost Pump Installation
Figure 401/28-22-41-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Tanks - Forward Fuel Boost Pump - Wiring Check
Figure 402/28-22-41-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



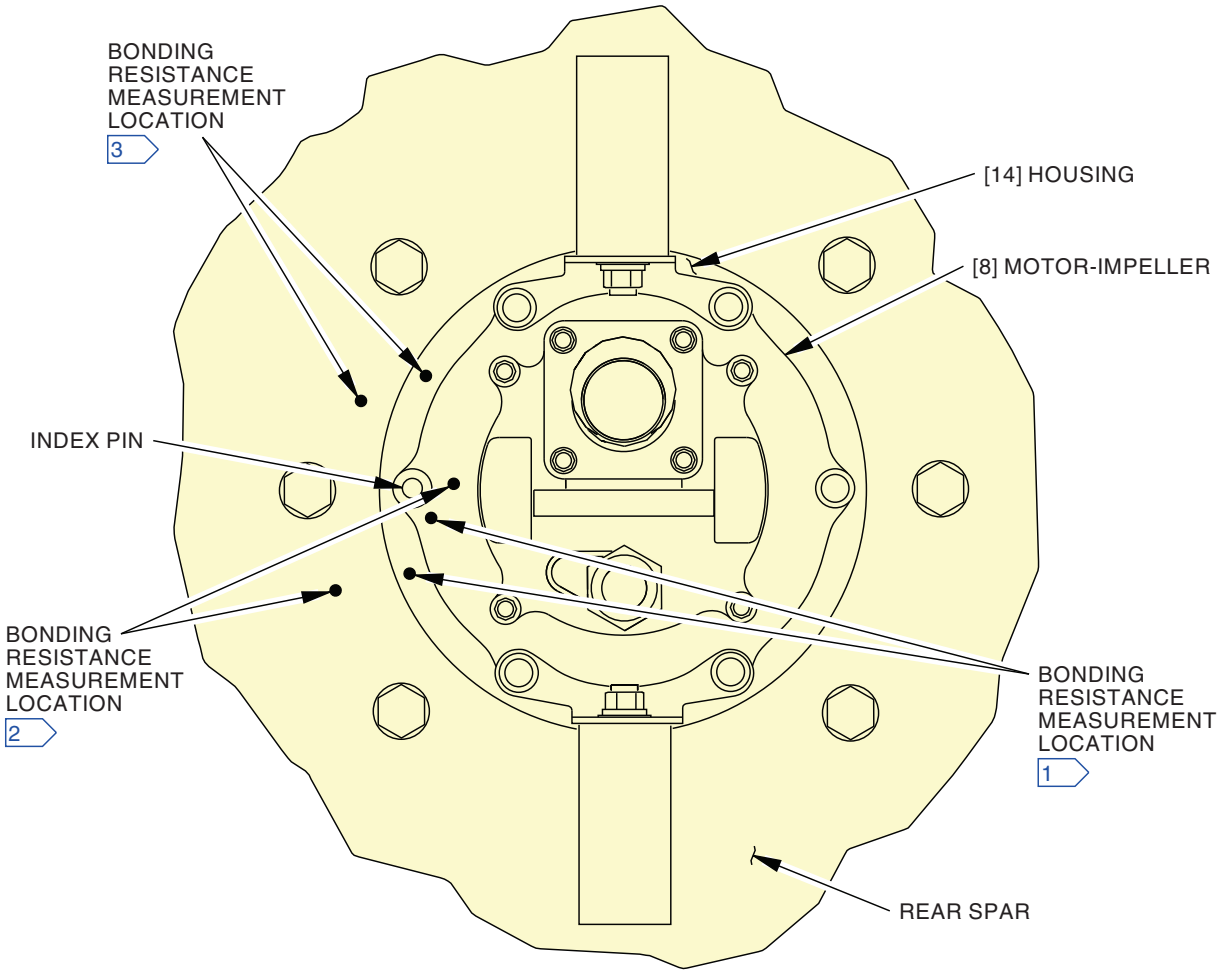
FORWARD FUEL BOOST PUMP

B

2413647 S00061533001_V1

Main Tanks - Forward Fuel Boost Pump - Wiring Check
Figure 402/28-22-41-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



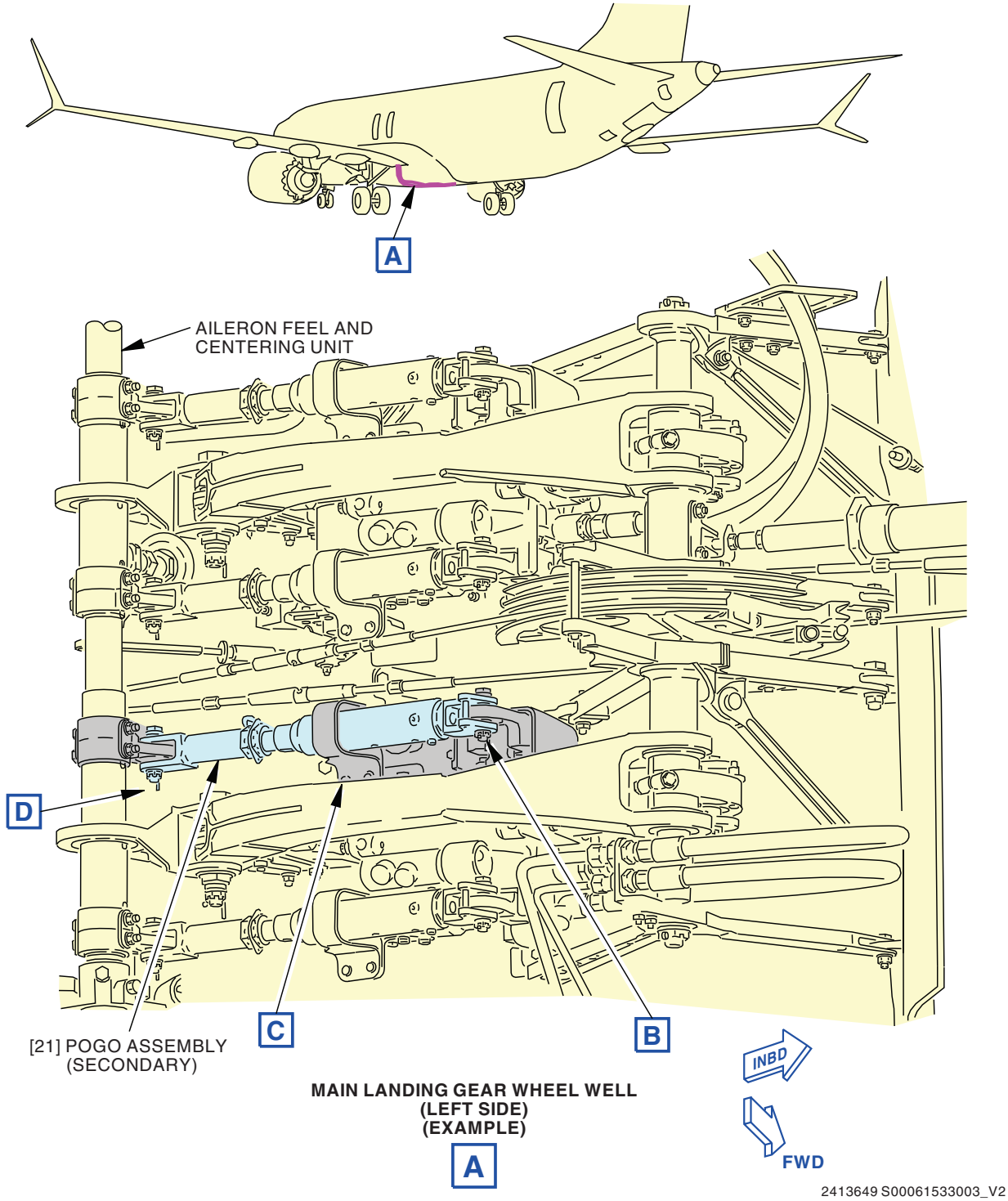
PROBE LOCATION (EXAMPLE)

1 BONDING RESISTANCE BETWEEN THE HOUSING AND MOTOR-IMPELLER.
2 BONDING RESISTANCE BETWEEN THE REAR SPAR AND MOTOR-IMPELLER.
3 BONDING RESISTANCE BETWEEN THE HOUSING AND REAR SPAR.

2413648 S00061533002_V1

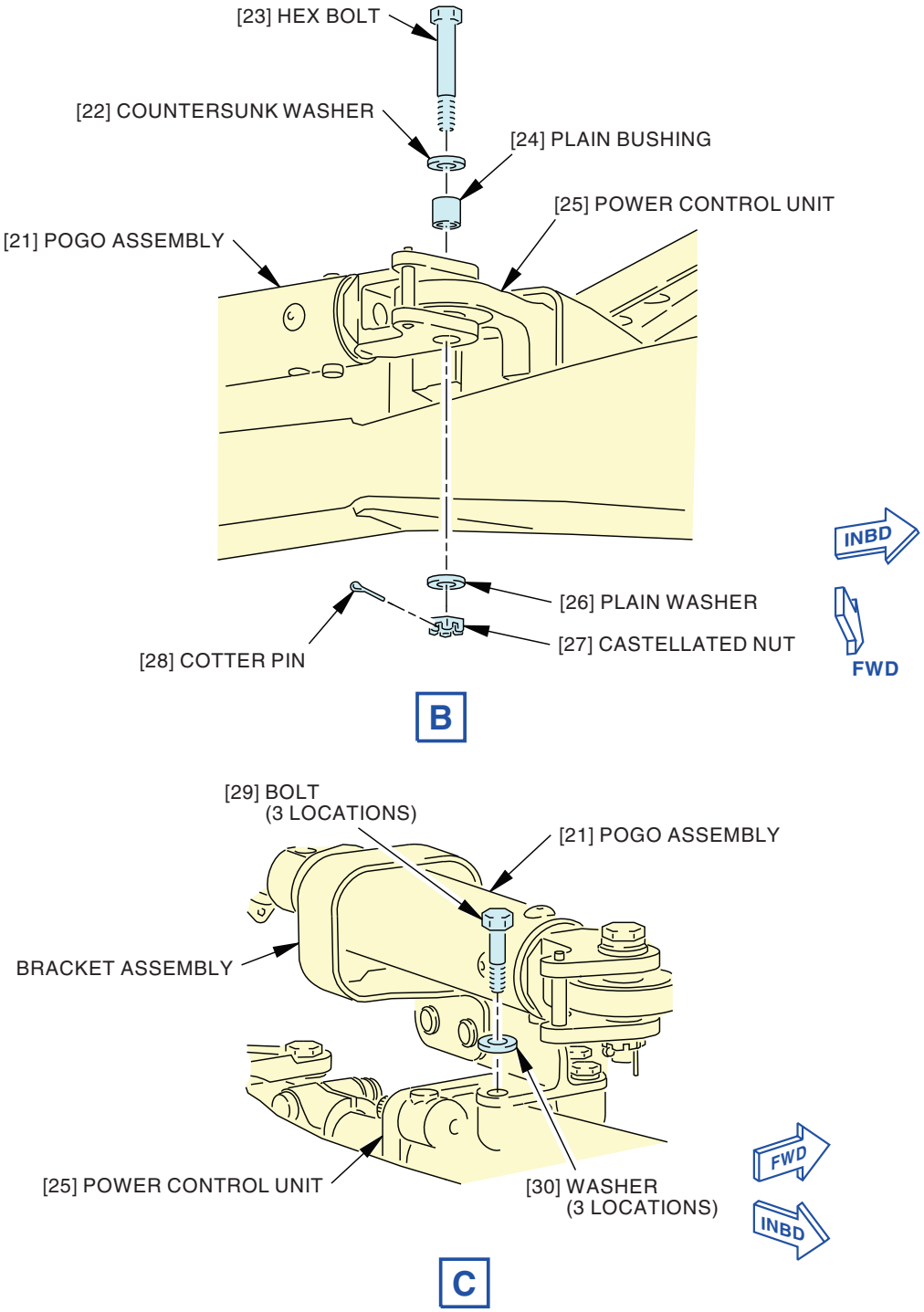
Main Tanks - Fuel Boost Pump Resistance Measurement - Probe Locations
Figure 403/28-22-41-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details



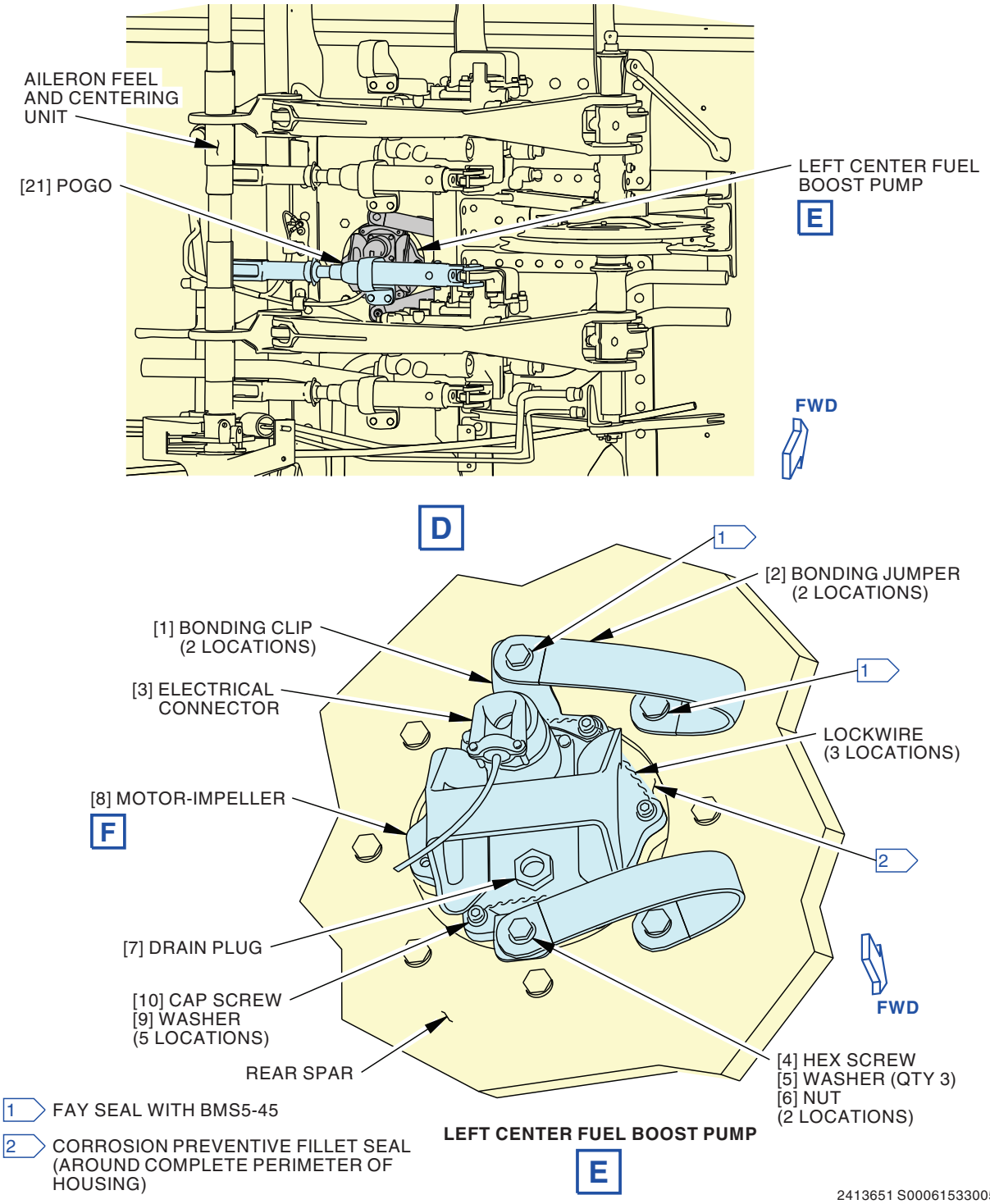
Center Tank - Left Center Fuel Boost Pump Installation
Figure 404/28-22-41-990-804 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



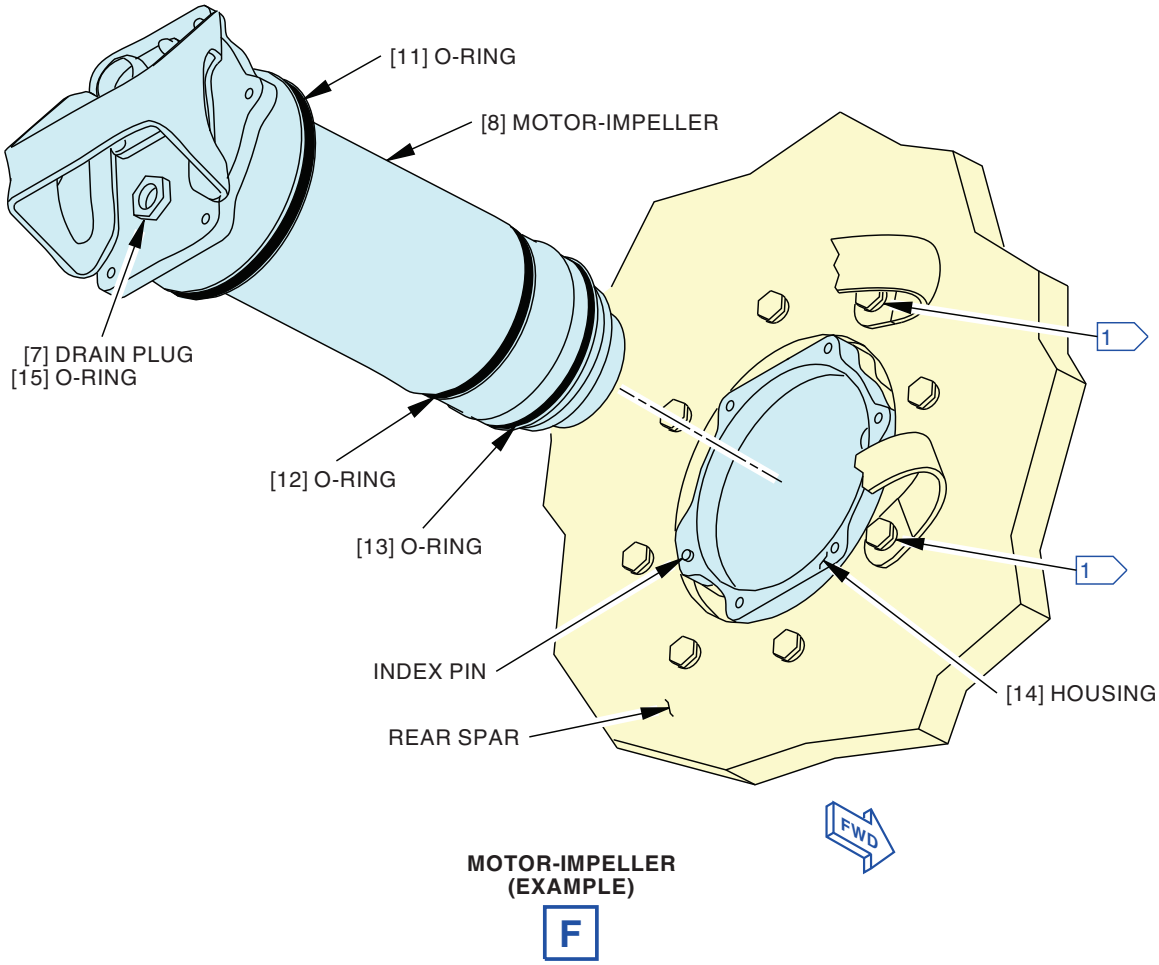
Center Tank - Left Center Fuel Boost Pump Installation
Figure 404/28-22-41-990-804 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Center Tank - Left Center Fuel Boost Pump Installation
Figure 404/28-22-41-990-804 (Sheet 3 of 4)

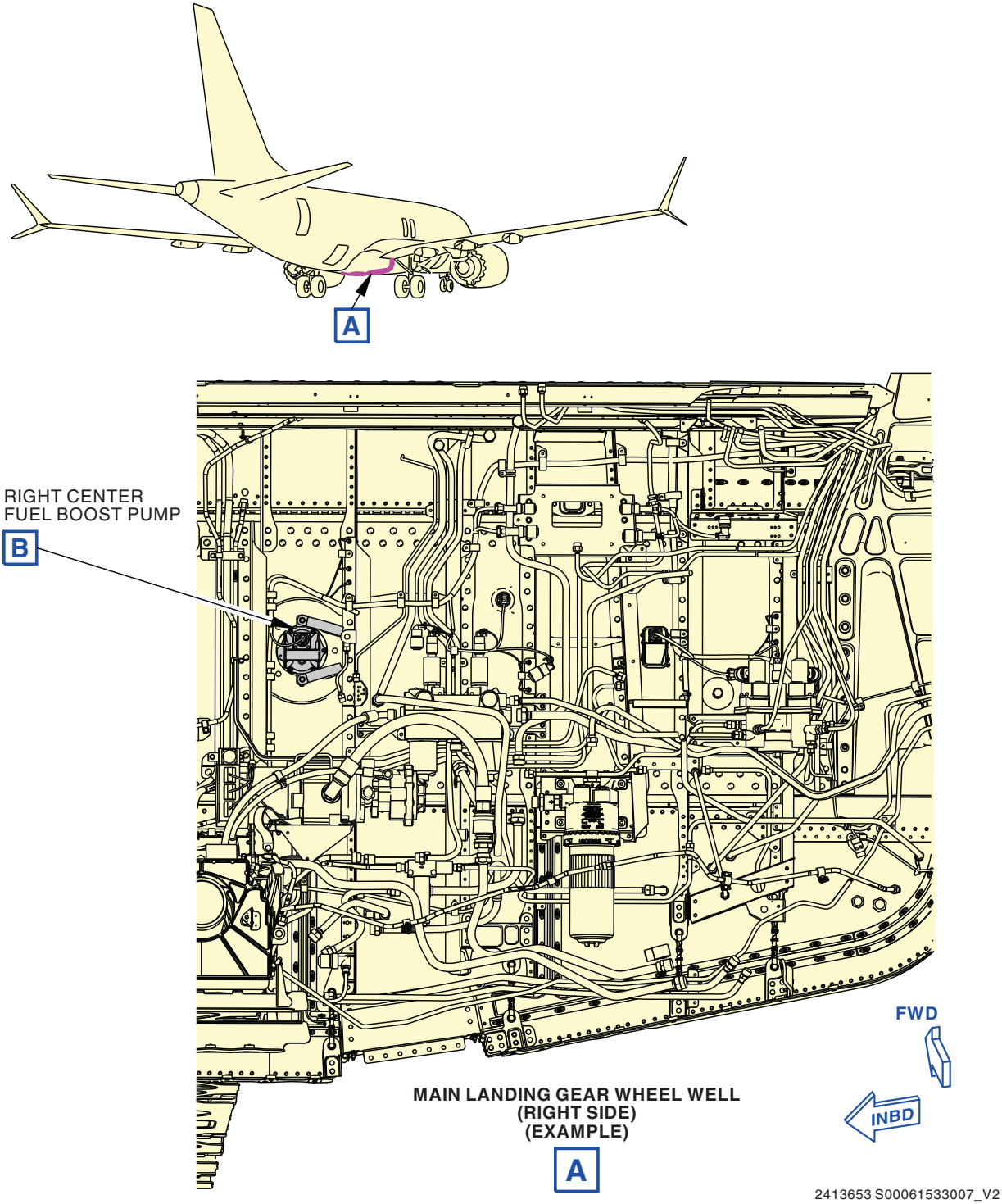
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2463497 S0000570718_V1

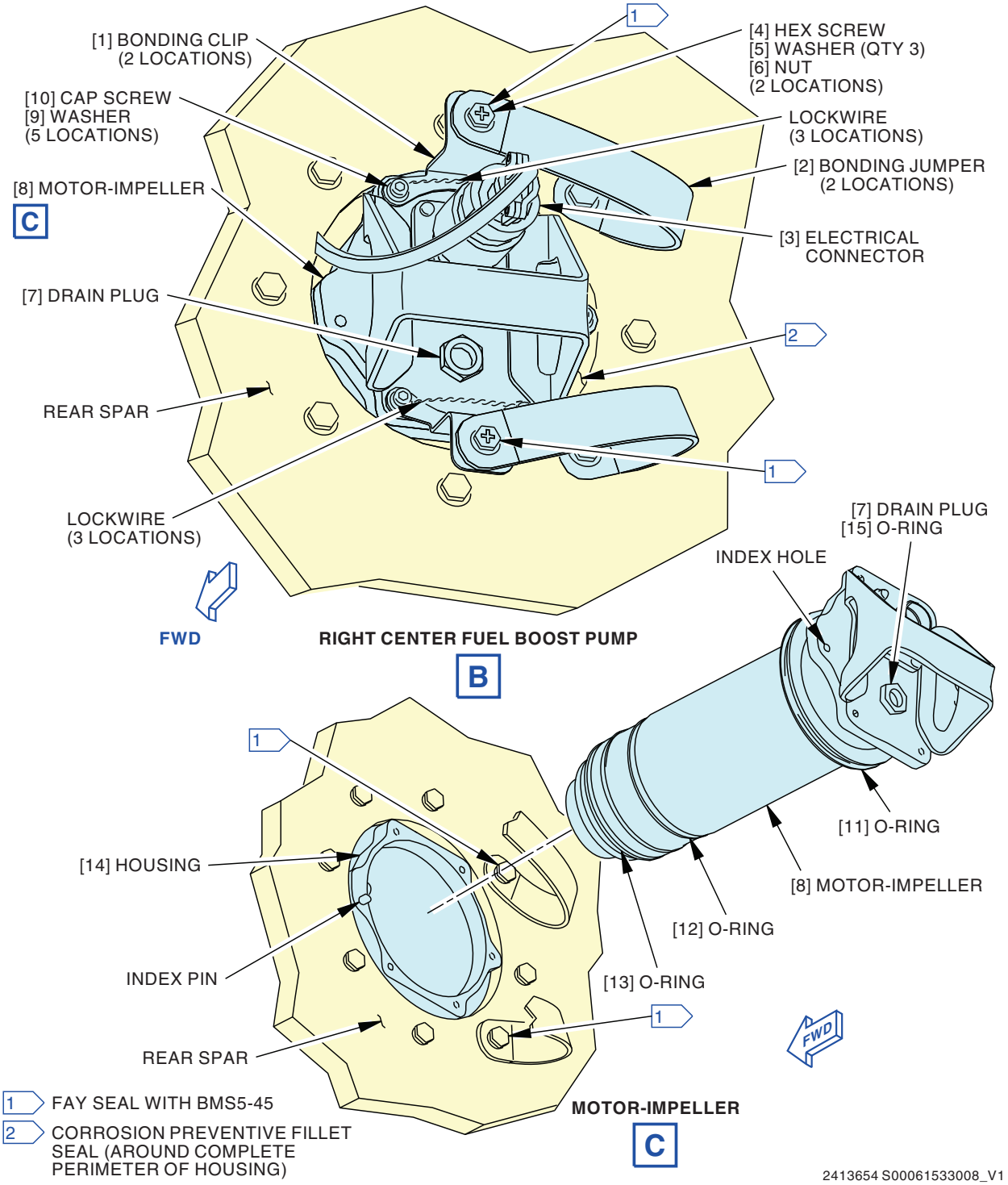
Center Tank - Left Center Fuel Boost Pump Installation
Figure 404/28-22-41-990-804 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



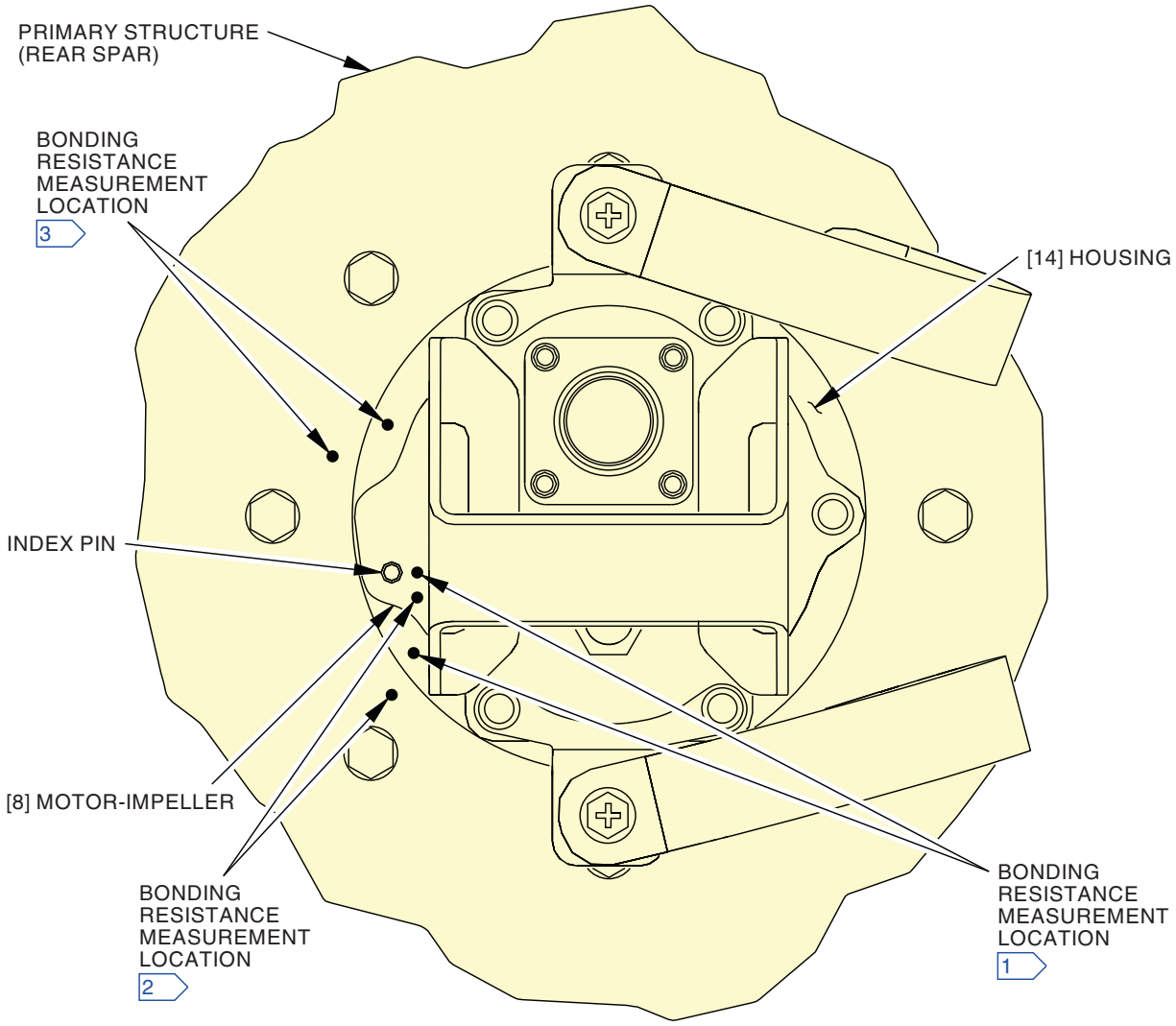
Center Tank - Right Center Fuel Boost Pump Installation
Figure 405/28-22-41-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Center Tank - Right Center Fuel Boost Pump Installation
Figure 405/28-22-41-990-806 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PROBE LOCATION
(EXAMPLE)

1. BONDING RESISTANCE BETWEEN THE HOUSING AND MOTOR-IMPELLER.
2. BONDING RESISTANCE BETWEEN THE REAR SPAR AND MOTOR-IMPELLER.
3. BONDING RESISTANCE BETWEEN THE HOUSING AND REAR SPAR.

2413652 S00061533006_V1

Center Tank - Center Fuel Boost Pump Resistance Measurement - Probe Locations
Figure 406/28-22-41-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-41-400-801

3. Fuel Boost Pump Motor-Impeller Installation

(Figure 401, Figure 402, Figure 403, Figure 404, Figure 405, Figure 406)

A. General

(1) The fuel boost pump motor-impeller is referred to as the motor-impeller in this task.

(2) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
24-22-00-860-804	Remove External Power (P/B 201)
27-11-00-820-802	Pogo and Power Control Unit (PCU) Adjustment (P/B 501)
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-22-41-710-801	Fuel Boost Pump - Operational Test (P/B 501)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
C00862	Coating - Chemical Conversion - Bonderite M-CR 600 Aero (Formerly Alodine 600)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
D00504	Grease - Petrolatum	VV-P-236

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G00291	Tape - Aluminum Foil, 3M 425	AMS-T-23397 / L-T-80
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G02272	Fuel - Turbine, Aviation (Grades JP-4, JP-5, JP-5/JP-8ST)	MIL-DTL-5624

E. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
551	Left Wing - Rear Spar To Landing Gear Support Beam
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
651	Right Wing - Rear Spar to Landing Gear Support Beam

F. Prepare for the Installation

SUBTASK 28-22-41-480-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 28-22-41-420-001

(2) Make sure that the wire bundle for the forward or aft fuel boost pump has the clamp installed (Figure 402).

NOTE: The clamp prevents the wire bundle from chafing.

SUBTASK 28-22-41-640-001

(3) Apply a thin layer of grease, D00504 to the new O-ring [15].

SUBTASK 28-22-41-420-002

(4) Install the O-ring [15] on the drain plug [7].

SUBTASK 28-22-41-420-003

(5) Install the drain plug [7] on the motor-impeller [8].

SUBTASK 28-22-41-420-004

(6) Tighten the drain plug [7] to 45 ±5 in-lb (5.1 ±0.6 N·m).

G. Fuel Boost Pump Motor-Impeller Installation

SUBTASK 28-22-41-100-001

(1) Do these steps to prepare the motor-impeller [8], housing [14], and rear spar for installation:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the sealant from the rear spar, do this task: Clean Bare, Clad, or Plated Metal with Solvent, TASK 20-10-34-110-802.

28-AWL-14: CDCCL

(b) Apply a fay surface bond between the mating surfaces of the motor-impeller [8] end-cap and the pump housing [14] (SWPM 20-20-00):

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

(c) Apply chemical conversion coating to the abraded fayed surfaces:

1) Apply 3M 425 Aluminum Foil Tape, G00291, around all surfaces that do not require chemical conversion.

2) Manually apply Bonderite M-CR 600 Aero coating, C00862, to the contact surfaces with a clean cotton wiper, G00034.

NOTE: Keep the surfaces wet, until a perceptible color develops. Color change takes about 2-3 minutes.

3) Air dry at room temperature.

NOTE: The maximum drying air temperature is 130°F (54°C).

SUBTASK 28-22-41-640-002

(2) Apply a thin layer of grease, D00504, to the new O-ring [11], new O-ring [12], and new O-ring [13].

SUBTASK 28-22-41-420-005

(3) Install the O-ring [11], O-ring [12], and O-ring [13] in the grooves of the motor-impeller [8].

SUBTASK 28-22-41-650-002

(4) Put 0.5 pt (0.24 l) to 1 pt (0.47 l) of fuel, G02272, into the two smaller slots in the upper side of the motor-impeller [8].

SUBTASK 28-22-41-980-001

(5) Turn the motor-impeller [8] to make sure that fuel touches the internal parts.

SUBTASK 28-22-41-820-001

(6) Align the index hole on the motor-impeller [8] with the index pin on the housing [14].

SUBTASK 28-22-41-420-006

(7) Put the motor-impeller [8] into the housing [14].

NOTE: Be careful not to damage the O-ring [11], O-ring [12], and O-ring [13] when you put the motor-impeller [8] into the housing [14].

SUBTASK 28-22-41-420-007

(8) Install a cap screw [10] and washer [9] in the mounting hole.

NOTE: Do not install in a mounting hole where the bonding clips [1] will be installed.

SUBTASK 28-22-41-100-002

(9) Do these steps to prepare the motor-impeller [8] and bonding clips [1] for installation:

(a) Remove the masking on the bonding clips [1].

NOTE: Only new bonding clips [1] have masking installed.

(b) If the bonding clip [1] is attached to the terminal of the bonding jumper [2], remove the bonding clip [1] from the terminal of the bonding jumper [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

28-AWL-14: CDCCL

(c) Apply a fay sealed fay surface bond between the mating surfaces of the end-cap on the motor-impeller [8] and each bonding clip [1] (SWPM 20-20-00):

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-400-001

(10) Install the cap screws [10], washers [9], and bonding clips [1] on the motor-impeller [8].

SUBTASK 28-22-41-860-004

28-AWL-14: CDCCL

(11) Measure the electrical bonding resistance between the end-cap of the motor-impeller [8], and the pump housing [14] (Figure 403 and Figure 406) (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-14: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-200-001

28-AWL-14: CDCCL

(12) Measure the electrical bonding resistance between the end-cap of the pump motor-impeller [8] and the rear spar (structure) (Figure 406) (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-14: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0015 ohm (1.5 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-41-860-005

28-AWL-14: CDCCL

(13) Measure the electrical bonding resistance between the end-cap of the motor-impeller [8] and each bonding clip [1] (Figure 403, Figure 406) (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-14: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0005 (0.5 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-380-001

(14) Apply Bonderite M-CR 600 Aero coating, C00862 to the abraded fayed surfaces:

SUBTASK 28-22-41-390-007

28-AWL-14: CDCCL

(15) If the bonding jumpers [2] are removed from the structure, apply a fay sealed fay surface bond between each terminal of the bonding jumper [2], washer [55], and the structure (rear spar) (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-400-002

(16) Install the terminals of the bonding jumpers [2], mounting bolts [56], and washers [55] to the rear spar (structure).

SUBTASK 28-22-41-200-003

(17) If the bonding jumpers [2] were removed from the structure, do this step:

28-AWL-14: CDCCL

(a) Prior to attaching the terminal of the bonding jumpers [2] to the bonding clips [1], measure the electrical bonding resistance between the loose end terminals of the bonding jumpers [2] and the rear spar (structure).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

1) Use an intrinsically safe approved bonding meter, COM-1550.

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28-AWL-14: CDCCL

2) Make sure that the electrical bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-200-002

(18) If the bonding jumpers [2] are not removed from the rear spar (structure), do this step:

28-AWL-14: CDCCL

(a) Prior to attaching the terminal of the bonding jumpers [2] to the bonding clips [1], measure the electrical bonding resistance between the loose end terminals of the bonding jumpers [2] and the rear spar (structure).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

1) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-14: CDCCL

2) Make sure that the electrical bonding resistance is 0.0010 ohm (1.0 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-390-002

28-AWL-14: CDCCL

(19) Apply a fay sealed fay surface bond between the terminal of the bonding jumper [2] and bonding clip [1] (two locations) (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-420-009

(20) Install the hex screws [4], washers [5], the terminals of the bonding jumpers [2], and the nuts [6] to each two bonding clips [1].

SUBTASK 28-22-41-200-004

28-AWL-14: CDCCL

(21) Make sure that the two bonding jumpers [2] are installed between the end-cap of the motor-impeller [8] and the structure.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

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SUBTASK 28-22-41-860-006

28-AWL-14: CDCCL

(22) Measure the electrical bonding resistance between the end-cap of the motor-impeller [8] and the airplane structure (Figure 403, Figure 406) (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-14: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0004 ohm (0.4 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-390-006

(23) Apply sealant, A50110 (alternate: sealant, A50155) to the bare fay surface on each bonding clip [1].

NOTE: For corrosion protection, use sufficient sealant on the bare areas.

SUBTASK 28-22-41-420-010

(24) Install the MS20995NC32 lockwire, G01912 between the cap screw [10] and drain plug [7].

NOTE: Put the lockwire on the cap screw [10] near the 3 O'clock position.

SUBTASK 28-22-41-420-029

(25) Install the MS20995NC32 lockwire, G01912 on the four cap screws [10].

SUBTASK 28-22-41-420-011

(26) Connect the electrical connector [3] to the motor-impeller [8].

H. Fuel Boost Pump Motor-Impeller Installation Test

SUBTASK 28-22-41-650-003

(1) Refuel the fuel tanks for the applicable fuel boost pump (TASK 12-11-00-650-803).

(a) If it is not possible to refuel the fuel tanks to the specified quantity, do one of the alternative procedures to prime the boost pump (TASK 28-22-41-420-801).

(b) Forward or aft fuel boost pumps, left wing:

1) No. 1 main tank – 500 lb (227 kg) minimum

(c) Forward or aft fuel boost pumps, right wing:

1) No. 2 main tank – 500 lb (227 kg) minimum

(d) Left or right center fuel boost pumps:

1) Center tank – 14,000 lb (6350 kg) minimum

SUBTASK 28-22-41-710-001

(2) For the applicable fuel boost pump, do this task: Fuel Boost Pump - Operational Test, TASK 28-22-41-710-801.

SUBTASK 28-22-41-210-002

(3) Make sure that there are no fuel leaks at the fuel boost pump.

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I. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-41-410-001

(1) For the left center fuel boost pump, install the pogo assembly [21] (Figure 404):

NOTE: Be careful not to change the length of the pogo assembly [21].

(a) Connect the bracket assembly to the power control unit [25] with three bolts [29] and washers [30].

(b) Apply grease, D00633 to the mating surfaces on the pogo assembly [21] and power control unit [25]

(c) Connect the pogo assembly [21] to the power control unit [25] with a hex bolt [23], countersunk washer [22], plain bushing [24], plain washer [26], castellated nut [27] and new cotter pin [28].

NOTE: Install with the hex bolt [23] and plain bushing [24] head up.

(d) Make sure that the rigging pin A/S-4, from the rigging pin kit, SPL-1585, can easily be removed and installed in the aileron bus drum (TASK 27-11-00-820-802).

(e) If the rig pin A/S-4 cannot be removed and installed, do this task: Pogo and Power Control Unit (PCU) Adjustment, TASK 27-11-00-820-802.

NOTE: The pogo assembly [21] length changed.

SUBTASK 28-22-41-860-007

(2) Remove electrical power, if necessary (TASK 24-22-00-860-804).

——— END OF TASK ———

TASK 28-22-41-420-801

4. Fuel Boost Pump and Override Pump Priming

(Figure 401, Figure 404, Figure 405)

A. General

(1) This task primes the fuel boost pump without adding a large quantity of fuel into the applicable fuel tank.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
28-22-41-710-801	Fuel Boost Pump - Operational Test (P/B 501)

C. Tools/Equipment

Reference	Description
STD-1054	Container - Fuel Resistant, 5-Gallon (19-Liter)

D. Consumable Materials

Reference	Description	Specification
G02272	Fuel - Turbine, Aviation (Grades JP-4, JP-5, JP-5/JP-8ST)	MIL-DTL-5624

E. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right

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(Continued)

Zone	Area
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam

F. Fuel Boost Pump and Override Pump Priming

SUBTASK 28-22-41-650-004

(1) Refuel the center or the No.1 and No. 2 fuel tank, as applicable (Pressure Refuel Procedure, TASK 12-11-00-650-803).

NOTE: These are the minimum fuel quantities to make sure that the fuel boost pumps are primed correctly.

(a) No. 1 main tank – 500 lb (227 kg) minimum

(b) No. 2 main tank – 500 lb (227 kg) minimum

(c) Center tank – 14,000 lb (6350 kg) minimum.

SUBTASK 28-22-41-710-006

(2) For the applicable fuel boost pump, do this task: Fuel Boost Pump - Operational Test, TASK 28-22-41-710-801.

(a) If the fuel boost pump is satisfactory, you primed the fuel boost pump correctly and no more steps are necessary.

(b) If the fuel boost pump is not satisfactory, then do the Fuel Boost Pump and Override Pump Priming Alternative 1.

G. Fuel Boost Pump and Override Pump Priming Alternative 1

SUBTASK 28-22-41-640-003

(1) Prime the fuel boost pump through the drain hole on the motor-impeller [8]:

(a) Remove the drain plug [7] from the applicable motor-impeller [8].

(b) Put approximately 1.5 pt (0.7 l) of fuel, G02272 into the drain hole.

SUBTASK 28-22-41-420-012

(2) Install the drain plug [7] on the motor-impeller [8].

SUBTASK 28-22-41-420-013

(3) Tighten the drain plug [7] to 45 ±5 in-lb (5 ±1 N·m).

SUBTASK 28-22-41-710-002

(4) For the applicable fuel boost pump, do this task: Fuel Boost Pump - Operational Test, TASK 28-22-41-710-801.

(a) If the fuel boost pump is satisfactory, then install the lockwire between the drain plug [7] and cap screw [10] (TASK 20-10-44-400-801).

NOTE: You primed the fuel boost pump correctly and no more steps are necessary.

(b) If the fuel boost pump is not satisfactory, then do the Fuel Boost Pump and Override Pump Priming Alternative 2.

H. Fuel Boost Pump and Override Pump Priming Alternative 2

SUBTASK 28-22-41-010-003

(1) Remove the motor-impeller [8] (TASK 28-22-41-000-801).

(a) Remove the drain plug [7].

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(b) Put the motor-impeller [8] in a 5-gallon (19-liter) fuel resistant container, STD-1054 of fuel, G02272.

NOTE: Keep the electrical connector on the motor-impeller [8] above the fuel.

(c) Install the drain plug [7].

SUBTASK 28-22-41-410-002

(2) Install the motor-impeller [8]:

(a) Hold the motor-impeller [8] in the correct direction with the open fuel slot on the top.

NOTE: This makes sure that the fuel does not drain from the motor-impeller [8] before installation.

(b) Make sure that the electrical connector on the motor-impeller [8] has no fuel.

(c) Immediately install the motor-impeller [8] (TASK 28-22-41-400-801).

(d) Tighten the drain plug [7] to 45 ±5 in-lb (5 ±1 N·m).

SUBTASK 28-22-41-710-003

(3) For the fuel boost pump, do this task: Fuel Boost Pump - Operational Test, TASK 28-22-41-710-801.

(a) Install the lockwire between the drain plug [7] and cap screw [10] (TASK 20-10-44-400-801).

NOTE: You primed the fuel boost pump correctly and no more steps are necessary.

——— END OF TASK ———

TASK 28-22-41-000-802

5. Fuel Boost Pump Housing Removal

(Figure 407)

A. General

(1) The fuel boost pump housing is referred to as the housing in this task.

(2) This task gives instruction to remove the housing.

B. References

Reference	Title
20-10-51-000-801	Flareless Tubing Assembly Removal (P/B 401)
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-22-15-000-803	Couplings Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-42-11-000-801	Fuel Boost Pump Fuel Pressure Switch Removal (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
551	Left Wing - Rear Spar To Landing Gear Support Beam
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
651	Right Wing - Rear Spar to Landing Gear Support Beam

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D. Access Panels

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
631AB	Center Tank Access Door - Wing Station 168

E. Prepare for the Removal

SUBTASK 28-22-41-480-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-22-41-650-005

(2) Defuel the center tank (TASK 28-26-00-650-801).

SUBTASK 28-22-41-650-006

(3) Defuel the No. 1 main tank for the following fuel boost pumps (TASK 28-26-00-650-801):

(a) Left forward fuel boost pump

(b) Left aft fuel boost pump

(c) Left center fuel boost pump.

SUBTASK 28-22-41-650-007

(4) Defuel the No. 2 main tank for the following fuel boost pumps (TASK 28-26-00-650-801):

(a) Right forward fuel boost pump

(b) Right aft fuel boost pump

(c) Right center fuel boost pump.

SUBTASK 28-22-41-860-008

(5) On the P5-2 Fuel System Module, turn the CROSS FEED switch to the closed position.

(a) Make sure that the VALVE OPEN light goes off for the fuel crossfeed valve.

SUBTASK 28-22-41-650-008

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(6) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Access the forward or aft fuel boost pump for the No. 1 main tank, open this access door:

(TASK 28-11-31-000-801)

Number	Name/Location
531AB	Center Tank Access Door - Wing Station 168

(b) Access the forward or aft fuel boost pump for the No. 2 main tank, remove this access door:

(TASK 28-11-31-000-801)

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Number	Name/Location
631AB	Center Tank Access Door - Wing Station 168

(c) Access the applicable center fuel boost pump, remove this access door:
(TASK 28-11-31-000-801)

Number	Name/Location
131AB	Center Tank Access

SUBTASK 28-22-41-010-004

(7) Remove the motor-impeller [8] from the housing [14] (TASK 28-22-41-000-801).

SUBTASK 28-22-41-860-009

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(8) Go into the fuel tank (TASK 28-11-00-910-802).

F. Fuel Boost Pump Housing Removal

SUBTASK 28-22-41-020-011

(1) Disconnect the scavenge pump pressure line [48] from the elbow [47] (TASK 20-10-51-000-801).

NOTE: The forward fuel boost pump on the right wing does not have a scavenge pump pressure line [48].

SUBTASK 28-22-41-020-012

(2) Disconnect the pressure switch line [49]:

(a) Disconnect the rigid coupling assembly [63] (TASK 28-22-15-000-803).

1) Remove and discard the O-ring [62].

(b) Make sure that fuel has drained from the pressure switch line [49].

NOTE: This prevents the collection of water in the pressure switch line [49]. Water can freeze and cause damage to the pressure switch line [49]. To prevent this type of damage, the pressure switch line [49] can be loosened at the pressure switch (TASK 28-42-11-000-801). A flush may be necessary to remove all water from the pressure switch line [49].

SUBTASK 28-22-41-020-013

(3) Disconnect the discharge check valve [41] from the housing [14]:

(a) Remove the four screws [67], washers [66], and nuts [61] from the discharge check valve [41].

(b) Remove the O-rings [65] and pressure switch plate adapter [64].

1) Discard the O-rings [65].

SUBTASK 28-22-41-020-014

(4) Disconnect the removal check valve [43] from the housing [14]:

(a) Remove the six screws [44], washers [45], and nuts [46] from the removal check valve [43].

(b) Remove the O-ring [42].

(c) Discard the O-ring [42].

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SUBTASK 28-22-41-020-015

(5) Fuel boost pumps in the No. 1 or No. 2 main tank, disconnect the discharge port line from the discharge port.

(a) Disconnect the flexible full coupling [50] (TASK 28-22-15-000-803).

1) Remove and discard the O-rings [52].

SUBTASK 28-22-41-020-022

CAUTION

DO NOT USE METAL TOOLS. METAL TOOLS CAN CAUSE DAMAGE TO THE PART YOU WILL DO WORK ON OR ADJACENT PARTS.

(6) Remove the fillet seal from the edge of the housing [14].

SUBTASK 28-22-41-020-016

(7) Remove the safety cables or lockwires from the mounting bolts [56] and mounting bolts [57].

SUBTASK 28-22-41-020-017

(8) Remove the mounting bolt [57], bonding jumper [2], washer [58], and flat washer [59] (two locations).

NOTE: The housing and center tank are held together with a fay seal. A non-metallic tool may be necessary to help remove the housing.

SUBTASK 28-22-41-020-018

(9) Hold the housing [14] from the fuel side of the rear spar.

SUBTASK 28-22-41-020-019

(10) Remove the six mounting bolts [56] and washers [55].

SUBTASK 28-22-41-020-020

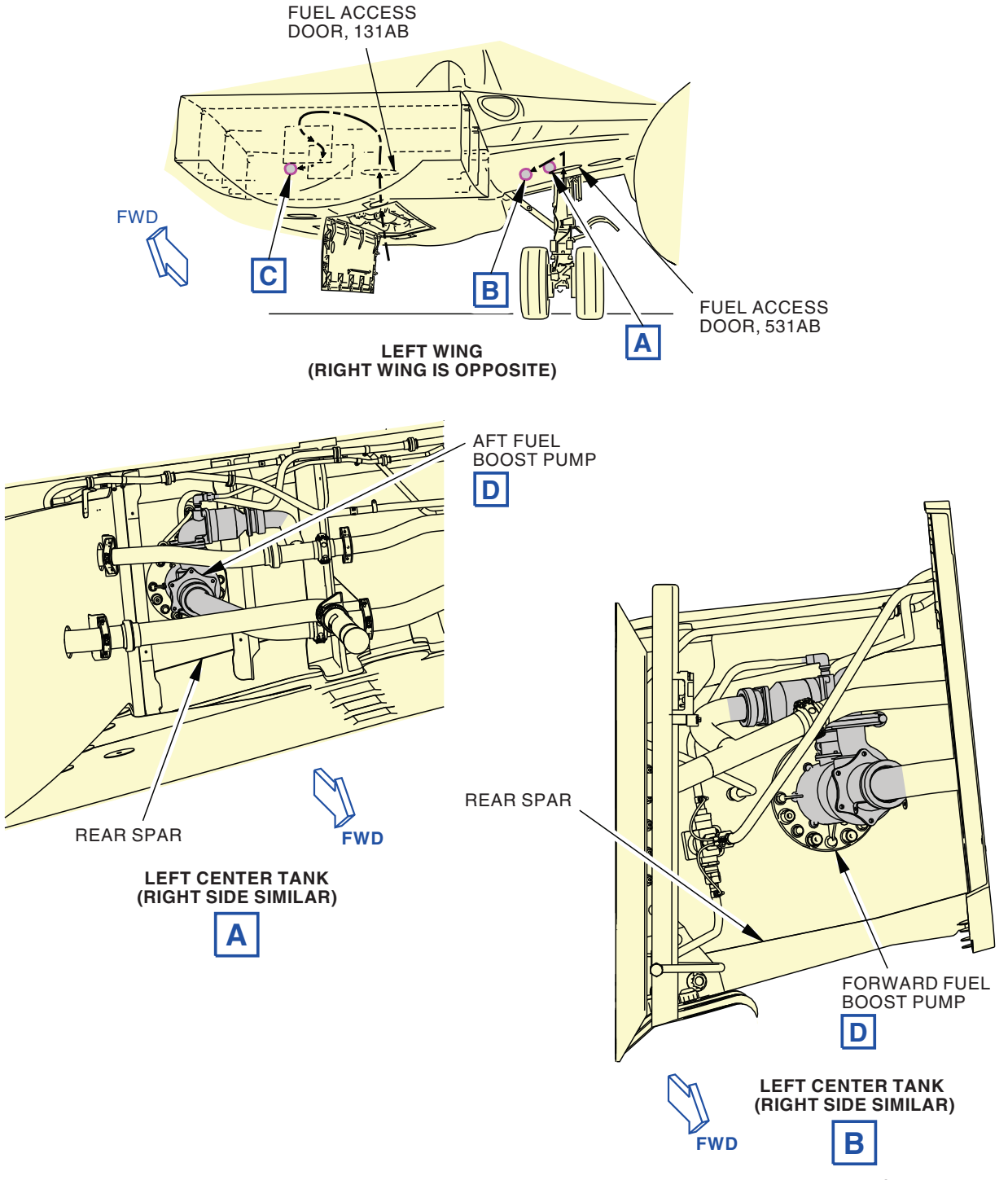
(11) Remove the Gask-O-Seal [60].

SUBTASK 28-22-41-020-021

(12) Remove the housing [14] from the center tank.

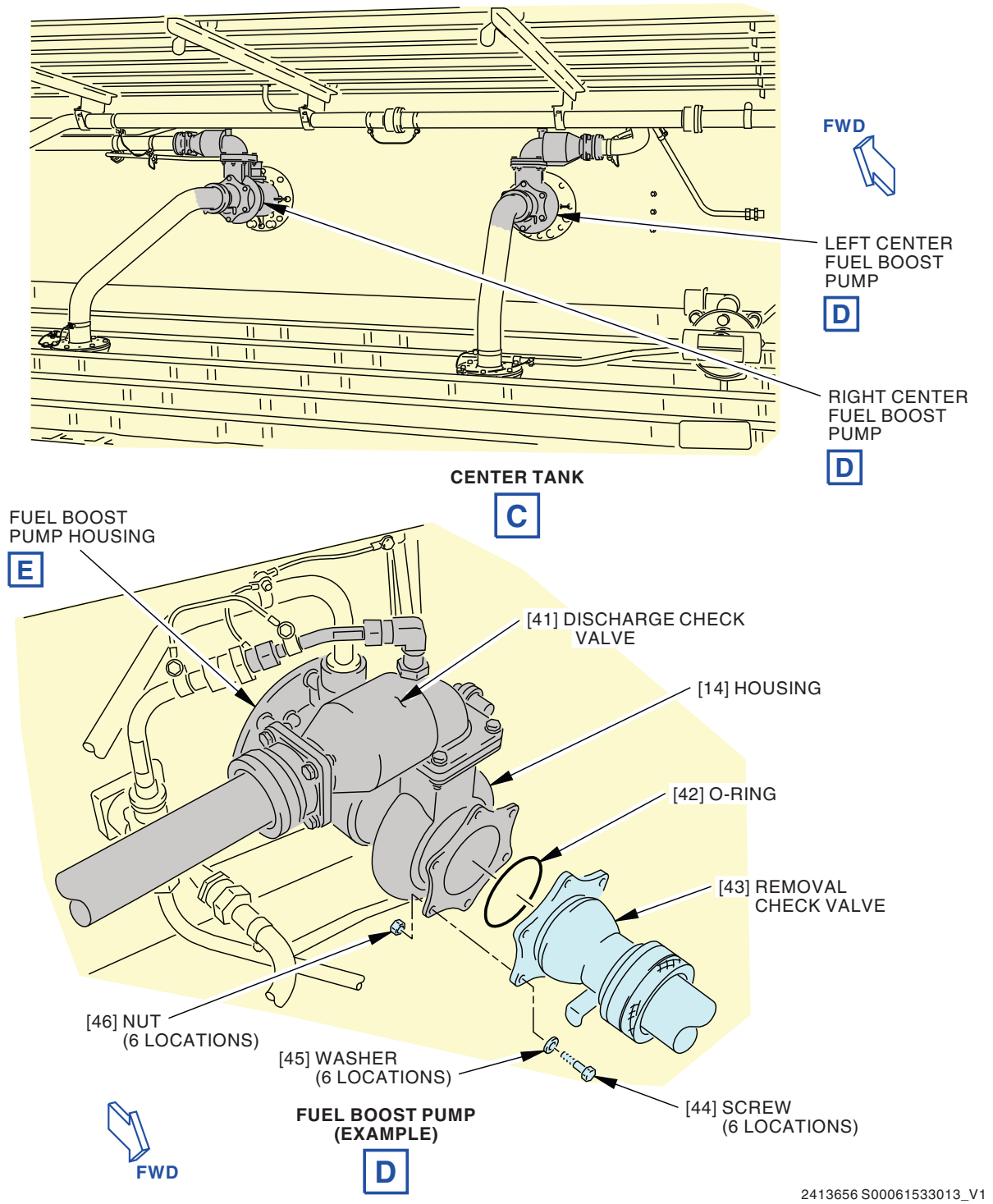
END OF TASK

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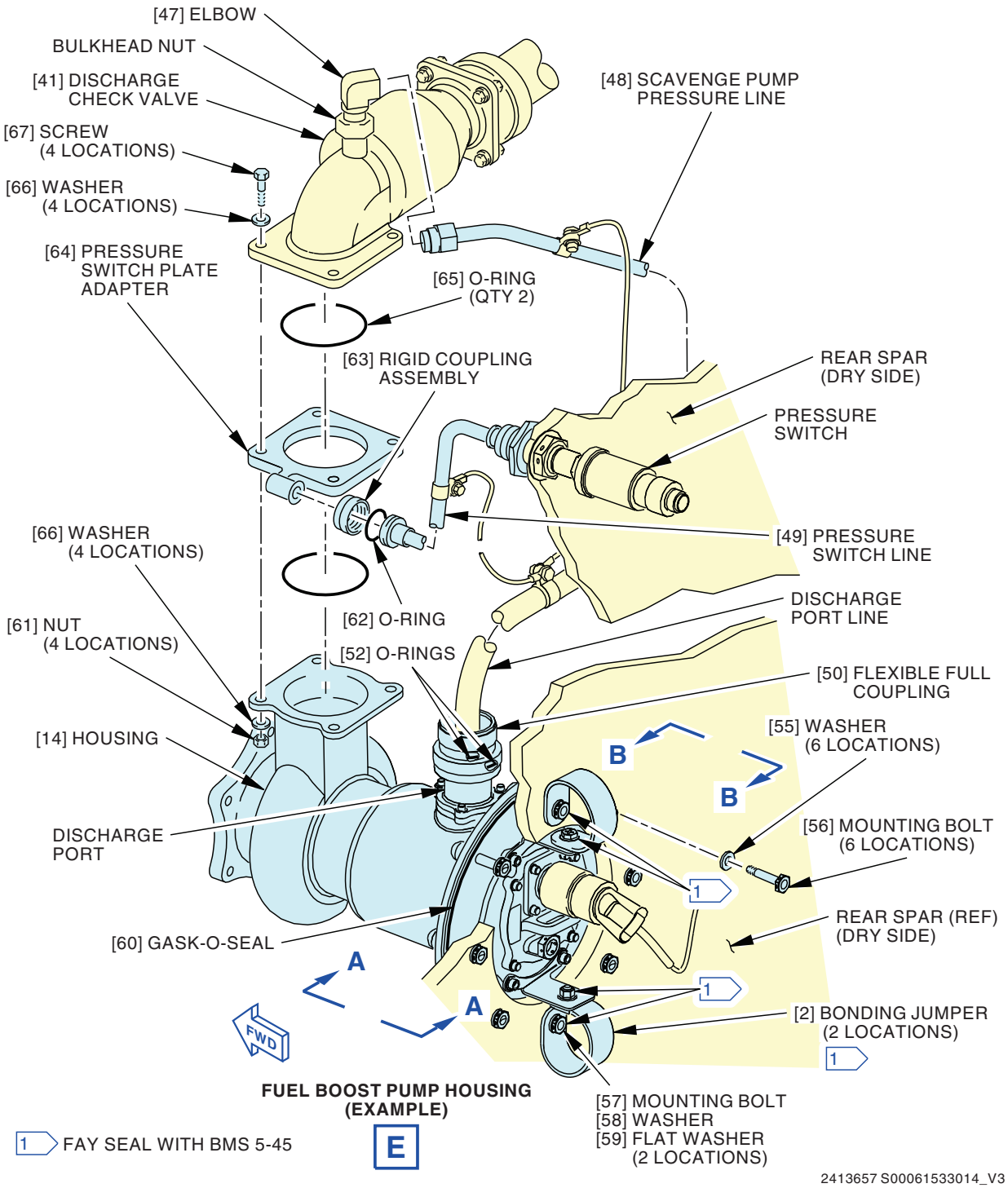
Fuel Boost Pump Housing Installation
Figure 407/28-22-41-990-807 (Sheet 1 of 4)

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Fuel Boost Pump Housing Installation
Figure 407/28-22-41-990-807 (Sheet 2 of 4)

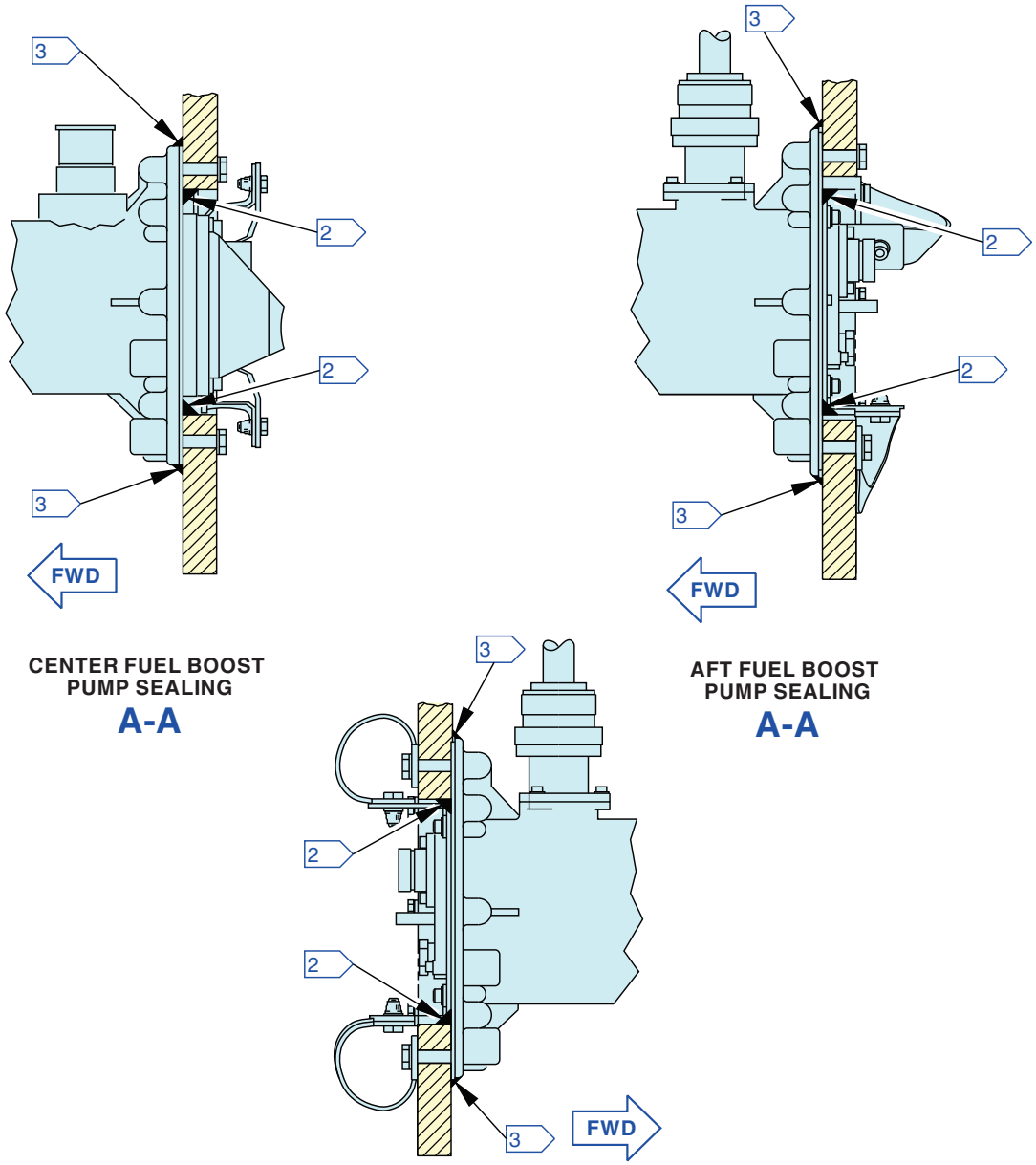
ECCN 9E991 BOEING PROPRIETARY - See title page for details



FUEL BOOST PUMP HOUSING (EXAMPLE)

Fuel Boost Pump Housing Installation
Figure 407/28-22-41-990-807 (Sheet 3 of 4)

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2 FILLET SEAL WITH BMS 5-142 OR BMS 5-45
3 FILLET SEAL WITH BMS 5-45

2413658 S00061533015_V1

Fuel Boost Pump Housing Installation
Figure 407/28-22-41-990-807 (Sheet 4 of 4)

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TASK 28-22-41-400-802

6. Fuel Boost Pump Housing Installation

(Figure 407)

A. General

(1) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

(2) The fuel boost pump housing is referred to as the housing in this task.

(3) This task gives instruction to install the housing.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
20-10-17-400-801	O-Rings Installation (P/B 401)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
20-10-51-400-801	Flareless Tubing Assembly Installation (P/B 401)
24-22-00-860-801	Supply Electrical Power (P/B 201)
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-11-00-300-803	Repair of Sealant Leaks in the Fuel Tank Structure (P/B 801)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
28-22-15-400-803	Couplings Installation (P/B 401)
51-22-00-390-806	Fillet Seal Application (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A00767	Sealant - Fuel Tank	BMS5-45
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II

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(Continued)

Reference	Description	Specification
A50153	Sealant - Fuel Tank - Class A-2	BMS5-45 Class A-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50337	Sealant - Fuel Tank	BMS5-45 Class B
D00504	Grease - Petrolatum	VV-P-236

E. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
551	Left Wing - Rear Spar To Landing Gear Support Beam
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
651	Right Wing - Rear Spar to Landing Gear Support Beam

F. Access Panels

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
631AB	Center Tank Access Door - Wing Station 168

G. Fuel Boost Pump Housing Installation

SUBTASK 28-22-41-100-004

28-AWL-14: CDCCL

(1) Apply a fay surface bond between the pump housing [14], the Gask-O-Seal [60] (gasket), and the rear spar (structure) inside the tank (SWPM 20-20-00):

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-820-002

(2) Align the housing [14] with the rear spar.

SUBTASK 28-22-41-420-014

(3) Put the Gask-O-Seal [60] (gasket) between the housing [14] and rear spar.

SUBTASK 28-22-41-420-015

(4) Apply sealant, A50337 (sealant, A50155, alternate) to wet install the mounting bolts [56] and washers [55] (six locations).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-41-860-010

28-AWL-14: CDCCL

(5) Measure the electrical bonding resistance between the pump housing [14] and the rear spar (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-14: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0010 ohm (1.0 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-420-016

28-AWL-14: CDCCL

(6) Apply a fay sealed fay surface bond between the terminal of the bonding jumper [2], flat washer [59], and the structure (rear spar) (SWPM 20-20-10).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-420-017

(7) Install the first bonding jumper [2] on the rear spar:

(a) Install the mounting bolt [57], washer [58], the terminal of the bonding jumper [2], and the flat washer [59] on the rear spar.

NOTE: The washer [58] is installed between the mounting bolt [57] and the terminal of the bonding jumper.

SUBTASK 28-22-41-760-001

28-AWL-14: CDCCL

(8) Measure the electrical bonding resistance between the loose end of the bonding jumper [2] terminal and the rear spar (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

28-AWL-14: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-420-018

28-AWL-14: CDCCL

(9) Apply a fay sealed fay surface bond between the terminal of the bonding jumper [2], flat washer [59], and the structure (rear spar) (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-420-019

(10) Install the second bonding jumper [2] on the rear spar:

(a) Install the mounting bolt [57], washer [58], the terminal of the bonding jumper [2], and flat washer [59] on the rear spar (SWPM 20-20-10).

NOTE: The washer [58] is installed between the mounting bolt [57] and bonding jumper.

SUBTASK 28-22-41-760-002

28-AWL-14: CDCCL

(11) Measure the electrical bonding resistance between the loose end of the bonding jumper [2] terminal and the rear spar (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-14: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-420-020

(12) Install the safety cables or lockwires on the six mounting bolts [56] and two mounting bolts [57] (TASK 20-10-44-400-801).

SUBTASK 28-22-41-390-003

(13) Apply a cap seal of sealant, A00767 (sealant, A02315, alternate) to the six mounting bolts [56].

SUBTASK 28-22-41-390-004

(14) Apply sealants between the housing [14] and the rear spar in the tank:

NOTE: Use sufficient sealant to cover un-primed areas of the structure for corrosion protection.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Apply a thin, narrow brushcoat of sealant, A50153, around the perimeter of the housing [14] in the tank (TASK 28-11-00-300-803).

NOTE: To prevent contamination, replace the brush and sealant containers in 6 in. (15 cm) intervals.

28-AWL-14: CDCCL

(b) Apply a fillet seal around the entire perimeter of the gasket (Gask-O-Seal [60]), the pump housing [14], and the rear spar (structure) in the tank (TASK 28-11-00-300-803).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-390-005

28-AWL-14: CDCCL

(15) Apply a fillet seal around the perimeter of the pump housing [14] and the rear spar (structure) out of the tank (TASK 51-22-00-390-806).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-14.

SUBTASK 28-22-41-100-006

(16) Apply a fay surface bond between the mating surfaces of the housing [14] and the removal check valve [43] (SWPM 20-20-00).

SUBTASK 28-22-41-420-021

(17) Connect the removal check valve [43] to the housing [14]:

(a) Lubricate the O-ring [42] with a thin layer of grease, D00504.

(b) Install the O-ring [42] in the groove of the flange on the removal check valve [43].

(c) Align the flanges of the removal check valve [43] and housing [14].

(d) Install the six screws [44], washers [45], and nuts [46].

SUBTASK 28-22-41-100-007

(18) Apply a fay surface bond between the mating surfaces of the housing [14], pressure switch plate adapter [64], and the discharge check valve [41] (SWPM 20-20-00).

SUBTASK 28-22-41-420-022

(19) Connect the discharge check valve [41]:

(a) Put the new O-ring [65] in the groove of the discharge check valve [41] (TASK 20-10-17-400-801).

(b) Put the new O-ring [65] in the groove of the pressure switch plate adapter [64] (TASK 20-10-17-400-801).

(c) Put the pressure switch plate adapter [64] between the housing [14] and the discharge check valve [41].

NOTE: The port faces the pressure switch line.

(d) Align the pressure switch plate adapter [64] with the housing [14] and discharge check valve [41].

(e) Install the four screws [67], washers [66], and nuts [61].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-41-760-003

(20) Measure the electrical bonding resistance between the pressure switch plate adapter [64] and the housing [14] (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the measured resistance is not more than the maximum bonding resistance value.

SUBTASK 28-22-41-760-004

(21) Measure the electrical bonding resistance between the discharge check valve [41] and the pressure switch plate adapter [64] (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the measured resistance is not more than the maximum bonding resistance value.

SUBTASK 28-22-41-420-023

(22) Connect the scavenge pump pressure line [48] to the elbow [47] (TASK 20-10-51-400-801).

NOTE: The forward fuel boost pump on the right wing does not have a elbow or a scavenge pump pressure line.

SUBTASK 28-22-41-420-024

(23) Install the rigid coupling assembly [63] to attach the pressure switch line [49] to the pressure switch plate adapter [64] (TASK 28-22-15-400-803).

NOTE: Do not install the safety cable or lockwire on the rigid coupling assembly.

(a) Install the new O-ring [62].

SUBTASK 28-22-41-420-025

(24) Install the safety cable or lockwire between the rigid coupling assembly [63] and bulkhead nut on the discharge check valve [41] (TASK 20-10-44-400-801):

(a) Aft fuel boost pump, left wing

(b) Forward fuel boost pump, right wing

(c) Left center fuel boost pump, center tank

(d) Right center fuel boost pump, center tank.

SUBTASK 28-22-41-420-026

(25) Install the safety cable or lockwire between the rigid coupling assembly [63] and the screw [44] for these locations (TASK 20-10-44-400-801):

(a) Aft fuel boost pump, right wing

(b) Forward fuel boost pump, left wing.

SUBTASK 28-22-41-420-027

(26) Install the safety cable or lockwire between the bulkhead nut and screw on the discharge check valve [41] for these locations (TASK 20-10-44-400-801):

(a) Aft fuel boost pump, right wing

(b) Forward fuel boost pump, left wing.

SUBTASK 28-22-41-420-028

(27) Forward or aft fuel boost pumps, connect the discharge port line to the discharge port on the housing [14].

(a) Install the flexible full coupling [50] on the discharge port line (TASK 28-22-15-400-803).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Installation Test

SUBTASK 28-22-41-410-003

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Install the applicable access doors:

(TASK 28-11-31-400-801)

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
631AB	Center Tank Access Door - Wing Station 168

SUBTASK 28-22-41-650-009

(2) Refuel the fuel tanks for the applicable fuel boost pump (TASK 12-11-00-650-803).

(a) Forward or aft fuel boost pumps, left wing:

1) No. 1 main tank – 500 lb (227 kg) minimum.

(b) Forward or aft fuel boost pumps, right wing:

1) No. 2 main tank – 500 lb (227 kg) minimum.

(c) Left or right center fuel boost pumps:

1) Center tank – 14,000 lb (6350 kg) minimum.

SUBTASK 28-22-41-210-003

(3) Make sure that the area around the housing [14] has no fuel leaks.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-41-410-004

(1) Install the fuel boost pump motor-impeller (TASK 28-22-41-400-801).

SUBTASK 28-22-41-860-011

(2) Supply electrical power, if necessary (TASK 24-22-00-860-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL BOOST PUMP - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks for the fuel boost pump system:

(1) Fuel Boost Pump - Operational Test

(2) Ground Fault Interrupter (GFI) - Operational Test

(3) Fuel Boost Pump Output Pressure Check

TASK 28-22-41-710-801

2. Fuel Boost Pump - Operational Test

A. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 28-22-41-860-030

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Do these steps for the applicable tanks:

(a) For the No. 1 main tank, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) For the aft fuel boost pump, remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT

2) For the forward fuel boost pump, remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD

(b) For the No. 2 main tank, do these steps:

1) For the aft fuel boost pump, remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT

2) For the forward fuel boost pump, remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD

(c) For the center tank, do these steps:

1) For the left center fuel boost pump, remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

2) For the right center fuel boost pump, remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-41-410-005

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-41-650-010

(3) Make sure that the fuel tanks, for the applicable fuel boost pump, have the minimum fuel quantity (Table 501).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 501/28-22-41-993-805 Fuel Tank - Minimum Fuel Quantity

FUEL TANK	FUEL QUANTITY
Center Tank	1000 lb (454 kg)
No. 1 Main Tank	500 lb (227 kg)
No. 2 Main Tank	500 lb (227 kg)

(a) To refuel the fuel tanks, do this task: Pressure Refuel Procedure, TASK 12-11-00-650-803.

E. Fuel Boost Pump Test

NOTE: To operate the fuel boost pumps, a person must be in the flight compartment. This person must continuously monitor the fuel quantity and the low pressure indication for the applicable fuel tank.

SUBTASK 28-22-41-860-016

WARNING

DO NOT OPERATE A FUEL PUMP IF THE AMBER PRESS LIGHT COMES ON AND STAYS ON. IT CAN CAUSE THE IGNITION OF THE FUEL FUMES IN THE TANK WHICH WILL CAUSE A FIRE OR EXPLOSION. FIRES AND EXPLOSIONS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Immediately set the applicable FUEL PUMPS switch to the OFF position if the LOW PRESSURE light comes on and stays on.

SUBTASK 28-22-41-860-017

(2) On the P5-2 Fuel System Module, set the applicable fuel boost pump switch to the ON position (Table 502).

Table 502/28-22-41-993-806 Fuel Boost Pump - Fuel Tank Location

FUEL BOOST PUMP LOCATION	FUEL TANK	SWITCH/CONTROL
Aft Fuel Boost Pump	No. 1 Main Tank	FUEL PUMPS AFT 1
Forward Fuel Boost Pump	No. 1 Main Tank	FUEL PUMPS FWD 1
Left Center Fuel Boost Pump	Center Tank	FUEL PUMPS CTR L
Right Center Fuel Boost Pump	Center Tank	FUEL PUMPS CTR R
Aft Fuel Boost Pump	No. 2 Main Tank	FUEL PUMPS AFT 2
Forward Fuel Boost Pump	No. 2 Main Tank	FUEL PUMPS FWD 2

SUBTASK 28-22-41-280-001

(3) Make sure that the applicable fuel boost pump operates.

NOTE: A person can hear the fuel boost pump while it operates.

SUBTASK 28-22-41-210-004

WARNING

DO NOT OPERATE A FUEL PUMP IF THE LOW PRESSURE LIGHT COMES ON AND STAYS ON. THIS CONDITION CAN CAUSE THE IGNITION OF THE FUEL FUMES IN THE FUEL TANK. A FIRE OR AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Make sure that the applicable LOW PRESSURE light on the P5-2 module goes off.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-41-860-018

(5) Set the applicable FUEL PUMPS switch to the OFF position.

SUBTASK 28-22-41-280-002

(6) Make sure that the applicable fuel boost pump does not operate.

SUBTASK 28-22-41-210-005

(7) For the forward or aft fuel boost pumps, make sure that the applicable LOW PRESSURE light comes on.

SUBTASK 28-22-41-210-006

(8) For the left or right center fuel boost pumps, make sure that the applicable LOW PRESSURE light stays off.

——— END OF TASK ———

TASK 28-22-41-720-801

3. Ground Fault Interrupter (GFI) - Operational Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task has one or more steps which are a means to satisfy Airworthiness Limitation Instruction (ALI) requirements. An ALI note will follow the step to which it applies. Any step or sub-step that precedes or follows an ALI identified step is not subject to the ALI requirement.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-20.

B. References

Reference	Title
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Test

SUBTASK 28-22-41-010-005

(1) To get access to Power Distribution Panel 1, P91, or Power Distribution Panel 2, P92, open this acces panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-41-860-020

(2) On the P5-2 Fuel System Module, set these switches to the OFF position:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) FUEL PUMPS 1 AFT

(b) FUEL PUMPS 1 FWD

(c) FUEL PUMPS CTR L

(d) FUEL PUMPS CTR R

(e) FUEL PUMPS 2 AFT

(f) FUEL PUMPS 2 FWD

SUBTASK 28-22-41-650-011

(3) Make sure that these fuel tanks for the applicable fuel boost pump has the minimum fuel quantity shown in Table 503.

Table 503/28-22-41-993-804 Fuel Tank - Minimum Fuel Quantity

FUEL TANK	FUEL QUANTITY
Center Tank	1000 lb (454 kg)
No. 1 Main Tank	500 lb (227 kg)
No. 2 Main Tank	500 lb (227 kg)

F. Ground Fault Interrupter (GFI) Test

NOTE: To operate the fuel boost pumps, a person must be in the flight compartment. This person must continuously monitor the fuel quantity and the low pressure indication for the applicable fuel tank.

SUBTASK 28-22-41-860-022

WARNING

DO NOT OPERATE A FUEL PUMP IF THE LOW PRESSURE LIGHT COMES ON AND STAYS ON. THIS CONDITION CAN CAUSE THE IGNITION OF THE FUEL FUMES IN THE FUEL TANK. A FIRE OR AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Immediately set the applicable FUEL PUMPS switch to the OFF position if the LOW PRESSURE light comes on and stays on.

SUBTASK 28-22-41-860-023

(2) Put the applicable fuel boost pump switch to the ON position on the P5-2 module (Table 504).

Table 504/28-22-41-993-802 Fuel Boost Pump - Fuel Tank Location

FUEL BOOST PUMP LOCATION	FUEL TANK	SWITCH/CONTROL
Aft Fuel Boost Pump	No. 1 Main Tank	FUEL PUMPS AFT 1
Forward Fuel Boost Pump	No. 1 Main Tank	FUEL PUMPS FWD 1
Left Center Fuel Boost Pump	Center Tank	FUEL PUMPS CTR L
Right Center Fuel Boost Pump	Center Tank	FUEL PUMPS CTR R
Aft Fuel Boost Pump	No. 2 Main Tank	FUEL PUMPS AFT 2
Forward Fuel Boost Pump	No. 2 Main Tank	FUEL PUMPS FWD 2

(a) Make sure that the applicable fuel boost pump operates.

NOTE: A person can hear the fuel boost pump while it operates.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-41-210-007

WARNING DO NOT OPERATE A FUEL PUMP IF THE LOW PRESSURE LIGHT COMES ON AND STAYS ON. THIS CONDITION CAN CAUSE THE IGNITION OF THE FUEL FUMES IN THE FUEL TANK. A FIRE OR AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Make sure that the LOW PRESSURE light goes off for the applicable fuel boost pump.

SUBTASK 28-22-41-860-024

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Push the applicable GFI relay TEST button in the P91 or P92 power distribution panel (Table 505).

NOTE: The GFI relays are found in the P91 and P92 panels.

NOTE: When an electrical fault occurs, the GFI circuit relays in the P91 and P92 panel detects the fault and opens the circuit. The RESET button on the top of the GFI relay shows a narrow white band when the circuit opens. The TEST button on the top of the GFI, when pressed, will also show the narrow white band.

NOTE: It is recommended that you use your finger to push the TEST button to prevent damage.

Table 505/28-22-41-993-803 Fuel Boost Pump - Relay Location

FUEL BOOST PUMP LOCATION	FUEL TANK	RELAY	RELAY PANEL LOCATION
Aft Fuel Boost Pump	No. 1 Main Tank	R18	Power Distribution Panel 2, P92
Forward Fuel Boost Pump	No. 1 Main Tank	R19	Power Distribution Panel 1, P91
Left Center Fuel Boost Pump	Center Tank	R54	Power Distribution Panel 1, P91
Right Center Fuel Boost Pump	Center Tank	R55	Power Distribution Panel 2, P92
Aft Fuel Boost Pump	No. 2 Main Tank	R20	Power Distribution Panel 1, P91
Forward Fuel Boost Pump	No. 2 Main Tank	R21	Power Distribution Panel 2, P92

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-41-210-012

28-AWL-20: ALI

(5) Make sure that the RESET button on the applicable GFI relay has moved out, and shows a white band.

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-20.

28-AWL-20: ALI

(a) For the No. 1 tank aft fuel boost pump, the relay is R18, found on the P92 panel.

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-20.

28-AWL-20: ALI

(b) For the No. 1 tank forward fuel boost pump, the relay is R19, found on the P91 panel.

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-20.

28-AWL-20: ALI

(c) For the left center tank fuel boost pump, the relay is R54, found on the P91 panel.

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-20.

28-AWL-20: ALI

(d) For the No. 2 aft fuel boost pump, the relay is R20, found on the P91 panel.

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-20.

28-AWL-20: ALI

(e) For the No. 2 forward fuel boost pump, the relay is R21, found on the P92 panel.

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-20.

28-AWL-20: ALI

(f) For the right center tank fuel boost pump, the relay is R55, found on the P92 panel.

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-20.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-41-280-005

28-AWL-20: ALI

(6) Listen and make sure that the applicable fuel boost pump does not operate.

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-20.

SUBTASK 28-22-41-210-008

(7) Make sure that the applicable LOW PRESSURE light comes on.

SUBTASK 28-22-41-860-025

(8) Set the applicable FUEL PUMPS switch to the OFF position.

SUBTASK 28-22-41-710-004

(9) To make sure that the applicable fuel boost pump does not operate when the GFI circuit disables the relay, do the subsequent steps:

(a) Set the applicable FUEL PUMPS switch to the ON position.

1) Make sure that the applicable fuel boost pump does not operate.

(b) Forward or aft fuel boost pumps, make sure that the applicable LOW PRESSURE light is on.

(c) Left or right center fuel boost pumps, make sure that the applicable LOW PRESSURE light comes on.

(d) Set the applicable FUEL PUMPS switch to the OFF position.

SUBTASK 28-22-41-860-026

(10) Push the RESET button on the applicable GFI relay.

(a) Make sure that the RESET button does not move back out, and the white band does not show.

SUBTASK 28-22-41-860-027

(11) Set the applicable FUEL PUMPS switch to the ON position.

SUBTASK 28-22-41-280-003

(12) Make sure that the applicable fuel boost pump operates.

SUBTASK 28-22-41-210-009

WARNING

DO NOT OPERATE A FUEL PUMP IF THE LOW PRESSURE LIGHT COMES ON AND STAYS ON. THIS CONDITION CAN CAUSE THE IGNITION OF THE FUEL FUMES IN THE FUEL TANK. A FIRE OR AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(13) Make sure that the applicable LOW PRESSURE light goes off.

SUBTASK 28-22-41-860-028

(14) Set the applicable FUEL PUMPS switch to the OFF position.

SUBTASK 28-22-41-280-004

(15) Make sure that the applicable fuel boost pump does not operate.

SUBTASK 28-22-41-210-010

(16) Forward or aft fuel boost pumps, make sure that the applicable LOW PRESSURE light comes on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-41-710-005

(17) Left or right center fuel boost pumps, make sure that the applicable LOW PRESSURE light stays off.

SUBTASK 28-22-41-860-029

(18) Repeat the GFI Test for the next fuel boost pump.

SUBTASK 28-22-41-410-006

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(19) Close the P91 and P92 power distribution panels.

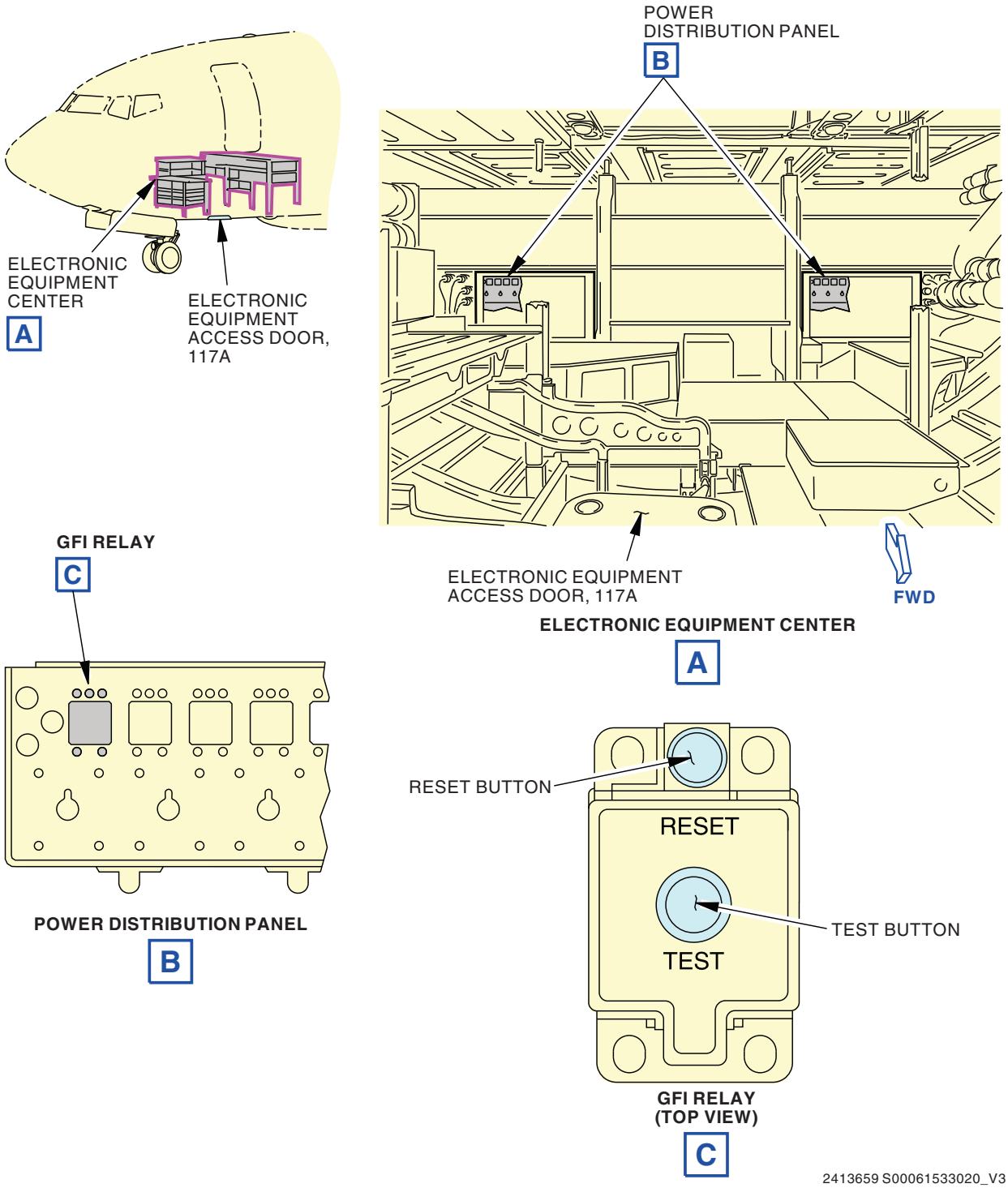
SUBTASK 28-22-41-410-007

(20) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ground Fault Interrupter (GFI) Relay Location
Figure 501/28-22-41-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-41-700-801

4. Fuel Boost Pump Output Pressure Check

A. General

(1) This inspection determines the output pressure of each pump. Low output pressure can lead to intermittent performance and fault messages.

B. References

Reference	Title
28-42-11-000-801	Fuel Boost Pump Fuel Pressure Switch Removal (P/B 401)
28-42-11-420-801	Fuel Boost Pump Fuel Pressure Switch Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-21846	Test Equipment - Pressure Output, Fuel Boost Pump Part #: A28018-14 Supplier: 81205 Opt Part #: A28018-1 Supplier: 81205

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Fuel Boost Pump Output Pressure Check

SUBTASK 28-22-41-020-023

(1) Remove the pressure switch for the applicable fuel boost pump (TASK 28-42-11-000-801).

SUBTASK 28-22-41-400-003

(2) Install the pressure output test equipment, SPL-21846, in place of the pressure switch that was removed.

SUBTASK 28-22-41-010-007

(3) To get access to the P91 and P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-41-860-031



WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

MAKE SURE THAT YOU DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Remove the safety tag and close the applicable circuit breakers for the applicable fuel boost pump:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-41-700-001

(5) Set the applicable fuel boost pump switch on the fuel management panel, on the P5 overhead panel, to the ON position.

(a) Make sure that there are no leaks prior to continuing.

1) If leaks are present, correct the issue before continuing with this task.

2) If no leaks are present, continue with the task.

SUBTASK 28-22-41-970-001

(6) Observe and record the pressure output of the applicable fuel boost pump.

Fuel Boost Pump Position	Pressure Range	Observed Pressure
Left Aft	18.0 psig (124 kPa) - 24.4 psig (168 kPa)	
Left Forward	18.0 psig (124 kPa) - 24.4 psig (168 kPa)	
Right Aft	18.0 psig (124 kPa) - 24.4 psig (168 kPa)	
Right Forward	18.0 psig (124 kPa) - 24.4 psig (168 kPa)	
Center Left	30.0 psig (207 kPa) - 40.5 psig (279 kPa)	
Center Right	30.0 psig (207 kPa) - 40.5 psig (279 kPa)	

SUBTASK 28-22-41-860-032

(7) Set the applicable fuel boost pump switch on the fuel management panel, on the P5 overhead panel, to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-41-860-033

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

MAKE SURE THAT YOU DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(8) Open the applicable circuit breaker and install safety tags for the applicable fuel boost pump:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-41-000-001

(9) Remove the pressure output test equipment, SPL-21846.

SUBTASK 28-22-41-420-030

(10) Install the removed pressure switch for the applicable fuel boost pump (TASK 28-42-11-420-801).

SUBTASK 28-22-41-800-002

(11) Repeat the Fuel Boost Pump Pressure Check procedure for the remaining fuel boost pumps.

SUBTASK 28-22-41-800-003

(12) Review the results.

NOTE: If the pressure measurements are out of range, these are the possible causes:

- The pump pressure is bypassing within the housing assembly.
- The pump pressure is bled off due to a leak internal to the fuel tank.
- The boost pump has mechanical degradation.
- Follow your company guidelines to determine your next maintenance action.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-41-410-008

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL SYSTEM MODULE, P5-2 - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Fuel System Module, P5-2 Removal

(2) Fuel System Module, P5-2 Installation.

TASK 28-22-43-020-801

2. Fuel System Module, P5-2 Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the fuel system module, P5-2.

B. References

Reference	Title
20-40-12 P/B 201	ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) DEVICES - MAINTENANCE PRACTICES
28-43-21-020-801	Fuel Temperature Indicator Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Removal

SUBTASK 28-22-43-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1
B	5	C00540	FUEL SPAR VALVE IND
B	7	C00361	FUEL CROSS FEED VALVE CONT
C	3	C01637	BOOST PMP CTR TNK L AUTO SHUT OFF-DC
C	4	C01471	FUEL SHUTOFF VALVES PWR PACK
C	6	C01472	FUEL SHUTOFF VALVES BUS
C	7	C01638	BOOST PMP CTR TNK R AUTO SHUT OFF-DC
C	11	C01275	MASTER CAUTION ANNUNCIATOR CONT 1
D	7	C01659	AT S-O/UCPO BST PMP CTR TNK L AC
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND
E	7	C01658	FUEL AT S-O/UCPO BST PMP CTR TNK R AC
E	11	C00313	INDICATOR MASTER DIM SECT 1
F	12	C00318	INDICATOR MASTER DIM SECT 6

SUBTASK 28-22-43-010-003

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-43-860-009

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-43-760-002

CAUTION

DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(4) Before you touch the fuel system module, P5-2 [1], do the procedure for devices that are sensitive to electrostatic discharge (PAGEBLOCK 20-40-12/201).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-43-010-002

(5) Open the P5 Overhead Panel.

(a) Loosen the quarter-turn fasteners on the bottom corners of the P5 Overhead Panel.

WARNING

HOLD THE PANEL TIGHTLY BEFORE YOU RELEASE THE LATCH. THE PANEL IS HEAVY. INJURIES CAN OCCUR IF IT HITS YOU.

(b) Release the P5 Overhead Panel safety latches.

F. Fuel System Module, P5-2 Removal

SUBTASK 28-22-43-020-001

(1) Loosen the six quarter-turn fasteners on the fuel system module, P5-2 [1].

SUBTASK 28-22-43-010-001

(2) Carefully move the fuel system module, P5-2 [1] from the P5 panel.

SUBTASK 28-22-43-020-002

(3) Disconnect the three connectors from the rear of the fuel system module, P5-2 [1].

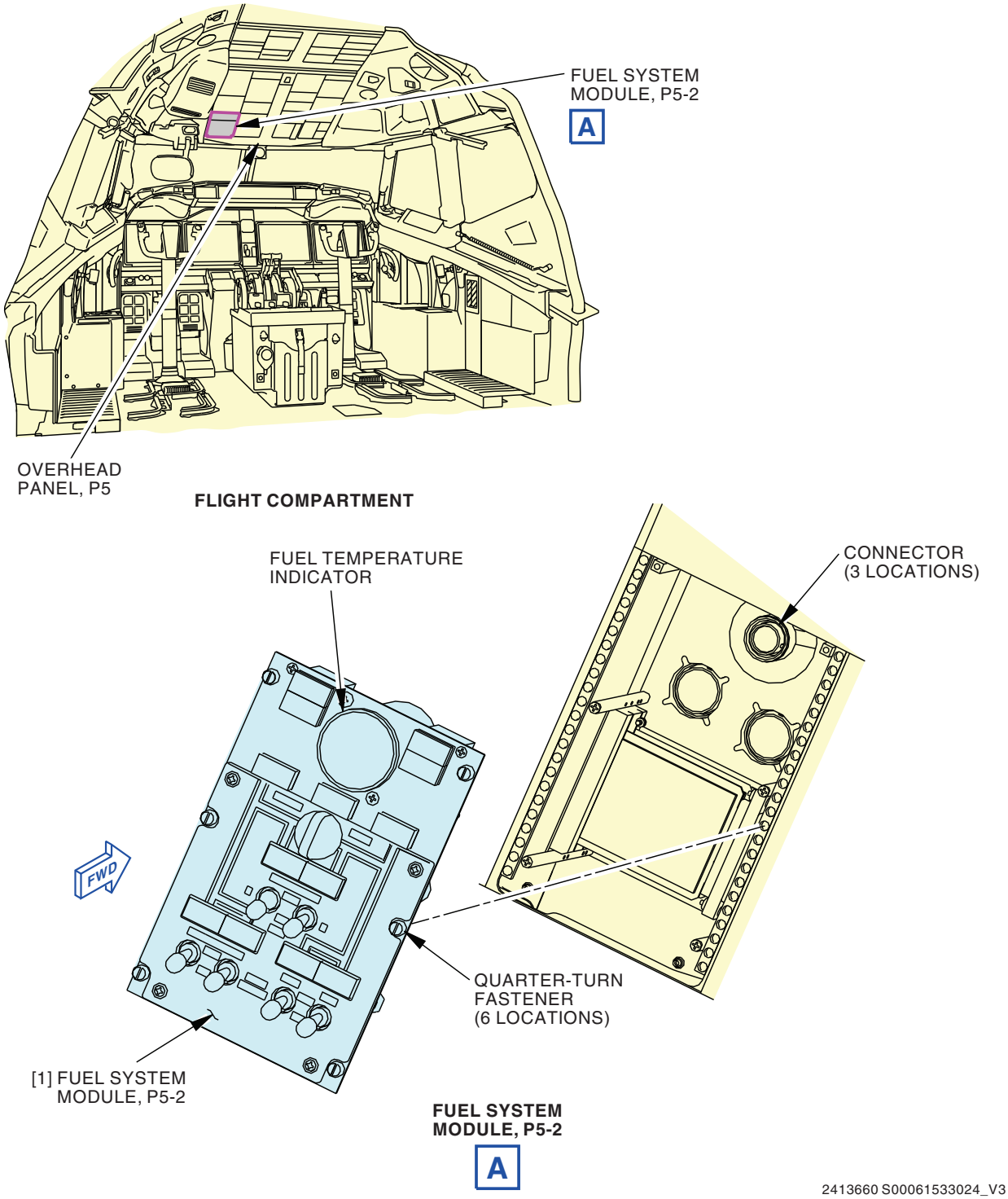
NOTE: The connectors are D626, D628, and D616.

SUBTASK 28-22-43-020-004

(4) Remove the fuel temperature indicator as necessary (TASK 28-43-21-020-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel System Module, P5-2 Installation
Figure 401/28-22-43-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-43-420-801

3. Fuel System Module, P5-2 Installation

(Figure 401)

A. General

(1) This task gives instructions for the installation of the fuel system module, P5-2.

B. References

Reference	Title
20-40-12 P/B 201	ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) DEVICES - MAINTENANCE PRACTICES
28-22-43-710-801	Fuel System Module, P5-2 - Operational Test (P/B 501)
28-43-21-020-802	Fuel Temperature Indicator Installation (P/B 401)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fuel system module, P5-2	28-22-43-01-050	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Installation

SUBTASK 28-22-43-760-003

CAUTION

DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(1) Before you touch the fuel system module, P5-2 [1], do the procedure for devices that are sensitive to electrostatic discharge (PAGEBLOCK 20-40-12/201).

SUBTASK 28-22-43-420-001

(2) Install the fuel temperature indicator as applicable (TASK 28-43-21-020-802).

G. Fuel System Module, P5-2 Installation

SUBTASK 28-22-43-420-003

(1) Align the fuel system module, P5-2 [1] in its position on the P5 Overhead Panel.

SUBTASK 28-22-43-420-005

(2) Tighten the six quarter-turn fasteners.

SUBTASK 28-22-43-420-004

(3) Connect the three connectors to the fuel system module, P5-2 [1].

NOTE: The connectors are D616, D626, and D628.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-43-410-002

(4) Close the P5 Overhead Panel, as follows:

(a) Raise the P5 Overhead Panel.

(b) Make sure that the safety latch is in the proper position.

(c) Tighten the quarter-turn fasteners on the bottom corners of the P5 Overhead Panel.

H. Fuel System Module, P5-2 Installation Test

SUBTASK 28-22-43-860-010

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-43-410-003

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-43-860-002

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1
B	5	C00540	FUEL SPAR VALVE IND

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	7	C00361	FUEL CROSS FEED VALVE CONT
C	3	C01637	BOOST PMP CTR TNK L AUTO SHUT OFF-DC
C	4	C01471	FUEL SHUTOFF VALVES PWR PACK
C	6	C01472	FUEL SHUTOFF VALVES BUS
C	7	C01638	BOOST PMP CTR TNK R AUTO SHUT OFF-DC
C	11	C01275	MASTER CAUTION ANNUNCIATOR CONT 1
D	7	C01659	AT S-O/UCPO BST PMP CTR TNK L AC
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND
E	7	C01658	FUEL AT S-O/UCPO BST PMP CTR TNK R AC
E	11	C00313	INDICATOR MASTER DIM SECT 1
F	12	C00318	INDICATOR MASTER DIM SECT 6

SUBTASK 28-22-43-710-001

(4) Do this task: Fuel System Module, P5-2 - Operational Test, TASK 28-22-43-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL SYSTEM MODULE, P5-2 - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) Fuel System Module, P5-2 - Operational Test.

TASK 28-22-43-710-801

2. Fuel System Module, P5-2 - Operational Test

(Figure 501)

A. References

Reference	Title
28-22-41-710-801	Fuel Boost Pump - Operational Test (P/B 501)
71-00-00-700-809-G00	Test No. 33 - Lamp Check Out Special Functions Test (P/B 501)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 28-22-43-860-004

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1
B	5	C00540	FUEL SPAR VALVE IND
B	7	C00361	FUEL CROSS FEED VALVE CONT
C	3	C01637	BOOST PMP CTR TNK L AUTO SHUT OFF-DC
C	4	C01471	FUEL SHUTOFF VALVES PWR PACK
C	6	C01472	FUEL SHUTOFF VALVES BUS
C	7	C01638	BOOST PMP CTR TNK R AUTO SHUT OFF-DC
C	11	C01275	MASTER CAUTION ANNUNCIATOR CONT 1
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND
E	11	C00313	INDICATOR MASTER DIM SECT 1
F	12	C00318	INDICATOR MASTER DIM SECT 6

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-43-010-004

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-43-860-011

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Make sure that these circuit breakers are closed:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD

SUBTASK 28-22-43-860-005

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

E. Fuel System Module Test

SUBTASK 28-22-43-710-002

(1) On the P5-2 module, make sure that these engine No. 1 lights are on:

(a) SPAR VALVE CLOSED

(b) ENG VALVE CLOSED.

SUBTASK 28-22-43-710-003

(2) On the forward overhead panel, P5, make sure that the ENGINE START 1 and 2 switches are in the OFF position.

SUBTASK 28-22-43-710-004

(3) On the control stand, P10, set the Engine 1 Start Lever to the IDLE position.

(a) Make sure that engine No. 1 SPAR VALVE CLOSED light comes on brightly while the valve goes from the closed to the open position.

(b) Make sure that engine No. 1 SPAR VALVE CLOSED light goes off in 6 seconds.

(c) Make sure that engine No. 1 ENG VALVE CLOSED light comes on brightly.

SUBTASK 28-22-43-710-005

(4) Set the Engine 1 Start Lever to the CUTOFF position.

(a) Make sure that engine No. 1 SPAR VALVE CLOSED light comes on.

(b) Make sure that engine No. 1 ENG VALVE CLOSED light comes on.

SUBTASK 28-22-43-710-006

(5) On the P5-2 module, make sure that these engine No. 2 lights are on:

(a) SPAR VALVE CLOSED

(b) ENG VALVE CLOSED.

SUBTASK 28-22-43-710-007

(6) On the control stand, P10, set the Engine 2 Start Lever to the IDLE position.

(a) Make sure that engine No. 2 SPAR VALVE CLOSED light comes on brightly while the valve goes from the closed to the open position.

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(b) Make sure that engine No. 2 SPAR VALVE CLOSED light for engine No. 2 goes off in 6 seconds.

(c) Make sure that engine No. 2 ENG VALVE CLOSED light comes on brightly.

SUBTASK 28-22-43-710-008

(7) Set the Engine 2 Start Lever to the CUTOFF position.

(a) Make sure that the engine No. 2 SPAR VALVE CLOSED light comes on.

(b) Make sure that the engine No. 2 ENG VALVE CLOSED light comes on.

SUBTASK 28-22-43-710-009

(8) Make sure that the fuel gage needle is not supported by the needle indicator stops.

SUBTASK 28-22-43-860-006

(9) Make sure that the fuel crossfeed VALVE OPEN light is off.

SUBTASK 28-22-43-710-010

(10) Set the CROSS FEED switch to the open position.

(a) Make sure that the fuel crossfeed VALVE OPEN light is on bright while the valve goes from the closed position to the open position.

(b) Make sure that the fuel crossfeed VALVE OPEN light dims in 6 seconds.

SUBTASK 28-22-43-710-011

(11) Set the CROSS FEED switch to the closed position.

(a) Make sure that the fuel crossfeed VALVE OPEN light goes off.

SUBTASK 28-22-43-710-012

(12) Make sure that the LOW PRESSURE lights are on for these fuel boost pumps:

(a) Main tank No. 1 - forward fuel boost pump

(b) Main tank No. 1 - aft fuel boost pump

(c) Main tank No. 2 - forward fuel boost pump

(d) Main tank No. 2 - aft fuel boost pump.

SUBTASK 28-22-43-710-013

(13) Do a test of the system master caution annunciator:

(a) Set the FUEL PUMPS CTR L switch to the ON position.

1) Make sure that the LOW PRESSURE light comes on for the FUEL PUMPS CTR L switch.

(b) After approximately 15 seconds, set the FUEL PUMPS CTR L switch to the OFF position.

(c) On the glareshield panel, P7, make sure that the FUEL annunciator light is on for the left system annunciator light.

(d) On the glareshield panel, P7, push the MASTER CAUTION pushbutton.

1) Make sure that the FUEL annunciator light goes off.

(e) On the glareshield panel, P7, push the FLT CONT/IRS/FUEL annunciator light pushbutton on the left system annunciator light.

1) Make sure that the FUEL annunciator light comes on.

(f) Set the FUEL PUMPS CTR R switch to the ON position.

1) Make sure that the LOW PRESSURE light comes on for the FUEL PUMPS CTR R switch.

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(g) After approximately 15 seconds, set the FUEL PUMPS CTR R switch to the OFF position.

(h) Make sure that the FUEL annunciator light is on.

(i) Push the MASTER CAUTION pushbutton.

1) Make sure that the FUEL annunciator light goes off.

(j) Push the FLT CONT/IRS/FUEL annunciator light pushbutton.

1) Make sure that the FUEL annunciator light comes on.

SUBTASK 28-22-43-710-014

(14) Do a test of the FILTER BYPASS (TASK 71-00-00-700-809-G00).

NOTE: This test will make sure that the FILTER BYPASS light operates correctly.

SUBTASK 28-22-43-860-007

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(15) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-43-410-004

(16) Close this access panel:

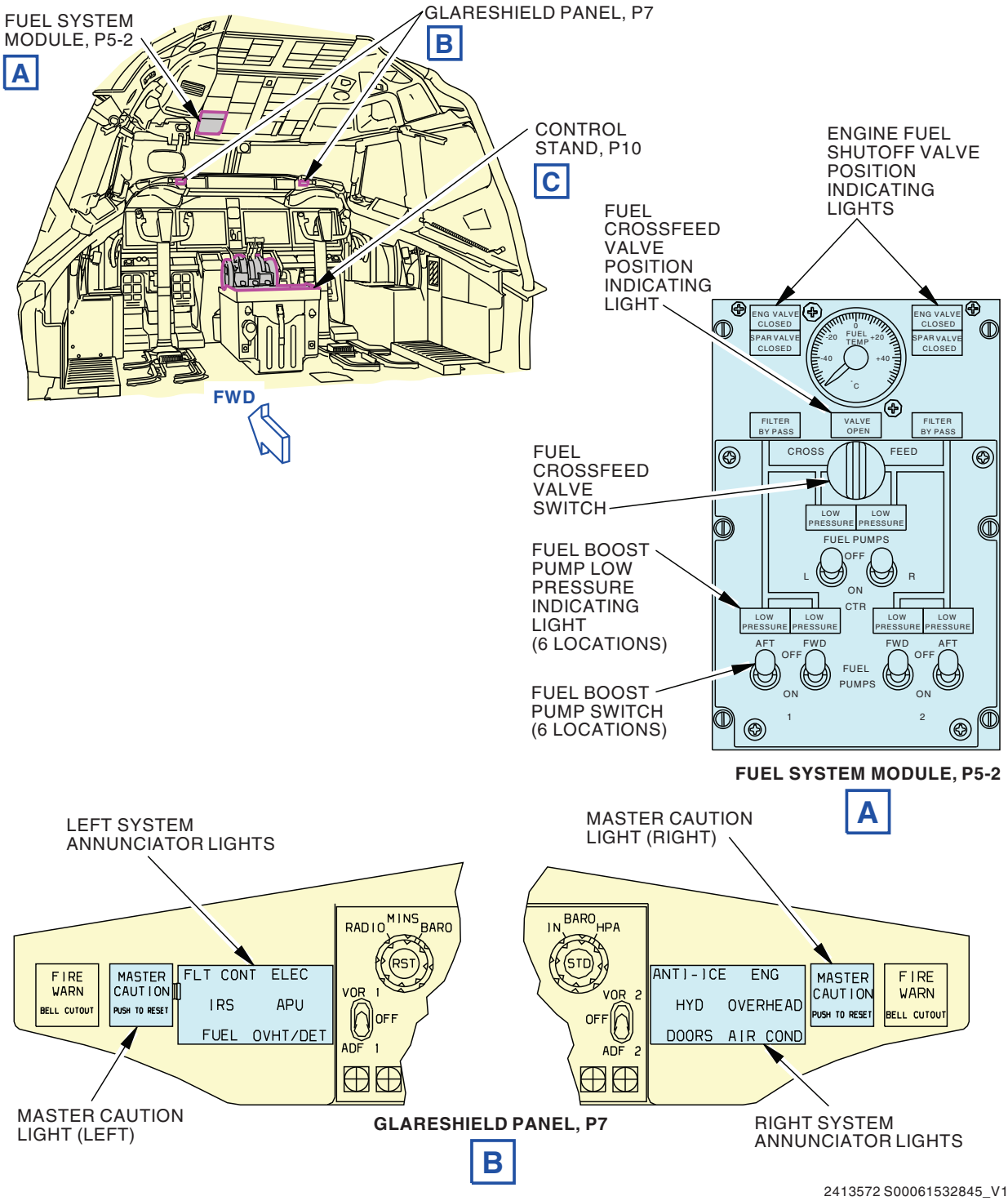
Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-43-760-001

(17) Do this task: Fuel Boost Pump - Operational Test, TASK 28-22-41-710-801.

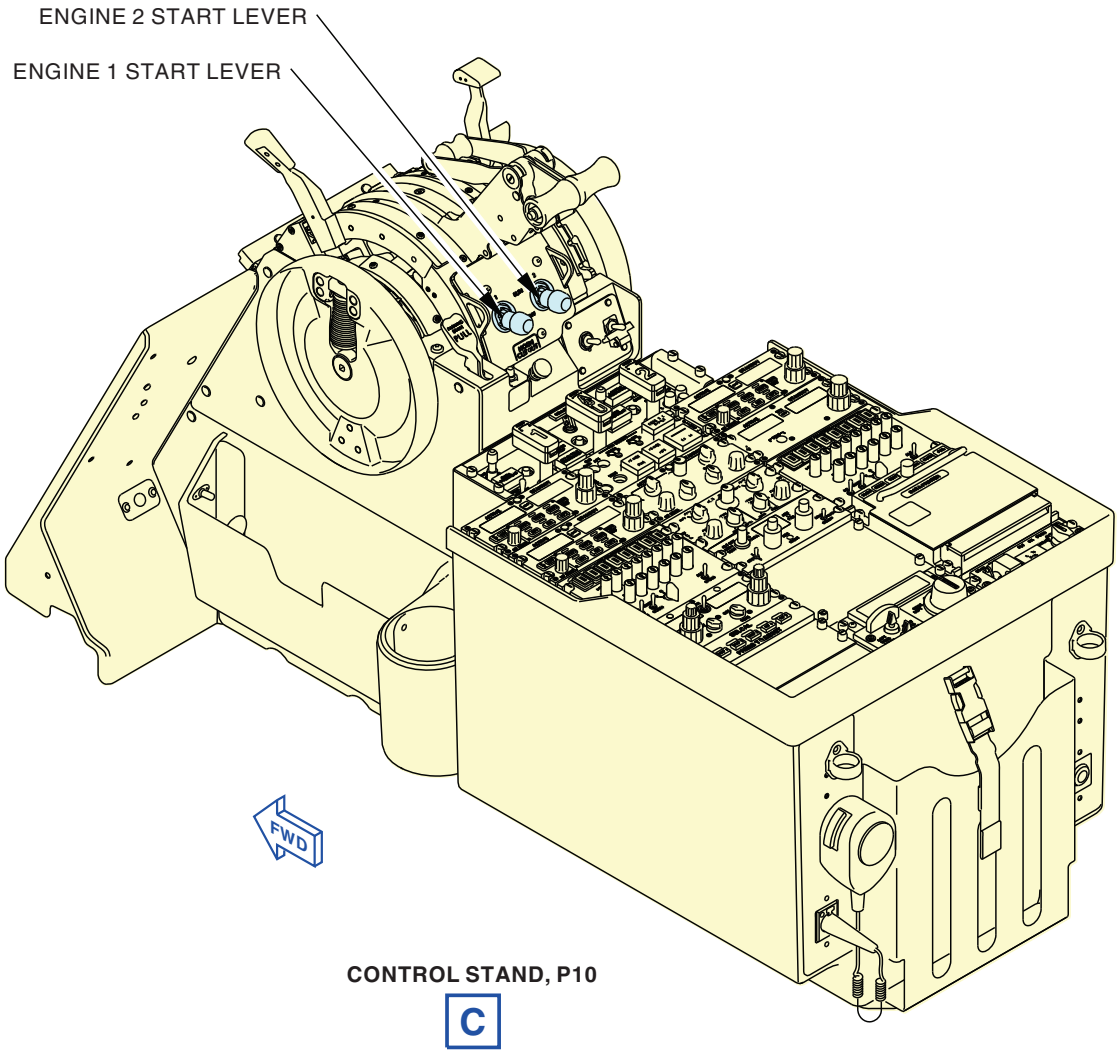
END OF TASK

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Engine Fuel Feed System
Figure 501/28-22-43-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Engine Fuel Feed System
Figure 501/28-22-43-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL BOOST PUMP REMOVAL CHECK VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the removal check valve

(2) An installation of the removal check valve.

B. The fuel boost pump removal check valve is referred to as the removal check valve in this procedure.

C. A removal check valve attaches to each of the six fuel boost pumps. All fuel boost pumps are installed on the rear spars. The removal check valve is designed to allow the removal of the motor-impeller without defueling the center tank or main tanks.

TASK 28-22-51-000-801

2. Removal Check Valve Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the removal check valve.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-22-15-000-803	Couplings Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
631AB	Center Tank Access Door - Wing Station 168

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E. Prepare for the Removal

SUBTASK 28-22-51-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-22-51-650-001

(2) Defuel the center tank (TASK 28-26-00-650-801).

SUBTASK 28-22-51-650-002

(3) Defuel the No.1 main tank for the removal check valves [3] (TASK 28-26-00-650-801):

(a) Left center fuel boost pump

(b) Forward or aft fuel boost pumps, on the left wing.

SUBTASK 28-22-51-650-003

(4) Defuel the No. 2 main tank for the removal check valves [3] (TASK 28-26-00-650-801):

(a) Right center fuel boost pump

(b) Forward or aft fuel boost pumps, on the right wing.

SUBTASK 28-22-51-860-001

(5) Set the CROSSFEED valve switch, on the P5-2 fuel system module, to the closed position.

(a) Make sure that the indicator light for the crossfeed valve position goes off.

SUBTASK 28-22-51-010-002

(6) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-51-860-002

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(7) Open the applicable circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-51-410-001

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(8) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Access the removal check valves for the No. 1 main tank, remove this access door: (TASK 28-11-31-000-801)

Number	Name/Location
531AB	Center Tank Access Door - Wing Station 168

(b) Access the removal check valves for the No. 2 main tank, remove this access door: (TASK 28-11-31-000-801)

Number	Name/Location
631AB	Center Tank Access Door - Wing Station 168

(c) Access the removal check valves for the center tank, remove this access door: (TASK 28-11-31-000-801)

Number	Name/Location
131AB	Center Tank Access

SUBTASK 28-22-51-410-002

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(9) Go into the fuel tank (TASK 28-11-00-910-802).

F. Removal Check Valve Removal

SUBTASK 28-22-51-000-001

(1) Remove lockwire between the flexible half coupling assembly [5] and screw [9] on the bracket of the removal check valve [3] (View E, Figure 401).

SUBTASK 28-22-51-020-007

(2) Remove the washer [11], spacer [10], and screw [9] from the bracket of the removal check valve [3].

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SUBTASK 28-22-51-020-001

(3) Disconnect the flexible half coupling assembly [5] between the removal check valve [3] and pump inlet line (TASK 28-22-15-000-803).

NOTE: The flexible half coupling assembly has two retaining rings.

(a) Remove and discard the O-ring [4].

SUBTASK 28-22-51-020-002

(4) Move the pump inlet line away from the removal check valve [3].

SUBTASK 28-22-51-020-003

(5) Remove the screws [6], washers [7], and nuts [8] from the flanges.

SUBTASK 28-22-51-020-004

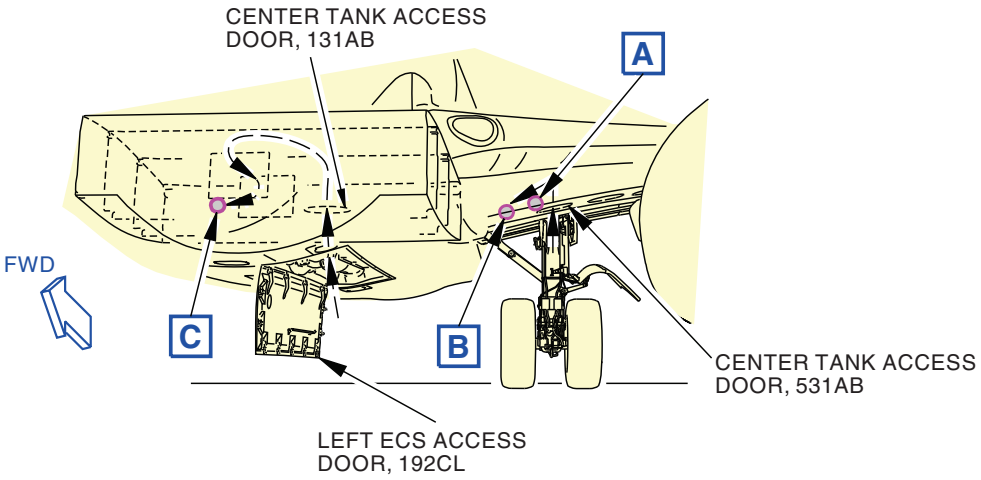
(6) Remove and discard the O-ring [2].

SUBTASK 28-22-51-020-005

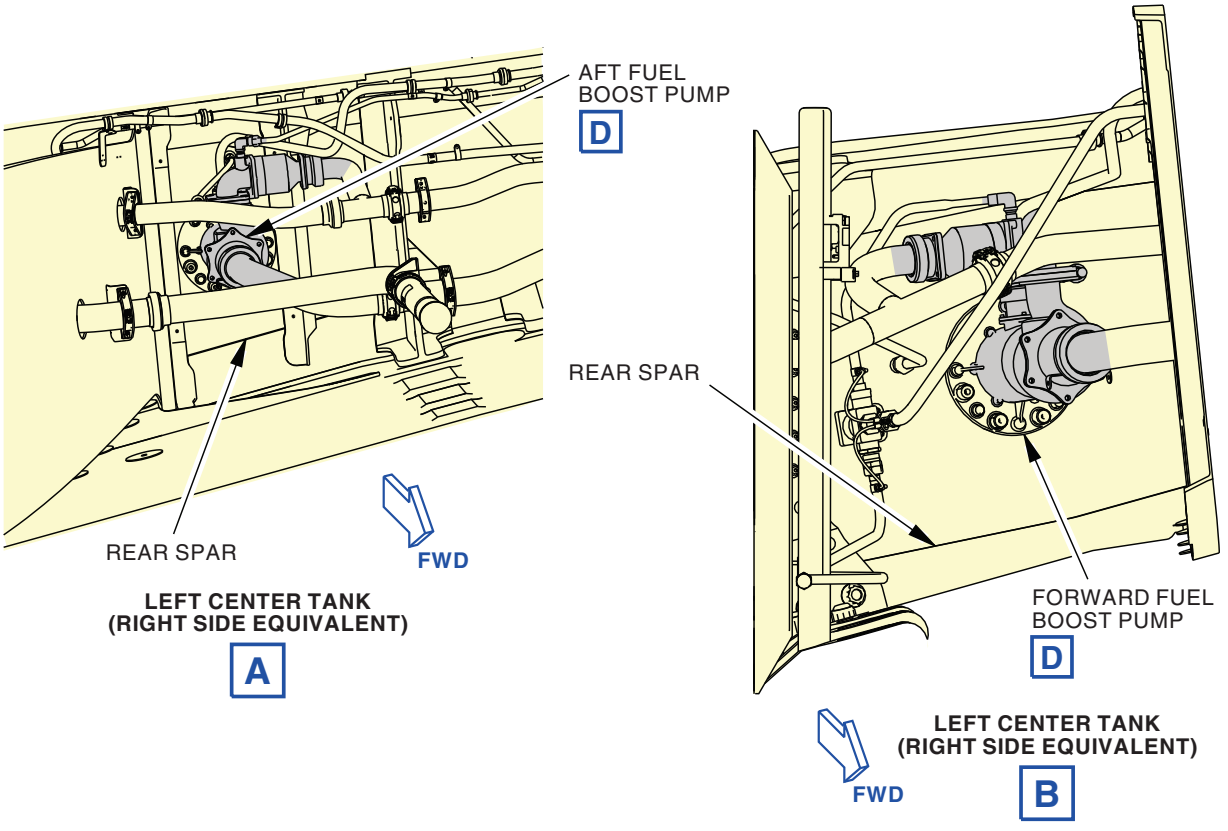
(7) Remove the removal check valve [3].

——— END OF TASK ———

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LEFT WING
(RIGHT WING IS OPPOSITE)



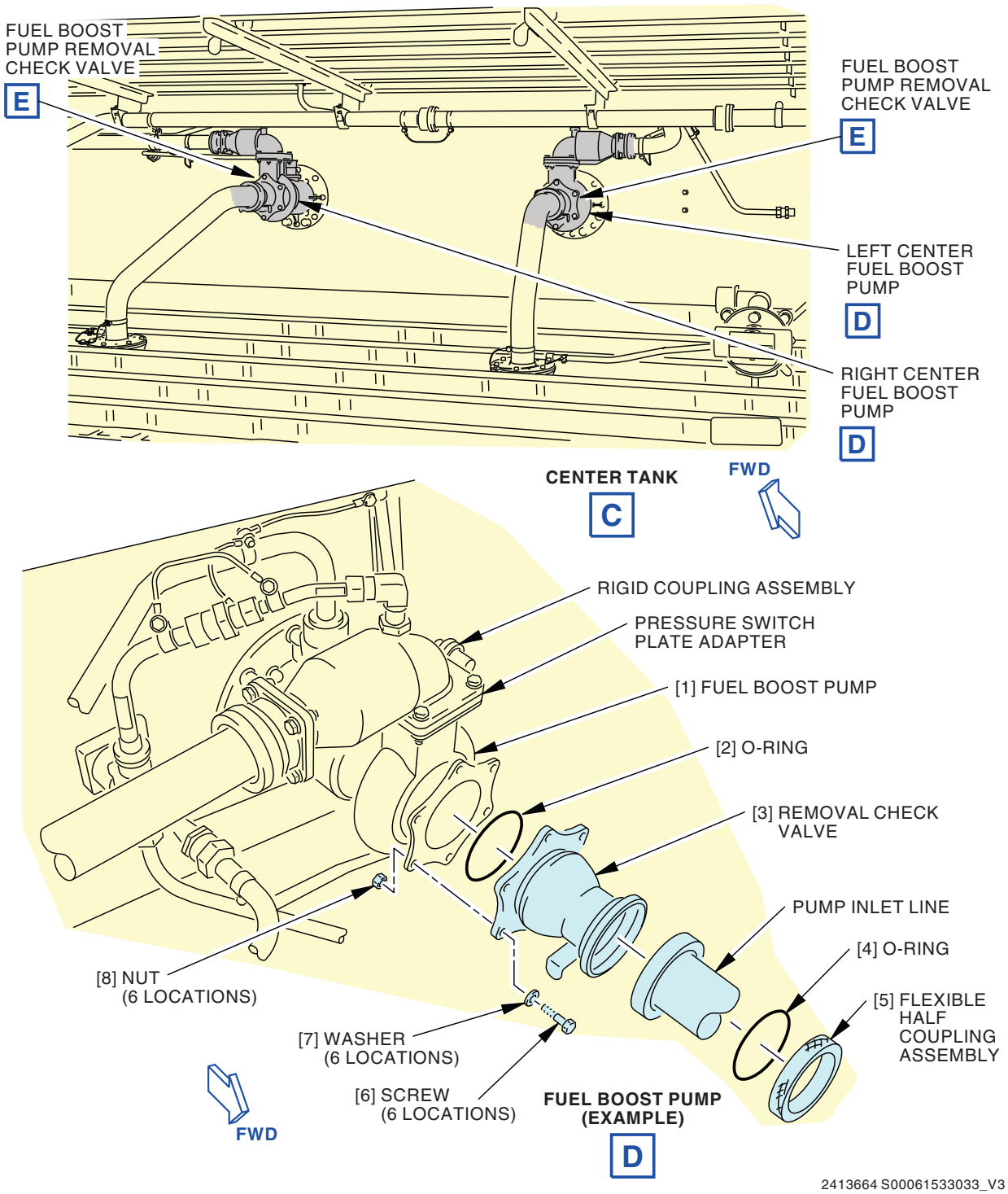
LEFT CENTER TANK
(RIGHT SIDE EQUIVALENT)

LEFT CENTER TANK
(RIGHT SIDE EQUIVALENT)

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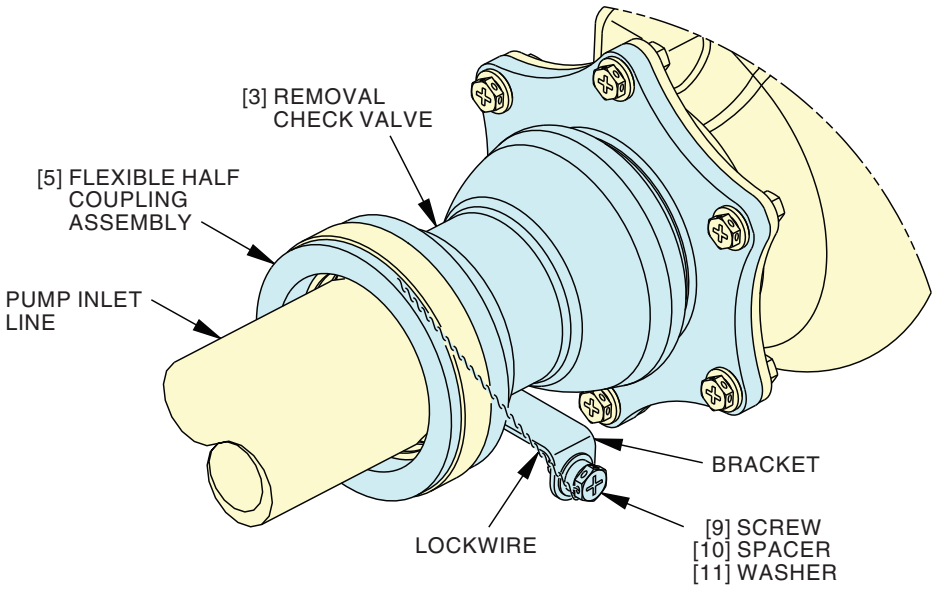
Boost Pump Removal Check Valve Installation
Figure 401/28-22-51-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Boost Pump Removal Check Valve Installation
Figure 401/28-22-51-990-801 (Sheet 2 of 3)

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FUEL BOOST PUMP REMOVAL CHECK VALVE (EXAMPLE)

E

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Boost Pump Removal Check Valve Installation
Figure 401/28-22-51-990-801 (Sheet 3 of 3)

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TASK 28-22-51-400-801

3. Removal Check Valve Installation

(Figure 401)

A. General

(1) This task gives instructions to install the removal check valve.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
28-22-15-400-803	Couplings Installation (P/B 401)
28-22-41-000-801	Fuel Boost Pump Motor-Impeller Removal (P/B 401)
28-22-41-400-801	Fuel Boost Pump Motor-Impeller Installation (P/B 401)
51-21-41-370-802	Apply Bonderite M-CR 600, Bonderite M-CR 1200, Bonderite M-CR 600RTU, or Bonderite M-CR 1200S Solution (P/B 701)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1769	Puller - Fuel Pump Part #: B28003-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C50315	Coating - Chemical Conversion - Bonderite M-CR 600 Aero (Formerly Alodine 600)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
D00504	Grease - Petrolatum	VV-P-236
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51293	Cotton Wiper - Knit, Process Cleaning Absorbent Wiper	BMS15-5 Class B

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Removal check valve	28-22-51-01-030	PTF ALL
		28-22-51-02-030	PTF ALL

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F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
631AB	Center Tank Access Door - Wing Station 168

H. Removal Check Valve Installation

SUBTASK 28-22-51-100-001

(1) Do these steps to prepare the new removal check valve [3] and fuel boost pump [1] for the installation:

(a) Clean the fay surfaces of the fuel boost pump [1] and removal check valve [3] (SWPM 20-20-00).

1) Use Cleaning Procedure 1 to manually clean the inlet flange fay surface on the removal check valve [3].

a) Use an abrasive pad to clean the flange fay surface.

b) Move the pad in a circular movement while you press with your thumb.

NOTE: Do not clean the O-ring groove.

2) Use Cleaning Procedure 1 to manually clean the inlet flange fay surface on the fuel boost pump [1].

a) Use an abrasive pad to clean the flange fay surface.

b) Move the pad in a circular movement while you press with your thumb.

3) Final Clean the fay surfaces of the fuel boost pump [1] and removal check valve [3].

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034 (recommended), or knit cotton wiper, G51293 (alternate), soaked with solvent, B00062, or solvent, B00184.

b) Rub dry with a clean cotton wiper.

c) Continue to clean and dry the surface until the cotton wiper stays clean.

(b) Manually apply Bonderite M-CR 600 Aero coating, C50315, to the abraded surfaces (TASK 51-21-41-370-802).

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SUBTASK 28-22-51-420-001

(2) Connect the removal check valve [3] to the fuel boost pump [1]:

(a) Lubricate the new O-ring [2] with grease, D00504.

(b) Install the O-ring [2] in the groove on the flange of the removal check valve [3].

(c) Align the flange of the removal check valve [3] with the flange of the fuel boost pump [1].

(d) Install the screws [6], washers [7], and nuts [8].

SUBTASK 28-22-51-420-002

(3) Install the flexible half coupling assembly [5] to attach the pump inlet line to the removal check valve [3] (TASK 28-22-15-400-803).

(a) Install new O-ring [4].

SUBTASK 28-22-51-420-004

(4) Install the washer [11], spacer [10], and screw [9] on the bracket of the removal check valve [3].

(a) Tighten the screw [9] to 33 in-lb (3.7 N·m) with the hand tool only.

1) Hold on the bracket with the adjustable wrench when torque the screw [9].

2) Make sure that the bracket does not twist after torque.

SUBTASK 28-22-51-420-005

(5) Install MS20995NC32 lockwire, G50347, between the flexible half coupling assembly [5] and screw [9] on the bracket of the removal check valve [3] (View E, Figure 401).

I. Removal Check Valve Installation Test

SUBTASK 28-22-51-010-001

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Install the applicable access panels:

(TASK 28-11-31-400-801)

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
631AB	Center Tank Access Door - Wing Station 168

SUBTASK 28-22-51-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-51-410-003

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-51-710-001

(4) To do an operational test of the applicable removal check valve [3] for the No. 1 main tank, do these steps:

(a) Refuel No.1 main tank to 300 lb (136 kg) minimum (TASK 12-11-00-650-803).

(b) Pull the motor-impeller out 0.6 in. (15.2 mm) (or to the MS35275-250 screws) from the fuel boost pump [1] (TASK 28-22-41-000-801).

NOTE: This allows the valve in the removal check valve to close.

1) Use the puller, SPL-1769, to pull out the motor-impeller.

WARNING

DO NOT LET FUEL SPILL. WHEN YOU REMOVE THE DRAIN PLUG, A SMALL QUANTITY OF FUEL WILL COME OUT OF THE MOTOR/IMPELLER UNIT. IF THERE IS A CONTINUOUS FLOW OF FUEL, THE VALVE IS NOT CLOSED. FUEL CAN CAUSE A FIRE OR AN EXPLOSION. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(c) Remove the drain plug.

1) Make sure that fuel does not continuously drain from the drain port.

(d) Install the drain plug.

(e) Install the motor-impeller (TASK 28-22-41-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-22-51-710-002

(5) To do an operational test of the applicable removal check valve [3] for the No. 2 main tank, do these steps:

(a) Refuel No. 2 main tank to 300 lb (136 kg) minimum (TASK 12-11-00-650-803).

(b) Pull the motor-impeller out 0.6 in. (15.2 mm) (or to the MS35275-250 screws) from the fuel boost pump [1] (TASK 28-22-41-000-801).

NOTE: This allows the valve in the removal check valve to close.

1) Use the puller, SPL-1769, to pull the motor-impeller out.

WARNING

DO NOT LET FUEL SPILL. WHEN YOU REMOVE THE DRAIN PLUG, A SMALL QUANTITY OF FUEL WILL COME OUT OF THE MOTOR/IMPELLER UNIT. IF THERE IS A CONTINUOUS FLOW OF FUEL, THE VALVE IS NOT CLOSED. FUEL CAN CAUSE A FIRE OR AN EXPLOSION. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(c) Remove the drain plug.

1) Make sure that fuel does not continuously drain from the drain port.

(d) Install the drain plug.

(e) Install the motor-impeller (TASK 28-22-41-400-801).

SUBTASK 28-22-51-710-003

(6) To do an operational test of the applicable removal check valve [3] for the center tank, do these steps:

(a) Refuel the center tank to 14,000 lb (6350 kg) minimum (TASK 12-11-00-650-803).

(b) Pull the motor-impeller out 0.6 in. (15.2 mm) (or to the MS35275-250 screws) from the fuel boost pump [1] (TASK 28-22-41-000-801).

NOTE: This allows the valve in the removal check valve to close.

WARNING

DO NOT LET FUEL SPILL. WHEN YOU REMOVE THE DRAIN PLUG, A SMALL QUANTITY OF FUEL WILL COME OUT OF THE MOTOR/IMPELLER UNIT. IF THERE IS A CONTINUOUS FLOW OF FUEL, THE VALVE IS NOT CLOSED. FUEL CAN CAUSE A FIRE OR AN EXPLOSION. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(c) Remove the drain plug.

1) Make sure that fuel does not continuously drain from the drain port.

(d) Install the drain plug.

(e) Install the motor-impeller (TASK 28-22-41-400-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN TANK FUEL BOOST PUMP BYPASS CHECK VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the main tank fuel boost pump bypass check valve

(2) An installation of the main tank fuel boost pump bypass check valve.

B. There are two fuel boost pump bypass check valves installed, one in each main tank. The fuel boost pump bypass check valves are located between stringers 4 and 5 and ribs No. 5 and 6.

TASK 28-22-61-000-801

2. Main Tank Fuel Boost Pump Bypass Check Valve Removal

(Figure 401)

A. General

(1) The main tank fuel boost pump bypass check valve is referred to as bypass check valve in this task.

(2) This task gives instructions to remove the bypass check valve.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-22-15-000-803	Couplings Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Access Panels

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216
632AB	Main Tank Access Door - Wing Station 216

E. Prepare for the Removal

SUBTASK 28-22-61-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-22-61-650-001

(2) Defuel the No. 1 or No. 2 main tank (TASK 28-26-00-650-801).

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SUBTASK 28-22-61-010-001

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove the applicable access door:

(TASK 28-11-11-000-801)

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216
632AB	Main Tank Access Door - Wing Station 216

SUBTASK 28-22-61-010-002

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Go into the fuel tank (TASK 28-11-00-910-802).

F. Bypass Check Valve Removal

SUBTASK 28-22-61-020-001

(1) Remove the screws [21], nuts [19], and washers [20] that attach the mounting bracket on the suction feed line [1] to the bracket.

SUBTASK 28-22-61-020-002

(2) Remove the screw [12], washers [13], washer [15], and nut [18] from the loop clamp [14] and loop clamp [16].

(a) Keep the loop clamp [14] and loop clamp [16] on the line and suction feed line [1].

SUBTASK 28-22-61-010-003

(3) Remove the hex head bolts [11], washers [10], washers [9], and nuts [8] that attach the suction feed line [1] to the bypass check valve [4].

SUBTASK 28-22-61-020-003

(4) Remove the rigid coupling assembly [5] (TASK 28-22-15-000-803).

(a) Discard the O-ring [7] and cable assembly [6].

SUBTASK 28-22-61-020-004

(5) Remove the bypass check valve [4].

SUBTASK 28-22-61-020-005

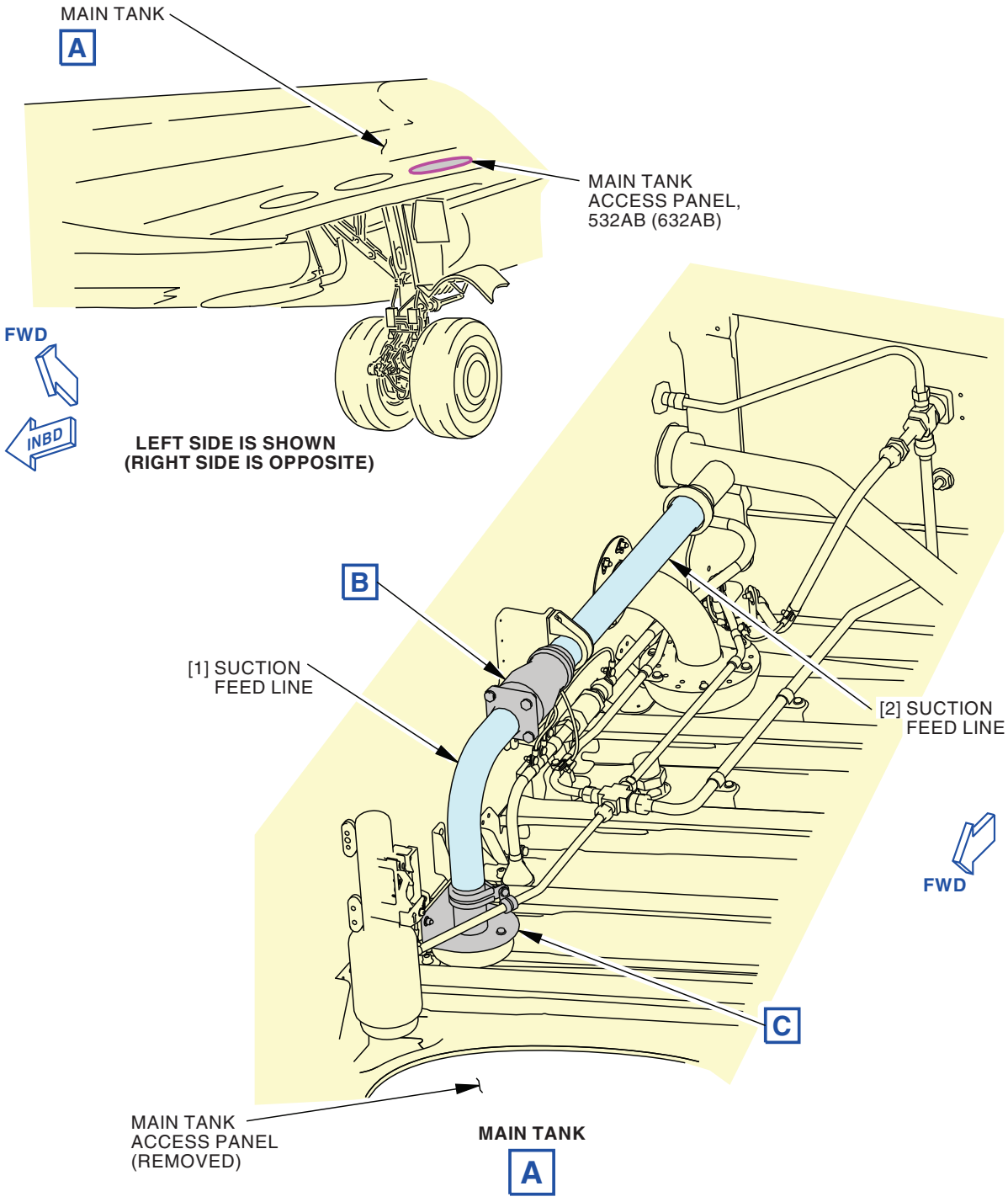
(6) Discard the O-ring [3].

SUBTASK 28-22-61-480-002

(7) Put covers on the open ends of the suction feed line [1] and suction feed line [2] to keep unwanted material out.

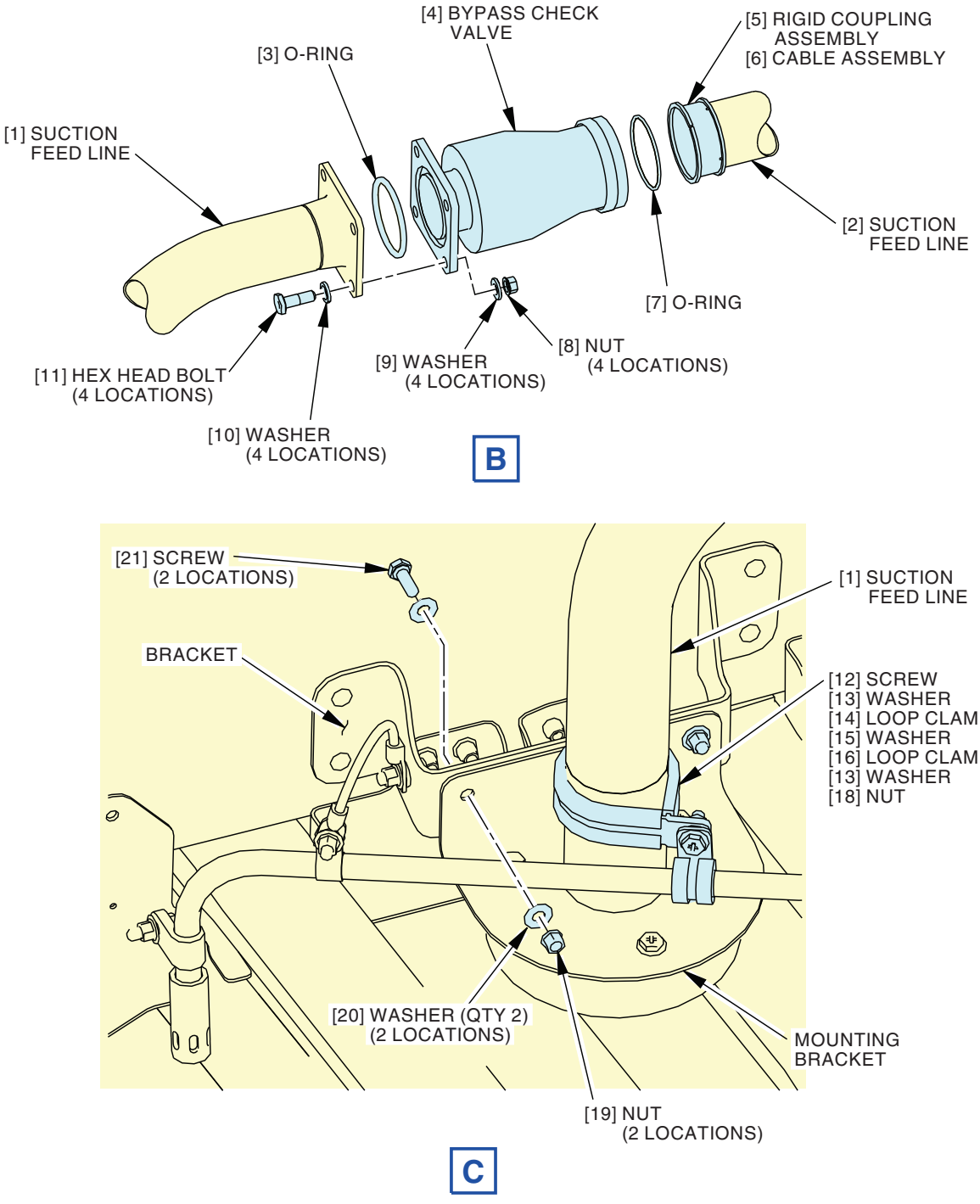
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Tank Fuel Boost Pump Bypass Check Valve Installation
Figure 401/28-22-61-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Main Tank Fuel Boost Pump Bypass Check Valve Installation
Figure 401/28-22-61-990-801 (Sheet 2 of 2)

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TASK 28-22-61-400-801

3. Main Tank Fuel Boost Pump Bypass Check Valve Installation

(Figure 401)

A. General

(1) The main tank fuel boost pump bypass check valve is referred to as bypass check valve in this task.

(2) This task gives instructions to install the bypass check valve.

B. References

Reference	Title
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
28-22-00-710-802	Engine Fuel Suction Feed - Operational Test (P/B 501)
28-22-15-400-803	Couplings Installation (P/B 401)
28-22-15-710-801	Engine Fuel Feed Manifold - Leak Test (P/B 601)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
D00504	Grease - Petrolatum	VV-P-236
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G51293	Cotton Wiper - Knit, Process Cleaning Absorbent Wiper	BMS15-5 Class B

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	O-ring	28-22-61-01-775	PTF ALL
		28-22-61-10-795	PTF ALL
4	Bypass check valve	28-22-61-01-770	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
4 (cont.)		28-22-61-10-790	PTF ALL
7	O-ring	28-22-61-01-555	PTF ALL
		28-22-61-10-560	PTF ALL

F. Location Zones

Zone	Area
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

G. Access Panels

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216
632AB	Main Tank Access Door - Wing Station 216

H. Prepare for the Installation

SUBTASK 28-22-61-080-001

(1) Remove the cover from the open end of the suction feed line [1].

SUBTASK 28-22-61-100-001

(2) Do these steps to prepare the flange on the bypass check valve [4], flange and mounting bracket on the suction feed line [1], and bracket for installation:

(a) Clean the fay surfaces of the aluminum flanges, aluminum brackets, and aluminum bypass check valve [4] (SWPM 20-20-00):

1) Use Cleaning Procedure 5 to clean the flange fay surface on the suction feed line [1] and the fasteners.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034, (alternate: knit cotton wiper, G51293) soaked with solvent, B00062 or solvent, B00184.

b) Use a new cotton wiper to dry the surfaces.

c) Continue to clean and dry the surface until the cotton wiper shows clean.

2) Use Cleaning Procedure 5 to clean the flange fay surface on the bypass check valve [4].

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a) Use a new cotton wiper, G00034, (alternate: knit cotton wiper, G51293) soaked with solvent, B00062 or solvent, B00184.

b) Use a new cotton wiper to dry the surfaces.

c) Continue to clean and dry the surface until the cotton wiper shows clean.

3) Use Cleaning Procedure 5 to clean the mounting bracket fay surface on the suction feed line [1].

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034, (alternate: knit cotton wiper, G51293) soaked with solvent, B00062 or solvent, B00184.

b) Use a new cotton wiper to dry the surfaces.

c) Continue to clean and dry the surface until the cotton wiper shows clean.

4) Use Cleaning Procedure 5 to clean the bracket fay surface that is attached to the rib.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034, (alternate: knit cotton wiper, G51293) soaked with solvent, B00062 or solvent, B00184.

b) Use a new cotton wiper to dry the surfaces.

c) Continue to clean and dry the surface until the cotton wiper shows clean.

SUBTASK 28-22-61-080-002

(3) Remove the cover from the open end of the suction feed line [2].

I. Bypass Check Valve Installation

SUBTASK 28-22-61-420-001

(1) Install the rigid coupling assembly [5] to secure the bypass check valve [4] to the suction feed line [2] (TASK 28-22-15-400-803).

(a) Install new O-ring [7].

SUBTASK 28-22-61-420-002

(2) Attach the bypass check valve [4] to the suction feed line [1] as follows:

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(a) Apply a thin layer of grease, D00504 to the new O-ring [3].

(b) Put the O-ring [3] in the groove of the bypass check valve [4].

(c) Align the bypass check valve [4] mounting holes with the suction feed line [1] mounting holes.

(d) Install the four hex head bolts [11], washers [10], and nuts [8] through the flanges to attach the bypass check valve [4] to the suction feed line [1].

(e) Tighten the four nuts [8] to 65 ±2 in-lb (7.3 ±0.3 N·m).

SUBTASK 28-22-61-420-003

(3) Install the two screws [21], four washers [20], and two nuts [19] to secure the mounting bracket on the suction feed line [1] to the bracket.

SUBTASK 28-22-61-420-004

(4) Tighten the two nuts [19] to 35 ±1 in-lb (4.0 ±0.2 N·m).

SUBTASK 28-22-61-860-001

(5) Measure the electrical bonding resistance between the flanges on the bypass check valve [4] and suction feed line [1] with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) Make sure that the electrical bonding resistance is 0.00550 ohm (5.50 milliohms) or less.

SUBTASK 28-22-61-860-002

(6) Measure the electrical bonding resistance between the mounting bracket on the suction feed line [1] and bracket with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) Make sure that the measured resistance is not more than the maximum bonding resistance value (SWPM 20-20-00).

SUBTASK 28-22-61-420-005

(7) Install the screw [12], washer [13], loop clamp [14], washer [15], loop clamp [16], washer [13], and nut [18] to secure the tube to the suction feed line [1].

J. Main Tank Fuel Boost Pump Bypass Check Valve Installation Test

SUBTASK 28-22-61-410-001

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Install these access doors:

(TASK 28-11-11-400-801)

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216
632AB	Main Tank Access Door - Wing Station 216

SUBTASK 28-22-61-710-001

(2) Do one of these tasks to test the bypass check valve [4] for leaks:

(a) Engine Fuel Feed Manifold - Leak Test, TASK 28-22-15-710-801.

(b) Engine Fuel Suction Feed - Operational Test, TASK 28-22-00-710-802.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL BOOST PUMP DISCHARGE CHECK VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Fuel Boost Pump Discharge Check Valve Removal

(2) Fuel Boost Pump Discharge Check Valve Installation.

TASK 28-22-71-000-801

2. Fuel Boost Pump Discharge Check Valve Removal

(Figure 401)

A. General

(1) The fuel boost pump discharge check valve is referred to as discharge check valve in this task.

(2) This task gives instructions to remove the discharge check valve.

B. References

Reference	Title
20-10-51-000-801	Flareless Tubing Assembly Removal (P/B 401)
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-22-15-000-803	Couplings Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-42-11-000-801	Fuel Boost Pump Fuel Pressure Switch Removal (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
631AB	Center Tank Access Door - Wing Station 168

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E. Prepare for the Removal

SUBTASK 28-22-71-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-22-71-650-001

(2) Defuel the center tank (TASK 28-26-00-650-801).

SUBTASK 28-22-71-860-001

(3) Defuel the No. 1 main tank to get access to the discharge check valves [17] in these locations (TASK 28-26-00-650-801):

(a) Left center fuel boost pump

(b) Forward or aft fuel boost pumps, on the left wing.

SUBTASK 28-22-71-650-002

(4) Defuel the No. 2 main tank to get access to the discharge check valves [17] in these locations (TASK 28-26-00-650-801):

(a) Right center fuel boost pump

(b) Forward or aft fuel boost pumps, on the right wing.

SUBTASK 28-22-71-860-002

(5) Set the CROSSFEED valve switch to the closed position on the P5-2 fuel system module.

(a) Make sure that the indicator light for the crossfeed valve position goes off.

SUBTASK 28-22-71-010-001

(6) To get access to the P91 and P2 panels, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-22-71-860-003

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

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(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(7) Open the applicable circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-71-940-001

WARNING

OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(8) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove the applicable access panels:

(TASK 28-11-31-000-801)

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
631AB	Center Tank Access Door - Wing Station 168

SUBTASK 28-22-71-940-002

WARNING

OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(9) Go into the fuel tank (TASK 28-11-00-910-802).

F. Discharge Check Valve Removal

SUBTASK 28-22-71-020-001

(1) Remove the elbow [1]:

NOTE: The forward fuel boost pump on the right wing does not have an elbow or scavenge pump pressure line.

(a) Disconnect the scavenge pump pressure line [2] from the elbow [1] on the discharge check valve [17] (TASK 20-10-51-000-801).

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(b) Remove and discard lockwire or safety cable between the rigid coupling assembly [13] and bulkhead nut [19], in these locations:

1) Forward fuel boost pump, left wing
2) Right center fuel boost pump, center tank
3) Forward fuel boost pump, right wing
4) Left center fuel boost pump, center tank.

(c) Remove and discard lockwire or safety cable between the bulkhead nut [19] and screw [4], in these locations:

1) Aft fuel boost pump, left wing
2) Aft fuel boost pump, right wing.

(d) Loosen the bulkhead nut [19].

(e) Remove the elbow [1].

(f) Remove the bulkhead nut [19] from the elbow [1].

(g) Remove and discard the O-ring [18].

SUBTASK 28-22-71-020-002

(2) Disconnect the pressure sensor line [9]:

(a) Remove and discard lockwire or safety cable between the rigid coupling assembly [13] and screw, on the removal check valve, in these locations:

1) Aft fuel boost pump, left wing
2) Aft fuel boost pump, right wing.

(b) Disconnect the rigid coupling assembly [13] (TASK 28-22-15-000-803).

1) Remove and discard the O-ring [10].

(c) Make sure that fuel has drained from the pressure sensor line [9].

NOTE: This prevents the collection of water in the pressure sensor line. Water can freeze and cause damage to the pressure sensor line. To prevent this type of damage, the pressure sensor line can be loosened at the pressure switch (TASK 28-42-11-000-801). A flush may be necessary to remove all water from the pressure sensor line.

SUBTASK 28-22-71-020-008

(3) Remove the screws [4], details [20], washers [6], and nuts [7] that attach the flange-adapter [3] to the discharge check valve [17] and pump discharge line.

SUBTASK 28-22-71-020-004

(4) Remove the screws [16], washers [15], and nuts [11] that attaches the pressure sensor plate adapter [14] and fuel boost pump housing [8] to the discharge check valve [17].

SUBTASK 28-22-71-020-005

(5) Remove the discharge check valve [17].

SUBTASK 28-22-71-020-006

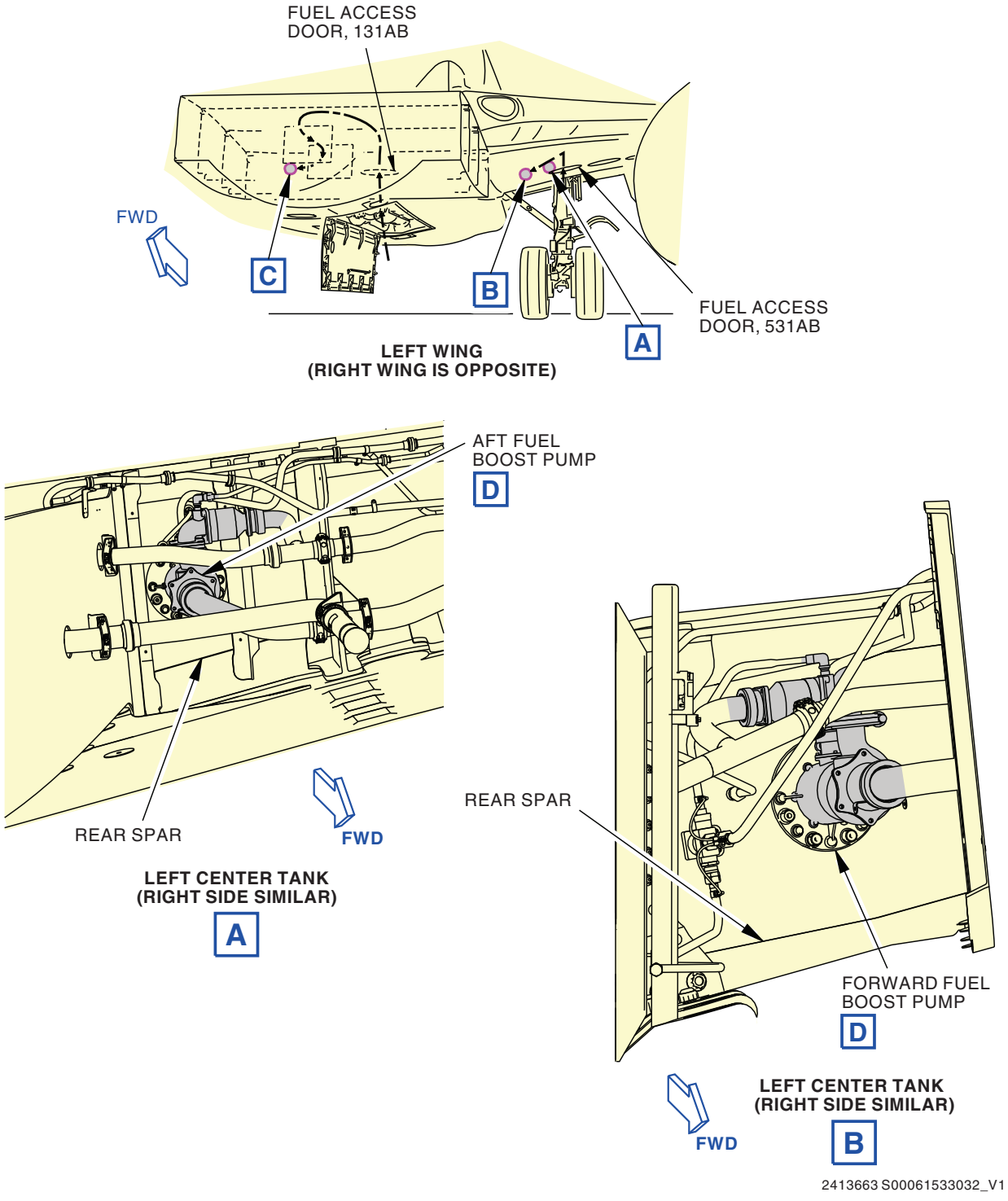
(6) Remove the pressure sensor plate adapter [14].

SUBTASK 28-22-71-020-007

(7) Remove and discard the O-rings [12] and O-ring [5].

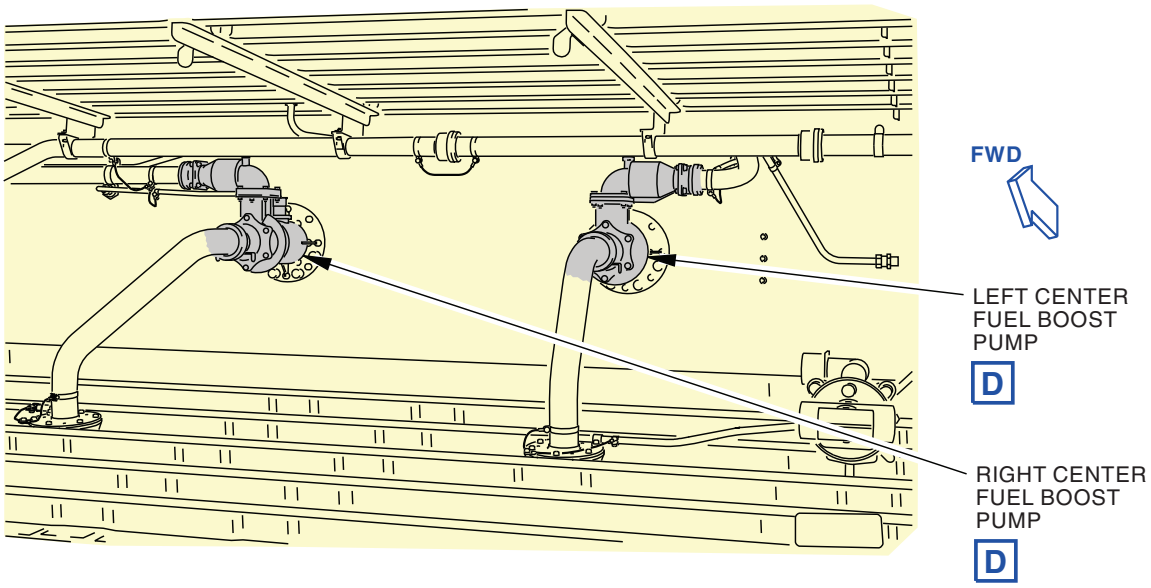
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Boost Pump Discharge Check Valve Installation
Figure 401/28-22-71-990-801 (Sheet 1 of 3)

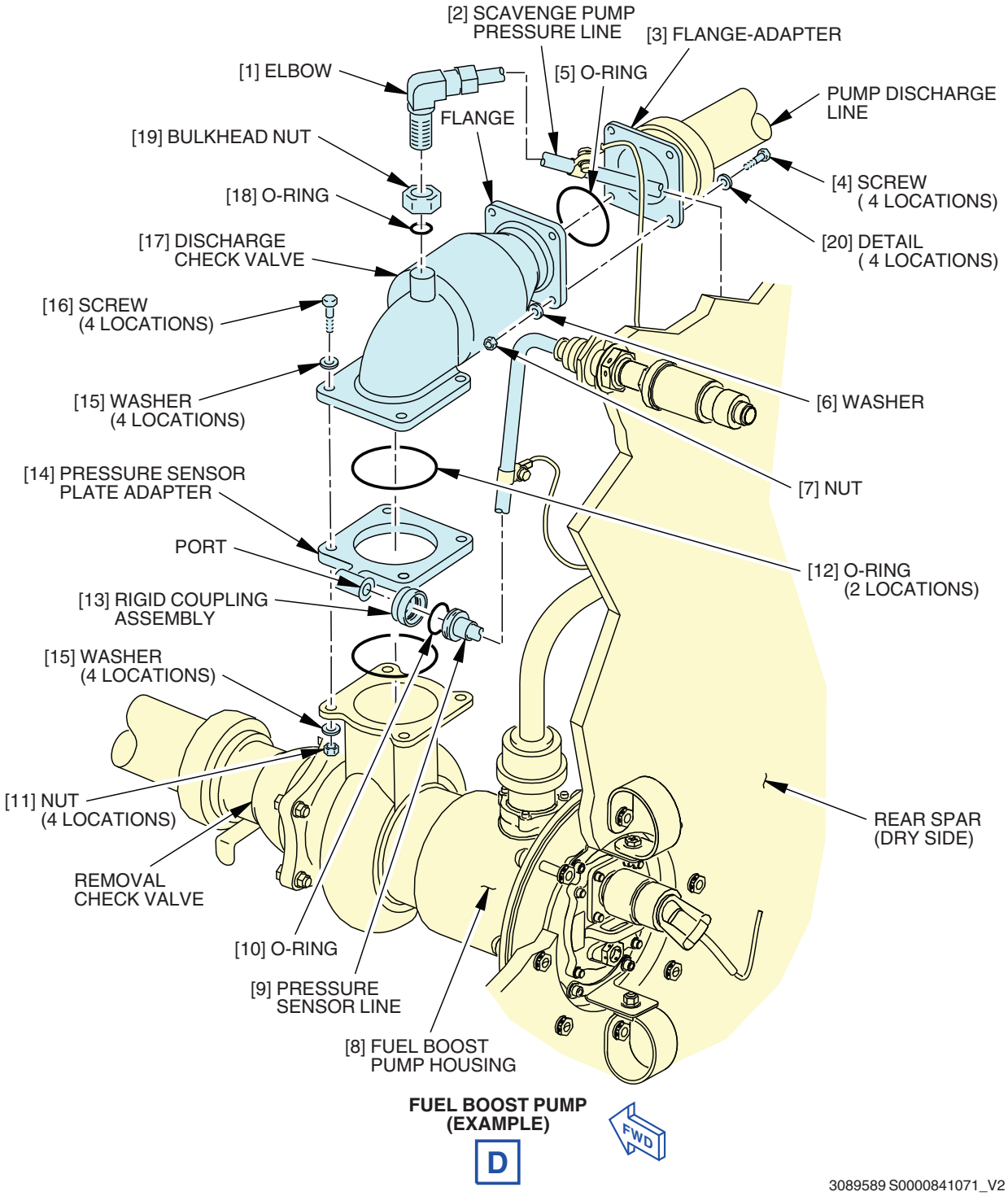
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Boost Pump Discharge Check Valve Installation
Figure 401/28-22-71-990-801 (Sheet 2 of 3)

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Boost Pump Discharge Check Valve Installation
Figure 401/28-22-71-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-22-71-400-801

3. Fuel Boost Pump Discharge Check Valve Installation

(Figure 401)

A. General

(1) This task gives instructions to install the discharge check valve.

(2) The fuel boost pump discharge check valve is referred to as discharge check valve in this task.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
20-10-44-400-802	Safety Cable Installation (P/B 401)
20-10-51-400-801	Flareless Tubing Assembly Installation (P/B 401)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
28-22-15-400-803	Couplings Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C50315	Coating - Chemical Conversion - Bonderite M-CR 600 Aero (Formerly Alodine 600)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51293	Cotton Wiper - Knit, Process Cleaning Absorbent Wiper	BMS15-5 Class B
G51372	Kit - Safety Cable, 321 CRES - 0.035 +/-0.003 Inch (1.0 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	O-ring	28-22-71-01-035	PTF ALL
		28-22-71-02-040	PTF ALL
		28-22-71-02-240	PTF ALL
10	O-ring	28-22-41-01-020	PTF ALL
		28-22-41-02-070	PTF ALL
12	O-ring	28-22-71-01-065	PTF ALL
		28-22-71-02-070	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
12 (cont.)		28-22-71-02-270	PTF ALL
17	Discharge check valve	28-22-71-01-060	PTF ALL
		28-22-71-02-065	PTF ALL
		28-22-71-02-265	PTF ALL
18	O-ring	28-22-13-01-110	PTF ALL
		28-22-71-02-100	PTF ALL
		28-22-71-02-295	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
631AB	Center Tank Access Door - Wing Station 168

G. Discharge Check Valve Installation

SUBTASK 28-22-71-100-001

(1) Do these steps to prepare the fuel boost pump housing [8], pressure sensor plate adapter [14], flange-adapter [3], and the discharge check valve [17] for installation:

(a) Clean the fay surfaces of the aluminum fuel boost pump housing [8], aluminum pressure sensor plate adapter [14], aluminum flange-adapter [3], and the aluminum discharge check valve [17] (SWPM 20-20-00):

1) Use Cleaning Procedure 1 to manually clean the discharge flange fay surface on the fuel boost pump housing [8].

a) Use an abrasive pad to clean the flange fay surface.

b) Move the pad in a circular movement while you press with your thumb.

2) Use Cleaning Procedure 1 to manually clean the fay surface on each side of the pressure sensor plate adapter [14].

a) Use an abrasive pad to clean the flange fay surface.

b) Move the pad in a circular movement while you press with your thumb.

NOTE: Do not clean the O-ring groove.

3) Use Cleaning Procedure 1 to manually clean the fay surfaces on the inlet and outlet flanges of the discharge check valve [17].

a) Use an abrasive pad to clean the flange fay surface.

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b) Move the pad in a circular movement while you press with your thumb.

NOTE: Do not clean the O-ring groove.

4) Use Cleaning Procedure 1 to manually clean the fay surface on the inlet flange of the flange-adapter [3].

a) Use an abrasive pad to clean the flange fay surface.

b) Move the pad in a circular movement while you press with your thumb.

5) Final Clean the fay surfaces of the fuel boost pump housing [8], pressure sensor plate adapter [14], flange-adapter [3], and discharge check valve [17].

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034, (alternate: knit cotton wiper, G51293) soaked with solvent, B00062, or solvent, B00184.

b) Rub dry with a clean cotton wiper.

c) Continue to clean and dry the surface until the cotton wiper stays clean.

WARNING

DO NOT GET ALODINE IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM ALODINE. PUT ON GOGGLES AND GLOVES WHEN YOU USE ALODINE. KEEP ALODINE AWAY FROM SPARKS, FLAME, AND HEAT. ALODINE IS POISONOUS AND FLAMMABLE, AND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Manually apply Bonderite M-CR 600 Aero coating, C50315, (alternate: coating, C00064) to the abraded surfaces on the pressure sensor plate adapter [14], discharge check valve [17], fuel boost pump housing [8], and the flange-adapter [3].

SUBTASK 28-22-71-420-001

(2) Install the discharge check valve [17]:

(a) Put the new O-ring [12] in the groove of the discharge check valve [17].

NOTE: Installed on the flange next to the pressure sensor plate adapter.

(b) Put the new O-ring [12] in the groove of the pressure sensor plate adapter [14].

(c) Put the new O-ring [5] in the groove of the discharge check valve [17].

NOTE: Installed on the flange next to the flange-adapter.

(d) Put the pressure sensor plate adapter [14] between the fuel boost pump housing [8] and the discharge check valve [17].

NOTE: The port faces the pressure sensor line.

(e) Align the discharge check valve [17] with the fuel boost pump housing [8] and the flange-adapter [3].

(f) Install the screws [16], washers [15], and nuts [11] to attach the pressure sensor plate adapter [14] and fuel boost pump housing [8] to the discharge check valve [17].

(g) Install the screws [4], details [20], washers [6], and nuts [7] to attach the flange-adapter [3] to the discharge check valve [17].

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SUBTASK 28-22-71-420-002

(3) Connect the scavenge pump pressure line [2] to the discharge check valve [17]:

NOTE: The forward fuel boost pump on the right wing does not have an elbow or scavenge pump pressure line.

(a) Install the bulkhead nut [19] on the elbow [1].

(b) Install the new O-ring [18] on the elbow [1].

(c) Loosely install the bulkhead nut [19] on the boss of the discharge check valve [17].

(d) Connect the scavenge pump pressure line [2] (TASK 20-10-51-400-801).

(e) Tighten the bulkhead nut [19].

SUBTASK 28-22-71-420-003

(4) Install the rigid coupling assembly [13] to attach the pressure sensor line [9] to the pressure sensor plate adapter [14] (TASK 28-22-15-400-803).

NOTE: Do not install the lockwire on the rigid coupling.

(a) Install the new O-ring [10].

SUBTASK 28-22-71-420-004

(5) Install MS20995NC32 lockwire, G01912, with the double-twist method, or safety cable kit, G51372, between the rigid coupling assembly [13] and bulkhead nut [19], in these locations (TASK 20-10-44-400-801 or TASK 20-10-44-400-802):

(a) Forward fuel boost pump, left wing

(b) Right center fuel boost pump, center tank

(c) Forward fuel boost pump, right wing

(d) Left center fuel boost pump, center tank.

SUBTASK 28-22-71-420-005

(6) Install MS20995NC32 lockwire, G01912, with the double-twist method, or safety cable kit, G51372, between the rigid coupling assembly [13] and screw, on the removal check valve, in these locations (TASK 20-10-44-400-801 or TASK 20-10-44-400-802):

(a) Aft fuel boost pump, left wing

(b) Aft fuel boost pump, right wing.

SUBTASK 28-22-71-420-006

(7) Install MS20995NC32 lockwire, G01912, with the double-twist method, or safety cable kit, G51372, between the bulkhead nut [19] and screw [4], in these locations (TASK 20-10-44-400-801 or TASK 20-10-44-400-802):

(a) Aft fuel boost pump, left wing

(b) Aft fuel boost pump, right wing.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 28-22-71-410-001

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Install the applicable access panels:

(TASK 28-11-31-400-801)

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
631AB	Center Tank Access Door - Wing Station 168

SUBTASK 28-22-71-860-004

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-22-71-410-002

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FUEL FEED SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has this task for the APU fuel feed system:

(1) APU Fuel Feed System - Operational Test

TASK 28-25-00-710-801

2. APU Fuel Feed System - Operational Test

(Figure 501, Figure 502)

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
551	Left Wing - Rear Spar To Landing Gear Support Beam

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 28-25-00-860-002

(1) On the P8-1 Engine and APU Fire Control Panel, make sure that the APU fire handle is in the NORMAL position.

SUBTASK 28-25-00-860-003

(2) On the P5 Overhead Panel, make sure that the APU master switch is in the OFF position.

SUBTASK 28-25-00-010-003

(3) To get access to EE bay, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-25-00-860-013

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-25-00-480-001

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(5) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

E. APU Fuel Feed System - Test

SUBTASK 28-25-00-860-004

(1) Make sure that the manual override handle on the actuator of the APU fuel shutoff valve is in the CLSD position (Figure 502).

NOTE: The actuator is on the rear spar in the main landing gear wheel well.

SUBTASK 28-25-00-860-005

(2) Make sure that these circuit breakers are closed:

Battery Shield, J9

Row	Col	Number	Name
A	5	C01340	BATTERY BUS

Standby Power Control Unit, M01720

Row	Col	Number	Name
B	4	C00169	SW HOT BAT BUS

SUBTASK 28-25-00-860-006

(3) On the P5 Overhead Panel, set the APU master switch to the ON position.

NOTE: The APU will start if the APU master switch is set to the START position.

SUBTASK 28-25-00-210-001

(4) Make sure that the manual override handle on the actuator of the APU fuel shutoff valve moves to the OPEN position in less than 10 seconds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-25-00-860-007

(5) On the P8-1 Engine and APU Fire Control Panel, do these steps:

(a) Under the APU fire handle, push the applicable manual override button.

CAUTION

DO NOT TURN THE FIRE HANDLE. YOU WILL CAUSE THE FIRE BOTTLES TO RELEASE THEIR CONTENTS IF YOU TURN THE HANDLE.

(b) Pull the APU fire handle.

(c) Make sure that the manual override handle on the APU fuel shutoff valve moves to the CLSD position in less than 30 seconds.

(d) On the P5-4 AC System, Generator and APU Module, make sure that the APU FAULT light comes on to show the fire shutdown.

CAUTION

DO NOT TURN THE FIRE HANDLE. YOU WILL CAUSE THE FIRE BOTTLES TO RELEASE THEIR CONTENTS IF YOU TURN THE HANDLE.

(e) Push the APU fire handle to the normal position.

SUBTASK 28-25-00-860-009

(6) Set the APU master switch to the OFF position and then set the APU master switch to ON position.

NOTE: The APU will start if the APU master switch is set to the START position.

SUBTASK 28-25-00-210-003

(7) Make sure that the manual override handle on the actuator of the APU fuel shutoff valve moves to the OPEN position in less than 10 seconds.

SUBTASK 28-25-00-210-002

(8) Make sure that the APU FAULT light goes off.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 28-25-00-860-008

(1) On the P5 Overhead Panel, set the APU master switch to the OFF position.

SUBTASK 28-25-00-210-004

(2) Make sure that the manual override handle on the APU fuel shutoff valve moves to the CLSD position in less than 30 seconds.

SUBTASK 28-25-00-860-017

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

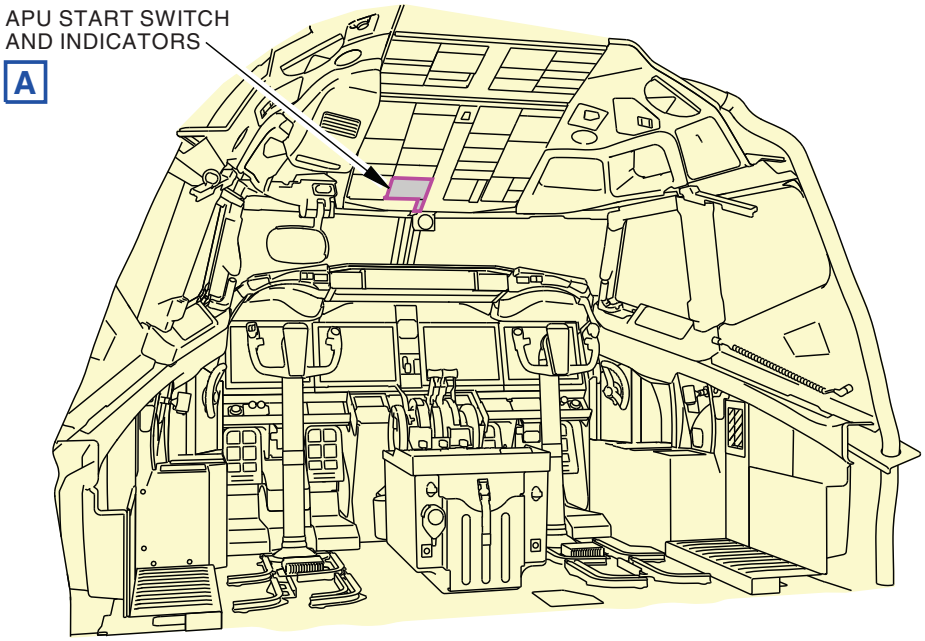
SUBTASK 28-25-00-410-003

(4) Close this access panel:

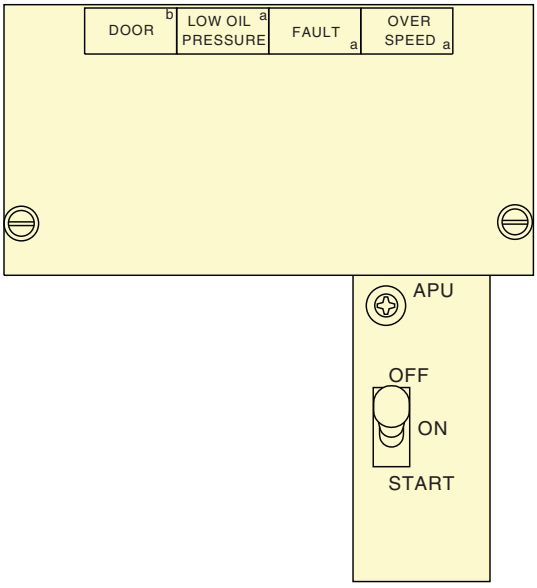
Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

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FLIGHT COMPARTMENT



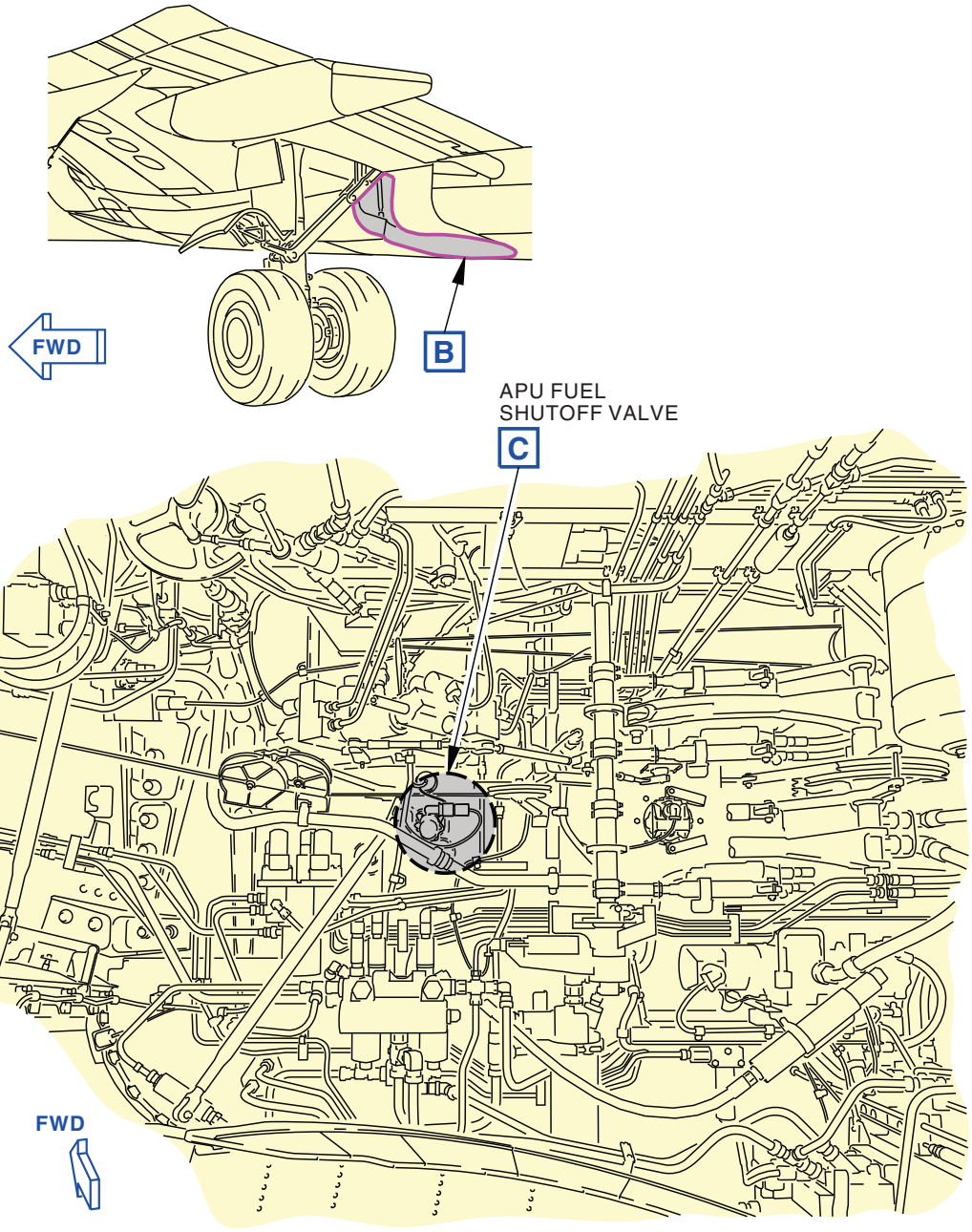
APU START SWITCH AND INDICATORS

A

2413668 S00061533050_V2

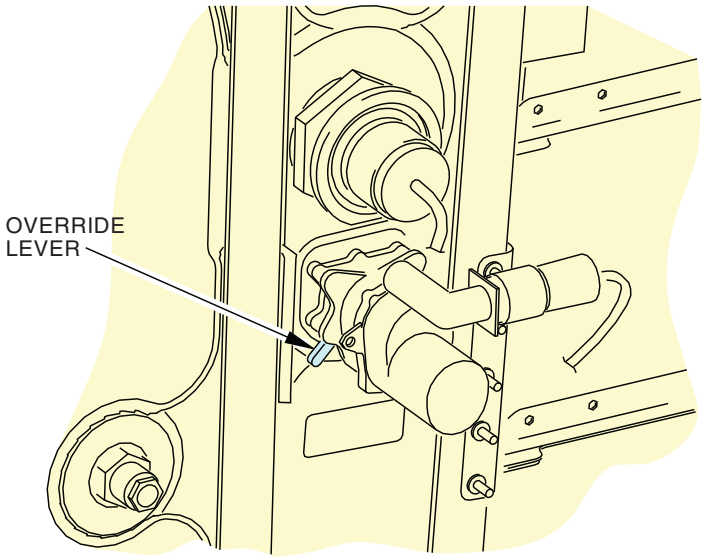
APU Fuel Feed System
Figure 501/28-25-00-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Fuel Feed System Test
Figure 502/28-25-00-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU FUEL SHUTOFF VALVE

C

2413670 S00061533052_V1

APU Fuel Feed System Test
Figure 502/28-25-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FUEL SHUTOFF VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) APU Fuel Shutoff Valve Actuator Assembly Removal

(2) APU Fuel Shutoff Valve Actuator Assembly Installation

(3) APU Shutoff Valve Operational Test

(4) APU Fuel Shutoff Valve Body Assembly Removal

(5) APU Fuel Shutoff Valve Body Assembly Installation.

B. The APU shutoff valve is installed on the rear spar in the main landing gear wheel well. It has these primary assemblies:

(1) APU shutoff valve actuator assembly

(2) APU shutoff valve body assembly

(3) Mounting adapter.

C. The APU shutoff valve actuator, V43, is also referred to as the APU fuel shutoff valve.

D. The APU fuel shutoff valve actuator assembly can be replaced without defueling the center tank. These procedures are necessary to replace the APU shutoff valve body:

(1) Removal of the APU shutoff valve actuator.

(2) Removal of the mounting adapter.

(3) Defueling the center tank.

TASK 28-25-02-000-801

2. APU Shutoff Valve Actuator Assembly Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives instructions to remove the Auxiliary Power Unit (APU) fuel shutoff valve actuator assembly.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

D. Prepare for the Removal

SUBTASK 28-25-02-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 28-25-02-480-001

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

E. APU Shutoff Valve Actuator Assembly Removal

SUBTASK 28-25-02-020-008

(1) Remove the lockwire from the four actuator mounting screws [3].

SUBTASK 28-25-02-020-001

(2) Disconnect the electrical connector [1] from the APU fuel shutoff valve actuator [2] (Figure 401).

SUBTASK 28-25-02-020-002

(3) Remove the screw [6], three washers [7], bonding jumper [9], and nut [8] from the APU fuel shutoff valve actuator [2].

SUBTASK 28-25-02-020-003

CAUTION

HOLD THE ACTUATOR ASSEMBLY UP TO KEEP ITS WEIGHT OFF THE SHAFT. ITS WEIGHT CAN CAUSE DAMAGE TO THE SHAFT.

(4) Hold the APU fuel shutoff valve actuator [2] and then remove the four actuator mounting screws [3].

SUBTASK 28-25-02-020-004

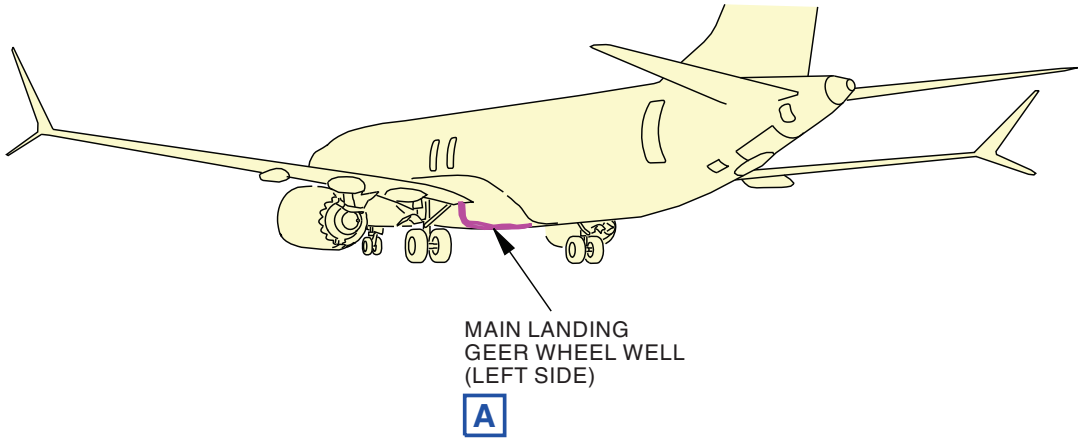
(5) Remove the APU fuel shutoff valve actuator [2].

(a) Carefully move the APU fuel shutoff valve actuator [2] away from the mounting adapter [5] until the output shaft is out of the mating part.

(b) Lift the APU fuel shutoff valve actuator [2] out of the airplane.

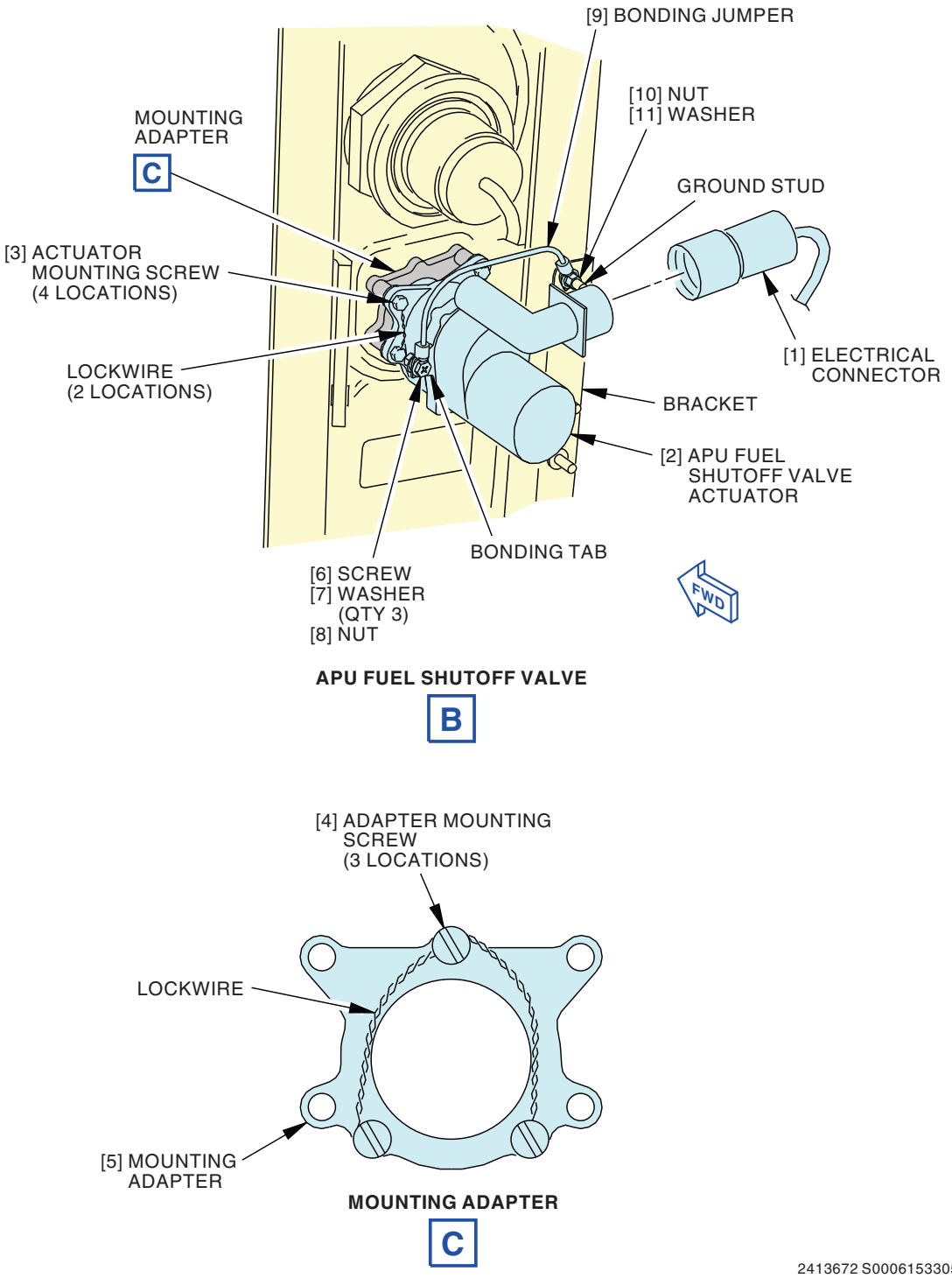
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Fuel Shutoff Valve Actuator Installation
Figure 401/28-25-02-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Fuel Shutoff Valve Actuator Installation
Figure 401/28-25-02-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-25-02-400-801

3. APU Shutoff Valve Actuator Assembly Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives instructions to install the Auxiliary Power Unit (APU) fuel shutoff valve actuator assembly.

(2) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
D00016	Grease - Aircraft, General Purpose, Wide Temperature Range	MIL-PRF-81322
G01505	Lockwire - Safety And Lock	NASM20995

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	APU fuel shutoff valve actuator	28-25-02-01-045	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

G. Prepare for the Installation

SUBTASK 28-25-02-100-003

(1) If the bonding jumper [9] is connected to the bracket on the stiffener, do these steps:

28-AWL-36: CDCCL

(a) Measure the electrical bonding resistance between the loose end of the bonding jumper [9] and the bracket on the stiffener (SWPM 20-20-10).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on CDCCLs.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

1) Use an intrinsically safe approved bonding meter, COM-1550 to measure the electrical bonding resistance.

28-AWL-36: CDCCL

2) Make sure that the electrical bonding resistance is 0.0020 ohm (2.0 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on CDCCLs.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

SUBTASK 28-25-02-100-001

(2) If the bonding jumper [9] is disconnected from the bracket on the stiffener, do these steps:

28-AWL-36: CDCCL

(a) Clean the fay surfaces on the terminal of the bonding jumper [9], fasteners, and the aluminum bracket on the stiffener (SWPM 20-20-10).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on CDCCLs.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

1) Do the solvent cleaning procedure for the surfaces of the bonding jumper [9], nut [10], washer [11], and the ground stud and bracket on the stiffener (SWPM 20-20-10).

28-AWL-36: CDCCL

(b) Apply sealant on the shank and threads on the ground stud.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on CDCCLs.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

1) Use sealant, A50110 or sealant, A50155.

28-AWL-36: CDCCL

(c) Apply sealant on the washer [11] and the terminal on the bonding jumper [9].

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on CDCCLs.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Use sealant, A50110 or sealant, A50155.

(d) Install the bonding jumper [9], washer [11], and nut [10] to the ground stud on the bracket (Figure 401).

(e) Tighten the nut [10] to 31.5 ±3.5 in-lb (3.6 ±0.4 N·m).

28-AWL-36: CDCCL

(f) Measure the electrical bonding resistance between the loose end of the bonding jumper [9] and the bracket on the stiffener (SWPM 20-20-10).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on CDCCLs.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

1) Use an intrinsically safe approved bonding meter, COM-1550 to measure the electrical bonding resistance.

28-AWL-36: CDCCL

2) Make sure that the electrical bonding resistance is 0.0015 ohm (1.5 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on CDCCLs.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

SUBTASK 28-25-02-940-001

(3) Apply a thin layer of the grease, D00016, to the keyway of the output shaft.

H. APU Shutoff Valve Actuator Assembly Installation

SUBTASK 28-25-02-420-001

CAUTION

HOLD THE ACTUATOR ASSEMBLY UP TO KEEP ITS WEIGHT OFF THE SHAFT. ITS WEIGHT CAN CAUSE DAMAGE TO THE SHAFT.

(1) Put the output shaft into the input shaft.

(a) Turn the output shaft until the keyway is aligned with the key in the input shaft.

(b) Carefully move the output shaft into the input shaft until the actuator mounting lugs touch the mounting adapter [5].

SUBTASK 28-25-02-420-002

(2) If the mounting holes on the APU fuel shutoff valve actuator [2] are not aligned with the mating holes, carefully turn the assembly to align the holes.

SUBTASK 28-25-02-420-003

(3) Install actuator mounting screws [3] on the APU fuel shutoff valve actuator [2].

(a) Tighten the actuator mounting screws [3] to 17.5 ±2.5 in-lb (2.0 ±0.3 N·m).

(b) Install lockwire, G01505, on each of the actuator mounting screws [3] (TASK 20-10-44-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-25-02-100-002

28-AWL-36: CDCCL

(4) Clean the fay surfaces on the terminal of the bonding jumper [9], fasteners, and the aluminum bonding tab on the APU fuel shutoff valve actuator [2] (SWPM 20-20-10).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on CDCCLs.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

(a) Do the abrasive cleaning procedure for the surfaces on the bonding tab or the APU fuel shutoff valve actuator [2] (SWPM 20-20-10).

(b) Do the solvent cleaning procedure for the surfaces of the bonding jumper [9], screw [6], three washers [7], and nut [8] (SWPM 20-20-10).

(c) Do the solvent cleaning procedure for the surfaces on the bonding tab of the APU fuel shutoff valve actuator [2] (SWPM 20-20-10).

SUBTASK 28-25-02-420-004

(5) Install the screw [6], three washers [7], and nut [8] to attach the bonding jumper [9] to the bonding tab on the APU fuel shutoff valve actuator [2].

SUBTASK 28-25-02-280-001

28-AWL-36: CDCCL

(6) Measure the electrical bonding resistance between the housing on the APU fuel shutoff valve actuator [2] and the terminal on the bonding jumper [9] (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on CDCCLs.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

(a) Use an intrinsically safe approved bonding meter, COM-1550 to measure the electrical bonding resistance.

28-AWL-36: CDCCL

(b) Make sure that the bonding resistance is 0.0010 ohm (1.0 milliohms) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on CDCCLs.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

SUBTASK 28-25-02-390-004

28-AWL-36: CDCCL

(7) Apply a fillet seal to the screw [6], three washers [7], nut [8], and the bonding jumper [9] on the APU fuel shutoff valve actuator [2].

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on CDCCLs.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

(a) Use sealant, A50110 or sealant, A50155.

SUBTASK 28-25-02-420-005

(8) Connect the electrical connector [1] to the APU fuel shutoff valve actuator [2].

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I. APU Shutoff Valve Actuator Assembly Installation Test

SUBTASK 28-25-02-860-002

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 28-25-02-710-001

(2) Do this task: APU Shutoff Valve Operational Test, TASK 28-25-02-710-801.

——— END OF TASK ———

TASK 28-25-02-710-801

4. APU Shutoff Valve Operational Test

(Figure 401)

A. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

B. APU Shutoff Valve Test

SUBTASK 28-25-02-860-003

(1) Make sure that these circuit breakers are closed:

Battery Shield, J9

Row	Col	Number	Name
A	5	C01340	BATTERY BUS

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

Standby Power Control Unit, M01720

Row	Col	Number	Name
B	4	C00169	SW HOT BAT BUS

SUBTASK 28-25-02-860-004

(2) On the P5-13 Electrical Meters, Battery, Cabin Utility and Inflight Entertainment Power Module, set the DC BAT switch to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-25-02-710-002

(3) On the P5 Overhead Panel, set the APU master switch to the ON position.

(a) Make sure the manual override handle on the APU fuel shutoff valve moves to the OPEN position.

SUBTASK 28-25-02-710-003

(4) Set the APU master switch to the OFF position.

(a) Make sure the manual override handle on the APU fuel shutoff valve moves to the CLOSED position in less than of 10 seconds.

SUBTASK 28-25-02-860-005

(5) Set the DC BAT switch to the OFF position.

——— END OF TASK ———

TASK 28-25-02-000-802

5. APU Shutoff Valve Body Assembly Removal

(Figure 401, Figure 402)

A. General

(1) This task gives instructions to remove the APU shutoff valve body assembly.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-22-00-730-802	Crossfeed Valve - Functional Test (P/B 501)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-26-00-650-802	Tank to Tank Fuel Transfer (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1054	Container - Fuel Resistant, 5-Gallon (19-Liter)

D. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

E. Access Panels

Number	Name/Location
131AB	Center Tank Access

F. Prepare for the Removal

SUBTASK 28-25-02-650-001

(1) Defuel the center fuel tank and the No.1 or transfer fuel out the tanks (Fuel Tank Defueling, TASK 28-26-00-650-801 or Tank to Tank Fuel Transfer, TASK 28-26-00-650-802).

SUBTASK 28-25-02-650-004

(2) If there is fuel in the No.2 tank, make sure that the crossfeed valve is closed and has no leakage (Crossfeed Valve - Functional Test, TASK 28-22-00-730-802).

(a) As an alternative, you can remove fuel from the No.2 tank (Fuel Tank Defueling, TASK 28-26-00-650-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-25-02-650-002

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(3) Do this task: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

SUBTASK 28-25-02-010-001

(4) Get access to the wing rear spar.

(a) Remove this access panel to get access to the center fuel tank:

(TASK 28-11-31-000-801)

Number	Name/Location
131AB	Center Tank Access

G. APU Shutoff Valve Body Assembly Removal

SUBTASK 28-25-02-020-005

(1) Do this task: APU Shutoff Valve Actuator Assembly Removal, TASK 28-25-02-000-801.

SUBTASK 28-25-02-020-006

(2) Put the 5-gallon (19-liter) fuel resistant container, STD-1054, below the APU shutoff valve body [23] and disconnect the APU fuel line from the unions [21].

SUBTASK 28-25-02-020-007

CAUTION DO NOT LET THE VALVE BODY ASSEMBLY FALL. YOU WILL CAUSE DAMAGE TO THE FUEL TANK SEAL.

(3) Do these steps to remove the APU shutoff valve body [23]:

(a) Remove sealant around the APU shutoff valve body [23] and the spar web.

(b) Two persons are necessary to do this task.

1) One person must be out of the tank.

2) One person must be in the tank.

(c) The person in the tank must hold the APU shutoff valve body [23] assembly up.

(d) The person out of the tank must remove the adapter mounting adapter mounting screws [4] that attach the mounting adapter [5] to the APU shutoff valve body [23].

(e) Move the APU shutoff valve body [23] away from the rear spar.

(f) Remove the mounting adapter [5].

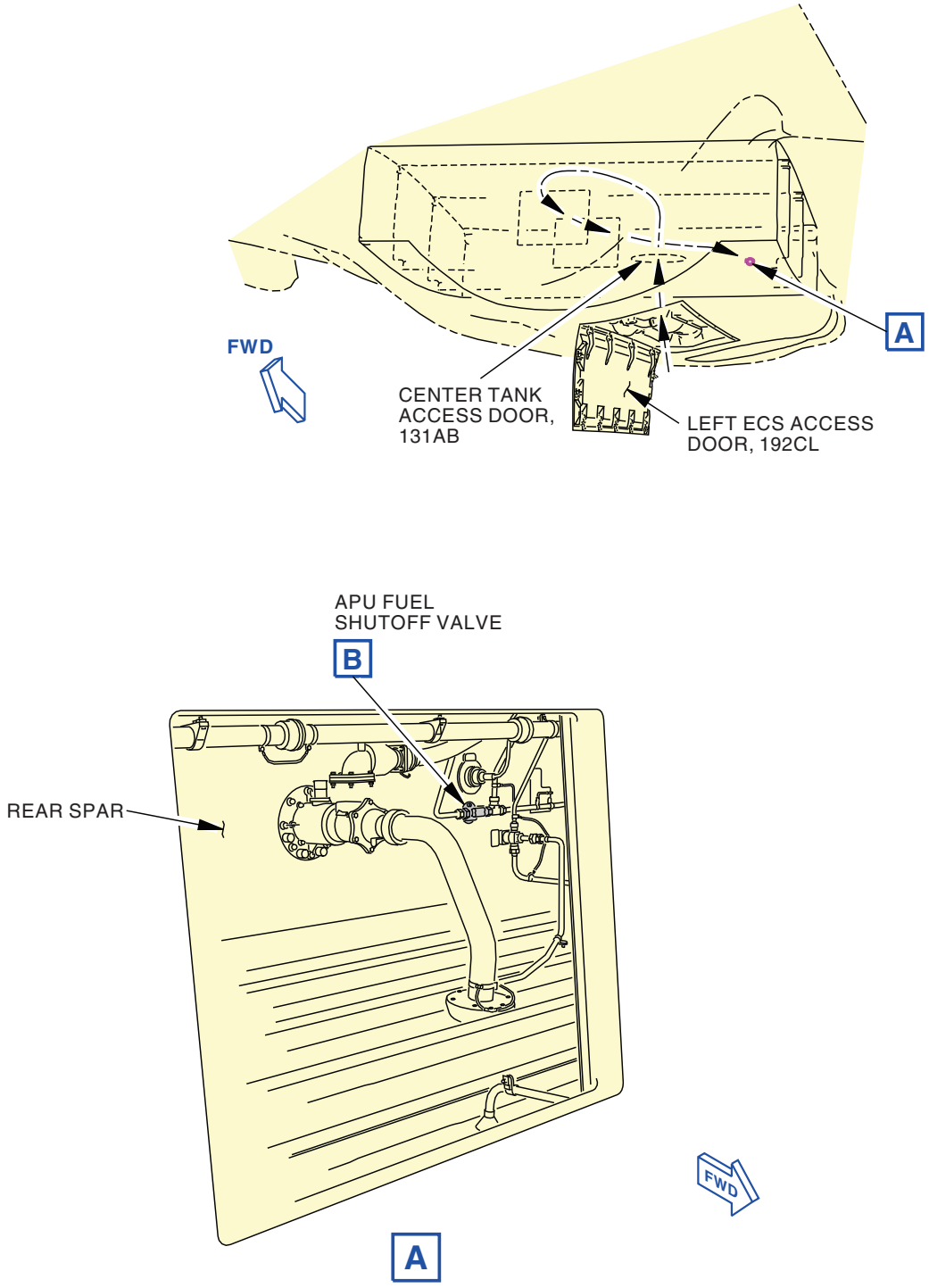
(g) Remove the unions [21] from the APU shutoff valve body [23].

(h) Remove and discard the O-rings [22].

(i) Remove and discard the O-ring [24] from the APU shutoff valve body [23].

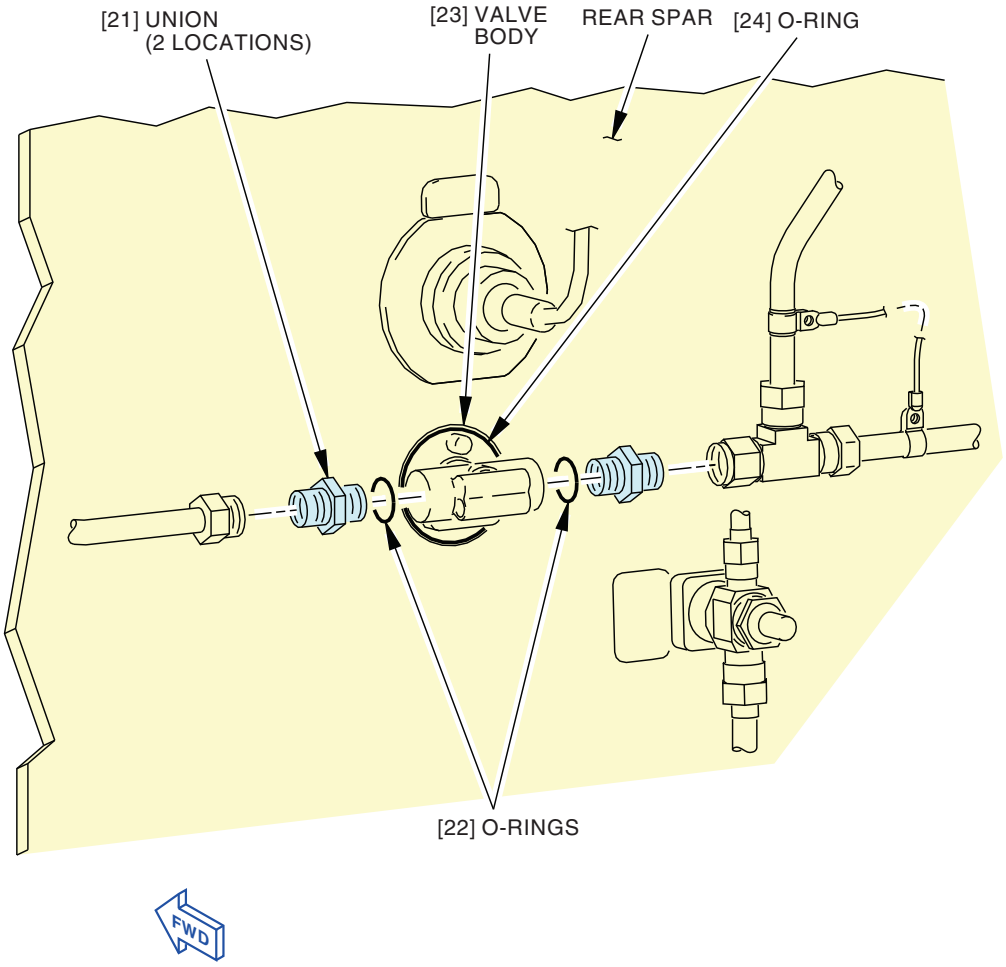
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Fuel Shutoff Valve Body Installation
Figure 402/28-25-02-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU FUEL SHUTOFF VALVE

B

2413674 S00061533062_V2

APU Fuel Shutoff Valve Body Installation
Figure 402/28-25-02-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-25-02-400-802

6. APU Shutoff Valve Body Assembly Installation

(Figure 401, Figure 402)

A. General

(1) This task gives instructions to install the APU fuel shutoff valve actuator assembly.

(2) One person out of the tank and one person in the tank are necessary for this task.

(3) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C50315	Coating - Chemical Conversion - Bonderite M-CR 600 Aero (Formerly Alodine 600)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
D00504	Grease - Petrolatum	VV-P-236
G01505	Lockwire - Safety And Lock	NASM20995

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(Continued)

Reference	Description	Specification
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
22	O-rings	28-25-04-01-095	PTF ALL
23	APU shutoff valve body	28-25-02-01-130	PTF ALL
24	O-ring	28-25-02-01-135	PTF ALL

F. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

G. Access Panels

Number	Name/Location
131AB	Center Tank Access

H. Prepare for the Installation

SUBTASK 28-25-02-160-001

28-AWL-36: CDCCL

(1) Prepare a fay surface bond between the APU shutoff valve body [23] and the rear spar inside the tank (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

SUBTASK 28-25-02-370-001

(2) Manually apply coating, C00064 or Bonderite M-CR 600 Aero coating, C50315 to the abraded areas on the rear spar and APU shutoff valve body [23].

SUBTASK 28-25-02-390-002

(3) Install new O-rings [22], lightly lubricated with fuel, on the unions [21].

SUBTASK 28-25-02-420-013

(4) Install the unions [21] in the APU shutoff valve body [23].

SUBTASK 28-25-02-640-001

(5) Install the new O-ring [24], lightly lubricated with grease, D00504, in the groove on the APU shutoff valve body [23].

I. APU Shutoff Valve Body Assembly Installation

SUBTASK 28-25-02-420-008

CAUTION

DO NOT LOOSEN THE O-RING OR PUSH IT OUT OF ITS CORRECT SHAPE. IF YOU CAUSE DAMAGE TO THE O-RING, A FUEL LEAK CAN EASILY OCCUR.

(1) Install the APU shutoff valve body [23], as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) The person in the tank must put the APU shutoff valve body [23] in the correct position between the fittings on the APU line.

(b) The person out of the tank must hold the APU shutoff valve body [23] up until it is installed.

(c) From outside the tank, install the mounting adapter [5].

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

1) Install the three adapter mounting screws [4] wet with Cor-Ban 27L Compound, G50237 , that attach the mounting adapter [5] to the APU shutoff valve body [23].

2) Tighten the three adapter mounting screws [4] to 30 ±5 in-lb (3 ±1 N·m).

SUBTASK 28-25-02-860-006

28-AWL-36: CDCCL

(2) From inside the tank, measure the electrical bonding resistance between the APU shutoff valve body [23] and the rear spar (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

(a) Use an intrinsically safe approved bonding meter, COM-1550 to measure the electrical bonding resistance.

28-AWL-36: CDCCL

(b) Make sure that the bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

SUBTASK 28-25-02-420-011

(3) From outside the tank, install the lockwire, G01505, on each of the adapter mounting screws [4] (TASK 20-10-44-400-801).

SUBTASK 28-25-02-420-009

(4) Connect the APU fuel line to the APU shutoff valve body [23].

SUBTASK 28-25-02-390-001

28-AWL-36: CDCCL

(5) Apply a fillet seal between the flange of the APU shutoff valve body [23] and the rear spar, in the tank.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-36.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that you apply sufficient sealant to cover the unprimed areas on the rear spar.

SUBTASK 28-25-02-390-003

(6) Apply a fillet seal of sealant, A02315, between the flange of the mounting adapter [5] and rear spar.

SUBTASK 28-25-02-410-002

(7) Install this access panel:

(TASK 28-11-31-400-801)

Number	Name/Location
131AB	Center Tank Access

SUBTASK 28-25-02-420-010

(8) Do this task: APU Shutoff Valve Actuator Assembly Installation, TASK 28-25-02-400-801.

J. APU Shutoff Valve Body Assembly Installation Test

SUBTASK 28-25-02-710-004

(1) If you did not do it as part of the actuator installation procedure, do this task: APU Shutoff Valve Operational Test, TASK 28-25-02-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FUEL SHUTOFF VALVE - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) A functional check of the bonding resistance between the structure and the Auxiliary Power Unit (APU) fuel shutoff valve actuator.

TASK 28-25-02-200-801

2. APU Fueling Shutoff Valve Actuator - Bonding Resistance Check

(Figure 601)

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-10	Replacement of Ground Studs and Bonding Jumper Installation

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

D. Prepare for the Check

SUBTASK 28-25-02-865-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-25-02-480-002

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

E. Electrical Bonding Measurement

SUBTASK 28-25-02-765-001

(1) Measure the electrical bonding resistance between the APU fuel shutoff valve actuator [1] (at the jumper bonding lug) and the spar (SWPM 20-20-00 and SWPM 20-20-10).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the resistance is 0.0010 ohm (1 milliohms) or less.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 28-25-02-865-002

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

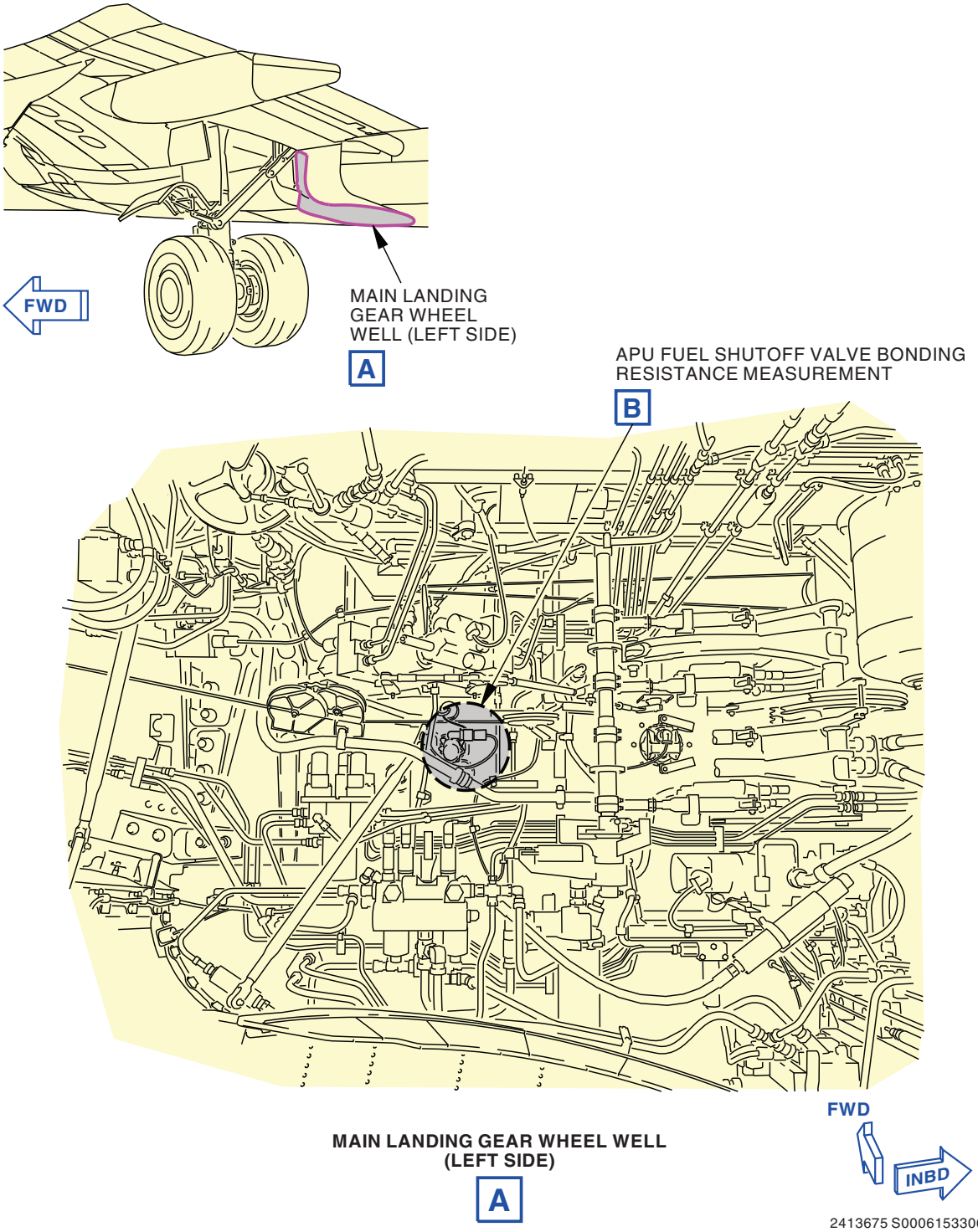
Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

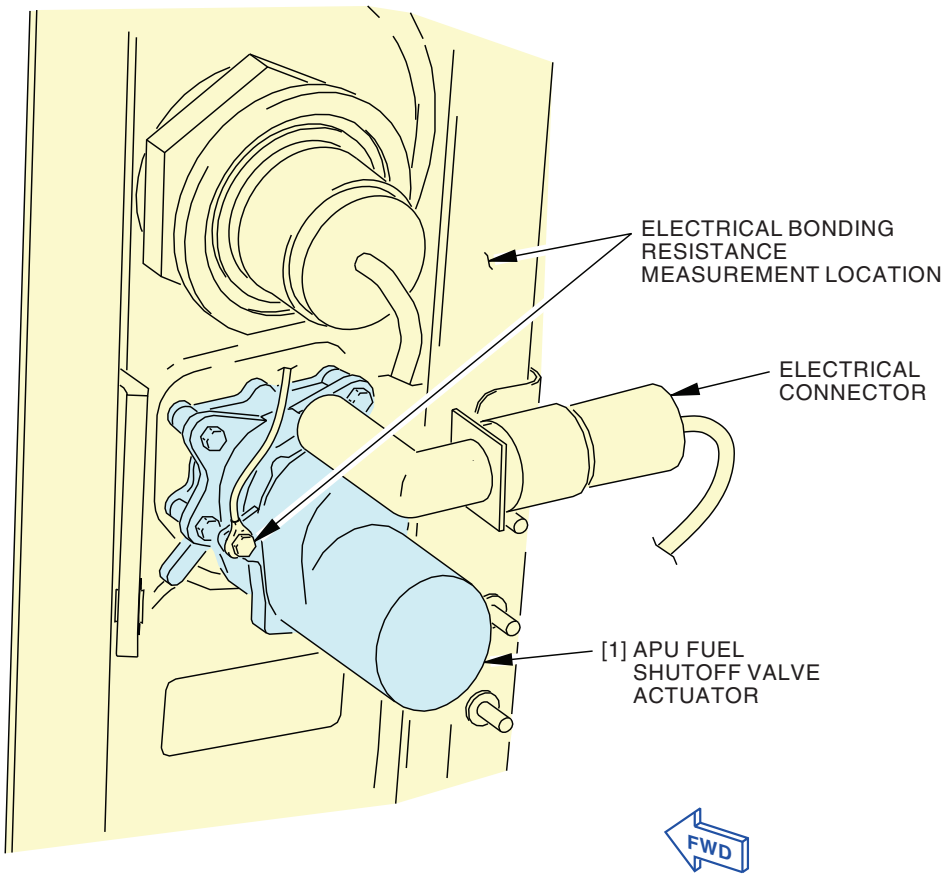
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Fueling Shutoff Valve Actuator - Bonding Resistance Check
Figure 601/28-25-02-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**APU FUEL SHUTOFF VALVE
BONDING RESISTANCE MEASUREMENT**

B

2413676 S00061533067_V1

**APU Fueling Shutoff Valve Actuator - Bonding Resistance Check
Figure 601/28-25-02-990-803 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FUEL FEED LINE - REMOVAL/INSTALLATION

1. General

A. The APU fuel feed line supplies fuel to the APU from the engine fuel feed line. If the engine fuel feed pumps are not operating, the APU fuel line supplies fuel to the APU from the suction inlet in the No. 1 tank.

B. The APU fuel feed line is referred to as the APU fuel line in this procedure.

C. To make the removal and the installation easier, the APU fuel line is divided into two parts.

(1) The part that goes from the bypass inlet in the No. 1 tank to the top skin of the wing center section.

NOTE: This part of the APU fuel line has the connection with the engine fuel feed line.

(2) The part that goes from the top skin of the wing center section to the APU firewall.

NOTE: This part of the APU fuel line has a shroud.

TASK 28-25-04-000-801

2. APU Fuel Line (No. 1 Tank and Center Tank) Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the APU fuel feed line [1A], APU fuel feed line [1B], APU fuel feed line [1C], APU fuel feed line [1D], APU fuel feed line [1E], APU fuel feed line [1F], APU fuel feed line [1G], and APU fuel feed line [1H] for No. 1 tank and center tank.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-22-15-000-801	Fuel Line, Fitting and Coupling Removal (P/B 401)
28-22-15-000-803	Couplings Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-26-00-650-802	Tank to Tank Fuel Transfer (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1054	Container - Fuel Resistant, 5-Gallon (19-Liter)

D. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50

E. Access Panels

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
531BB	Center Tank Access Door - Wing Station 192

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(Continued)

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216

F. Prepare for the Removal

SUBTASK 28-25-04-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-25-04-860-002

WARNING DO ALL THE SAFETY PROCEDURES TO DEFUEL THE TANK AND TO GO INTO IT. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR IF YOU DO NOT FOLLOW THE SAFETY PROCEDURES.

(2) Defuel the No. 1 tank and center tank or transfer all the fuel from the No. 1 tank and the center tank (TASK 28-26-00-650-801 or TASK 28-26-00-650-802).

SUBTASK 28-25-04-010-001

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) For the APU fuel lines in the main tank, remove this access door:

(TASK 28-11-11-000-801)

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216

(b) For the APU fuel lines in the center tank, remove these access doors:

(TASK 28-11-31-000-801)

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
531BB	Center Tank Access Door - Wing Station 192

G. APU Fuel Line (No. 1 Tank and Center Tank) Removal

SUBTASK 28-25-04-020-018

(1) Remove the APU fuel feed line [1A] (View A, Figure 401).

(a) Remove the screws [34], washers [35], and nuts [36] to disconnect the bonding jumpers [30] from the APU fuel feed line [1A] (View K, Figure 401).

(b) Remove the screws [32] and washers [33] that hold the APU fuel feed line [1A] in the clamps [27] (TASK 28-22-15-000-801) (View J, Figure 401).

(c) Disconnect the APU fuel feed line [1A] from the bulkhead fitting (View H, Figure 401).

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1) Remove the union [38].

2) Remove and discard the O-ring [39].

(d) Disconnect the APU fuel feed line [1A] from the APU shutoff valve [5] (View G, Figure 401).

1) Remove the union [8].

2) Remove and discard the O-ring [9].

(e) Drain the remaining fuel from the APU fuel feed lines into the 5-gallon (19-liter) fuel resistant container, STD-1054.

SUBTASK 28-25-04-020-013

(2) Remove the APU fuel feed line [1B] (View A, Figure 401).

(a) Remove the screw [34], washers [35], and nut [36] to disconnect the bonding jumper [30] from the APU fuel feed line [1B] (View K, Figure 401).

(b) Disconnect the APU fuel feed line [1B] from the tee [31] (View G, Figure 401).

(c) Disconnect the APU fuel feed line [1B] from the check valve [4] (View F, Figure 401).

1) Remove the union [2].

2) Remove and discard the O-ring [3].

(d) Drain the remaining fuel from the APU fuel feed lines into 5-gallon (19-liter) fuel resistant container, STD-1054.

SUBTASK 28-25-04-020-019

(3) Remove the APU fuel feed line [1C] (View A, Figure 401).

(a) Remove the screws [34], washers [35], and nuts [36] to disconnect the bonding jumpers [30] from the APU fuel feed line [1C] (View K, Figure 401).

(b) Remove the screw [32] and washers [33] that hold the APU fuel feed line [1C] in the clamp [27] (TASK 28-22-15-000-801) (View J, Figure 401).

(c) Disconnect the APU fuel feed line [1C] from the tee [31] (View G, Figure 401).

(d) Remove the tee [31] from the APU shutoff valve [5]:

1) Remove the union [7].

2) Remove and discard the O-ring [6]

(e) Disconnect the APU fuel feed line [1C] from the tube end union on APU fuel feed line [1D].

(f) Drain the remaining fuel from the APU fuel feed lines into the 5-gallon (19-liter) fuel resistant container, STD-1054.

SUBTASK 28-25-04-020-020

(4) Remove the APU fuel feed line [1D] (View B, Figure 401).

(a) Remove the screws [32] and washers [33] that hold the APU fuel feed line [1D] in the clamps [27] (TASK 28-22-15-000-801) (View J, Figure 401).

(b) Disconnect the rigid coupling [28] between the APU fuel feed line [1D] and APU fuel feed line [1E] (TASK 28-22-15-000-803).

1) Remove and discard the O-ring [29].

(c) Drain the remaining fuel from the APU fuel feed lines into the 5-gallon (19-liter) fuel resistant container, STD-1054.

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SUBTASK 28-25-04-020-021

(5) Remove the APU fuel feed line [1E] (View C, Figure 401).

(a) Remove the screw [34], washers [35], and nut [36] to disconnect the bonding jumper [30] from the APU fuel feed line [1E] (View K, Figure 401).

(b) Remove the screw [32] and washers [33] that hold the APU fuel feed line [1E] in the clamp [27] (TASK 28-22-15-000-801) (View J, Figure 401).

(c) Disconnect the rigid coupling [28] between the APU fuel feed line [1E] and APU fuel feed line [1F] (TASK 28-22-15-000-803).

1) Remove and discard the O-ring [29].

(d) Drain the remaining fuel from the APU fuel feed lines into the 5-gallon (19-liter) fuel resistant container, STD-1054.

SUBTASK 28-25-04-020-022

(6) Remove the APU fuel feed line [1F] (View D, Figure 401).

(a) Remove the screw [34], washers [35], and nut [36] to disconnect the bonding jumper [30] from the APU fuel feed line [1F] (View K, Figure 401).

(b) Remove the screws [32] and washers [33] that hold the APU fuel feed line [1F] in the clamps [27] (TASK 28-22-15-000-801) (View J, Figure 401).

(c) Disconnect the APU fuel feed line [1F] from the bulkhead fitting (View L, Figure 401).

(d) Drain the remaining fuel from the APU fuel feed lines into the 5-gallon (19-liter) fuel resistant container, STD-1054.

SUBTASK 28-25-04-020-023

(7) Remove the APU fuel feed line [1G] (View E, Figure 401).

(a) Remove the screws [34], washers [35], and nuts [36] to disconnect the bonding jumpers [30] from the APU fuel feed line [1G] (View K, Figure 401).

(b) Remove the screw [32] and washers [33] that hold the APU fuel feed line [1G] in the clamp [27] (TASK 28-22-15-000-801) (View J, Figure 401).

(c) Disconnect the APU fuel feed line [1G] from the check valve [4] (View F, Figure 401).

(d) Drain the remaining fuel from the APU fuel feed lines into the 5-gallon (19-liter) fuel resistant container, STD-1054.

SUBTASK 28-25-04-020-024

(8) Remove the APU fuel feed line [1H] (View E, Figure 401).

(a) Remove the screw [34], washers [35], and nut [36] to disconnect the bonding jumper [30] from the APU fuel feed line [1H] (View K, Figure 401).

(b) Disconnect the APU fuel feed line [1H] from the bulkhead fitting (View L, Figure 401).

1) Remove the nut [45] and washer [44].

2) Remove the union [42] and O-ring [43].

a) Discard the O-ring [43].

(c) Disconnect the APU fuel feed line [1H] from the check valve [4] (View F, Figure 401).

1) Remove the union [2].

2) Remove and discard the O-ring [3].

(d) Drain the remaining fuel from the APU fuel feed lines into the 5-gallon (19-liter) fuel resistant container, STD-1054.

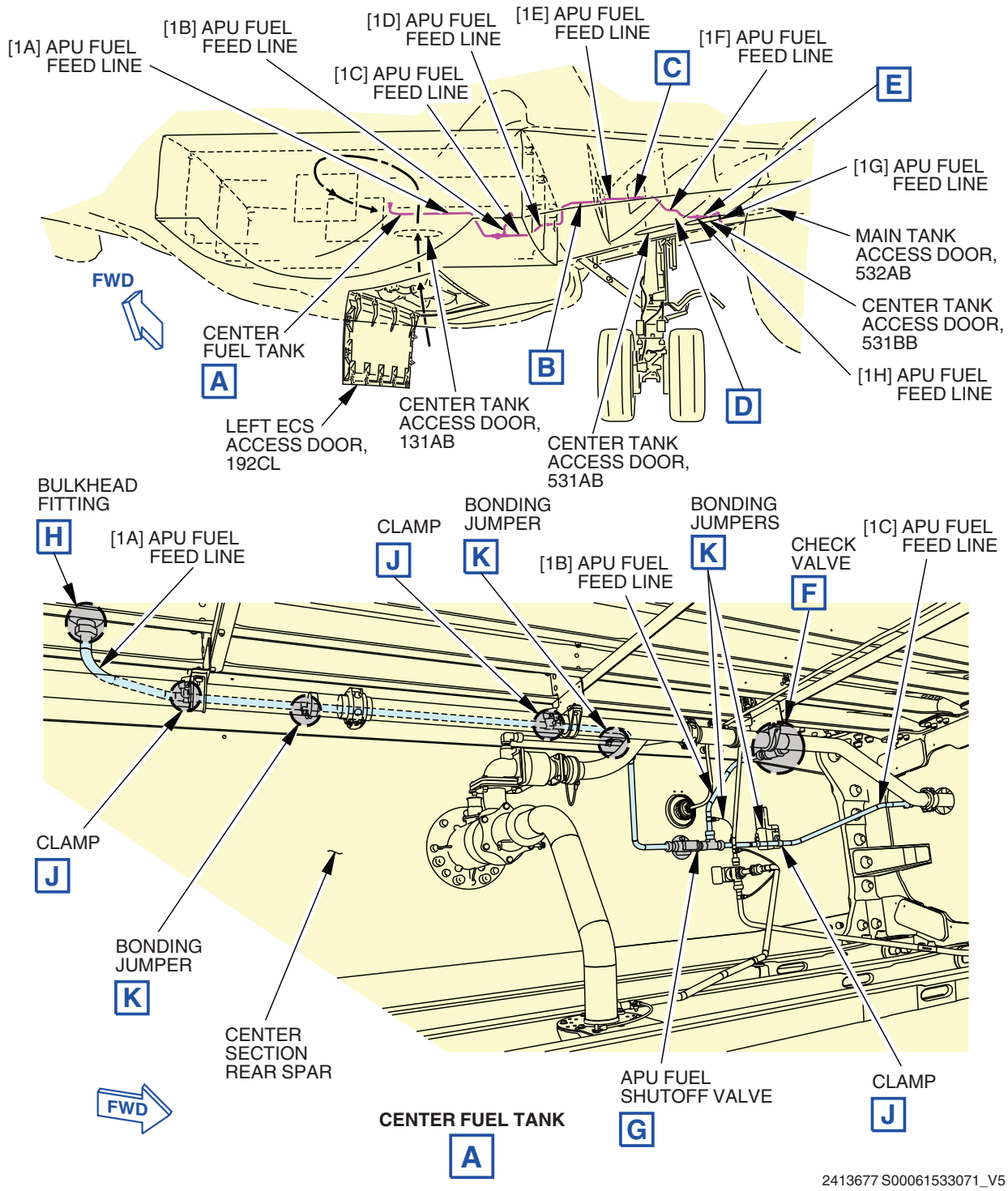
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SUBTASK 28-25-04-020-004

(9) Remove the APU fuel feed lines from the tank.

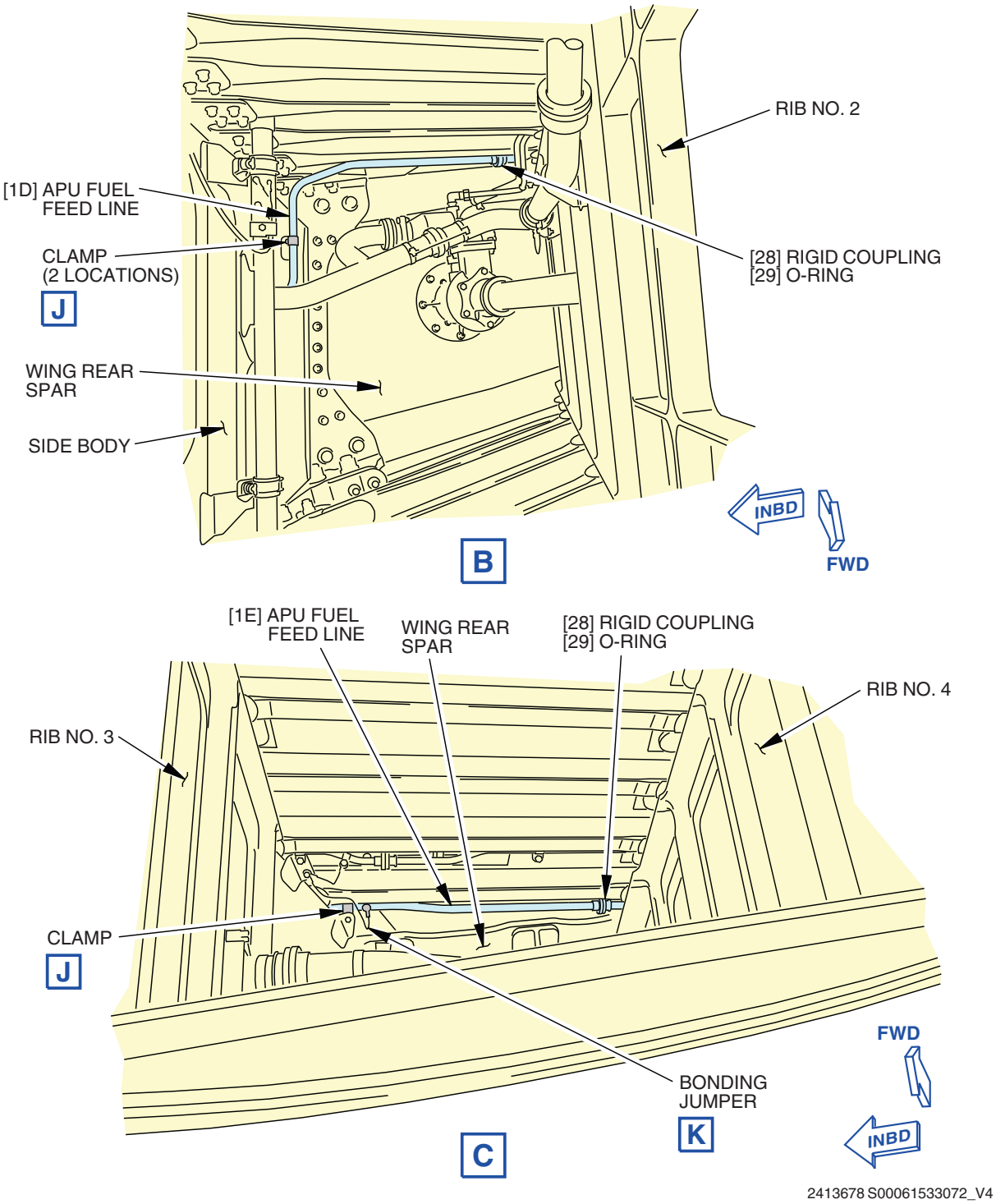
END OF TASK

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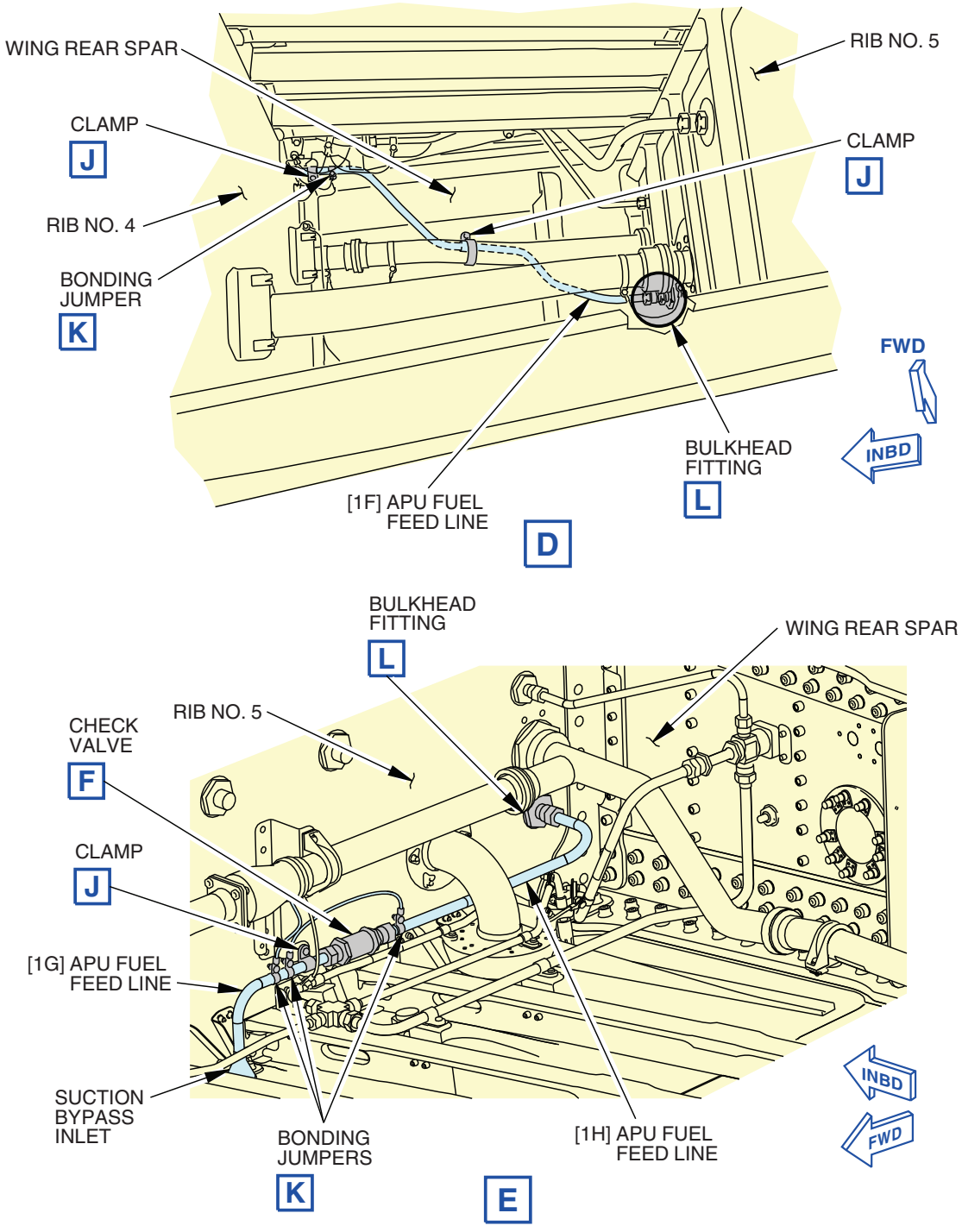
APU Fuel Feed Line Installation
Figure 401/28-25-04-990-801 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



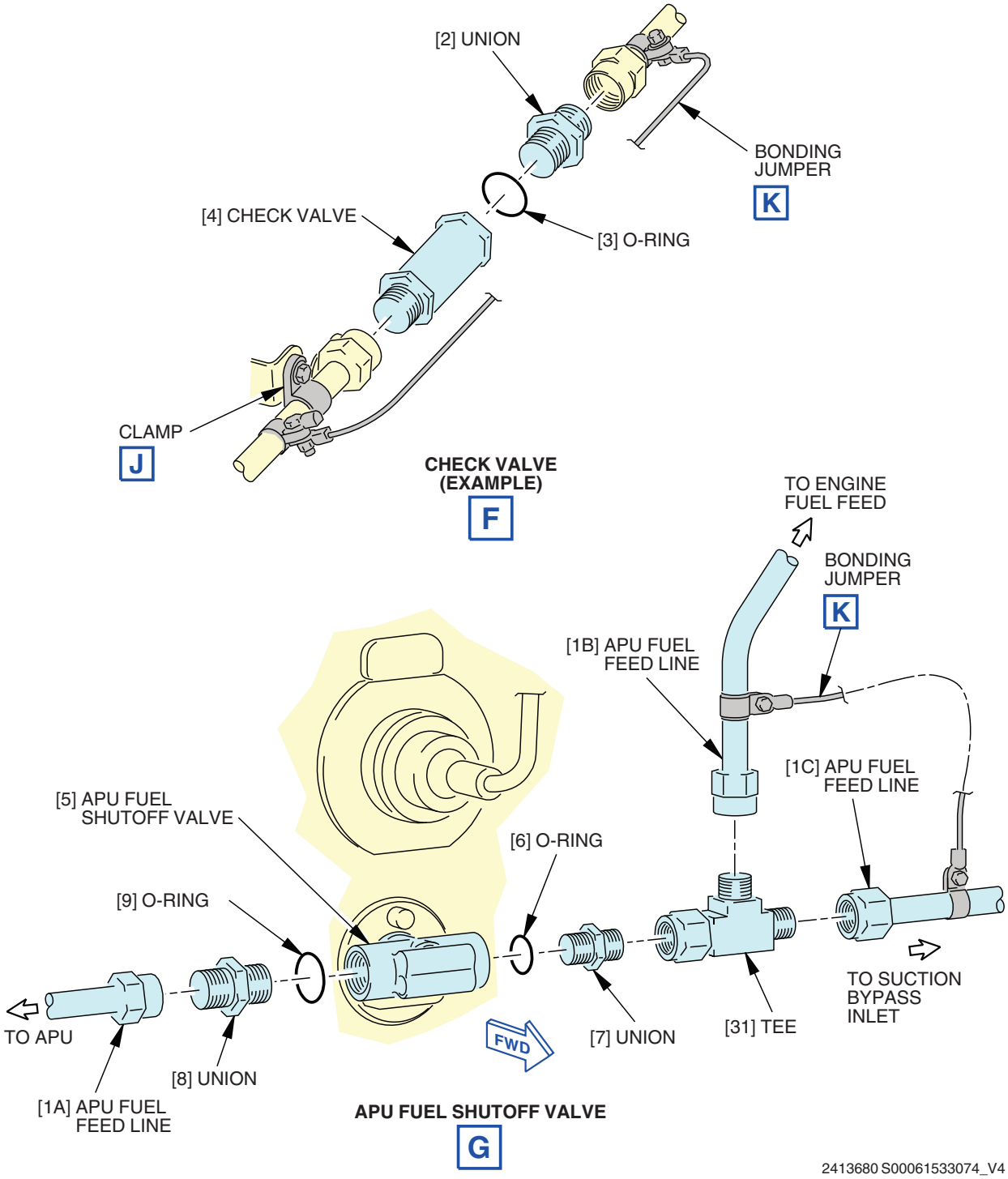
APU Fuel Feed Line Installation
Figure 401/28-25-04-990-801 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



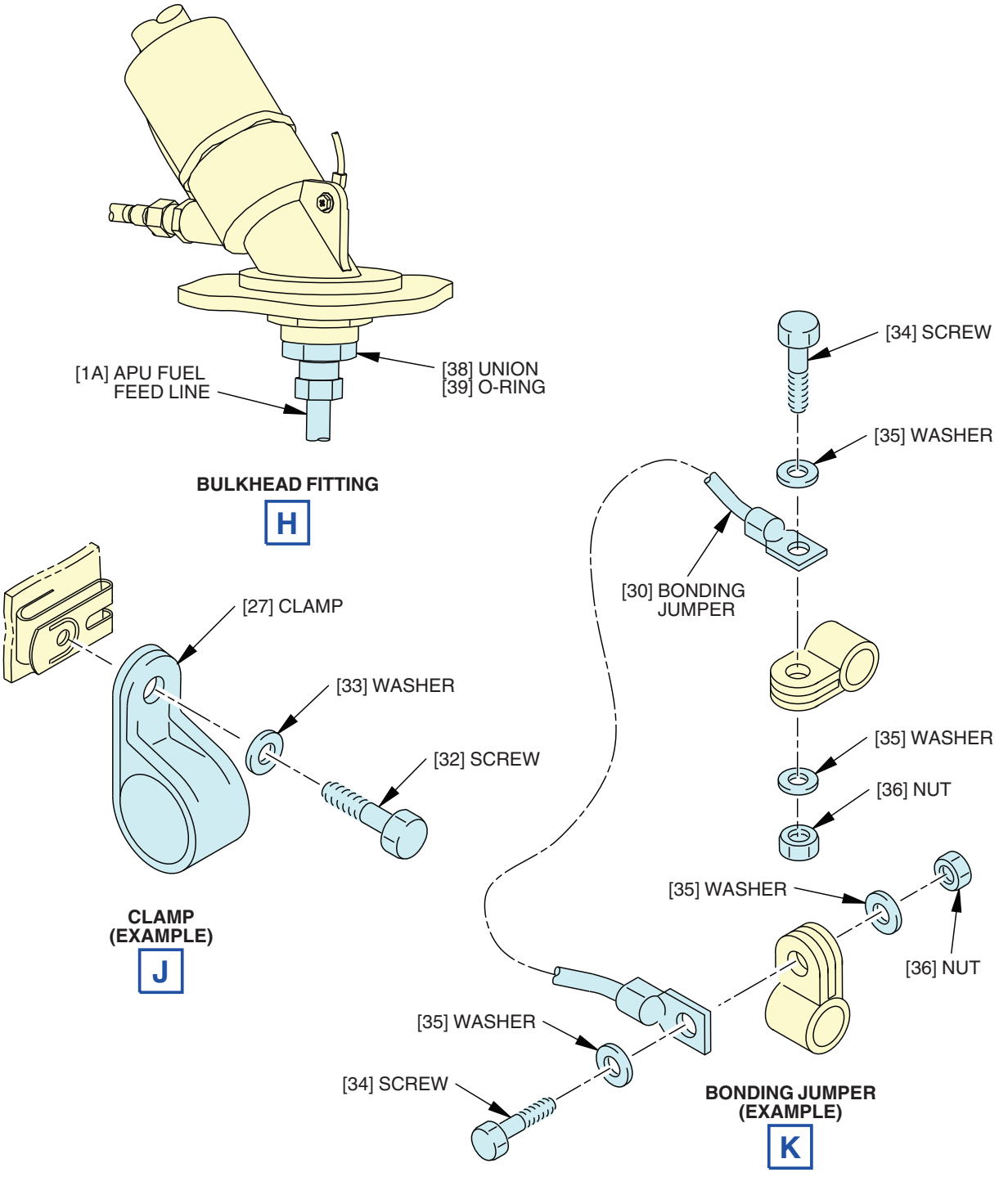
APU Fuel Feed Line Installation
Figure 401/28-25-04-990-801 (Sheet 3 of 6)

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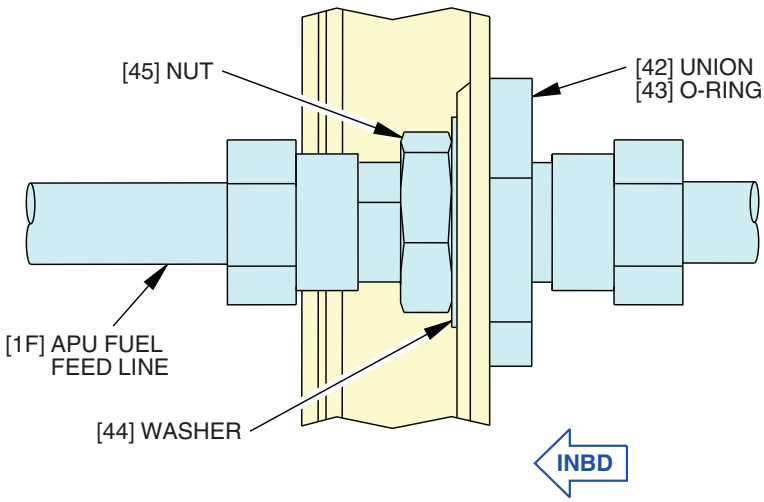
APU Fuel Feed Line Installation
Figure 401/28-25-04-990-801 (Sheet 4 of 6)

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APU Fuel Feed Line Installation
Figure 401/28-25-04-990-801 (Sheet 5 of 6)

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BULKHEAD FITTING

L

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APU Fuel Feed Line Installation
Figure 401/28-25-04-990-801 (Sheet 6 of 6)

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TASK 28-25-04-400-801

3. APU Fuel Feed Line (No. 1 Tank and Center Tank) Installation

(Figure 401)

A. General

(1) This task gives the instruction to Install the APU fuel feed line [1A], APU fuel feed line [1B], APU fuel feed line [1C], APU fuel feed line [1D], APU fuel feed line [1E], APU fuel feed line [1F], APU fuel feed line [1G], and APU fuel feed line [1H] loosely in their positions in the tank.

B. References

Reference	Title
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
28-22-15-400-801	Fuel Line, Fitting and Coupling Installation (P/B 401)
28-22-15-400-803	Couplings Installation (P/B 401)
51-21-41-370-802	Apply Bonderite M-CR 600, Bonderite M-CR 1200, Bonderite M-CR 600RTU, or Bonderite M-CR 1200S Solution (P/B 701)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A50337	Sealant - Fuel Tank	BMS5-45 Class B
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C50475	Coating - Chemical Conversion - Bonderite M-CR 600 Aero	BAC5719 Type II Class D
D00504	Grease - Petrolatum	VV-P-236
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1A	APU fuel feed line	28-25-04-01-105	PTF ALL
1B	APU fuel feed line	28-25-04-01-115	PTF ALL
1C	APU fuel feed line	28-25-04-01-110	PTF ALL
1D	APU fuel feed line	28-25-04-05-185	PTF ALL
1E	APU fuel feed line	28-25-04-05-170	PTF ALL
1F	APU fuel feed line	28-25-04-05-150	PTF ALL
1G	APU fuel feed line	28-25-04-05-145	PTF ALL
1H	APU fuel feed line	28-25-04-05-405	PTF ALL
3	O-ring	28-25-04-05-090	PTF ALL
6	O-ring	28-25-04-01-095	PTF ALL
9	O-ring	28-25-04-01-095	PTF ALL
29	O-ring	28-25-04-05-135	PTF ALL
39	O-ring	28-25-04-01-085	PTF ALL
43	O-ring	28-25-04-05-120	PTF ALL

F. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50

G. Access Panels

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
531BB	Center Tank Access Door - Wing Station 192
532AB	Main Tank Access Door - Wing Station 216

H. APU Fuel Line (No. 1 Tank and Center Tank) Installation

SUBTASK 28-25-04-420-014

(1) Install the APU fuel feed lines in the tank.

SUBTASK 28-25-04-420-015

(2) Install the APU fuel feed line [1H] (View E, Figure 401).

(a) Connect the APU fuel feed line [1H] to the check valve [4] (View F, Figure 401).

1) Apply a thin layer of grease, D00504, to the new O-ring [3].

2) Put the O-ring [3] on the union [2].

3) Install the union [2] on the check valve [4].

CAUTION

DO NOT USE TOOLS TO TIGHTEN THE FITTINGS. DAMAGE TO THE FITTING COMPONENTS COULD OCCUR. THE ADJACENT FITTINGS COULD BECOME LOOSE AND CAUSE DAMAGE.

(b) Connect the APU fuel feed line [1H] to the bulkhead fitting (View L, Figure 401).

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1) Abrasive clean the mating surface of the rib where the union [42] mates with rib (SWPM 20-20-00).

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

2) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the rib and union [42].
3) Apply coating, C00064, or Bonderite M-CR 600 Aero coating, C50475, to the abraded mating surfaces of the rib (TASK 51-21-41-370-802).
4) Apply a thin layer of grease, D00504, to the new O-ring [43].
5) Put the O-ring [43] in the groove of the union [42].
6) Install the union [42] into its position.
7) Install the washer [44] and nut [45].
8) Tighten the nut [45] to 500 ±50 in-lb (56.5 ±5.6 N·m).

(c) Connect the bonding jumper [30] to the APU fuel feed line [1H] (View K, Figure 401).

WARNING

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1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the APU fuel feed line [1H], clamp for the bonding jumper [30], and terminal lug of the bonding jumper [30].
2) Install the screw [34], washers [35], and nut [36].
3) Tighten the nut [36] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 28-25-04-420-016

(3) Install the APU fuel feed line [1G] (View E, Figure 401).

(a) Install the screw [32] and washers [33] that hold the APU fuel feed line [1G] in the clamp [27] (TASK 28-22-15-400-801) (View J, Figure 401).

1) Do not tighten the screw [32].

(b) Connect the APU fuel feed line [1G] to the check valve [4] (View F, Figure 401).

(c) Connect the bonding jumpers [30] to the APU fuel feed line [1G] (View K, Figure 401).

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WARNING DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the APU fuel feed line [1G], clamps for the bonding jumpers [30], and terminal lugs of the bonding jumpers [30].
2) Install the screws [34], washers [35], and nuts [36].
3) Tighten the nuts [36] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 28-25-04-420-017

(4) Install the APU fuel feed line [1F] (View D, Figure 401).

(a) Install the screws [32] and washers [33] that hold the APU fuel feed line [1F] in the clamps [27] (TASK 28-22-15-400-801) (View J, Figure 401).

1) Do not tighten the screws [32].

CAUTION DO NOT USE TOOLS TO TIGHTEN THE FITTINGS. DAMAGE TO THE FITTING COMPONENTS COULD OCCUR. THE ADJACENT FITTINGS COULD BECOME LOOSE AND CAUSE DAMAGE.

(b) Connect the APU fuel feed line [1F] to the bulkhead fitting (View L, Figure 401).

(c) Connect the bonding jumper [30] to the APU fuel feed line [1F] (View K, Figure 401).

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the APU fuel feed line [1F], clamp for the bonding jumper [30], and terminal lug of the bonding jumper [30].
2) Install the screw [34], washers [35], and nut [36].
3) Tighten the nut [36] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 28-25-04-420-018

(5) Install the APU fuel feed line [1E] (View C, Figure 401).

(a) Install the screw [32] and washers [33] that hold the APU fuel feed line [1E] in the clamp [27] (TASK 28-22-15-400-801) (View J, Figure 401).

1) Do not tighten the screw [32].

(b) Connect the rigid coupling [28] between the APU fuel feed line [1E] and APU fuel feed line [1F] (TASK 28-22-15-400-803).

1) Install the new O-ring [29].

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(c) Connect the bonding jumper [30] to the APU fuel feed line [1E] (View K, Figure 401).

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the APU fuel feed line [1E], clamp for the bonding jumper [30], and terminal lug of the bonding jumper [30].
2) Install the screw [34], washers [35], and nut [36].
3) Tighten the nut [36] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 28-25-04-420-019

(6) Install the APU fuel feed line [1D] (View B, Figure 401).

(a) Install the screws [32] and washers [33] that hold the APU fuel feed line [1D] in the clamps [27] (TASK 28-22-15-400-801) (View J, Figure 401).

1) Do not tighten the screw [32].

(b) Connect the rigid coupling [28] between the APU fuel feed line [1D] and APU fuel feed line [1E] (TASK 28-22-15-400-803).

1) Install the new O-ring [29].

SUBTASK 28-25-04-420-020

(7) Install the APU fuel feed line [1C] (View A, Figure 401).

(a) Install the screw [32] and washers [33] that hold the APU fuel feed line [1C] in the clamp [27] (TASK 28-22-15-400-801) (View J, Figure 401).

1) Do not tighten the screw [32].

(b) Install the tee [31] to the APU shutoff valve [5] (View G, Figure 401).

1) Apply a thin layer of grease, D00504, to the new O-ring [6].
2) Put the O-ring [6] on the union [7].
3) Install the union [7] on the APU shutoff valve [5].
4) Install the tee [31] to the union [7].

(c) Connect the APU fuel feed line [1C] to the tee [31] (View G, Figure 401).

(d) Connect the APU fuel feed line [1C] to the tube end union on APU fuel feed line [1D].

(e) Connect the bonding jumpers [30] to the APU fuel feed line [1C] (View K, Figure 401).

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WARNING DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the APU fuel feed line [1C], clamps for the bonding jumpers [30], and terminal lugs of the bonding jumpers [30].
2) Install the screws [34], washers [35], and nuts [36].
3) Tighten the nuts [36] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 28-25-04-420-021

(8) Install the APU fuel feed line [1B] (View A, Figure 401).
   (a) Connect the APU fuel feed line [1B] to the tee [31] (View G, Figure 401).
   (b) Connect the APU fuel feed line [1B] to the check valve [4] (View F, Figure 401).
      1) Apply a thin layer of grease, D00504, to the new O-ring [3].
      2) Put the O-ring [3] on the union [2].
      3) Install the union [2] on the check valve [4].
   (c) Connect the bonding jumper [30] to the APU fuel feed line [1B] (View K, Figure 401).

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the APU fuel feed line [1B], clamp for the bonding jumper [30], and terminal lug of the bonding jumper [30].
2) Install the screw [34], washers [35], and nut [36].
3) Tighten the nut [36] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 28-25-04-420-022

(9) Install the APU fuel feed line [1A] (View A, Figure 401).
   (a) Install the screws [32] and washers [33] that hold the APU fuel feed line [1A] in the clamps [27] (TASK 28-22-15-400-801) (View J, Figure 401).
      1) Do not tighten the screws [32].

CAUTION DO NOT USE TOOLS TO TIGHTEN THE FITTINGS. DAMAGE TO THE FITTING COMPONENTS COULD OCCUR. THE ADJACENT FITTINGS COULD BECOME LOOSE AND CAUSE DAMAGE.

   (b) Connect the APU fuel feed line [1A] to the bulkhead fitting (View H, Figure 401).
      1) Apply a thin layer of grease, D00504, to the new O-ring [39].

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2) Put the O-ring [39] on the union [38].

3) Install the union [38] into its position.

4) Connect the APU fuel feed line [1A] to the union [38].

(c) Connect the APU fuel feed line [1A] to the APU shutoff valve [5] (View G, Figure 401).

1) Apply a thin layer of grease, D00504, to the new O-ring [9].

2) Put the O-ring [9] on the union [8].

3) Install the union [8] on the APU shutoff valve [5].

4) Connect the APU fuel feed line [1A] to the union [8].

(d) Connect the bonding jumpers [30] to the APU fuel feed line [1A] (View K, Figure 401).

WARNING

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1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the APU fuel feed line [1A], clamps for the bonding jumpers [30], and terminal lugs of the bonding jumpers [30].

2) Install the screws [34], washers [35], and nuts [36].

3) Tighten the nuts [36] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 28-25-04-420-026

(10) Tighten the screws [32] for the APU fuel feed lines (TASK 28-22-15-400-801).

SUBTASK 28-25-04-420-027

(11) Measure the electrical bonding resistance between the bonding jumpers [30] and APU fuel feed lines (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the bonding resistance is 0.0009 ohm (0.9 milliohms) or less.

SUBTASK 28-25-04-760-001

(12) Measure the electrical bonding resistance between the union [42] and airplane structure (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the bonding resistance is 0.0010 ohm (1.0 milliohm) or less.

SUBTASK 28-25-04-390-001

(13) Apply a fillet seal with sealant, A50337, around the union [42] and nut [45].

SUBTASK 28-25-04-420-028

(14) Remove all of the tools and equipment from the fuel tanks.

I. APU Fuel Feed Line (No. 1 Tank and Center Tank) Installation Test

SUBTASK 28-25-04-790-001

(1) Do this task: APU Fuel Line Leak Test (with air), TASK 28-25-04-790-801.

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J. Put Airplane Back to Its Usual Condition

SUBTASK 28-25-04-410-007

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) For the APU fuel lines in the main tank, install this access door:

(TASK 28-11-11-400-801)

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216

(b) For the APU fuel lines in the center tank, install these access doors:

(TASK 28-11-31-400-801)

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
531BB	Center Tank Access Door - Wing Station 192

——— END OF TASK ———

TASK 28-25-04-000-802

4. APU Fuel Line (Center Wing Section to APU Firewall) Removal

(Figure 402)

A. General

(1) This task gives instructions to remove the APU Fuel Line (Center Wing Section to APU Firewall).

B. References

Reference	Title
24-22-00-860-804	Remove External Power (P/B 201)
25-27-15-000-801	Carpet Removal (P/B 401)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1054	Container - Fuel Resistant, 5-Gallon (19-Liter)

D. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Removal

SUBTASK 28-25-04-010-002

(1) Remove the aisle carpets at station 660 (immediately aft of the overwing emergency exit) (TASK 25-27-15-000-801).

SUBTASK 28-25-04-010-003

(2) Remove the floor panel at Station 660 (immediately aft of the overwing emergency exit) to get access to the interconnect fitting on the center tank (TASK 53-21-00-000-801).

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SUBTASK 28-25-04-860-014

(3) Do this task: Remove External Power, TASK 24-22-00-860-804.

SUBTASK 28-25-04-860-007

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

G. APU Fuel Line (Center Wing Section to APU Firewall) Removal

SUBTASK 28-25-04-020-005

(1) Loosen the forward shroud fitting [10] from the adapter fitting [14] on the center tank (View A, Figure 402).

SUBTASK 28-25-04-020-006

WARNING MAKE SURE THAT YOU KEEP THE ELECTRICAL POWER OFF WHEN YOU DISCONNECT OR CONNECT THE FUEL, THE HYDRAULIC, AND THE ELECTRICAL LINES. IF YOU DO NOT, YOU CAN CAUSE INJURY TO PERSONNEL AND DAMAGE EQUIPMENT.

WARNING YOU MUST ELECTRICALLY GROUND AND BOND THE AIRPLANE BEFORE YOU DO MAINTENANCE ON THE FUEL SYSTEM. IF YOU DO NOT GROUND AND BOND THE AIRPLANE, STATIC ELECTRICITY CAN CAUSE A FIRE OR EXPLOSION.

(2) Do these steps to disconnect the flexible hose [26] from the adapter fitting [14] of the center tank:

(a) Loosen the forward shroud fitting [10] until it is against the reducing ferrule on the aft APU shroud.

(b) Put a 5-gallon (19-liter) fuel resistant container, STD-1054, below the drain mast for the aft APU shroud.

(c) Loosen the reducer [13] on the flexible hose [26] to disconnect the flexible hose [26] from adapter fitting [14] of the center tank.

(d) Move the flexible hose [26] from the adapter fitting [14].

(e) Remove the O-ring [11] and O-ring [12] from the shroud and discard them.

(f) Remove the O-ring [15] from the reducer [13] and discard it.

SUBTASK 28-25-04-010-004

(3) Open this access door to get access to the aft APU shroud:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 28-25-04-020-007

(4) Disconnect the aft section of the aft shroud [17] from the aft flexible shroud [20] installed in the forward side of the APU firewall (View C, Figure 402).

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SUBTASK 28-25-04-020-008

(5) Move the aft flexible shroud [20] back to get access to the fitting on the forward side of the APU firewall.

SUBTASK 28-25-04-020-027

(6) Disconnect the flexible hose [26] from the fitting on the forward side of the APU firewall.

SUBTASK 28-25-04-020-009

(7) Remove the aft section of the aft shroud [17] (View B, Figure 402).

SUBTASK 28-25-04-020-010

(8) Do these steps to remove the flexible hose [26] from the aft shroud [17]:

(a) Hold the exposed part of the flexible hose [26] firmly.

(b) Pull the flexible hose [26] in the aft direction until it is completely removed from the shroud.

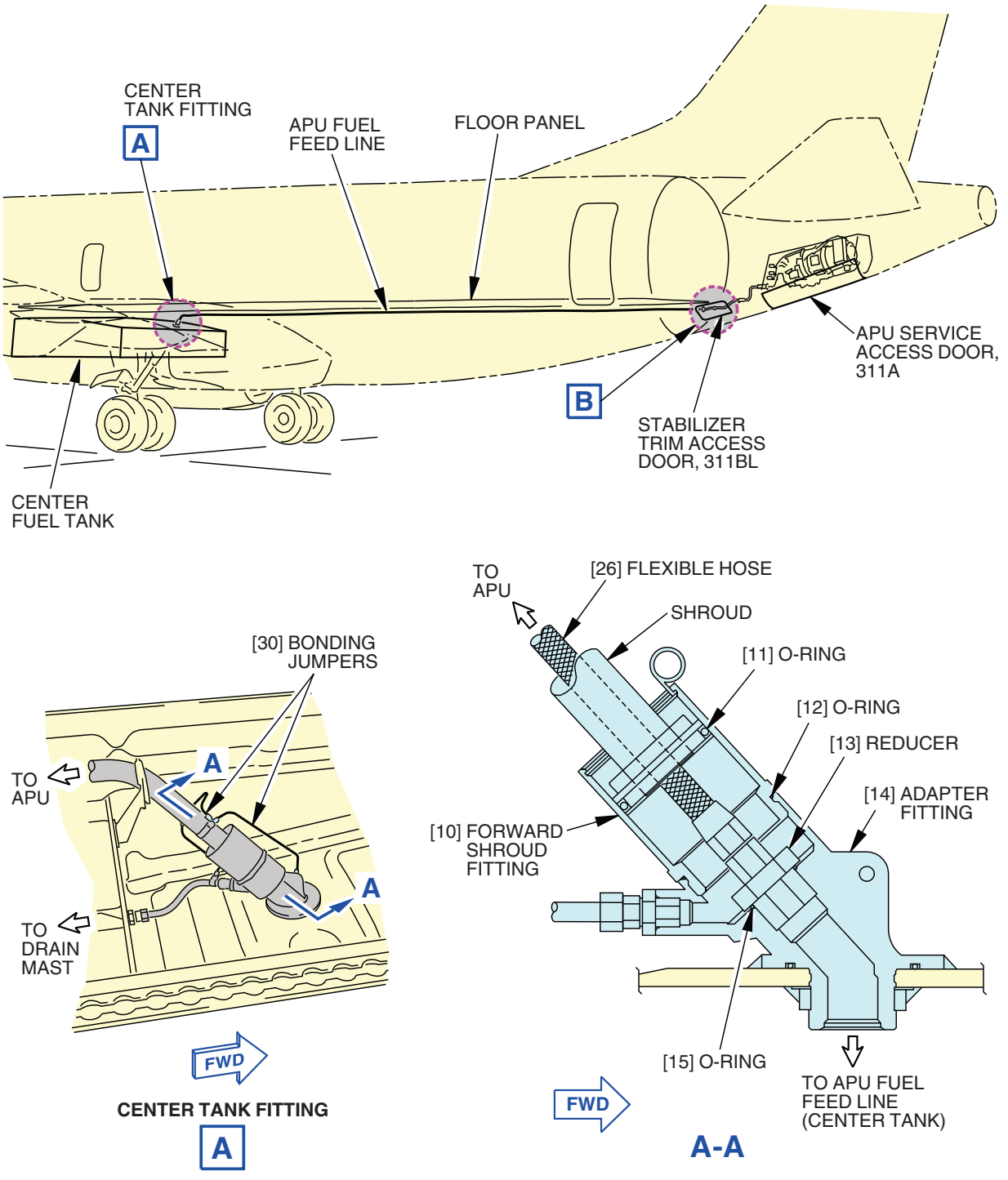
NOTE: This step removes the complete length of the flexible hose which you disconnected from the center tank fitting at the forward end. It may be necessary to twist and turn the flexible hose in the shroud to get it to move freely.

(c) Move the flexible hose [26] through the opening for this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

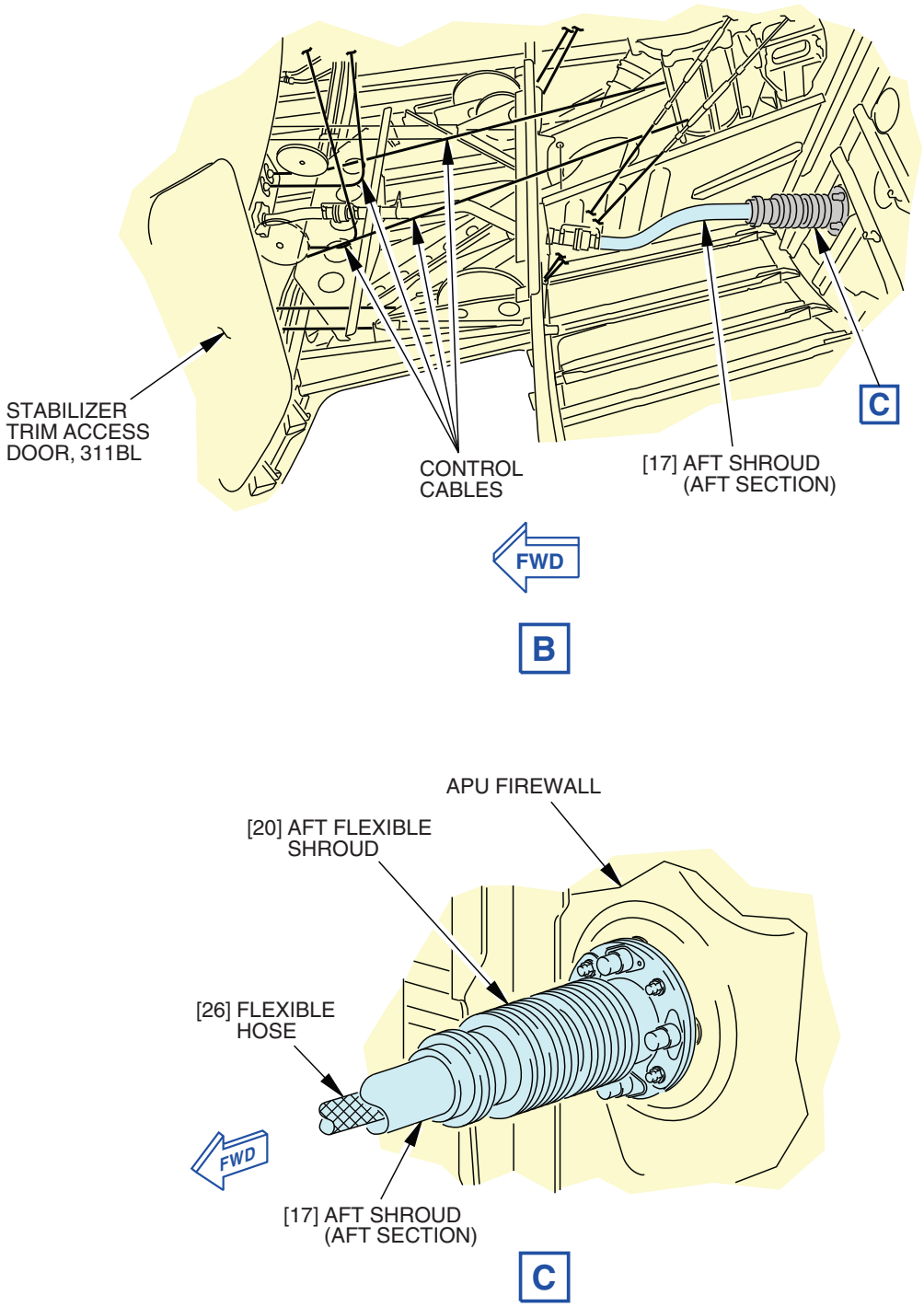
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Fuel Feed Line Installation
Figure 402/28-25-04-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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APU Fuel Feed Line Installation
Figure 402/28-25-04-990-802 (Sheet 2 of 2)

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TASK 28-25-04-400-802

5. APU Fuel Line (Center Wing Section to APU Firewall) Installation

(Figure 402)

A. General

(1) This task gives instructions to install the APU Fuel Line (Center Wing Section to APU Firewall).

B. References

Reference	Title
25-27-15-400-801	Carpet - Installation (P/B 401)
28-22-15-400-801	Fuel Line, Fitting and Coupling Installation (P/B 401)
28-25-05-790-801	APU Fuel Line Shroud and Drain Line - Leak Test (P/B 501)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00504	Grease - Petrolatum	VV-P-236

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
11	O-ring	28-25-05-01-135	PTF ALL
12	O-ring	28-25-05-01-155	PTF ALL
15	O-ring	28-25-05-01-185	PTF ALL
26	Flexible hose	28-25-04-10-005	PTF ALL
		28-25-05-01-090	PTF ALL

E. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. APU Fuel Line (Center Wing Section to APU Firewall) Installation

SUBTASK 28-25-04-560-001

(1) Put the flexible hose [26] through this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 28-25-04-420-008

(2) Put the forward end of the flexible hose [26] into the aft end of the aft shroud [17] (View B, Figure 402).

SUBTASK 28-25-04-420-009

(3) Continue to put the fuel line into the main shroud until the fuel line is against the adapter fitting [14] (View A, Figure 402).

SUBTASK 28-25-04-020-011

(4) Pull the forward shroud fitting [10] away from the adapter fitting [14].

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SUBTASK 28-25-04-420-010

(5) Do these steps to connect the flexible hose [26] (View A, Figure 402).

(a) Apply a thin layer of grease, D00504, to the new O-ring [11] and new O-ring [12].

(b) Put the O-ring [11] and O-ring [12] on the shroud.

(c) If the reducer [13] is not already installed in the flexible hose [26] with a new O-ring [15], do these steps:

1) Apply a thin layer of grease, D00504, to the new O-ring [15].

2) Put the O-ring [15] on the reducer [13].

3) Install the reducer [13] on the end of the flexible hose [26].

(d) Put the flexible hose [26] and the shroud into the forward shroud fitting [10].

(e) Connect the flexible hose [26] to the adapter fitting [14] with the reducer [13].

1) Do not attach the forward shroud fitting [10] until you complete the leak check for the fuel line.

(f) Tighten the reducer [13] in the adapter fitting [14].

SUBTASK 28-25-04-020-012

(6) Do these steps to install the aft section of the aft shroud [17] (View B, Figure 402):

(a) Put the aft section of the aft shroud [17] on the aft end of the flexible hose [26].

(b) Connect the aft section of the aft shroud [17] to the adjacent section of the APU shroud (TASK 28-22-15-400-801).

(c) Push the aft flexible shroud [20] aft to get access to the fitting at the APU firewall for the flexible hose [26].

(d) Connect the aft end of flexible hose [26] to the fitting on the forward side of the APU firewall (Figure 402).

1) Do not attach the aft shroud [17] to the aft flexible shroud [20] until you complete the leak check of the fuel line.

(e) Tighten the fitting on the forward side of the APU firewall to 280 ±14 in-lb (31.6 ±1.6 N·m).

SUBTASK 28-25-04-790-002

(7) Do this task: APU Fuel Line Leak Test (with air), TASK 28-25-04-790-801.

SUBTASK 28-25-04-420-011

(8) Connect the forward shroud fitting [10] to the adapter fitting [14] (View A, Figure 402).

(a) Tighten the forward shroud fitting [10] to 700 ±70 in-lb (79.1 ±7.9 N·m).

SUBTASK 28-25-04-420-012

(9) Connect the coupling of the aft section of the aft shroud [17] to the aft flexible shroud [20] (View C, Figure 402).

SUBTASK 28-25-04-790-003

(10) Do this task: APU Fuel Line Shroud and Drain Line - Leak Test, TASK 28-25-05-790-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 28-25-04-410-003

(1) Install the floor panel at STA 660 (TASK 53-21-00-400-801).

SUBTASK 28-25-04-410-004

(2) Install the aisle carpets (TASK 25-27-15-400-801).

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SUBTASK 28-25-04-860-009

(3) Close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 28-25-04-410-005

(4) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

TASK 28-25-04-790-801

6. APU Fuel Line Leak Test (with air)

(Figure 403)

A. General

(1) This procedure does an air pressure test of the APU fuel feed line from the APU to the bypass inlet in the No. 1 tank.

B. Tools/Equipment

Reference	Description
STD-1201	Gauge - Pressure, 0-75 PSIG (0-518 KPa)
STD-3940	Air Source - Regulated, Dry Filtered, 0-150 psig

C. Access Panels

Number	Name/Location
315A	APU Cowl Door

D. Pressurize the APU Fuel Line with Air and Do a Check for Leaks

SUBTASK 28-25-04-860-011

(1) Manually open the APU fuel shutoff valve with the override lever.

SUBTASK 28-25-04-010-005

(2) Open this access panel:

Number	Name/Location
315A	APU Cowl Door

SUBTASK 28-25-04-860-012

(3) Disconnect the APU flex line at the APU service compartment near the APU firewall.

SUBTASK 28-25-04-860-013

(4) Install a regulated air source, STD-3940, and pressure gauge, STD-1201, on the fitting at the APU firewall.

SUBTASK 28-25-04-790-004

(5) Pressurize the APU fuel feed line to a maximum of 40 psig (3 kg/cm^2).

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SUBTASK 28-25-04-790-005

(6) Keep the pressure on the fuel line for a minimum of 5 minutes.

SUBTASK 28-25-04-790-006

(7) Make sure that the pressure does not decrease.

SUBTASK 28-25-04-790-007

(8) If there is leakage look for indications of leakage and make the necessary repairs.

SUBTASK 28-25-04-790-008

(9) Do these steps to put the airplane back to its usual condition:

(a) Remove the pressure from the APU fuel feed line.

(b) Remove the regulated air source, STD-3940, and pressure gauge, STD-1201, from the APU flex line.

(c) Connect the APU flex line to the fitting at the APU firewall.

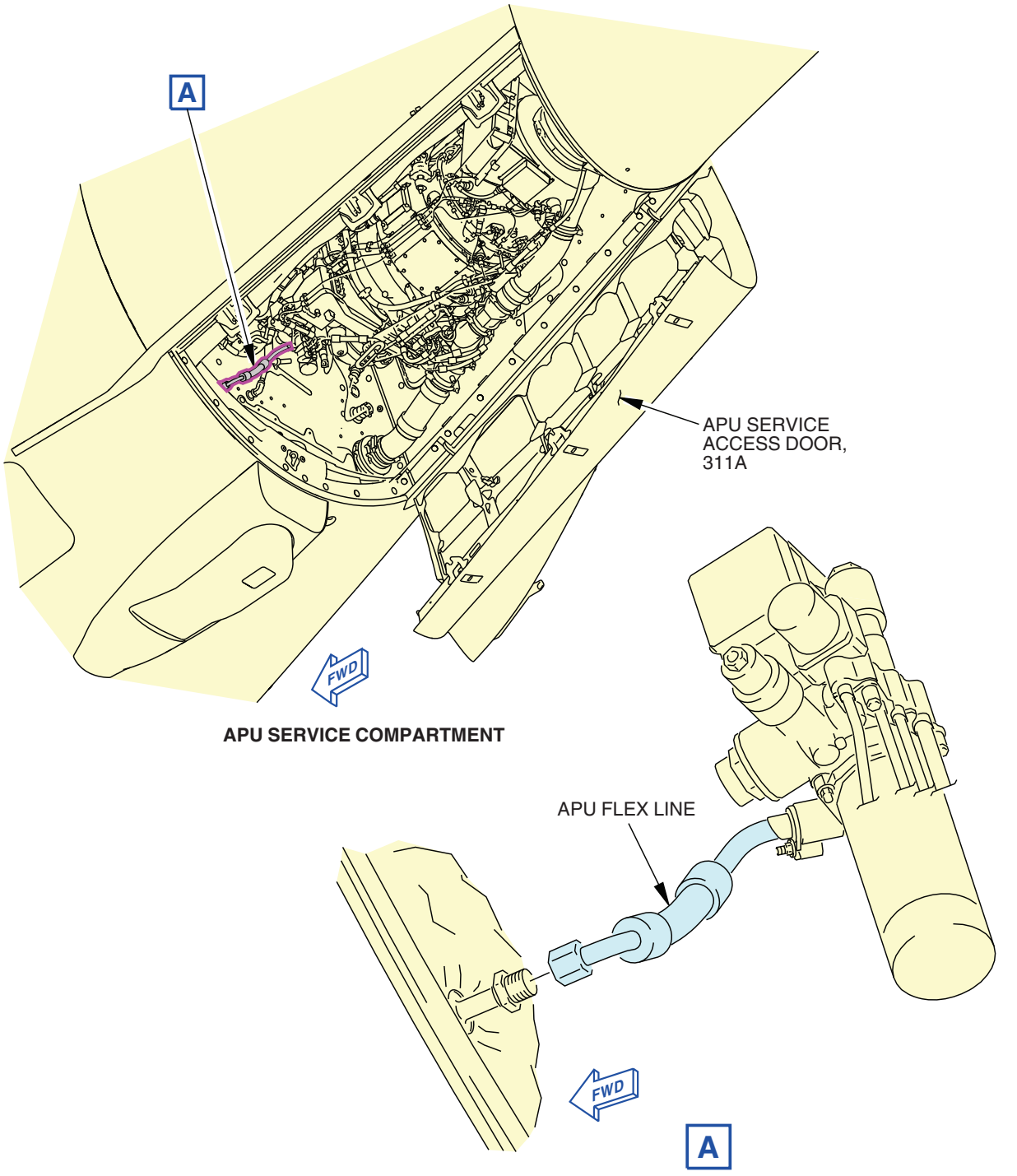
SUBTASK 28-25-04-410-006

(10) Close this access panel:

Number	Name/Location
315A	APU Cowl Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Fuel Line Leak Test
Figure 403/28-25-04-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FUEL FEED LINE SHROUD - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Main APU Fuel Feed Line Shroud Removal

(2) Main APU Feed Fuel Line Shroud Installation

(3) Aft APU Feed Fuel Line Shroud Removal

(4) Aft APU Feed Fuel Line Shroud Installation.

B. The APU fuel feed line shroud is referred to as the APU shroud.

C. The APU shroud is routed from the forward end of the APU shroud to the adapter fitting on the upper panel of the center fuel tank. The aft end of the APU shroud ends at the flexible shroud attached to the forward side of the APU firewall (View A, Figure 401).

D. The APU shroud is divided in two parts. The part that goes from the top of the center tank to the forward side of the pressure bulkhead is referred to as the main shroud. The part that goes from the aft side of the pressure bulkhead to the APU firewall is referred to as the aft shroud.

TASK 28-25-05-000-801

2. Main APU Fuel Feed Line Shroud Removal

(Figure 401)

A. General

(1) This task gives the instructions for the removal of the main APU fuel feed line shroud.

(2) The APU fuel feed line shroud is referred to as the APU shroud.

B. References

Reference	Title
20-10-44-000-801	Lockwire, Cotter Pins, and Lockrings - Removal (P/B 401)
25-27-15-000-801	Carpet Removal (P/B 401)
28-22-15-000-803	Couplings Removal (P/B 401)
28-25-04-000-802	APU Fuel Line (Center Wing Section to APU Firewall) Removal (P/B 401)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-6646	Container - Fuel Resistant

D. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

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F. Prepare for the Removal

SUBTASK 28-25-05-040-001

(1) On the P5 Overhead Panel, put the APU master switch to the OFF position.

SUBTASK 28-25-05-910-001

(2) Put a DO NOT OPERATE tag, STD-858 on the APU master switch.

SUBTASK 28-25-05-040-002

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 28-25-05-020-021

(4) Remove the aisle carpets between Station 616 and Station 1016 (TASK 25-27-15-000-801).

SUBTASK 28-25-05-020-022

(5) Remove the floor panels between Station 616 and Station 1016 (TASK 53-21-00-000-801).

SUBTASK 28-25-05-010-003

(6) To get access to the aft shroud, open this access panel:

(View B, Figure 401).

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Main APU Fuel Feed Line Shroud Removal

SUBTASK 28-25-05-020-037

(1) Loosen the forward shroud fitting [10] until it is against the ferrule [3] on the APU shroud tube assembly.

SUBTASK 28-25-05-020-001

WARNING

MAKE SURE YOU KEEP THE ELECTRICAL POWER OFF WHEN YOU DISCONNECT OR CONNECT THE FUEL, THE HYDRAULIC, AND THE ELECTRICAL LINES. IF YOU DO NOT, YOU CAN CAUSE INJURY TO PERSONNEL WITH THE ACCIDENTAL APPLICATION OF PRESSURIZED FLUIDS, ENERGIZING OF ELECTRICAL CIRCUITS, OR FIRE.

WARNING

YOU MUST ELECTRICALLY GROUND AND BOND THE AIRPLANE BEFORE YOU DO MAINTENANCE ON THE FUEL SYSTEM. IF YOU DO NOT GROUND AND BOND THE AIRPLANE, STATIC ELECTRICITY CAN CAUSE A FIRE OR EXPLOSION.

(2) Remove the flexible hose [11] from the main APU shroud [1].

NOTE: If you are scheduled to remove the flexible hose [11], refer to the applicable procedure (TASK 28-25-04-000-802).

(a) Disconnect the flexible hose [11] from the center tank adapter fitting [7] (View A-A, Figure 401).

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1) Put a fuel resistant container, STD-6646, below the APU shroud drain mast.

NOTE: A small quantity of fuel will spill from the flexible hose [11] and drain from the drain mast for the APU shroud.

SUBTASK 28-25-05-020-009

(3) Remove the O-ring [5] from the forward shroud fitting [10].

(a) Discard the O-ring [5].

SUBTASK 28-25-05-820-004

(4) Move the flexible hose [11] and the main APU shroud [1] from the adapter fitting [7].

SUBTASK 28-25-05-020-010

(5) Remove the O-ring [4] from the main APU shroud [1].

(a) Discard the O-ring [4].

SUBTASK 28-25-05-020-024

(6) Remove the safety cable or lockwire from the coupling nut assembly at the APU firewall (TASK 20-10-44-000-801) (View D, Figure 401).

SUBTASK 28-25-05-020-025

(7) Loosen the coupling nut assembly.

SUBTASK 28-25-05-020-026

(8) Disconnect the aft APU shroud [15] from the aft flexible shroud [20] at the APU firewall.

SUBTASK 28-25-05-020-012

(9) Remove the O-ring [26] from the aft APU shroud [15].

(a) Discard the O-ring [26].

SUBTASK 28-25-05-820-001

(10) Pull the aft flexible shroud [20] back toward the APU firewall.

SUBTASK 28-25-05-020-013

(11) Disconnect the flexible hose coupling [35] at the APU firewall.

SUBTASK 28-25-05-020-015

(12) Pull the flexible hose [11] through the main APU shroud [1].

SUBTASK 28-25-05-020-027

(13) Stow the flexible hose [11] in the aft compartment.

SUBTASK 28-25-05-020-014

(14) Remove the main APU shroud [1] from the slip joint fitting [16] on the pressure bulkhead as follows:

(a) Remove the six bolts [18] and six washers [19] that attach the two welded shroud tubes [37] to the pressure bulkhead (Figure 401).

(b) Discard the two gaskets [17].

(c) Remove the main APU shroud [1] from the slip joint fitting [16] on the pressure bulkhead.

(d) Remove the O-ring [25] from the aft end fitting of the main APU shroud [1].

(e) Discard the O-ring [25].

SUBTASK 28-25-05-020-002

(15) Remove the main APU shroud [1] from the floor beams.

(a) Remove the loop clamps and clamps that attach the main APU shroud [1] to the airplane structure.

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(b) Remove the bonding components.

1) Remove the clamp [30], screw [31], two washers [32], nut [33] and bonding jumper [34] from the main APU shroud [1].

(c) Remove the applicable couplings (TASK 28-22-15-000-803).

SUBTASK 28-25-05-020-016

(16) Pull the main APU shroud [1] forward through the floor beams.

SUBTASK 28-25-05-550-001

(17) Turn the main APU shroud [1] until it is clear of the control cables.

SUBTASK 28-25-05-550-002

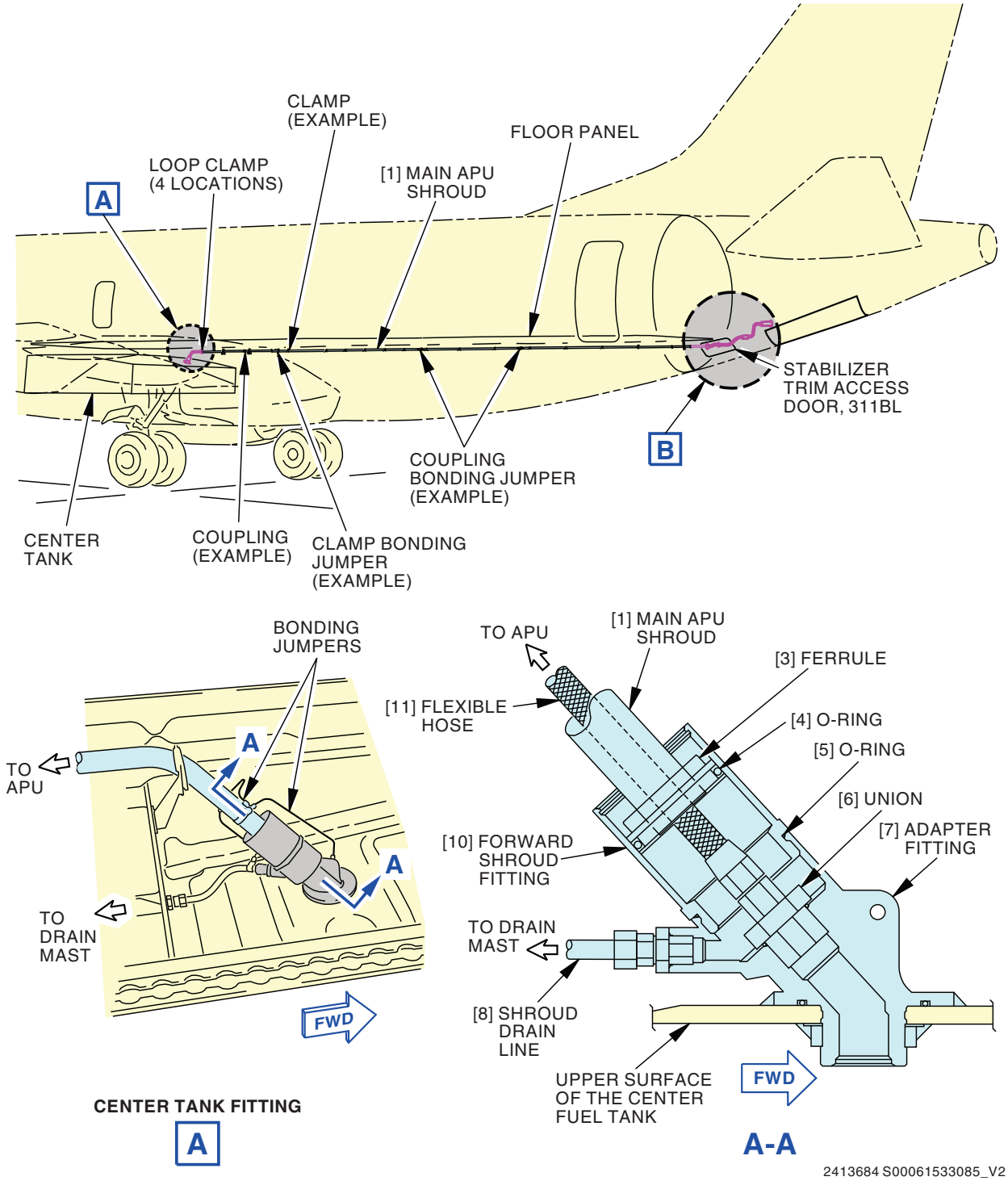
(18) Lift the main APU shroud [1] until it is clear of the floor panels.

SUBTASK 28-25-05-020-003

(19) Remove the main APU shroud [1] from the airplane through the aft entry door.

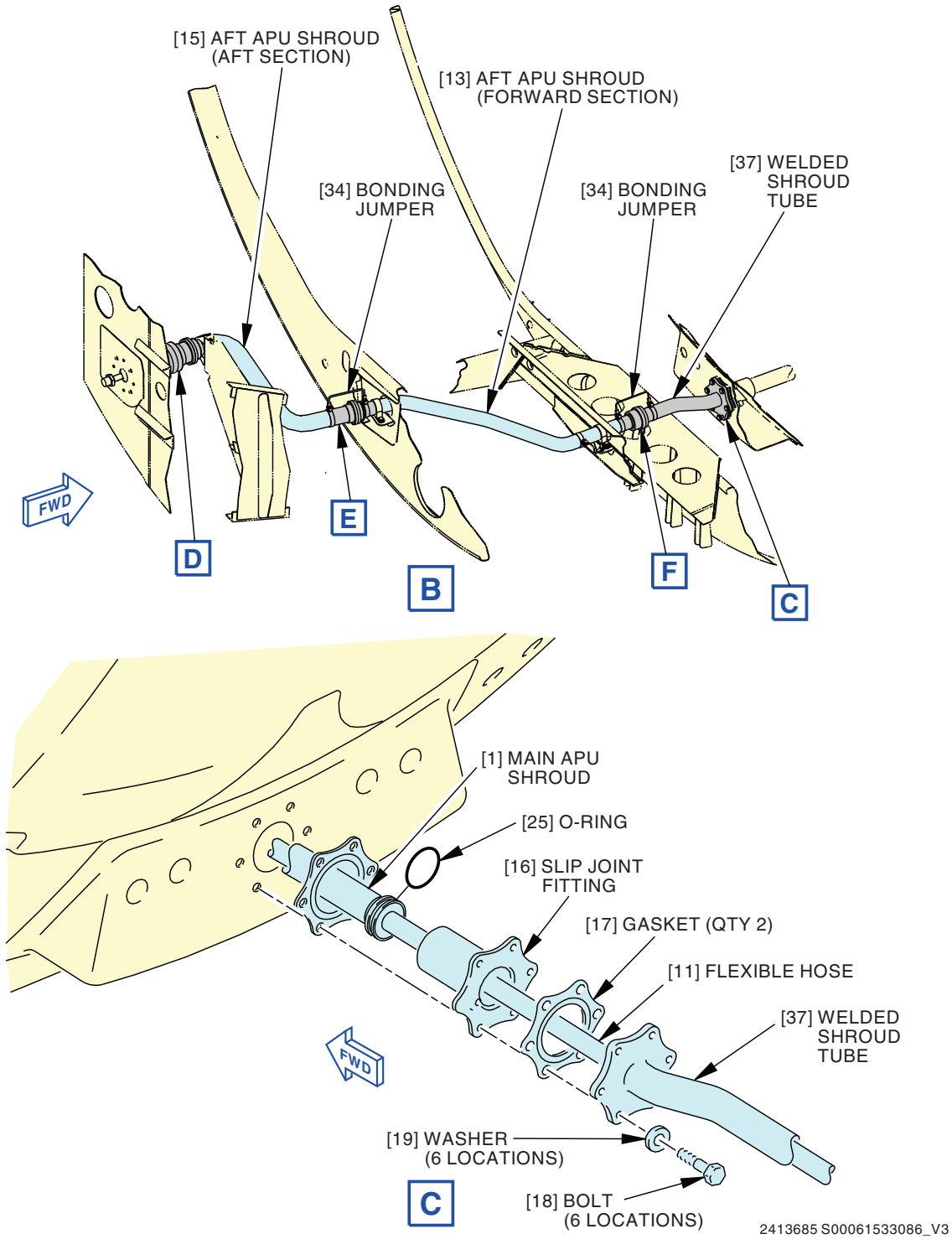
——— END OF TASK ———

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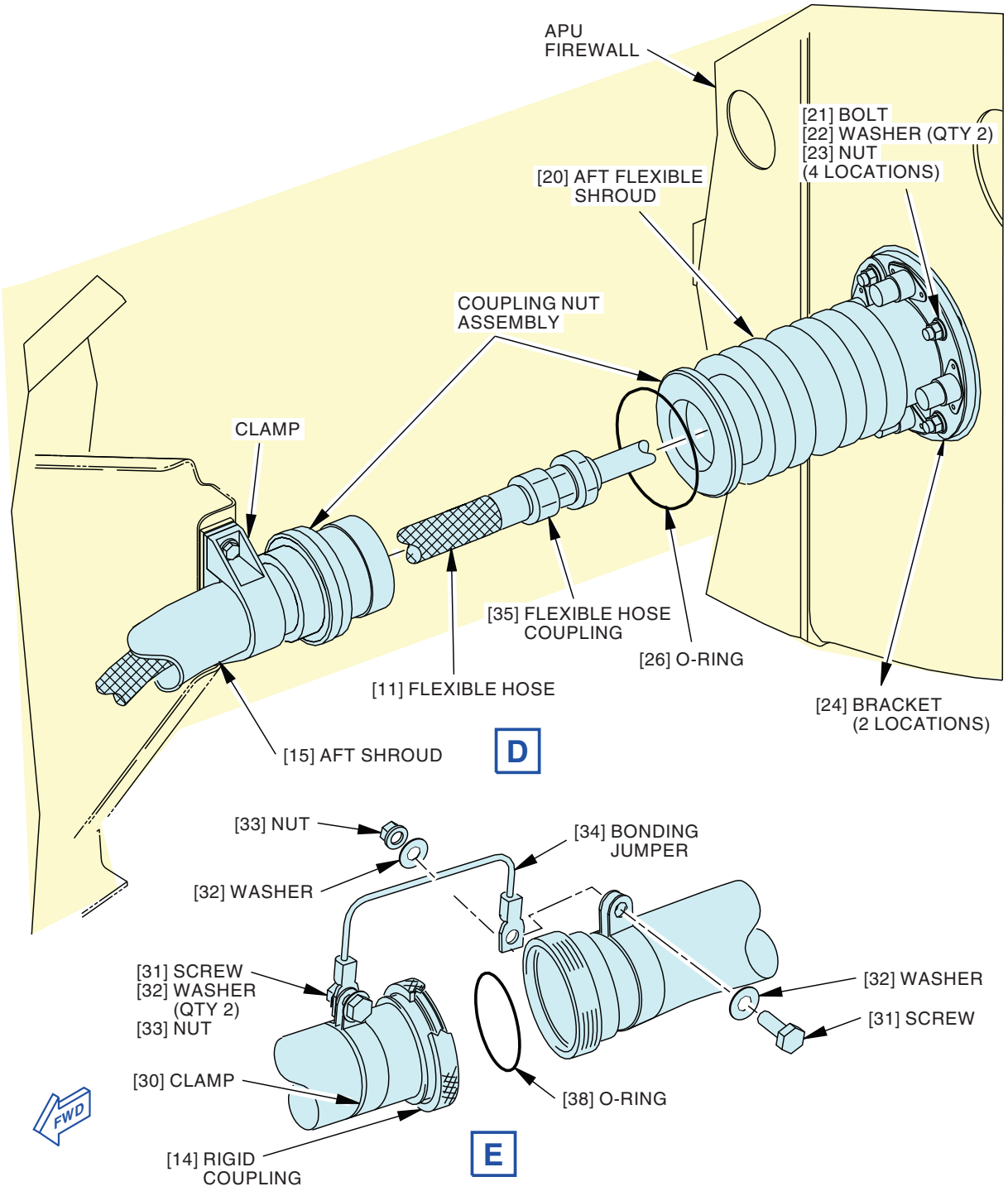
APU Fuel Feed Line Shroud Installation
Figure 401/28-25-05-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



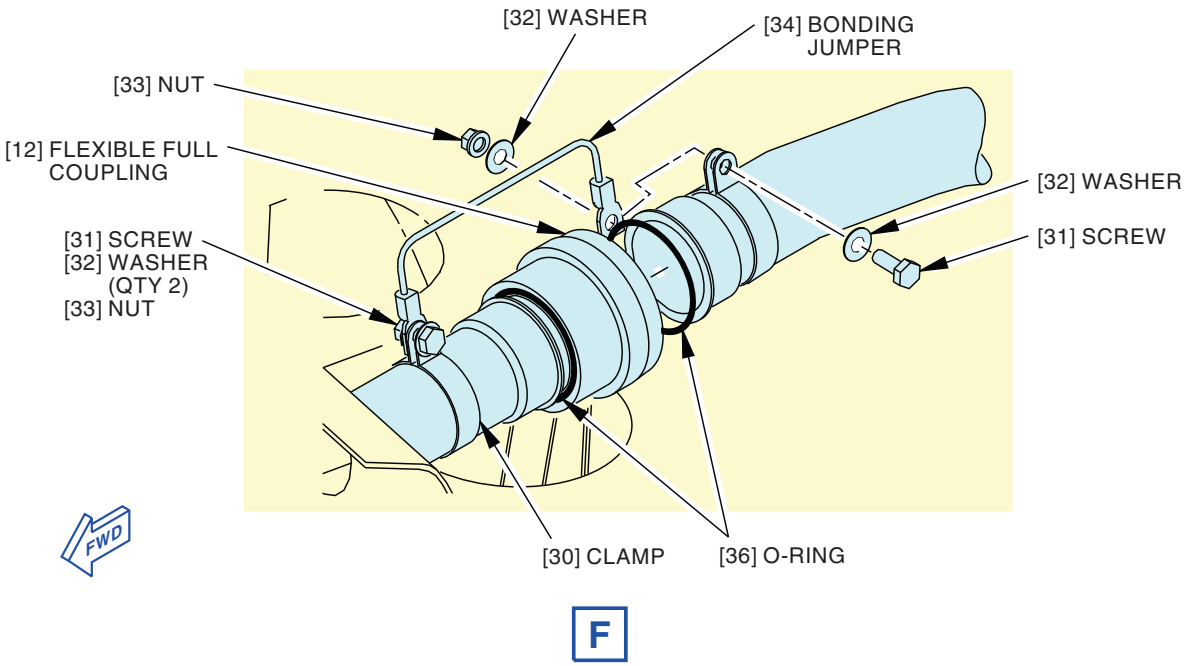
APU Fuel Feed Line Shroud Installation
Figure 401/28-25-05-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Fuel Feed Line Shroud Installation
Figure 401/28-25-05-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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APU Fuel Feed Line Shroud Installation
Figure 401/28-25-05-990-801 (Sheet 4 of 4)

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TASK 28-25-05-400-801

3. Main APU Fuel Feed Line Shroud Installation

(Figure 401)

A. General

(1) This task gives instructions for the installation of the main APU fuel feed line shroud.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
20-50-11-910-801	Standard Torque Values (P/B 201)
25-27-15-400-801	Carpet - Installation (P/B 401)
28-22-15-400-801	Fuel Line, Fitting and Coupling Installation (P/B 401)
28-22-15-400-803	Couplings Installation (P/B 401)
28-25-04-790-801	APU Fuel Line Leak Test (with air) (P/B 401)
28-25-05-790-801	APU Fuel Line Shroud and Drain Line - Leak Test (P/B 501)
51-22-00-390-806	Fillet Seal Application (P/B 201)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)
SWPM 20-20-00 Paragraph 21	MAXIMUM PERMITTED RESISTANCE OF ELECTRICAL BONDS
SWPM 20-20-00 Paragraph 8	Clamp and Tube or Conduit Bonds
SWPM 20-20-10	Replacement of Ground Studs and Bonding Jumper Installation

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
C50475	Coating - Chemical Conversion - Bonderite M-CR 600 Aero	BAC5719 Type II Class D

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(Continued)

Reference	Description	Specification
D00230	Oil - Lubricating, Aircraft Piston Engines (Non-Dispersant Mineral Oil)	SAE J1966
D00504	Grease - Petrolatum	VV-P-236
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50375	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule)	BACC13AT3K, AMS 5689

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Main APU shroud	28-25-05-01-090	PTF ALL
		28-25-05-10-125	PTF ALL
		28-25-05-10-130	PTF ALL
		28-25-05-10-140	PTF ALL
4	O-ring	28-25-05-01-135	PTF ALL
5	O-ring	28-25-05-01-155	PTF ALL
13	Aft APU shroud	28-25-05-20-200	PTF ALL
15	Aft APU shroud	28-25-05-20-210	PTF ALL
17	Gasket	28-25-05-20-225	PTF ALL
25	O-ring	28-25-05-10-120	PTF ALL
26	O-ring	28-25-05-20-180	PTF ALL

F. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left

G. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

H. Main APU Fuel Feed Line Shroud Installation

SUBTASK 28-25-05-420-001

(1) Load the main APU shroud [1] through the aft entry door.

(a) Feed the aft end of the main APU shroud [1] through the opening in the passenger cabin floor above the rear spar of the center wing section.

(b) Move the main APU shroud [1] aft through the floor beams.

(c) Align the main APU shroud [1] in the correct position.

SUBTASK 28-25-05-420-003

(2) Install the main APU shroud [1] in the slip joint fitting [16] on the pressure bulkhead fuel pan as follows (View C, Figure 401):

(a) Lightly coat a new O-ring [25] with grease, D00504.

(b) Install the new O-ring [25] on the aft end fitting of the main APU shroud [1].

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(c) Align the slip joint fitting [16], the two new gaskets [17], welded shroud tube [37] and the main APU shroud [1] to the pressure bulkhead fuel pan (View C, Figure 401).

(d) Install the six bolts [18] and washers [19] for the slip joint fitting [16] and the new gaskets [17].

(e) Tighten the bolts [18] on the welded shroud tube [37] .

SUBTASK 28-25-05-420-004

(3) Install the forward shroud fitting [10] as follows:

(a) Lightly coat a new O-ring [4] with grease, D00504.

(b) Install the new O-ring [4] on the ferrule [3].

(c) Install a new O-ring [5] on the forward shroud fitting [10].

(d) Install the forward shroud fitting [10] on the adapter fitting [7] next to the center tank.

(e) Tighten the forward shroud fitting [10] to 222 ±18 in-lb (25 ±2 N·m).

(f) Install the bonding jumpers [34] as follows:

1) Clean the fay surfaces of the adapter fitting [7], washers [32], and terminal of the bonding jumper [34] (SWPM 20-20-10).

2) Connect the bonding jumper to the adapter fitting [7] with a screw [31], three washers [32], and a nut [33].

3) Tighten the nut [33] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

4) Measure the electrical bonding resistance between the terminal on the bonding jumper [34] and the adapter fitting [7] with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-10).

a) Make sure that the electrical bonding resistance between the terminal on the bonding jumper [34] and the adapter fitting [7] is 0.0010 ohm (1.0 milliohms) or less.

5) Apply a fillet seal of sealant, A02315, to the bonding jumper [34] and fasteners.

6) Refinish the abraded surface of adapter fitting [7].

a) Apply coating, C00064, or Bonderite M-CR 600 Aero coating, C50475.

b) Apply one layer of primer, C00259.

7) Clean the fay surfaces of the main APU shroud [1], clamp [30], and terminals of the bonding jumpers [34] (SWPM 20-20-00 Paragraph 8).

8) Connect the bonding jumpers [34] to the main APU shroud [1] and clamp [30] with a screw [31], two washers [32], and a nut [33].

9) Tighten the nut [33] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

10) Measure the electrical bonding resistance between the terminal on the bonding jumper [34] and the main APU shroud [1] with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00 Paragraph 21).

a) Make sure the electrical bonding resistance between the terminal on the bonding jumper [34] and the main APU shroud [1] is 0.0009 ohm (0.9 milliohms) or less.

11) Apply a fillet seal of sealant, A02315, on both edges of the clamp band.

12) Apply sealant, A02315, to fill the void between the clamp tabs.

13) Apply a fillet seal of sealant, A02315, to the bonding jumpers [34] and fasteners.

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a) Make sure that the gap between the fastener and clamp band is filled with sealant to prevent control cable hang-up.

SUBTASK 28-25-05-420-005

(4) Install the loop clamps for the main APU shroud [1] (TASK 28-22-15-400-801).

(a) Loosely install the clamps [30].

SUBTASK 28-25-05-420-049

(5) Install the applicable couplings for the main APU shroud [1] (TASK 28-22-15-400-803).

SUBTASK 28-25-05-010-004

(6) Clean the fay surfaces of the main APU shroud [1], clamp [30] and the terminal of the bonding jumper [34] (3 locations) (SWPM 20-20-00, Paragraph 8).

SUBTASK 28-25-05-420-024

(7) Install the clamp [30], screw [31], two washers [32], nut [33] and bonding jumper [34] to three locations on the main APU shroud [1].

(a) Install each clamp [30] aft of the forward shroud fitting [10] at a 90 +20 / -0°arc (1.6 +0.4 / -0.0 rad) from the main APU shroud [1] (2 locations).

(b) Tighten the nut [33] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 28-25-05-760-001

(8) Measure the bonding resistance between the main APU shroud [1] and the structure of the airplane with an intrinsically safe approved bonding meter, COM-1550 (3 locations) (SWPM 20-20-00).

(a) Make sure that the bonding resistance from the main APU shroud [1] to the structure is 0.0009 ohm (0.9 milliohm) or less.

SUBTASK 28-25-05-420-050

(9) Tighten the loop clamps and clamps.

SUBTASK 28-25-05-390-001

(10) Apply a fillet seal of sealant, A02315, around the clamp [30] and the main APU shroud [1] (TASK 51-22-00-390-806).

SUBTASK 28-25-05-390-005

(11) Apply sealant, A02315, to fill the void between the clamp tabs.

SUBTASK 28-25-05-390-006

(12) Apply a fillet seal of sealant, A02315, to the bonding jumpers [34] and fasteners.

SUBTASK 28-25-05-420-006

(13) Connect the flexible hose [11].

(a) Feed the flexible hose [11] through the aft APU shroud [13].

(b) Move the flexible hose [11] in the forward direction until it touches the center tank adapter fitting [7].

SUBTASK 28-25-05-020-017

(14) Remove the forward shroud fitting [10] from the center tank adapter fitting [7].

SUBTASK 28-25-05-600-001

(15) Lightly coat a new O-ring [5] with grease, D00504.

SUBTASK 28-25-05-420-026

(16) Install the new O-ring [5] on the forward shroud fitting [10].

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SUBTASK 28-25-05-600-002

(17) Lubricate the threads with oil, D00230, on the forward end of the flexible hose [11].

SUBTASK 28-25-05-420-025

(18) Connect the flexible hose coupling [35] to the aft flexible shroud [20] at the APU firewall (View D, Figure 401).

SUBTASK 28-25-05-420-036

(19) Tighten the flexible hose coupling [35] to 266 in-lb (30 N·m) - 294 in-lb (33.2 N·m).

NOTE: Do not attach the flange for the aft flexible shroud [20] until a fuel leak check is complete.

SUBTASK 28-25-05-790-001

(20) Do this task: APU Fuel Line Leak Test (with air), TASK 28-25-04-790-801.

SUBTASK 28-25-05-600-004

(21) Lightly coat a new O-ring [26] with grease, D00504.

SUBTASK 28-25-05-420-028

(22) Install the new O-ring [26] on the aft APU shroud [15].

SUBTASK 28-25-05-420-037

(23) Connect the coupling nut assembly on the aft APU shroud [15] to the aft flexible shroud [20] at the APU firewall (View D, Figure 401).

SUBTASK 28-25-05-420-038

(24) Tighten the coupling nut assembly (TASK 20-50-11-910-801).

SUBTASK 28-25-05-420-040

(25) Install safety cable kit, G50375 or MS20995C32 lockwire, G01048 on the coupling nut assembly (TASK 20-10-44-400-801).

SUBTASK 28-25-05-420-008

(26) Install the forward shroud fitting [10] (View A-A, Figure 401).

(a) Align the forward shroud fitting [10] in the center tank adapter fitting [7].

(b) Tighten the forward shroud fitting [10] to 660 in-lb (75 N·m) - 730 in-lb (82 N·m).

(c) Install the snap-ring [2] on the forward shroud fitting [10].

SUBTASK 28-25-05-790-002

(27) Do this task: APU Fuel Line Shroud and Drain Line - Leak Test, TASK 28-25-05-790-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 28-25-05-410-001

(1) Install the floor panels (TASK 53-21-00-400-801).

SUBTASK 28-25-05-410-002

(2) Install the aisle carpets (TASK 25-27-15-400-801).

SUBTASK 28-25-05-410-003

(3) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

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SUBTASK 28-25-05-440-001

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 28-25-05-440-002

(5) On the P5 Overhead Panel, put the APU master switch in the ON position.

——— END OF TASK ———

TASK 28-25-05-000-802

4. Aft APU Fuel Feed Line Shroud Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the Aft APU Fuel Feed Line Shroud.

(2) The APU fuel feed line shroud is referred to as the APU shroud.

B. References

Reference	Title
20-10-44-000-801	Lockwire, Cotter Pins, and Lockrings - Removal (P/B 401)
28-22-15-000-801	Fuel Line, Fitting and Coupling Removal (P/B 401)
28-22-15-000-803	Couplings Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-6646	Container - Fuel Resistant

D. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Removal

SUBTASK 28-25-05-040-003

(1) On the P5 Overhead Panel, put the APU master switch to the OFF position.

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SUBTASK 28-25-05-040-004

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 28-25-05-010-002

(3) Open this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Aft APU Fuel-Feed Line Shroud Removal

SUBTASK 28-25-05-020-004

WARNING

MAKE SURE YOU KEEP THE ELECTRICAL POWER OFF WHEN YOU DISCONNECT OR CONNECT THE FUEL, THE HYDRAULIC, AND THE ELECTRICAL LINES. IF YOU DO NOT, YOU CAN CAUSE INJURY TO PERSONNEL WITH THE ACCIDENTAL APPLICATION OF PRESSURIZED FLUIDS, ENERGIZING OF ELECTRICAL CIRCUITS, OR FIRE.

WARNING

YOU MUST ELECTRICALLY GROUND AND BOND THE AIRPLANE BEFORE YOU DO MAINTENANCE ON THE FUEL SYSTEM. IF YOU DO NOT GROUND AND BOND THE AIRPLANE, STATIC ELECTRICITY CAN CAUSE A FIRE OR EXPLOSION.

(1) Disconnect the flexible hose [11].

(a) Remove the safety cable or the lockwire from the coupling nut assembly (TASK 20-10-44-000-801).

(b) Loosen the coupling nut assembly.

(c) Disconnect the aft APU shroud [15] from the aft flexible shroud [20] at the APU firewall.

(d) Discard the O-ring [26].

(e) Push the aft flexible shroud [20] aft to get access to the flexible hose coupling [35].

(f) Put a fuel resistant container, STD-6646, below the flexible hose coupling [35].

(g) Disconnect the flexible hose coupling [35] at the APU firewall.

SUBTASK 28-25-05-020-030

(2) Remove the clamp [30], screw [31], two washers [32], nut [33] and bonding jumper [34] from the aft APU shroud [13].

SUBTASK 28-25-05-020-034

(3) Remove the flexible full coupling [12] that connects the aft APU shroud [13] to the welded shroud tube [37] (TASK 28-22-15-000-803).

(a) Discard the two O-rings [36].

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SUBTASK 28-25-05-020-032

(4) Remove the clamp [30], screw [31], two washers [32], nut [33] and bonding jumper [34] from the aft APU shroud [15].

SUBTASK 28-25-05-020-035

(5) Remove the rigid coupling [14] that connects the aft end of the aft APU shroud [13] to the aft APU shroud [15] (TASK 28-22-15-000-803).

(a) Discard the O-ring [38].

SUBTASK 28-25-05-020-005

(6) Remove the aft APU shroud [13] and aft APU shroud [15].

(a) Remove the six bolts [18] and washers [19] that connect the slip joint fitting [16] to the aft pressure bulkhead (View C, Figure 401).

(b) Remove the APU shroud clamps that hold the aft APU shroud [13] and the aft APU shroud [15] (TASK 28-22-15-000-801), (View B, Figure 401).

SUBTASK 28-25-05-020-020

(7) Move the aft APU shroud [13], aft APU shroud [15], slip joint fitting [16], welded shroud tube [37], O-ring [25] and the gaskets [17] off of the flexible hose [11].

(a) Discard the two gaskets [17] and O-ring [25].

SUBTASK 28-25-05-010-005

(8) Remove the shroud and the gaskets from the airplane through this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

TASK 28-25-05-400-802

5. Aft APU Fuel Feed Line Shroud Installation

(Figure 401)

A. General

(1) This task gives instructions to install the Aft APU Fuel Feed Line Shroud.

(2) The APU fuel feed line shroud is referred to as the APU shroud.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
20-50-11-910-801	Standard Torque Values (P/B 201)
28-22-15-400-801	Fuel Line, Fitting and Coupling Installation (P/B 401)
28-22-15-400-803	Couplings Installation (P/B 401)
28-25-04-790-801	APU Fuel Line Leak Test (with air) (P/B 401)
28-25-05-790-801	APU Fuel Line Shroud and Drain Line - Leak Test (P/B 501)
51-22-00-390-806	Fillet Seal Application (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
D00230	Oil - Lubricating, Aircraft Piston Engines (Non-Dispersant Mineral Oil)	SAE J1966
D00504	Grease - Petrolatum	VV-P-236
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50375	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule)	BACC13AT3K, AMS 5689

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
13	Aft APU shroud	28-25-05-20-200	PTF ALL
15	Aft APU shroud	28-25-05-20-210	PTF ALL
17	Gasket	28-25-05-20-225	PTF ALL
25	O-ring	28-25-05-10-120	PTF ALL
26	O-ring	28-25-05-20-180	PTF ALL
36	O-ring	28-25-05-01-085	PTF ALL
		28-25-05-10-115	PTF ALL
		28-25-05-20-160	PTF ALL
38	O-ring	28-25-05-20-170	PTF ALL

F. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left

G. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

H. Aft APU Fuel Feed Line Shroud Installation

SUBTASK 28-25-05-420-009

(1) Install the aft APU shroud [13].

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(a) Put the two new gaskets [17], slip joint fitting [16], and the welded shroud tube [37] on the flexible hose [11] (View C, Figure 401).

1) Align the new gaskets [17], slip joint fitting [16], and the welded shroud tube [37] in their position at the aft pressure bulkhead.

2) Put the aft APU shroud [13] over the flexible hose [11] in the forward direction until it touches the pressure bulkhead.

SUBTASK 28-25-05-420-043

(2) Put one retaining ring on the ferrule of the aft APU shroud [13] and welded shroud tube [37].

SUBTASK 28-25-05-820-007

(3) Align the aft APU shroud [13] and welded shroud tube [37].

SUBTASK 28-25-05-420-030

(4) Install the flexible full coupling [12] that connects the aft APU shroud [13] to the welded shroud tube [37] (TASK 28-22-15-400-803).

(a) Install the two new O-rings [36].

SUBTASK 28-25-05-100-002

(5) Clean the fay surfaces of the aft APU shroud [13], clamp [30] and the terminal of the bonding jumper [34] (SWPM 20-20-00, Paragraph 8).

SUBTASK 28-25-05-640-004

(6) Lightly coat a new O-ring [25] with grease, D00504.

SUBTASK 28-25-05-420-047

(7) Install the new O-ring [25]

SUBTASK 28-25-05-820-002

(8) Align the slip joint fitting [16], two new gaskets [17], new O-ring [25], welded shroud tube [37] and the aft APU shroud [13] (View C, Figure 401).

SUBTASK 28-25-05-420-031

(9) Install the six bolts [18] and washers [19] to the slip joint fitting [16] and the welded shroud tube [37] (Figure 401).

SUBTASK 28-25-05-420-044

(10) Tighten the six bolts [18] on the welded shroud tube [37].

SUBTASK 28-25-05-420-032

(11) Install the clamp [30], screw [31], two washers [32], nut [33] and bonding jumper [34] to the aft APU shroud [13].

(a) Tighten the nut [33] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 28-25-05-760-002

(12) Measure the bonding resistance between the aft APU shroud [13] and the structure of the airplane with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) Make sure that the bonding resistance from the aft APU shroud [13] to the structure is 0.0009 ohm (0.9 milliohm) or less.

SUBTASK 28-25-05-390-002

(13) Apply a fillet seal of sealant, A02315, around the clamp [30] and the aft APU shroud [13] (TASK 51-22-00-390-806).

SUBTASK 28-25-05-390-007

(14) Apply sealant, A02315, to fill the void between the clamp tabs.

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SUBTASK 28-25-05-390-008

(15) Apply a fillet seal of sealant, A02315, to the bonding jumpers [34] and fasteners.

SUBTASK 28-25-05-820-005

(16) Put the aft APU shroud [15] over the flexible hose [11] until the flexible hose [11] and the aft APU shroud [15] can be reconnected at the aft flexible shroud [20].

SUBTASK 28-25-05-820-006

(17) Put a retaining ring on the ferrule to hold the nut on the APU shroud.

SUBTASK 28-25-05-820-003

(18) Align the aft APU shroud [13] and aft APU shroud [15].

SUBTASK 28-25-05-420-033

(19) Install the rigid coupling [14] that connects the aft end of the aft APU shroud [13] to the aft APU shroud [15] (TASK 28-22-15-400-803).

(a) Install the new O-ring [38].

SUBTASK 28-25-05-100-001

(20) Clean the fay surfaces of the aft APU shroud [15], clamp [30] and the terminal of the bonding jumper [34] (SWPM 20-20-00, Paragraph 8).

SUBTASK 28-25-05-420-034

(21) Install the clamp [30], screw [31], two washers [32], nut [33] and bonding jumper [34] to the aft APU shroud [15].

(a) Tighten the nut [33] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 28-25-05-760-003

(22) Measure the bonding resistance between the aft APU shroud [15] and the structure of the airplane with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) Make sure that the bonding resistance from the aft APU shroud [15] to the structure is 0.0009 ohm (0.9 milliohm) or less.

SUBTASK 28-25-05-390-003

(23) Apply a fillet seal of sealant, A02315, around the clamp [30] and the aft APU shroud [15] (TASK 51-22-00-390-806).

SUBTASK 28-25-05-390-009

(24) Apply sealant, A02315, to fill the void between the clamp tabs.

SUBTASK 28-25-05-390-010

(25) Apply a fillet seal of sealant, A02315, to the bonding jumpers [34] and fasteners.

SUBTASK 28-25-05-420-014

(26) Install the clamps for the aft APU shroud [13] and the aft APU shroud [15] (TASK 28-22-15-400-801).

(a) Loosely attach the clamps to the pressure bulkhead.

SUBTASK 28-25-05-420-010

(27) Lubricate the threads with oil, D00230, on the aft end of the flexible hose [11].

SUBTASK 28-25-05-420-015

(28) Connect the flexible hose [11] at the APU firewall.

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SUBTASK 28-25-05-420-016

(29) Tighten the flexible hose coupling [35] to 266 in-lb (30 N·m) - 294 in-lb (33.2 N·m).

NOTE: Do not attach the aft APU shroud [15] to the aft flexible shroud [20] until a leak fuel check is done.

SUBTASK 28-25-05-790-003

(30) Do this task: APU Fuel Line Leak Test (with air), TASK 28-25-04-790-801.

SUBTASK 28-25-05-420-011

(31) Lightly coat the new O-ring [26] with grease, D00504.

SUBTASK 28-25-05-420-017

(32) Install a new O-ring [26] on the aft APU shroud [15].

SUBTASK 28-25-05-420-018

(33) Connect the coupling nut assembly on the aft APU shroud [15] to the aft flexible shroud [20] at the APU firewall.

SUBTASK 28-25-05-420-019

(34) Tighten the coupling nut assembly (TASK 20-50-11-910-801).

SUBTASK 28-25-05-420-020

(35) Install safety cable kit, G50375 or MS20995C32 lockwire, G01048 on the coupling nut assembly (TASK 20-10-44-400-801).

SUBTASK 28-25-05-790-004

(36) Do this task: APU Fuel Line Shroud and Drain Line - Leak Test, TASK 28-25-05-790-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 28-25-05-410-004

(1) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 28-25-05-440-003

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 28-25-05-440-004

(3) On the P5 Overhead Panel, put the APU master switch to the ON position.

——— END OF TASK ———

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APU FUEL FEED LINE SHROUD - ADJUSTMENT/TEST

1. General

A. This procedure has the following test:

(1) APU Fuel Line Shroud and Drain Line - Leak Test.

TASK 28-25-05-790-801

2. APU Fuel Line Shroud and Drain Line - Leak Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1782	Test Hose - Leak and Pressure, APU Fuel Line Shroud Part #: A28005-82 Supplier: 81205 Opt Part #: A28005-42 Supplier: 81205 Opt Part #: A28005-50 Supplier: 81205
SPL-1783	Adapter - Leak Test, APU Fuel Line Shroud Part #: A28005-48 Supplier: 81205
STD-1234	Air Source - Regulated, Dry Filtered, Compressed 90-130 PSIG (6.0 CFM)

C. Location Zones

Zone	Area
194	Lower Wing-To-Body Fairing - Aft of Wheel Well

D. Pressurize the APU Shroud and Do a Check for Leaks

SUBTASK 28-25-05-400-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-25-05-420-042

(2) Connect the adapter, SPL-1783, and test hose, SPL-1782, on the drain mast.

SUBTASK 28-25-05-420-012

(3) Connect the compressed (90-130 PSIG, 6.0 CFM) dry filtered regulated air source, STD-1234, to the test hose, SPL-1782.

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SUBTASK 28-25-05-780-001

WARNING BE CAREFUL WHEN YOU USE HIGH PRESSURE AIR AS A TEST MEDIUM. HIGH PRESSURE AIR CAN CAUSE AIRFRAME DAMAGE AND CAN CAUSE INJURY.

CAUTION DO NOT PRESSURIZE THE SHROUD MORE THAN 20 PSIG (34.75 PSIA). IF YOU PRESSURIZE TO MORE THAN 20 PSIG (34.75 PSIA), DAMAGE TO THE SHROUD CAN OCCUR.

(4) Apply pressure slowly and continuously to the APU shroud and drain line system until the pressure is 18.0 ±0.5 psig (124.1 ±3.5 kPa).

(a) Close the air pressure source.

(b) Wait for a minimum of 1 minute to make sure that the pressure is stable.

(c) Make sure that the pressure is stable at 18.0 ±0.5 psig (124.1 ±3.5 kPa).

(d) Record the pressure.

(e) Monitor the pressure for 5 - 6 minutes.

(f) Record the pressure.

(g) Make sure that the pressure does not decrease more than 0.05 psig (0.34 kPa) after the 5 - 6 minutes.

SUBTASK 28-25-05-780-002

(5) Decrease the pressure slowly and continuously to 0 psig (0 kPa).

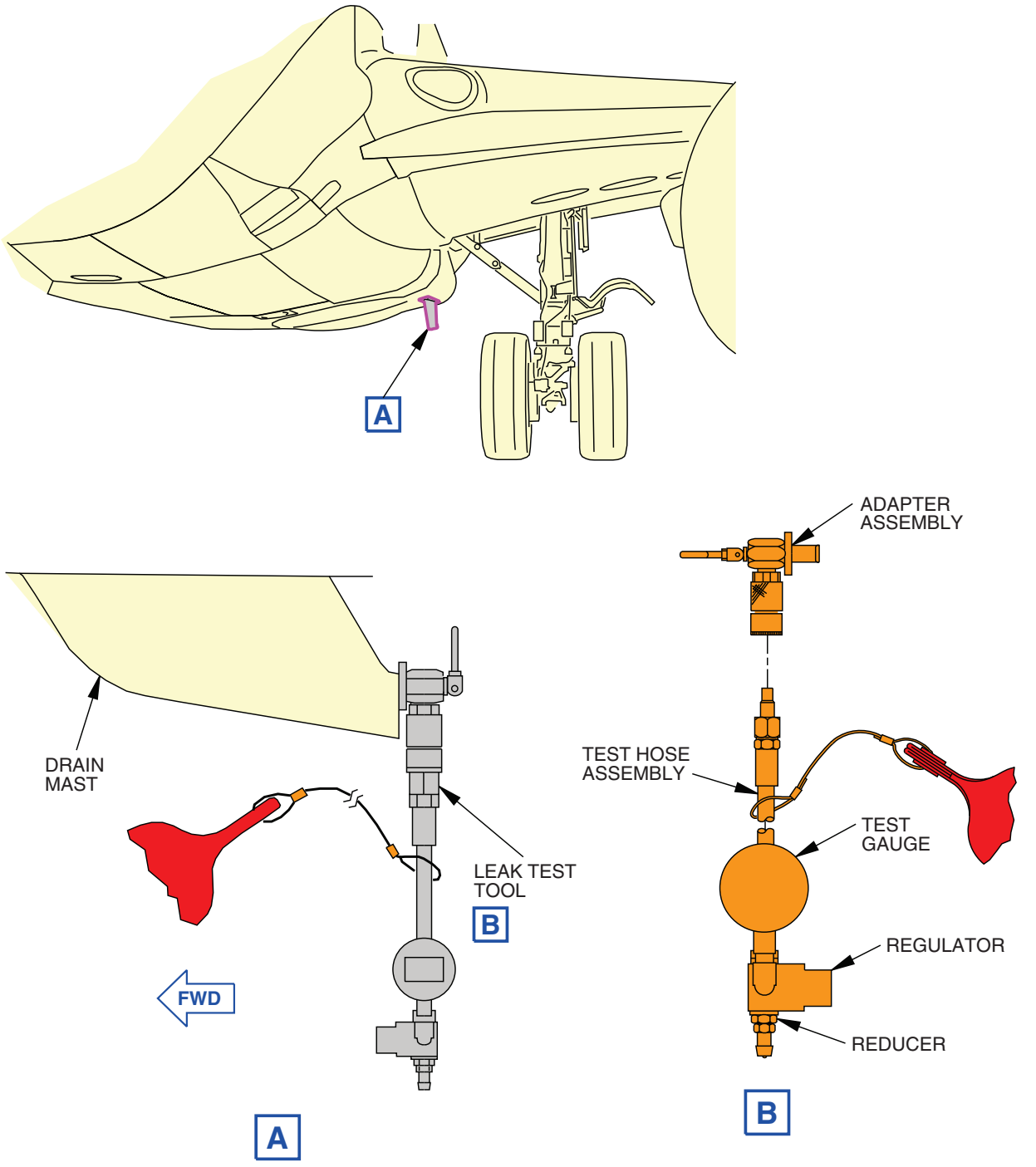
SUBTASK 28-25-05-080-001

(6) Remove the compressed (90-130 PSIG, 6.0 CFM) dry filtered regulated air source, STD-1234, from the test hose, SPL-1782.

(7) Remove the adapter, SPL-1783, and test hose, SPL-1782, from the drain mast.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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APU Shroud Leak Test Equipment
Figure 501/28-25-05-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU CHECK VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) The removal of the APU check valve

(2) The installation of the APU check valve.

TASK 28-25-06-000-801

2. APU Check Valve Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the APU check valve.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-22-00-730-802	Crossfeed Valve - Functional Test (P/B 501)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1054	Container - Fuel Resistant, 5-Gallon (19-Liter)

D. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50

E. Access Panels

Number	Name/Location
131AB	Center Tank Access
532AB	Main Tank Access Door - Wing Station 216

F. Prepare for the Removal

SUBTASK 28-25-06-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-25-06-650-001

(2) Defuel the center fuel tank and the No. 1 tank or transfer fuel out of the tanks (TASK 28-26-00-650-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-25-06-860-001

(3) If there is fuel in the No. 2 tank, make sure that the crossfeed valve is closed and has no leakage (TASK 28-22-00-730-802).

(a) As an alternative, remove fuel from the No. 2 tank (TASK 28-26-00-650-801).

SUBTASK 28-25-06-010-004

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) For the APU check valve in the main tank, remove this access door:

(TASK 28-11-11-000-801)

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216

(b) For the APU check valve in the center tank, remove this access door:

(TASK 28-11-31-000-801)

Number	Name/Location
131AB	Center Tank Access

G. APU Check Valve Removal

SUBTASK 28-25-06-020-001

(1) Remove the APU check valve [2].

(a) Loosen the couplings on the APU fuel feed line [1] and APU fuel feed line [5].

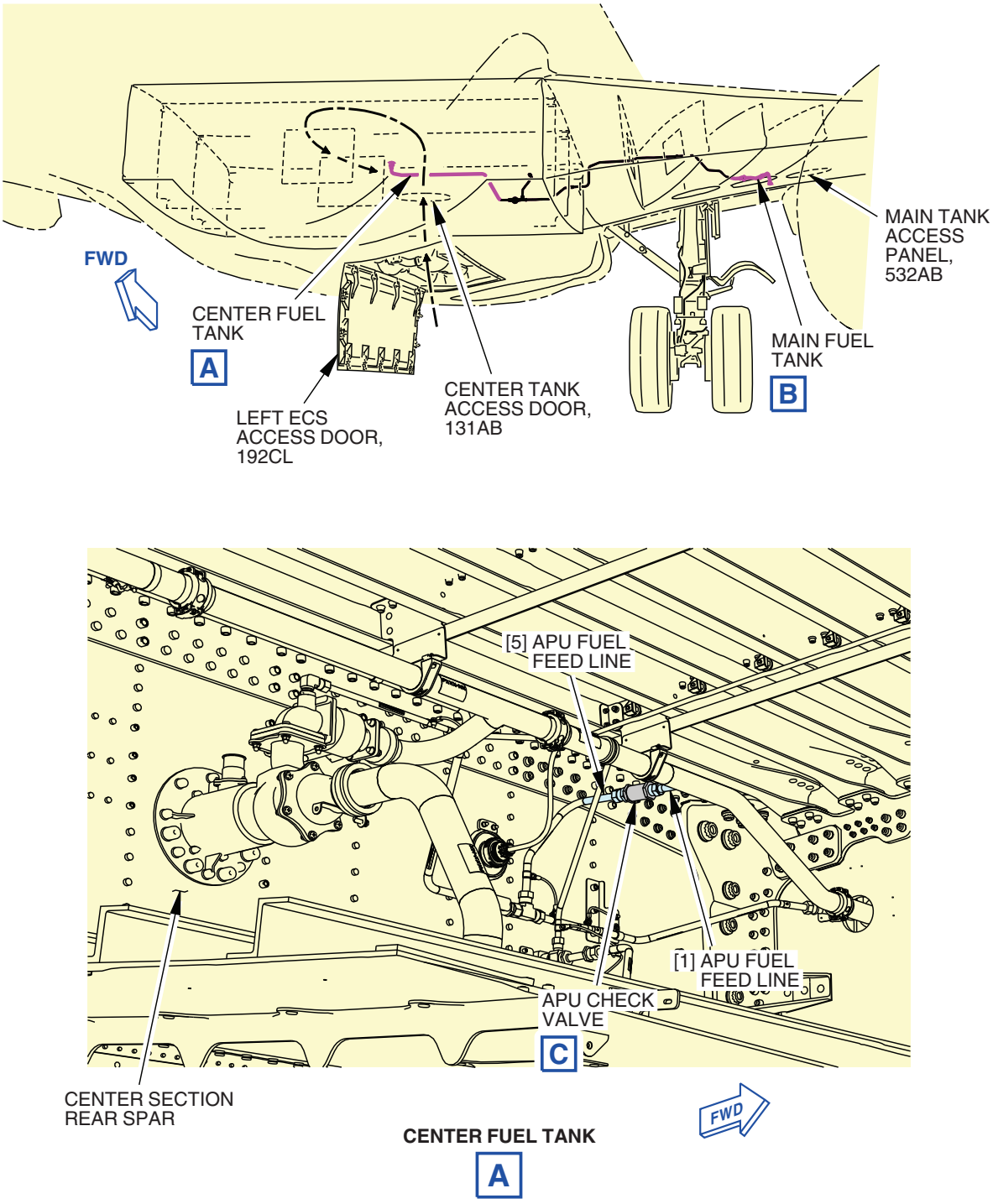
(b) Remove the union [3] and O-ring [4].

(c) Discard the O-ring [4].

(d) Drain the remaining fuel from the APU fuel feed line [1] and APU fuel feed line [5] into the 5-gallon (19-liter) fuel resistant container, STD-1054.

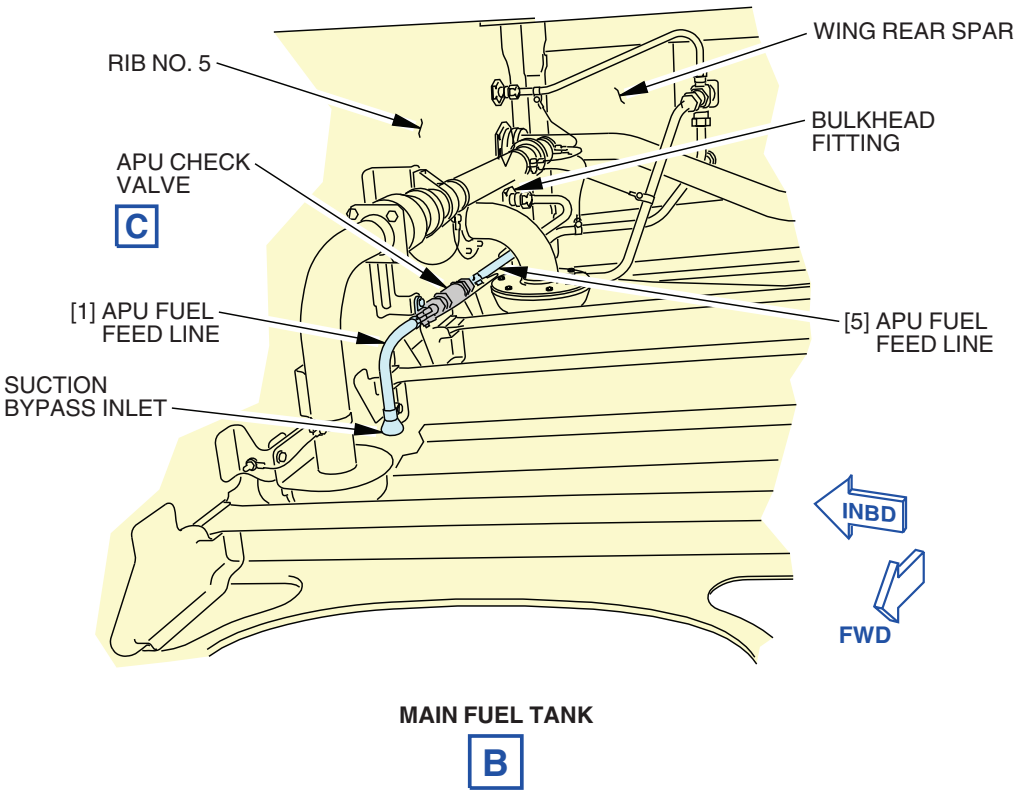
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Check Valve Installation
Figure 401/28-25-06-990-801 (Sheet 1 of 3)

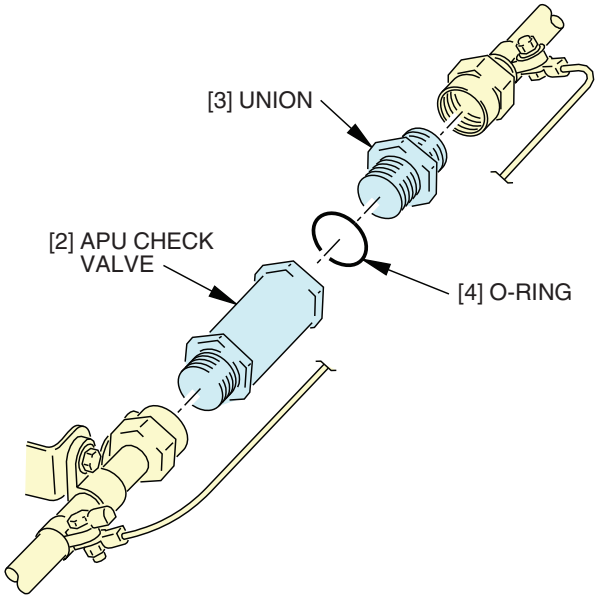
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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APU Check Valve Installation
Figure 401/28-25-06-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU CHECK VALVE (EXAMPLE)

C

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APU Check Valve Installation
Figure 401/28-25-06-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-25-06-400-801

3. APU Check Valve Installation

(Figure 401)

A. General

(1) This task gives instructions to install the APU check valve [2].

B. References

Reference	Title
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
49-31-00-700-802	APU Fuel Supply Flow Check (P/B 201)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	APU check valve	28-25-04-05-100	PTF ALL
		28-25-06-01-020	PTF ALL
4	O-ring	28-25-04-05-090	PTF ALL
		28-25-06-01-015	PTF ALL

D. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left

E. Access Panels

Number	Name/Location
131AB	Center Tank Access

F. APU Check Valve Installation

SUBTASK 28-25-06-420-001

(1) Put the APU check valve [2], new O-ring [4], and union [3] in their position between the two ends of the APU fuel feed lines [1].

(a) Use new O-ring [4].

SUBTASK 28-25-06-420-002

(2) Make sure that the arrow on the APU check valve [2] points in the aft direction (away from the engine fuel feed manifold).

SUBTASK 28-25-06-420-003

(3) Attach the APU check valve [2] to the ends of the APU fuel feed line [1] with the couplings on both ends.

(a) Tighten each of the couplings to a torque of 280 ±14 in-lb (32 ±2 N·m).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 28-25-06-420-004

(1) Do the applicable steps to close the center tank (TASK 28-11-00-410-801).

SUBTASK 28-25-06-010-003

(2) Do this task: Center Tank Access Door - Installation, TASK 28-11-31-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Install this access door for the center wing section:

Number	Name/Location
131AB	Center Tank Access

SUBTASK 28-25-06-710-001

(3) Do this task: APU Fuel Supply Flow Check, TASK 49-31-00-700-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FUEL FEED LINE SHROUD DRAIN MAST - REMOVAL/INSTALLATION

1. General

A. This procedure contains two tasks:

(1) A removal of the APU fuel feed line shroud drain mast

(2) An installation of the APU fuel feed line shroud drain mast.

B. The APU fuel feed line shroud drain mast is referred to as the drain mast in this procedure.

TASK 28-25-07-020-801

2. APU Fuel Feed Line Shroud Drain Mast Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the APU Fuel Feed Line Shroud Drain Mast.

B. References

Reference	Title
28-22-15-000-803	Couplings Removal (P/B 401)

C. Location Zones

Zone	Area
194	Lower Wing-To-Body Fairing - Aft of Wheel Well

D. Access Panels

Number	Name/Location
194DL	Aft Wing To Body Fairing Panel

E. Prepare for the Removal

SUBTASK 28-25-07-010-001

(1) Remove this access panel:

Number	Name/Location
194DL	Aft Wing To Body Fairing Panel

F. APU Fuel Feed Line Shroud Drain Mast Removal

SUBTASK 28-25-07-000-001

(1) Remove the clamp [12], screw [13], washers [14], nut [15], bonding jumper [16], and bonding jumper [19] from the mast hydraulic vent line.

SUBTASK 28-25-07-000-002

(2) Remove the clamp [7], screw [8], washers [9], nut [10], and bonding jumper [11] from the mast APU fuel drain line.

SUBTASK 28-25-07-020-007

(3) Remove the screw [20], washers [21], nut [22], and bonding jumper [19] from the bonding lug of the drain mast assembly [4].

SUBTASK 28-25-07-020-001

(4) Disconnect the coupling [6] from the APU fuel shroud drain line [5] (TASK 28-22-15-000-803).

(a) Remove and discard the two O-rings [18] from the coupling [6].

SUBTASK 28-25-07-020-002

(5) Disconnect the hydraulic vent line [1] and the APU fuel shroud drain line [5] from the drain mast fittings.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the breather plug [17] from the coupling [6].

SUBTASK 28-25-07-020-005

(6) Loosen the four bolts [3] that attach the drain mast assembly [4] to the airplane structure.

SUBTASK 28-25-07-020-006

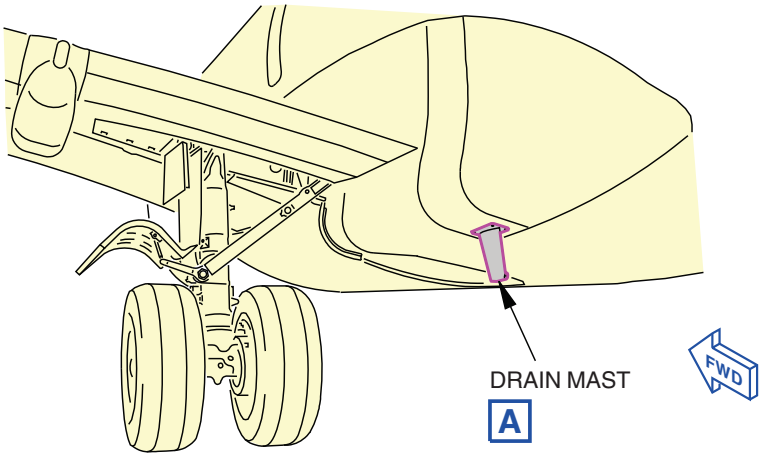
(7) Carefully separate the drain mast assembly [4] from the fairing with a sealant removal tool.

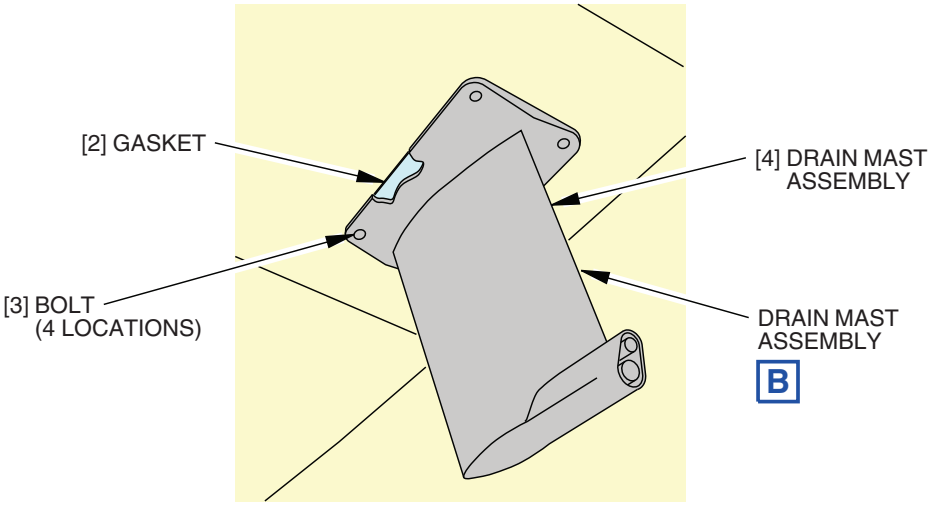
(a) Discard the gasket [2].

(b) Remove the sealant on the drain mast assembly [4].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



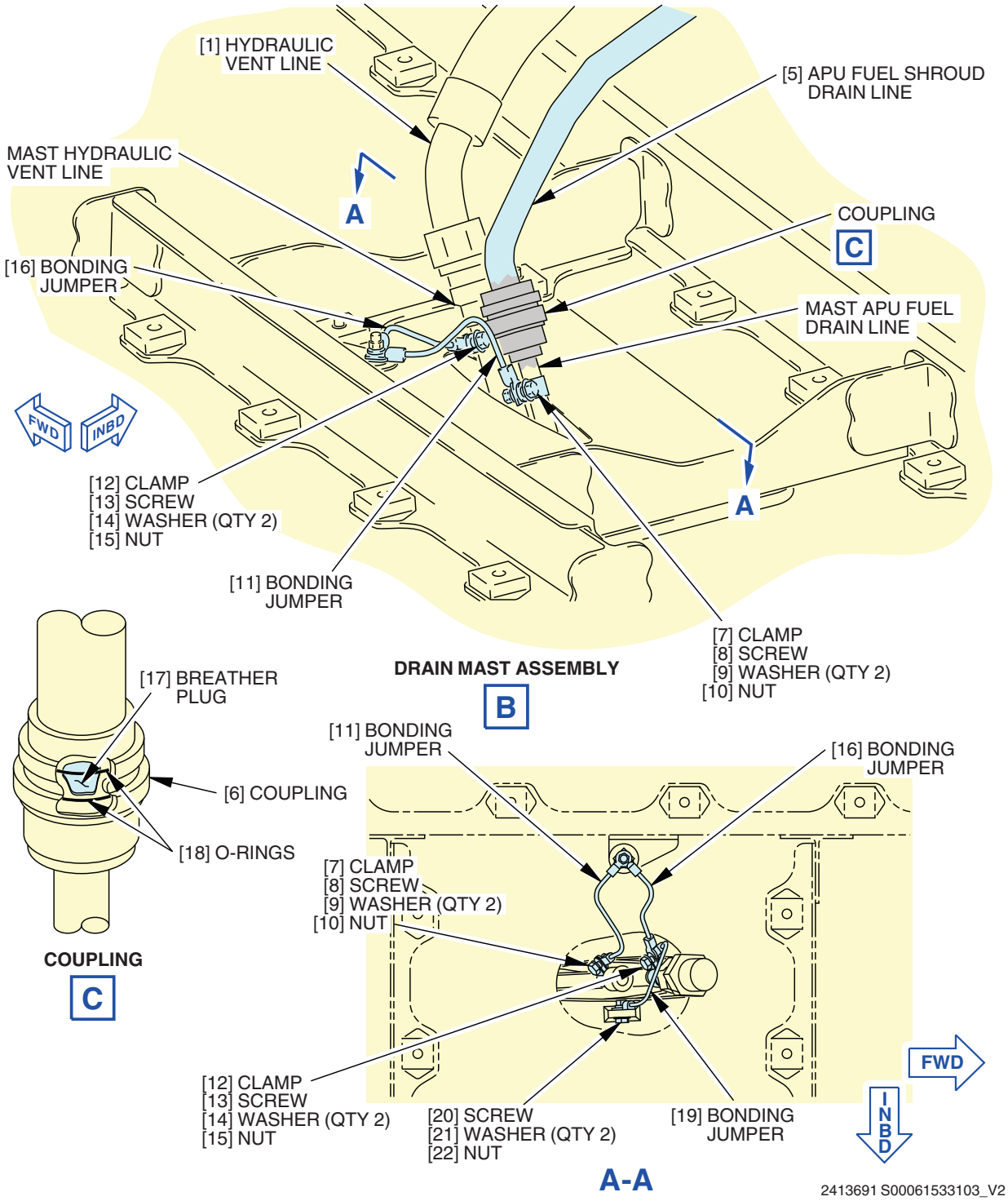


DRAIN MAST
A

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APU Shroud Drain Mast Installation
Figure 401/28-25-07-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Shroud Drain Mast Installation
Figure 401/28-25-07-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-25-07-420-801

3. APU Fuel Feed Line Shroud Drain Mast Installation

(Figure 401)

A. General

(1) This task gives instructions to install the APU Fuel Feed Line Shroud Drain Mast.

B. References

Reference	Title
28-22-15-400-803	Couplings Installation (P/B 401)
28-25-05-790-801	APU Fuel Line Shroud and Drain Line - Leak Test (P/B 501)
51-22-00-390-806	Fillet Seal Application (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-00 Paragraph 8	Clamp and Tube or Conduit Bonds

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Gasket	28-25-07-01-050	PTF ALL
4	Drain mast assembly	28-25-07-01-055	PTF ALL
18	O-ring	29-21-52-08-500	PTF ALL

F. Location Zones

Zone	Area
194	Lower Wing-To-Body Fairing - Aft of Wheel Well

G. Access Panels

Number	Name/Location
194DL	Aft Wing To Body Fairing Panel

H. APU Fuel Feed Line Shroud Drain Mast Installation

SUBTASK 28-25-07-420-001

(1) Do these steps to install the drain mast assembly [4] in its position:

(a) Install a new gasket [2] on the drain mast assembly [4].

(b) Put the drain mast assembly [4] with the gasket [2] in its position on the fairing.

(c) Apply a thin layer of compound, C00528, to the all areas of the holes.

(d) Immediately install and then tighten the bolts [3].

(e) Make sure that the head of each bolt [3] is as near to flush as possible to the drain mast assembly [4].

NOTE: A positive flushness between the bolts and drain mast assembly is not permitted.

SUBTASK 28-25-07-420-002

(2) Connect the hydraulic vent line [1] to the mast hydraulic vent line.

SUBTASK 28-25-07-420-003

(3) Do these steps to connect the APU fuel shroud drain line [5] to the drain mast assembly [4]:

(a) Install the breather plug [17] in the APU fuel shroud drain line [5].

(b) Connect the coupling [6] for the APU fuel shroud drain line [5] and mast APU fuel drain line (TASK 28-22-15-400-803).

1) Install the two new O-rings [18].

SUBTASK 28-25-07-100-001

(4) Use Cleaning Procedure 1 to manually clean the mating surfaces of the mast APU fuel drain line, mast hydraulic vent line, and bonding lug of the drain mast assembly [4] (SWPM 20-20-00).

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SUBTASK 28-25-07-100-002

WARNING	DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(5) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the mast APU fuel drain line, mast hydraulic vent line, bonding lug of the drain mast assembly [4], washers [21], clamp [7], clamp [12], terminal lug of the bonding jumper [11], terminal lug of the bonding jumper [16], and terminal lug of the bonding jumper [19].

SUBTASK 28-25-07-390-003

(6) Apply sealant, A50110, sealant, A50155, sealant, A50231, or sealant, A50296, to the shank and threads of the screw [20] and to the mating surfaces of the screw [20], washers [21], nut [22], and bonding jumper [19].

SUBTASK 28-25-07-420-006

(7) Install the screw [20], washers [21], and nut [22] to attach the bonding jumper [19] to the bonding lug of the drain mast assembly [4].

(a) Tighten the nut [22] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

(b) Make sure that the sealant squeeze-out is continuous around all the interfaces of each component.

SUBTASK 28-25-07-420-004

(8) Install the clamp [7], screw [8], washers [9], nut [10], and bonding jumper [11] to the mast APU fuel drain line.

(a) Tighten the nut [10] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 28-25-07-420-005

(9) Install the clamp [12], screw [13], washers [14], nut [15], bonding jumper [16], and bonding jumper [19] to the mast hydraulic vent line.

(a) Tighten the nut [15] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 28-25-07-760-001

(10) Measure the electrical bonding resistance between the bonding jumper [16] and mast hydraulic vent line (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the resistance is 0.0009 ohm (0.9 milliohms) or less.

SUBTASK 28-25-07-760-002

(11) Measure the electrical bonding resistance between the bonding jumper [11] and mast APU fuel drain line (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the resistance is 0.0009 ohm (0.9 milliohms) or less.

SUBTASK 28-25-07-760-003

(12) Measure the electrical bonding resistance between the bonding jumper [19] and bonding lug of the drain mast assembly [4] (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the resistance is 0.0010 ohm (1 milliohms) or less.

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SUBTASK 28-25-07-760-004

(13) Measure the electrical bonding resistance between the airplane structure and bonding lug of the drain mast assembly [4] (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the resistance is 0.0025 ohm (2.5 milliohms) or less.

SUBTASK 28-25-07-390-001

(14) Apply a fillet seal of sealant, A50337, sealant, A50231, or sealant, A50359, around the clamp [7] and clamp [12] (SWPM 20-20-00 Paragraph 8).

SUBTASK 28-25-07-390-002

(15) Apply a fillet seal of sealant, A02315, around the flange of the drain mast assembly [4] (TASK 51-22-00-390-806).

SUBTASK 28-25-07-790-001

(16) Do this task: APU Fuel Line Shroud and Drain Line - Leak Test, TASK 28-25-05-790-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 28-25-07-010-002

(1) Install this access panel:

Number	Name/Location
194DL	Aft Wing To Body Fairing Panel

——— END OF TASK ———

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DEFUELING - MAINTENANCE PRACTICES

1. General

A. This procedure has three tasks:

(1) Precautions and Limits for the Defuel Operation

(2) Fuel Tank Defueling

(3) Tank to Tank Fuel Transfer

NOTE: For tank to tank fuel transfer, refer to steps to remove (purge) air from the engine fuel feed system (Fuel Tank Closure, TASK 28-11-00-410-801).

B. The heat exchanger for the hydraulic system can become too hot when it is above the surface of the fuel.

(1) Do not operate an engine driven or electric motor driven pump in the hydraulic system if the No. 1 main tank or the No. 2 main tank has less than 1675 lb (760 kg) of fuel (TASK 29-11-00-860-805).

C. All fuel tanks can be defueled at the same time or one at a time. If you defuel only one fuel tank, make sure the fuel and the fuel fumes in the other fuel tanks do not go into the fuel tank you defueled.

D. Stop the defueling procedure when thunderstorms or lightning are in a 10 mi (16 km) radius of the immediate area.

E. Strong wind conditions can cause a build-up of static electricity. Large charges of static electricity can develop on support equipment while parked as a result of the movement of dust particles and air currents during strong wind conditions. Strong wind conditions also cause the unwanted movement of items or equipment which can hit the airplane or injure persons. Wind gusts can damage the airplane structure. Defueling procedures should stop if strong wind conditions are present.

F. You must carefully do all of these safety precautions:

(1) The safety precautions for defueling in this manual

(2) The safety precautions that the airport tells you to do

(3) The safety precautions that the airline tells you to do.

G. If you do not obey these safety precautions, a fire or an accident can easily occur.

H. Low Pressure light Illumination During Defueling and Fuel Transfer

(1) Very high flow conditions can occur during defueling and fuel transfer operations. The high flow condition can decrease the back-pressure in the refuel manifold. The lack of back-pressure can reduce the output pressure of the fuel pumps to a level where the LOW PRESSURE light comes on. During these high flow conditions the LOW PRESSURE light does not indicate that a pump is running dry or that the pump has failed.

NOTE: Back-pressure is in relation to any configuration that can cause low fuel flow to the pumps. A low back-pressure condition can occur even if there is a significant amount of fuel in the tanks during defueling or transferring fuel from tank to tank.

(2) To monitor pump performance during high flow conditions it is necessary to monitor the pump low pressure indication and the fuel quantity indication. If the pump LOW PRESSURE light comes on (steady on) and the fuel quantity for the fuel tank does not continue to decrease, it is possible that the pump is running dry or that the pump has failed. The pump switch must be selected off and remain off until you can evaluate and repair the cause of the no-transfer condition.

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TASK 28-26-00-600-801

2. Precautions and Limits for the Defuel Operation

A. General

(1) If the center tank has more than 1000 lb (454 kg) of fuel, the No. 1 main tank and No. 2 main tank must be filled fully.

NOTE: This requirement is applicable to the fuel configuration for flight. A given quantity of fuel can be added to the center tank or transferred from the No. 1 or No. 2 main tank for ground maintenance (TASK 28-26-00-650-802).

B. References

Reference	Title
49-11-00-710-801	APU Operation Limits (P/B 201)
49-11-00-860-801	APU Starting and Operation (P/B 201)
49-11-00-860-802	APU Usual Shutdown (P/B 201)
49-11-00-860-803	APU Emergency Shutdown (P/B 201)

C. Location Zones

Zone	Area
500	Left Wing
600	Right Wing

D. APU Operations During Defueling Limits and Precautions

SUBTASK 28-26-00-860-001

(1) If the Auxiliary Power Unit (APU) is operating during refueling, do these steps:

WARNING OBEY THE PRECAUTIONS FOR APU OPERATION WHEN YOU OPERATE THE APU DURING THE FUELING OPERATION OF THE AIRPLANE. IF YOU OPERATE THE APU INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(a) Obey the limits and operation for the operation of the APU (TASK 49-11-00-710-801, TASK 49-11-00-860-801).

(b) The APU can be started during defueling if the start is an initial start or a restart after a normal shutdown.

(c) The APU can be shutdown (manual or automatic) during the defueling operation.

WARNING IF THERE IS A PROTECTIVE AUTOMATIC SHUTDOWN OF THE APU OR A FAILURE TO START CONDITION, DO NOT TRY TO START THE APU DURING THE FUELING OPERATION OF THE AIRPLANE. IF YOU TRY TO START THE APU, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(d) If there is a protective automatic shutdown or failure to start condition on the APU, do one of these steps:

1) Complete the defuel operation before the APU is tried to start again.

2) Stop the defuel operation.

a) Disconnect the fuel hose(s) from the airplane fueling adapter(s) before the APU is started again.

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WARNING STOP THE FUELING OPERATION AND FOLLOW THE SAFETY PRECAUTIONS IF AN APU FIRE OCCURS DURING THE FUELING OPERATION OF THE AIRPLANE. IF YOU DO NOT OBEY, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(e) If an APU fire occurs, do these steps in this sequence:

1) Stop the defueling operation.
2) Make sure that the APU shuts down automatically.
   a) If it does not shut down automatically, do this task: APU Emergency Shutdown, TASK 49-11-00-860-803.
3) Discharge the APU fire bottles (TASK 49-11-00-860-803).
4) Tell persons onboard the airplane and Airport Fire Services.

WARNING STOP THE FUELING OPERATION AND FOLLOW THE SAFETY PRECAUTIONS IF YOU SPILL FUEL DURING THE FUELING OPERATION OF THE AIRPLANE. IF YOU DO NOT OBEY, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(f) If fuel spillage occurs, do these steps:

1) Stop the defueling operation.
2) Tell a person onboard the airplane.
3) Unload the APU and shut it down (TASK 49-11-00-860-802).
   a) Do not start the APU until the spilled fuel is removed and there is no further risk of spilled fuel or vapors.

WARNING MAKE SURE THAT FUELING VEHICLES ARE NOT PARKED IN THE EXHAUST FLOW OF THE ENGINES OF THIS AIRPLANE, OR ADJACENT AIRPLANES. THE HOT EXHAUST CAN CAUSE A FIRE OR EXPLOSION.

(g) Make sure that fueling vehicles are in a position that prevents risk of coming in the path of the APU exhaust stream.

NOTE: The APU exhaust stream can interfere with the fueling vehicles for other airplanes.

(h) Make sure that the exhaust stream from the APU does not go into a flammable vapor zone off airplane (for example, an adjacent aircraft during refueling).

SUBTASK 28-26-00-910-001

(2) If an adjacent aircraft is operating its APU, do this step:

(a) Make sure that the APU exhaust stream does not go into a flammable vapor zone (for example, the fuel vents or area around the fueling station and fueling vehicles).

END OF TASK

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TASK 28-26-00-650-801

3. Fuel Tank Defueling

(Figure 201, Figure 202)

A. General

(1) There are three different procedures that you can use to defuel the fuel tanks:

(a) Defuel through the pressure fueling receptacle with the airplane fuel boost pumps and the defueling (suction) pump on the refuel truck. This procedure defuels some or all of the fuel tanks at the same time. It is the fastest procedure to defuel the fuel tanks.

(b) Defuel through the pressure fueling receptacle with the airplane fuel boost pumps only.

(c) Defuel through the pressure fueling receptacle with the defueling (suction) pump on the fuel truck only. This procedure can be used only on the No. 2 main tank. Use this procedure only if it is necessary. It is a very slow procedure to be used only in emergencies.

(2) With pressure defueling (fuel boost pumps operate), or with pressure and suction together, you can defuel all of the fuel tanks to the level of the sump drain valves.

(3) When the defueling valve handle is hard to operate, you can lubricate or replace the mechanism.

B. References

Reference	Title
12-11-00-650-807	Drain the Fuel from the Sumps after Defueling (P/B 301)
20-40-11-760-801	Electrical Bonding (P/B 201)
20-40-11-910-801	Static Grounding (P/B 201)
24-22-00-860-801	Supply Electrical Power (P/B 201)
28-21-11-000-801	Fueling Receptacle Removal (P/B 401)
28-21-11-400-801	Fueling Receptacle Installation (P/B 401)
28-26-11 P/B 401	DEFUELING VALVE - REMOVAL/INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
33-18-00-710-801	Master Dim and Test - Operational Test (P/B 201)
49-11-00-860-801	APU Starting and Operation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
D50248	Oil - Instrument, Aircraft, Low Volatility	MIL-PRF-6085 (Supersedes MIL-L-6085)

D. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

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E. Access Panels

Number	Name/Location
621EB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door

F. Prepare for Procedure

SUBTASK 28-26-00-860-063

(1) The airplane usually must have the correct attitude (0-degree roll, +0.18 degree pitch, nose up) to defuel it correctly.

SUBTASK 28-26-00-650-009

(2) To completely defuel the center tank, do these steps:

(a) Adjust the airplane to 0-degree roll, +0.18 degrees pitch, nose up, if it is possible.

NOTE: This will minimize the quantity of fuel that remains in the center tank.

(b) Make sure that the No. 1 main tank has sufficient capacity available for the fuel to be scavenged from the center tank.

NOTE: The scavenge systems operates only when the No. 1 main tank fuel quantity is less than 4100 lb (1860 kg).

(c) Operate the forward No. 1 fuel boost pump to supply the motive flow during fuel scavenging.

G. Defuel through the Pressure Fueling Receptacle With the Airplane Fuel Boost Pumps and the Defueling (Suction) Pump on the Fuel Truck

NOTE: It is necessary to operate the fuel boost pumps.

SUBTASK 28-26-00-860-002

WARNING OBEY THE SAFETY PRECAUTIONS. IF YOU DO NOT OBEY THE SAFETY PRECAUTIONS, YOU CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Read and obey the precautions (TASK 28-26-00-600-801).

SUBTASK 28-26-00-860-003

CAUTION DO NOT OPERATE THE HYDRAULIC SYSTEM IF THERE IS LESS THAN A SUFFICIENT QUANTITY OF FUEL IN THE APPLICABLE TANK. DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do not operate an engine-driven or electric motor-driven pump in the hydraulic system if the No.1 or the No. 2 main tank has less than 1675 lb (760 kg) of fuel (TASK 29-11-00-860-805).

NOTE: It is not necessary to operate the engine-driven or electric motor-driven hydraulic pumps for this task.

SUBTASK 28-26-00-860-004

(3) Bond the fueling vehicle to the airplane (TASK 20-40-11-760-801).

SUBTASK 28-26-00-860-005

(4) Connect the electrical power to the airplane (TASK 24-22-00-860-801).

(a) If you are not scheduled to defuel the No. 1 main tank fully, you can use the APU to supply electrical power (TASK 49-11-00-860-801)

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1) You must have a minimum amount quantity of fuel in the No. 1 main tank to use the APU.

NOTE: The APU can be shutdown while defueling.

SUBTASK 28-26-00-860-006

(5) Make sure that all circuit breakers for the fuel system on the P6 panel are closed.

SUBTASK 28-26-00-710-001

(6) Do the master dim and test system operational test (TASK 33-18-00-710-801).

(a) Make sure that the fuel pump low pressure lights are operative.

SUBTASK 28-26-00-860-007

(7) If you are scheduled to defuel the No. 1 main tank or the center tank, do these steps:

(a) On the P5-2 Fuel System Module, set the CROSS FEED switch to the open position.

(b) On the P15 Refuel Panel, make sure that the TANK 1 or CENTER TANK VALVE POSITION LIGHTS is on dim (blue).

SUBTASK 28-26-00-010-001

(8) Open this access panel:

Number	Name/Location
621GB	Refuel Valve Door

SUBTASK 28-26-00-020-001

(9) Remove the cap on the fueling receptacle.

SUBTASK 28-26-00-480-001



WARNING MAKE SURE THAT THERE IS NO TENSION ON THE HOSE. THE HOSE MUST HANG FREELY FROM THE REFUEL ADAPTER, WITH NO FORCE ON IT. TENSION ON THE HOSE CAN CAUSE DAMAGE TO THE FUEL RECEPTACLE AND CAUSE THE HOSE TO DISCONNECT. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(10) Do an inspection of the fueling receptacle before you connect the defueling hose nozzle:

(a) Make sure that the fueling receptacle is clean, not damaged and there are no fuel leaks.

1) Make sure that the screws, lugs, and slots are not loose or damaged.

(b) If there is a problem with the fueling receptacle, replace the fueling receptacle (TASK 28-21-11-000-801 and TASK 28-21-11-400-801).



CAUTION MAKE SURE THAT THE HOSE NOZZLE AND HOSE-END CONTROL VALVE ARE IN THE CORRECT CONFIGURATION. REFER TO THE SUPPLIER'S MAINTENANCE MANUAL FOR PROCEDURES. IF YOU DO NOT, IT COULD CAUSE DAMAGE TO THE AIRPLANE.

(c) Connect defueling hose nozzle to the fueling receptacle.

SUBTASK 28-26-00-860-008

(11) Open the shutoff handle on the defueling hose nozzle.

NOTE: It is recommended that the defueling nozzle has a filter screen.

SUBTASK 28-26-00-860-009

(12) Make sure that the two spar valves are closed.

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SUBTASK 28-26-00-860-010

(13) On the P15 Refuel Panel, make sure that the switches for the fueling shutoff valves are in the CLOSED position.

SUBTASK 28-26-00-860-011

(14) On the P15 Refuel Panel, make sure that all of the VALVE POSITION LIGHTS are off.

SUBTASK 28-26-00-970-001

(15) Make sure that you know the correct quantity of fuel in the fuel tanks.

(a) Record the fuel quantity for the fuel tank or the tanks that you are scheduled to defuel.

(b) Hold the test switch on the P15 Refuel Panel in the TEST GAUGES position for one second.

NOTE: The test blanks the fueling indicators display for two seconds, then all LED segments go on for two seconds. This sequence continues as long as you hold the TEST SWITCH. If you hold the TEST SWITCH for more than 20 seconds, the test mode will time out and the indicator will go back to its usual operating mode. If an internal fault is found during the test, the indicator will show Ind FAIL.

(c) Release the switch.

(d) Make sure that the indicator goes back to its initial value with a tolerance of ± 10 pounds (± 10 kilograms).

SUBTASK 28-26-00-010-002

(16) Open this access panel:

(Figure 202)

Number	Name/Location
621EB	Lower Leading Edge Access Panel

SUBTASK 28-26-00-860-012

(17) Set the defueling valve handle to the open position.

(a) If the defueling valve handle is hard to operate, lubricate the mechanism with aircraft instrument oil, D50248.

(b) If that does not decrease the force necessary to move the defueling valve handle, remove and replace the defueling valve handle (PAGEBLOCK 28-26-11/401).

SUBTASK 28-26-00-860-013

(18) On the P5-2 Fuel System Module, set the CROSS FEED switch to the open position.

NOTE: If you are scheduled to defuel the No. 2 main tank (right side), it is not necessary to open or to close the fuel crossfeed valve.

(a) On the P15 Refuel Panel, make sure that the TANK 2 VALVE POSITION LIGHTS is on dim (blue).

SUBTASK 28-26-00-860-014

(19) Obey these fuel pump limitations during the pump operation:

(a) To operate the fuel pumps, you must be in the flight compartment to continuously monitor the fuel quantity and the low pressure indication in the applicable tank.

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WARNING DO NOT OPERATE A FUEL PUMP IF THE LOW PRESSURE LIGHT COMES ON AND STAYS ON. THIS CONDITION CAN CAUSE THE IGNITION OF THE FUEL FUMES IN THE FUEL TANK. A FIRE OR AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Immediately set the applicable fuel pump switch(es) to the OFF position if the LOW PRESSURE light comes on and stays on unless there is a reduced back-pressure condition.

NOTE: When you defuel or transfer with minimum back-pressure to the pumps, it is possible for the LOW PRESSURE light(s) to come on. In this case, monitor fuel tank quantity and set the pump switch to the OFF position if the fuel tank quantity is not changing.

1) If one LOW PRESSURE light comes on because of a reduced back-pressure condition, do these steps:
   a) Put the applicable fuel boost pump switch to the OFF position and continue defueling.
   b) If the remaining LOW PRESSURE light for the fuel boost pump switch for the fuel tank comes on, monitor the fuel tank quantity.
   c) If the fuel quantity for the fuel tank does not change, then immediately set the fuel boost pump switch to the OFF position.
   d) For the No. 1 or No. 2 main tank, do not operate the main fuel boost pumps with fuel quantities less than 100 lb (45 kg).
   e) If the automatic shutoff system for the center tank fuel boost pump is disabled, do not operate the center tank fuel boost pumps with fuel quantities less than 2000 lb (907 kg).
2) If two LOW PRESSURE lights for the fuel boost pump switches in the fuel tank come on at the same time, do these steps:
   a) For the No. 1 or No. 2 main tanks, put the FUEL PUMP 1 AFT or the FUEL PUMP 2 AFT switch to the OFF position and continue to monitor the fuel tank quantity.
   b) For the center tank, put one of the fuel boost pump switches to the OFF position and continue to monitor the fuel tank quantity.
   c) If the fuel quantity in the fuel tank does not change, immediately put the remaining fuel boost pump switch to the OFF position.
   d) For the No. 1 or No. 2 main tank, do not operate the main fuel boost pumps with fuel quantities less than 100 lb (45 kg).
   e) If the automatic shutoff system for the center tank fuel boost pump is disabled, do not operate the center tank fuel boost pumps with fuel quantities less than 2000 lb (907 kg).

(c) If the automatic shutoff system for the center fuel boost pump turns the center fuel boost pump(s) off because of a reduced back-pressure condition, and there are no passengers on board or have at least 2000 lb (907 kg) in the center tank, do these steps:

1) On the P5-2 Fuel System Module, set the switches for all main tank fuel boost pumps to the OFF position.
   a) FUEL PUMPS 1 FWD

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b) FUEL PUMPS 1 AFT

c) FUEL PUMPS 2 FWD

d) FUEL PUMPS 2 AFT

2) Open the indicated circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	3	C01637	BOOST PMP CTR TNK L AUTO SHUT OFF-DC
C	7	C01638	BOOST PMP CTR TNK R AUTO SHUT OFF-DC
D	7	C01659	AT S-O/UCPO BST PMP CTR TNK L AC
E	7	C01658	FUEL AT S-O/UCPO BST PMP CTR TNK R AC

3) Continue from where you were last in the fuel transfer procedure.

(d) Do these steps to defuel an airplane with passengers on board:

1) Do not let the fuel quantity in a main fuel tank become less than 2000 lb (907 kg).

2) If the automatic shutoff system for the center tank fuel boost pump is disabled, do not let the fuel quantity in the center tank become less than 2000 lb (907 kg).

SUBTASK 28-26-00-860-015

(20) On the P5-2 Fuel System Module, set the switches for the fuel boost pumps to the positions shown in this table:

Table 201/28-26-00-993-801

	SWITCH POSITION FOR DEFUELING		
	MAIN TANK NO. 1	MAIN TANK NO. 2	CENTER TANK
FUEL PUMPS AFT 1	ON	OFF	OFF
FUEL PUMPS FWD 1	ON	OFF	OFF
FUEL PUMPS AFT 2	OFF	ON	OFF
FUEL PUMPS FWD 2	OFF	ON	OFF
FUEL PUMPS L CTR	OFF	OFF	ON
FUEL PUMPS R CTR	OFF	OFF	ON

SUBTASK 28-26-00-650-001

(21) Start the defueling pump on the refuel truck.

SUBTASK 28-26-00-650-002

(22) Defuel the applicable tank until the refuel quantity indicator shows the specified quantity of fuel in the fuel tank.

(a) You can also defuel the fuel tank until the applicable fuel boost pump LOW PRESSURE light goes on.

(b) Set the applicable fuel boost pump switches to the OFF position.

SUBTASK 28-26-00-860-016

(23) On the P5-2 Fuel System Module, set the CROSS FEED switch to the closed position.

(a) On the P15 Refuel Panel, make sure that the VALVE POSITION LIGHTS is bright when the valve position changes.

(b) Make sure that the VALVE POSITION LIGHTS position light goes off when the valve is closed.

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SUBTASK 28-26-00-860-017

(24) Stop the defueling pump on the fuel truck.

SUBTASK 28-26-00-860-018

(25) Set the defueling valve handle to the closed position.

(a) If the defueling valve handle is hard to operate, lubricate the mechanism with aircraft instrument oil, D50248.

(b) If that does not decrease the force necessary to move the defueling valve handle, remove and replace the defueling valve handle (PAGEBLOCK 28-26-11/401).

SUBTASK 28-26-00-860-019

(26) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	3	C01637	BOOST PMP CTR TNK L AUTO SHUT OFF-DC
C	7	C01638	BOOST PMP CTR TNK R AUTO SHUT OFF-DC
D	7	C01659	AT S-O/UCPO BST PMP CTR TNK L AC
E	7	C01658	FUEL AT S-O/UCPO BST PMP CTR TNK R AC

SUBTASK 28-26-00-410-001

(27) Close this access panel:

Number	Name/Location
621EB	Lower Leading Edge Access Panel

SUBTASK 28-26-00-860-020

(28) Close the fuel shutoff handle on the hose nozzle.

SUBTASK 28-26-00-080-001

(29) Disconnect the nozzle of the defueling hose from the fueling receptacle.

SUBTASK 28-26-00-080-002

(30) Disconnect the bonding cable that you connected between the fueling source and the airplane (TASK 20-40-11-760-801).

SUBTASK 28-26-00-420-001

(31) Put the cap on the fueling receptacle.

SUBTASK 28-26-00-410-002

(32) Close this access panel:

Number	Name/Location
621GB	Refuel Valve Door

SUBTASK 28-26-00-680-001

(33) Do this task: Drain the Fuel from the Sumps after Defueling, TASK 12-11-00-650-807.

H. Defuel through Pressure Fueling Receptacle With the Airplane Fuel Boost Pump Only

SUBTASK 28-26-00-860-022

WARNING OBEY THE SAFETY PRECAUTIONS. IF YOU DO NOT OBEY THE SAFETY PRECAUTIONS, YOU CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Read and obey the precautions (TASK 28-26-00-600-801).

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SUBTASK 28-26-00-860-023

CAUTION DO NOT OPERATE THE HYDRAULIC SYSTEM IF THERE IS LESS THAN A SUFFICIENT QUANTITY OF FUEL IN THE APPLICABLE TANK. DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do not operate an engine-driven or electric motor-driven pump in the hydraulic system if the No.1 or the No. 2 main tank has less than 1675 lb (760 kg) of fuel (TASK 29-11-00-860-805).

NOTE: It is not necessary to operate the engine-driven or electric motor-driven hydraulic pumps for this task.

SUBTASK 28-26-00-480-002

(3) Electrically ground the airplane and the fueling vehicle (TASK 20-40-11-910-801).

SUBTASK 28-26-00-860-024

(4) Bond the fueling source to the airplane (TASK 20-40-11-760-801).

SUBTASK 28-26-00-860-025

(5) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

(a) If you are not scheduled to defuel the No. 1 main tank fully, you can use the APU to supply electrical power (TASK 49-11-00-860-801)

1) You must have a minimum amount quantity of fuel in the No. 1 main tank to use the APU.

NOTE: The APU can be shutdown while defueling.

SUBTASK 28-26-00-860-026

(6) Make sure that all circuit breakers for the fuel system on the P6 panel are closed.

SUBTASK 28-26-00-860-027

(7) If you are scheduled to defuel the No. 1 main tank or the center tank, do these steps:

(a) On the P5-2 Fuel System Module, set the CROSS FEED switch to the open position.

(b) On the P15 Refuel Panel, make sure that the TANK 1 or CENTER TANK VALVE POSITION LIGHTS is dim (blue).

SUBTASK 28-26-00-010-003

(8) Open this access panel:

Number	Name/Location
621GB	Refuel Valve Door

SUBTASK 28-26-00-020-002

(9) Remove the cap on the fueling receptacle.

SUBTASK 28-26-00-480-003

WARNING MAKE SURE THAT THERE IS NO TENSION ON THE HOSE. THE HOSE MUST HANG FREELY FROM THE REFUEL ADAPTER, WITH NO FORCE ON IT. TENSION ON THE HOSE CAN CAUSE DAMAGE TO THE FUEL RECEPTACLE AND CAUSE THE HOSE TO DISCONNECT. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(10) Do an inspection of the fueling receptacle before you connect the defueling hose nozzle:

(a) Make sure that the fueling receptacle is clean, not damaged and there are no fuel leaks.

1) Make sure that the screws, lugs, and slots are not loose or damaged.

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(b) If there is a problem with the fueling receptacle, replace the fueling receptacle (TASK 28-21-11-000-801 and TASK 28-21-11-400-801).

CAUTION

MAKE SURE THAT THE HOSE NOZZLE AND HOSE-END CONTROL VALVE ARE IN THE CORRECT CONFIGURATION. REFER TO THE SUPPLIER'S MAINTENANCE MANUAL FOR PROCEDURES. IF YOU DO NOT, IT COULD CAUSE DAMAGE TO THE AIRPLANE.

(c) Connect defueling hose nozzle to the fueling receptacle.

SUBTASK 28-26-00-860-028

(11) Open the shutoff handle on the defueling hose nozzle.

NOTE: It is recommended that the defueling nozzle has a filter screen.

SUBTASK 28-26-00-860-029

(12) Make sure that the spar valves are closed.

SUBTASK 28-26-00-860-030

(13) On the P15 Refuel Panel, make sure that the switches for the fueling shutoff valves are in the CLOSED position.

SUBTASK 28-26-00-860-031

(14) On the P15 Refuel Panel, make sure that the VALVE POSITION LIGHTS are off.

SUBTASK 28-26-00-970-002

(15) Make sure that you know the correct quantity of fuel in the fuel tanks.

(a) Write the fuel quantity for the fuel tank or the tanks that you are scheduled to defuel.

(b) Hold the test switch on the P15 Refuel Panel in the TEST GAUGES position for one second.

NOTE: The test blanks the fueling indicators display for two seconds, then all LED segments go on for two seconds. This sequence continues as long as you hold the TEST SWITCH. If you hold the TEST SWITCH for more than 20 seconds, the test mode will time out and the indicator will go back to its usual operating mode. If an internal fault is found during the test, the indicator will show Ind FAIL.

(c) Release the switch.

(d) Make sure that the indicator goes back to its initial value with a tolerance of ± 10 pounds (± 10 kilograms).

SUBTASK 28-26-00-010-004

(16) Open this access panel:

Number	Name/Location
621EB	Lower Leading Edge Access Panel

(Figure 202).

SUBTASK 28-26-00-860-032

(17) Set the defueling valve handle to the open position.

(a) If the defueling valve handle is hard to operate, lubricate the mechanism with aircraft instrument oil, D50248.

(b) If that does not decrease the force necessary to move the defueling valve handle, remove and replace the defueling valve handle (PAGEBLOCK 28-26-11/401).

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SUBTASK 28-26-00-860-033

(18) On the P5-2 Fuel System Module, set the CROSS FEED switch to the open position.

NOTE: If you are scheduled to defuel the No. 2 main tank (right side), it is not necessary to open or to close the fuel crossfeed valve.

(a) Make sure that the TANK 2 VALVE POSITION LIGHTS is on dim (blue).

SUBTASK 28-26-00-860-034

WARNING

DO NOT OPERATE A FUEL PUMP IF THE LOW PRESSURE LIGHT COMES ON AND STAYS ON. THIS CONDITION CAN CAUSE THE IGNITION OF THE FUEL FUMES IN THE FUEL TANK. A FIRE OR AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(19) To operate the fuel pumps, you must be in the flight compartment to continuously monitor the fuel quantity and the low pressure indication in the applicable tank.

(a) Immediately set the applicable fuel pump switch(es) to the OFF position if the LOW PRESSURE light comes on and stays on unless there is a reduced back-pressure condition.

NOTE: When you defuel or transfer fuel with minimum back-pressure to the pumps, it is possible for the LOW PRESSURE light(s) to come on. In this case, monitor fuel tank quantity and set the pump switch to the OFF position if the fuel tank quantity is not changing.

1) If one LOW PRESSURE light comes on because of a reduced back-pressure condition, do these steps:

a) Put the applicable fuel boost pump switch to the OFF position and continue defueling.

b) If the remaining LOW PRESSURE light for the fuel boost pump switch for the fuel tank comes on, monitor the fuel tank quantity.

c) If the fuel quantity for the fuel tank does not change, then immediately set the fuel boost pump switch to the OFF position.

d) For the No. 1 or No. 2 main tank, do not operate the main fuel boost pumps with fuel quantities less than 100 lb (45 kg).

e) If the automatic shutoff system for the center tank fuel boost pump is disabled, do not operate the center tank fuel boost pumps with fuel quantities less than 2000 lb (907 kg).

2) If two LOW PRESSURE lights for the fuel boost pump switches in the fuel tank come on at the same time, do these tasks:

a) For the No. 1 or No. 2 main tanks, put the Fuel Pump 1 AFT or the Fuel Pump 2 AFT switch to the OFF position and continue to monitor the fuel tank quantity.

b) For the center tank, put one of the fuel boost pump switches to the OFF position and continue to monitor the fuel tank quantity.

c) If the fuel quantity in the fuel tank does not change, immediately put the remaining fuel boost pump switch to the OFF position.

d) For the No. 1 or No. 2 main tank, do not operate the main fuel boost pumps with fuel quantities less than 100 lb (45 kg).

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e) If the automatic shutoff system for the center tank fuel boost pump is disabled, do not operate the center tank fuel boost pumps with fuel quantities less than 2000 lb (907 kg).

(b) If the automatic shutoff system for the center fuel boost pump turns the center fuel boost pump(s) off because of a reduced back-pressure condition, and there are no passengers on board or have at least 2000 lb (907 kg) in the center tank, do these steps:

1) On the P5-2 Fuel System Module, set the switches for all main tank fuel boost pumps to the OFF position.

a) FUEL PUMPS 1 FWD

b) FUEL PUMPS 1 AFT

c) FUEL PUMPS 2 FWD

d) FUEL PUMPS 2 AFT

2) Open the indicated circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	3	C01637	BOOST PMP CTR TNK L AUTO SHUT OFF-DC
C	7	C01638	BOOST PMP CTR TNK R AUTO SHUT OFF-DC
D	7	C01659	AT S-O/UCPO BST PMP CTR TNK L AC
E	7	C01658	FUEL AT S-O/UCPO BST PMP CTR TNK R AC

3) Continue from where you were last in the fuel transfer procedure.

(c) Do these steps to defuel an airplane with passengers on board:

1) Do not let the fuel quantity in a main fuel tank become less than 2000 lb (907 kg).

2) If the automatic shutoff system for the center tank fuel boost pump is disabled, do not let the fuel quantity in the center fuel tank become less than 2000 lb (907 kg).

SUBTASK 28-26-00-860-035

(20) On the P5-2 Fuel System Module, set the switches for the fuel boost pumps to the positions shown in this table:

Table 202/28-26-00-993-802

	SWITCH POSITION FOR DEFUELING		
	TANK NO. 1	TANK NO. 2	CENTER TANK
FUEL PUMPS AFT 1	ON	OFF	OFF
FUEL PUMPS FWD 1	ON	OFF	OFF
FUEL PUMPS AFT 2	OFF	ON	OFF
FUEL PUMPS FWD 2	OFF	ON	OFF
FUEL PUMPS L CTR	OFF	OFF	ON
FUEL PUMPS R CTR	OFF	OFF	ON

SUBTASK 28-26-00-650-003

(21) Defuel the applicable tank until the refuel quantity indicator shows the specified quantity of fuel in the fuel tank.

SUBTASK 28-26-00-860-036

(22) On the P5-2 Fuel System Module, set the CROSS FEED switch to the closed position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-26-00-860-037

(23) Set the defueling valve handle to the closed position.

(a) If the defueling valve handle is hard to operate, lubricate the mechanism with aircraft instrument oil, D50248.

(b) If that does not decrease the force necessary to move the defueling valve handle, remove and replace the defueling valve handle (PAGEBLOCK 28-26-11/401).

SUBTASK 28-26-00-860-038

(24) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	3	C01637	BOOST PMP CTR TNK L AUTO SHUT OFF-DC
C	7	C01638	BOOST PMP CTR TNK R AUTO SHUT OFF-DC
D	7	C01659	AT S-O/UCPO BST PMP CTR TNK L AC
E	7	C01658	FUEL AT S-O/UCPO BST PMP CTR TNK R AC

SUBTASK 28-26-00-410-003

(25) Close this access panel:

Number	Name/Location
621EB	Lower Leading Edge Access Panel

SUBTASK 28-26-00-860-039

(26) Close the fuel shutoff handle on the hose nozzle.

SUBTASK 28-26-00-080-003

(27) Disconnect the nozzle of the defueling hose from the fueling receptacle.

SUBTASK 28-26-00-420-002

(28) Put the cap on the fueling receptacle.

SUBTASK 28-26-00-080-004

(29) Disconnect the bonding cable that you connected between the fueling source and the airplane (TASK 20-40-11-760-801).

SUBTASK 28-26-00-680-002

(30) Do this task: Drain the Fuel from the Sumps after Defueling, TASK 12-11-00-650-807.

I. Defuel the No. 2 Main Tank through Pressure Fueling Receptacles With the Fuel Truck Defueling (Suction) Pump Only

NOTE: This defueling procedure can only be used on the No. 2 main tank. Use it only in an emergency when you cannot use electrical power on the airplane. It is recommended that the defueling nozzle has a filter screen.

NOTE: The recommended procedures for defueling are given above: (1) Defuel through the Pressure Fueling Receptacle With the Airplane Fuel Boost Pumps and the Defueling (Suction) Pump on the Fuel Truck (2) Defuel through Pressure Fueling Receptacle With the Airplane Fuel Boost Pump Only.

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SUBTASK 28-26-00-860-041

CAUTION

DO NOT OPERATE THE HYDRAULIC SYSTEM IF THERE IS LESS THAN A SUFFICIENT QUANTITY OF FUEL IN THE APPLICABLE TANK. DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do not operate an engine-driven or electric motor-driven pump in the hydraulic system if the No. 1 or the No. 2 main tank has less than 1675 lb (760 kg) of fuel (TASK 29-11-00-860-805).

NOTE: It is not necessary to operate the engine-driven or electric motor-driven hydraulic pumps for this task.

SUBTASK 28-26-00-860-042

(2) Electrically ground the airplane and the fuel truck (TASK 20-40-11-910-801).

SUBTASK 28-26-00-010-005

(3) Open this access panel:

(Figure 202)

Number	Name/Location
621GB	Refuel Valve Door

SUBTASK 28-26-00-020-003

(4) Remove the cap on the fueling receptacle.

SUBTASK 28-26-00-480-004

WARNING

MAKE SURE THAT THERE IS NO TENSION ON THE HOSE. THE HOSE MUST HANG FREELY FROM THE REFUEL ADAPTER, WITH NO FORCE ON IT. TENSION ON THE HOSE CAN CAUSE DAMAGE TO THE FUEL RECEPTACLE AND CAUSE THE HOSE TO DISCONNECT. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Do an inspection of the fueling receptacle before you connect the defueling hose nozzle:

(a) Make sure that the fueling receptacle is clean, not damaged and there are no fuel leaks.

1) Make sure that the screws, lugs, and slots are not loose or damaged.

(b) If there is a problem with the fueling receptacle, replace the fueling receptacle (TASK 28-21-11-000-801 and TASK 28-21-11-400-801).

CAUTION

MAKE SURE THAT THE HOSE NOZZLE AND HOSE-END CONTROL VALVE ARE IN THE CORRECT CONFIGURATION. REFER TO THE SUPPLIER'S MAINTENANCE MANUAL FOR PROCEDURES. IF YOU DO NOT, IT COULD CAUSE DAMAGE TO THE AIRPLANE.

(c) Connect defueling hose nozzle to the fueling receptacle.

SUBTASK 28-26-00-860-043

(6) Open the shutoff handle on the hose nozzle.

SUBTASK 28-26-00-010-006

(7) Open this access panel:

(Figure 202)

Number	Name/Location
621EB	Lower Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-26-00-860-044

(8) Set the defueling valve handle to the open position.

(a) If the defueling valve handle is hard to operate, lubricate the mechanism with aircraft instrument oil, D50248.

(b) If that does not decrease the force necessary to move the defueling valve handle, remove and replace the defueling valve handle (PAGEBLOCK 28-26-11/401).

SUBTASK 28-26-00-860-045

(9) On the P5-2 Fuel System Module, make sure that the CROSS FEED switch is in the closed position.

SUBTASK 28-26-00-650-004

(10) Start the defueling operation.

NOTE: If it is not easy to get fuel to flow from the airplane, use the fuel truck to pump the airplane lines full of fuel (pressure-prime) then start the defueling pumps on the fuel truck.

(a) Use the pump on the fuel truck to fill the airplane fuel lines with fuel (pressure prime).

(b) Start the defueling pumps on the fuel truck.

SUBTASK 28-26-00-650-005

(11) Continue to defuel the airplane until the fuel measuring sticks show the correct quantity of fuel in the No. 2 main tank.

(a) You can also continue to defuel the airplane until the flowmeter on the fuel truck shows no fuel flow.

SUBTASK 28-26-00-650-006

(12) Stop the defueling pump on the fuel truck.

SUBTASK 28-26-00-860-046

(13) Set the defueling valve handle to the closed position.

(a) If the defueling valve handle is hard to operate, lubricate the mechanism with aircraft instrument oil, D50248.

(b) If that does not decrease the force necessary to move the defueling valve handle, remove and replace the defueling valve handle (PAGEBLOCK 28-26-11/401).

SUBTASK 28-26-00-410-004

(14) Close this access panel:

Number	Name/Location
621EB	Lower Leading Edge Access Panel

SUBTASK 28-26-00-860-047

(15) Close the shutoff handle on the nozzle of the fuel hose.

SUBTASK 28-26-00-080-005

(16) Disconnect the defueling hose nozzle.

SUBTASK 28-26-00-420-003

(17) Put the cap on the fueling receptacle.

SUBTASK 28-26-00-080-006

(18) Disconnect the bonding cable that you connected between the fueling source and the airplane (TASK 20-40-11-760-801).

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SUBTASK 28-26-00-080-007

(19) Disconnect the ground cables from the fueling source and the airplane (TASK 20-40-11-910-801).

SUBTASK 28-26-00-410-005

(20) Close this access panel:

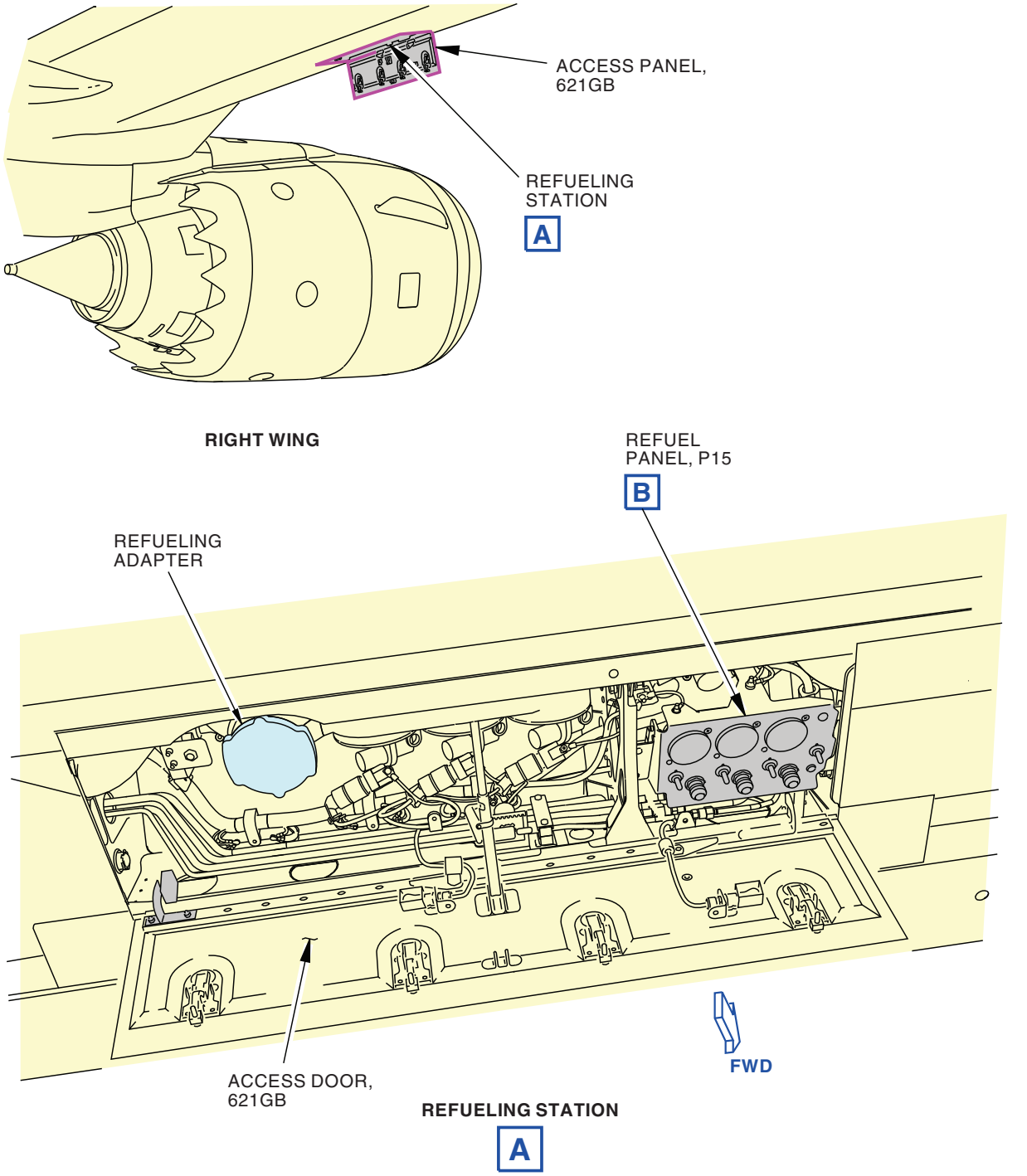
Number	Name/Location
621GB	Refuel Valve Door

SUBTASK 28-26-00-650-007

(21) If you are scheduled to defuel the fuel tanks completely, do this task: TASK 12-11-00-650-807.

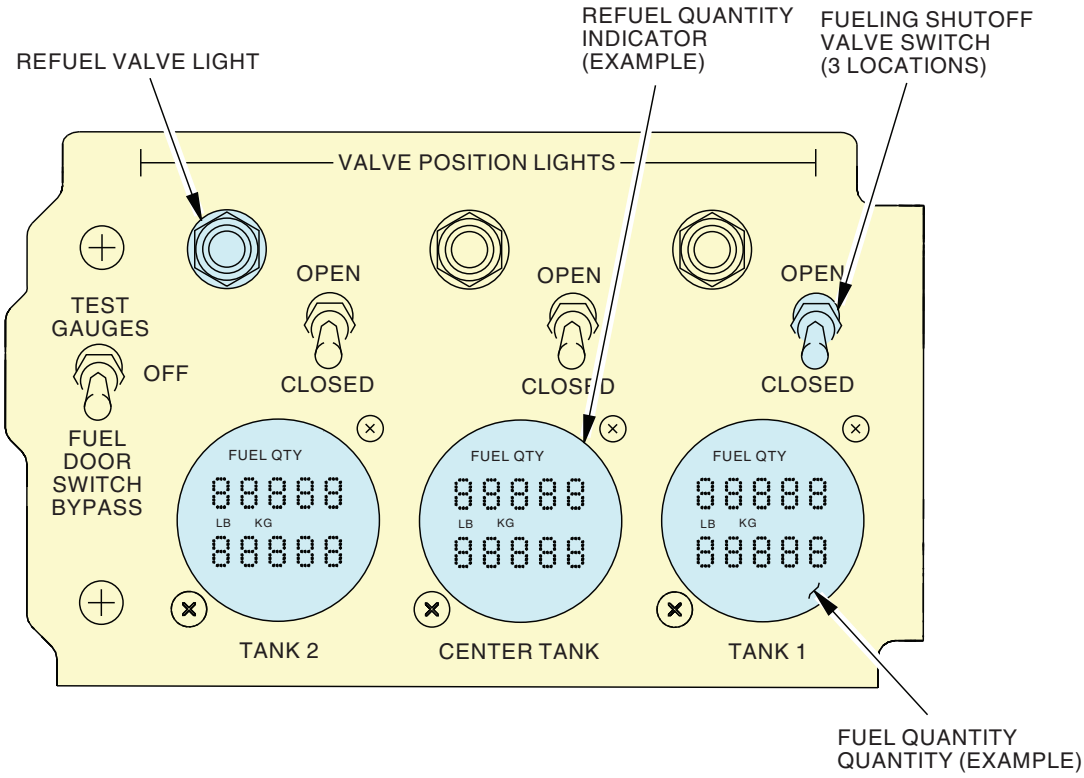
——— END OF TASK ———

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Refueling Station Component Location
Figure 201/28-26-00-990-801 (Sheet 1 of 2)

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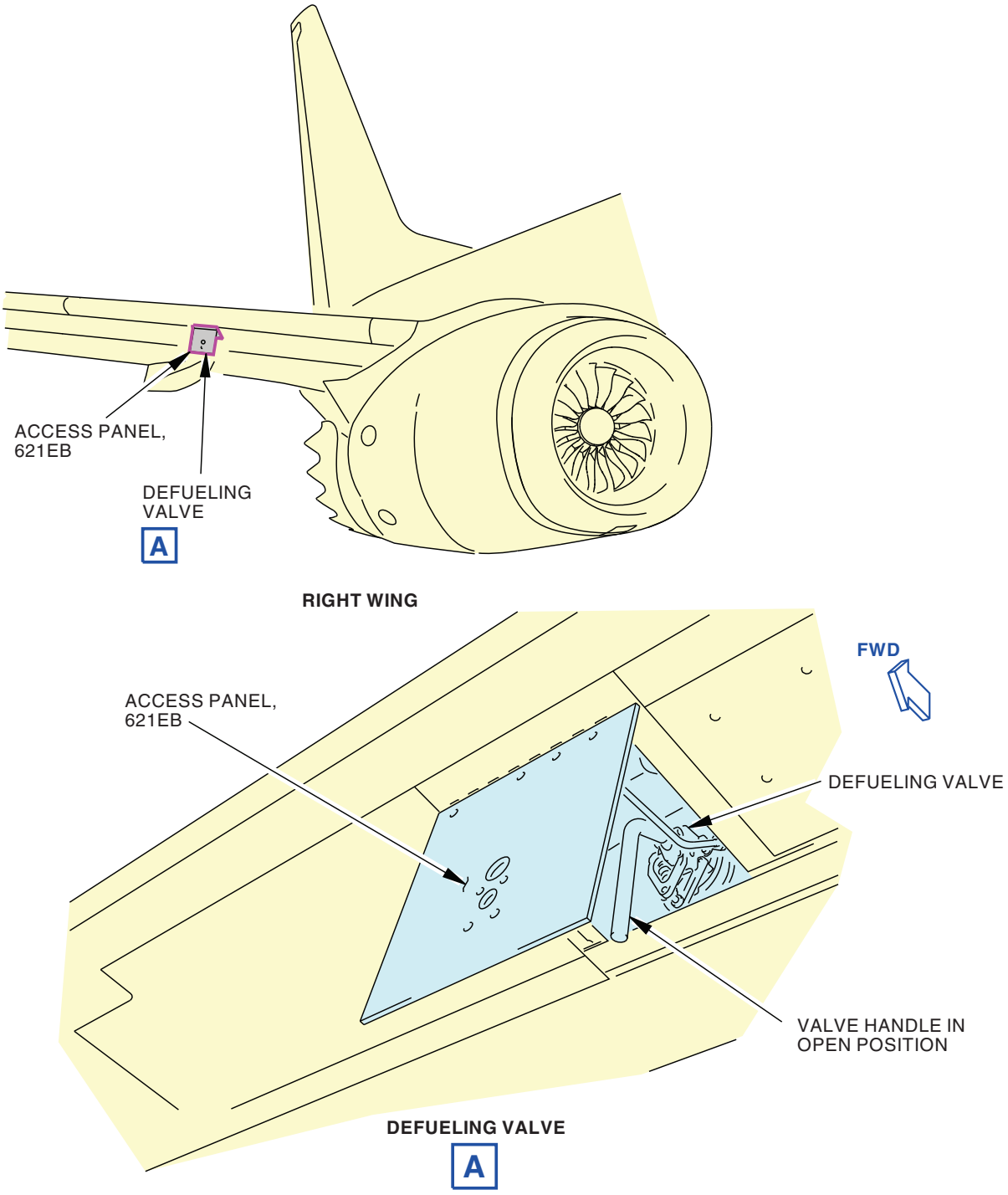
REFUEL PANEL, P15

B

2413576 S00061532849_V3

Refueling Station Component Location
Figure 201/28-26-00-990-801 (Sheet 2 of 2)

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Defueling Valve Component Location
Figure 202/28-26-00-990-802

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TASK 28-26-00-650-802

4. Tank to Tank Fuel Transfer

A. General

(1) If it is necessary to defuel only one of the fuel tanks, you can move the fuel from that fuel tank to a different fuel tank. For example, you can use this procedure when you must repair a fuel leak. The procedure is almost the same as the procedure to defuel the fuel tank with the fuel boost pumps.

(2) Monitor the fuel quantity in each fuel tank when you move the fuel from one fuel tank to a different fuel tank.

(3) When the defueling valve handle is hard to operate, you can lubricate or replace the mechanism.

(4) An unexpected fuel transfer from the center tank to the No. 1 main tank can occur during maintenance.

(a) To prevent fuel transfer from the center tank to the No. 1 main tank through the fuel scavenge system, do not put more than 23,000 lb (10,433 kg) of fuel in the center tank when the No. 1 main tank is empty for maintenance.

NOTE: The fuel transfer will stop with less than 25,000 lb (11,340 kg) of fuel, however, it is recommended to use 23,000 lb (10,433 kg) for variations in airplane pitch and roll attitude while maintenance is done.

B. References

Reference	Title
12-11-00-650-807	Drain the Fuel from the Sumps after Defueling (P/B 301)
24-22-00-860-801	Supply Electrical Power (P/B 201)
28-26-11 P/B 401	DEFUELING VALVE - REMOVAL/INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
33-18-00-710-801	Master Dim and Test - Operational Test (P/B 201)
49-11-00-860-801	APU Starting and Operation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
D50248	Oil - Instrument, Aircraft, Low Volatility	MIL-PRF-6085 (Supersedes MIL-L-6085)

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
621EB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Tank to Tank Fuel Transfer

SUBTASK 28-26-00-860-048

WARNING OBEY THE SAFETY PRECAUTIONS. IF YOU DO NOT OBEY THE SAFETY PRECAUTIONS, YOU CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Read and obey the precautions (TASK 28-26-00-600-801).

SUBTASK 28-26-00-860-049

CAUTION DO NOT OPERATE THE HYDRAULIC SYSTEM IF THERE IS LESS THAN A SUFFICIENT QUANTITY OF FUEL IN THE APPLICABLE TANK. DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do not operate an engine-driven or electric motor-driven pump in the hydraulic system if the No. 1 or the No. 2 main tank has less than 1675 lb (760 kg) of fuel (TASK 29-11-00-860-805).

NOTE: It is not necessary to operate the engine-driven or electric motor-driven hydraulic pumps for this task.

SUBTASK 28-26-00-860-050

(3) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

(a) If you are not scheduled to defuel the No. 1 main tank fully, you can supply power with the APU (TASK 49-11-00-860-801).

1) Do a check for minimum fuel quantities and other data before you start the APU.

NOTE: The APU can be shutdown while defueling.

SUBTASK 28-26-00-860-051

(4) On the P6 panel, make sure that all of the circuit breakers for the fuel system are closed.

SUBTASK 28-26-00-710-002

(5) Do the master dim and test system operational test (TASK 33-18-00-710-801).

NOTE: Make sure that the fuel pump low pressure lights operate.

SUBTASK 28-26-00-860-052

(6) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

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SUBTASK 28-26-00-860-053

(7) If you are scheduled to transfer fuel out of the No. 2 main tank, make sure that the CROSS FEED switch is in the closed position.

SUBTASK 28-26-00-860-054

(8) If you are scheduled to transfer fuel out of the No. 1 main tank or the center tank, set the CROSS FEED switch to the open position.

SUBTASK 28-26-00-010-007

(9) Open this access panel for the refuel station:

(Figure 201)

Number	Name/Location
621GB	Refuel Valve Door

SUBTASK 28-26-00-710-003

(10) On the P15 Refuel Panel, set the fueling shutoff valve switches to the OPEN position to make sure that the VALVE POSITION LIGHTS operate correctly.

SUBTASK 28-26-00-860-055

(11) Find the fueling shutoff valve switch for the fuel tank that will get the fuel that you are scheduled to transfer.

(a) Set that fueling shutoff valve switch to the OPEN position.

(b) Make sure that all of the other fueling shutoff valve switches are in the CLOSED position.

SUBTASK 28-26-00-010-009

(12) Open this access panel:

Number	Name/Location
621EB	Lower Leading Edge Access Panel

SUBTASK 28-26-00-010-008

(13) Set the defueling valve handle to the open position.

(a) If the defueling valve handle is hard to operate, lubricate the mechanism with aircraft instrument oil, D50248.

(b) If that does not decrease the force necessary to move the defueling valve handle, remove and replace the defueling valve handle (PAGEBLOCK 28-26-11/401).

SUBTASK 28-26-00-860-056

(14) Obey these fuel pump limitations during the pump operation:

(a) To operate the fuel pumps, you must be in the flight compartment to continuously monitor the fuel quantity and the low pressure indication in the applicable fuel tank.

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WARNING DO NOT OPERATE A FUEL PUMP IF THE LOW PRESSURE LIGHT COMES ON AND STAYS ON. THIS CONDITION CAN CAUSE THE IGNITION OF THE FUEL FUMES IN THE FUEL TANK. A FIRE OR AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Immediately set the applicable fuel pump switch(es) to the OFF position if the LOW PRESSURE light comes on and stays on unless there is a reduced back-pressure condition.

NOTE: When you defuel or transfer fuel with minimum back-pressure to the pumps, it is possible for the LOW PRESSURE light(s) to come on. In this case, monitor the fuel tank quantity and set the pump switch to the OFF position if the fuel tank quantity is not changing.

1) If one LOW PRESSURE light comes on because of a reduced back-pressure condition, do these steps:
   a) Put the applicable fuel boost pump switch to the OFF position and continue defueling.
   b) If the remaining LOW PRESSURE light for the fuel boost pump switch for the fuel tank comes on, monitor the fuel tank quantity.
   c) If the fuel quantity for the fuel tank does not change, then immediately set the fuel pump switch(es) to the OFF position.
   d) For the No. 1 or No. 2 main tank, do not operate the main boost pumps with fuel quantities less than 100 lb (45 kg).
   e) If the automatic shutoff system for the center tank fuel boost pump is disabled, do not operate the center tank fuel boost pumps with fuel quantities less than 2000 lb (907 kg).
2) If two LOW PRESSURE lights for the fuel boost pump switches in the fuel tank come on at the same time, do these steps:
   a) For the No. 1 or No. 2 main tanks, put the Fuel Pump 1 AFT or the Fuel Pump 2 AFT switch to the OFF position and continue to monitor the fuel tank quantity.
   b) For the center tank, put one of the fuel boost pump switches to the OFF position and continue to monitor the fuel tank quantity.
   c) If the fuel quantity in the fuel tank does not change, immediately put the remaining fuel boost pump switch to the OFF position.
   d) For the No. 1 or No. 2 main tank, do not operate the main boost pumps with fuel quantities less than 100 lb (45 kg).
   e) If the automatic shutoff system for the center tank fuel boost pump is disabled, do not operate the center tank fuel boost pumps with fuel quantities less than 2000 lb (907 kg).
3) If the center tank fuel boost pump automatic shutoff system turns off the center tank fuel boost pump(s) because of a reduced back-pressure condition, refer to Table 203 to do these steps:

   NOTE: The table contains a list of configurations into which you can put the airplane. Configuration A shows the basic conditions to use when the automatic shutoff system does not cause problems.

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Table 203/28-26-00-993-803

	CONFIGURATION						
	A	B	C	D	E	F	G
FUEL PUMPS AFT 1	OFF	ON	OFF	OFF	ON	OFF	OFF
FUEL PUMPS FWD 1	OFF	ON	OFF	OFF	ON	OFF	OFF
FUEL PUMPS AFT 2	OFF	OFF	ON	OFF	OFF	ON	OFF
FUEL PUMPS FWD 2	OFF	OFF	ON	OFF	OFF	ON	OFF
FUEL PUMPS L CTR	ON	ON	ON	ON	ON	ON	ON
FUEL PUMPS R CTR	ON	ON	ON	ON	ON	ON	ON
DEFUELING VALVE*[1]	OPEN	OPEN	OPEN	½	½	½	OPEN
Circuit Breaker C01637*[2]	X	X	X	X	X	X	O
Circuit Breaker C01638*[2]	X	X	X	X	X	X	O
Circuit Breaker C01639*[2]	X	X	X	X	X	X	O
Circuit Breaker C01640*[2]	X	X	X	X	X	X	O

*[1] ½ = “half-open/half-closed”

*[2] X = “closed”, O = “open”

a) Turn on the fuel boost pumps for the fuel tank that will receive the fuel.

NOTE: Turning on the main tank fuel boost pumps will increase back-pressure to the center tank fuel boost pumps.

<1> If you must transfer fuel from the center tank into the No. 1 main tank, put the airplane in configuration B.

<2> If you must transfer fuel from the center tank into the No. 2 main tank, put the airplane in configuration C.

<3> Continue from where you were last in the fuel transfer procedure.

b) If the main tank fuel boost pumps did not solve the problem, do these steps to put the airplane in configuration D:

<1> On the P5 Overhead Panel, set the switches for the applicable main tank fuel boost pumps to the OFF position.

<2> Manually set the defueling valve to a half-open/half-closed position.

<3> Continue from where you were last in the fuel transfer procedure.

c) If adjusting the defueling valve did not solve the problem, do these steps:

<1> Keep the defueling valve in the half-open/half-closed position.

<2> On the P5 Overhead Panel, set the applicable fuel pump switches to the positions indicated in Table 203.

<a> If you must transfer fuel from the center tank into the No. 1 main tank, put the airplane in configuration E.

<b> If you must transfer fuel from the center tank into the No. 2 main tank, put the airplane in configuration F.

<3> Continue from where you were last in the fuel transfer procedure.

d) If the automatic shutoff system continues to prevent the fuel transfer, do these steps to put the airplane in configuration G:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<1> On the P5 Overhead Panel, set the switches for all main tank fuel boost pumps to the OFF position.

<2> Set the defueling valve handle to the open position.

<a> If the defueling valve handle is hard to operate, lubricate the mechanism with aircraft instrument oil, D50248.

<b> If that does not decrease the force necessary to move the defueling valve handle, remove and replace the defueling valve handle (PAGEBLOCK 28-26-11/401).

<3> Open the indicated circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	3	C01637	BOOST PMP CTR TNK L AUTO SHUT OFF-DC
C	7	C01638	BOOST PMP CTR TNK R AUTO SHUT OFF-DC
D	7	C01659	AT S-O/UCPO BST PMP CTR TNK L AC
E	7	C01658	FUEL AT S-O/UCPO BST PMP CTR TNK R AC

<4> Continue from where you were last in the fuel transfer procedure.

(c) Do these steps to transfer fuel with passengers on board:

WARNING YOU MUST OBEY THE FUEL SAFETY PRECAUTIONS WHILE PASSENGERS ARE IN THE AIRCRAFT. MAKE SURE THAT THE FUEL LEVEL DOES NOT BECOME LESS THAN 2000 LB (907 KG). IF YOU DO NOT OBEY THE SAFETY PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

1) Do not let the fuel quantity in a main tank become less than 2000 lb (907 kg).

WARNING YOU MUST OBEY THE FUEL SAFETY PRECAUTIONS WHILE PASSENGERS ARE IN THE AIRCRAFT. MAKE SURE THAT THE FUEL LEVEL DOES NOT BECOME LESS THAN 2000 LB (907 KG). IF YOU DO NOT OBEY THE SAFETY PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

2) If the automatic shutoff system for the center tank fuel boost pump is disabled, do not let the fuel quantity in the center tank become less than 2000 lb (907 kg).

SUBTASK 28-26-00-860-057

(15) To move fuel out of a fuel tank, use this table to set the fuel boost pump switch positions to remove the fuel from that fuel tank:

Table 204/28-26-00-993-804

FUEL BOOST PUMP SWITCH	SWITCH POSITION FOR DEFUELING		
	MAIN TANK NO. 1	MAIN TANK NO. 2	CENTER TANK
FUEL PUMPS AFT 1	ON	OFF	OFF
FUEL PUMPS FWD 1	ON	OFF	OFF
FUEL PUMPS AFT 2	OFF	ON	OFF

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Table 204/28-26-00-993-804 (Continued)

FUEL BOOST PUMP SWITCH	SWITCH POSITION FOR DEFUELING		
	MAIN TANK NO. 1	MAIN TANK NO. 2	CENTER TANK
FUEL PUMPS FWD 2	OFF	ON	OFF
FUEL PUMPS L CTR	OFF	OFF	ON
FUEL PUMPS R CTR	OFF	OFF	ON

SUBTASK 28-26-00-650-008

(16) Continue to move the fuel until the refuel quantity indicator shows the necessary quantity of remaining fuel in the fuel tank.

(a) You can also monitor the applicable low pressure light to show when the applicable fuel tank is empty.

(b) When the applicable low pressure lights come on, the applicable fuel tank is empty.

(c) Set the applicable fuel boost pump switches to the OFF position.

SUBTASK 28-26-00-860-058

(17) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	3	C01637	BOOST PMP CTR TNK L AUTO SHUT OFF-DC
C	7	C01638	BOOST PMP CTR TNK R AUTO SHUT OFF-DC
D	7	C01659	AT S-O/UCPO BST PMP CTR TNK L AC
E	7	C01658	FUEL AT S-O/UCPO BST PMP CTR TNK R AC

SUBTASK 28-26-00-860-059

(18) On the P15 Refuel Panel, set the fueling shutoff valve switches to the CLOSED position.

SUBTASK 28-26-00-410-007

(19) Close this access panel for the refuel station:

Number	Name/Location
621GB	Refuel Valve Door

SUBTASK 28-26-00-860-060

(20) Set the defueling valve handle to the closed position.

(a) If the defueling valve handle is hard to operate, lubricate the mechanism with aircraft instrument oil, D50248.

(b) If that does not decrease the force necessary to move the defueling valve handle, remove and replace the defueling valve handle (PAGEBLOCK 28-26-11/401).

SUBTASK 28-26-00-410-008

(21) Close this access panel for the defueling valve:

Number	Name/Location
621EB	Lower Leading Edge Access Panel

SUBTASK 28-26-00-860-061

(22) On the P5-2 Fuel System Module, set the CROSS FEED switch to the closed position if you opened it before this step.

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SUBTASK 28-26-00-860-062

(23) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 28-26-00-680-003

(24) If you are scheduled to drain the sump fuel, do this task: TASK 12-11-00-650-807.

——— END OF TASK ———

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DEFUELING VALVE - REMOVAL/INSTALLATION

1. General

A. The defueling valve has these three primary assemblies:
   (1) The actuator handle assembly
   (2) The adapter shaft
   (3) The valve body

B. This procedure has these tasks:
   (1) The removal actuator handle assembly
   (2) The installation of the actuator handle assembly
   (3) The removal of the adapter shaft
   (4) The installation of the adapter shaft
   (5) The alignment of the adapter shaft
   (6) The removal of the valve body
   (7) The installation of the valve body

C. You must fully defuel and purge the No. 2 main tank to remove the adapter shaft or the valve body.

TASK 28-26-11-010-801

2. Defueling Valve Handle Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the defueling valve handle.

B. References

Reference	Title
20-10-44-000-801	Lockwire, Cotter Pins, and Lockrings - Removal (P/B 401)

C. Location Zones

Zone	Area
621	Right Wing - Leading Edge to Front Spar
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Access Panels

Number	Name/Location
621EB	Lower Leading Edge Access Panel

E. Prepare for the Removal

SUBTASK 28-26-11-010-001

(1) Open this access panel to get access to the defueling valve handle:

Number	Name/Location
621EB	Lower Leading Edge Access Panel

NOTE: The access door is on the forward side of the front spar and on the outboard side of the No. 2 engine.

SUBTASK 28-26-11-860-001

(2) Make sure that the defueling valve handle [6] is in the CLOSED position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Defueling Valve Handle Removal

SUBTASK 28-26-11-020-001

(1) Remove the lockwire from the four screws [1] (TASK 20-10-44-000-801).

SUBTASK 28-26-11-020-009

(2) Remove the sealant from the four screws [1] and washers [2].

SUBTASK 28-26-11-020-002

(3) Remove the four screws [1] and washers [2] that attach the defueling valve handle [6] to the valve index plate [4].

SUBTASK 28-26-11-020-003

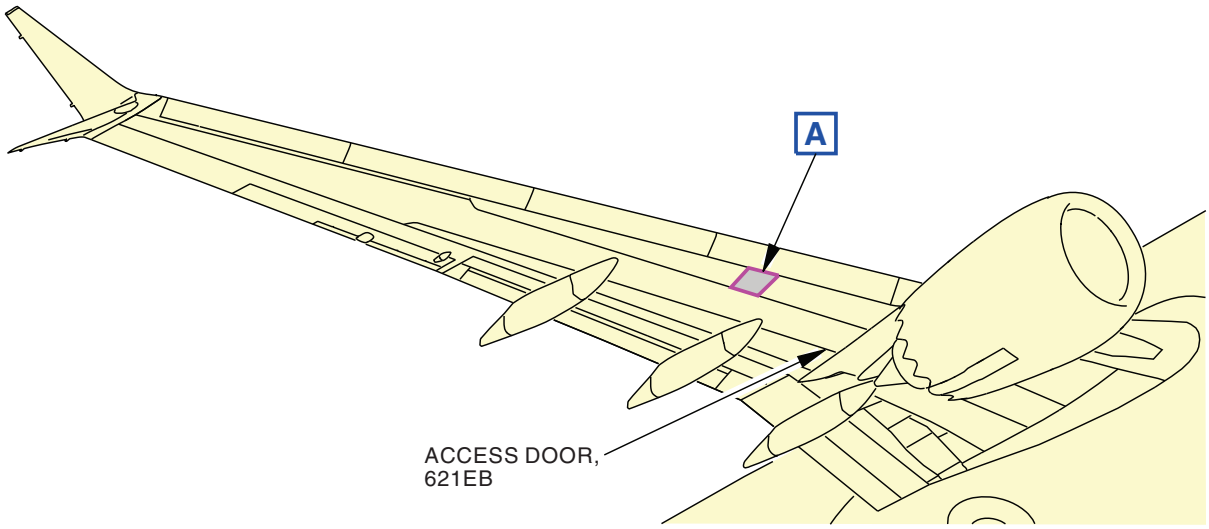
(4) Carefully remove the defueling valve handle [6] from the opening on the front spar.

SUBTASK 28-26-11-420-001

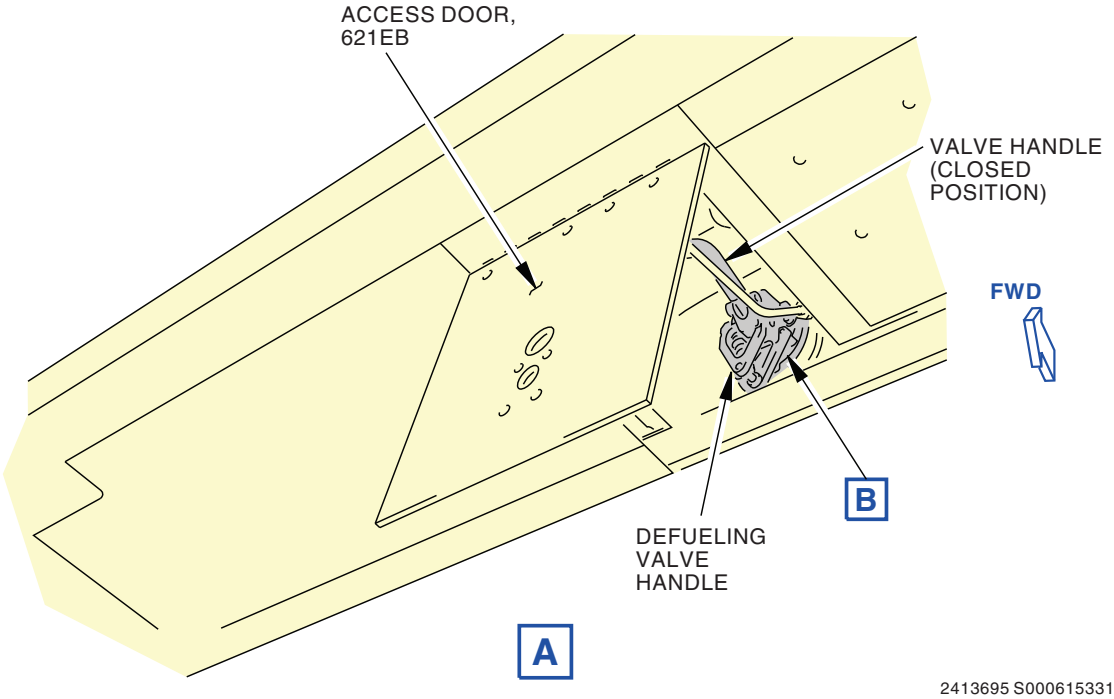
(5) Make sure that the adapter shaft did not move if you are not scheduled to remove the adapter shaft.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

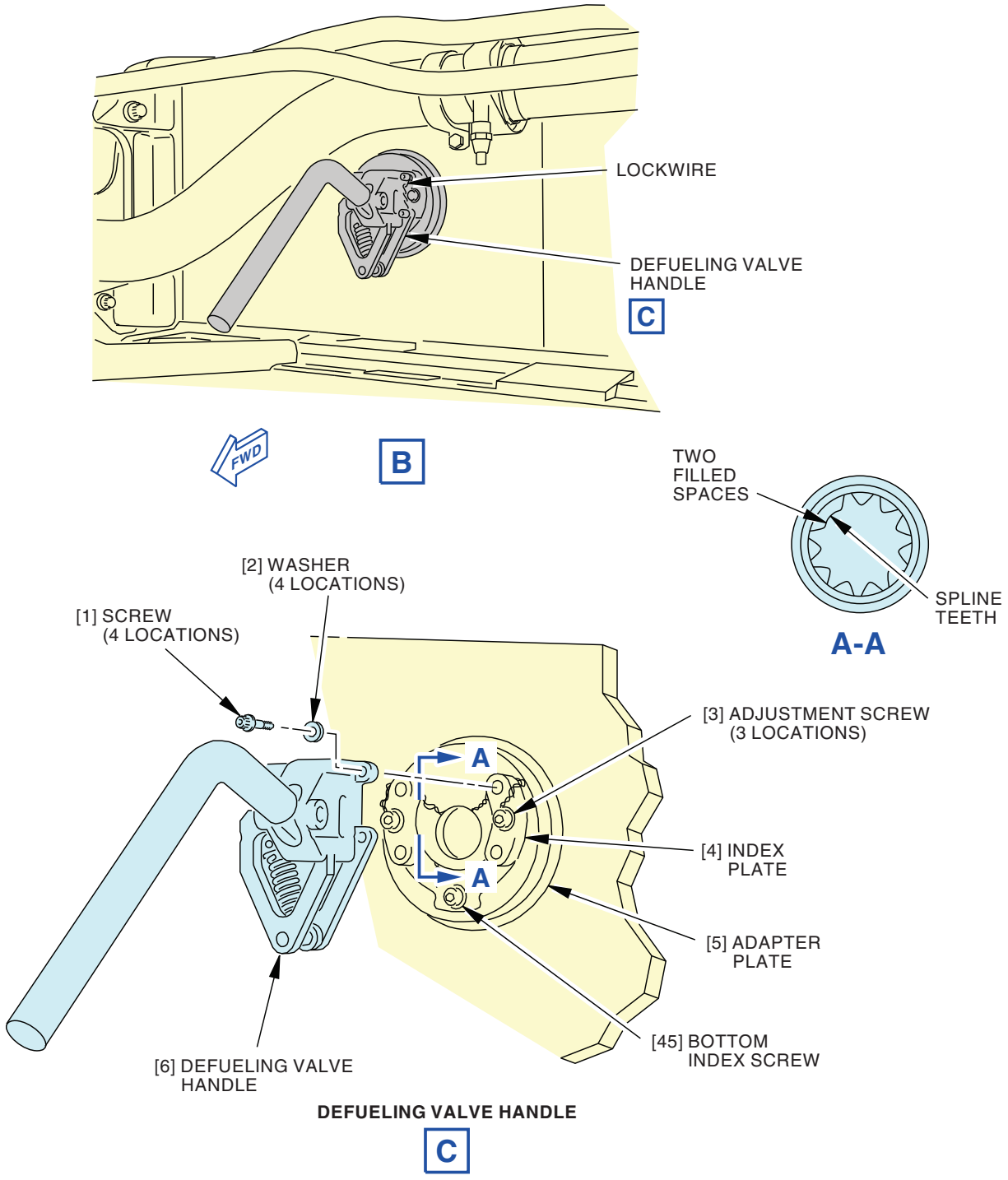


RIGHT WING



Defueling Valve Handle Installation
Figure 401/28-26-11-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Defueling Valve Handle Installation
Figure 401/28-26-11-990-801 (Sheet 2 of 2)

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TASK 28-26-11-400-801

3. Defueling Valve Handle Installation

(Figure 401)

A. General

(1) This task gives instructions to install the defueling valve handle.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
51-22-00-390-808	Fastener Seal Application (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G01505	Lockwire - Safety And Lock	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
6	Defueling valve handle	28-26-11-01-010	PTF ALL

E. Location Zones

Zone	Area
621	Right Wing - Leading Edge to Front Spar
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

F. Access Panels

Number	Name/Location
621EB	Lower Leading Edge Access Panel

G. Defueling Valve Handle Installation

SUBTASK 28-26-11-860-002

(1) Move the handle on the defueling valve handle [6] to the CLOSED position.

SUBTASK 28-26-11-420-002

(2) Align the mating shaft on the defueling valve handle [6] with the two teeth spaces on the adapter shaft (View A-A, Figure 401).

NOTE: The mounting feet on the defueling valve handle [6] automatically align with the mounting points on the index plate [4].

SUBTASK 28-26-11-420-003

(3) Attach the defueling valve handle [6] to the valve index plate [4] with the screws [1].

(a) Install the four screws [1] and washers [2].

1) Tighten the four screws [1] to 20 in-lb (2 N·m).

(b) Install the lockwire, G01505, on the screws [1] (TASK 20-10-44-400-801).

SUBTASK 28-26-11-390-004

(4) Apply sealant, A00247 to the four screws [1] and washers [2] (TASK 51-22-00-390-808).

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H. Put Airplane Back to Its Usual Condition

SUBTASK 28-26-11-410-001

(1) Close this access panel:

Number	Name/Location
621EB	Lower Leading Edge Access Panel

——— END OF TASK ———

TASK 28-26-11-010-802

4. Defueling Valve - Adapter Shaft Removal

(Figure 402)

A. General

(1) This task gives instructions to remove the defueling valve - adapter shaft.

(2) Two people minimum are required to do this procedure.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-26-00-650-802	Tank to Tank Fuel Transfer (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
COM-4746	Handle - Sealant Removal Tool Part #: 310/1 (for SkyScraper) Supplier: F6892 Part #: SHR0272-T (for PnJ scraper) Supplier: 7RKH2 Part #: ST982LH-F-3 (for ST982LF) Supplier: 81205 Opt Part #: SHW0215 (for PnJ scraper) Supplier: 7RKH2

D. Location Zones

Zone	Area
621	Right Wing - Leading Edge to Front Spar
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

E. Access Panels

Number	Name/Location
621EB	Lower Leading Edge Access Panel
632DB	Main Tank Access Door - Wing Station 313

F. Prepare for the Removal

SUBTASK 28-26-11-860-003

(1) On the P5-2 Fuel System Module, turn the CROSS FEED switch to the closed position.

(a) Make sure that the fuel crossfeed valve light is off.

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SUBTASK 28-26-11-860-004

WARNING MAKE SURE THAT YOU COMPLETE ALL OF THE DEFUEL SAFETY PROCEDURES BEFORE YOU GO INTO THE FUEL TANK. DAMAGE AND INJURIES TO PERSONS CAN OCCUR.

(2) Remove fuel from the No. 2 main tank and the center tank (TASK 28-26-00-650-801 or TASK 28-26-00-650-802).

SUBTASK 28-26-11-860-005

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(3) Do this task: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access door:

(TASK 28-11-11-000-801).

Number	Name/Location
632DB	Main Tank Access Door - Wing Station 313

SUBTASK 28-26-11-010-004

(4) Open this access panel to get access to the defueling valve adapter:

Number	Name/Location
621EB	Lower Leading Edge Access Panel

NOTE: The access panel is on the forward side of the front spar and on the outboard side of the No. 2 engine.

G. Defueling Valve - Adapter Shaft Removal

SUBTASK 28-26-11-010-005

(1) Do these steps on the outside of the fuel tank:

(a) Remove the defueling valve handle [6] (TASK 28-26-11-010-801).

(b) Use a sealant removal tool to remove the sealant from the periphery of the valve adapter plate [5] (View A-A, Figure 402).

1) Use a sealant removal tool, COM-2481 and sealant removal tool handle, COM-4746 or equivalent.

SUBTASK 28-26-11-010-006

WARNING DO THE PURGING AND FUEL TANK ENTRY PROCEDURE. FAILURE TO FOLLOW THE PURGING AND FUEL TANK ENTRY PROCEDURE COULD CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Have the second person go in the fuel tank to the defueling valve [25] location (TASK 28-11-00-910-802) (View A, Figure 402):

(a) Go to the valve adapter [11] location (View B, Figure 402).

(b) Remove the sealant from the bolts [22] and the valve adapter [11] front spar penetration (View A-A, Figure 402).

(c) Remove the bolts [22] and washers [21] that attach the valve adapter plate [5] to the front spar.

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(d) Disengage the adapter shaft from the defueling valve [25] (View C, Figure 402).

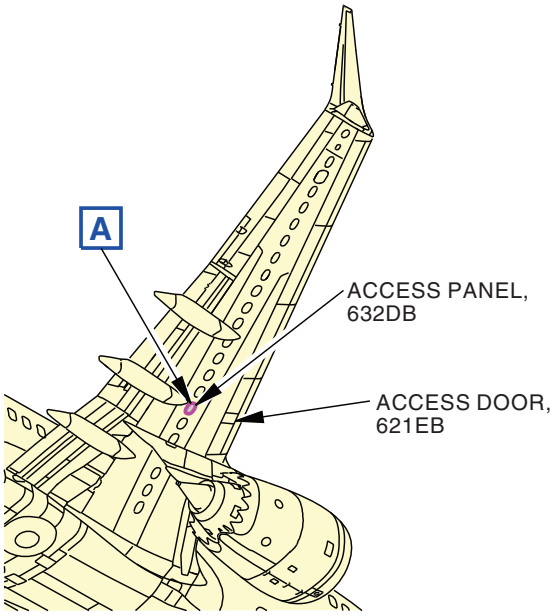
SUBTASK 28-26-11-010-007

(3) Have the person outside remove the adapter shaft through the hole in the front spar.

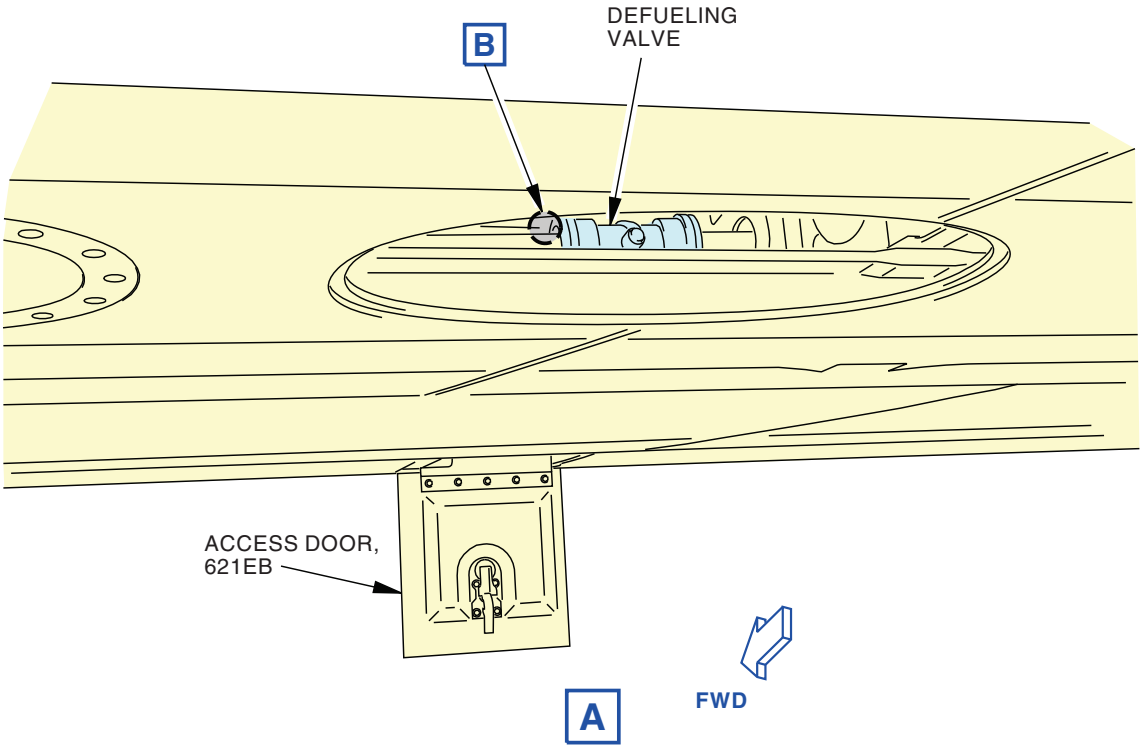
(a) Discard the O-ring [27].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



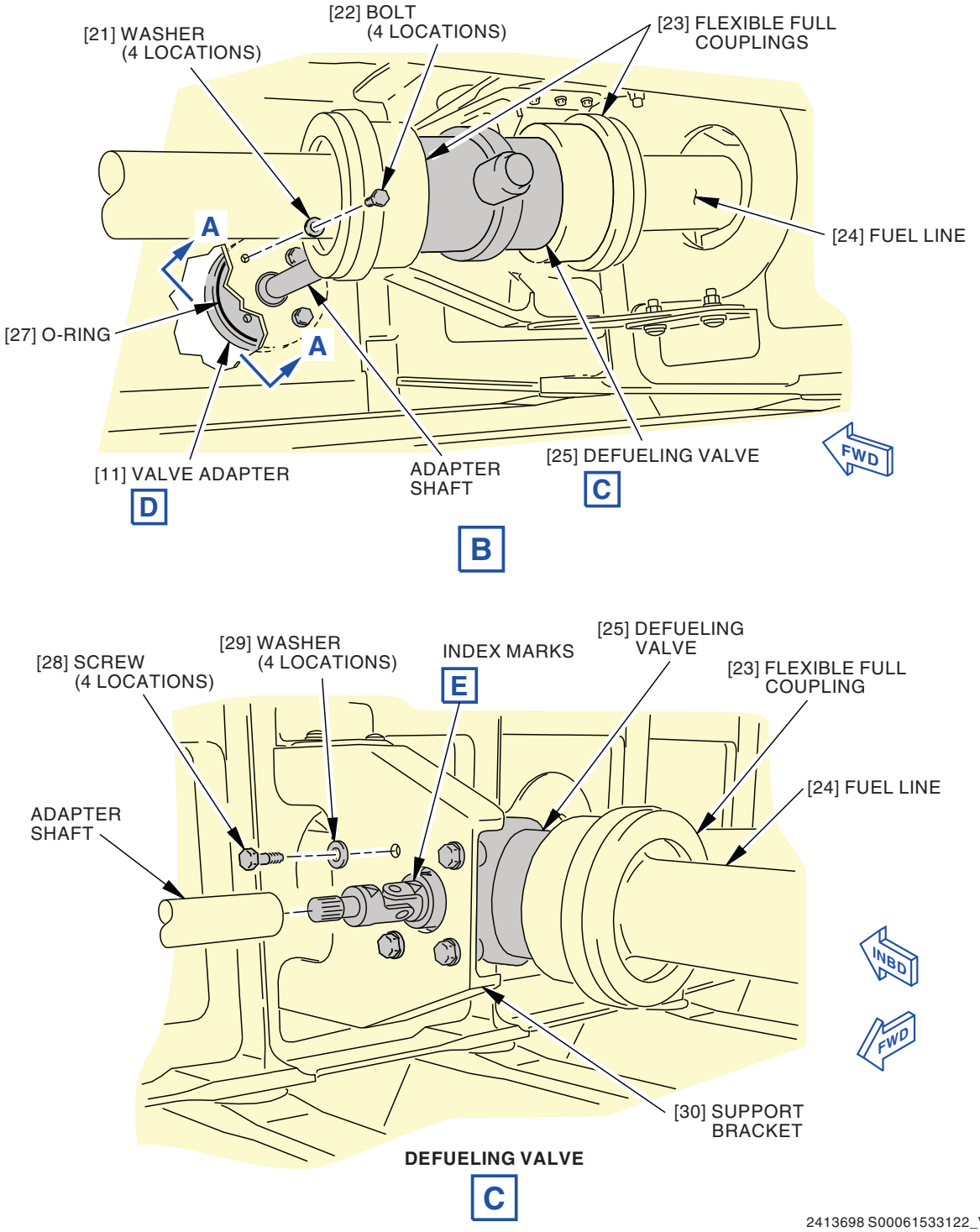
RIGHT WING



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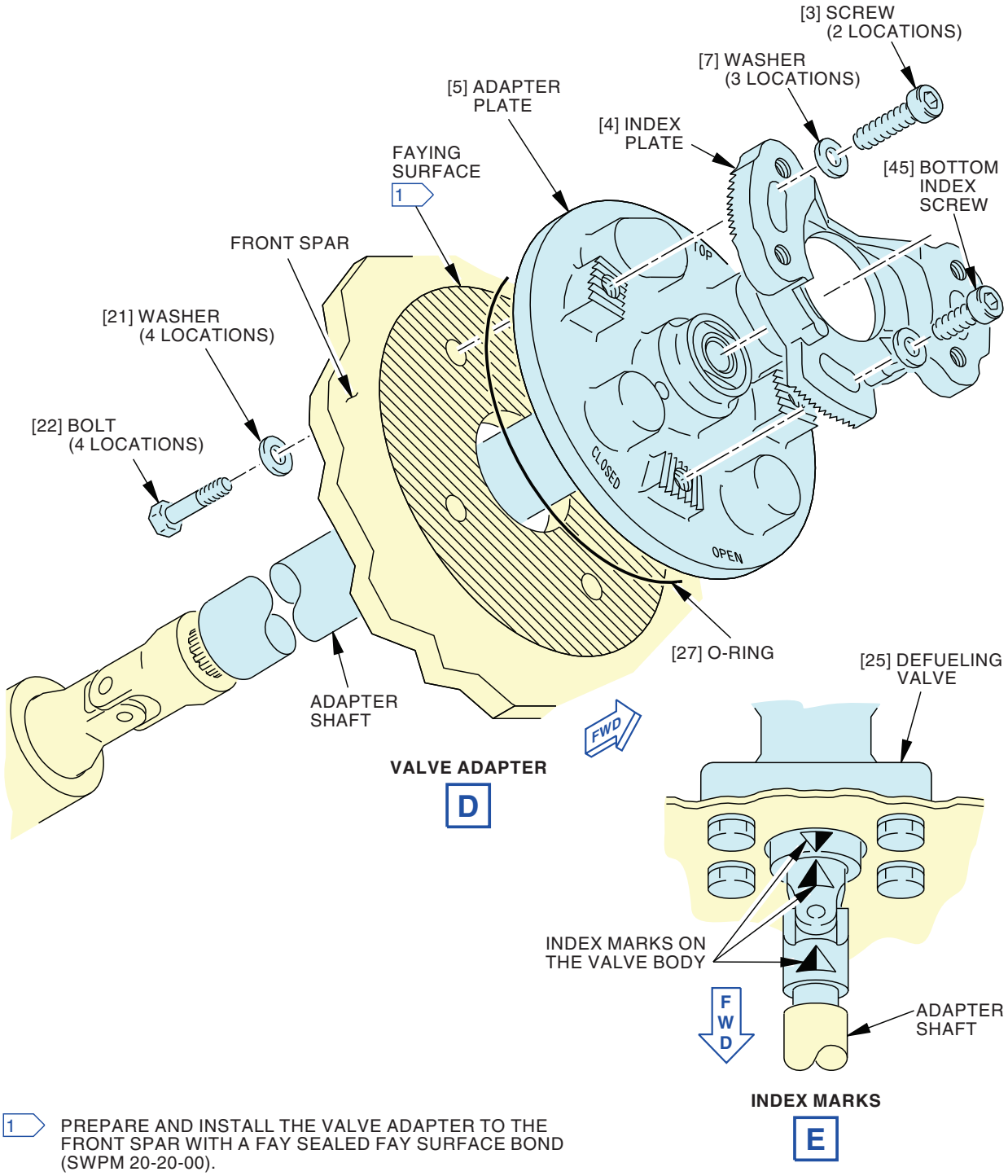
Defueling Valve Installation
Figure 402/28-26-11-990-802 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



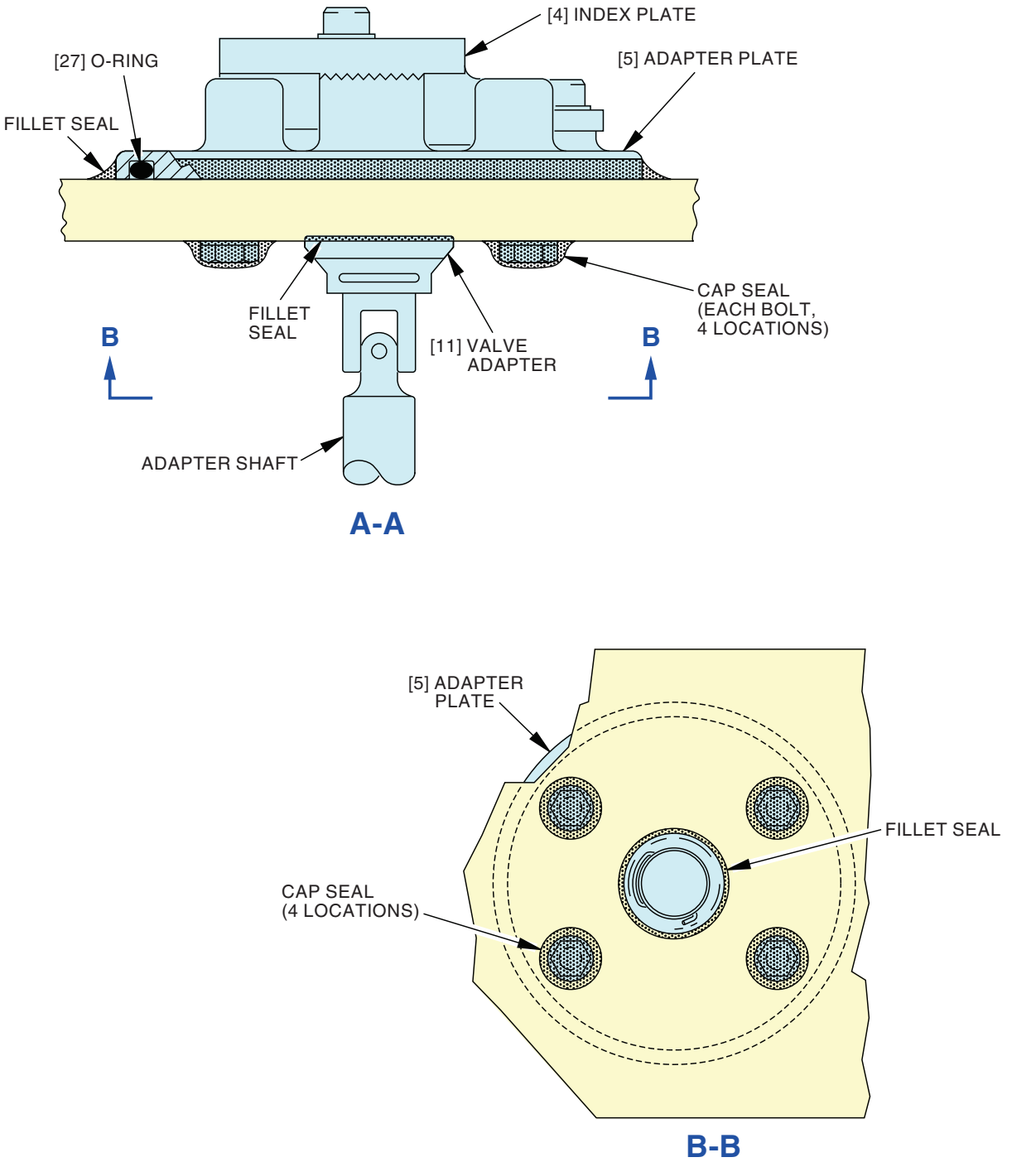
Defueling Valve Installation
Figure 402/28-26-11-990-802 (Sheet 2 of 4)

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Defueling Valve Installation
Figure 402/28-26-11-990-802 (Sheet 3 of 4)

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Defueling Valve Installation
Figure 402/28-26-11-990-802 (Sheet 4 of 4)

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TASK 28-26-11-410-801

5. Defueling Valve - Adapter Shaft Installation

(Figure 402)

A. General

(1) This task gives instructions to install the defueling valve - adapter shaft.

(2) The adapter shaft is used to connect the defueling valve handle to the defueling valve. The adapter shaft consists of these components:

(a) Adapter plate (with index plate installed)

(b) Adapter shaft (with U-joint connection).

(3) Two people minimum are required to do this procedure.

(4) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-31.

B. References

Reference	Title
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-00-600-801	Apply the Corrosion Resistant Finish (Topcoat) (P/B 701)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
51-21-41-370-802	Apply Bonderite M-CR 600, Bonderite M-CR 1200, Bonderite M-CR 600RTU, or Bonderite M-CR 1200S Solution (P/B 701)
51-22-00-390-806	Fillet Seal Application (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

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(Continued)

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
COM-4746	Handle - Sealant Removal Tool Part #: 310/1 (for SkyScraper) Supplier: F6892 Part #: SHR0272-T (for PnJ scraper) Supplier: 7RKH2 Part #: ST982LH-F-3 (for ST982LF) Supplier: 81205 Opt Part #: SHW0215 (for PnJ scraper) Supplier: 7RKH2
STD-123	Brush - Soft Bristle

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A00767	Sealant - Fuel Tank	BMS5-45
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00307	Coating - Corrosion Resistant Finish For Integral Fuel Tanks	BMS10-20 Type II
C50033	Coating - Chemical Conversion - Bonderite M-CR 1200S Aero (Formerly Alodine 1200)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
D00504	Grease - Petrolatum	VV-P-236

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(Continued)

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
25	Defueling valve	28-26-11-05-065	PTF ALL
26	Adapter shaft	28-26-11-05-010	PTF ALL
27	O-ring	28-26-11-05-055	PTF ALL

F. Location Zones

Zone	Area
621	Right Wing - Leading Edge to Front Spar
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

G. Access Panels

Number	Name/Location
621EB	Lower Leading Edge Access Panel
632DB	Main Tank Access Door - Wing Station 313

H. Prepare for Installation

SUBTASK 28-26-11-100-001

(1) Remove the sealant and clean the front spar, valve adapter plate [5], and fasteners.

CAUTION

USE APPROVED SEALANT REMOVAL TOOLS TO REMOVE THE SEALANT. IF YOU USE A TOOL THAT IS NOT APPROVED, YOU CAN CAUSE DAMAGE TO THE COMPONENT SURFACES AND FINISHES.

(a) To remove the sealant from the front spar use a sealant removal tool, COM-2481 and sealant removal tool handle, COM-4746 or equivalent.

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE. OBEY THE MATERIAL SAFETY DATA SHEETS (MSDS) FOR SOLVENTS. OBEY LOCAL REGULATIONS FOR THE CORRECT PROCEDURES TO USE OR DISCARD SOLVENTS. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) To remove the remaining sealant, grease, oil, dirt, and other contamination from the components, use one of these brushes/pads soaked with solvent, B00184 or solvent, B00148:

1) soft bristle brush, STD-123

2) cotton wiper, G00034

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Defueling Valve - Adapter Shaft Installation

SUBTASK 28-26-11-100-002

28-AWL-31: CDCCL

(1) Apply a fay sealed fay surface bond of sealant, A00767 between the valve adapter plate [5] and the front spar outside of the fuel tank (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-31.

SUBTASK 28-26-11-420-004

(2) Prepare the adapter shaft and the valve adapter plate [5] for installation.

(a) Apply a thin layer of grease, D00504, on the new O-ring [27].

(b) Install the O-ring [27] in the O-ring groove of the valve adapter plate [5].

(c) Put the valve adapter shaft [26] through the hole in the front spar.

SUBTASK 28-26-11-420-005

(3) Have the second person hold the valve adapter shaft [26] (View A, Figure 402).

SUBTASK 28-26-11-420-006

(4) Install the valve adapter plate [5].

(a) Align the index marks on the defueling valve [25] and the valve adapter shaft [26] (View E, Figure 402).

NOTE: A small misalignment of the index marks is satisfactory.

(b) Put the valve adapter shaft [26] spline in the spline on the defueling valve [25] (View C, Figure 402).

(c) Make sure that the shoulder of the valve adapter plate [5] does not bind in the opening of the front spar.

(d) Install the four bolts [22] and washers [21] to attach the valve adapter plate [5] to the front spar.

SUBTASK 28-26-11-220-001

28-AWL-31: CDCCL

(5) Measure the electrical bonding resistance between the valve adapter plate [5] and spar outside the tank (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-31.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-31: CDCCL

(b) Make sure that the bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-31.

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SUBTASK 28-26-11-390-001

(6) On the inside of the fuel tank, apply sealant, A00767, to the components below (View B-B, Figure 402):

28-AWL-31: CDCCL

(a) Apply a cap seal to each of the four bolts [22] (TASK 51-22-00-390-806).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-31.

28-AWL-31: CDCCL

(b) Apply a fillet seal around the edge of the hole and on the stationary part of the valve adapter plate [5].

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-31.

1) Make sure that the sealant does not touch the adapter shaft or other parts of the valve adapter plate [5] that turn.

SUBTASK 28-26-11-390-002

(7) On the spar outside of the fuel tank, apply a fillet seal of sealant, A00247, around the periphery of the valve adapter plate [5] (TASK 51-22-00-390-806) (View A-A, Figure 402).

SUBTASK 28-26-11-800-001

(8) Apply protective finish coating, C00064 or coating, C00307 to the bare metal areas of the front spar (TASK 28-11-00-600-801).

(a) Re-apply the protective coating to the front spar at all bare metal areas not covered by the fillet seal, and where the electrical bonding probe removed the finishes.

SUBTASK 28-26-11-800-002

(9) Apply Bonderite M-CR 1200S Aero coating, C50033, to all bare metal areas of the valve adapter plate [5] (TASK 51-21-41-370-802).

(a) Re-apply the protective coating to the valve adapter plate [5] where the electrical bonding probe removed the finish.

SUBTASK 28-26-11-410-002

(10) Do this task: Defueling Valve - Adapter Shaft Alignment, TASK 28-26-11-820-801.

J. Put Airplane Back to Its Usual Condition

SUBTASK 28-26-11-410-003

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Install this access door:

(TASK 28-11-11-400-801).

Number	Name/Location
632DB	Main Tank Access Door - Wing Station 313

Close this access panel:

Number	Name/Location
621EB	Lower Leading Edge Access Panel

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——— END OF TASK ———

TASK 28-26-11-820-801

6. Defueling Valve - Adapter Shaft Alignment

(Figure 403)

A. References

Reference	Title
20-10-44-000-801	Lockwire, Cotter Pins, and Lockrings - Removal (P/B 401)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1587	Wrench - Torque, 30 in-lbs, +/-5% accuracy Part #: 5718A42 Supplier: 3A054 Opt Part #: TE3FUA Supplier: 55719
SPL-1771	Alignment Equipment - Fuel Shutoff Valve Part #: B28009-1 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Location Zones

Zone	Area
621	Right Wing - Leading Edge to Front Spar
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

E. Prepare for the Procedure

SUBTASK 28-26-11-010-013

(1) Do this task: Defueling Valve Handle Removal, TASK 28-26-11-010-801

SUBTASK 28-26-11-010-014

(2) Remove the valve index plate [4] from the valve adapter plate [5].

(a) Remove the lockwires from the top two screws [3] and the bottom index screw [45] (TASK 20-10-44-000-801).

(b) Remove the top two screws [3] from the valve index plate [4].

(c) Remove the bottom index screw [45].

(d) Carefully disassemble the valve index plate [4] from the valve adapter plate [5].

NOTE: The two plates are bonded with faying surface sealant.

SUBTASK 28-26-11-480-001

(3) Install the alignment tool [44] (valve alignment equipment, SPL-1771) in the valve adapter plate [5] (View B, Figure 403):

(a) Position and align the serrated surfaces on the valve index plate [4] with the valve adapter plate [5].

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(b) Loosely install the bottom index screw [45] in the valve index plate [4].

(c) Install the alignment tool [44].

(d) Align the alignment tool [44] guide pins with the two screw holes and attach to the valve index plate [4].

(e) Make sure that the spline on the alignment tool [44] engages with the shaft spline.

F. Defueling Valve - Adapter Shaft Alignment (Alignment Tool with Handle)

SUBTASK 28-26-11-820-001

(1) To align the adapter shaft on the adapter plate [5], do these steps:

NOTE: This procedure finds the point where the internal valve butterfly engages the valve seal. The torque necessary to turn the shaft increases as the butterfly engages the seal, then decreases again as the butterfly leaves the other side of the seal. Once the edge of the seal is found, the valve shaft will then be turned to center the butterfly on the seal.

(a) Align the tool handle in the counterclockwise position.

(b) Carefully turn the alignment tool handle clockwise until the torque necessary to turn the handle increases.

(c) If you cannot find where the torque increases, move the adapter shaft counterclockwise to its initial position.

(d) Turn the alignment tool [44] clockwise until the toque necessary to turn the handle increases, then immediately stop turning the handle.

(e) If you cannot find where the torque increases, go to the next procedure.

(f) Read the scale [42] on the alignment tool [44].

NOTE: This is the point where the valve butterfly starts to engage the edge of the seal.

(g) Turn the handle 13 degrees clockwise to center the valve seal.

NOTE: If the handle was turned more than 13 degrees, do not turn the handle back in the counterclockwise direction again. Start the procedure again with the handle in the counterclockwise position.

SUBTASK 28-26-11-480-003

(2) Tighten the lockscrew [43] to keep the alignment tool [44] in its position on the valve adapter plate [5].

NOTE: The lockscrew [43] keeps the adapter shaft calibrated while the valve index plate [4] is aligned.

SUBTASK 28-26-11-820-008

(3) Turn the valve index plate [4] until the actuator mounting holes in the valve index plate [4] align with the alignment screws [41].

SUBTASK 28-26-11-820-009

(4) Install the two alignment screws [41] to secure the valve index plate [4] to the valve adapter plate [5].

SUBTASK 28-26-11-480-004

(5) Tighten the bottom index screw [45] to a torque of 22 ±1 in-lb (2.5 ±0.1 N·m).

SUBTASK 28-26-11-080-003

(6) Disengage the two alignment screws [41].

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SUBTASK 28-26-11-080-004

(7) Loosen the lockscrew [43].

SUBTASK 28-26-11-080-005

(8) Carefully remove the alignment tool [44].

SUBTASK 28-26-11-930-001

(9) Use a fine point felt pen to apply a reference mark between the valve index plate [4] and the valve adapter plate [5] at one of the serrated attachment points.

SUBTASK 28-26-11-420-015

(10) Install the top two screws [3] in the valve index plate [4].

SUBTASK 28-26-11-480-005

(11) Tighten the top two screws [3] to 22 ±1 in-lb (2.5 ±0.1 N·m).

SUBTASK 28-26-11-420-016

(12) Install MS20995C32 lockwire, G01048 on the screws [3] and bottom index screw [45] (TASK 20-10-44-400-801).

SUBTASK 28-26-11-820-003

(13) Tighten the lockscrew [43] on the alignment tool [44] to keep the tool in its position on the adapter shaft (in its position).

G. Defueling Valve - Adapter Shaft Alignment (Alignment Tool with Wrench)

SUBTASK 28-26-11-820-002

(1) To align the adapter shaft on the valve adapter plate [5], do these steps:

NOTE: A torque wrench (0-30 lbs), COM-1587 is necessary to do this procedure.

(a) Put the torque wrench (0-30 lbs), COM-1587 in the base of the handle.

(b) Carefully turn the torque wrench (0-30 lbs), COM-1587, on the alignment tool handle, clockwise until the torque necessary to turn the handle increases.

(c) Write down the maximum torque that you applied, while you turned the torque wrench (0-30 lbs), COM-1587 in the clockwise position.

(d) Align the adapter shaft back to the initial counterclockwise position.

(e) Turn the torque wrench (0-30 lbs), COM-1587 clockwise at approximately the same rate.

(f) Stop at the position where the torque increases to approximately two-thirds of the maximum torque that you wrote.

(g) Read the scale [42] on the alignment tool [44] at this position.

NOTE: This is the position where valve butterfly starts to engage the edge of the seal.

(h) Turn the torque wrench (0-30 lbs), COM-1587 13 degrees clockwise to center the valve seal.

NOTE: If you turn the handle more than 13 degrees, do not turn the handle counterclockwise. Start the procedure again with the tool handle in the counterclockwise position.

SUBTASK 28-26-11-480-006

(2) Tighten the lockscrew [43] to keep the alignment tool [44] in its position on the valve adapter plate [5]

NOTE: The lockscrew [43] keeps the adapter shaft calibrated while the valve index plate [4] is aligned.

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SUBTASK 28-26-11-820-004

(3) Turn the index plate [4] until the actuator mounting holes in the valve index plate [4] are aligned with the two alignment screws [41].

SUBTASK 28-26-11-820-005

(4) Install the two alignment screws [41] to secure the valve index plate [4] to the valve adapter plate [5].

SUBTASK 28-26-11-420-007

(5) Tighten the bottom index screw [45] to a torque of 22 ±1 in-lb (2.5 ±0.1 N·m).

SUBTASK 28-26-11-020-004

(6) Disengage the two alignment screws [41].

SUBTASK 28-26-11-080-001

(7) Loosen the lockscrew [43].

SUBTASK 28-26-11-080-006

(8) Carefully remove the alignment tool [44].

SUBTASK 28-26-11-930-002

(9) Use a fine point felt pen to apply a reference mark between the valve index plate [4] and the valve adapter plate [5] at one of the serrated attachment points.

SUBTASK 28-26-11-420-008

(10) Install the top two screws [3] in the valve index plate [4].

SUBTASK 28-26-11-480-007

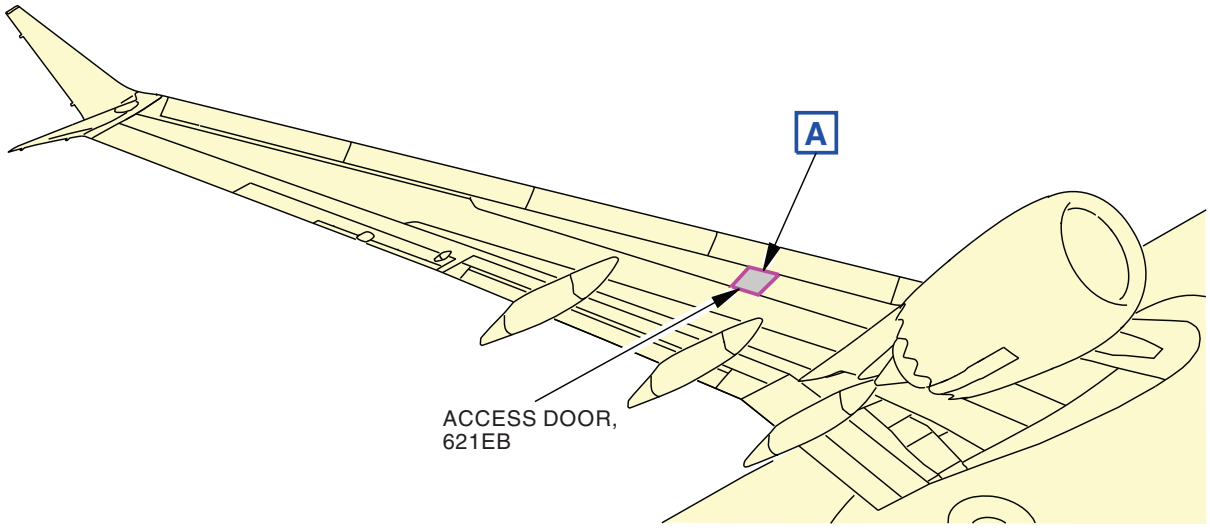
(11) Tighten the top two screws [3] to 22 ±1 in-lb (2.5 ±0.1 N·m).

SUBTASK 28-26-11-420-009

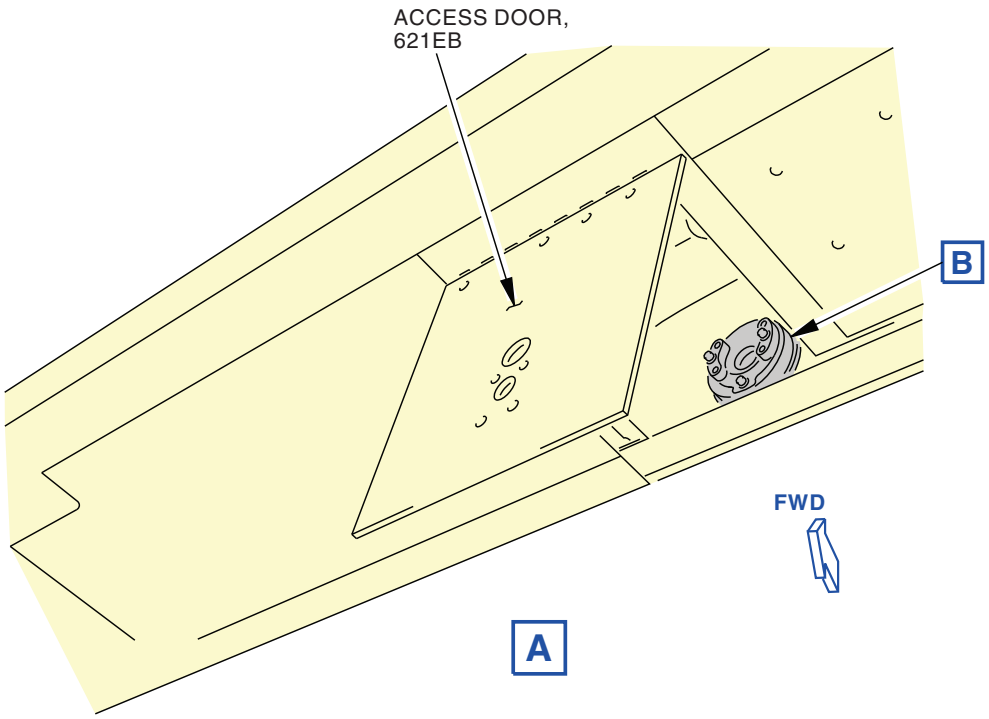
(12) Install MS20995C32 lockwire, G01048 on the screws [3] and bottom index screw [45] (TASK 20-10-44-400-801).

——— END OF TASK ———

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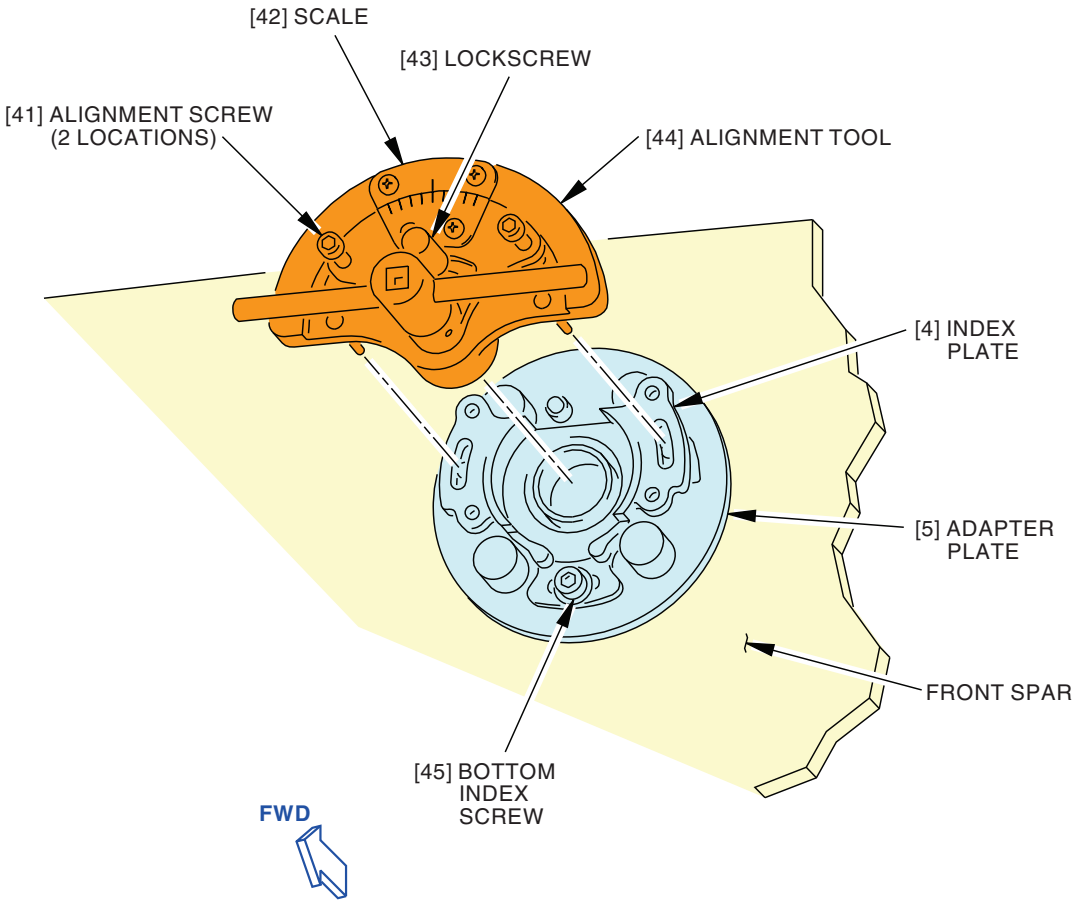
RIGHT WING



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Defueling Valve Alignment
Figure 403/28-26-11-990-803 (Sheet 1 of 2)

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Defueling Valve Alignment
Figure 403/28-26-11-990-803 (Sheet 2 of 2)

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TASK 28-26-11-020-801

7. Defueling Valve - Valve Body Removal

(Figure 402)

A. General

(1) This task gives instructions to remove the defueling valve - valve body.

(2) The valve body is referred to as the defueling valve [25] in this task.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-22-15-000-803	Couplings Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)

C. Location Zones

Zone	Area
621	Right Wing - Leading Edge to Front Spar
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Access Panels

Number	Name/Location
632DB	Main Tank Access Door - Wing Station 313

E. Prepare for the Removal

SUBTASK 28-26-11-860-006

WARNING DO ALL THE SAFETY PROCEDURES TO DEFUEL THE TANK AND TO GO INTO IT. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR IF YOU DO NOT FOLLOW THE SAFETY PROCEDURES.

(1) Do this task: Fuel Tank Defueling, TASK 28-26-00-650-801.

SUBTASK 28-26-11-860-007

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(2) Do this task: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

SUBTASK 28-26-11-010-015

(3) Remove this access door:

(TASK 28-11-11-000-801).

Number	Name/Location
632DB	Main Tank Access Door - Wing Station 313

F. Defueling Valve - Valve Body Removal

SUBTASK 28-26-11-010-011

(1) Go into the No. 2 main tank and find the defueling valve [25].

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SUBTASK 28-26-11-020-005

(2) Disconnect the flexible full coupling [23] from each side of the defueling valve [25] (TASK 28-22-15-000-803).

SUBTASK 28-26-11-020-006

(3) Hold the defueling valve [25] and remove the screws [28] and washers [29] that attach the defueling valve [25] to the support bracket [30].

SUBTASK 28-26-11-020-007

(4) Disengage the defueling valve [25] from the valve adapter shaft [26].

SUBTASK 28-26-11-020-008

(5) Remove the defueling valve [25].

SUBTASK 28-26-11-480-002

(6) Put covers on the holes in the fuel line [24] to keep unwanted materials out.

——— END OF TASK ———

TASK 28-26-11-420-801

8. Defueling Valve - Valve Body Installation

(Figure 402)

A. General

(1) This task gives instructions to install the defueling valve - valve body.

(2) The valve body is referred to as the defueling valve in this task.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
28-11-00-600-801	Apply the Corrosion Resistant Finish (Topcoat) (P/B 701)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
28-22-15-400-803	Couplings Installation (P/B 401)
28-26-00-650-802	Tank to Tank Fuel Transfer (P/B 201)
51-22-00-390-806	Fillet Seal Application (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
C00307	Coating - Corrosion Resistant Finish For Integral Fuel Tanks	BMS10-20 Type II

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
25	Defueling valve	28-26-11-05-065	PTF ALL

F. Location Zones

Zone	Area
621	Right Wing - Leading Edge to Front Spar
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

G. Access Panels

Number	Name/Location
632DB	Main Tank Access Door - Wing Station 313

H. Defueling Valve - Valve Body Installation

SUBTASK 28-26-11-080-002

(1) Remove the covers from the fuel line.

SUBTASK 28-26-11-820-006

(2) Align the index marks on the defueling valve [25] and the valve adapter shaft [26].

NOTE: A small misalignment of the index marks is satisfactory.

SUBTASK 28-26-11-420-010

(3) Engage the defueling valve [25] in the valve adapter shaft [26].

(a) Make sure that the two teeth spaces on the spline of the valve adapter shaft [26] align with the teeth spaces in the rotor on the defueling valve [25].

(b) Make sure that the spline on the valve adapter shaft [26] is correctly engaged with the spline on the rotor of the defueling valve [25].

SUBTASK 28-26-11-420-014

(4) Apply a fillet sealed fay surface bond between the defueling valve [25] and the support bracket [30] (SWPM 20-20-00, Paragraph 2).

(a) Apply sealant, A50110 to the fay surface of the defueling valve [25].

SUBTASK 28-26-11-820-007

(5) Align the defueling valve [25] with the support bracket [30].

SUBTASK 28-26-11-420-011

(6) Install the washers [29] and screws [28] through the support bracket [30] to attach to the defueling valve [25].

(a) Tighten the screws [28] to 35 ±1 in-lb (4 ±0 N·m).

SUBTASK 28-26-11-420-013

(7) Measure the bonding resistance between the defueling valve [25] and the support bracket [30] with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) Make sure that the bonding resistance is 0.01 ohm (10 milliohms) or less.

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SUBTASK 28-26-11-390-003

(8) Apply fillet seals of sealant, A50110 over the heads of the screws [28] on the support bracket [30] (TASK 51-22-00-390-806).

SUBTASK 28-26-11-800-003

(9) Apply protective finish coating, C00307 to the bare metal areas of the support bracket [30] not covered by the fillet seal, and where the electrical bonding probe removed the finishes (TASK 28-11-00-600-801).

SUBTASK 28-26-11-420-012

(10) Connect the flexible full couplings [23] that attach the defueling valve [25] to the fuel line [24] (TASK 28-22-15-400-803).

I. Defueling Valve - Valve Body Installation Test

SUBTASK 28-26-11-010-012

(1) Install this access door:

(TASK 28-11-11-400-801).

Number	Name/Location
632DB	Main Tank Access Door - Wing Station 313

SUBTASK 28-26-11-710-001

(2) To do an operational test of the defueling valve [25], do these steps:

(a) Add a minimum of 300 lb (136 kg) to the No. 1 main tank (TASK 12-11-00-650-803).

(b) Make sure the defueling valve [25] is in the closed position.

1) Transfer a minimum of 100 lb (45 kg) from the No. 1 main tank to the No. 2 main tank (TASK 28-26-00-650-802).

(c) Monitor the fuel quantity indicators for the No. 1 and the No. 2 main tanks for five minutes.

(d) Make sure on the P5-2 Fuel System Module that all fuel boost pumps are OFF and the fuel crossfeed valve is closed.

(e) Make sure that there is no fuel transfer as shown by the fuel quantity indicating system.

——— END OF TASK ———

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FUEL QUANTITY INDICATING SYSTEM (FQIS) - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks for the fuel quantity indicating system (FQIS):

(1) Fuel Quantity Indicating System - Operational Test

(2) Fuel Quantity Indicating System (FQIS) with the Fuel Measuring Sticks - Comparison Check

(3) Fuel Quantity Indicating System - Dry Capacitance Test

B. The operational test makes sure the FQIS operates correctly.

C. The comparison check makes sure the fuel quantity shown agrees with the correct quantity of fuel in the fuel tanks.

D. The dry capacitance test makes sure that the total capacitance for the tank units and the compensators are in the permitted limits for the FQIS precision.

TASK 28-41-00-710-801

2. Fuel Quantity Indicating System - Operational Test

(Figure 501, Figure 502)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure contains:

(a) A check of the fuel quantity indicators, on the P15 refuel panel.

(b) A fuel quantity Built-In-Test Equipment (BITE) procedure to be done from the Control Display Unit (CDU) in the flight compartment.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
621	Right Wing - Leading Edge to Front Spar

C. Access Panels

Number	Name/Location
621GB	Refuel Valve Door

D. Check of the Fuel Quantity Indicators

SUBTASK 28-41-00-010-001

(1) Open this access panel:

Number	Name/Location
621GB	Refuel Valve Door

SUBTASK 28-41-00-210-001

(2) Make sure that the display on each refuel quantity indicator shows the quantity of fuel in each fuel tank.

SUBTASK 28-41-00-710-001

(3) On the P15 refuel panel, select and hold the fueling indication test switch to the TEST GAUGES position.

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(a) Make sure that all of the refuel quantity indicators operate correctly.

NOTE: The test blanks the fueling indicators display for two seconds, then all Light Emitting Diode (LED) segments go on for two seconds. This sequence continues as long as you hold the test switch. If you hold the test switch for more than 20 seconds, the test mode will time out and the indicator will go back to its usual operating mode. If an internal fault is found during the test, the indicator will show Ind FAIL.

(b) Release the fueling indication test switch.

SUBTASK 28-41-00-210-002

(4) Make sure that the display on each refuel quantity indicator shows the quantity of fuel in each fuel tank.

SUBTASK 28-41-00-410-001

(5) Close this access panel:

Number	Name/Location
621GB	Refuel Valve Door

E. Fuel Quantity BITE Test

SUBTASK 28-41-00-740-001

(1) Do the BITE procedure for the Fuel Quantity Indicating System (FQIS).

(a) If you are not at the CDU Initial Reference Index (INIT/REF INDEX), or one of the FQIS BITE menus, do these steps:

1) Push the INIT REF function key.

2) Push the line select key next to the INDEX prompt.

NOTE: This makes the CDU Initial Reference Index (INIT/REF INDEX) show.

(b) On the CDU, push the line select key next to the MAINT prompt on the CDU Initial Reference Index.

NOTE: This brings you to the MAINT BITE INDEX.

(c) Push the line select key next to the FQIS prompt.

(d) Do these steps to look for maintenance messages in CURRENT STATUS:

1) Push the line select key next to the CURRENT STATUS prompt.

2) Refer to the IFIM and do the applicable procedure(s) for each maintenance message.

a) If there is more than one page of faults in CURRENT STATUS, use the NEXT PAGE key and the PREV PAGE key to see all of the maintenance messages.

NOTE: The number of pages of faults in CURRENT STATUS are shown in the upper right corner of the display. If a fault shows the message FAULT NO LONGER PRESENT, the fault was corrected while the CURRENT STATUS display was on.

3) If NO PRESENT FAULTS shows on the CURRENT STATUS display, continue.

NOTE: No current faults are found in the system.

4) Push the line select key next to the INDEX prompt.

NOTE: This will bring you back to the FQIS BITE TEST main menu.

(e) Do these steps to do the FQIS ground test:

1) Push the line select key next to the GROUND TEST prompt.

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a) Push the line select key next to the YES prompt to make sure that you want to do the ground test.

2) When the ground test is complete, the display will show GROUND TEST COMPLETE PASS or FAIL.

3) If the display shows GROUND TEST COMPLETE PASS, there are no faults found in the FQIS.

NOTE: If faults show in CURRENT STATUS, keep a record of the faults for later fault isolation of intermittent faults. The corrective action given for each fault found in CURRENT FAULTS can also be used to correct intermittent faults that occur frequently.

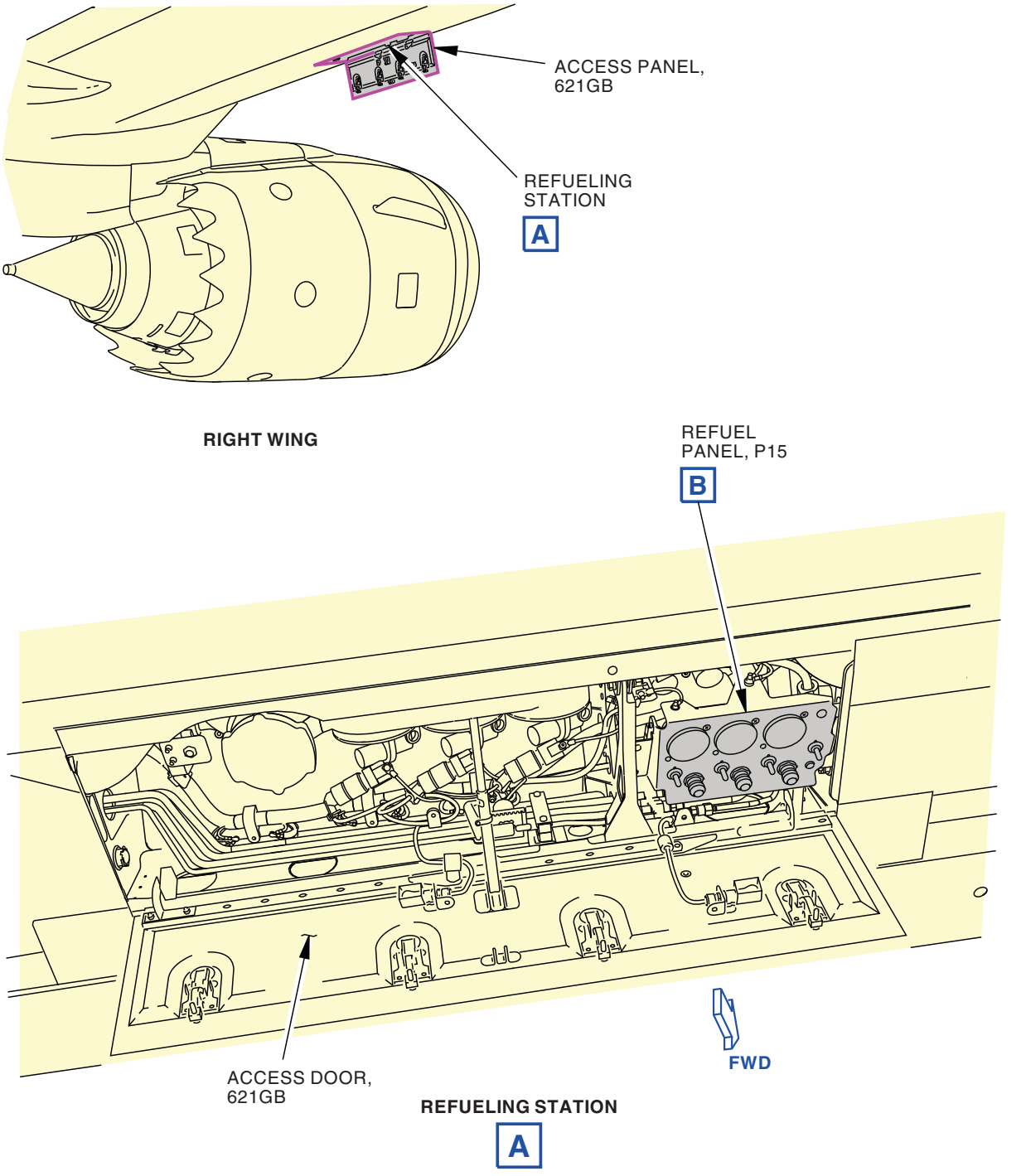
4) If the display shows GROUND TEST COMPLETE FAIL, refer to the IFIM and do the applicable procedure(s) for each maintenance message.

a) If there is more than one page of faults in GROUND TEST FAULTS, use the NEXT PAGE key and the PREV PAGE key to see all of the maintenance messages.

NOTE: The number of pages of faults in GROUND TEST FAULTS are shown in the upper right corner of the display.

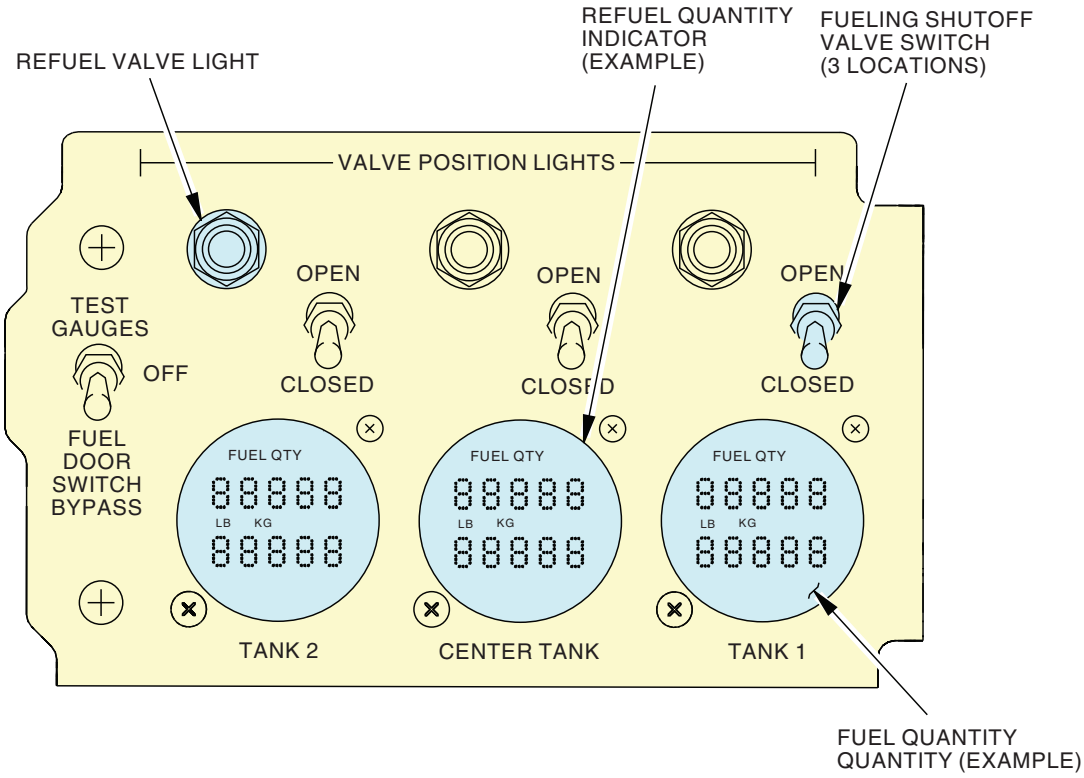
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Refuel Panel
Figure 501/28-41-00-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



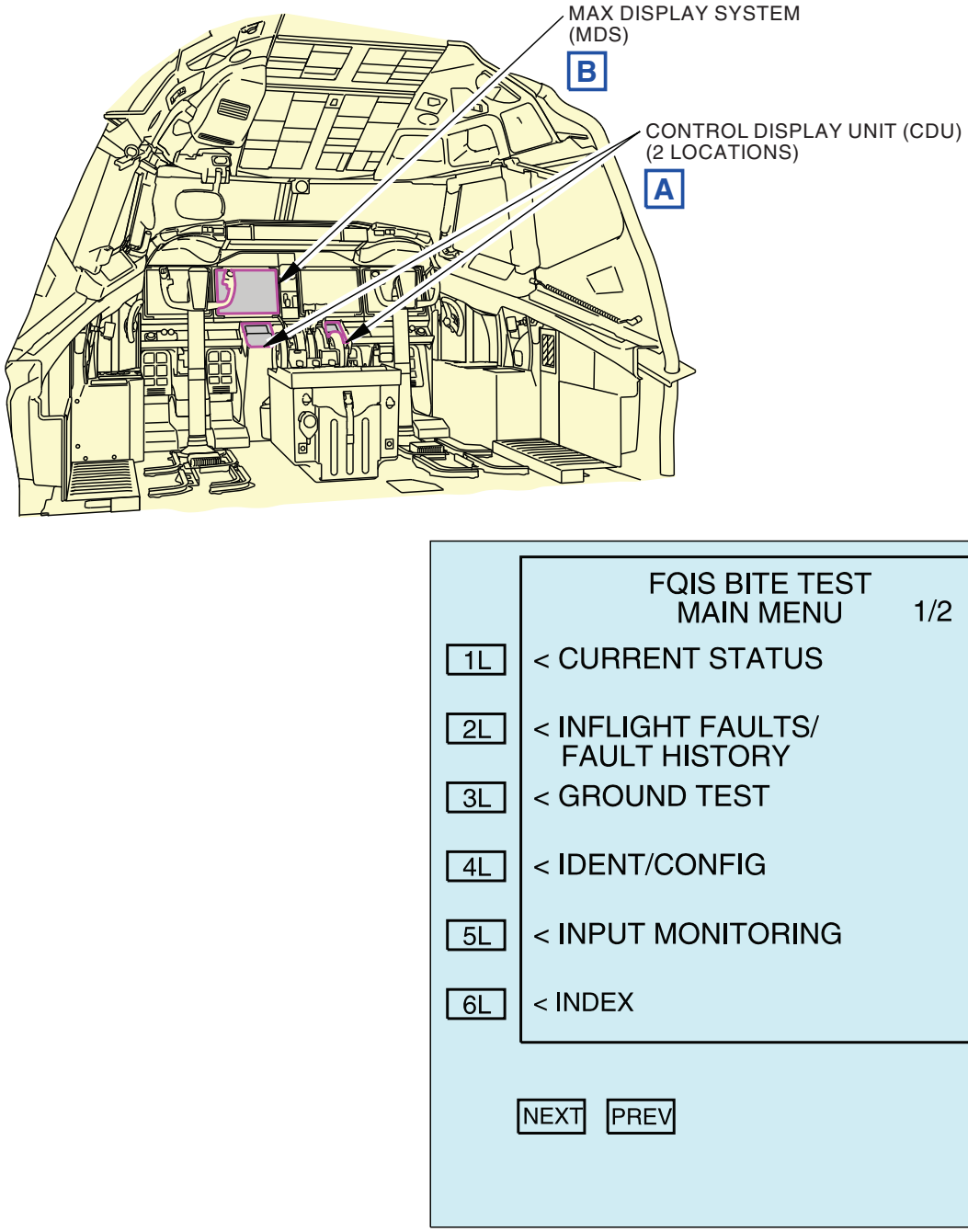
REFUEL PANEL, P15

B

2413576 S00061532849_V3

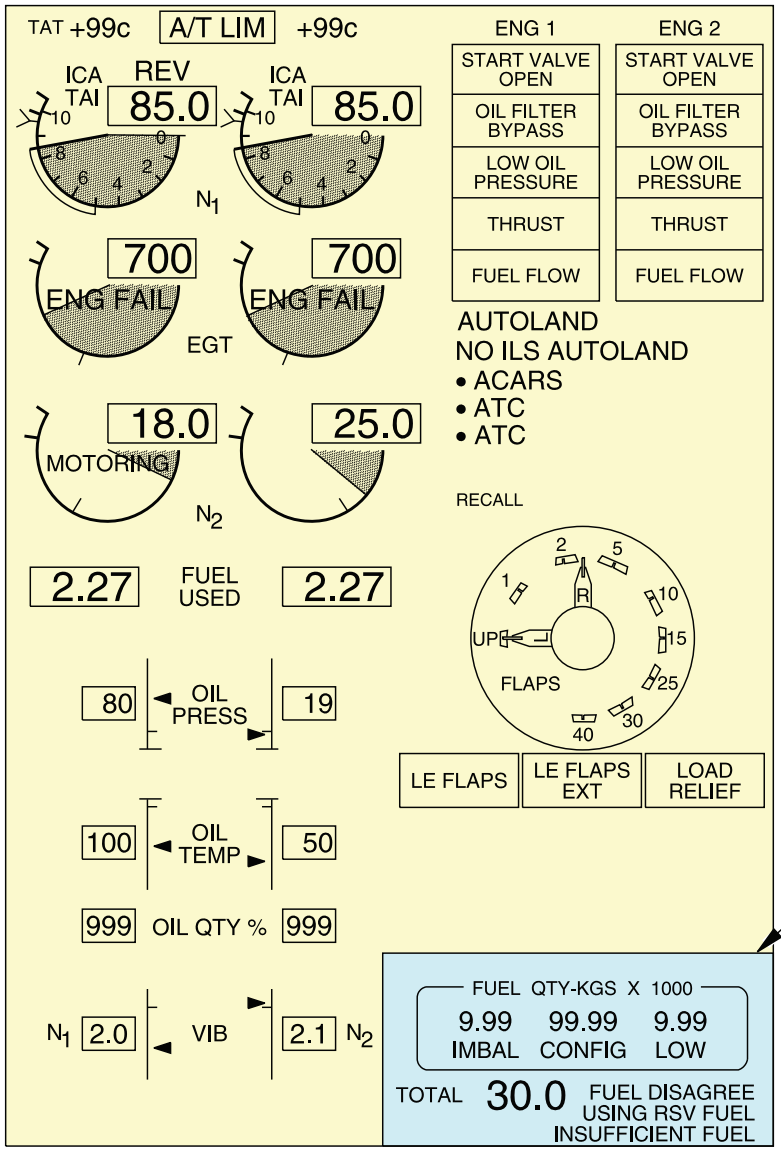
Refuel Panel
Figure 501/28-41-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Quantity Indicating System Test
Figure 502/28-41-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



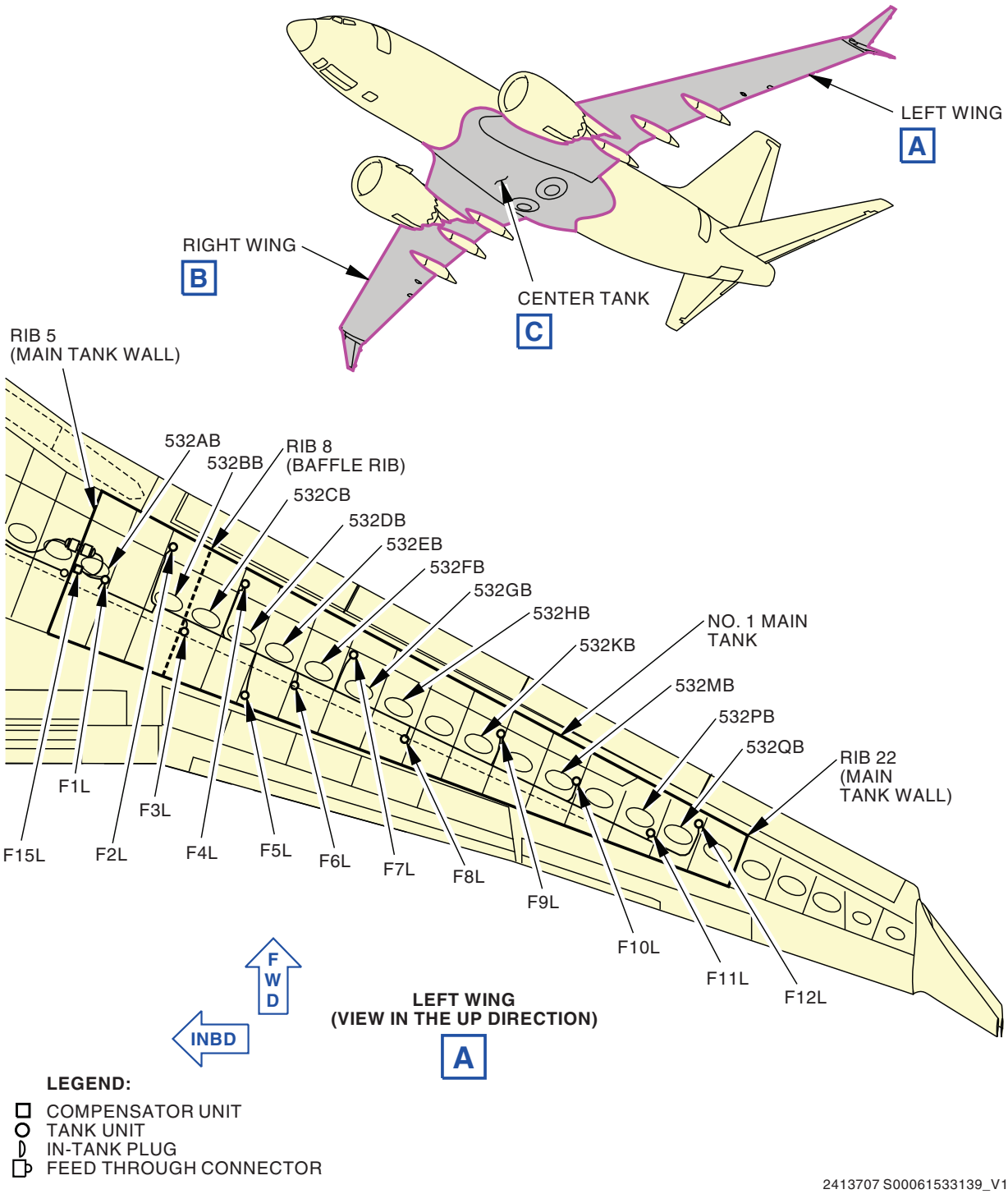
MAX DISPLAY SYSTEM (MDS)

B

2413428 S00061532562_V1

Fuel Quantity Indicating System Test
Figure 502/28-41-00-990-802 (Sheet 2 of 2)

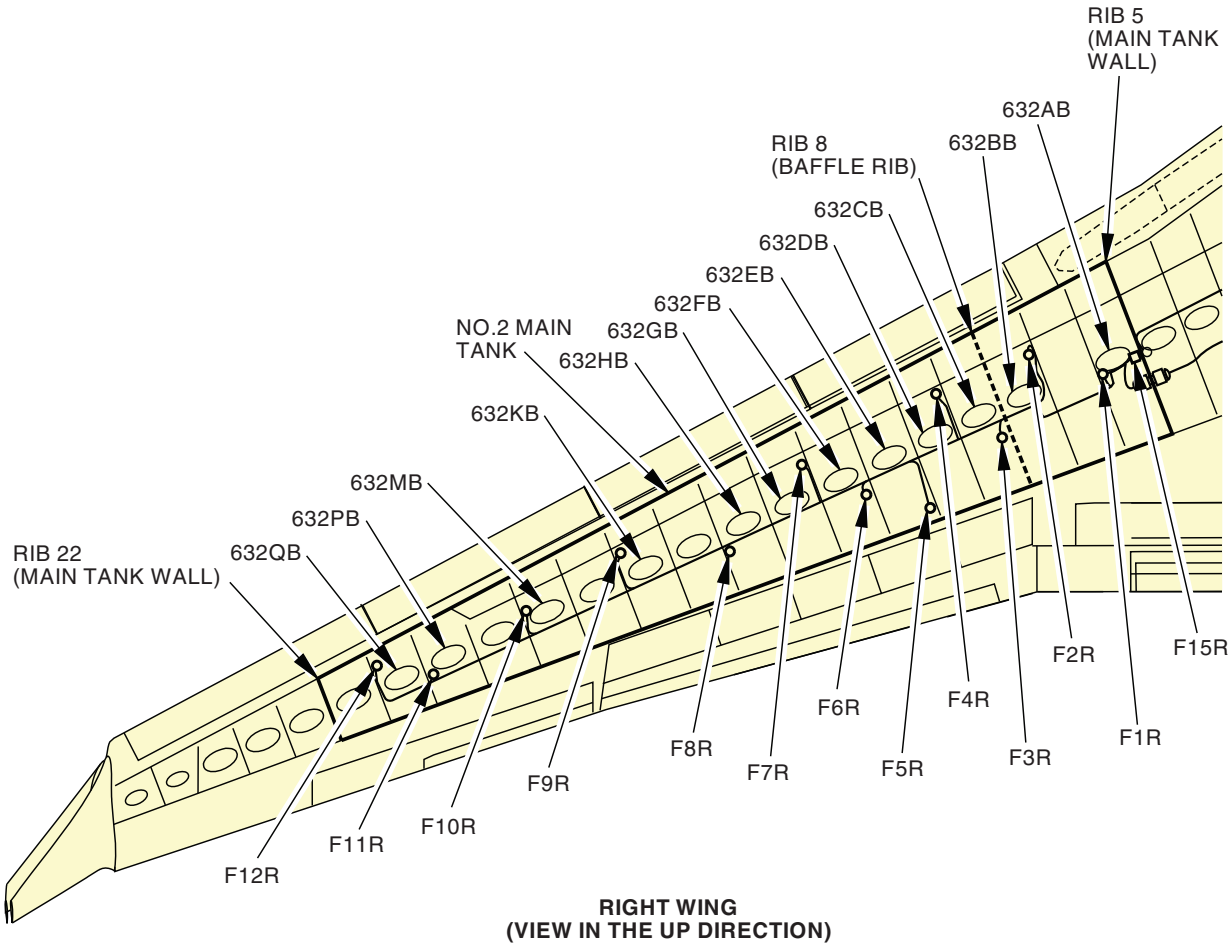
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413707 S00061533139_V1

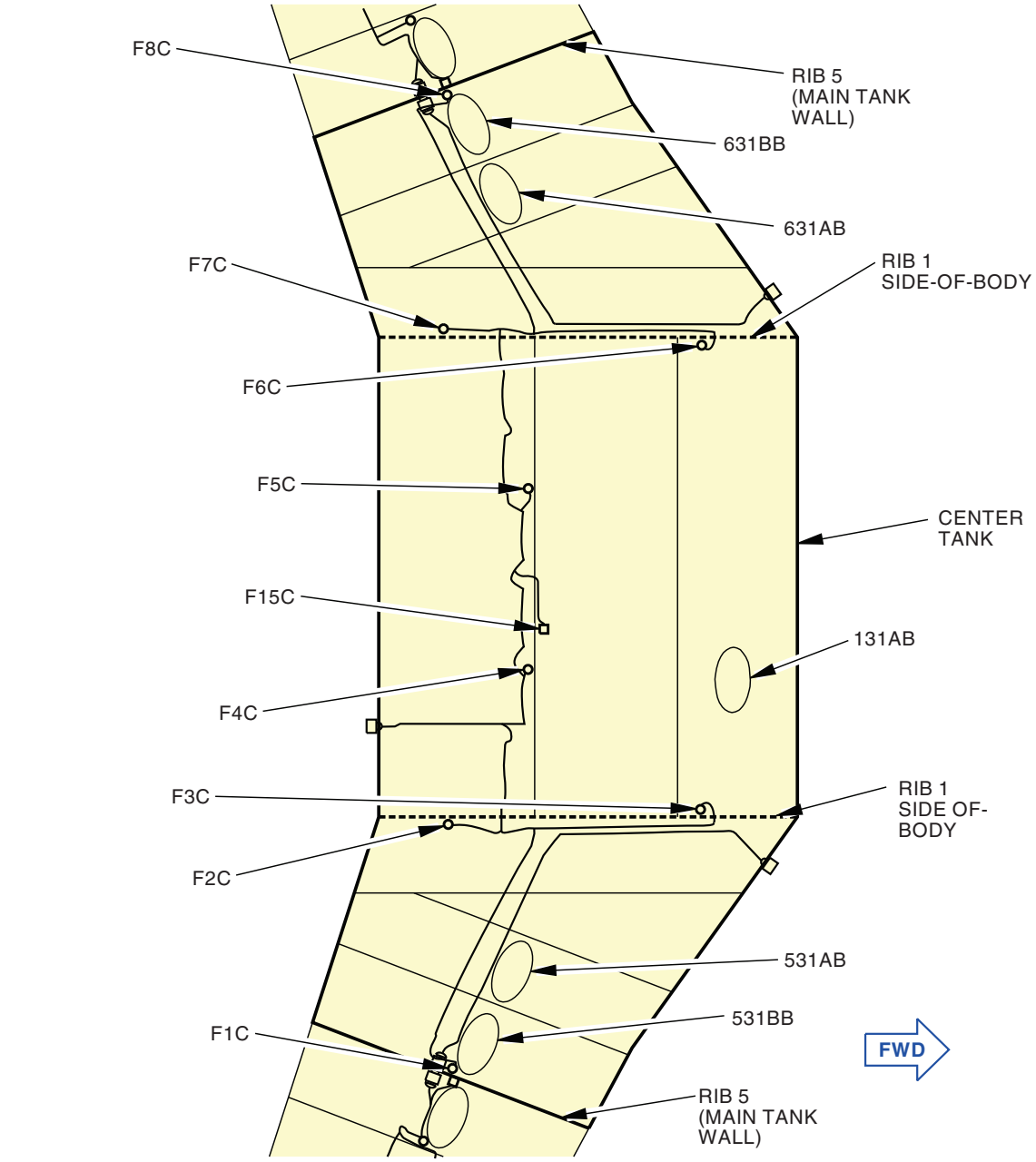
Fuel Quantity Indicating System
Figure 503/28-41-00-990-803 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Quantity Indicating System
Figure 503/28-41-00-990-803 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Quantity Indicating System
Figure 503/28-41-00-990-803 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-41-00-720-801

3. Fuel Quantity Indicating System (FQIS) with the Fuel Measuring Sticks - Comparison Check

A. General

(1) This procedure contains:

(a) A comparison check of the FQIS with the fuel measuring sticks.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Access Panels

Number	Name/Location
621GB	Refuel Valve Door

E. Fuel Quantity Indicating System (FQIS) with the Fuel Measuring Sticks - Comparison Check

SUBTASK 28-41-00-010-002

(1) Open this access panel:

Number	Name/Location
621GB	Refuel Valve Door

SUBTASK 28-41-00-650-001

(2) Make sure that there is a minimum of 3000 lb (1361 kg) of fuel in the center tank and a minimum of 1000 lb (454 kg) of fuel in the main tanks.

(a) Refuel the fuel tank (if it is necessary) (TASK 12-11-00-650-803).

SUBTASK 28-41-00-970-001

(3) Use the fuel measuring sticks to calculate the fuel quantity for each fuel tank.

SUBTASK 28-41-00-220-001

(4) Make sure that the difference between the calculated fuel quantity (with the fuel measuring sticks) for each tank and the fuel quantity shown on the fuel quantity indicators on the P15 Refuel Panel and the MAX display system in the flight compartment is in the subsequent range:

(a) Main tank ± 350 lbs

(b) Main tank ± 160 kgs

(c) Center tank ± 1160 lbs

(d) Center tank ± 520 kgs

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SUBTASK 28-41-00-410-002

(5) Close this access panel:

Number	Name/Location
621GB	Refuel Valve Door

——— END OF TASK ———

TASK 28-41-00-730-801

4. Fuel Quantity Indicating System - Dry Capacitance Test

A. General

(1) The fuel quantity processor unit will do a capacitance test of the fuel tank units and compensators.

NOTE: This test is a dry capacitance test (no fuel in the tank).

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
12-11-00-650-807	Drain the Fuel from the Sumps after Defueling (P/B 301)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-26-00-650-802	Tank to Tank Fuel Transfer (P/B 201)
28-41-21-710-801	Tank and Compensator Units - Insulation Resistance and Dry Capacitance Check (P/B 501)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Dry Capacitance Test in the Main Equipment Center (No Fuel in the Tank)

SUBTASK 28-41-00-740-002

(1) Do this task: Fuel Quantity Indicating System - Operational Test, TASK 28-41-00-710-801.

(a) If the BITE test shows faults, then do the corrective procedure for the fault shown.

(b) If the BITE test does not show faults, then continue.

SUBTASK 28-41-00-650-002

(2) Defuel the applicable fuel tank or transfer fuel out of the applicable tank (TASK 28-26-00-650-801 or TASK 28-26-00-650-802).

SUBTASK 28-41-00-650-003

(3) Drain the fuel from the applicable fuel tank (TASK 12-11-00-650-807).

SUBTASK 28-41-00-730-001

(4) On the Control Display Unit (CDU), push the line-select key adjacent to the MAINT prompt on the CDU Initial Reference Index.

NOTE: This brings you to the MAINT BITE INDEX.

(a) Push the line select key adjacent to the FQIS prompt.

(b) Push the line select key adjacent to the INPUT MONITORING prompt.

(c) Push the NEXT key to get to the second INPUT MONITORING page.

(d) Make sure that the quantities shown below TANK UNIT CAP are in this range for the applicable tank:

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1) No. 1 Main Tank and No. 2 Main Tank: 422.00 pf maximum/403.70 pf minimum

2) Center Tank: 756.40 pf maximum/737.40 pf minimum

(e) Make sure that the quantities shown below COMPENSATOR CAP are in this range for the applicable tank:

1) 59.2 pf maximum/57.6 pf minimum

NOTE: These values apply to the No. 1 main tank, No. 2 main tank, and the center tank.

SUBTASK 28-41-00-730-002

(5) If the capacitances are not in the correct range for empty tank capacitances, do this task: Fuel Quantity Indicating System - Operational Test, TASK 28-41-00-710-801.

(a) If the BITE test shows faults, then do the corrective action for the fault shown.

(b) If the BITE test does not show faults, then do the tank unit and compensator test from the wing spar to isolate the problem (TASK 28-41-21-710-801).

SUBTASK 28-41-00-650-004

(6) If no more tests are necessary, refuel the applicable tank (TASK 12-11-00-650-803).

——— END OF TASK ———

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TANK AND COMPENSATOR UNITS - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Tank Unit or Compensator Unit Removal

(2) Tank Unit or Compensator Unit Installation.

B. The access to the tank units or compensator units is different for each unit.

TASK 28-41-21-000-801

2. Tank Unit or Compensator Unit - Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the tank unit or the compensator unit.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-41-21-110-801	Tank Unit or Compensator Unit Cleaning (P/B 701)

C. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Access Panels

Number	Name/Location
131AB	Center Tank Access
532AB	Main Tank Access Door - Wing Station 216
632AB	Main Tank Access Door - Wing Station 216

E. Prepare for the Removal

SUBTASK 28-41-21-860-002

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) To access the applicable compensator, remove the applicable access doors:

(TASK 28-11-31-000-801 or TASK 28-11-11-000-801)

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Number	Name/Location
131AB	Center Tank Access
532AB	Main Tank Access Door - Wing Station 216
632AB	Main Tank Access Door - Wing Station 216

(b) To access the tank units, remove the applicable access doors (TASK 28-11-31-000-801 or TASK 28-11-11-000-801).

SUBTASK 28-41-21-860-013

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	3	C00032	FUEL FUELING CONT
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

F. Tank Unit or Compensator Unit Removal

SUBTASK 28-41-21-020-001

(1) Remove the captive screw [18], plain washer [19], and relief clamp [20] from the terminal block.

SUBTASK 28-41-21-020-002

(2) Disconnect the electrical cable from the terminal block:

(a) Remove the self-locking insert [14], washer [15], and HI-Z wire [16].

(b) Remove the self-locking insert [24], washer [25], and HI-Z shield wire [23].

(c) Remove the self-locking insert [22], washer [21], and LO-Z wire [17].

(d) Remove the electrical cable from the terminal block.

SUBTASK 28-41-21-020-003

(3) Remove the two screws [10] and washers [11] that secures the fixed mounting bracket [12] to the structure.

SUBTASK 28-41-21-020-004

(4) Remove the two screws [10] and washers [11] that secures the rotating mounting bracket [13] to the structure.

SUBTASK 28-41-21-020-005

CAUTION

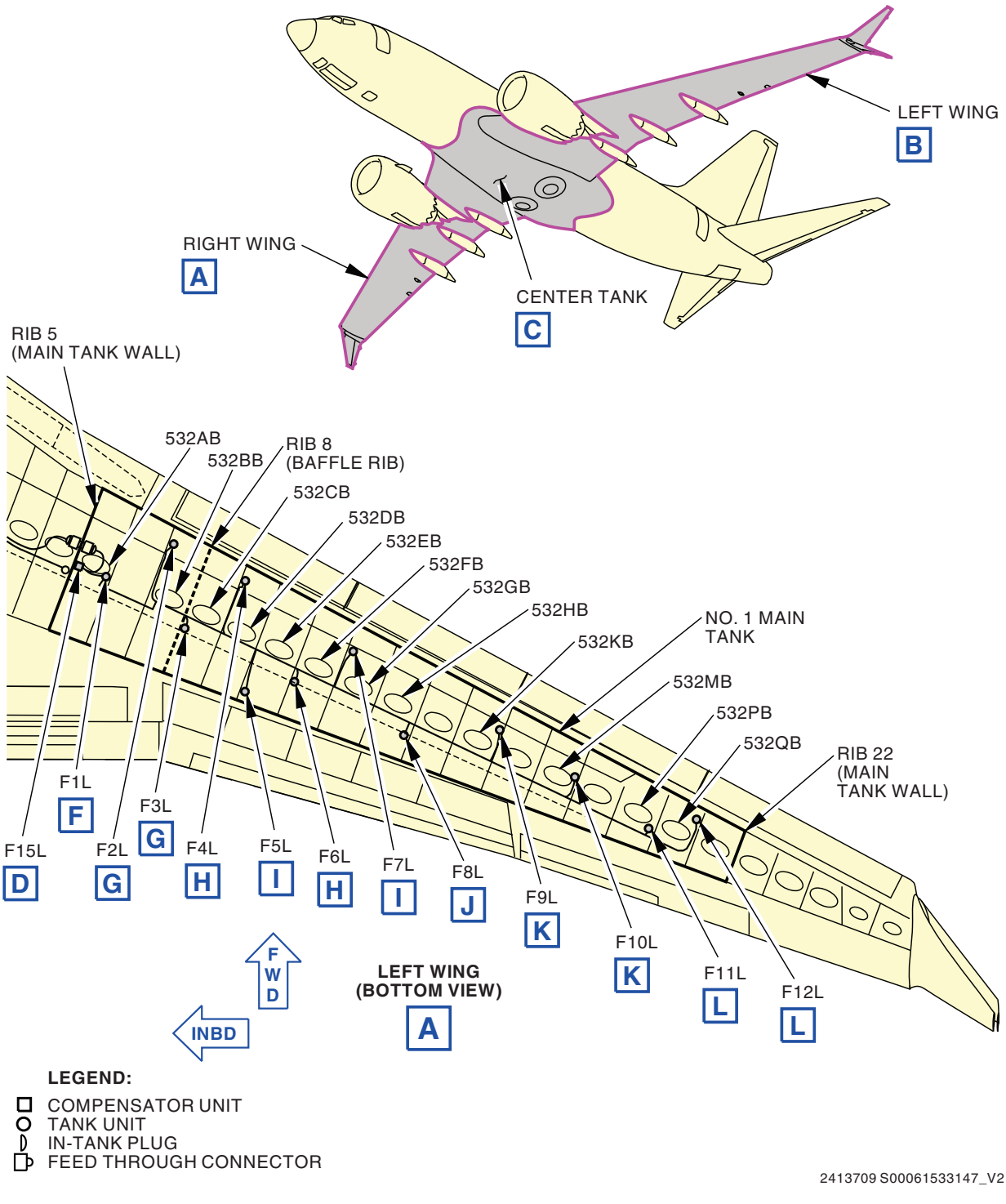
MOVE THE TANK UNITS AND THE COMPENSATOR UNITS CAREFULLY. YOU CAN CAUSE DAMAGE TO THE PROPERTIES OF THE TANK UNITS AND THE COMPENSATOR UNIT.

(5) Remove the compensator unit [1], tank unit [2], tank unit [3], tank unit [4], tank unit [5], tank unit [6], tank unit [7], tank unit [8] , or tank unit [9].

(a) If it is necessary, clean the tank unit [2], tank unit [3], tank unit [4], tank unit [5], tank unit [6], tank unit [7], tank unit [8], tank unit [9], or compensator unit [1], do this task: Tank Unit or Compensator Unit Cleaning, TASK 28-41-21-110-801.

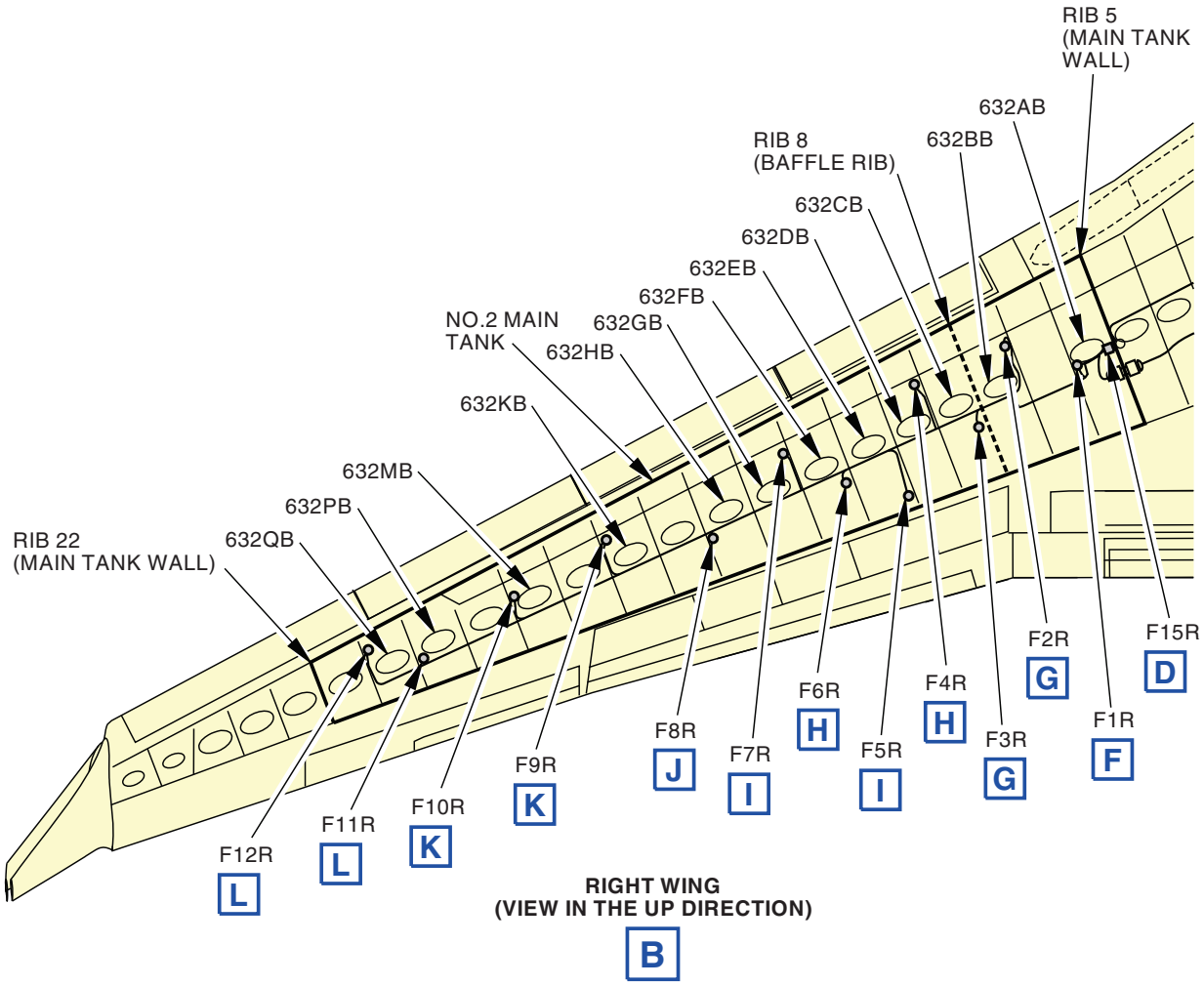
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Tank and Compensator Unit Installation
Figure 401/28-41-21-990-801 (Sheet 1 of 8)

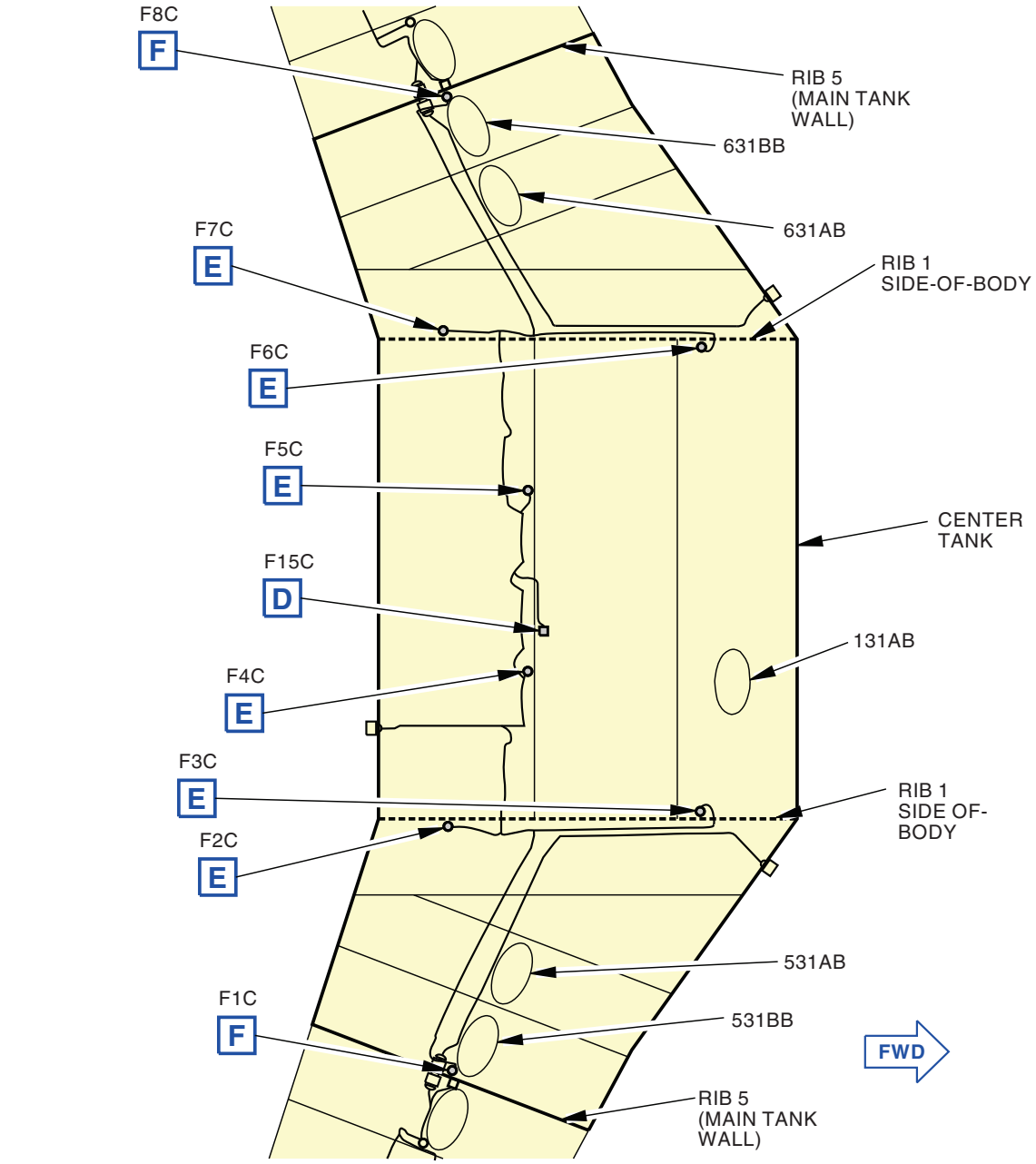
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413710 S00061533148_V2

Tank and Compensator Unit Installation
Figure 401/28-41-21-990-801 (Sheet 2 of 8)

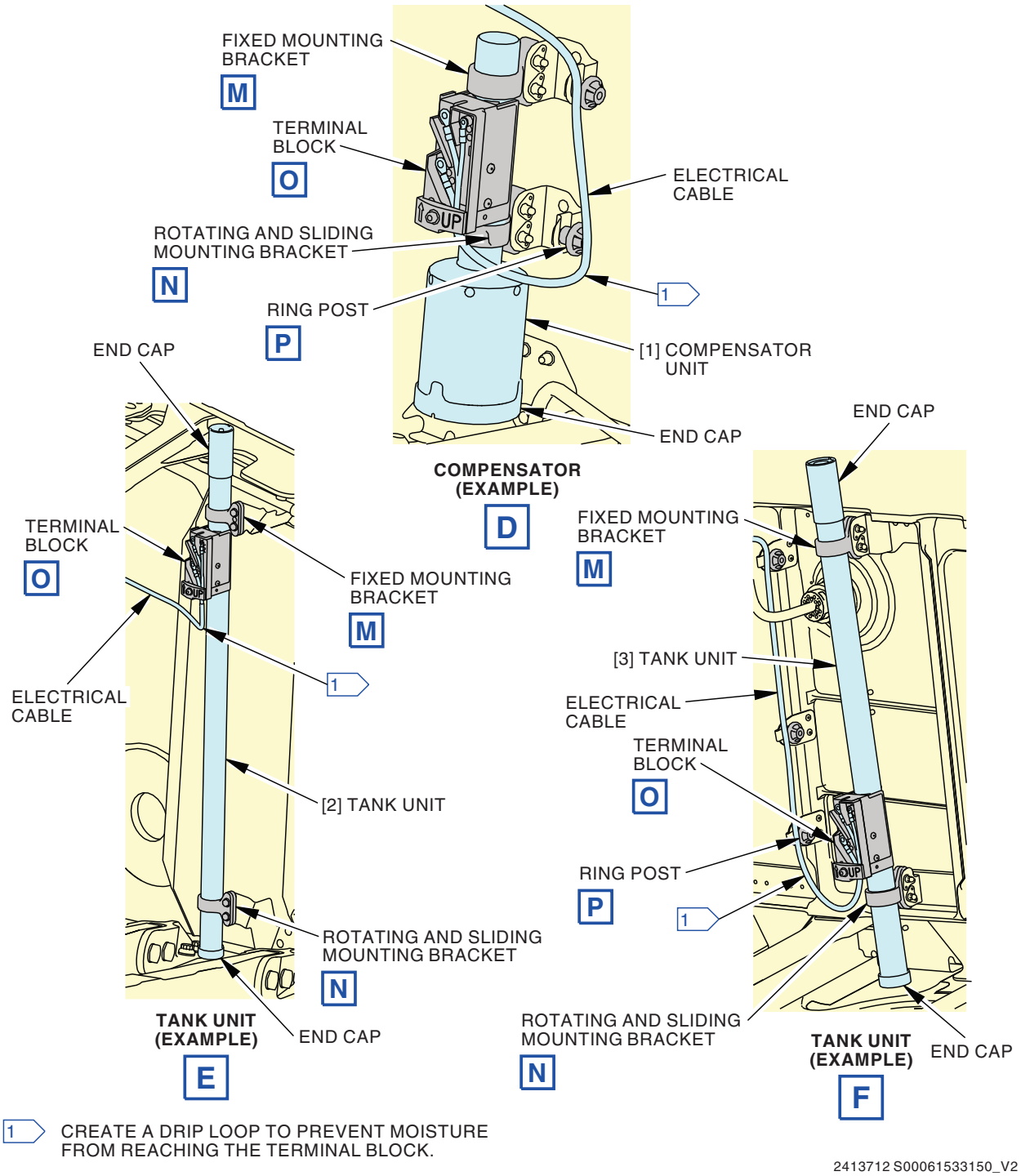
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413711 S00061533149_V2

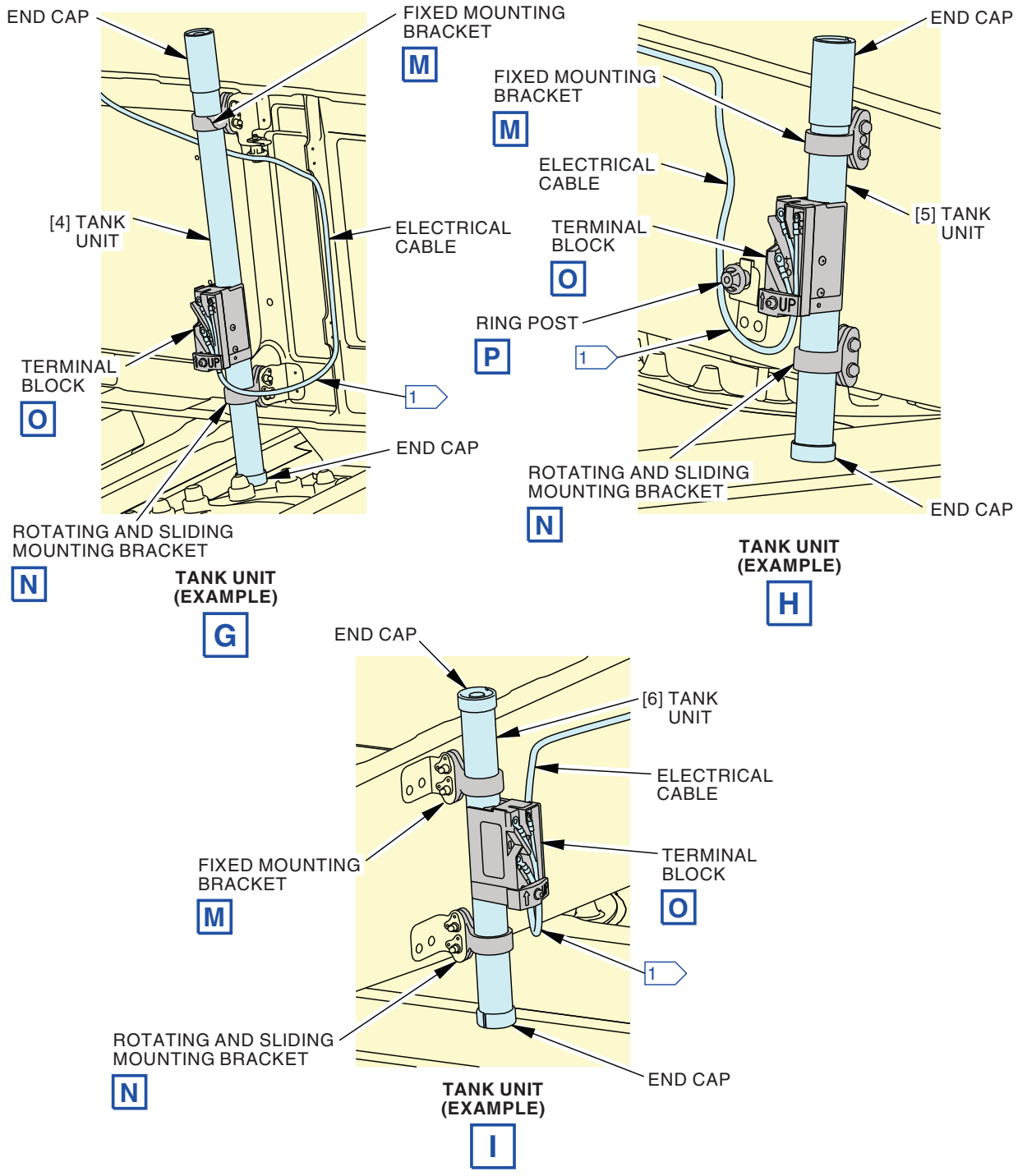
Tank and Compensator Unit Installation
Figure 401/28-41-21-990-801 (Sheet 3 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Tank and Compensator Unit Installation
Figure 401/28-41-21-990-801 (Sheet 4 of 8)

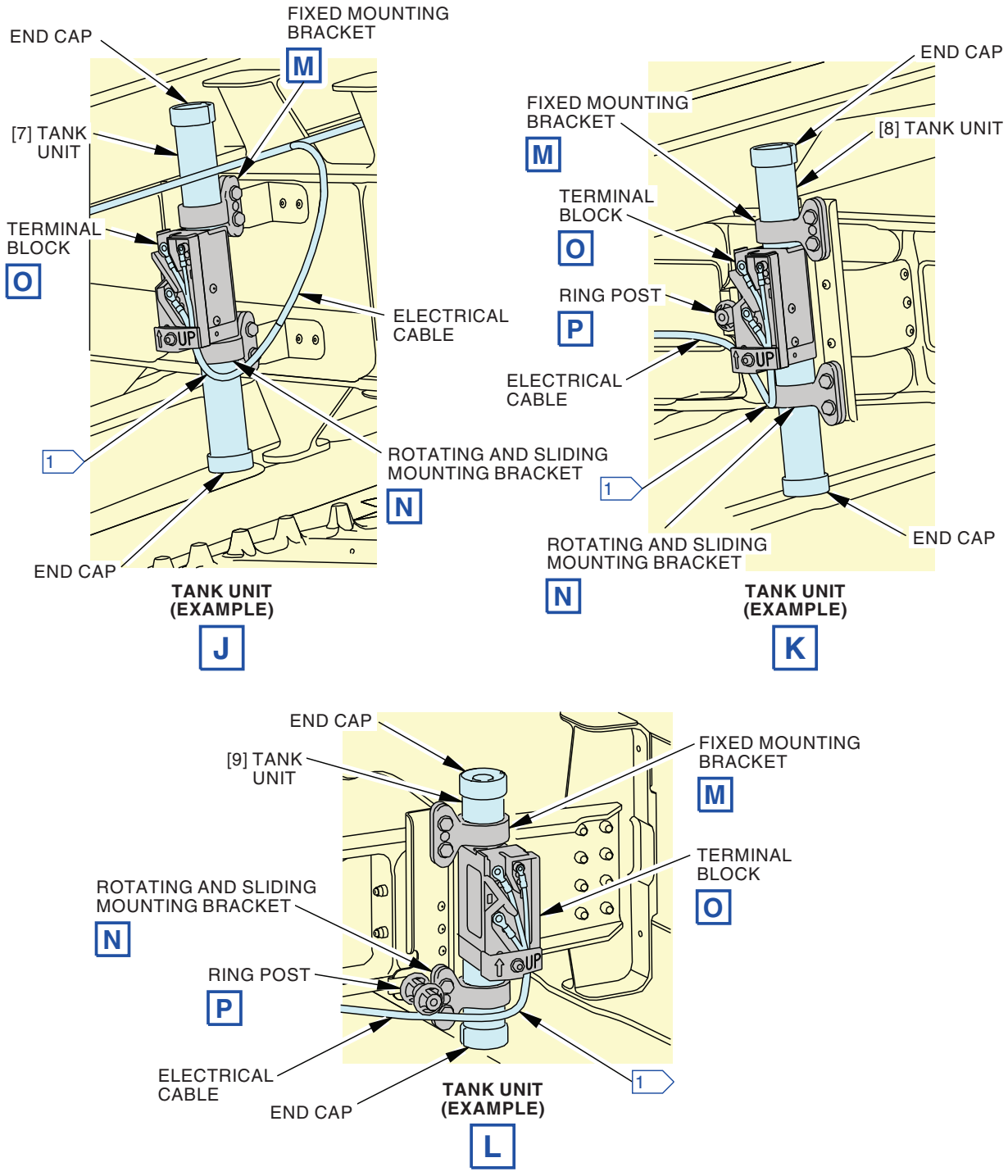
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Tank and Compensator Unit Installation
Figure 401/28-41-21-990-801 (Sheet 5 of 8)

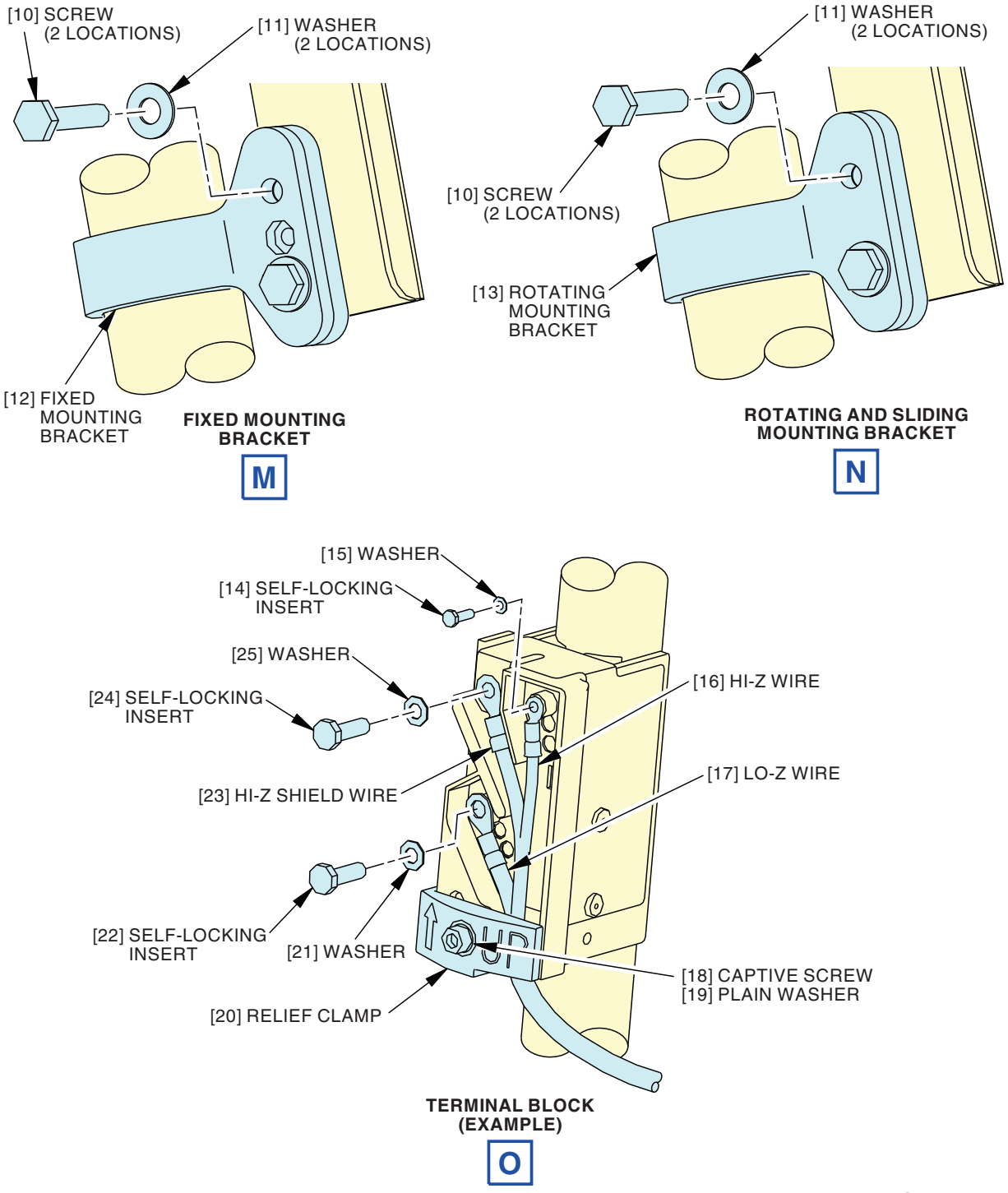
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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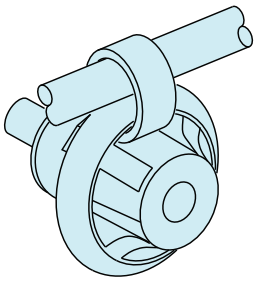
Tank and Compensator Unit Installation
Figure 401/28-41-21-990-801 (Sheet 6 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



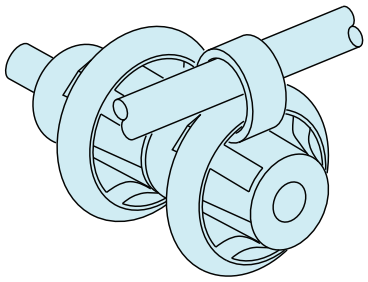
Tank and Compensator Unit Installation
Figure 401/28-41-21-990-801 (Sheet 7 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



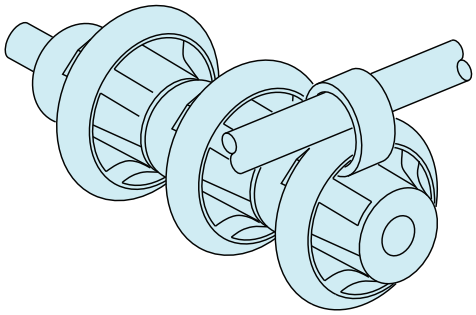
SINGLE RING POST
(EXAMPLE)

P



DOUBLE RING POST
(EXAMPLE)

P



TRIPLE RING POST
(EXAMPLE)

P

2504433 S0000587281_V1

Tank and Compensator Unit Installation
Figure 401/28-41-21-990-801 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-41-21-400-801

3. Tank Unit or Compensator Unit - Installation

(Figure 401)

A. General

(1) This task has one or more steps which are controlled by Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09 and 28-AWL-30.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
28-41-00-730-801	Fuel Quantity Indicating System - Dry Capacitance Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Compensator unit	28-41-21-01-015	PTF ALL
		28-41-21-05-020	PTF ALL
		28-41-21-10-020	PTF ALL
2	Tank unit	28-41-21-01-020	PTF ALL
		28-41-21-01-025	PTF ALL
		28-41-21-05-025	PTF ALL
		28-41-21-10-025	PTF ALL
3	Tank unit	28-41-21-05-030	PTF ALL
		28-41-21-05-035	PTF ALL
		28-41-21-10-030	PTF ALL
		28-41-21-10-035	PTF ALL
4	Tank unit	28-41-21-05-040	PTF ALL
		28-41-21-05-045	PTF ALL
		28-41-21-10-040	PTF ALL
		28-41-21-10-045	PTF ALL
5	Tank unit	28-41-21-05-050	PTF ALL
		28-41-21-05-060	PTF ALL
		28-41-21-10-050	PTF ALL
		28-41-21-10-060	PTF ALL
6	Tank unit	28-41-21-05-055	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
6 (cont.)		28-41-21-05-065	PTF ALL
		28-41-21-10-055	PTF ALL
		28-41-21-10-065	PTF ALL
7	Tank unit	28-41-21-05-070	PTF ALL
		28-41-21-10-070	PTF ALL
8	Tank unit	28-41-21-05-075	PTF ALL
		28-41-21-05-080	PTF ALL
		28-41-21-10-075	PTF ALL
		28-41-21-10-080	PTF ALL
9	Tank unit	28-41-21-05-085	PTF ALL
		28-41-21-10-085	PTF ALL

D. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

E. Tank Unit Installation

SUBTASK 28-41-21-980-001

(1) Put the applicable tank unit in its position.

NOTE: The tank unit [2], tank unit [3], tank unit [4], tank unit [5], tank unit [6], tank unit [7], tank unit [8], and tank unit [9] is referred to as applicable tank unit in this task.

(a) Make sure that the arrow, on the terminal block, points to the upper wing skin.

28-AWL-30: CDCCL

(b) Make sure that the applicable tank unit has the end caps (upper and lower) installed.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-30.

SUBTASK 28-41-21-420-001

(2) Close the fixed mounting bracket [12] with your hand, around the applicable tank unit.

NOTE: The fixed mounting bracket [12] needs to be replaced when the bracket cannot be closed with your hand.

(a) Make sure that the tab inside the fixed mounting bracket [12] is seated in the orientation hole on the applicable tank unit.

(b) Make sure that the terminal block face points away from the spanwise beams, ribs, or other structures.

(c) Install the screws [10] and washers [11] to secure the applicable tank unit to the bracket.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-21-420-002

(3) Close the rotating mounting bracket [13] with your hand around the applicable tank unit.

NOTE: The rotating mounting bracket [13] needs to be replaced when the bracket cannot be closed with your hand.

(a) Install the screws [10] and washers [11] to attach the applicable tank unit to the bracket.

SUBTASK 28-41-21-220-001

(4) Do a check for these gaps:

28-AWL-30: CDCCL

(a) Make sure that the gap between the tank unit end caps and the upper and lower fuel tank structure is 0.2 in. (5 mm) or more.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-30.

28-AWL-30: CDCCL

(b) Make sure that the gap between the side of the tank unit and the fuel tank structure is 0.25 in. (6.4 mm) or more.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-30.

SUBTASK 28-41-21-420-003

(5) Connect the electrical cable to the terminal block (SWPM 20-10-12, SWPM 20-10-19).

(a) Attach the relief clamp [20], electrical cable, captive screw [18], and plain washer [19] loosely to the terminal block.

(b) Adjust the electrical cable, accounting for slack in all directions, to maintain a clearance of 0.5 in. (12.7 mm) between the electrical cable and components or structure.

(c) When a clearance of 0.5 in. (12.7 mm) between the electrical cable and components or structure is not possible, do this step:

28-AWL-09: CDCCL

1) Make sure the clearance between the wires and structure, accounting for slack in all directions, is a minimum of 0.250 in. (6.4 mm).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

(d) Tighten the captive screw [18] to 18.5 ±1.5 in-lb (2.09 ±0.17 N·m).

SUBTASK 28-41-21-420-004

(6) Install the self-locking insert [14], washer [15], and HI-Z wire [16] to the terminal block.

(a) Tighten the self-locking insert [14] to 13.5 ±1.5 in-lb (1.53 ±0.17 N·m).

(b) Make sure that the self-locking insert [14] holds the terminal tightly and the HI-Z wire [16] does not turn.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-21-420-005

(7) Install the self-locking insert [24], washer [25], and HI-Z shield wire [23] to the terminal block.

(a) Tighten the self-locking insert [24] to 17.5 ±2.5 in-lb (1.98 ±0.29 N·m).

(b) Make sure that the self-locking insert [24] holds the terminal tightly and the HI-Z shield wire [23] does not turn.

SUBTASK 28-41-21-420-006

(8) Install the self-locking insert [22], washer [21], and LO-Z wire [17] to the terminal block.

(a) Tighten the LO-Z wire [17] to 30 ±5 in-lb (3.4 ±0.6 N·m).

(b) Make sure that the self-locking insert [22] holds the terminal tightly and the LO-Z wire [17] does not turn.

SUBTASK 28-41-21-210-001

(9) Make sure that the electrical cable has a drip loop.

NOTE: With no drip loop, water drops can collect on the electrical cable and fall on the electrical terminals. The drip loop prevents damage to the tank unit.

F. Compensator Unit Installation

SUBTASK 28-41-21-980-002

(1) Put the applicable compensator unit [1] in its position.

(a) Make sure that the arrow, on the terminal block, points to the upper wing skin.

28-AWL-30: CDCCL

(b) Make sure that the lower end-cap is installed on the compensator unit [1].

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-30.

SUBTASK 28-41-21-420-007

(2) Close the fixed mounting bracket [12] with your hand, around the compensator unit [1].

NOTE: The fixed mounting bracket [12] needs to be replaced when the bracket cannot be closed with your hand.

(a) Make sure that the tab on the fixed mounting bracket [12] is seated correctly in the orientation hole on the compensator unit [1].

(b) Make sure that the terminal block points away from the rib or spanwise beam.

(c) Install the screws [10] and washers [11] to attach the compensator unit [1] to the bracket.

SUBTASK 28-41-21-420-008

(3) Close the rotating mounting bracket [13] with your hand, around the compensator unit [1].

NOTE: The rotating mounting bracket [13] needs to be replaced when the bracket cannot be closed with your hand.

(a) Install the screws [10] and washers [11] to attach the compensator unit [1] to the bracket.

SUBTASK 28-41-21-220-002

(4) Do a check of these gaps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

28-AWL-30: CDCCL

(a) Make sure that the gap between the side of the compensator unit [1] and the fuel tank structure is 0.2 in. (5 mm) or more.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-30.

28-AWL-30: CDCCL

(b) Make sure that the gap between the lower end-cap of the compensator unit [1] and the fuel tank structure is 0.25 in. (6.4 mm) or more.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-30.

SUBTASK 28-41-21-420-009

(5) Connect the electrical cable to the terminal block (SWPM 20-10-12, SWPM 20-10-19).

(a) Attach the relief clamp [20], electrical cable, captive screw [18], and plain washer [19] loosely to the terminal block.

(b) Adjust the electrical cable slack to maintain a clearance of 0.5 in. (12.7 mm) between the electrical cable and components or structure.

28-AWL-09: CDCCL

(c) If the clearance of 0.5 in. (12.7 mm) between the electrical cable and components or structure is not possible, do this step:

1) Make sure the clearance between the wires and structure or component, accounting for slack in all directions, is a minimum of 0.250 in. (6.4 mm).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

(d) Tighten the captive screw [18] to 18.5 ±1.5 in-lb (2.09 ±0.17 N·m).

SUBTASK 28-41-21-420-010

(6) Install the self-locking insert [14], washer [15], and HI-Z wire [16] to the terminal block.

(a) Tighten the self-locking insert [14] to 13.5 ±1.5 in-lb (1.53 ±0.17 N·m).

(b) Make sure that the self-locking insert [14] holds the terminal tightly and the HI-Z wire [16] does not turn.

SUBTASK 28-41-21-420-011

(7) Install the self-locking insert [24], washer [25], and HI-Z shield wire [23] to the terminal block.

(a) Tighten the self-locking insert [24] to 17.5 ±2.5 in-lb (1.98 ±0.29 N·m).

(b) Make sure that the self-locking insert [24] holds the terminal tightly and the HI-Z shield wire [23] does not turn.

SUBTASK 28-41-21-420-012

(8) Install the self-locking insert [22], washer [21], and LO-Z wire [17] to the terminal block.

(a) Tighten the LO-Z wire [17] to 30 ±5 in-lb (3.4 ±0.6 N·m).

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(b) Make sure that the self-locking insert [22] holds the terminal tightly and the LO-Z wire [17] does not turn.

SUBTASK 28-41-21-210-002

(9) Make sure that the electrical cable has a drip loop.

NOTE: With no drip loop, water drops can collect on the electrical cable and fall on the electrical terminals. The drip loop prevents damage to the compensator unit [1].

G. Tank Unit or Compensator Unit Installation Test

SUBTASK 28-41-21-860-003

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	3	C00032	FUEL FUELING CONT
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

SUBTASK 28-41-21-860-004

(2) Do this task: Fuel Quantity Indicating System - Dry Capacitance Test, TASK 28-41-00-730-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 28-41-21-410-001

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Install the applicable access doors (TASK 28-11-11-400-801 or TASK 28-11-31-400-801).

SUBTASK 28-41-21-650-001

(2) If it is necessary, do this task: Pressure Refuel Procedure, TASK 12-11-00-650-803.

——— END OF TASK ———

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TANK AND COMPENSATOR UNITS - ADJUSTMENT/TEST

1. General

A. This procedure measures the capacitance of each tank and compensator unit. It also measures the insulation resistance of the related wiring.

B. This procedure has these tasks:

(1) Tank and Compensator Units - Insulation Resistance and Dry Capacitance Check.

(2) Tank and Compensator Units - Wet Capacitance Check.

(3) Tank and Compensator Units - Fuel Quantity Indicating System Health Check.

TASK 28-41-21-710-801

2. Tank and Compensator Units - Insulation Resistance and Dry Capacitance Check

(Figure 501, Figure 502)

A. References

Reference	Title
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-41-41-000-801	Bussing Plug Removal (P/B 401)
28-41-41-400-801	Bussing Plug Installation (P/B 401)
28-41-42-020-801	FQIS Spar Penetration Connector Removal (P/B 401)
28-41-42-420-801	FQIS Spar Penetration Connector Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1770	Tester - Fuel Gauge System, Includes V.T.O. Capability Part #: 22160649 Supplier: 51190 Part #: 361-012-001 Supplier: 26055 Part #: PSD60-2R Supplier: 51190 Opt Part #: PSD40-1 Supplier: 89305
COM-1780	Cables - Adapter, Fuel Quantity Indicating System (FQIS) Tester Part #: PSD737-1 Supplier: 41364
COM-15149	Adapter - Cable, FQIS Part #: 112-00007 Supplier: 21844 Part #: PSD40-524B Supplier: 41364
COM-15150	Adapter - Cable FQIS
COM-19223	Tester - Fuel Gauge System, Intrinsically Safe with Frequency Selector Part #: PSD60-2R Supplier: 51190

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(Continued)

Reference	Description
COM-23749	Test Set - AC Fuel Quantity System, PSD90-3AC (Intrinsically Safe) Part #: 22160649 Supplier: 51190

C. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Prepare for the Test

SUBTASK 28-41-21-730-001

(1) Use system tester, COM-1770, tester, COM-19223, or tester, COM-23749, with the cables, COM-1780.

NOTE: The cables, COM-1780, is referred to as test harness in this task.

SUBTASK 28-41-21-800-001

(2) The table below supplies more data about the approved Fuel Quantity Indicating System (FQIS) test units and adapter equipment:

Table 501/28-41-21-993-801

Test Unit	Adapter Equipment
PSD60-2R, PSD60-2, PSD60-1, PSD40, or 22160649 (PSD90-3AC)	cables, COM-1780
Gull 361-012-001	cables, COM-1780 with FQIS cable adapter, COM-15149
Barfield 101-00540 MOD8000	FQIS adapter cable, COM-15150 or cables, COM-1780

SUBTASK 28-41-21-650-002

(3) Defuel the applicable fuel tanks (TASK 28-26-00-650-801).

SUBTASK 28-41-21-860-005

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) To test the tank or compensator units for the main tanks, extend the leading edge flaps (TASK 27-81-00-860-803).

WARNING YOU MUST CAREFULLY DO THE STEPS IN THE TASK BELOW TO INSTALL THE LE FLAP SAFETY LOCKS. THE LE FLAPS CAN MOVE QUICKLY IF YOU DO NOT INSTALL THE SAFETY LOCKS CORRECTLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Install safety locks on the leading edge flaps (TASK 27-81-00-480-801).

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SUBTASK 28-41-21-860-006

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	3	C00032	FUEL FUELING CONT
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

E. Tank and Compensator Units Insulation Resistance and Dry Capacitance Test

SUBTASK 28-41-21-020-006

(1) Disconnect the bussing plug from the applicable fuel tank feed-through connector (TASK 28-41-41-000-801).

SUBTASK 28-41-21-480-001

(2) Connect the adapter plug on the test harness to the feed-through connector.

SUBTASK 28-41-21-480-002

(3) Do one of these steps to connect the test harness to the system tester (Figure 501).

(a) For the system tester, COM-1770, or tester, COM-19223, do these steps:

1) Connect the alligator style clamp labeled AIRFRAME GROUND on the test harness to the aircraft ground/structure.

NOTE: The bonding bracket or stud used by the FQIS harness near the feed-through connector is a good location. A good ground is possible through the sealant, paint, or primer that might be present.

2) Connect the banana jack connector, P5, on the test harness to the connector, J7, on the tester labeled CHASSIS.

3) Connect the connector, P3, labeled Hi-Z on the test harness to the connector, J5, on the tester labeled TANK UNITS Hi-Z.

4) To test the applicable tank units, do these steps:

a) Connect the connector, P2, labeled Lo-Z on the test harness to the connector, P4, labeled TANK UNITS LO-Z on the tester.

b) On the test harness, select the applicable tank unit to measure.

NOTE: To measure the total capacitance of the tank units, select TOTAL on the test harness.

5) To test the applicable compensator units, do these steps:

NOTE: The following step to move cables is not necessary to test the COMP when using cables, COM-1780. It is only necessary to select COMP on the cables, COM-1780.

a) Connect the connector, P2, labeled Lo-Z on the test harness to the connector, P6, labeled TANK UNITS COMP on the tester.

b) On the test harness, select COMP.

6) Connect the connector, P1, labeled P1 on the test harness to the feed-through connector on the rear or front spar.

7) Setup the system tester:

NOTE: No adjustment of the frequency is necessary.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Set the POWER switch to the ON position.

NOTE: Three minutes is necessary for the system tester to stabilize.

b) Set the CAPACITANCE MEASUREMENT MODE switch to the MODE A position.

(b) For the system tester, COM-23749, do these steps:

1) Connect the alligator style clamp identified as AIRFRAME GROUND on the test harness to the aircraft ground/structure.

NOTE: The bonding bracket or stud used by the FQIS harness near the feed-through connector is a good location. A good ground is possible through the sealant, paint or primer that is present.

2) Connect the banana jack connector, P5, on the test harness to the connector on the tester identified as CHASSIS.

3) Connect the connector, P3, identified as HI-Z on the test harness to the MEASURE TANKS TU HI-Z plug on the tester.

4) To test the applicable tank units, do these steps:

a) Connect the connector, P2, identified as LO-Z on the test harness the MEASURE TANKS TU LO-Z plug on the tester.

b) On the test harness's selector box, select the applicable tank unit to measure.

c) To measure the total capacitance of the tank units, select TOTAL on the test harness.

5) Connect the connector, P1, on the test harness to the feed-through connector on the rear or front spar.

SUBTASK 28-41-21-710-006

(4) Measure the insulation resistance of each tank unit and compensator unit (Table 502, Table 503).

(a) Make sure that the resistance is not less than the value shown in Table 502 or Table 503.

(b) For the system tester, COM-1770, or tester, COM-19223, do these steps:

1) Set the Megger Mode Switch to 3-WIRE on the tester, COM-1770, and tester, COM-19223.

(c) For the system tester, COM-23749, do these steps:

1) On the test harness, select the applicable tank unit or compensator to measure.

2) Turn the POWER/TYPE SELECT switch knob to AC CAP.

3) Turn the FUNCTION SELECT switch knob to MEGGER.

4) Set the MEG MODE/SIMULATE SELECT switch knob to 3-WIRE.

5) Set the MEGGER SELECT switch to measure and record each of the circuits:

6) Set MEGGER SELECT switch to these positions for Table 503:

a) LO-Z/HI-Z or COMP/HI-Z

b) GND/HI-Z

c) LO-Z/COMP or COMP/SHIELD.

7) Set MEGGER SELECT switch to these positions for Table 502:

a) LO-Z/HI-Z or COMP/HI-Z

b) GND/HI-Z

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c) LO-Z/COMP or COMP/SHIELD.

Table 502/28-41-21-993-831 Insulation Resistance - Center Tank

COMPONENT (CENTER TANK)	PIN NUMBER*[1]	INSULATION RESISTANCE (MEGOHMS)		
		A to B	A to GRD	B to SHLD
F15C	3 AND 1	1000	10	4.9
F1C	6 AND 1	250	10	4.9
F2C	7 AND 1	250	10	4.9
F3C	8 AND 1	250	10	4.9
F4C	9 AND 1	250	10	4.9
F5C	10 AND 1	250	10	4.9
F6C	11 AND 1	250	10	4.9
F7C	12 AND 1	250	10	4.9
F8C	13 AND 1	250	10	4.9

*[1] Pin numbers correspond to numbers on the test harness and pin numbers on the airplane connector receptacle. A to B, A to GRD, and B to SHLD correspond to the system tester switch positions in Table 504.

Table 503/28-41-21-993-832 Insulation Resistance - Main Tank

COMPONENT NO. 1 (L) OR NO. 2 (R) FUEL TANK	PIN NUMBER*[1]	INSULATION RESISTANCE (MEGOHMS)		
		A to B	A to GRD	B to SHLD
F15 (L,R)	3 AND 1	1000	10	4.9
F1 (L,R)	6 AND 1	250	10	4.9
F2 (L,R)	7 AND 1	250	10	4.9
F3 (L,R)	8 AND 1	250	10	4.9
F4 (L,R)	9 AND 1	250	10	4.9
F5 (L,R)	10 AND 1	250	10	4.9
F6 (L,R)	11 AND 1	250	10	4.9
F7 (L,R)	12 AND 1	250	10	4.9
F8 (L,R)	13 AND 1	250	10	4.9
F9 (L,R)	14 AND 1	250	10	4.9
F10 (L,R)	15 AND 1	250	10	4.9
F11 (L,R)	16 AND 1	250	10	4.9
F12 (L,R)	17 AND 1	250	10	4.9

*[1] Pin numbers correspond to numbers on the test harness and pin numbers on the airplane connector receptacle. A to B, A to GRD, and B to SHLD correspond to the system tester switch positions in Table 504.

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Table 504/28-41-21-993-833 Tester Jack Location

GULL TESTER	SIMMONDS (PSD TESTER)
A TO B = COAX TO TANK OR COMP	A TO B = LO-Z TO HI-Z OR COMP TO HI-Z
A TO GRD = COAX TO SHIELD	A TO GRD = HI-Z TO GND
B TO SHLD = TANK OR COMP TO SHIELD	B TO SHLD = LO-Z TO SHIELD OR COMP TO SHIELD

SUBTASK 28-41-21-710-002

(5) Measure the capacitance of each tank unit and compensator unit (Table 505, Table 506).

Table 505/28-41-21-993-807 Empty Capacitance - Center Tank

COMPONENT	PIN NUMBER *[1]	EMPTY CAPACITANCE (PICOFARADS)
F15C	3 AND 1	57.60 – 58.70
F1C	6 AND 1	50.1 – 51.6
F2C	7 AND 1	78.1 – 80.2
F3C	8 AND 1	141.9 – 145.3
F4C	9 AND 1	98.6 – 101.1
F5C	10 AND 1	98.6 – 101.1
F6C	11 AND 1	141.9 – 145.3
F7C	12 AND 1	78.1 – 80.2
F8C	13 AND 1	50.1 – 51.6

*[1] Pin numbers correspond to numbers on the test harness and pin numbers on the airplane connector receptacle.

Table 506/28-41-21-993-804 Empty Capacitance - Main Tank

COMPONENT (NO. 1 (L) OR NO. 2 (R) MAIN TANK)	PIN NUMBER *[1]	EMPTY CAPACITANCE (PICOFARADS)
F15 (L,R)	3 AND 1	57.60 - 58.70
F1 (L,R)	6 AND 1	63.4 – 65.2
F2 (L,R)	7 AND 1	40.0 – 41.5
F3 (L,R)	8 AND 1	48.8 – 50.3
F4 (L,R)	9 AND 1	26.2 – 27.7
F5 (L,R)	10 AND 1	31.6 – 33.1
F6 (L,R)	11 AND 1	35.6 – 37.1
F7 (L,R)	12 AND 1	34.2 – 35.7
F8 (L,R)	13 AND 1	36.6 – 38.1
F9 (L,R)	14 AND 1	33.2 – 34.7
F10 (L,R)	15 AND 1	28.7 – 30.2
F11 (L,R)	16 AND 1	12.7 – 14.2
F12 (L,R)	17 AND 1	12.7 – 14.2

*[1] Pin numbers correspond to numbers on the test harness and pin numbers on the airplane connector receptacle.

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SUBTASK 28-41-21-080-001

(6) Disconnect the adapter plug from the feed-through connector.

SUBTASK 28-41-21-080-002

(7) Disconnect the AIRFRAME GROUND alligator clamp.

SUBTASK 28-41-21-410-002

(8) Carefully connect the bussing plug to the feed-through connector (TASK 28-41-41-400-801).

(a) If the feed-through connector is damaged, replace the feed-through connector (TASK 28-41-42-020-801 or TASK 28-41-42-420-801).

NOTE: The connector pins and sockets are easily damaged.

SUBTASK 28-41-21-710-003

(9) Do the test again for the other fuel tanks.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 28-41-21-080-003

(1) Do one of these steps to disconnect the system tester.

(a) For the system tester, COM-1770, or tester, COM-19223:

1) Set the POWER switch to the OFF position.
2) Disconnect the connector, P1.
3) Disconnect the connector, P2.
4) Disconnect the connector, Hi-Z.
5) Disconnect the banana jack connector, P5.

(b) For the system tester, COM-23749:

1) Set the POWER/TYPE SELECT switch to POWER OFF.
2) Disconnect the connector, P1.
3) Disconnect the connector, P2.
4) Disconnect the connector, Hi-Z.
5) Disconnect the banana jack connector, P5.

SUBTASK 28-41-21-860-007

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	3	C00032	FUEL FUELING CONT
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

SUBTASK 28-41-21-080-004

WARNING YOU MUST CAREFULLY DO THE STEPS IN THE TASK BELOW TO REMOVE THE LE FLAP SAFETY LOCK. THE LE FLAPS CAN MOVE QUICKLY IF YOU DO NOT DO THE STEPS CORRECTLY. THIS CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(3) Remove the leading edge flap locks (TASK 27-81-00-080-801).

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SUBTASK 28-41-21-860-008

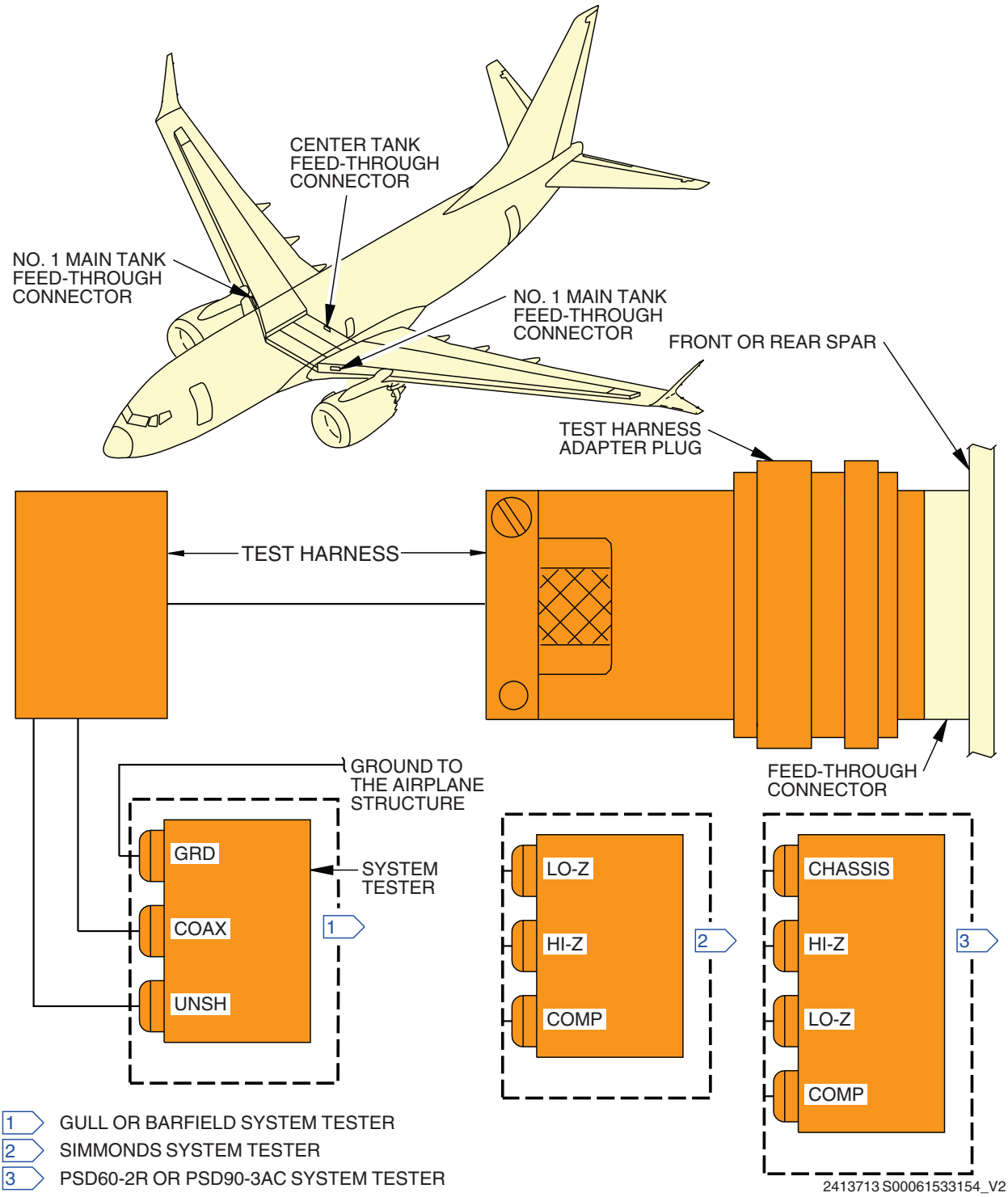
WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Retract the leading edge flaps (TASK 27-81-00-860-804).

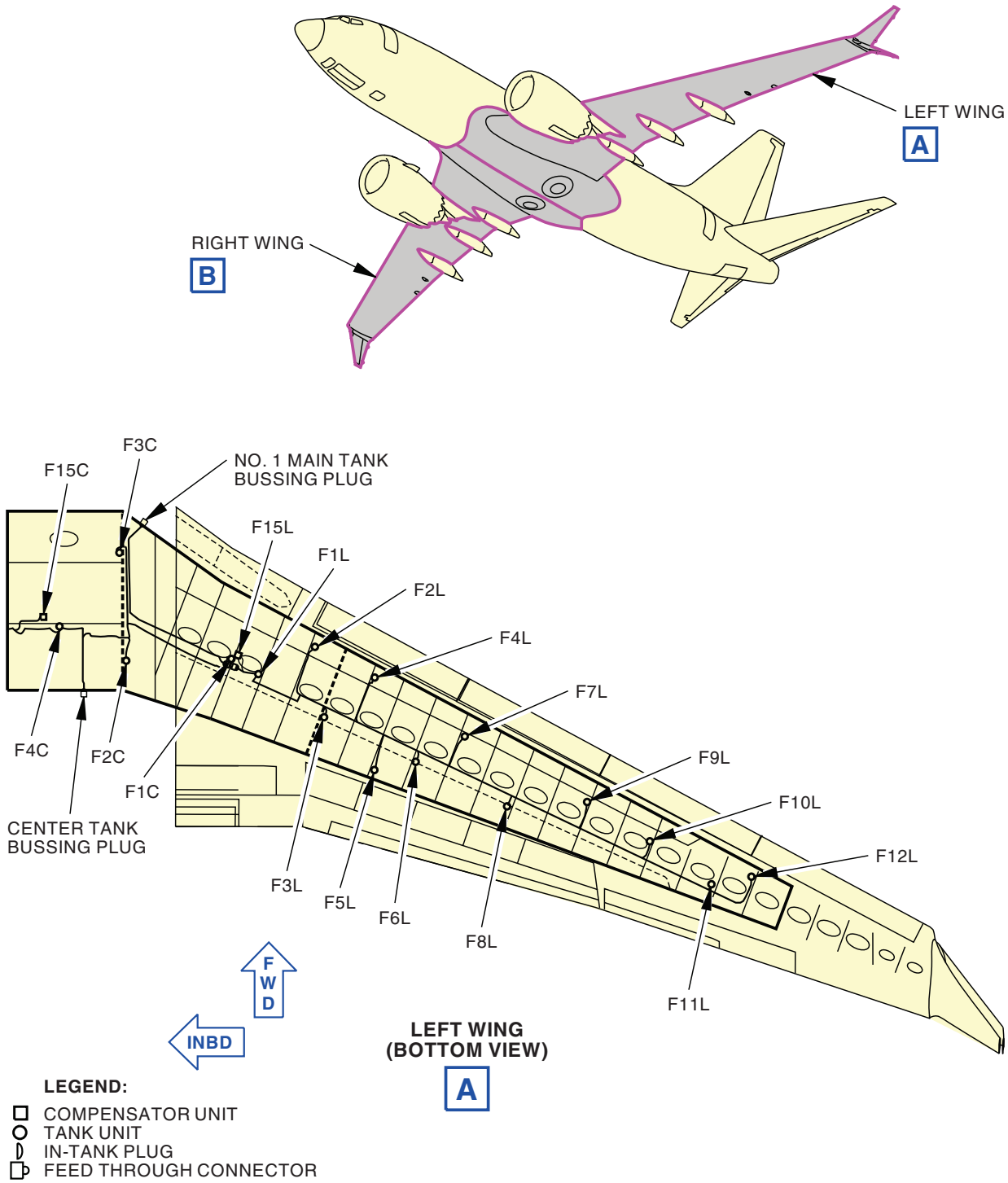
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Tank Unit and Compensator Unit Test Connections
Figure 501/28-41-21-990-802

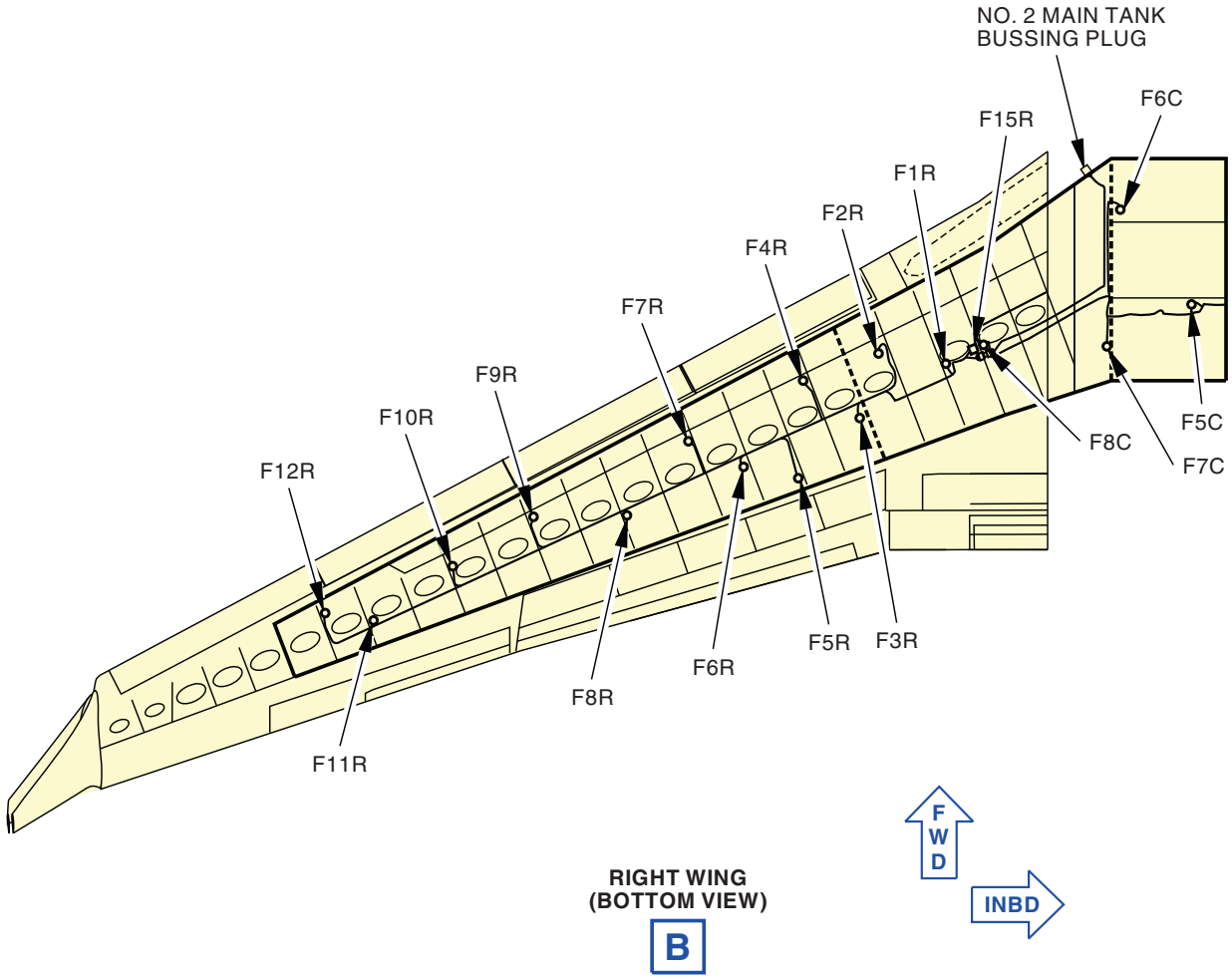
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413714 S00061533155_V2

Tank Unit and Compensator Unit Test
Figure 502/28-41-21-990-803 (Sheet 1 of 2)

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LEGEND:

- COMPENSATOR UNIT
- TANK UNIT
- IN-TANK PLUG
- FEED THROUGH CONNECTOR

2413715 S00061533156_V3

Tank Unit and Compensator Unit Test
Figure 502/28-41-21-990-803 (Sheet 2 of 2)

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TASK 28-41-21-710-802

3. Tank and Compensator Units - Wet Capacitance Check

(Figure 501, Figure 502)

A. General

(1) The wet capacitance check is an alternative procedure to test the tank and compensator units with fuel. In isolated conditions, it is possible that problems can appear with a full fuel tank that does not appear with an empty fuel tank.

(2) The capacitance read by the tank or compensator unit changes with fuel properties and fuel temperature. The capacitance values supplied in this procedure are approximate and can help you identify possible faults.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
28-41-41-000-801	Bussing Plug Removal (P/B 401)
28-41-41-400-801	Bussing Plug Installation (P/B 401)
28-41-42-020-801	FQIS Spar Penetration Connector Removal (P/B 401)
28-41-42-420-801	FQIS Spar Penetration Connector Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1770	Tester - Fuel Gauge System, Includes V.T.O. Capability Part #: 22160649 Supplier: 51190 Part #: 361-012-001 Supplier: 26055 Part #: PSD60-2R Supplier: 51190 Opt Part #: PSD40-1 Supplier: 89305
COM-1780	Cables - Adapter, Fuel Quantity Indicating System (FQIS) Tester Part #: PSD737-1 Supplier: 41364
COM-19223	Tester - Fuel Gauge System, Intrinsically Safe with Frequency Selector Part #: PSD60-2R Supplier: 51190
COM-23749	Test Set - AC Fuel Quantity System, PSD90-3AC (Intrinsically Safe) Part #: 22160649 Supplier: 51190

D. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5

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(Continued)

Zone	Area
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

E. Prepare for the Test

SUBTASK 28-41-21-730-002

(1) Use system tester, COM-1770, tester, COM-19223, or tester, COM-23749, with the cables, COM-1780.

NOTE: The cables, COM-1780, is referred to as test harness in this procedure.

SUBTASK 28-41-21-650-003

(2) Add fuel to the applicable fuel tank (fill the fuel tank to its full capacity) (TASK 12-11-00-650-803).

SUBTASK 28-41-21-860-009

WARNING KEEP ALL PERSONNEL AND EQUIPMENT CLEAR OF THE LEADING EDGE FLAPS. THE LEADING EDGE FLAPS CAN MOVE QUICKLY. THE MOVEMENT OF THE LEADING EDGE FLAPS CAN CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(3) To test the tank or compensator units for the main tanks, extend the leading edge flaps (TASK 27-81-00-860-803).

WARNING YOU MUST CAREFULLY DO THE STEPS IN THE TASK BELOW TO INSTALL THE LE FLAP SAFETY LOCKS. THE LE FLAPS CAN MOVE QUICKLY IF YOU DO NOT INSTALL THE SAFETY LOCKS CORRECTLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Install safety locks on the leading edge flaps (TASK 27-81-00-480-801).

SUBTASK 28-41-21-860-010

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	3	C00032	FUEL FUELING CONT
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

F. Tank and Compensator Units Wet Capacitance Check

SUBTASK 28-41-21-010-001

(1) Disconnect the applicable bussing plug from the applicable fuel tank feed-through connector (TASK 28-41-41-000-801).

SUBTASK 28-41-21-480-004

(2) Connect the adapter plug on the test harness to the feed-through connector (Figure 501).

SUBTASK 28-41-21-480-005

(3) Do one of these steps to connect the test harness to the system tester (Figure 501).

(a) For the system tester, COM-1770, or tester, COM-19223, do these steps:

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1) Connect the alligator style clamp labeled AIRFRAME GROUND on the test harness to the aircraft ground/structure.

   NOTE: The bonding bracket or stud used by the Fuel Quantity Indicating System (FQIS) harness near the FQIS feed-through connector is a good location if you can get a good ground through any sealant, paint or primer that might be present.

2) Connect the banana jack connector, P5, on the test harness to the connector, J7, on the tester identified as CHASSIS.

3) Connect the connector, P3, labeled HI-Z on the test harness to the connector, J5, on the tester identified as TANK UNITS HI-Z.

4) To test the applicable tank units, do these steps:

   a) Connect the connector, P2, identified as LO-Z on the test harness to the connector, P4, identified as TANK UNITS LO-Z on the tester.

   b) On the test harness's selector box, select the applicable tank unit to measure.

   c) To measure the total capacitance of the tank units, select TOTAL on the test harness.

5) Connect the connector, P1, on the test harness to the FQIS feed-through connector on the rear or front spar.

6) To test the applicable compensator units, do these steps:

   NOTE: The following step to move cables is not necessary to test the COMP when using cables, COM-1780. It is only necessary to select COMP on the cables, COM-1780.

   a) Connect the connector, P2, identified as LO-Z on the tester cables to the connector, P6, identified as COMP on the tester.

   b) On the tester cables, select COMP.

      NOTE: Use capacitance measurement MODE A.

      NOTE: No adjustment of the frequency is necessary.

7) Set the POWER switch to ON.

   NOTE: Allow three minutes for the tester to stabilize.

(b) For the system tester, COM-23749, do these steps:

1) Connect the alligator style clamp labeled AIRFRAME GROUND on the test harness to the aircraft ground/structure.

   NOTE: The bonding bracket or stud used by the FQIS harness near the feed-through connector is a good location. A good ground is possible through the sealant, paint, or primer.

2) Connect the banana jack connector, P5, on the test harness to the connector on the tester labeled CHASSIS.

3) Connect the connector, P3, labeled HI-Z on the test harness to the MEASURE TANKS TU HI-Z plug on the tester.

4) Connect the connector, P1, on the test harness to the feed-through connector on the rear or front spar.

5) To test the applicable tank units, do these steps:

   a) Connect the connector, P2, labeled LO-Z on the test harness the MEASURE TANKS TU LO-Z plug on the tester.

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b) On the test harness's selector box, select the applicable tank unit to measure.

c) To measure the total capacitance of the tank units, select TOTAL on the test harness.

d) Turn the POWER/TYPE SELECT switch knob to AC CAP.

SUBTASK 28-41-21-710-004

(4) Measure the capacitance of each tank unit and compensator unit (Table 507 and Table 508).

NOTE: Wet capacitance values change with fuel properties and fuel temperature. The capacitance values supplied in the table below are the approximate tank unit and compensator capacitances when these units are fully covered with fuel. These values must only be used with caution and to identify further fault isolation.

Table 507/28-41-21-993-805 Wet Capacitance - Main Tank

COMPONENT NO. 1 (L) OR NO. 2 (R) MAIN TANK	PIN NUMBER*[1]	WET CAPACITANCE (PICOFARADS)
F15 (L,R)	3 AND 1	122.5
F1 (L,R)	6 AND 1	136.9
F2 (L,R)	7 AND 1	86.9
F3 (L,R)	8 AND 1	105.1
F4 (L,R)	9 AND 1	57.6
F5 (L,R)	10 AND 1	69.1
F6 (L,R)	11 AND 1	77.6
F7 (L,R)	12 AND 1	74.7
F8 (L,R)	13 AND 1	79.8
F9 (L,R)	14 AND 1	72.5
F10 (L,R)	15 AND 1	62.9
F11 (L,R)	16 AND 1	25.0*[2]
F12 (L,R)	17 AND 1	18.8*[2]

*[1] Pin numbers correspond to numbers on the test harness and pin numbers on the airplane feed-through connector.

*[2] Tank unit is not completely covered in fuel at Volumetric Shutoff (VSO). This is an approximate wet capacitance value for the fuel level at VSO and can be different.

Table 508/28-41-21-993-808 Wet Capacitance - Center Tank

COMPONENT	PIN NUMBER*[1]	WET CAPACITANCE (PICOFARADS)
F15C	3 AND 1	122.5
F1C	6 AND 1	107.8
F2C	7 AND 1	167.8
F3C	8 AND 1	304.4
F4C	9 AND 1	211.6
F5C	10 AND 1	211.6
F6C	11 AND 1	304.4

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Table 508/28-41-21-993-808 Wet Capacitance - Center Tank (Continued)

COMPONENT	PIN NUMBER*[1]	WET CAPACITANCE (PICOFARADS)
F7C	12 AND 1	167.8
F8C	13 AND 1	107.8

*[1] Pin numbers correspond to numbers on the test harness and pin numbers on the airplane feed-through connector.

SUBTASK 28-41-21-080-005

(5) Disconnect the adapter plug from the feed-through connector.

SUBTASK 28-41-21-080-006

(6) Disconnect the AIRFRAME GROUND alligator clamp.

SUBTASK 28-41-21-480-006

(7) Carefully connect the bussing plug to the feed-through connector (TASK 28-41-41-400-801).

(a) If the feed-through connector is damaged, replace the feed-through connector (TASK 28-41-42-020-801, TASK 28-41-42-420-801).

NOTE: The connector pins and sockets are easily damaged.

SUBTASK 28-41-21-710-005

(8) Do the procedure again for the other fuel tanks.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 28-41-21-980-004

(1) Do one of these steps to disconnect the system tester.

(a) For the system tester, COM-1770, or tester, COM-19223:

1) Set the POWER switch to the OFF position.
2) Disconnect the connector, P1.
3) Disconnect the connector, P2.
4) Disconnect the connector, Hi-Z.
5) Disconnect the banana jack connector, P5.

(b) For the system tester, COM-23749:

1) Set the POWER/TYPE SELECT switch to POWER OFF.
2) Disconnect the connector, P1.
3) Disconnect the connector, P2.
4) Disconnect the connector, Hi-Z.
5) Disconnect the banana jack connector, P5.

SUBTASK 28-41-21-860-011

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	3	C00032	FUEL FUELING CONT
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

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SUBTASK 28-41-21-080-007

WARNING YOU MUST CAREFULLY DO THE STEPS IN THE TASK BELOW TO REMOVE THE LE FLAP SAFETY LOCK. THE LE FLAPS CAN MOVE QUICKLY IF YOU DO NOT DO THE STEPS CORRECTLY. THIS CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(3) Remove the leading edge flap locks (TASK 27-81-00-080-801).

SUBTASK 28-41-21-860-012

WARNING KEEP ALL PERSONNEL AND EQUIPMENT CLEAR OF THE LEADING EDGE FLAPS. THE LEADING EDGE FLAPS CAN MOVE QUICKLY. THE MOVEMENT OF THE LEADING EDGE FLAPS CAN CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(4) Retract the leading edge flaps (TASK 27-81-00-860-804).

——— END OF TASK ———

TASK 28-41-21-710-803

4. Tank and Compensator Units - Fuel Quantity Indicating System Health Check

(Figure 501, Figure 502)

A. General

(1) This task contains steps to inspect the tank fuel probes, compensator units, and Fuel Quantity Indicating System (FQIS) wiring harness and connectors for contamination and degradation that affects the capacitance measurement ability of the system.

(2) Water, Foreign Object Debris (FOD), and Microbiological Growth (MBG) in a fuel tank can lead to significant interruptions to the FQIS and corrosion of an aircraft structure. Hydrocarbon fuels are subject to bacterial and fungal growth during storage, transportation, and use. The organisms that feed on these hydrocarbons not only grow in small quantities of water present in the fuel systems, but also grow in the hydrocarbon fuel itself. Microbes can live for months, and any subsequent presence of free water may lead to rapid population increase. These micro organisms will feed on the boundary layers that exist between the water and fuel. Since water is heavier than fuel, it will eventually migrate to the lower inboard areas of the fuel tanks. If not addressed, secretions of the microbes will start to eat away the corrosion protection layers of FQIS components, wiring harnesses, structure, and lead to material corrosion damage.

(3) MBG or fungus contamination, will appear as a dirty slime that can stain the fuel tank primer and/or tank sealant. This FQIS Health Check will detect water, FOD, and MBG that typically occurs over time, but also after extended maintenance and storage.

(4) In order to mitigate future MBG reoccurrences and subsequent FQIS intermittent operation, sump the drains, use an approved fuel biocide protection additive, test the fuel regularly in accordance with maintenance manual recommendations and perform proactive FQIS health checks. While Boeing cannot guarantee prevention of MBG occurring due to the many outstanding variables, we do believe these recommendations will help mitigate any future MBG issues.

(5) This check is performed using intrinsically safe equipment to facilitate testing with any level of fuel in the tank. This enhances the ability to uncover issues that occur only in-flight and also eliminates the need to drain, sump, and purge the tanks prior to testing.

NOTE: A tank entry is necessary if the test fails and in-tank equipment or wire harness require replacement.

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B. References

Reference	Title
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-41-00-710-801	Fuel Quantity Indicating System - Operational Test (P/B 501)
28-41-21-000-801	Tank Unit or Compensator Unit - Removal (P/B 401)
28-41-21-400-801	Tank Unit or Compensator Unit - Installation (P/B 401)
28-41-41-000-801	Bussing Plug Removal (P/B 401)
28-41-41-400-801	Bussing Plug Installation (P/B 401)
28-41-42-020-801	FQIS Spar Penetration Connector Removal (P/B 401)
28-41-42-420-801	FQIS Spar Penetration Connector Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1780	Cables - Adapter, Fuel Quantity Indicating System (FQIS) Tester Part #: PSD737-1 Supplier: 41364
COM-19223	Tester - Fuel Gauge System, Intrinsically Safe with Frequency Selector Part #: PSD60-2R Supplier: 51190
COM-23749	Test Set - AC Fuel Quantity System, PSD90-3AC (Intrinsically Safe) Part #: 22160649 Supplier: 51190

D. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

E. Prepare for the Test

SUBTASK 28-41-21-730-003

WARNING MAKE SURE THAT YOU USE AN INTRINSICALLY SAFE METER IN AN UNCONTROLLED TANK. AN ALTERNATIVE TOOL MUST BE INTRINSICALLY SAFE. IF YOU DO NOT OBEY, A FIRE OR EXPLOSION CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Use one of these system testers:

- tester, COM-19223 with cables, COM-1780.

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- tester, COM-23749 with cables, COM-1780.

NOTE: The cables, COM-1780, are referred to as the test harness in this procedure.

SUBTASK 28-41-21-650-004

(2) This test may be performed at any fuel level.

(a) If the health check is performed to troubleshoot an issue, the ability to uncover issues may be increased by duplicating the conditions leading to the event by filling or transferring fuel to the tank at the level reported when the problem occurred.

SUBTASK 28-41-21-860-014

WARNING KEEP ALL PERSONNEL AND EQUIPMENT CLEAR OF THE LEADING EDGE FLAPS. THE LEADING EDGE FLAPS CAN MOVE QUICKLY. THE MOVEMENT OF THE LEADING EDGE FLAPS CAN CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(3) Extend the leading edge flaps to test the tank or compensator units for the main tanks (TASK 27-81-00-860-803).

WARNING YOU MUST CAREFULLY DO THE STEPS IN THE TASK BELOW TO INSTALL THE LE FLAP SAFETY LOCKS. THE LE FLAPS CAN MOVE QUICKLY IF YOU DO NOT INSTALL THE SAFETY LOCKS CORRECTLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Install safety locks on the leading edge flaps (TASK 27-81-00-480-801).

SUBTASK 28-41-21-860-015

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	3	C00032	FUEL FUELING CONT
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

F. Tank and Compensator Units - Fuel Quantity Indicating System Health Check

SUBTASK 28-41-21-010-002

(1) Disconnect the applicable bussing plug from the applicable fuel tank feed-through connector (TASK 28-41-41-000-801).

SUBTASK 28-41-21-480-007

(2) Connect the adapter plug on the test harness to the feed-through connector (Figure 501).

SUBTASK 28-41-21-480-008

(3) Do one of these steps to connect the test harness to the system tester (Figure 501).

(a) For the system tester, COM-19223, do these steps:

1) Connect the alligator style clamp identified as AIRFRAME GROUND on the test harness to the aircraft ground/structure.

NOTE: The bonding bracket or stud used by the FQIS harness near the feed-through connector is a good location. A good ground is possible through the sealant, paint, or primer.

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2) Connect the banana jack connector, P5, on the test harness to the connector, J7, on the tester identified as CHASSIS.

3) Connect the connector, P3, identified as HI-Z on the test harness to the connector, J5, on the tester identified as TANK UNIT HI-Z.

4) Test the applicable tank units, do these steps:

a) Connect the connector, P2, identified as LO-Z on the test harness to the connector, P4, identified as TANK UNIT LO-Z on the tester.

b) On the test harness's selector box, select the applicable tank unit to measure.

<1> If it is necessary, select TOTAL on the test harness to measure the total capacitance of the tank units.

5) Connect the connector, P1, on the test harness to the feed-through connector on the rear or front spar.

(b) For the system tester, COM-23749, do these steps:

1) Connect the alligator style clamp identified as AIRFRAME GROUND on the test harness to the aircraft ground/structure.

NOTE: The bonding bracket or stud used by the FQIS harness near the feed-through connector is a good location. A good ground is possible through the sealant, paint or primer that is present.

2) Connect the banana jack connector, P5, on the test harness to the connector on the tester identified as CHASSIS.

3) Connect the connector, P3, identified as HI-Z on the test harness to the MEASURE TANKS TU HI-Z plug on the tester.

4) To test the applicable tank units, do these steps:

a) Connect the connector, P2, identified as LO-Z on the test harness the MEASURE TANKS TU LO-Z plug on the tester.

b) On the test harness's selector box, select the applicable tank unit to measure.

c) To measure the total capacitance of the tank units, select TOTAL on the test harness.

5) Connect the connector, P1, on the test harness to the feed-through connector on the rear or front spar.

SUBTASK 28-41-21-480-009

(4) Do one of these steps to setup the system tester to do the Fuel Quantity Indicating System Health Check:

NOTE: Adjustment of the frequency will be necessary to do this health check.

(a) For the system tester, COM-19223, do these steps:

1) Set the POWER switch to the ON position and wait for 3 minutes.

NOTE: Three minutes is necessary for the system tester to stabilize.

2) Set the CAPACITANCE MEASUREMENT MODE switch to the MODE A position.

3) Use the Frequency Select Switch to set the frequency to 5000Hz, push the knob down to enter.

4) Set the Function Select Switch to MEASUREMENT EXT.

5) Set the Megger Mode Switch to 3-WIRE.

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6) Set the Megger Select to TU.

NOTE: You can ignore the CAP SIMULATOR settings.

(b) For the system tester, COM-23749, do these steps:

1) Set the POWER/TYPE SELECT switch knob to the PROBE CHECK position.

2) Set the FUNCTION SELECT switch knob to the MEAS EXT - TU position.

NOTE: You can ignore the MEG MODE/SIMULATE SELECT switch knob, and the MEGGER SELECT switch knob when you are using the PROBE CHECK function.

SUBTASK 28-41-21-710-007

(5) Do one of these steps to measure the capacitance of each tank unit and compensator unit at 5000HZ and 500Hz, and record the data (Table 509, Table 512).

NOTE: Each compensator or tank unit has a PROBE SELECT switch position. See Table 509, Table 511, and Table 512 for the correct compensator or tank unit position on the switch. The switch positions will not be the same for all the fuel tanks. Some of the PROBE SELECT positions are not used.

(a) For the system tester, COM-19223, do this step:

1) Push the frequency select knob to change the frequency.

(b) For the system tester, COM-23749, do these steps:

1) Measure the difference in capacitance of each tank unit using the PROBE CHECK function.

2) On the test harness, select the applicable tank unit or compensator to measure.

3) Record the measured percent error on the applicable table.

NOTE: The percent error is displayed in the upper left corner of the display.

NOTE: It is not necessary to measure or record the capacitance values at 500 Hz or at 5000 Hz. The PROBE CHECK function measures the capacitance at multiple frequencies and calculates the percent error automatically.

NOTE: If necessary, push PWR INT or cycle power OFF/ON to enable the frequency change display.

SUBTASK 28-41-21-710-008

(6) Do one of these steps to measure the insulation resistance of the tank units:

NOTE: During testing for each unit, the value can fluctuate and be slow to climb. Allow up to 30 seconds for the value to settle, if necessary.

NOTE: Tester reading as -0r- is Out of Range.

NOTE: The frequency setting does not affect the IR test.

(a) For the system tester, COM-19223, do this step:

1) Measure the insulation resistance of the tank units and compensator units by rotating the tester Megger Select switch knob to measure and record each of the circuits as indicated in Table 510, Table 511, and Table 513.

(b) For the system tester, COM-23749, do these steps:

1) On the test harness, select the applicable tank unit or compensator to measure.

2) Turn the POWER/TYPE SELECT switch knob to AC CAP.

3) Turn the FUNCTION SELECT switch knob to MEGGER.

4) Set the MEG MODE/SIMULATE SELECT switch knob to 3-WIRE.

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5) Set the MEGGER SELECT switch to measure and record each of the circuits:
   a) Set MEGGER SELECT switch to these positions for Table 510:
      <1> LO-Z/HI-Z or COMP/HI-Z
      <2> GND/HI-Z
      <3> LO-Z/COMP or COMP/SHIELD
   b) Set MEGGER SELECT switch to these positions for Table 511:
      <1> LO-Z/HI-Z or COMP/HI-Z
      <2> GND/HI-Z
      <3> LO-Z/COMP or COMP/SHIELD
   c) Set MEGGER SELECT switch to these positions for Table 513:
      <1> LO-Z/HI-Z or COMP/HI-Z
      <2> GND/HI-Z
      <3> LO-Z/COMP or COMP/SHIELD

SUBTASK 28-41-21-750-004

(7) Do one of these steps to check the measured capacitance values for the tank units and compensator:
   (a) For the system tester, COM-19223, do these steps:
      1) Make sure that the measured capacitance values for the tank units and compensator are the following:
         a) Do not vary by more than 10% of the 500Hz reading when the frequency is lowered to 500Hz.
         b) Are within the limits given in Table 509, and Table 512 at 5000Hz.
   (b) For the system tester, COM-23749, do this step:
      1) Make sure that the percent (%) difference measured in PROBE CHECK is less than 10%.

         NOTE: The difference in percent is shown in the top left corner of the display using probe check mode.

         NOTE: Probe Check mode varies the frequency from 5000 Hz to 500 Hz for each measurement, and will display the difference in ρF.

SUBTASK 28-41-21-750-001

(8) If the capacitance value for a tank unit is out of tolerance for the steps above, further fault isolate by doing the following:
   (a) De-fuel the fuel tank for entry (TASK 28-26-00-650-801).
   (b) Disconnect the respective tank unit or compensator and retest.
      1) If the test values are within limits, the respective tank unit or compensator is defective or contaminated.
         a) Remove and replace component and retest (TASK 28-41-21-000-801, TASK 28-41-21-400-801).
      2) If the test still varies by more than 10% with the components disconnected, continue to fault isolate the defective harness segment by shaking the harness while monitoring the tester fluctuations.

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a) Verify faulty wiring harness bundle by disconnecting suspect segments and retesting.

b) If isolated to the connector, clean or replace and retest.

3) If the capacitance values of the wire bundle segment continues to be out of tolerance, remove and replace or repair the in-tank FQIS wire bundle and retest after the corrective action to ensure the test values are within limits (Standard Wiring Practices Manual (SWPM)).

SUBTASK 28-41-21-750-003

(9) Make sure that the measured insulation resistance for each circuit is greater than the minimum insulation resistance as indicated in Table 510, Table 511, and Table 513.

SUBTASK 28-41-21-750-002

(10) If the insulation resistance value was out of tolerance, further fault isolate by doing the following:

(a) De-fuel the fuel tank for entry (TASK 28-26-00-650-801).

(b) Disconnect the respective tank unit or compensator and retest.

1) If the insulation resistance values are in range, the respective tank unit or compensator is defective or contaminated.

a) Remove and replace component and retest (TASK 28-41-21-000-801, TASK 28-41-21-400-801).

2) If the insulation resistance test still fails with the components disconnected, continue to fault isolate the defective harness segment by shaking the harness while monitoring the tester fluctuations.

a) Verify faulty wiring harness bundle by disconnecting suspect segments and retesting.

b) If isolated to the connector, clean or replace and retest.

3) If the insulation resistance values of the wire bundle segment continues to be out of tolerance, remove and replace or repair the in-tank FQIS wire bundle and retest after the corrective action to ensure the test values are within limits (SWPM).

Table 509/28-41-21-993-815 Health Range Capacitance - Main Tank

COMPONENT NO. 1 (L) OR NO. 2 (R) MAIN TANK	PIN NUMBER *[1]	CAPACITANCE (PICOFARADS)		MEASURED CAPACITANCE AT 5000 Hz (PICOFARADS)		MEASURED CAPACITANCE AT 500 Hz (PICOFARADS)	
		EMPTY	MAX WET	L TANK	R TANK	L TANK	R TANK
F15 (L,R)	3 AND 1	57.6 - 58.7	122.5				
F1 (L,R)	6 AND 1	63.4 - 65.2	136.9				
F2 (L,R)	7 AND 1	40.0 - 41.5	86.9				
F3 (L,R)	8 AND 1	48.8 - 50.3	105.1				
F4 (L,R)	9 AND 1	26.2 - 27.7	57.6				
F5 (L,R)	10 AND 1	31.6 - 33.1	69.1				
F6 (L,R)	11 AND 1	35.6 - 37.1	77.6				
F7 (L,R)	12 AND 1	34.2 - 35.7	74.7				

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Table 509/28-41-21-993-815 Health Range Capacitance - Main Tank (Continued)

COMPONENT NO. 1 (L) OR NO. 2 (R) MAIN TANK	PIN NUMBER *[1]	CAPACITANCE (PICOFARADS)		MEASURED CAPACITANCE AT 5000 Hz (PICOFARADS)		MEASURED CAPACITANCE AT 500 Hz (PICOFARADS)	
		EMPTY	MAX WET	L TANK	R TANK	L TANK	R TANK
F8 (L,R)	13 AND 1	36.6 - 38.1	79.8				
F9 (L,R)	14 AND 1	33.2 - 34.7	72.5				
F10 (L,R)	15 AND 1	28.7 - 30.2	62.9				
F11 (L,R)	16 AND 1	12.7 - 14.2	25.0 *[2]				
F12 (L,R)	17 AND 1	12.7 - 14.2	18.8 *[2]				

*[1] Pin numbers correspond to numbers on the test harness and pin numbers on the airplane feed-through connector.

*[2] Tank unit is not completely covered in fuel at volumetric shutoff (VSO). This is an approximate wet capacitance value for the fuel level at VSO and can be different.

Table 510/28-41-21-993-817 Health Range Insulation Resistance - Left Tank

COMP	PIN NUMBER	LO-Z to HI-Z*[1] (250 Min) or Comp to HI-Z*[1] (1000 Min) (Megaohms)	HI-Z to SHLD*[1] (10 Min) (Megaohms)	LO-Z to SHLD or Comp to SHLD*[1] (4.9 Min - 6 Max) (Megaohms)
F15 (L)	3 AND 1			
F1 (L)	6 AND 1			
F2 (L)	7 AND 1			
F3 (L)	8 AND 1			
F4 (L)	9 AND 1			
F5 (L)	10 AND 1			
F6 (L)	11 AND 1			
F7 (L)	12 AND 1			
F8 (L)	13 AND 1			
F9 (L)	14 AND 1			
F10 (L)	15 AND 1			
F11 (L)	16 AND 1			
F12 (L)	17 AND 1			

*[1] Units in Megaohms.

Table 511/28-41-21-993-819 Health Range Insulation Resistance - Right Tank

COMP	PIN NUMBER	LO-Z to HI-Z*[1](250 Min) or Comp to HI-Z*[1] (1000 Min) (Megaohms)	HI-Z to SHLD*[1] (10 Min) (Megaohms)	LO-Z to SHLD or Comp to SHLD*[1] (4.9 Min - 6 Max) (Megaohms)
F15 (R)	3 AND 1			

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Table 511/28-41-21-993-819 Health Range Insulation Resistance - Right Tank (Continued)

COMP	PIN NUMBER	LO-Z to HI-Z*[1](250 Min) or Comp to HI-Z*[1] (1000 Min) (Megaohms)	HI-Z to SHLD*[1] (10 Min) (Megaohms)	LO-Z to SHLD or Comp to SHLD*[1] (4.9 Min - 6 Max) (Megaohms)
F1 (R)	6 AND 1			
F2 (R)	7 AND 1			
F3 (R)	8 AND 1			
F4 (R)	9 AND 1			
F5 (R)	10 AND 1			
F6 (R)	11 AND 1			
F7 (R)	12 AND 1			
F8 (R)	13 AND 1			
F9 (R)	14 AND 1			
F10 (R)	15 AND 1			
F11 (R)	16 AND 1			
F12 (R)	17 AND 1			

*[1] Units in Megaohms.

Table 512/28-41-21-993-816 Health Range Capacitance - Center Tank

COMP	PIN NUMBER *[1]	CAPACITANCE (PICOFARADS)		MEASURED CAPACITANCE AT 5000Hz (PICOFARADS)		MEASURED CAPACITANCE AT 500 Hz (PICOFARADS)	
		EMPTY	MAX WET				
F15 (C)	3 AND 1	57.6 - 58.7	122.5				
F1 (C)	6 AND 1	50.1 - 51.6	107.8				
F2 (C)	7 AND 1	78.1 - 80.2	167.8				
F3 (C)	8 AND 1	141.9 - 145.3	304.4				
F4 (C)	9 AND 1	98.6 - 101.1	211.6				
F5 (C)	10 AND 1	98.6 - 101.1	211.6				
F6 (C)	11 AND 1	141.9 - 145.3	304.4				
F7 (C)	12 AND 1	78.1 - 80.2	167.8				
F8 (C)	13 AND 1	50.1 - 51.6	107.8				

*[1] Pin numbers correspond to numbers on the test harness and pin numbers on the airplane feed-through connector.

Table 513/28-41-21-993-818 Health Range Insulation Resistance - Center Tank

COMP	PIN NUMBER	LO-Z to HI-Z*[1] (250 Min) or Comp to HI-Z*[1] (1000 Min) (Megaohms)	HI-Z to SHLD*[1] (10 Min) (Megaohms)	LO-Z to SHLD or Comp to SHLD*[1] (4.9 Min - 6 Min) (Megaohms)
F15 (C)	3 AND 1			

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Table 513/28-41-21-993-818 Health Range Insulation Resistance - Center Tank (Continued)

COMP	PIN NUMBER	LO-Z to HI-Z*[1] (250 Min) or Comp to HI-Z*[1] (1000 Min) (Megaohms)	HI-Z to SHLD*[1] (10 Min) (Megaohms)	LO-Z to SHLD or Comp to SHLD*[1] (4.9 Min - 6 Min) (Megaohms)
F1 (C)	6 AND 1			
F2 (C)	7 AND 1			
F3 (C)	8 AND 1			
F4 (C)	9 AND 1			
F5 (C)	10 AND 1			
F6 (C)	11 AND 1			
F7 (C)	12 AND 1			
F8 (C)	13 AND 1			

*[1] Units in Megaohms.

SUBTASK 28-41-21-080-008

(11) Disconnect the adapter plug from the feed-through connector.

SUBTASK 28-41-21-080-009

(12) Disconnect the AIRFRAME GROUND alligator clamp.

SUBTASK 28-41-21-420-013

(13) Carefully connect the bussing plug to the feed-through connector (TASK 28-41-41-400-801).

(a) If the feed-through connector is damaged, replace the feed-through connector (TASK 28-41-42-020-801, TASK 28-41-42-420-801).

NOTE: The connector pins and sockets are easily damaged.

SUBTASK 28-41-21-700-001

(14) If it is necessary, do the procedure again for the other fuel tanks.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 28-41-21-980-006

(1) For the system tester, COM-19223, set the POWER switch of the tester to the OFF position.

SUBTASK 28-41-21-980-007

(2) For the system tester, COM-23749, set the POWER/TYPE SELECT switch on the tester to POWER OFF.

SUBTASK 28-41-21-860-016

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	3	C00032	FUEL FUELING CONT
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

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SUBTASK 28-41-21-280-001

(4) Do the FQIS Built-In-Test Equipment (BITE) Test procedure at the Control Display Unit (CDU) in the flight deck and make sure that the test passes (TASK 28-41-00-710-801).

SUBTASK 28-41-21-080-010

WARNING

YOU MUST CAREFULLY DO THE STEPS IN THE TASK BELOW TO INSTALL THE LE FLAP SAFETY LOCKS. THE LE FLAPS CAN MOVE QUICKLY IF YOU DO NOT INSTALL THE SAFETY LOCKS CORRECTLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Remove the leading edge flap locks (TASK 27-81-00-080-801).

SUBTASK 28-41-21-860-017

WARNING

KEEP ALL PERSONNEL AND EQUIPMENT CLEAR OF THE LEADING EDGE FLAPS. THE LEADING EDGE FLAPS CAN MOVE QUICKLY. THE MOVEMENT OF THE LEADING EDGE FLAPS CAN CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(6) Retract the leading edge flaps (TASK 27-81-00-860-804).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TANK AND COMPENSATOR UNITS - CLEANING/PAINTING

1. General

A. This procedure contains this task:

(1) Tank unit or compensator unit cleaning.

TASK 28-41-21-110-801

2. Tank Unit or Compensator Unit Cleaning

A. References

Reference	Title
28-41-21-000-801	Tank Unit or Compensator Unit - Removal (P/B 401)
28-41-21-400-801	Tank Unit or Compensator Unit - Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-123	Brush - Soft Bristle
STD-1158	Container - 20 Gallon (76 Liter)

C. Consumable Materials

Reference	Description	Specification
B50138	Cleaner - Brulin Formula 815GD	
G02418	Water - De-ionized	

D. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

E. Tank Unit or Compensator Unit Cleaning

SUBTASK 28-41-21-010-003

(1) Remove the applicable tank unit or the compensator (TASK 28-41-21-000-801).

SUBTASK 28-41-21-110-001



WARNING MAKE SURE YOU USE APPROVED SAFETY EQUIPMENT FOR YOUR FACE, HANDS, AND ARMS WHEN YOU DO WORK NEAR HEATED WATER. THIS WILL HELP PREVENT INJURY TO PERSONS.

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(WARNING PRECEDES)

WARNING USE RUBBER GLOVES AND EYE PROTECTION WHEN YOU HANDLE LIQUID DETERGENT. LIQUID DETERGENT AND ANTI-FREEZE SOLVENT HAVE LOW FLASHPOINTS. KEEP AWAY FROM HEAT AND AN OPEN FLAME. MAKE SURE THAT THERE IS SUFFICIENT AIRFLOW. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT COULD OCCUR.

(2) Fill the 20 gallon (76 liter) container, STD-1158, with heated de-ionized water, G02418, (140°F (60°C) - 160°F (71°C)) and Brulin Formula 815GD Cleaner, B50138, or equivalent.

NOTE: The concentration of Brulin Formula 815GD Cleaner, B50138, to water shall be 8 to 10% by volume.

SUBTASK 28-41-21-110-002

(3) Fill different 20 gallon (76 liter) container, STD-1158, with unheated de-ionized water, G02418, (48°F (9°C) - 88°F (31°C)) to allow for rinsing the Tank Unit.

SUBTASK 28-41-21-110-003

WARNING MAKE SURE YOU USE APPROVED SAFETY EQUIPMENT FOR YOUR FACE, HANDS, AND ARMS WHEN YOU DO WORK NEAR HEATED WATER. THIS WILL HELP PREVENT INJURY TO PERSONS.

(4) Fill another 20 gallon (76 liter) container, STD-1158, with heated de-ionized water, G02418, (140°F (60°C) - 180°F (82°C)) for rinsing the Tank Unit.

SUBTASK 28-41-21-410-003

WARNING USE RUBBER GLOVES AND EYE PROTECTION WHEN YOU HANDLE LIQUID DETERGENT. LIQUID DETERGENT AND ANTI-FREEZE SOLVENT HAVE LOW FLASHPOINTS. KEEP AWAY FROM HEAT AND AN OPEN FLAME. MAKE SURE THAT THERE IS SUFFICIENT AIRFLOW. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT COULD OCCUR.

(5) Put the tank unit into the Brulin Formula 815GD Cleaner, B50138, for 2 - 5 minutes.

(a) Shake tank unit or compensator in the tank.

(b) Remove all contamination with the soft bristle brush, STD-123.

(c) Do not make scratches in the tank unit or the compensator.

SUBTASK 28-41-21-010-004

(6) Remove the tank unit from the Brulin Formula 815GD Cleaner, B50138, and let drain for a minimum of 4 seconds, maximum of 60 seconds.

SUBTASK 28-41-21-410-004

(7) Put the tank unit into the 20 gallon (76 liter) container, STD-1158, of unheated water for a minimum of 30 seconds.

(a) Shake the tank unit to promote rinsing.

NOTE: Regularly change the de-ionized water, G02418, in the Clean-Water Tank.

SUBTASK 28-41-21-010-005

(8) Remove the tank unit from unheated flush bath and let water drain for a minimum of 5 seconds into the bath.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-21-410-005

(9) Put the tank unit into the 20 gallon (76 liter) container, STD-1158, of heated water for a minimum of 30 seconds.

(a) Shake the tank unit to promote rinsing.

SUBTASK 28-41-21-010-006

(10) Remove the tank unit from heated flush bath and let water drain for a minimum of 5 seconds into the bath.

SUBTASK 28-41-21-410-006

WARNING DO NOT SUPPLY AIR AT A PRESSURE OF 50 PSIG (354 KPA) OR MORE AND/OR 450 F (232 C) OR MORE TO THE PNEUMATIC SYSTEM. IF YOU SUPPLY TOO MUCH PRESSURE OR TEMPERATURE, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(11) Blow dry the tank unit with compressed air (50 psig (344.7 kPa)) and air dry 2 ±0.5 hours at 48°F (9°C) - 88°F (31°C) or bake at 130°F (54°C) - 150°F (66°C) for 1 hour.

SUBTASK 28-41-21-410-007

(12) Install the tank unit or compensator unit (TASK 28-41-21-400-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TANK AND COMPENSATOR UNIT BUSSING PLUG - REMOVAL/INSTALLATION

1. General

A. There are three tank and compensator unit bussing plugs, one for each fuel tank. The removal and installation procedures for each plug are the same, only the location is different. Two bussing plugs are installed on the front spar behind the leading edge flaps and one on the rear spar in the wheel well area (Figure 401).

TASK 28-41-41-000-801

2. Bussing Plug Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the Bussing Plug.

B. References

Reference	Title
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
511	Left Wing - Leading Edge To Front Spar
611	Right Wing - Leading Edge to Front Spar

D. Prepare for the Removal

SUBTASK 28-41-41-710-002

(1) Do this step to get access to the bussing plug for No. 1 tank or No. 2 tank located on the front spar:

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Operate the LE flaps to the Full Extend position (TASK 27-81-00-860-803).

WARNING MAKE SURE THAT YOU INSTALL THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET TO PREVENT ACCIDENTAL OPERATION. THE LEADING EDGE FLAP AND SLAT CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

1) Do this task: Leading Edge Flap and Slat Locks Installation, TASK 27-81-00-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-41-480-002

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-41-41-860-002

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	3	C00032	FUEL FUELING CONT
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

E. Bussing Plug Removal

SUBTASK 28-41-41-020-001

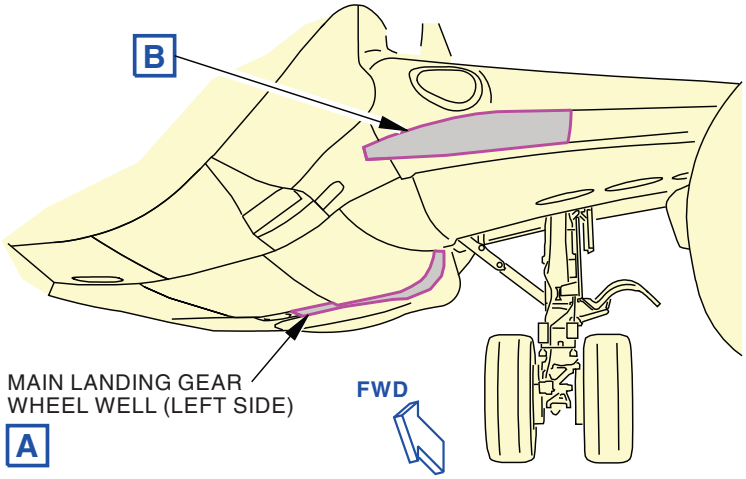
(1) Loosen the nut [4] to disconnect the ground wire [3].

SUBTASK 28-41-41-020-003

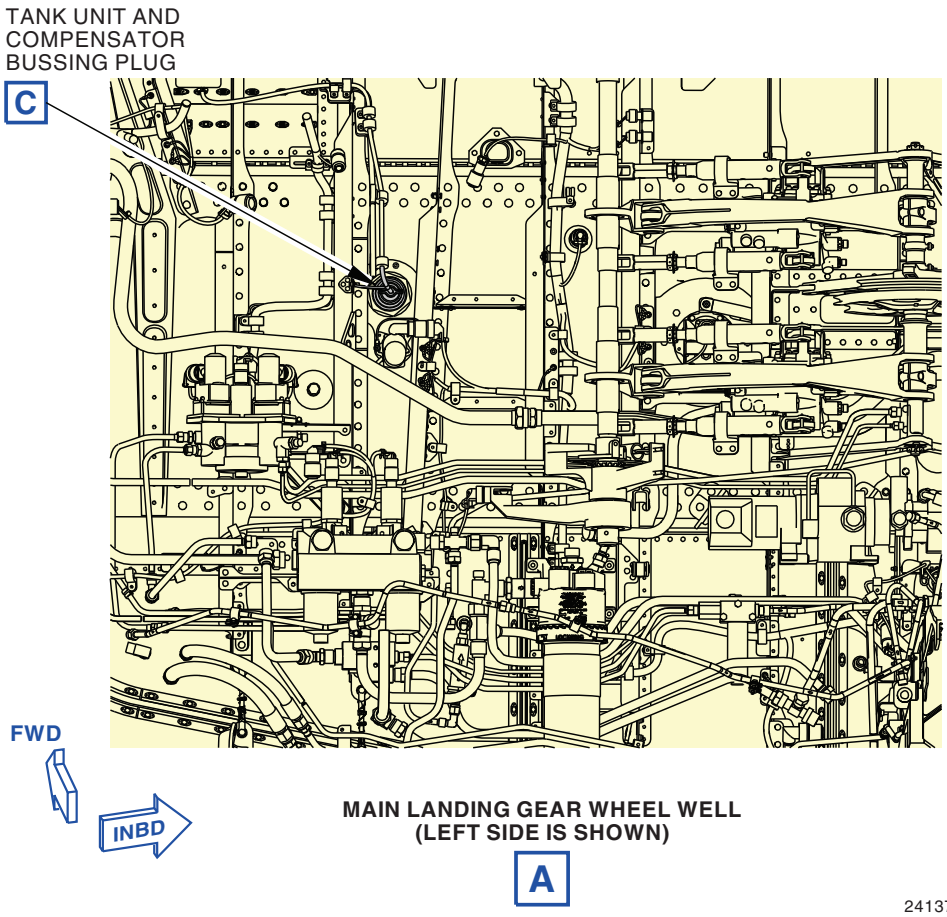
(2) Carefully remove the bussing plug [2] from the penetration connector [1] (SWPM 20-61-11).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS EQUIVALENT)



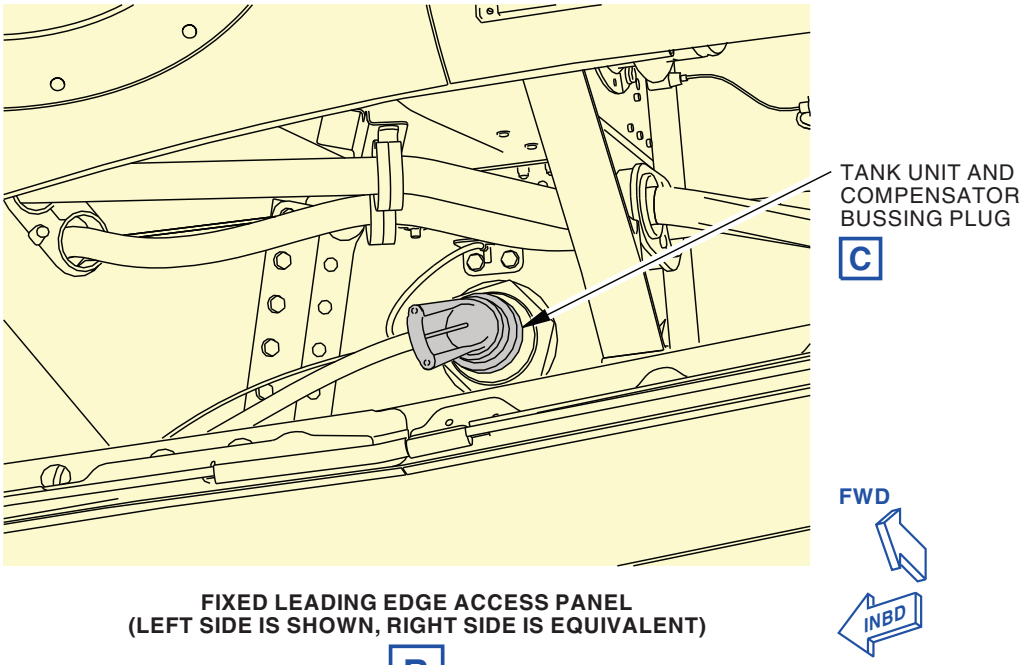
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN)

A

2413716 S00061533161_V1

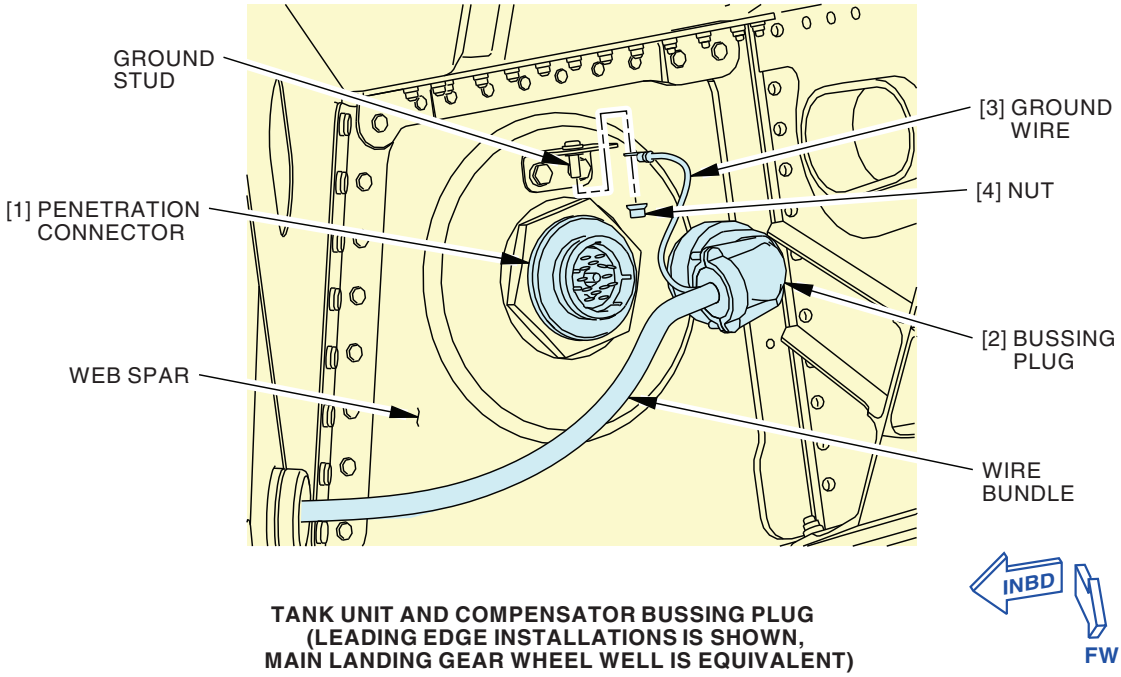
Tank Unit and Compensator Bussing Plug Installation
Figure 401/28-41-41-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FIXED LEADING EDGE ACCESS PANEL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

B



TANK UNIT AND COMPENSATOR BUSSING PLUG
(LEADING EDGE INSTALLATIONS IS SHOWN,
MAIN LANDING GEAR WHEEL WELL IS EQUIVALENT)

C

2413717 S00061533162_V1

Tank Unit and Compensator Bussing Plug Installation
Figure 401/28-41-41-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-41-41-400-801

3. Bussing Plug Installation

(Figure 401)

A. General

(1) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-04.

(2) This task gives instructions to install the Bussing Plug.

B. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-41-00-710-801	Fuel Quantity Indicating System - Operational Test (P/B 501)
SWPM 20-20-10	Replacement of Ground Studs and Bonding Jumper Installation

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50142	Sealant - Fuel Tank	BMS5-45 Class B-1/2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
511	Left Wing - Leading Edge To Front Spar
611	Right Wing - Leading Edge to Front Spar

F. Procedure

SUBTASK 28-41-41-110-001

(1) Clean the machined surface on the rear of the bussing plug and seal that touches the surface around the receptacle with solvent, B00083.

SUBTASK 28-41-41-160-001

(2) Rub the surfaces dry with a clean, cotton wiper, G00034.

SUBTASK 28-41-41-420-001

(3) Install the bussing plug [2] into the penetration connector [1] (SWPM 20-20-00, SWPM 20-60-06, SWPM 20-61-11).

(a) Tighten the bussing plug [2] to 95 in-lb (11 N·m) - 105 in-lb (12 N·m).

(b) For the center tank (connector D11316), inspect, clean and seal the bussing plug [2] (TASK 20-60-07-910-802).

(c) Make sure that the FQIS wire bundle is shielded and the shield ground is terminated (SWPM 20-10-15, SWPM 20-20-00).

SUBTASK 28-41-41-420-003

(4) For the main tanks and center tank do these steps to install the ground wire [3] to the structure:

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE. OBEY THE MATERIAL SAFETY DATA SHEETS (MSDS) FOR SOLVENTS. OBEY LOCAL REGULATIONS FOR THE CORRECT PROCEDURES TO USE OR DISCARD SOLVENTS. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Final clean the contact surfaces with a cotton wiper, G00034, soaked with alcohol, B00130, or solvent, B00184.

1) Rub dry with a clean, dry cotton wiper, G00034.

2) Continue to clean and dry the surface until the dry cotton wiper, G00034, stays clean.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Apply a thin continuous layer of sealant to both surfaces of the ground wire [3] and the two washers.

1) Use BMS 5-45 Class B-2 sealant, A50110, BMS 5-45 Class C sealant, A50155, BMS 5-95 Class B sealant, A50231, or BMS 5-95 Class C sealant, A50296, for the faying surface seal that is outside the tank.

2) Make sure that the sealant layer is approximately 0.005 in. (0.127 mm) thick.

(c) Install the nut [4], washer, and ground wire [3] to the structure.

(d) Tighten the nut [4] to 31.5 ±3.5 in-lb (4 ±1 N·m).

SUBTASK 28-41-41-760-001

28-AWL-04: CDCCL

(5) Measure the electrical bonding resistance from terminal lug on the ground wire [3] (shield pigtail) to the ground stud bracket for the applicable tank side FQIS connector (D16352, D16354, or D11316) (SWPM 20-20-10).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-04.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-04: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0010 ohm (1.0 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-04.

SUBTASK 28-41-41-390-001

(6) Apply sealant, A50142, over the nut [4] and the bonding jumper terminal (SWPM 20-20-10).

NOTE: It is satisfactory to encapsulate the ground stud head.

SUBTASK 28-41-41-860-003

(7) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	3	C00032	FUEL FUELING CONT
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

SUBTASK 28-41-41-080-001

WARNING

REMOVE THE LOCK FROM LEADING EDGE FLAP ACTUATOR CAREFULLY TO PREVENT ACCIDENTAL OPERATION OF THE LEADING EDGE FLAPS. THE LEADING EDGE FLAPS CAN MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(8) Do this task: Leading Edge Flap and Slat Locks Removal, TASK 27-81-00-080-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-41-710-003

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(9) Operate the LE flaps to the 0 detent and stowed (LE Flaps retracted) position (TASK 27-81-00-860-804).

SUBTASK 28-41-41-710-001

(10) Do this task: Fuel Quantity Indicating System - Operational Test, TASK 28-41-00-710-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FQIS SPAR PENETRATION CONNECTOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) FQIS Spar Penetration Connector Removal

(2) FQIS Spar Penetration Connector Installation.

TASK 28-41-42-020-801

2. FQIS Spar Penetration Connector Removal

(Figure 401)

A. General

(1) This task give instructions to remove the FQIS Spar Penetration Connector.

B. References

Reference	Title
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-41-41-000-801	Bussing Plug Removal (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
511	Left Wing - Leading Edge To Front Spar
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
611	Right Wing - Leading Edge to Front Spar
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5

D. Access Panels

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
631AB	Center Tank Access Door - Wing Station 168

E. Prepare for the Removal

SUBTASK 28-41-42-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	3	C00032	FUEL FUELING CONT
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-42-860-002

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. IF YOU DO NOT OBEY, THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Extend the leading edge slats (TASK 27-81-00-860-803).

SUBTASK 28-41-42-860-004

WARNING MAKE SURE THAT YOU INSTALL THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET TO PREVENT ACCIDENTAL OPERATION. THE LEADING EDGE FLAP AND SLAT CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Lock the leading edge slats (TASK 27-81-00-480-801).

SUBTASK 28-41-42-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(4) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-41-42-010-002

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Open the applicable access panels:

(TASK 28-11-31-000-801)

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
631AB	Center Tank Access Door - Wing Station 168

SUBTASK 28-41-42-020-001

(6) Disconnect the bussing plug [1] (TASK 28-41-41-000-801).

F. FQIS Spar Penetration Connector Removal

SUBTASK 28-41-42-020-002

(1) Remove the lockwire from the jamnut [3] and bolt [4].

SUBTASK 28-41-42-020-006

(2) Loosen the jamnut [3] from the spar penetration connector [2].

SUBTASK 28-41-42-020-003

(3) Remove the jamnut [3] and washer [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-42-020-004

(4) Remove the FQIS wire harness [7] from the spar penetration connector [2].

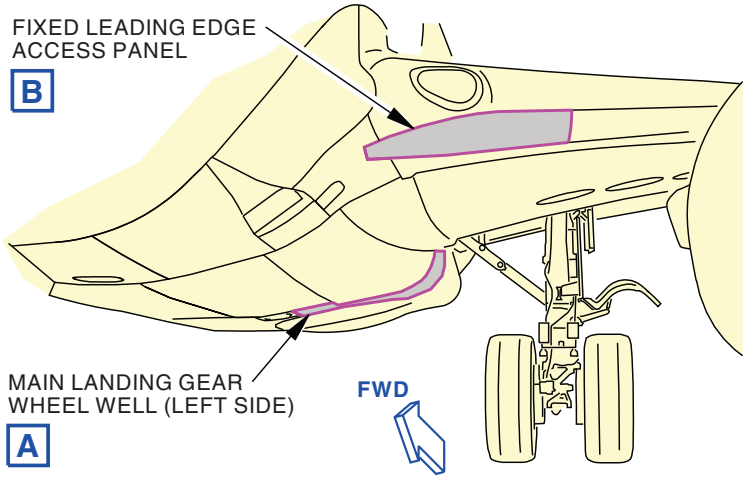
SUBTASK 28-41-42-020-005

(5) Remove the spar penetration connector [2] and O-ring [6] from the spar.

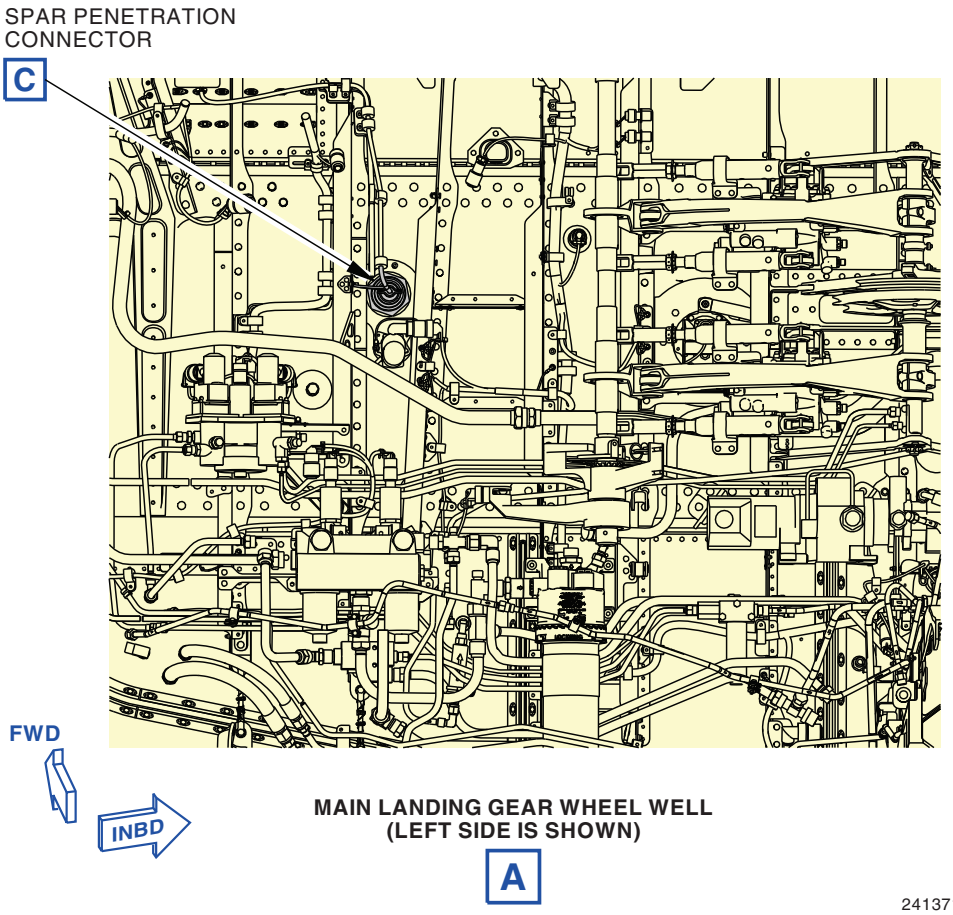
(a) Discard the O-ring [6].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS EQUIVALENT)



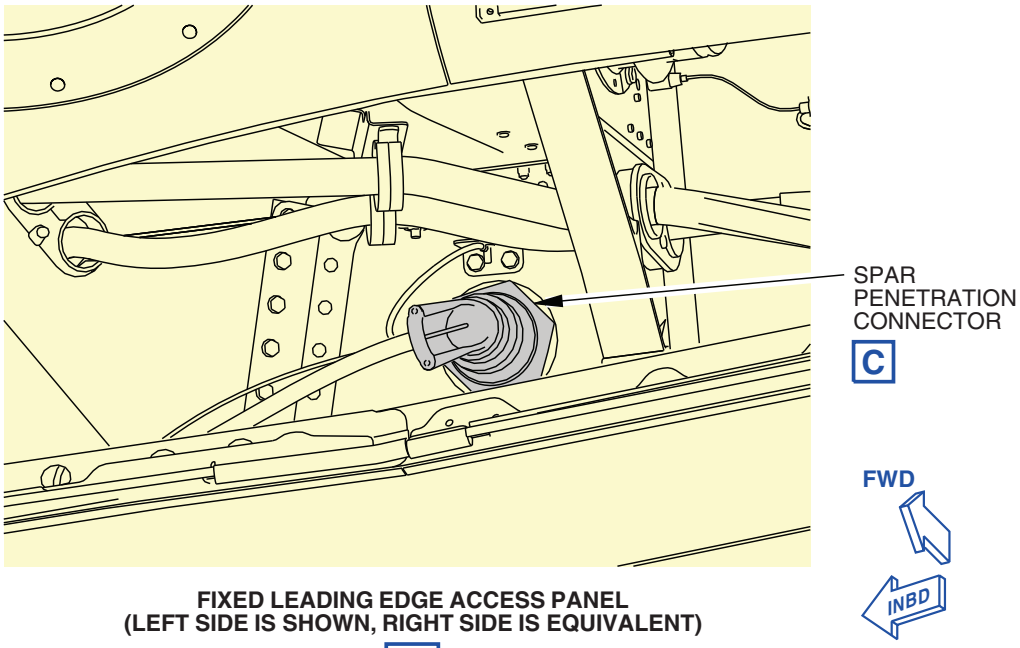
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN)

A

2413718 S00061533167_V1

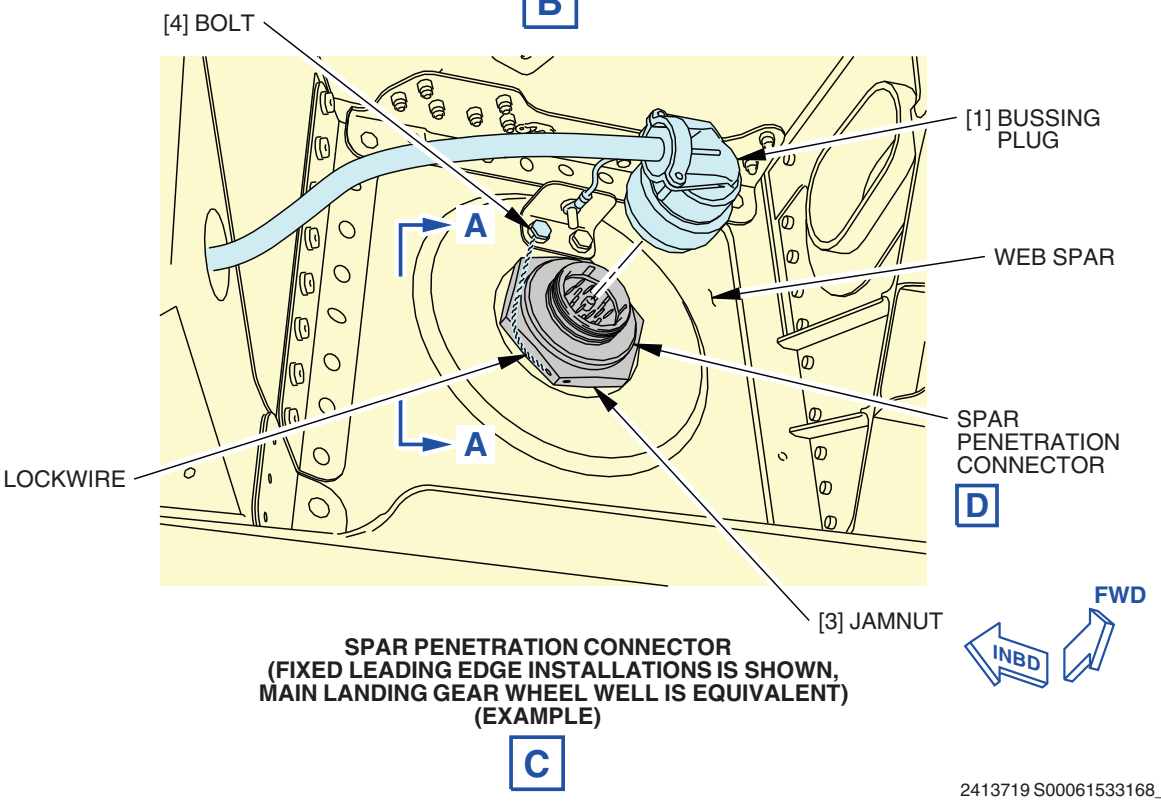
FQIS Penetration Connector Installation
Figure 401/28-41-42-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FIXED LEADING EDGE ACCESS PANEL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

B



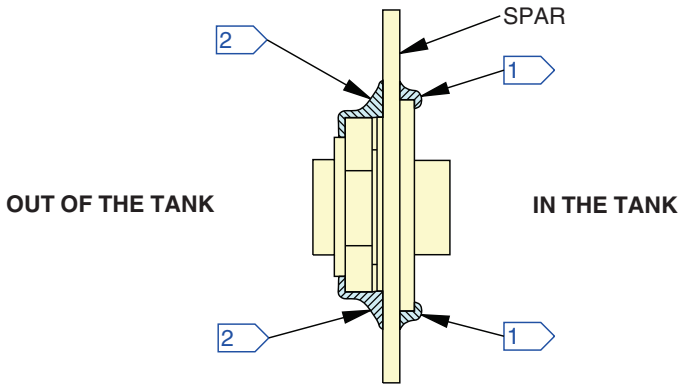
SPAR PENETRATION CONNECTOR
(FIXED LEADING EDGE INSTALLATIONS IS SHOWN,
MAIN LANDING GEAR WHEEL WELL IS EQUIVALENT)
(EXAMPLE)

C

2413719 S00061533168_V2

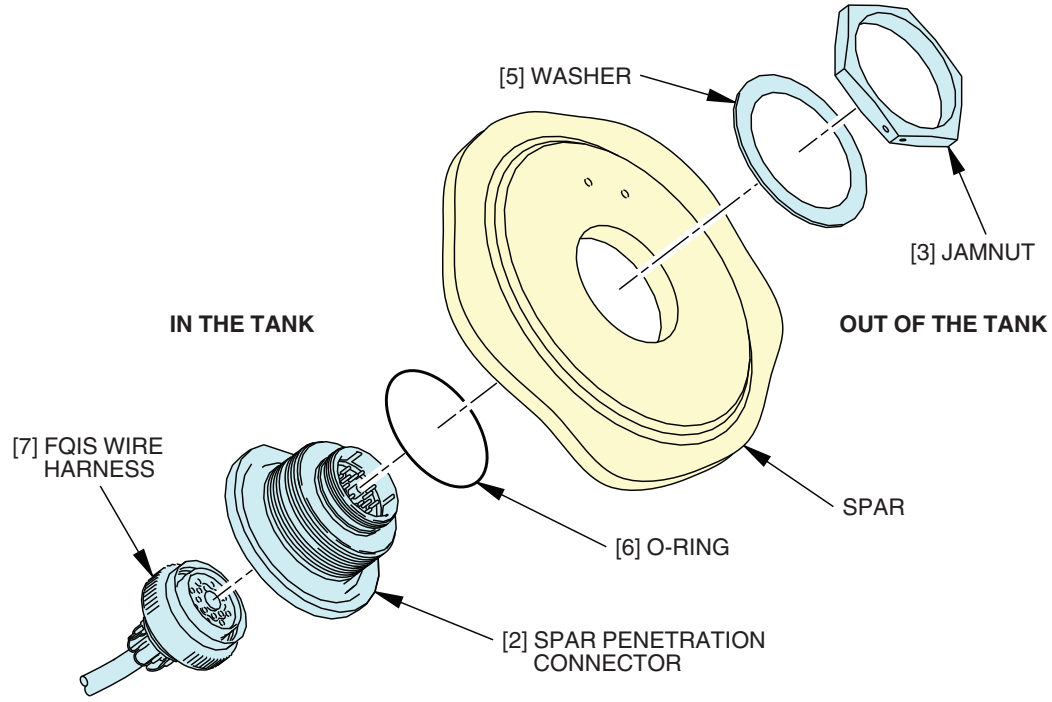
FQIS Penetration Connector Installation
Figure 401/28-41-42-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FQIS SPAR PENETRATION CONNECTOR SEALING

A-A



SPAR PENETRATION CONNECTOR (EXAMPLE)

D

1 FILLET SEAL WITH BMS 5-45
2 FILLET SEAL WITH BMS 5-95

2413720 S00061533169_V2

FQIS Penetration Connector Installation
Figure 401/28-41-42-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-41-42-420-801

3. FQIS Spar Penetration Connector Installation

(Figure 401)

A. General

(1) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

B. References

Reference	Title
12-11-00-650-803	Pressure Refuel Procedure (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-11-00-300-803	Repair of Sealant Leaks in the Fuel Tank Structure (P/B 801)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-00-790-804	Fuel Tank Closure Leak Check (P/B 601)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
28-41-00-730-801	Fuel Quantity Indicating System - Dry Capacitance Test (P/B 501)
28-41-41-400-801	Bussing Plug Installation (P/B 401)
51-22-00-390-806	Fillet Seal Application (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A00767	Sealant - Fuel Tank	BMS5-45
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Spar penetration connector	28-11-51-20-030	PTF ALL
		28-11-51-21-080	PTF ALL
		28-11-51-22-080	PTF ALL
6	O-ring	28-11-51-20-015	PTF ALL
		28-11-51-21-060	PTF ALL
		28-11-51-22-060	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
511	Left Wing - Leading Edge To Front Spar
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
611	Right Wing - Leading Edge to Front Spar
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5

G. Access Panels

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
631AB	Center Tank Access Door - Wing Station 168

H. FQIS Spar Penetration Connector Installation

SUBTASK 28-41-42-100-001

(1) Clean the fay surfaces of the aluminum spar (abrasive), the flange (solvent) on the spar penetration connector [2], and the washer [5] (SWPM 20-20-00, Paragraphs 2 and 3).

SUBTASK 28-41-42-420-001

(2) Install a new O-ring [6] in the O-ring groove of the spar penetration connector [2].

SUBTASK 28-41-42-390-002

(3) Apply a thin layer of sealant, A00767, to one side of the washer [5].

SUBTASK 28-41-42-420-009

(4) Align the washer [5] to the spar with the sealant facing the spar (dry side).

(a) Carefully remove excess sealant in the spar hole.

SUBTASK 28-41-42-420-002

(5) Install the spar penetration connector [2] to the spar, as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

28-AWL-09: CDCCL

(a) Apply a fay surface bond on the spar penetration connector [2] and the spar in the fuel tank (TASK 28-11-00-300-803, SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

1) Use sealant, A00767.

(b) Install the spar penetration connector [2] in the spar.

(c) Install the jamnut [3] to the spar penetration connector [2].

SUBTASK 28-41-42-420-010

(6) Tighten the jamnut [3] to 325 in-lb (37 N·m) to 350 in-lb (40 N·m).

SUBTASK 28-41-42-765-001

28-AWL-09: CDCCL

(7) Measure the electrical bonding resistance between the spar penetration connector [2] and the structure (spar) in the fuel tank (SWPM 20-20-00).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-09: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0010 ohm (1.0 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

SUBTASK 28-41-42-420-005

(8) Install the lockwire between the jamnut [3] and bolt [4] (TASK 20-10-44-400-801).

SUBTASK 28-41-42-390-001

28-AWL-09: CDCCL

(9) Apply a fillet seal around the spar penetration connector [2] in the fuel tank (TASK 51-22-00-390-806).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

(a) Use sealant, A00767.

SUBTASK 28-41-42-860-005

(10) Manually apply coating, C00064,on the bare metal surfaces on the spar in the fuel tank.

SUBTASK 28-41-42-860-006

(11) Apply primer, C00259, on the bare metal surfaces on the spar in the fuel tank.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-42-420-006

(12) Connect the FQIS wire harness [7] to the spar penetration connector [2].

SUBTASK 28-41-42-420-007

28-AWL-09: CDCCL

(13) Apply a fillet seal around the spar penetration connector [2] out of the fuel tank (TASK 51-22-00-390-806).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

(a) Use sealant, A00247.

SUBTASK 28-41-42-860-007

(14) Manually apply coating, C00064, on the bare metal surfaces on the spar in the fuel tank.

SUBTASK 28-41-42-860-008

(15) Apply primer, C00259, on the bare metal surfaces on the spar in the fuel tank.

SUBTASK 28-41-42-211-001

(16) Inspect the bussing plug [1] (TASK 20-60-07-910-802).

SUBTASK 28-41-42-420-008

(17) Connect the bussing plug [1] to the spar penetration connector [2] (TASK 28-41-41-400-801).

I. FQIS Spar Penetration Connector Installation Test

SUBTASK 28-41-42-410-001

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Close the applicable access panels:

(TASK 28-11-31-400-801)

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
631AB	Center Tank Access Door - Wing Station 168

SUBTASK 28-41-42-860-003

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

SUBTASK 28-41-42-820-001

(3) Do the fuel quantity indicating system test (TASK 28-41-00-730-801).

SUBTASK 28-41-42-650-003

(4) Refuel the applicable fuel tank (TASK 12-11-00-650-803).

SUBTASK 28-41-42-790-001

(5) Monitor the applicable spar penetration connector [2] for leakage for a minimum of one hour (TASK 28-11-00-790-804).

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J. Put the Airplane Back to Its Usual Condition

SUBTASK 28-41-42-080-001

(1) Remove and then retract the leading edge slats (TASK 27-81-00-080-801, TASK 27-81-00-860-804).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FQIS TANK-TO-TANK FEED-THROUGH CONNECTOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) FQIS Tank-to-Tank Feed-Through Connector Removal

(2) FQIS Tank-to-Tank Feed-Through Connector Installation.

TASK 28-41-43-000-801

2. FQIS Tank-to-Tank Feed-Through Connector Removal

A. References

Reference	Title
20-10-44-000-801	Lockwire, Cotter Pins, and Lockrings - Removal (P/B 401)
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

C. Access Panels

Number	Name/Location
531BB	Center Tank Access Door - Wing Station 192
532AB	Main Tank Access Door - Wing Station 216
631BB	Center Tank Access Door - Wing Station 192
632AB	Main Tank Access Door - Wing Station 216

D. Prepare for the Removal

SUBTASK 28-41-43-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

SUBTASK 28-41-43-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

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SUBTASK 28-41-43-010-001

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do this task for the applicable fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Open the applicable access panels:

(TASK 28-11-31-000-801, TASK 28-11-11-000-801)

Number	Name/Location
531BB	Center Tank Access Door - Wing Station 192
532AB	Main Tank Access Door - Wing Station 216
631BB	Center Tank Access Door - Wing Station 192
632AB	Main Tank Access Door - Wing Station 216

E. FQIS Tank-to-Tank Feed-Through Connector Removal

SUBTASK 28-41-43-020-001

(1) Disconnect the FQIS connector [1] from the tank-to-tank feed-through connector [7].

SUBTASK 28-41-43-020-002

(2) Disconnect the FQIS connector [2] from the tank-to-tank feed-through connector [7].

SUBTASK 28-41-43-020-007

(3) Remove sealant from the jamnut [5] on the main tank side.

SUBTASK 28-41-43-020-003

(4) Remove the lockwire between the jamnut [5] and bolt [6] (TASK 20-10-44-000-801).

SUBTASK 28-41-43-020-004

(5) Remove the jamnut [5] and washer [4] from the tank-to-tank feed-through connector [7].

SUBTASK 28-41-43-020-008

(6) Remove sealant from the connector on the center tank side of the tank-to-tank feed-through connector [7].

SUBTASK 28-41-43-020-005

(7) Remove the tank-to-tank feed-through connector [7].

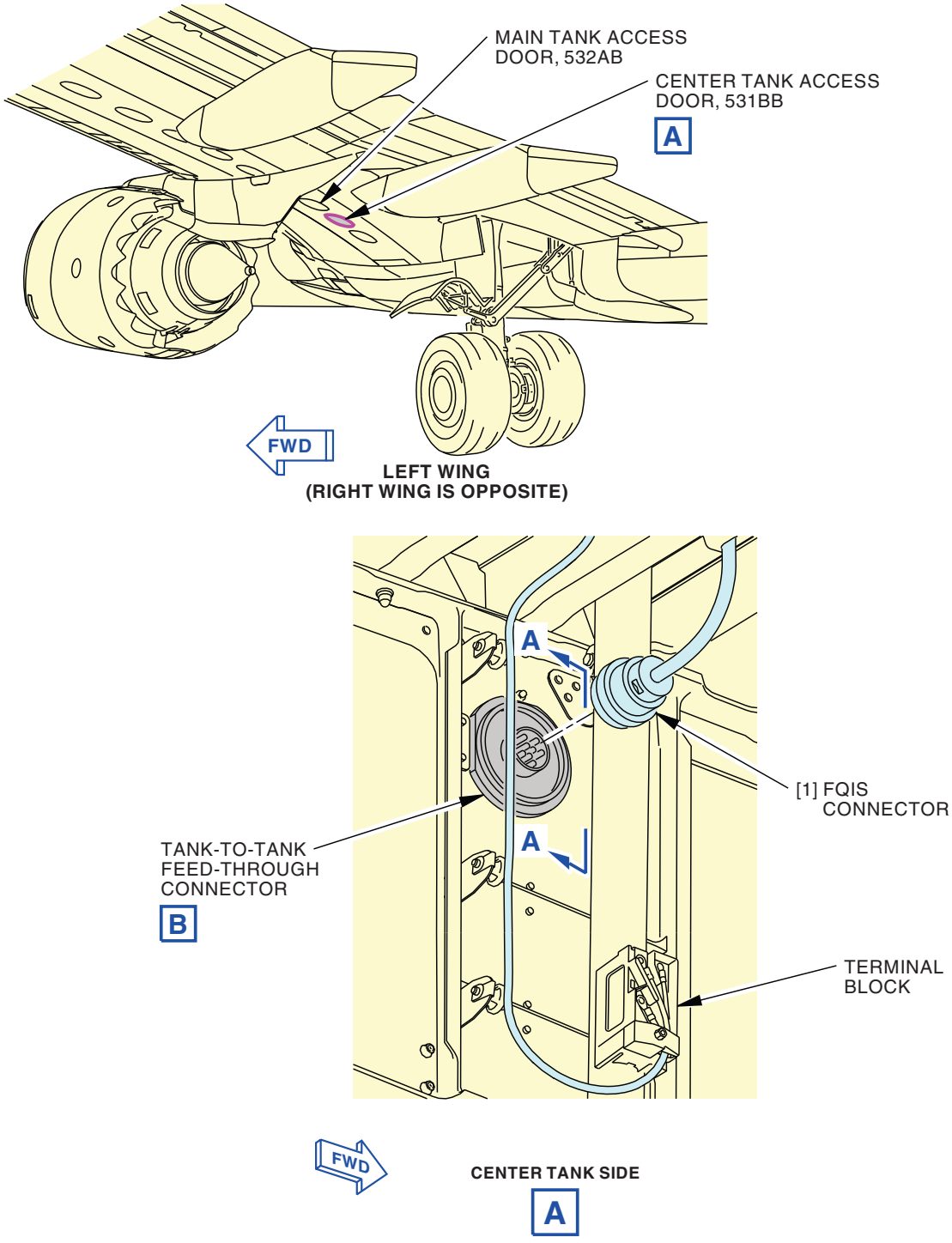
SUBTASK 28-41-43-020-006

(8) Remove the O-ring [3] from the tank-to-tank feed-through connector [7].

(a) Discard the O-ring [3]

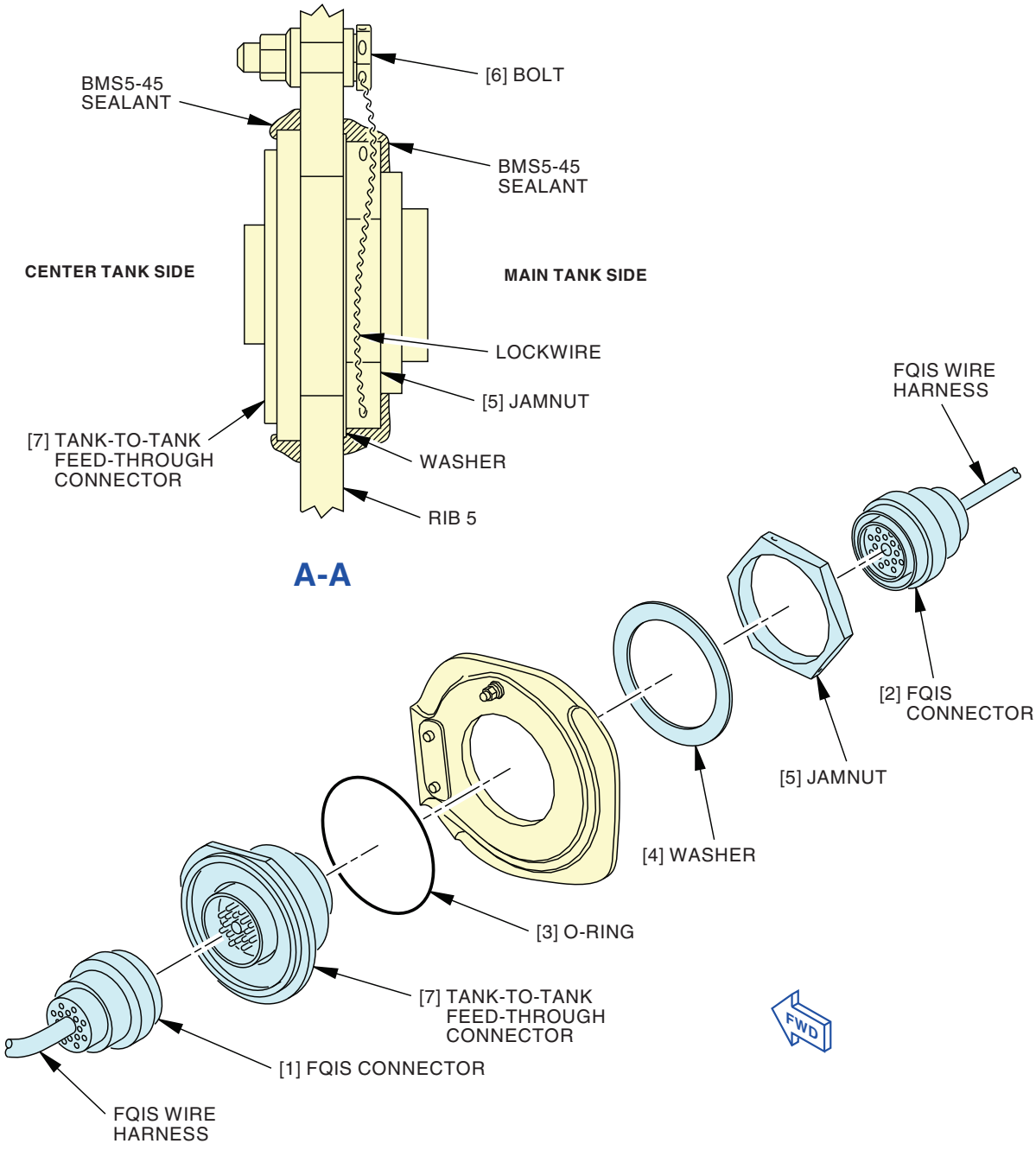
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FQIS Rib Penetration Connector Installation
Figure 401/28-41-43-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FQIS Rib Penetration Connector Installation
Figure 401/28-41-43-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-41-43-400-801

3. FQIS Tank-to-Tank Feed-Through Connector Installation

A. General

(1) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-00-790-801	Fuel Leak Detection Procedures (P/B 601)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-41-00-710-801	Fuel Quantity Indicating System - Operational Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A00767	Sealant - Fuel Tank	BMS5-45
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	O-ring	28-11-51-21-060	PTF ALL
		28-11-51-22-060	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Tank-to-tank feed-through connector	28-11-51-21-070	PTF ALL
		28-11-51-22-070	PTF ALL

F. Location Zones

Zone	Area
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

G. Access Panels

Number	Name/Location
531BB	Center Tank Access Door - Wing Station 192
532AB	Main Tank Access Door - Wing Station 216
631BB	Center Tank Access Door - Wing Station 192
632AB	Main Tank Access Door - Wing Station 216

H. FQIS Rib Penetration Connector Installation

SUBTASK 28-41-43-120-001

(1) Clean the fay surfaces of the washer [4] and outboard side of Rib 5 (SWPM 20-20-00, Paragraphs 2 and 3).

SUBTASK 28-41-43-390-001

28-AWL-09: CDCCL

(2) Apply a fay surface bond between the flange on the tank-to-tank feed-through connector [7], and the inboard side of Rib 5 (SWPM 20-20-00, Paragraphs 2 and 3).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

SUBTASK 28-41-43-420-001

(3) Install the new O-ring [3] in the O-ring groove on the tank-to-tank feed-through connector [7].

SUBTASK 28-41-43-390-002

(4) Apply a thin layer of sealant, A00767, to one side of the washer [4].

SUBTASK 28-41-43-820-001

(5) Align the washer [4] with the rib hole in the main tank.

(a) Make sure that the sealant side of the washer is inboard.

(b) Carefully remove sealant in the rib hole.

SUBTASK 28-41-43-420-002

(6) Install the tank-to-tank feed-through connector [7] in the rib.

SUBTASK 28-41-43-420-003

(7) Install the jamnut [5] to the tank-to-tank feed-through connector [7].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-43-420-004

(8) Tighten the jamnut [5] to 337.5 ±12.5 in-lb (38.1 ±1.4 N·m).

SUBTASK 28-41-43-860-002

28-AWL-09: CDCCL

(9) Measure the electrical bonding resistance between the tank-to-tank feed-through connector [7] and the structure (rib 5) (SWPM 20-20-00, Paragraph 21).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

NOTE: Measure the bonding resistance from the inboard side of Rib 5.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

28-AWL-09: CDCCL

(b) Make sure that the electrical bonding resistance is 0.0010 ohm (1.0 milliohm) or less.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

SUBTASK 28-41-43-420-005

(10) Install the lockwire between the jamnut [5] and bolt [6] (TASK 20-10-44-400-801).

SUBTASK 28-41-43-390-003

28-AWL-09: CDCCL

(11) Apply a fillet seal around the tank-to-tank feed-through connector [7] with sealant, A00767, inboard of Rib 5.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

SUBTASK 28-41-43-390-004

(12) Manually apply coating, C00064, on the bare metal surfaces on the rib in the center tank.

SUBTASK 28-41-43-390-005

(13) Apply primer, C00259, on the bare metal surfaces on the rib in the center tank.

SUBTASK 28-41-43-420-006

(14) Connect the FQIS connector [1] to the tank-to-tank feed-through connector [7].

NOTE: See SWPM 20-60-06 for more details.

(a) Tighten the FQIS connector [1] to 95 in-lb (11 N·m) - 105 in-lb (12 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-43-390-006

28-AWL-09: CDCCL

(15) Apply a fillet seal around the tank-to-tank feed-through connector [7] and the jamnut [5] with sealant, A00767, outboard of Rib 5.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

SUBTASK 28-41-43-390-007

(16) Manually apply coating, C00064, on the bare metal surfaces on the rib in the main tank.

SUBTASK 28-41-43-390-008

(17) Apply primer, C00259, on the bare metal surfaces on the rib in the main tank.

SUBTASK 28-41-43-420-007

(18) Connect the FQIS connector [2] to the tank-to-tank feed-through connector [7].

(a) Tighten the FQIS connector [2] to 95 in-lb (11 N·m) - 105 in-lb (12 N·m).

I. FQIS Tank-to-Tank Feed-Through Connector Installation Test

SUBTASK 28-41-43-710-001

(1) Do the air pressure and bubble solution (back-blowing) test (TASK 28-11-00-790-801).

NOTE: This test is done between the center tank and main tank.

SUBTASK 28-41-43-710-002

(2) Do this task: Fuel Quantity Indicating System - Operational Test, TASK 28-41-00-710-801.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 28-41-43-410-001

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Close the applicable access panels:

(TASK 28-11-31-000-801, TASK 28-11-11-400-801)

Number	Name/Location
531BB	Center Tank Access Door - Wing Station 192
532AB	Main Tank Access Door - Wing Station 216
631BB	Center Tank Access Door - Wing Station 192
632AB	Main Tank Access Door - Wing Station 216

SUBTASK 28-41-43-860-003

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL TANK BULKHEAD (SPAR) RECEPTACLE WIRE HARNESS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) FQIS Wire Harness Removal

(2) FQIS Wire Harness Installation.

B. This task replaces the FQIS wire harness in each fuel tank.

TASK 28-41-44-000-801

2. FQIS Wire Harness Removal

(Figure 401)

A. General

(1) The FQIS wire harness is referred to as the wire harness in this task.

B. References

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-11-11-000-803	Main Tank Inner Access Door Removal (P/B 401)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)

C. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Access Panels

Number	Name/Location
532AZ	Main Tank Inner Access at Rib 6
532BZ	Main Tank Inner Access at Rib 6
632AZ	Main Tank Inner Access at Rib 6
632BZ	Main Tank Inner Access at Rib 6

E. Prepare for the Removal

SUBTASK 28-41-44-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-44-860-002

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(2) Do this task for the applicable fuel tanks: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) For the No. 1 or No. 2 main tank, remove the access doors (TASK 28-11-11-000-801).

(b) For the center tank, remove the access doors (TASK 28-11-31-000-801).

SUBTASK 28-41-44-010-001

(3) Areas between rib No. 6 and No. 7, remove the applicable access panels:

Number	Name/Location
532AZ	Main Tank Inner Access at Rib 6
532BZ	Main Tank Inner Access at Rib 6
632AZ	Main Tank Inner Access at Rib 6
632BZ	Main Tank Inner Access at Rib 6

NOTE: Only one access panel can be removed with the engine installed.

(a) Do this task: Main Tank Inner Access Door Removal, TASK 28-11-11-000-803.

F. FQIS Wire Harness Removal

SUBTASK 28-41-44-020-001

(1) Disconnect the in-tank plug from the feed-through connector.

SUBTASK 28-41-44-020-002

(2) Disconnect the in-tank plug from the tank-to-tank feed through connector.

SUBTASK 28-41-44-020-003

(3) Disconnect the wire harness [1] from each terminal block on the tank units and compensator units:

(a) Remove the self-locking insert [4], washer [5], and HI-Z wire [6].

(b) Remove the self-locking insert [11], washer [10], and LO-Z wire [7].

(c) Remove the self-locking insert [13], washer [14], and HI-Z shield wire [12].

(d) Loosen the captive screw [8] on the relief clamp [9].

(e) Remove the wire harness [1] from the terminal block.

SUBTASK 28-41-44-020-004

(4) Cut all self-locking plastic straps from the stringer clips [2] and ring-post supports [3] (SWPM 20-10-12).

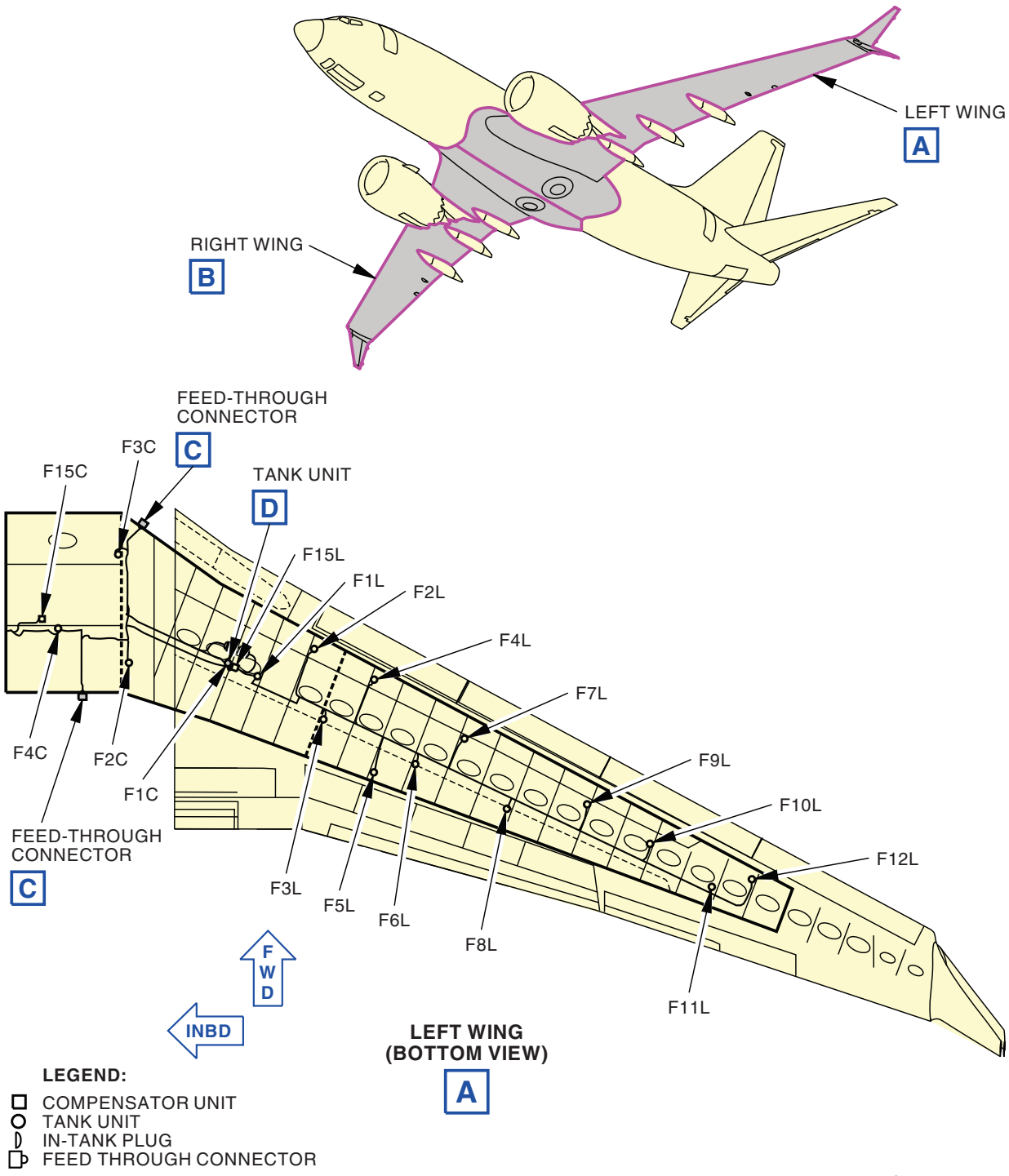
(a) Discard all self-locking plastic straps.

SUBTASK 28-41-44-020-005

(5) Remove the wire harness [1] from the fuel tank.

END OF TASK

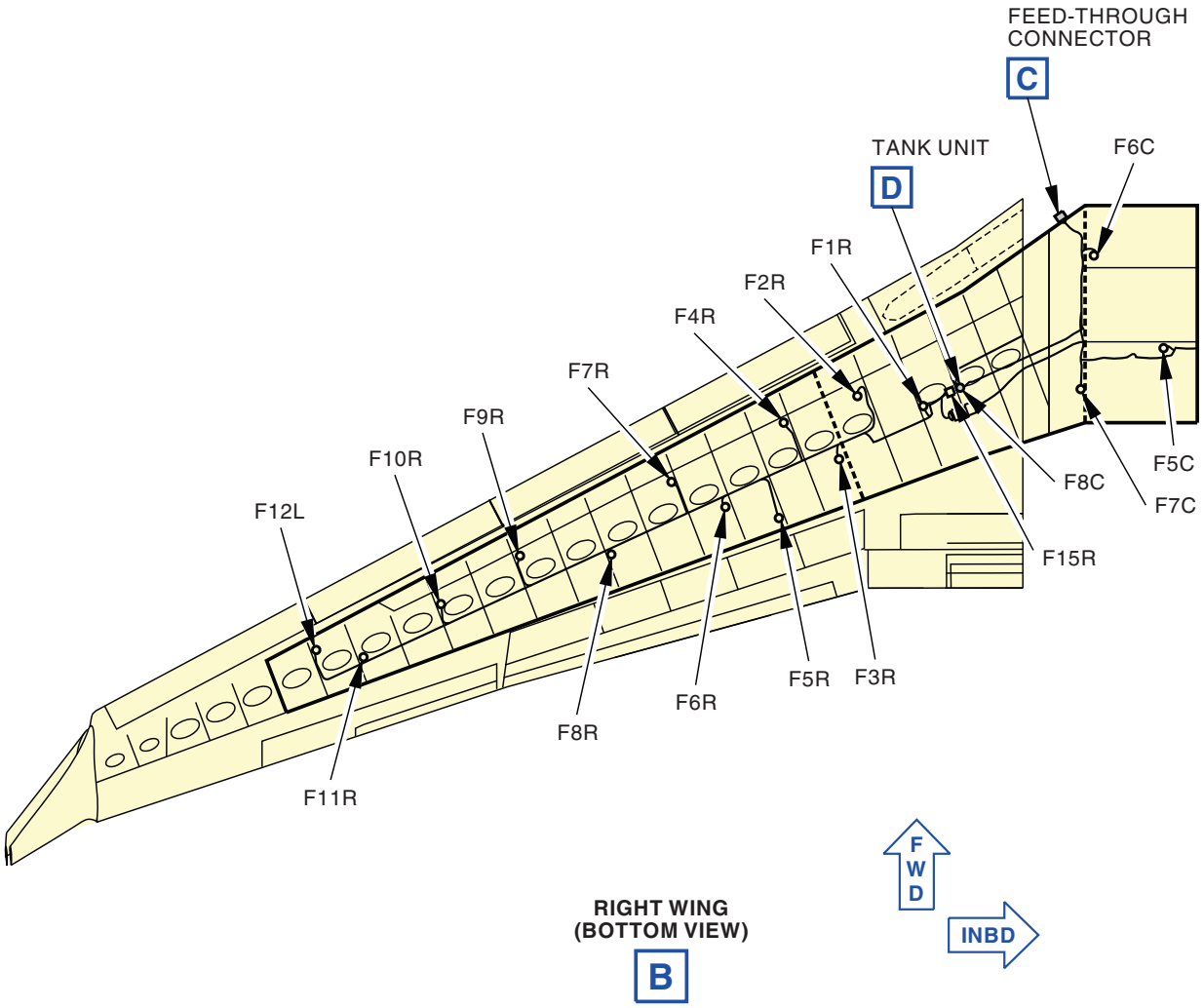
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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FQIS Wire Harness Installation
Figure 401/28-41-44-990-801 (Sheet 1 of 5)

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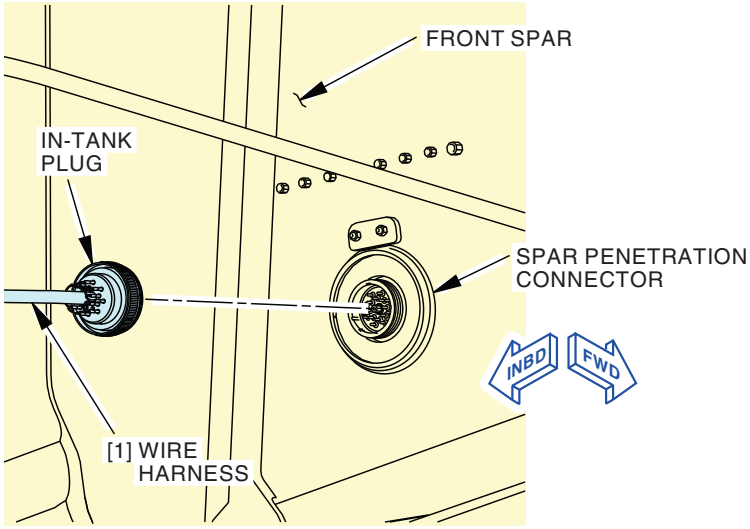
LEGEND:

- COMPENSATOR UNIT
- TANK UNIT
- IN-TANK PLUG
- FEED THROUGH CONNECTOR

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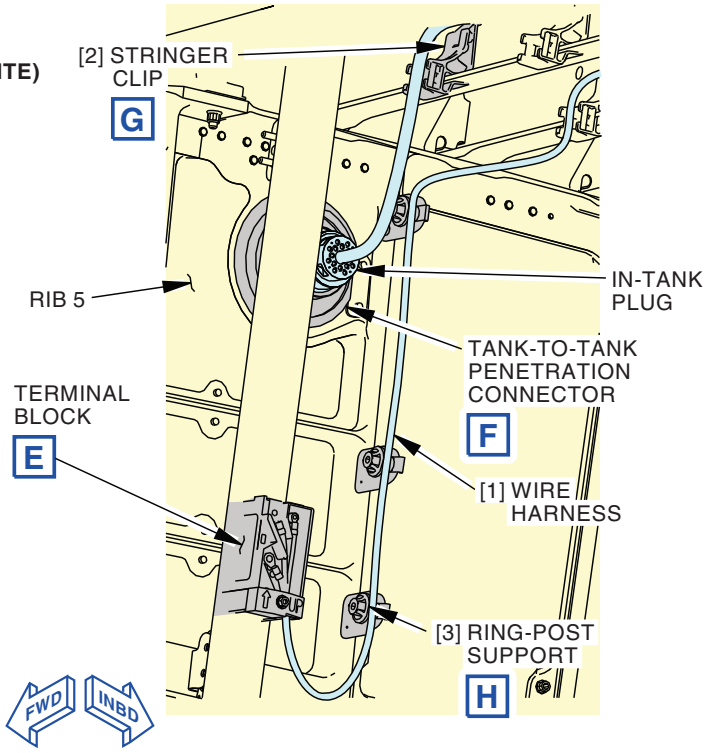
FQIS Wire Harness Installation
Figure 401/28-41-44-990-801 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SPAR FEED THROUGH CONNECTOR
(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)
(EXAMPLE)

C



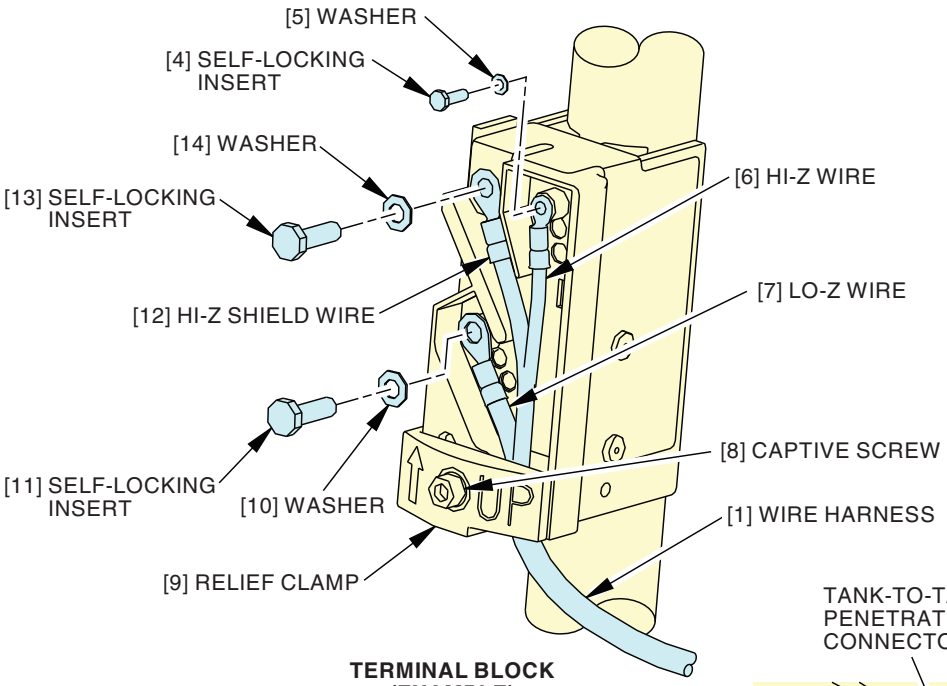
TANK-TO-TANK FEED THROUGH CONNECTOR AT RIB 5
(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)
(EXAMPLE)

D

2413724 S00061533177_V3

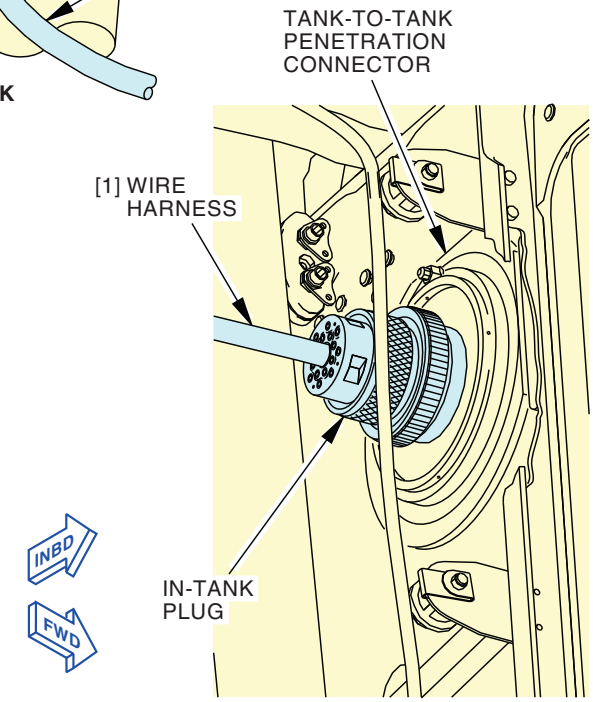
FQIS Wire Harness Installation
Figure 401/28-41-44-990-801 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TERMINAL BLOCK (EXAMPLE)

E



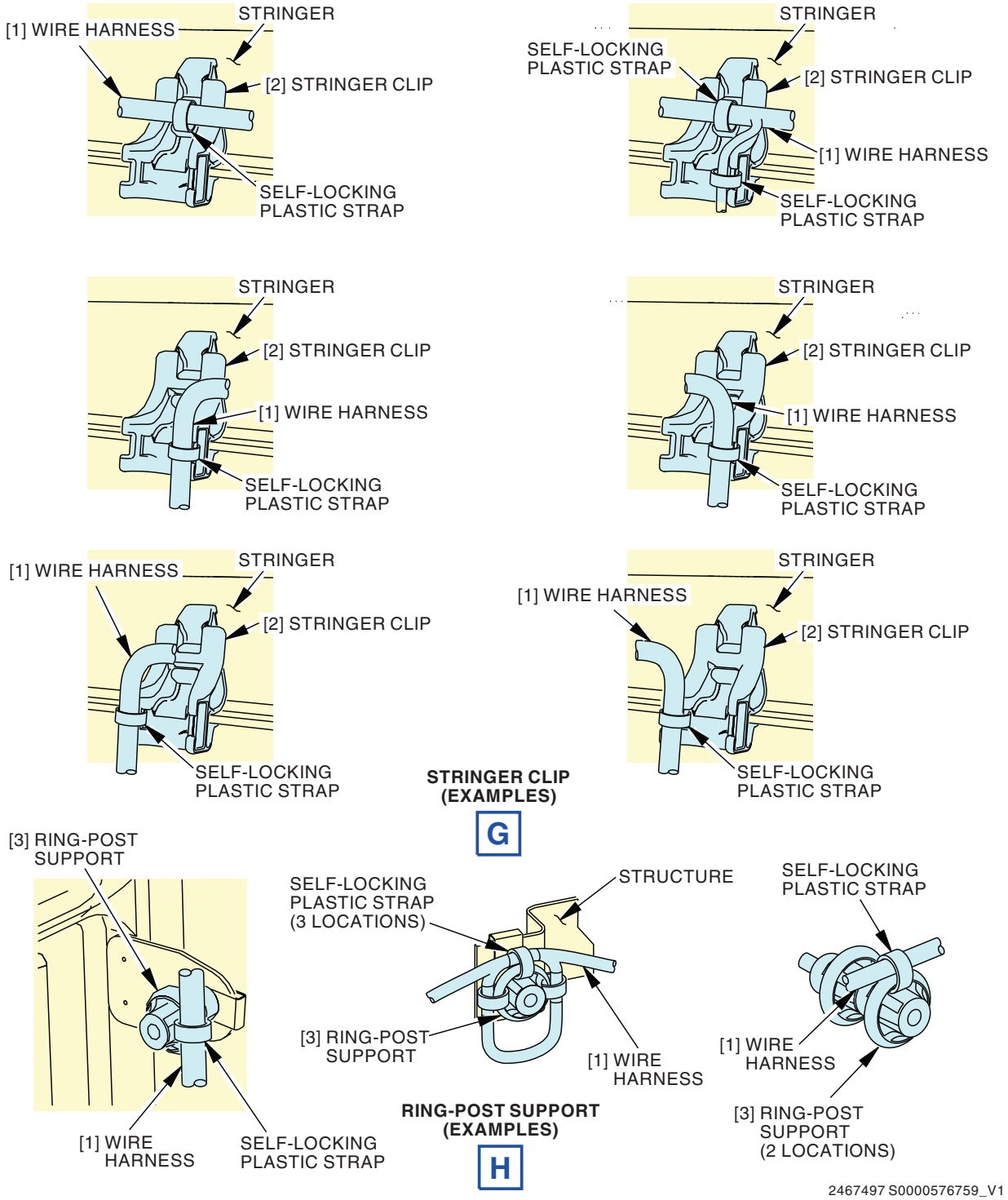
TANK-TO-TANK PENETRATION CONNECTOR AT RIB 5 (RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE) (EXAMPLE)

F

2413725 S00061533178_V3

FQIS Wire Harness Installation
Figure 401/28-41-44-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FQIS Wire Harness Installation
Figure 401/28-41-44-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-41-44-400-801

3. FQIS Wire Harness Installation

(Figure 401)

A. General

(1) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

(b) This task includes the steps to install the FQIS wire harness.

B. References

Reference	Title
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
28-11-11-400-804	Main Tank Inner Access Door Installation (P/B 401)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
28-41-00-730-801	Fuel Quantity Indicating System - Dry Capacitance Test (P/B 501)

C. Consumable Materials

Reference	Description	Specification
G51043	Strap - Plastic, Wire Harness Tie, Self-locking, 8.25 Inches (209.55 mm) Long	BACS38W3
G51074	Strap - Plastic, Wire Harness Tie, Self-locking, 7.25 Inches (184.15 mm) Long	BACS38W1
G51075	Strap - Plastic, Wire Harness Tie, Self-locking, 7.50 Inches (190.50 mm) Long	BACS38W2
G51076	Strap - Plastic, Wire Harness Tie, Self-locking, 10.50 Inches (264.16 mm) Long	BACS38W4

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Wire harness	28-11-51-20-035	PTF ALL
		28-11-51-21-095	PTF ALL
		28-11-51-22-095	PTF ALL
		28-11-51-23-075	PTF ALL
		28-11-51-24-080	PTF ALL

E. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

F. Access Panels

Number	Name/Location
532AZ	Main Tank Inner Access at Rib 6
532BZ	Main Tank Inner Access at Rib 6
632AZ	Main Tank Inner Access at Rib 6
632BZ	Main Tank Inner Access at Rib 6

G. Prepare for the Installation

SUBTASK 28-41-44-210-001

28-AWL-09: CDCCL

(1) Make sure to maintain the existing (or newly approved) wire harness [1] routing, clamping, and nylon support rods.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

SUBTASK 28-41-44-210-002

28-AWL-09: CDCCL

(2) Make sure that the wire harness [1], clamps, and nylon support rods are not damaged.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

SUBTASK 28-41-44-210-003

28-AWL-09: CDCCL

(3) Make sure that the wires are not chafed.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

H. FQIS Wire Harness Installation

SUBTASK 28-41-44-420-001

(1) Connect the in-tank plug to the feed-through connector.

NOTE: See SWPM 20-60-06 for more details.

(a) Tighten the in-tank plug to 95 in-lb (11 N·m) - 105 in-lb (12 N·m).

SUBTASK 28-41-44-420-002

(2) Route the wire harness [1] in the fuel tank through these locations (SWPM 20-10-11):

- Compensator units
- Tank units

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Tank-to-tank feed-through connector.

SUBTASK 28-41-44-420-003

(3) Attach the wire harness [1] to the stringer clips [2].

(a) Hold the wire harness [1] against the stringer clip [2].

(b) Loosely tie the self-locking plastic strap to the stringer clip [2] and wire harness [1].

NOTE: The strap, G51074, strap, G51075, strap, G51043, and strap, G51076, are referred to as self-locking plastic strap in this task.

SUBTASK 28-41-44-420-004

(4) Attach the wire harness [1] to the ring-post supports [3].

(a) Hold the wire harness [1] against the ring-post support [3].

(b) Loosely tie the self-locking plastic strap to the ring-post support [3].

SUBTASK 28-41-44-420-005

(5) Adjust the wire harnesses [1] to make sure that the wires do not touch the structure.

(a) Move the wire harness [1] to make sure that it is equally spaced along the wire harness [1] routing.

(b) Keep the wire harness [1] 0.5 in. (12.7 mm) away from the structure.

28-AWL-09: CDCCL

(c) Make sure the clearance between the wire harness [1] and the structure, accounting for slack in all directions, is a minimum of 0.250 in. (6.4 mm).

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-09.

SUBTASK 28-41-44-420-006

(6) Connect the in-tank plug to the tank-to-tank feed-through connector.

NOTE: See SWPM 20-60-06 for more details.

(a) Tighten the in-tank plug to 95 in-lb (11 N·m) - 105 in-lb (12 N·m).

SUBTASK 28-41-44-420-007

(7) Connect the wire harnesses [1] to the terminal block.

(a) Install the self-locking insert [13], washer [14], and HI-Z shield wire [12] to the terminal block.

1) Tighten the self-locking insert [13] to 17.5 ±2.5 in-lb (1.98 ±0.29 N·m).

2) Make sure that the self-locking insert [13] holds the terminal tightly and HI-Z shield wire [12] does not turn.

(b) Install the self-locking insert [4], washer [5], and HI-Z wire [6] to the terminal block.

1) Tighten the self-locking insert [4] to 13.5 ±1.5 in-lb (1.53 ±0.17 N·m).

2) Make sure that the self-locking insert [4] holds the terminal tightly and HI-Z wire [6] does not turn.

(c) Install the self-locking insert [11], washer [10], and LO-Z wire [7] to the terminal block.

1) Tighten the self-locking insert [11] to 30 ±5 in-lb (3.4 ±0.6 N·m).

2) Make sure that the self-locking insert [11] holds the terminal tightly and LO-Z wire [7] does not turn.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. FQIS Wire Harness Installation Test

SUBTASK 28-41-44-410-001

(1) Install these access panels:

Number	Name/Location
532AZ	Main Tank Inner Access at Rib 6
532BZ	Main Tank Inner Access at Rib 6
632AZ	Main Tank Inner Access at Rib 6
632BZ	Main Tank Inner Access at Rib 6

(a) Do this task: Main Tank Inner Access Door Installation, TASK 28-11-11-400-804.

SUBTASK 28-41-44-410-002

(2) Close the fuel tanks (TASK 28-11-00-410-801).

(a) Install the access doors for the No. 1 or No. 2 main tanks (TASK 28-11-11-400-801).

(b) Install the access doors for the center tank (TASK 28-11-31-400-801).

SUBTASK 28-41-44-860-003

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

SUBTASK 28-41-44-860-004

(4) Do this task: Fuel Quantity Indicating System - Dry Capacitance Test, TASK 28-41-00-730-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL TANK BULKHEAD (SPAR) RECEPTACLE WIRE HARNESS - INSPECTION/CHECK

1. General

A. This procedure contains tasks to visually inspect the Fuel Quantity Indicating System (FQIS) wiring and components for each fuel tank.

B. This procedure has these tasks:

(1) FQIS Wiring and Component - Visual Inspection

(2) FQIS, No. 1 Main Tank - Inspection

(3) FQIS, No. 2 Main Tank - Inspection

(4) FQIS, Center Tank - Inspection

TASK 28-41-44-210-801

2. FQIS Wiring and Component Visual Inspection

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task has one or more steps which are a means to satisfy Airworthiness Limitation Instruction (ALI) requirements. An ALI note will follow the step to which it applies. Any step or sub-step that precedes or follows an ALI identified step is not subject to the ALI requirement.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-33.

B. References

Reference	Title
28-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)

C. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Wiring Examination

SUBTASK 28-41-44-210-004

28-AWL-33: ALI

(1) Examine the wiring for the following conditions:

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-33.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

28-AWL-33: ALI

(a) Wire harness, clamps, and nylon support rod damage.

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-33.

(b) Insulation is not abraded, cracked or over-stressed.

28-AWL-33: ALI

(c) Clearance between wires and structure, accounting for slack in all directions, is a minimum of 0.250 in. (6.4 mm).

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-33.

28-AWL-33: ALI

(d) Wires that are chafed.

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 28-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-33.

(e) Wires are not routed correctly.

(f) Conductors or shields that are broken or exposed.

E. Bussing Plug and Connector Examination

SUBTASK 28-41-44-210-043

(1) Examine the tank unit and compensator bussing plugs on the front spar between side-of-body rib and rib 2 for the following conditions:

(a) Tank unit and compensator bussing plugs that are lose or damaged.

(2) Examine the FQIS connectors at the inboard and outboard sides of rib 5 for the following conditions:

(a) Tank-to-tank FQIS connectors that are lose or damaged.

(3) Examine the tank unit and compensator bussing plug in the center tank located at the rear spar for the following conditions:

Tank unit and compensator bussing plug that is lose or damaged.

F. Compensator and Tank Unit Examination

SUBTASK 28-41-44-210-005

(1) Examine the compensators and tank units for damage and condition as follows:

(a) Wires attached to the terminals incorrectly.

(b) Wiring to the terminals that are damaged or incorrectly routed.

(c) An end cap is missing.

(d) Clearance from the structure that is not sufficient.

(e) Mounting brackets and hardware that are loose.

(f) Terminals that are bent.

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(g) Electrical connectors and seals that are damaged, worn, or show signs of fuel leakage.

——— END OF TASK ———

TASK 28-41-44-280-801

3. FQIS, No. 1 Main Tank - Inspection

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-803	Main Tank Inner Access Door Removal (P/B 401)
28-11-11-400-804	Main Tank Inner Access Door Installation (P/B 401)

B. Location Zones

Zone	Area
434	Engine 1 - Aft Strut Fairing
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50

C. Access Panels

Number	Name/Location
434CL	Aft Strut Fairing, Left Access To Fuel Door, Strut 1
532AB	Main Tank Access Door - Wing Station 216
532AZ	Main Tank Inner Access at Rib 6
532BB	Main Tank Access Door - Wing Station 265
532BZ	Main Tank Inner Access at Rib 6
532CB	Main Tank Access Door - Wing Station 290
532DB	Main Tank Access Door - Wing Station 313
532EB	Main Tank Access Door - Wing Station 337
532FB	Main Tank Access Door - Wing Station 367
532GB	Main Tank Access Door - Wing Station 390
532HB	Main Tank Access Door - Wing Station 417
532JB	Main Tank Access Door - Wing Station 443
532KB	Main Tank Access Door - Wing Station 470
532LB	Main Tank Access Door - Wing Station 496
532MB	Main Tank Access Door - Wing Station 523
532NB	Main Tank Access Door - Wing Station 549
532PB	Main Tank Access Door - Wing Station 576
532QB	Main Tank Access Door - Wing Station 602
532RB	Main Tank Access Door - Wing Station 629

D. No. 1 Main Tank Between Rib No. 5 and No. 7 - Inspection

NOTE: Rib No. 5, also known as the main tank wall.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-44-860-005

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216

SUBTASK 28-41-44-480-001

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-010-002

(3) To access areas between rib No. 6 and No. 7, remove only one of these access panels on rib No. 6 (unless the engine is removed):

Number	Name/Location
532AZ	Main Tank Inner Access at Rib 6
532BZ	Main Tank Inner Access at Rib 6

(a) Do this task: Main Tank Inner Access Door Removal, TASK 28-11-11-000-803.

SUBTASK 28-41-44-210-006

(4) For all FQIS wiring and components between Rib No. 5 and No. 7, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

E. No. 1 Main Tank Between Rib No. 7 and No. 8 - Inspection

SUBTASK 28-41-44-010-003

(1) Remove this access panel:

Number	Name/Location
434CL	Aft Strut Fairing, Left Access To Fuel Door, Strut 1

SUBTASK 28-41-44-860-006

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(2) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532BB	Main Tank Access Door - Wing Station 265

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-44-480-002

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-007

(4) For all FQIS wiring and components between Rib No. 7 and No. 8, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

F. No. 1 Main Tank Between Rib No. 8 and No. 9 - Inspection

SUBTASK 28-41-44-860-007

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532CB	Main Tank Access Door - Wing Station 290

SUBTASK 28-41-44-480-003

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-008

(3) For all FQIS wiring and components between Rib No. 8 and No. 9, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

G. No. 1 Main Tank Between Rib No. 9 and No. 10 - Inspection

SUBTASK 28-41-44-860-008

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532DB	Main Tank Access Door - Wing Station 313

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-44-480-004

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-009

(3) For all FQIS wiring and components between Rib No. 9 and No. 10, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

H. No. 1 Main Tank Between Rib No. 10 and No. 11 - Inspection

SUBTASK 28-41-44-860-009

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532EB	Main Tank Access Door - Wing Station 337

SUBTASK 28-41-44-480-005

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-010

(3) For all FQIS wiring and components between Rib No. 10 an Rib No. 11, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

I. No. 1 Main Tank Between Rib No. 11 and No. 12 - Inspection

SUBTASK 28-41-44-860-010

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532FB	Main Tank Access Door - Wing Station 367

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-44-480-006

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-011

(3) For all FQIS wiring and components between Rib No. 11 and No. 12, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

J. No. 1 Main Tank Between Rib No. 12 and No. 13 - Inspection

SUBTASK 28-41-44-860-011

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532GB	Main Tank Access Door - Wing Station 390

SUBTASK 28-41-44-480-007

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-012

(3) For all FQIS wiring and components between Rib No. 12 and No. 13, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

K. No. 1 Main Tank Between Rib No. 13 and No. 14 - Inspection

SUBTASK 28-41-44-860-012

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532HB	Main Tank Access Door - Wing Station 417

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-44-480-008

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-013

(3) For all FQIS wiring and components between Rib No. 13 and No. 14, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

L. No. 1 Main Tank Between Rib No. 14 and No. 15 - Inspection

SUBTASK 28-41-44-860-013

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532JB	Main Tank Access Door - Wing Station 443

SUBTASK 28-41-44-480-009

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-014

(3) For all FQIS wiring and components between Rib No. 14 and No. 15, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

M. No. 1 Main Tank Between Rib No. 15 and No. 16 - Inspection

SUBTASK 28-41-44-860-014

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532KB	Main Tank Access Door - Wing Station 470

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-44-480-010

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-015

(3) For all FQIS wiring and components between Rib No. 15 and No. 16, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

N. No. 1 Main Tank Between Rib No. 16 and No. 17 - Inspection

SUBTASK 28-41-44-860-015

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532LB	Main Tank Access Door - Wing Station 496

SUBTASK 28-41-44-480-011

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-016

(3) For all FQIS wiring and components between Rib No. 16 and No. 17, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

O. No. 1 Main Tank Between Rib No. 17 and No. 18 - Inspection

SUBTASK 28-41-44-860-016

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532MB	Main Tank Access Door - Wing Station 523

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-44-480-012

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-017

(3) For all FQIS wiring and components between Rib No. 17 and No. 18, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

P. No. 1 Main Tank Between Rib No. 18 and No. 19 - Inspection

SUBTASK 28-41-44-860-017

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532NB	Main Tank Access Door - Wing Station 549

SUBTASK 28-41-44-480-013

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-018

(3) For all FQIS wiring and components between Rib No. 18 and No. 19, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

Q. No. 1 Main Tank Between Rib No. 19 and No. 20 - Inspection

SUBTASK 28-41-44-860-018

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532PB	Main Tank Access Door - Wing Station 576

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-44-480-014

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-019

(3) For all FQIS wiring and components between Rib No. 19 and No. 20, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

R. No. 1 Main Tank Between Rib No. 20 and No. 21 - Inspection

SUBTASK 28-41-44-860-019

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532QB	Main Tank Access Door - Wing Station 602

SUBTASK 28-41-44-480-015

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-020

(3) For all FQIS wiring and components between Rib No. 20 and No. 21, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

S. No. 1 Main Tank Between Rib No. 21 and No. 22 - Inspection

SUBTASK 28-41-44-860-020

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
532RB	Main Tank Access Door - Wing Station 629

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-44-480-016

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-021

(3) For all FQIS wiring and components between Rib No. 21 and No. 22, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801

T. Put the Airplane Back to Its Usual Condition

SUBTASK 28-41-44-410-003

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) If necessary, close these access panels:

Number	Name/Location
532AZ	Main Tank Inner Access at Rib 6
532BZ	Main Tank Inner Access at Rib 6

1) Do this task: Main Tank Inner Access Door Installation, TASK 28-11-11-400-804.

(b) Close the applicable access panels if needed:

Number	Name/Location
434CL	Aft Strut Fairing, Left Access To Fuel Door, Strut 1
532AB	Main Tank Access Door - Wing Station 216
532BB	Main Tank Access Door - Wing Station 265
532CB	Main Tank Access Door - Wing Station 290
532DB	Main Tank Access Door - Wing Station 313
532EB	Main Tank Access Door - Wing Station 337
532FB	Main Tank Access Door - Wing Station 367
532GB	Main Tank Access Door - Wing Station 390
532HB	Main Tank Access Door - Wing Station 417
532JB	Main Tank Access Door - Wing Station 443
532KB	Main Tank Access Door - Wing Station 470
532LB	Main Tank Access Door - Wing Station 496
532MB	Main Tank Access Door - Wing Station 523
532NB	Main Tank Access Door - Wing Station 549
532PB	Main Tank Access Door - Wing Station 576
532QB	Main Tank Access Door - Wing Station 602
532RB	Main Tank Access Door - Wing Station 629

END OF TASK

TASK 28-41-44-280-802

4. FQIS, No. 2 Main Tank - Inspection

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
28-11-00-410-801	Fuel Tank Closure (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-803	Main Tank Inner Access Door Removal (P/B 401)
28-11-11-400-804	Main Tank Inner Access Door Installation (P/B 401)

B. Location Zones

Zone	Area
444	Engine 2 - Aft Strut Fairing
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

C. Access Panels

Number	Name/Location
444CR	Aft Strut Fairing, Right Access To Fuel Door, Strut 2
632AB	Main Tank Access Door - Wing Station 216
632AZ	Main Tank Inner Access at Rib 6
632BB	Main Tank Access Door - Wing Station 265
632BZ	Main Tank Inner Access at Rib 6
632CB	Main Tank Access Door - Wing Station 290
632DB	Main Tank Access Door - Wing Station 313
632EB	Main Tank Access Door - Wing Station 337
632FB	Main Tank Access Door - Wing Station 367
632GB	Main Tank Access Door - Wing Station 390
632HB	Main Tank Access Door - Wing Station 417
632JB	Main Tank Access Door - Wing Station 443
632KB	Main Tank Access Door - Wing Station 470
632LB	Main Tank Access Door - Wing Station 496
632MB	Main Tank Access Door - Wing Station 523
632NB	Main Tank Access Door - Wing Station 549
632PB	Main Tank Access Door - Wing Station 576
632QB	Main Tank Access Door - Wing Station 602
632RB	Main Tank Access Door - Wing Station 629

D. No. 2 Main Tank Between Rib No. 5 and No. 7 - Inspection

NOTE: Rib No. 5, also known as the main tank wall.

SUBTASK 28-41-44-860-021

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632AB	Main Tank Access Door - Wing Station 216

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-41-44-480-017

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-010-004

(3) To access areas between rib No. 6 and No. 7, remove only one of these access panels on rib No. 6 (unless the engine is removed):

Number	Name/Location
632AZ	Main Tank Inner Access at Rib 6
632BZ	Main Tank Inner Access at Rib 6

(a) Do this task: Main Tank Inner Access Door Removal, TASK 28-11-11-000-803.

SUBTASK 28-41-44-210-022

(4) For all FQIS wiring and components between Rib No. 5 and No. 7, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

E. No. 2 Main Tank Between Rib No. 7 and No. 8 - Inspection

SUBTASK 28-41-44-010-005

(1) Open this access panel:

Number	Name/Location
444CR	Aft Strut Fairing, Right Access To Fuel Door, Strut 2

SUBTASK 28-41-44-860-022

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(2) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632BB	Main Tank Access Door - Wing Station 265

SUBTASK 28-41-44-480-018

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-023

(4) For all FQIS wiring and components between Rib No. 7 and No. 8, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. No. 2 Main Tank Between Rib No. 8 and No. 9 - Inspection

SUBTASK 28-41-44-860-023

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632CB	Main Tank Access Door - Wing Station 290

SUBTASK 28-41-44-480-019

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-024

(3) For all FQIS wiring and components between Rib No. 8 and No. 9, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

G. No. 2 Main Tank Between Rib No. 9 and No. 10 - Inspection

SUBTASK 28-41-44-860-024

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632DB	Main Tank Access Door - Wing Station 313

SUBTASK 28-41-44-480-020

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-025

(3) For all FQIS wiring and components between Rib No. 9 and No. 10, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. No. 2 Main Tank Between Rib No. 10 and No. 11 - Inspection

SUBTASK 28-41-44-860-025

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632EB	Main Tank Access Door - Wing Station 337

SUBTASK 28-41-44-480-021

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-026

(3) For all FQIS wiring and components between Rib No. 10 and No. 11, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

I. No. 2 Main Tank Between Rib No. 11 and No. 12 - Inspection

SUBTASK 28-41-44-860-026

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632FB	Main Tank Access Door - Wing Station 367

SUBTASK 28-41-44-480-022

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-027

(3) For all FQIS wiring and components between Rib No. 11 and No. 12, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

J. No. 2 Main Tank Between Rib No. 12 and No. 13 - Inspection

SUBTASK 28-41-44-860-027

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632GB	Main Tank Access Door - Wing Station 390

SUBTASK 28-41-44-480-023

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-028

(3) For all FQIS wiring and components between Rib No. 12 and No. 13, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

K. No. 2 Main Tank Between Rib No. 13 and No. 14 - Inspection

SUBTASK 28-41-44-860-028

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632HB	Main Tank Access Door - Wing Station 417

SUBTASK 28-41-44-480-024

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-029

(3) For all FQIS wiring and components between Rib No. 13 and No. 14, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

L. No. 2 Main Tank Between Rib No. 14 and No. 15 - Inspection

SUBTASK 28-41-44-860-029

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632JB	Main Tank Access Door - Wing Station 443

SUBTASK 28-41-44-480-025

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-030

(3) For all FQIS wiring and components between Rib No. 14 and No. 15, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

M. No. 2 Main Tank Between Rib No. 15 and No. 16 - Inspection

SUBTASK 28-41-44-860-030

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632KB	Main Tank Access Door - Wing Station 470

SUBTASK 28-41-44-480-026

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-031

(3) For all FQIS wiring and components between Rib No. 15 and No. 16, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

N. No. 2 Main Tank Between Rib No. 16 and No. 17 - Inspection

SUBTASK 28-41-44-860-031

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632LB	Main Tank Access Door - Wing Station 496

SUBTASK 28-41-44-480-027

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-032

(3) For all FQIS wiring and components between Rib No. 16 and No. 17, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

O. No. 2 Main Tank Between Rib No. 17 and No. 18 - Inspection

SUBTASK 28-41-44-860-032

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632MB	Main Tank Access Door - Wing Station 523

SUBTASK 28-41-44-480-028

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-033

(3) For all FQIS wiring and components between Rib No. 17 and No. 18, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

P. No. 2 Main Tank Between Rib No. 18 and No. 19 - Inspection

SUBTASK 28-41-44-860-033

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632NB	Main Tank Access Door - Wing Station 549

SUBTASK 28-41-44-480-029

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-034

(3) For all FQIS wiring and components between Rib No. 18 and No. 19, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

Q. No. 2 Main Tank Between Rib No. 19 and No. 20 - Inspection

SUBTASK 28-41-44-860-034

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632PB	Main Tank Access Door - Wing Station 576

SUBTASK 28-41-44-480-030

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-035

(3) For all FQIS wiring and components between Rib No. 19 and No. 20, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

R. No. 2 Main Tank Between Rib No. 20 and No. 21 - Inspection

SUBTASK 28-41-44-860-035

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632QB	Main Tank Access Door - Wing Station 602

SUBTASK 28-41-44-480-031

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-036

(3) For all FQIS wiring and components between Rib No. 20 and No. 21, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

S. No. 2 Main Tank Between Rib No. 21 and No. 22 - Inspection

SUBTASK 28-41-44-860-036

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the fuel tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
632RB	Main Tank Access Door - Wing Station 629

SUBTASK 28-41-44-480-032

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-037

(3) For all FQIS wiring and components between Rib No. 21 and No. 22, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

T. Put the Airplane Back to Its Usual Condition

SUBTASK 28-41-44-410-004

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If necessary, close these access panels:

Number	Name/Location
632AZ	Main Tank Inner Access at Rib 6
632BZ	Main Tank Inner Access at Rib 6

1) Do this task: Main Tank Inner Access Door Installation, TASK 28-11-11-400-804.

(b) Close the applicable access panels if needed:

Number	Name/Location
444CR	Aft Strut Fairing, Right Access To Fuel Door, Strut 2
632AB	Main Tank Access Door - Wing Station 216
632BB	Main Tank Access Door - Wing Station 265
632CB	Main Tank Access Door - Wing Station 290
632DB	Main Tank Access Door - Wing Station 313
632EB	Main Tank Access Door - Wing Station 337
632FB	Main Tank Access Door - Wing Station 367
632GB	Main Tank Access Door - Wing Station 390
632HB	Main Tank Access Door - Wing Station 417
632JB	Main Tank Access Door - Wing Station 443
632KB	Main Tank Access Door - Wing Station 470
632LB	Main Tank Access Door - Wing Station 496
632MB	Main Tank Access Door - Wing Station 523
632NB	Main Tank Access Door - Wing Station 549
632PB	Main Tank Access Door - Wing Station 576
632QB	Main Tank Access Door - Wing Station 602
632RB	Main Tank Access Door - Wing Station 629

——— END OF TASK ———

TASK 28-41-44-280-803

5. FQIS, Center Tank - Inspection

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
28-11-00-410-801	Fuel Tank Closure (P/B 201)
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)

B. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5

C. Access Panels

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
531BB	Center Tank Access Door - Wing Station 192
631AB	Center Tank Access Door - Wing Station 168

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
631BB	Center Tank Access Door - Wing Station 192

D. Left Wing Center Tank Between Rib No. 4 and No. 5 - Inspection

SUBTASK 28-41-44-860-037

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the center tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
531BB	Center Tank Access Door - Wing Station 192

SUBTASK 28-41-44-480-033

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-038

(3) For all FQIS wiring and components between Rib No. 4 and No. 5, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

E. Left Wing Center Tank Between Rib No. 1 and No. 4 - Inspection

NOTE: Rib No. 1, also known as side-of-body.

SUBTASK 28-41-44-860-038

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the center tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
531AB	Center Tank Access Door - Wing Station 168

SUBTASK 28-41-44-480-034

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-039

(3) For all FQIS wiring and components between Rib No. 1 and No. 4, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Right Wing Center Tank Between Rib No. 4 and No. 5 - Inspection

SUBTASK 28-41-44-860-039

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the center tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
631BB	Center Tank Access Door - Wing Station 192

SUBTASK 28-41-44-480-035

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-040

(3) For all FQIS wiring and components between Rib No. 4 and No. 5, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

G. Right Wing Center Tank Between Rib No. 1 and No. 4 - Inspection

NOTE: Rib No. 1, also known as side-of-body.

SUBTASK 28-41-44-860-040

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the center tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
631AB	Center Tank Access Door - Wing Station 168

SUBTASK 28-41-44-480-036

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-041

(3) For all FQIS wiring and components between Rib No. 1 and No. 4, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Fuselage Center Tank - Inspection

SUBTASK 28-41-44-860-041

WARNING OBEY THE PURGING AND FUEL TANK ENTRY PRECAUTIONS. IF YOU IGNORE THE PRECAUTIONS, THE CONDITIONS CAN KILL PERSONNEL, CAUSE INJURIES, AND CAUSE DAMAGE TO EQUIPMENT.

(1) Do this task for the center tank: Purging and Fuel Tank Entry, TASK 28-11-00-910-802.

(a) Remove this access panel:

Number	Name/Location
131AB	Center Tank Access

SUBTASK 28-41-44-480-037

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Go into the fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-41-44-210-042

(3) For all FQIS wiring and components in the fuselage center tank, do this task: FQIS Wiring and Component Visual Inspection, TASK 28-41-44-210-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 28-41-44-410-005

(1) Do this task: Fuel Tank Closure, TASK 28-11-00-410-801.

(a) Close the applicable access panels if needed:

Number	Name/Location
131AB	Center Tank Access
531AB	Center Tank Access Door - Wing Station 168
531BB	Center Tank Access Door - Wing Station 192
631AB	Center Tank Access Door - Wing Station 168
631BB	Center Tank Access Door - Wing Station 192

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

REFUEL QUANTITY INDICATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

PTF ALL; AIRPLANES WITHOUT ALOFT AUXILIARY TANK (STC TM-9000-AMMS AND TM-9001-AMMS)

(1) Refuel Quantity Indicator Removal

(2) Refuel Quantity Indicator Installation

PTF ALL

B. When you remove a refuel quantity indicator, it is recommended that you inspect the wire bundles for wire chafing behind the P15 Refuel Panel. To check the wire bundles, you need to remove the P15 Refuel Panel.

PTF ALL; AIRPLANES WITHOUT ALOFT AUXILIARY TANK (STC TM-9000-AMMS AND TM-9001-AMMS)

TASK 28-41-61-000-801

2. Refuel Quantity Indicator Removal

(Figure 401)

A. General

(1) This task gives you instructions to remove the refuel quantity indicator.

B. References

Reference	Title
20-40-12 P/B 201	ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) DEVICES - MAINTENANCE PRACTICES
28-21-82-000-801	Refuel Panel, P15 Removal (P/B 401)

C. Location Zones

Zone	Area
621	Right Wing - Leading Edge to Front Spar

D. Access Panels

Number	Name/Location
621GB	Refuel Valve Door

E. Prepare for the Removal

SUBTASK 28-41-61-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	3	C00032	FUEL FUELING CONT
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

SUBTASK 28-41-61-010-001

(2) Open this access panel:

Number	Name/Location
621GB	Refuel Valve Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITHOUT ALOFT AUXILIARY TANK (STC TM-9000-AMMS AND TM-9001-AMMS) (Continued)

SUBTASK 28-41-61-010-002

(3) Remove the refuel panel [3] (TASK 28-21-82-000-801).

NOTE: The removal of the refuel panel is recommended, but not necessary. With the refuel panel [3] removed, you can do a visual check of the wire bundle behind the refuel panel [3].

F. Refuel Quantity Indicator Removal

SUBTASK 28-41-61-760-001

CAUTION

DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(1) Before you touch the refuel quantity indicator [4], do the procedure for devices that are sensitive to electrostatic discharge (PAGEBLOCK 20-40-12/201).

SUBTASK 28-41-61-020-003

(2) Loosen the adjustment screw [2] of the applicable refuel quantity indicator [4].

SUBTASK 28-41-61-480-001

(3) Put a tag or mark on the applicable electrical connector [1] and the refuel quantity indicator [4].

SUBTASK 28-41-61-020-002

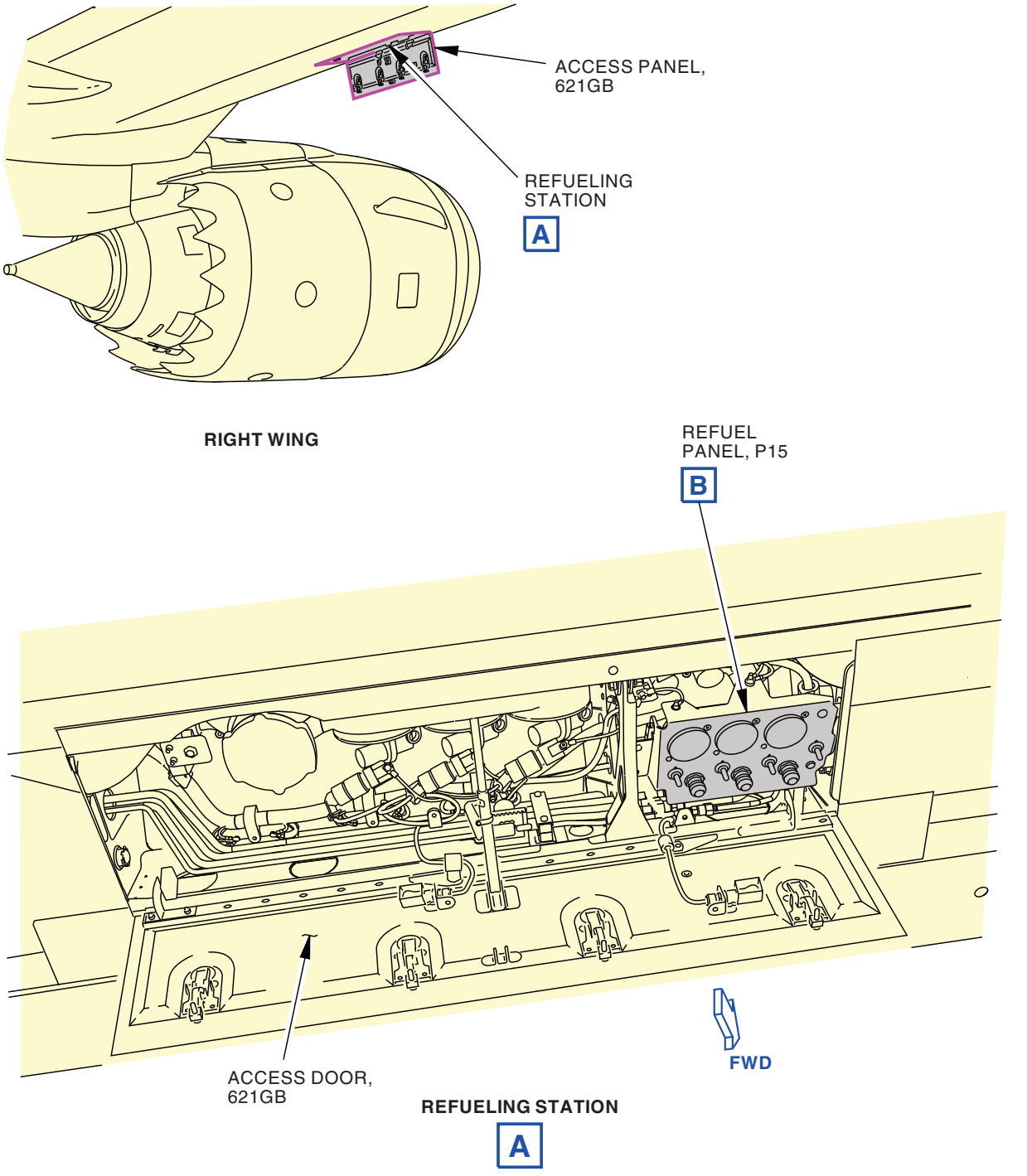
(4) Disconnect the applicable electrical connector [1] from the rear of the refuel quantity indicator [4].

SUBTASK 28-41-61-020-001

(5) Remove the refuel quantity indicator [4] through the forward side of the refuel panel [3].

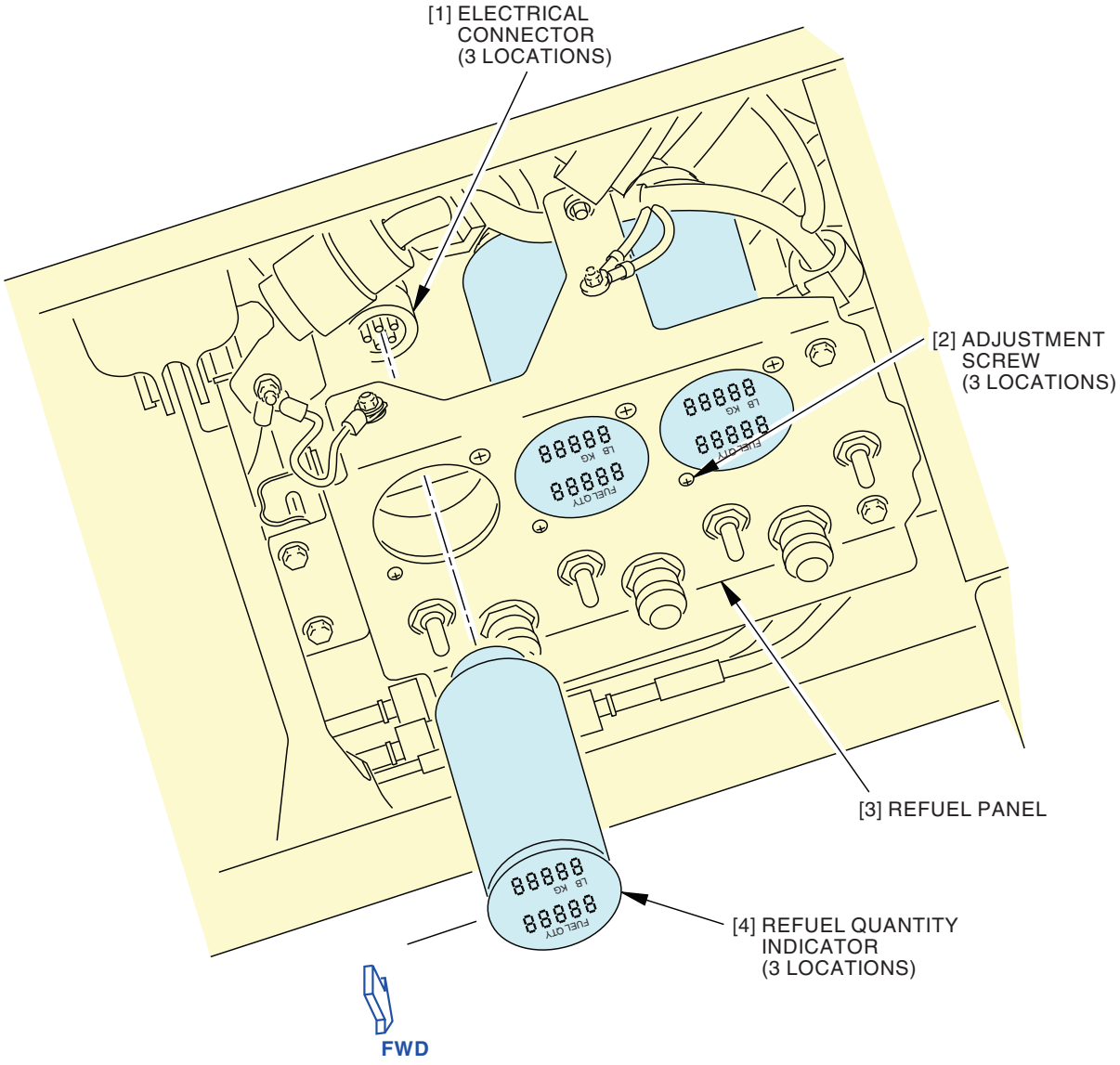
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Refuel Quantity Indicators Installation
Figure 401/28-41-61-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



REFUEL PANEL, P15

B

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Refuel Quantity Indicators Installation
Figure 401/28-41-61-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-41-61-400-801

3. Refuel Quantity Indicator Installation

(Figure 401)

A. References

Reference	Title
20-40-12 P/B 201	ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) DEVICES - MAINTENANCE PRACTICES
28-21-82-400-801	Refuel Panel, P15 Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Refuel quantity indicator	28-41-61-01-005	PTF ALL

C. Location Zones

Zone	Area
621	Right Wing - Leading Edge to Front Spar

D. Access Panels

Number	Name/Location
621GB	Refuel Valve Door

E. Refuel Quantity Indicator Installation

SUBTASK 28-41-61-760-002

CAUTION

DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(1) Before you touch the refuel quantity indicator [4], do the procedure for devices that are sensitive to electrostatic discharge (PAGEBLOCK 20-40-12/201).

SUBTASK 28-41-61-420-001

(2) Install the refuel quantity indicator [4] into the refuel panel [3].

SUBTASK 28-41-61-420-003

(3) Connect the electrical connector [1] to the refuel quantity indicator [4].

NOTE: Make sure that the temporary tags or marks match before you connect the electrical connector [1] to the refuel quantity indicator [4].

SUBTASK 28-41-61-080-001

(4) Remove the temporary tags or marks from the electrical connector [1] and refuel quantity indicator [4].

SUBTASK 28-41-61-820-001

(5) Align the face of the refuel quantity indicator [4] with the refuel panel [3].

SUBTASK 28-41-61-420-002

(6) Tighten the adjustment screw [2] to secure the refuel quantity indicator [4] to the refuel panel [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITHOUT ALOFT AUXILIARY TANK (STC TM-9000-AMMS AND TM-9001-AMMS) (Continued)

SUBTASK 28-41-61-410-002

(7) Install the refuel panel [3] (TASK 28-21-82-400-801).

F. Refuel Quantity Indicator Test

SUBTASK 28-41-61-860-002

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	3	C00032	FUEL FUELING CONT
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

SUBTASK 28-41-61-710-001

(2) On the P15 Refuel Panel, hold the fueling indication test switch to the TEST GAUGES position.

(a) Make sure that each refuel quantity indicator [4] operates correctly.

NOTE: The test blanks the displays on the three refuel quantity indicators [4] for two seconds, then all LED segments go on for two seconds. This sequence continues as long as you hold the fueling indication test switch to the TEST GAUGES position.

If you hold the fueling indication test switch for more than 20 seconds, the test mode times out and the refuel quantity indicators [4] goes back to its usual operating mode. If an internal fault is found during the test, the indicator will show Ind FAIL.

SUBTASK 28-41-61-860-003

(3) Release the fueling indication test switch.

SUBTASK 28-41-61-410-001

(4) Close this access panel:

Number	Name/Location
621GB	Refuel Valve Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL QUANTITY PROCESSOR UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the Fuel Quantity Processor Unit (FQPU)

(2) An installation of the FQPU.

B. The FQPU is installed in the forward cargo bay.

TASK 28-41-81-000-801

2. Fuel Quantity Processor Unit Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the fuel quantity processor unit.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-803	Conductive Dust Cap and Connector Cover Installation (P/B 201)

C. Location Zones

Zone	Area
122	Forward Cargo Compartment - Right

D. Access Panels

Number	Name/Location
821	Forward Cargo Door

E. Prepare for the Removal

SUBTASK 28-41-81-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

SUBTASK 28-41-81-010-001

(2) Open this access panel:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 28-41-81-020-001

(3) Remove the washer head screws [2] from the shroud [1].

SUBTASK 28-41-81-010-002

(4) Remove the shroud [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Fuel Quantity Processor Unit Removal

SUBTASK 28-41-81-910-001

CAUTION

BE CAREFUL WHEN YOU REMOVE OR INSTALL THE FQIS PROCESSOR. DAMAGE TO THE FQIS CONNECTOR PINS CAN EASILY OCCUR.

CAUTION

DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(1) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 28-41-81-020-002

(2) Disconnect the electrical connectors from the fuel quantity processor unit [5].

SUBTASK 28-41-81-420-004

(3) Install protective caps on the electrical connectors and fuel quantity processor unit [5] (TASK 20-40-12-400-803).

SUBTASK 28-41-81-020-003

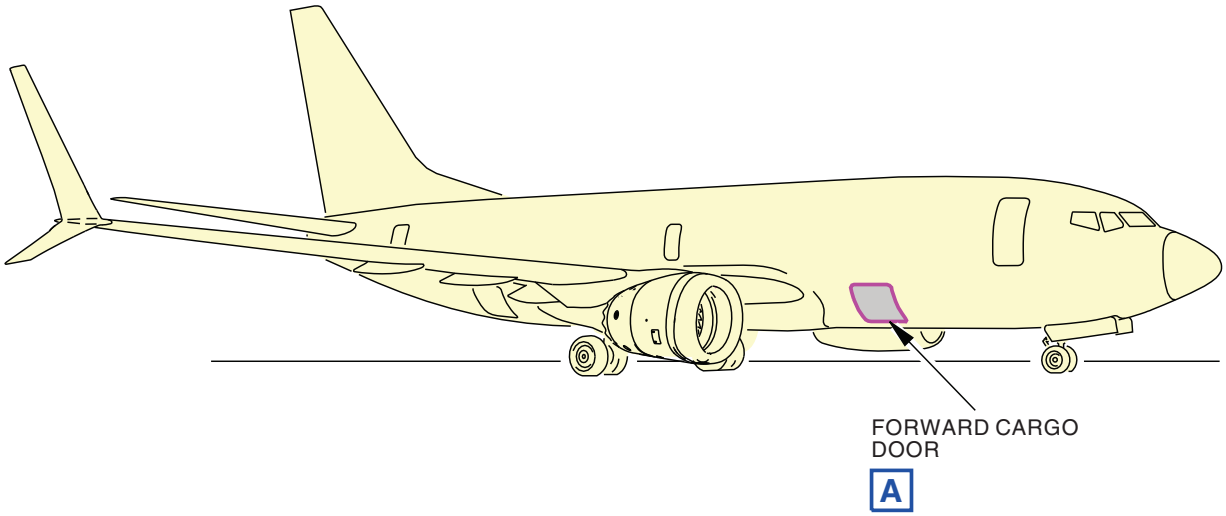
(4) Remove the fuel quantity processor unit [5], do these steps:

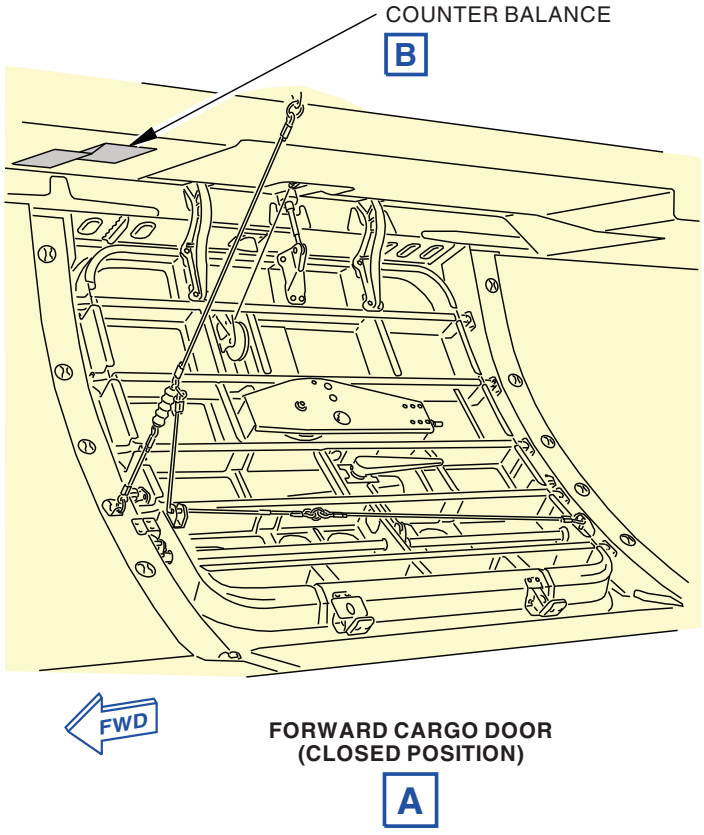
(a) Hold the fuel quantity processor unit [5], remove the bolts [3] and washers [4] from the FQPU mounting bracket legs [7].

(b) Remove the fuel quantity processor unit [5].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

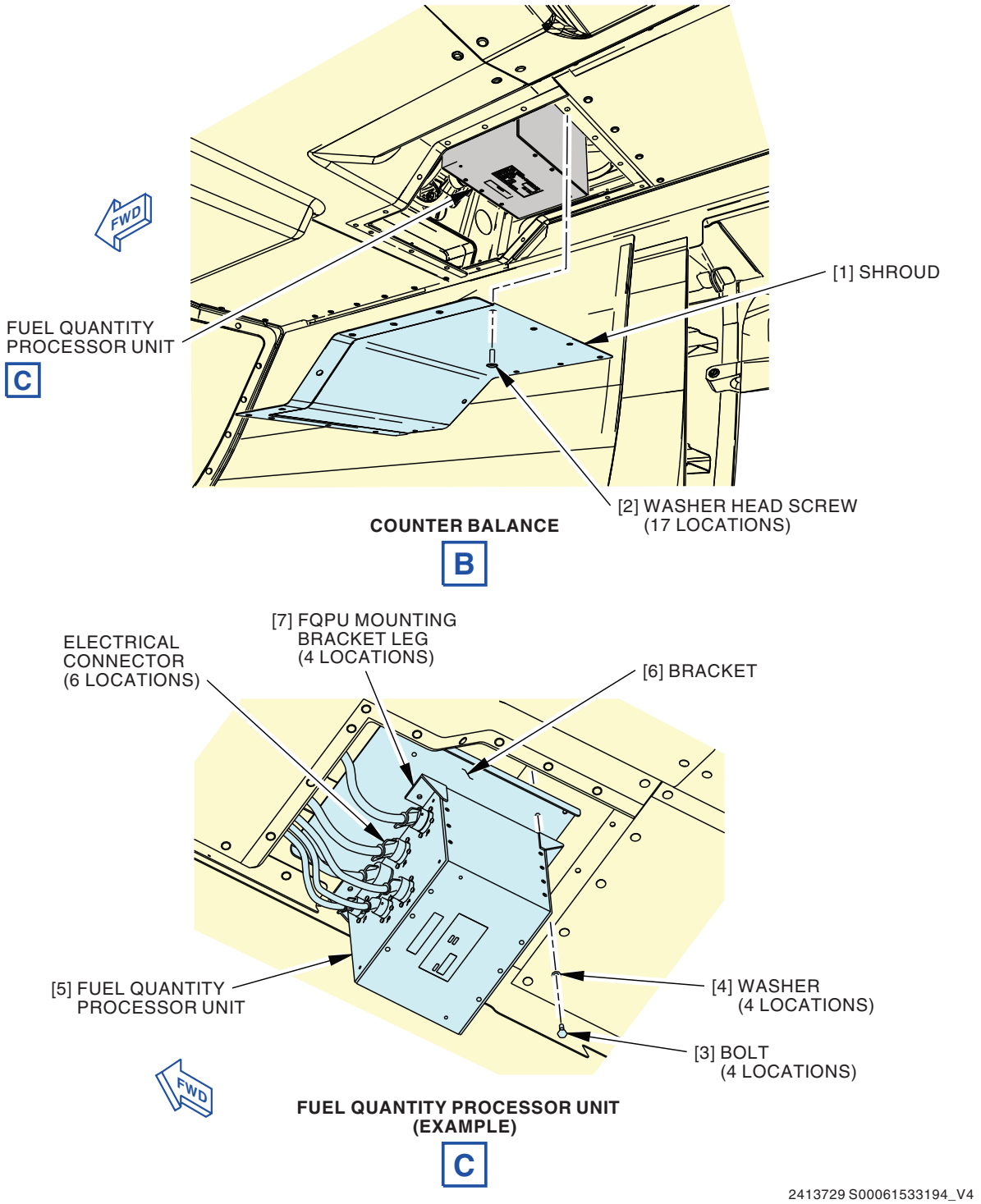




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Fuel Quantity Processor Unit Installation
Figure 401/28-41-81-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Quantity Processor Unit Installation
Figure 401/28-41-81-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-41-81-400-801

3. Fuel Quantity Processor Unit Installation

(Figure 401)

A. General

(1) This task gives instructions to install the fuel quantity processor unit.

B. References

Reference	Title
20-40-12-000-803	Conductive Dust Cap and Connector Cover Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
20-40-12-400-803	Conductive Dust Cap and Connector Cover Installation (P/B 201)
28-41-00-710-801	Fuel Quantity Indicating System - Operational Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
G00291	Tape - Aluminum Foil, 3M 425	AMS-T-23397 / L-T-80
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50398	Pad - Adbrasive, Scotch-Brite Type S, Abrasive Pad	
G50492	Pad - 3M Scotch Brite 7448 Ultra Fine Pad	MIL-A-9962A Type III Grade AAA
G51293	Cotton Wiper - Knit, Process Cleaning Absorbent Wiper	BMS15-5 Class B
G51677	Cloth - 100% Synthetic or Blended Synthetic, Cotton or Cellulose Material	AMS3819 Class 1, 2, or 4, Grade A or B, Form 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Fuel quantity processor unit	28-41-81-01-055	PTF ALL
		28-41-81-01-100	PTF ALL

F. Location Zones

Zone	Area
122	Forward Cargo Compartment - Right

G. Access Panels

Number	Name/Location
821	Forward Cargo Door

H. Fuel Quantity Processor Unit Installation

SUBTASK 28-41-81-020-006

(1) Remove the protective caps from the electrical connectors and fuel quantity processor unit [5] (TASK 20-40-12-000-803).

(a) Make sure that the connector sockets on the electrical connectors are not damaged.

(b) Make sure that the pins on the fuel quantity processor unit [5] that go into the sockets on the electrical connectors are not damaged.

SUBTASK 28-41-81-910-004

CAUTION

BE CAREFUL WHEN YOU REMOVE OR INSTALL THE FQIS PROCESSOR. DAMAGE TO THE FQIS CONNECTOR PINS CAN EASILY OCCUR.

CAUTION

DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(2) Do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 28-41-81-420-005

(3) Install protective caps on the electrical connectors and fuel quantity processor unit [5] (TASK 20-40-12-400-803).

SUBTASK 28-41-81-110-001

(4) Do these steps to prepare the FQPU mounting bracket legs [7] and bracket [6] for installation:

(a) Clean the fay surfaces of the FQPU mounting bracket legs [7] and bracket [6] (SWPM 20-20-00).

1) Use Cleaning Procedure 1 to manually clean the entire bracket [6] fay surface.

a) Use a very fine Scotch-Brite Type S pad, G50398, or Scotch-Brite 7448 pad, G50492.

b) Move the pad in a linear movement while you press with your thumb.

2) Use Cleaning Procedure 1 to manually clean the fay surface on the FQPU mounting bracket legs [7].

a) Use a very fine Scotch-Brite Type S pad, G50398, or Scotch-Brite 7448 pad, G50492.

b) Move the pad in a linear movement while you press with your thumb.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<1> Clean the surface 1/16 in. (2 mm) - 3/16 in. (5 mm) more outside of the fay surface.

SUBTASK 28-41-81-110-003

(5) If it is necessary, do these steps to refinish the abraded surfaces:

(a) Apply 3M 425 Aluminum Foil Tape, G00291, to the bracket [6] where the FQPU mounting bracket legs [7] and bracket [6] mate.

(b) Apply 3M 425 Aluminum Foil Tape, G00291, to the bracket [6] abraded surface 1/16 in. (2 mm) - 3/16 in. (5 mm) more outside of the fay surface.

(c) Manually apply coating, C00064, with a clean cloth, G51677, to the abraded surfaces that are not masked.

1) Keep the surfaces wet, until a perceptible color develops.

NOTE: Color change takes about 2-3 minutes.

(d) Air dry at room temperature.

NOTE: The maximum drying air temperature is 130°F (54°C).

(e) Remove 3M 425 Aluminum Foil Tape, G00291, from the bracket [6].

(f) Final clean the fay surfaces of the FQPU mounting bracket legs [7] and bracket [6].

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

1) Use a new cloth, G51677 (recommended), or knit cotton wiper, G51293 (alternate), soaked with solvent, B00062, or solvent, B00184.

2) Rub dry with a clean dry cotton wiper.

3) Continue to clean and dry the surface until the cotton wiper stays clean.

(g) Apply a thin coat of corrosion inhibiting material, G50136, to both mating faying surfaces of the FQPU mounting bracket legs [7] or bracket [6].

SUBTASK 28-41-81-410-001

(6) Hold the fuel quantity processor unit [5], install the bolts [3] and washers [4].

SUBTASK 28-41-81-420-003

(7) Tighten the bolts [3] to 35 ±5 in-lb (4.0 ±0.6 N·m).

(a) Make sure that the corrosion inhibiting material has a continuous squeeze out between the FQPU mounting bracket legs [7] and bracket [6].

NOTE: Verification of corrosion inhibiting material squeeze out is not required in areas where corrosion inhibiting material cannot be seen.

SUBTASK 28-41-81-110-005

(8) Apply a thin coat of corrosion inhibiting material, G50136, to exposed abraded areas.

(a) If it is necessary, use excess squeeze out corrosion inhibiting material to cover exposed abraded area.

SUBTASK 28-41-81-220-001

(9) Measure the electrical bonding resistance between the FQPU mounting bracket leg [7] and bracket [6] (SWPM 20-20-00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the electrical bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

SUBTASK 28-41-81-020-007

(10) Remove the protective caps from the electrical connectors and fuel quantity processor unit [5] (TASK 20-40-12-000-803).

SUBTASK 28-41-81-420-001

(11) Connect the electrical connectors to the fuel quantity processor unit [5].

I. Fuel Quantity Processor Unit Installation Test

SUBTASK 28-41-81-860-002

(1) Close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	4	C01441	FUEL FUELING IND
A	5	C00398	FUEL QTY 2
A	6	C00397	FUEL QTY 1

SUBTASK 28-41-81-740-001

(2) Do this task: Fuel Quantity Indicating System - Operational Test, TASK 28-41-00-710-801.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 28-41-81-420-002

(1) Install the shroud [1]:

(a) Hold and align the shroud [1].

(b) Loosely install the washer head screws [2] in a star pattern.

(c) Tighten the washer head screws [2] to 20 ±3 in-lb (2.3 ±0.4 N·m).

SUBTASK 28-41-81-410-002

(2) Close this access panel:

Number	Name/Location
821	Forward Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL BOOST PUMP FUEL PRESSURE SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Fuel Boost Pump Fuel Pressure Switch Removal

(2) Fuel Boost Pump Fuel Pressure Switch Installation.

B. There is one fuel pressure switch installed on these fuel boost pumps:

(1) Left aft fuel boost pump

(2) Left forward fuel boost pump

(3) Left center fuel boost pump

(4) Right center fuel boost pump

(5) Right forward fuel boost pump

(6) Right aft fuel boost pump.

TASK 28-42-11-000-801

2. Fuel Boost Pump Fuel Pressure Switch Removal

(Figure 401 or Figure 402)

A. General

(1) The fuel boost pump fuel pressure switch is referred to as the fuel pressure switch in this task.

(2) This task gives instructions to remove the fuel pressure switch.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Main Tank Fuel Pressure Switches Removal

SUBTASK 28-42-11-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-42-11-860-001

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	11	C00313	INDICATOR MASTER DIM SECT 1
F	12	C00318	INDICATOR MASTER DIM SECT 6

SUBTASK 28-42-11-010-005

(3) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-42-11-860-018

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Do these steps:

(a) For the aft fuel pressure switch, open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT

(b) For the forward fuel pressure switch, open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD

SUBTASK 28-42-11-010-001

(5) Get access to the applicable fuel pressure switch [1] inboard of the main landing gear.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Center Tank Fuel Pressure Switch Removal

SUBTASK 28-42-11-210-017

WARNING

MAKE SURE THAT THE QUANTITY OF FUEL IS LESS THAN THE SPECIFIED MAXIMUM. IF THE QUANTITY OF FUEL IS MORE THAN THE SPECIFIED MAXIMUM, FUEL CAN DRAIN FROM THE FUEL TANK. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Make sure that the fuel quantity in the center tank is less than 11,400 lb (5171 kg).

NOTE: The MDS Engine Display shows the fuel quantity on the center display unit (P2).

SUBTASK 28-42-11-480-004

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 28-42-11-860-014

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	11	C00313	INDICATOR MASTER DIM SECT 1
F	12	C00318	INDICATOR MASTER DIM SECT 6

SUBTASK 28-42-11-010-004

(4) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 28-42-11-860-015

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

H. Main Tank Fuel Pressure Switches Removal

SUBTASK 28-42-11-020-001

(1) Disconnect the electrical connector [3] from the fuel pressure switch [1].

SUBTASK 28-42-11-010-006

(2) Remove sealant around the nut on the boss and safety cable [4] connection on the rear spar, out of the fuel tank.

(a) Use sealant removal tool, COM-2481, or equivalent.

SUBTASK 28-42-11-020-002

(3) Remove the safety cable [4] from the fuel pressure switch [1].

PTF ALL POST SB 737-28-1370

SUBTASK 28-42-11-020-014

(4) Remove the sealing boot [5].

PTF ALL

SUBTASK 28-42-11-020-003

(5) Remove the fuel pressure switch [1] from the fuel adapter on the rear spar.

NOTE: When the forward fuel pressure switch is removed, fuel can leak from the fuel adapter. No more than 20 drops in one minute is permitted. When the fuel leaks more than 20 drops in one minute, replace the fuel adapter fitting.

SUBTASK 28-42-11-020-004

(6) Remove and discard the O-ring [2].

I. Center Tank Fuel Pressure Switch Removal

SUBTASK 28-42-11-020-009

(1) Disconnect the electrical connector [3] from the fuel pressure switch [1].

SUBTASK 28-42-11-010-007

(2) Remove sealant around the nut on the boss and safety cable [4] connection on the rear spar, out of the fuel tank.

(a) Use sealant removal tool, COM-2481, or equivalent.

SUBTASK 28-42-11-020-010

(3) Remove the safety cable [4] from the fuel pressure switch [1].

SUBTASK 28-42-11-020-011

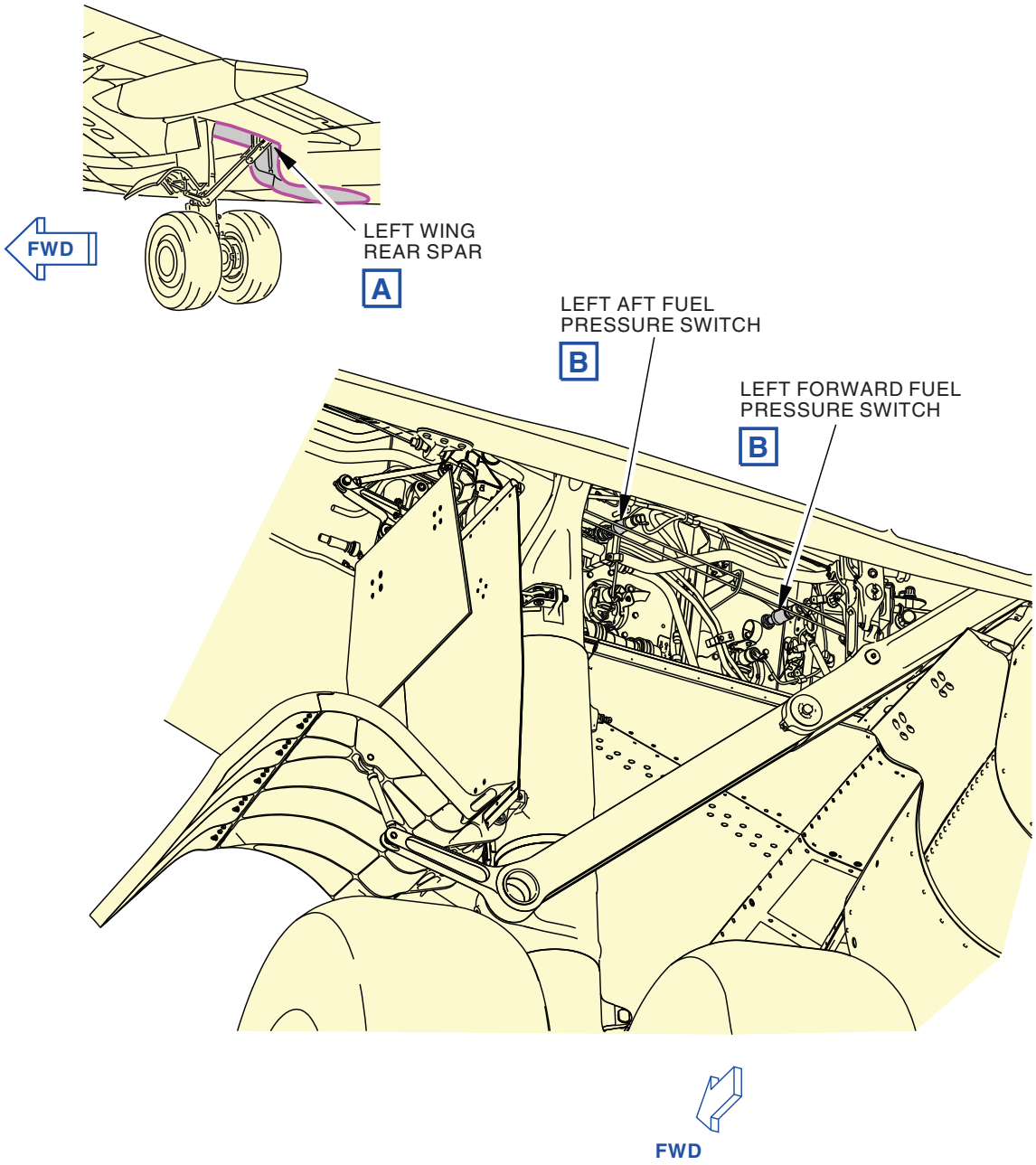
(4) Remove the fuel pressure switch [1] from the fuel adapter on the rear spar.

SUBTASK 28-42-11-020-012

(5) Remove and discard the O-ring [2].

END OF TASK

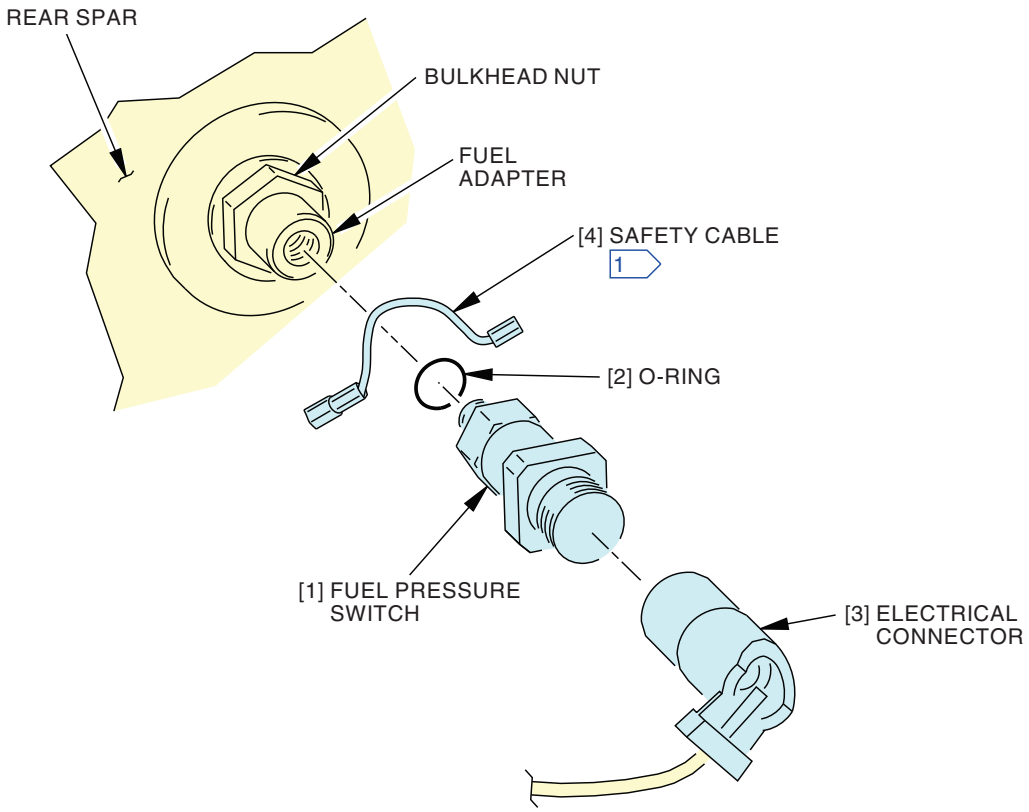
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413731 S00061533201_V1

Main Tanks Fuel Boost Pump Pressure Switch Installation
Figure 401/28-42-11-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FUEL PRESSURE SWITCH (EXAMPLE)

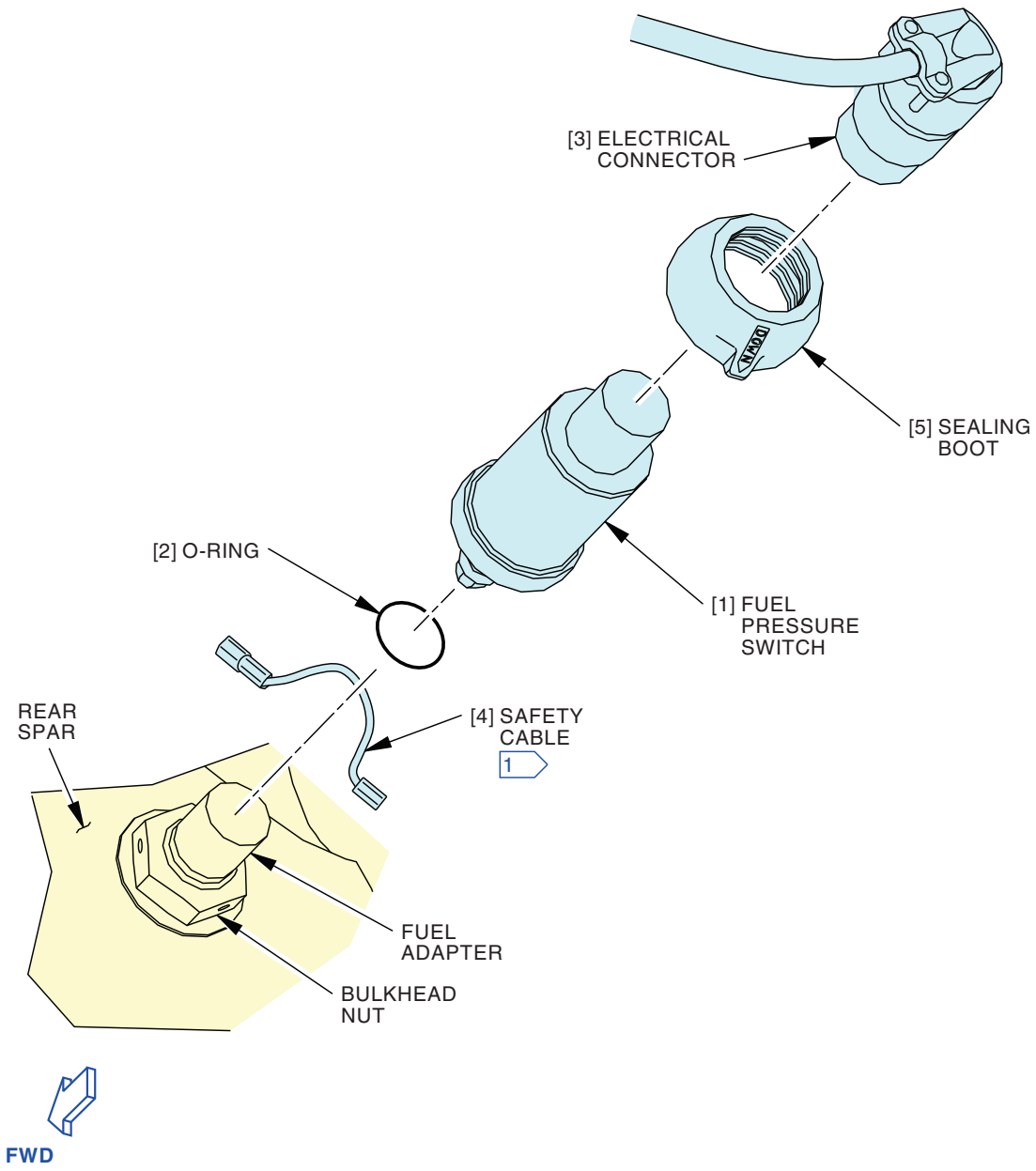
B

1 SAFETY CABLE PREFERRED, LOCKWIRE ALTERNATE.

2413732 S00061533202_V2

Main Tanks Fuel Boost Pump Pressure Switch Installation
Figure 401/28-42-11-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FUEL PRESSURE SWITCH (EXAMPLE)

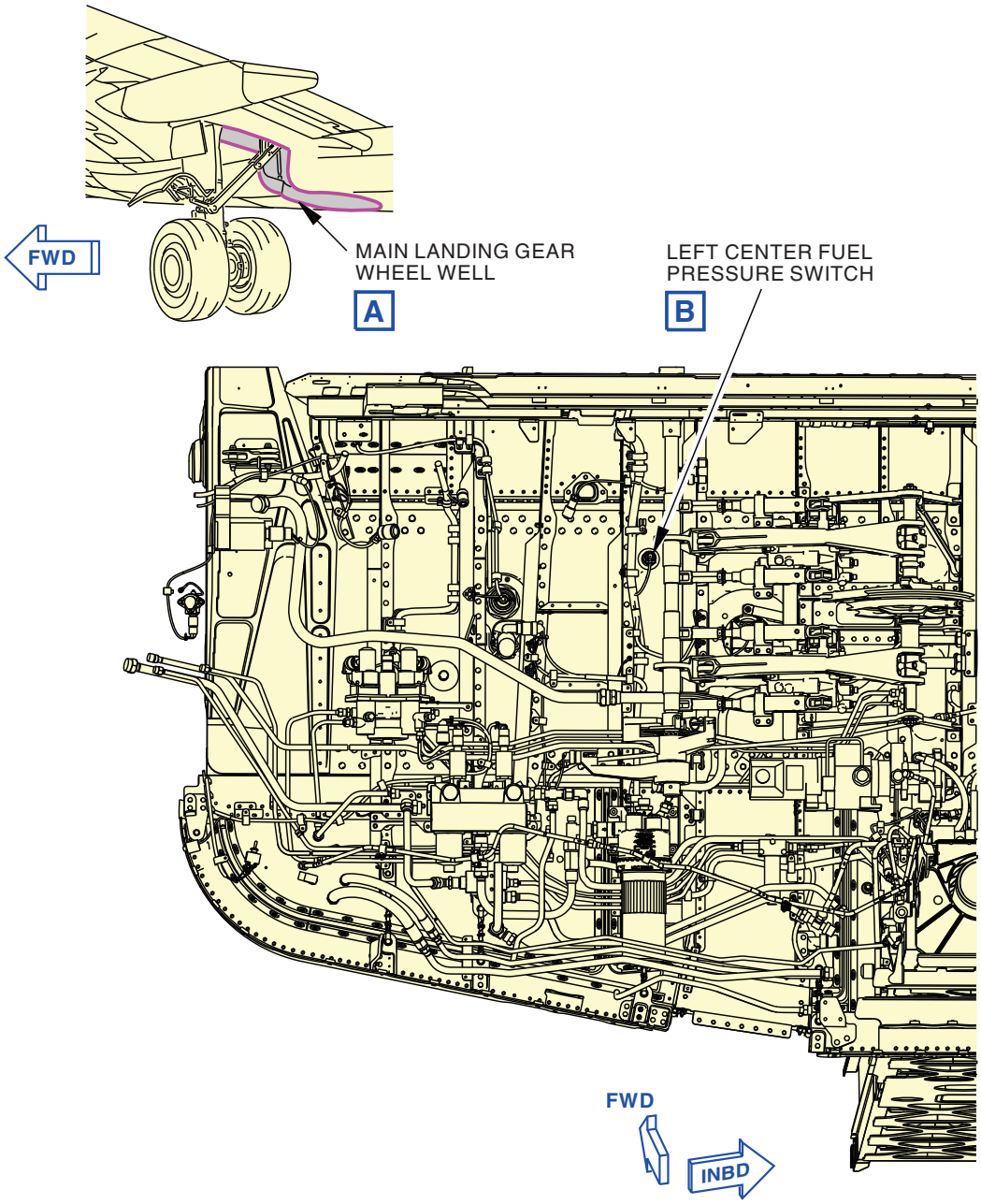
B

1 SAFETY CABLE PREFERRED, LOCKWIRE ALTERNATE.

2968218 S0000747620_V1

Main Tanks Fuel Boost Pump Pressure Switch Installation
Figure 401/28-42-11-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



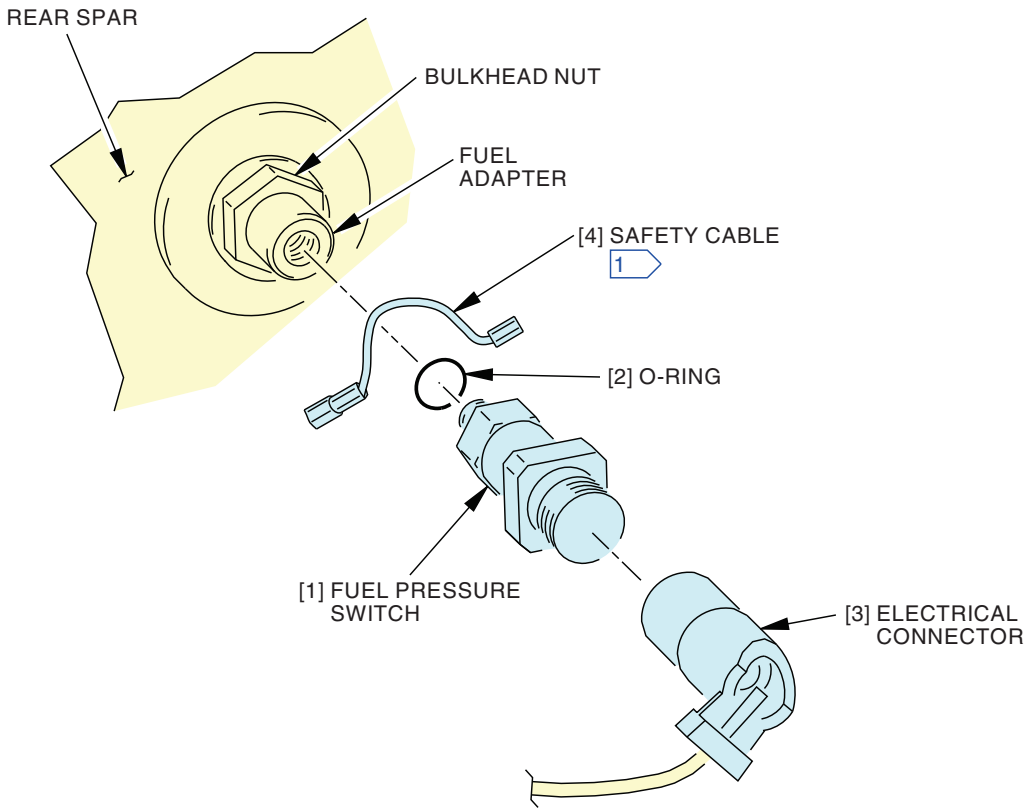
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS ALMOST THE SAME)

A

2413730 S00061533200_V1

Center Tank Fuel Boost Pump Pressure Switch Installation
Figure 402/28-42-11-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FUEL PRESSURE SWITCH (EXAMPLE)

B

1 SAFETY CABLE PREFERRED, LOCKWIRE ALTERNATE.

2413732 S00061533202_V2

Center Tank Fuel Boost Pump Pressure Switch Installation
Figure 402/28-42-11-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-42-11-420-801

3. Fuel Boost Pump Fuel Pressure Switch Installation

(Figure 401 or Figure 402)

A. General

(1) The fuel boost pump fuel pressure switch is referred to as the fuel pressure switch in this task.

(2) This task gives instructions to install the fuel pressure switch.

B. References

Reference	Title
20-10-17-400-801	O-Rings Installation (P/B 401)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
D00504	Grease - Petrolatum	VV-P-236
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50170	Compound - Corrosion Inhibiting Compound, Soft Film, Exterior Use - Ardrox AV25	
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	
G51372	Kit - Safety Cable, 321 CRES - 0.035 +/-0.003 Inch (1.0 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fuel pressure switch	28-42-11-01-005	PTF ALL
		28-42-11-02-010	PTF ALL
2	O-ring	28-42-11-01-020	PTF ALL
		28-42-11-02-015	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
621GB	Refuel Valve Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Main Tank Pressure Switches Installation

SUBTASK 28-42-11-640-001

(1) Apply a thin layer of grease, D00504, to the new O-ring [2].

SUBTASK 28-42-11-420-001

(2) Install the O-ring [2] on the fuel pressure switch [1] (TASK 20-10-17-400-801).

SUBTASK 28-42-11-420-002

(3) Install the fuel pressure switch [1] to the fuel adapter on the rear spar.

(a) Tighten the fuel pressure switch [1] to 30 in-lb (3.4 N·m).

PTF ALL POST SB 737-28-1370

SUBTASK 28-42-11-210-015

(4) Make sure that surface of the fuel pressure switch [1] and nearest area are dry.

(a) If it is necessary, remove any moisture.

NOTE: It is necessary to keep pressure switch and area dry during installation.

SUBTASK 28-42-11-410-002

(5) Install the sealing boot [5] on the fuel pressure switch [1] with drain hole in downward position.

NOTE: Larger end of the sealing boot must be sealed and flush with the bottom of the pressure switch housing and smaller end of the sealing boot must be seated in the housing grooves.

NOTE: It is not necessary for the sealing boot drain hole to line up with the fuel pressure switch vent hole.

PTF ALL

SUBTASK 28-42-11-420-003

(6) Install the safety cable [4] (MS20995NC32 lockwire, G01912, or safety cable kit, G51372) from the fuel pressure switch [1] to the bulkhead nut (TASK 20-10-44-400-801).

SUBTASK 28-42-11-910-001

(7) Apply the sealant, A02315, fully around the nut on the boss.

SUBTASK 28-42-11-210-004

(8) Examine the electrical connector [3] for corrosion.

(a) If there is corrosion, correct the problem (SWPM 20-60-03).

SUBTASK 28-42-11-390-001

(9) Apply D5026NS or ZC-026 compound, G50171, to the electrical connector [3] (SWPM 20-60-03).

SUBTASK 28-42-11-420-004

(10) Connect the electrical connector [3] to the fuel pressure switch [1].

NOTE: Corrosion inhibiting compound is applied to the exterior of the mated electrical connectors after the operational test is completed.

H. Center Tank Fuel Pressure Switch Installation

SUBTASK 28-42-11-640-003

(1) Apply a thin layer of grease, D00504, to the new O-ring [2].

SUBTASK 28-42-11-420-009

(2) Install the O-ring [2] on the fuel pressure switch [1] (TASK 20-10-17-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-42-11-420-010

CAUTION: DO NOT INSTALL THE SEALING BOOT ON THE PRESSURE SWITCH OF THE CENTER TANK. IF YOU DO NOT OBEY, DAMAGE TO THE PRESSURE SWITCH CAN OCCUR.

(3) Install the fuel pressure switch [1] to the fuel adapter on the rear spar.

(a) Tighten the fuel pressure switch [1] to 30 in-lb (3.4 N·m).

SUBTASK 28-42-11-420-011

(4) Install the safety cable [4] (MS20995NC32 lockwire, G01912, or safety cable kit, G51372) from the fuel pressure switch [1] to the bulkhead nut (TASK 20-10-44-400-801).

SUBTASK 28-42-11-910-002

(5) Apply the sealant, A02315, fully around the nut on the boss.

SUBTASK 28-42-11-210-010

(6) Examine the electrical connector [3] for corrosion.

(a) If there is corrosion, correct the problem (SWPM 20-60-03).

SUBTASK 28-42-11-390-003

(7) Apply D5026NS or ZC-026 compound, G50171, to the electrical connector [3] (SWPM 20-60-03).

SUBTASK 28-42-11-420-012

(8) Connect the electrical connector [3] to the fuel pressure switch [1].

NOTE: Corrosion inhibiting compound is applied to the exterior of the mated electrical connectors after the operational test is completed.

I. Main Tank Fuel Pressure Switches Installation Test

NOTE: Put a person in the flight compartment to operate the fuel boost pumps and to monitor the fuel quantity and LOW PRESSURE indicators.

SUBTASK 28-42-11-860-016

WARNING: WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING: DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) For the aft fuel pressure switch, remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	3	C00828	FUEL BOOST PUMP TANK 2 AFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	1	C00826	FUEL BOOST PUMP TANK 1 AFT

(b) For the forward fuel pressure switch, remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	1	C00827	FUEL BOOST PUMP TANK 1 FWD

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	3	C00829	FUEL BOOST PUMP TANK 2 FWD

SUBTASK 28-42-11-860-008

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	11	C00313	INDICATOR MASTER DIM SECT 1
F	12	C00318	INDICATOR MASTER DIM SECT 6

SUBTASK 28-42-11-210-018

(3) Make sure that the main tank has a minimum of 500 lb (227 kg) of fuel.

SUBTASK 28-42-11-210-008

(4) On the P5-2 Fuel System Module, make sure that the LOW PRESSURE light for the No. 1 or No. 2 forward/aft fuel boost pump comes on.

SUBTASK 28-42-11-860-009

WARNING DO NOT OPERATE A FUEL PUMP IF THE LOW PRESSURE LIGHT COMES ON AND STAYS ON. THIS CONDITION CAN CAUSE THE IGNITION OF THE FUEL FUMES IN THE FUEL TANK. A FIRE OR AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) If the LOW PRESSURE light comes on and stays on, immediately set the applicable FUEL PUMPS switch to the OFF position.

SUBTASK 28-42-11-720-003

(6) On the P5-2 Fuel System Module, set the FUEL PUMPS FWD/FUEL PUMPS AFT 1 or 2 switch to the ON position.

(a) Make sure that the applicable LOW PRESSURE light goes off in 15 seconds or less.

SUBTASK 28-42-11-210-009

(7) Make sure that there is no fuel leakage at the fuel pressure switch [1].

SUBTASK 28-42-11-720-004

(8) Set the FUEL PUMPS FWD/FUEL PUMPS AFT 1 or 2 switch to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable LOW PRESSURE light comes on.

J. Center Tank Fuel Pressure Switch Installation Test

NOTE: Put a person in the flight compartment to operate the fuel boost pumps and to monitor the fuel quantity and LOW PRESSURE indicators.

SUBTASK 28-42-11-860-017

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	5	C00845	FUEL BOOST PUMP CTR TANK LEFT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	5	C00846	FUEL BOOST PUMP CTR TANK RIGHT

SUBTASK 28-42-11-860-011

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	11	C00313	INDICATOR MASTER DIM SECT 1
F	12	C00318	INDICATOR MASTER DIM SECT 6

SUBTASK 28-42-11-010-003

(3) Open this access panel:

Number	Name/Location
621GB	Refuel Valve Door

SUBTASK 28-42-11-210-011

(4) Make sure that the center tank has a minimum of 3000 lb (1361 kg) of fuel.

SUBTASK 28-42-11-210-012

(5) On the P5-2 Fuel System Module, make sure that the LOW PRESSURE light for the left or right center tank fuel boost pump is off.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-42-11-860-012

WARNING DO NOT OPERATE A FUEL PUMP IF THE LOW PRESSURE LIGHT COMES ON AND STAYS ON. THIS CONDITION CAN CAUSE THE IGNITION OF THE FUEL FUMES IN THE FUEL TANK. A FIRE OR AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(6) Immediately set the applicable FUEL PUMPS switch to the OFF position if the LOW PRESSURE light comes on and stays on.

SUBTASK 28-42-11-720-005

(7) On the P5-2 Fuel System Module, set the FUEL PUMPS CTR L or R switch to the ON position.

(a) Make sure that the applicable LOW PRESSURE light goes off in 15 seconds or less.

SUBTASK 28-42-11-210-013

(8) Make sure that there is no fuel leakage at the fuel pressure switch [1].

SUBTASK 28-42-11-720-006

(9) Set the FUEL PUMPS CTR L or R switch to the OFF position.

(a) Make sure that the applicable LOW PRESSURE light is off.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 28-42-11-390-004

(1) Apply Ardrox AV25 compound, G50170, to the mated electrical connector [3] completely, until the compound drips from the connector (SWPM 20-60-03).

(a) Do not remove the excess Ardrox AV25 compound, G50170.

SUBTASK 28-42-11-410-001

(2) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
621GB	Refuel Valve Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL TEMPERATURE INDICATING SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Fuel Temperature Indicating System Test (Master Thermometer)

(2) Fuel Temperature Indicator Functional Test (Spare Temperature Indicator)

(3) Fuel Temperature Indicator Functional Test (Decade Resistor)

(4) Fuel Temperature Bulb Resistance Test.

B. The fuel temperature indicating system test (master thermometer) does a test of the fuel temperature bulb.

(1) To do the test for the fuel temperature bulb, you measure the temperature at the bulb with a master thermometer. You then compare the value on the temperature indicator with the value on the master thermometer.

C. The fuel temperature indicator functional test (spare temperature indicator) does a test of the fuel temperature indicator.

(1) To do the test for the fuel temperature indicator, you compare the value on the airplane temperature indicator with the value on a spare temperature indicator.

NOTE: The spare temperature indicator is not installed on the airplane, but is equipment necessary to do the test.

D. The functional test for the temperature indicator (decade resistor) can be used as an alternative to this task: Fuel Temperature Indicator Functional Test (Spare Temperature Indicator), TASK 28-43-00-720-801.

(1) To do the functional test for the temperature indicator, you stimulate the temperature bulb at three different temperatures with a decade resistor.

TASK 28-43-00-710-801

2. Fuel Temperature Indicating System Test (Master Thermometer)

(Figure 501, Figure 502)

A. Tools/Equipment

Reference	Description
STD-1336	Thermometer - Digital, 0-150 +/- 2 Degrees F, with a Probe, Thermocouple or Equivalent

B. Location Zones

Zone	Area
211	Flight Compartment - Left
551	Left Wing - Rear Spar To Landing Gear Support Beam

C. Access Panels

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the System Test

SUBTASK 28-43-00-860-002

(1) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

SUBTASK 28-43-00-860-003

(2) Permit the temperature to become stable for 10 minutes.

SUBTASK 28-43-00-010-001

(3) Open this access panel:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

E. Fuel Temperature Indicating System Test

SUBTASK 28-43-00-970-001

(1) Make a record of the values shown on the fuel temperature indicator on the P5 panel.

SUBTASK 28-43-00-020-001

(2) Remove the fuel temperature bulb.

NOTE: The fuel temperature bulb is on the aft side of the aft spar. It is on the outboard side of the left main landing gear.

(a) Turn the locking cap for the temperature bulb to release the bulb from the pin.

(b) Pull the bulb from the housing.

SUBTASK 28-43-00-970-002

(3) Measure the temperature in the bulb housing.

(a) Put the 0-150 +/- 2 degrees F digital thermometer, STD-1336, in the bulb housing.

(b) Permit the temperature to become stable.

(c) Make a record of the temperature shown on the thermometer.

SUBTASK 28-43-00-970-003

(4) Compare the value on the 0-150 +/- 2 degrees F digital thermometer, STD-1336, and the value on the fuel temperature indicator.

(a) Make sure the two values are the same with a tolerance of ± 5°F (3°C).

SUBTASK 28-43-00-420-001

(5) Put the fuel temperature bulb back in the housing.

SUBTASK 28-43-00-860-004

(6) Open this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

SUBTASK 28-43-00-710-001

(7) Monitor the value shown on the fuel temperature indicator.

(a) Make sure the pointer on the indicator goes to the bottom of the scale.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-43-00-860-005

(8) Close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

F. Put the Airplane Back to Its Usual Condition

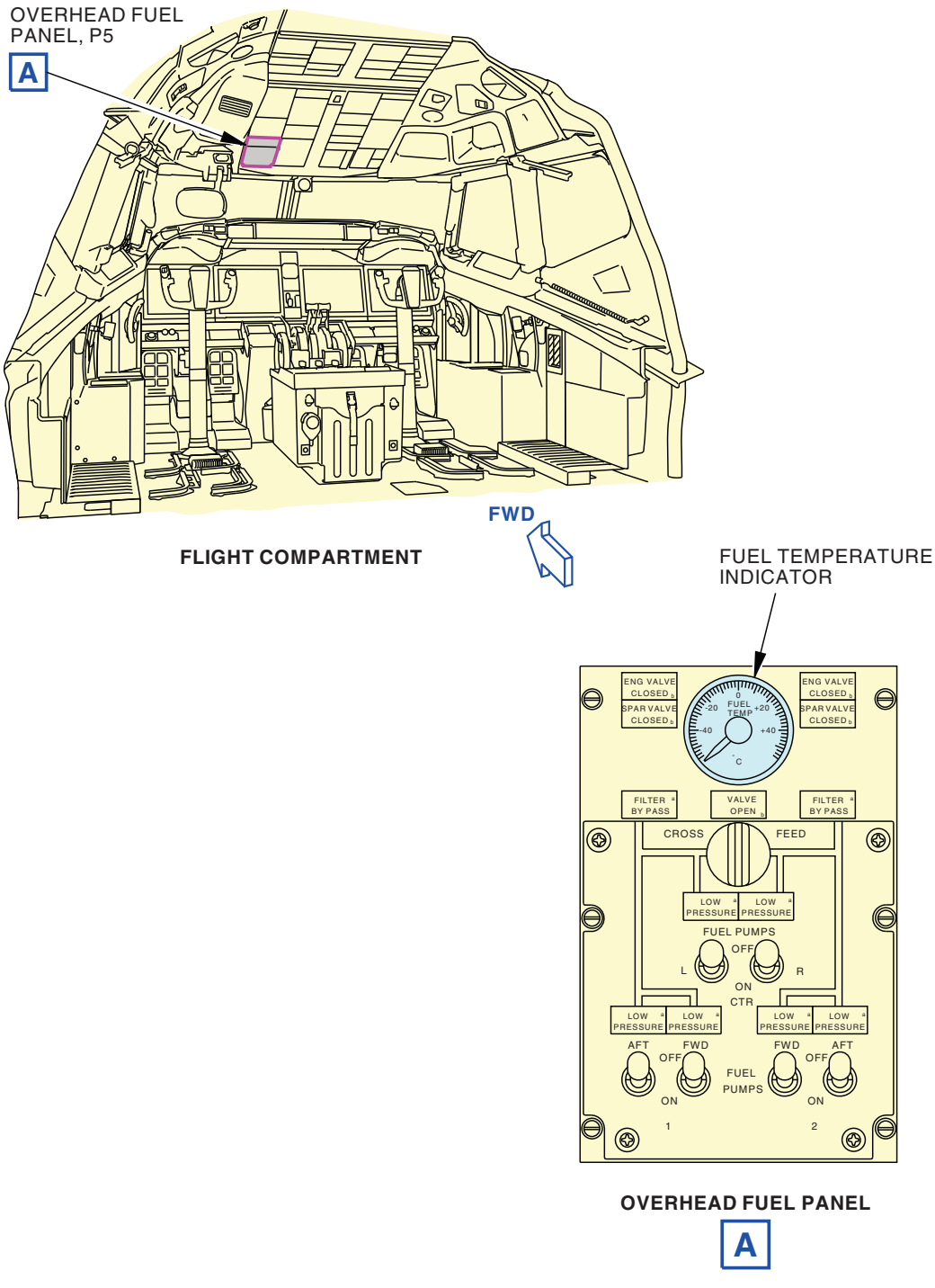
SUBTASK 28-43-00-410-001

(1) Close this access panel:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

——— END OF TASK ———

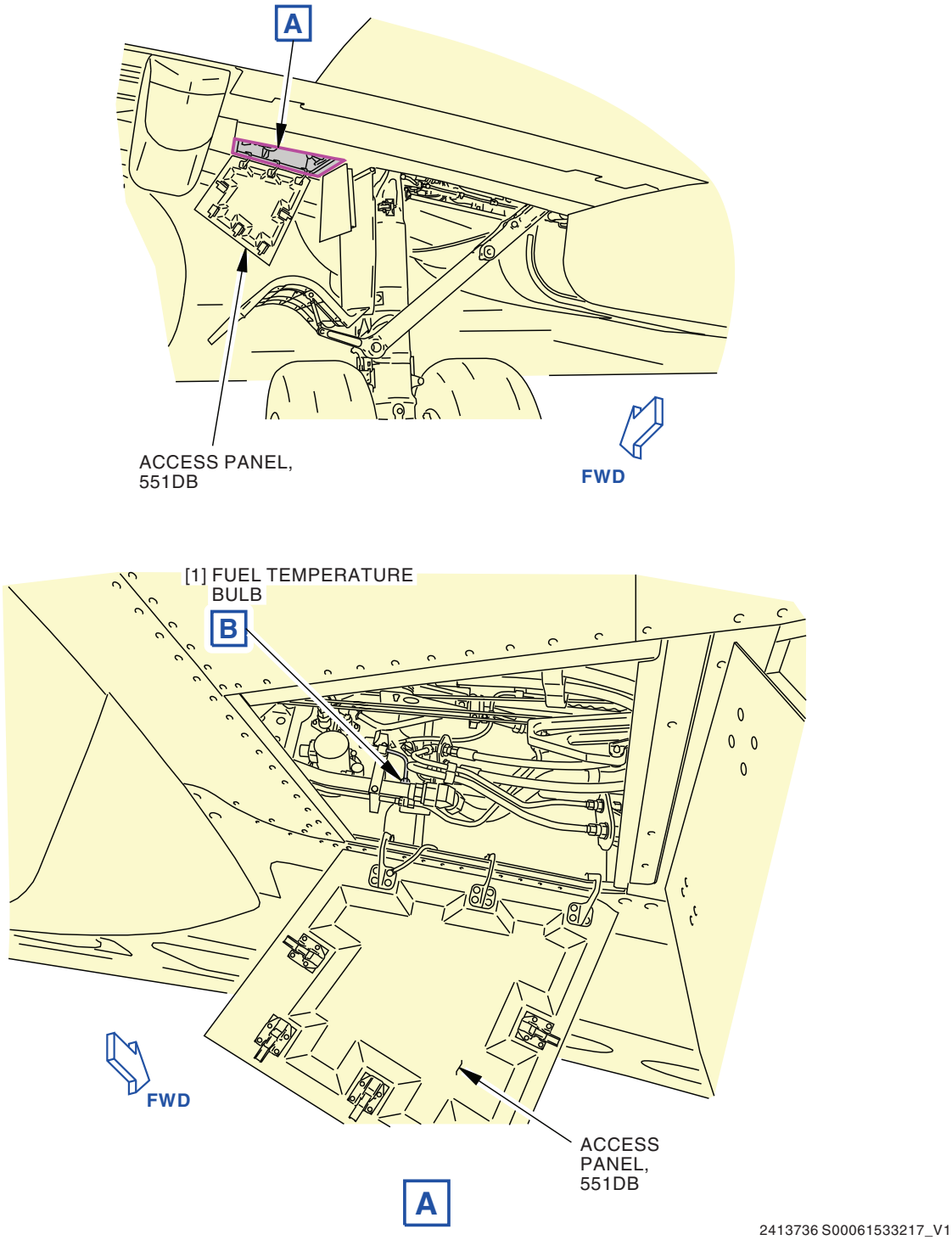
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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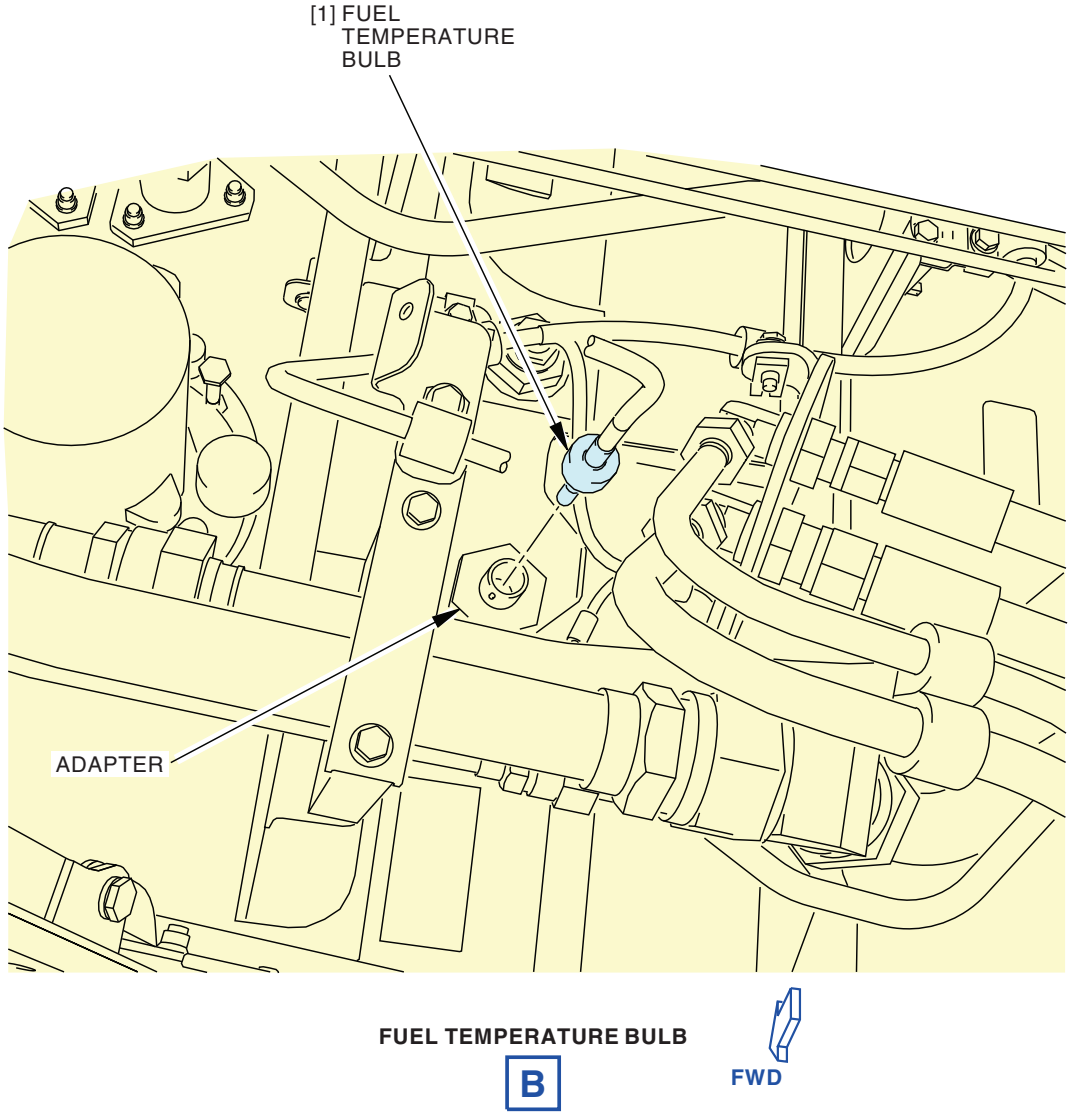
Fuel Temperature Indicator
Figure 501/28-43-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Temperature Bulb Installation
Figure 502/28-43-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fuel Temperature Bulb Installation
Figure 502/28-43-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-43-00-720-801

3. Fuel Temperature Indicator Functional Test (Spare Temperature Indicator)

(Figure 501, Figure 502).

A. General

(1) As an alternative to this task, you can do a test for the fuel temperature indication with this task: Fuel Temperature Indicator Functional Test (Decade Resistor), TASK 28-43-00-710-802.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
551	Left Wing - Rear Spar To Landing Gear Support Beam

C. Prepare for the Functional Test

SUBTASK 28-43-00-860-008

(1) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

SUBTASK 28-43-00-860-009

(2) Permit the temperature to become stable for 10 minutes.

D. Fuel Temperature Indicator Functional Test

SUBTASK 28-43-00-970-004

(1) Make a record of the values shown on the fuel temperature indicator.

SUBTASK 28-43-00-860-010

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

SUBTASK 28-43-00-020-002

(3) Disconnect the electrical connector from the fuel temperature indicator on the airplane.

SUBTASK 28-43-00-420-002

(4) Connect the electrical connector to a spare fuel temperature indicator.

NOTE: This fuel temperature indicator is not installed on the airplane. A new or certified fuel temperature indicator is used to make sure that the fuel temperature indicator installed on the airplane operates correctly.

SUBTASK 28-43-00-860-011

(5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

SUBTASK 28-43-00-970-005

(6) Compare the value shown on the spare indicator with the value from the fuel temperature indicator that you recorded before this step.

(a) These values must be the same within a tolerance of ±9°F (±5°C).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-43-00-860-012

(7) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

SUBTASK 28-43-00-020-003

(8) Disconnect the spare fuel temperature indicator.

SUBTASK 28-43-00-420-003

(9) Connect the fuel temperature indicator on the airplane to the electrical connector.

SUBTASK 28-43-00-860-013

(10) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

SUBTASK 28-43-00-970-006

(11) Make sure the value shown on the fuel temperature indicator is the same as it was before you did the test.

SUBTASK 28-43-00-860-014

(12) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

SUBTASK 28-43-00-720-001

(13) Monitor the value shown on the fuel temperature indicator.

(a) The pointer on the indicator must go back to its usual position at the bottom of the scale.

SUBTASK 28-43-00-860-015

(14) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

——— END OF TASK ———

TASK 28-43-00-710-802

4. Fuel Temperature Indicator Functional Test (Decade Resistor)

A. General

(1) The functional test can be used as an alternative to the task: Fuel Temperature Indicator Functional Test (Spare Temperature Indicator), TASK 28-43-00-720-801.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-9597	Box - Decade, Resistance (11,111.1 Ohms total, 0.01 ohm/step) Part #: 1433-20 Supplier: 62015 Part #: RDS-63-B Supplier: 0BHF9

C. Location Zones

Zone	Area
193	Lower Wing-To-Body Fairing - Wheel Well
212	Flight Compartment - Right

D. Prepare for the Procedure

SUBTASK 28-43-00-860-017

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

SUBTASK 28-43-00-020-004

(2) At the left wheel well, disconnect the connector, D46028P, from the panel, AL0675B, position 4.

SUBTASK 28-43-00-860-018

(3) Set the decade resistance box, COM-9597, to 77.39 ohms.

SUBTASK 28-43-00-420-004

(4) Connect the high terminal of the decade resistance box, COM-9597, to connector, D46028J, pin 16.

SUBTASK 28-43-00-420-005

(5) Connect the low terminal of the decade resistance box, COM-9597, to connector, D46028J, pin 15 and connector, D46028P, pin 15.

SUBTASK 28-43-00-860-020

(6) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

E. Procedure

SUBTASK 28-43-00-210-001

(1) At the fuel temperature indicator, located on the P5 overhead panel, make sure that the fuel temperature indicator displays -40 ±13°F (-40 ±7°C).

SUBTASK 28-43-00-860-021

(2) Set the decade resistance box, COM-9597, to 90.38 ohms.

SUBTASK 28-43-00-210-002

(3) At the fuel temperature indicator, located on the P5 overhead panel, make sure the fuel temperature indicator displays 32 ±13°F (0 ±7°C).

SUBTASK 28-43-00-860-022

(4) Set the decade resistance box, COM-9597, to 104.6 ohms.

SUBTASK 28-43-00-210-003

(5) At the fuel temperature indicator, located on the P5 overhead panel, make sure the fuel temperature indicator displays 104 ±13°F (40 ±7°C).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Put the Airplane Back to Its Usual Condition

SUBTASK 28-43-00-860-023

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

SUBTASK 28-43-00-020-005

(2) Disconnect the decade resistance box, COM-9597, from connector, D46028J, and connector, D46028P.

SUBTASK 28-43-00-420-006

(3) At the left wheel well, re-connect the connector, D46028P, to the panel, AL0675B, position 4.

SUBTASK 28-43-00-860-024

(4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

——— END OF TASK ———

TASK 28-43-00-760-801

5. Fuel Temperature Bulb Resistance Test

(Figure 501, Figure 502)

A. References

Reference	Title
WDM 28-42-11	Wiring Diagram Manual

B. Tools/Equipment

Reference	Description
STD-1336	Thermometer - Digital, 0-150 +/- 2 Degrees F, with a Probe, Thermocouple or Equivalent

C. Location Zones

Zone	Area
211	Flight Compartment - Left
551	Left Wing - Rear Spar To Landing Gear Support Beam

D. Access Panels

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

E. Procedure

SUBTASK 28-43-00-860-027

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-43-00-010-002

(2) Open this access panel:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

SUBTASK 28-43-00-020-006

(3) Do these steps to remove the fuel temperature bulb:

NOTE: The fuel temperature bulb is on the aft side of the aft spar. It is on the outboard side of the left main landing gear.

(a) Turn the locking cap for the temperature bulb to release the bulb from the pin.

(b) Pull the bulb from the housing.

SUBTASK 28-43-00-970-007

(4) Measure the temperature in the bulb housing.

(a) Put the 0-150 +/- 2 degrees F digital thermometer, STD-1336 in the bulb housing.

(b) Permit the temperature to become stable.

(c) Write the temperature shown on the 0-150 +/- 2 degrees F digital thermometer, STD-1336.

SUBTASK 28-43-00-420-009

(5) Do these steps to install the temperature bulb again:

(a) Hold the temperature bulb by the locking cap.

(b) Put the fuel temperature bulb into the bulb housing.

(c) Push the fuel temperature bulb into the housing until it touches the pin.

(d) Turn the locking cap to lock it around the pin.

SUBTASK 28-43-00-020-007

(6) Disconnect the connector, D616, from the fuel temperature indicator in the flight compartment (WDM 28-42-11).

SUBTASK 28-43-00-020-008

(7) Disconnect the ground GD476-DC on the A leg of the temperature sensor wiring (WDM 28-42-11).

SUBTASK 28-43-00-970-008

(8) Measure the resistance from pin 3 to pin 4 on the connector D616.

SUBTASK 28-43-00-970-009

(9) Use the subsequent table of sensor temperature and resistance values to compare the resistance measurement with the temperature of the sensor.

NOTE: The resistance measured by this procedure includes the resistance of the wiring between the indicator and the sensor, approximately 0.8 ohms for 100 feet (30.5 meters) of wiring.

Table 501/28-43-00-993-801

TEMPERATURE (degrees Celsius)	RESISTANCE (ohms)
-30	80.56 + 0.40
-20	83.77 + 0.40

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 501/28-43-00-993-801 (Continued)

TEMPERATURE (degrees Celsius)	RESISTANCE (ohms)
-10	87.04 + 0.40
0	90.38 + 0.40
10	93.80 + 0.40
20	97.31 + 0.40
30	100.91 + 0.40
40	104.60 + 0.40
50	108.39 + 0.40
60	112.28 + 0.40

SUBTASK 28-43-00-420-007

(10) Connect the connector, D616, to the fuel temperature indicator in the flight compartment again (WDM 28-42-11).

SUBTASK 28-43-00-420-008

(11) Connect the ground GD476-DC on the A leg of the temperature sensor wiring to ground again (WDM 28-42-11).

SUBTASK 28-43-00-410-002

(12) Close this access panel:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

SUBTASK 28-43-00-860-028

(13) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL TEMPERATURE BULB - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) A task to remove the fuel temperature bulb

(2) A task to install the fuel temperature bulb.

TASK 28-43-11-000-801

2. Fuel Temperature Bulb Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the fuel temperature bulb [1].

B. Location Zones

Zone	Area
561	Left Wing - Rear Spar to Trailing Edge, Outboard Of Inboard Flap, Inboard of Fixed Trailing Edge

C. Access Panels

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

D. Prepare for the Removal

SUBTASK 28-43-11-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

E. Fuel Temperature Bulb Removal

SUBTASK 28-43-11-010-001

(1) Open this access panel:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

SUBTASK 28-43-11-020-001

(2) Cut the wires that go to the fuel temperature bulb [1].

NOTE: The fuel temperature bulb is on the aft side of the aft spar. It is immediately outboard of the left main landing gear.

SUBTASK 28-43-11-020-002

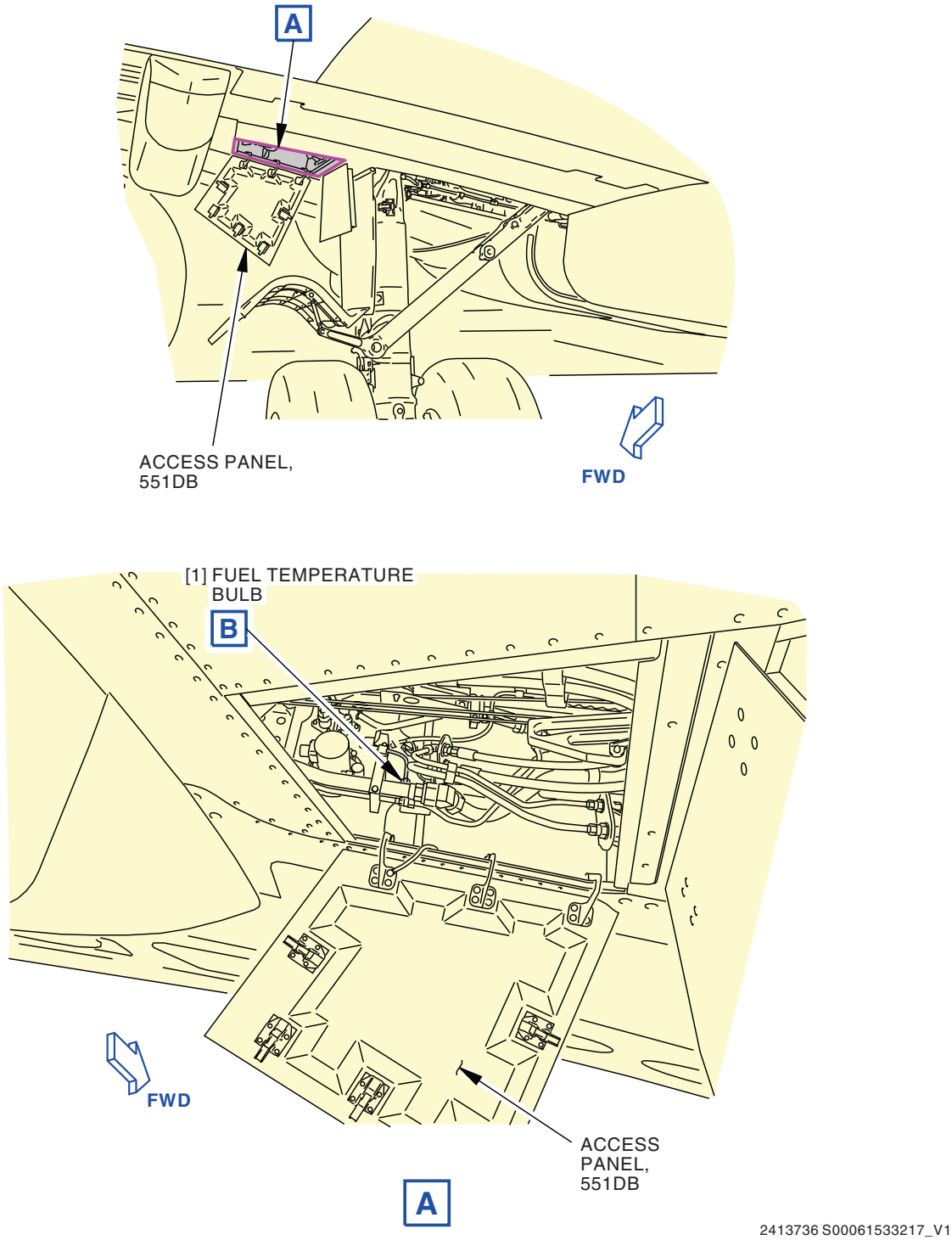
(3) Push the fuel temperature bulb [1] into the adapter with a maximum force of 4.0 lbf (17.8 N).

SUBTASK 28-43-11-020-003

(4) Turn the bulb until it is free and pull the fuel temperature bulb [1] out of the adapter.

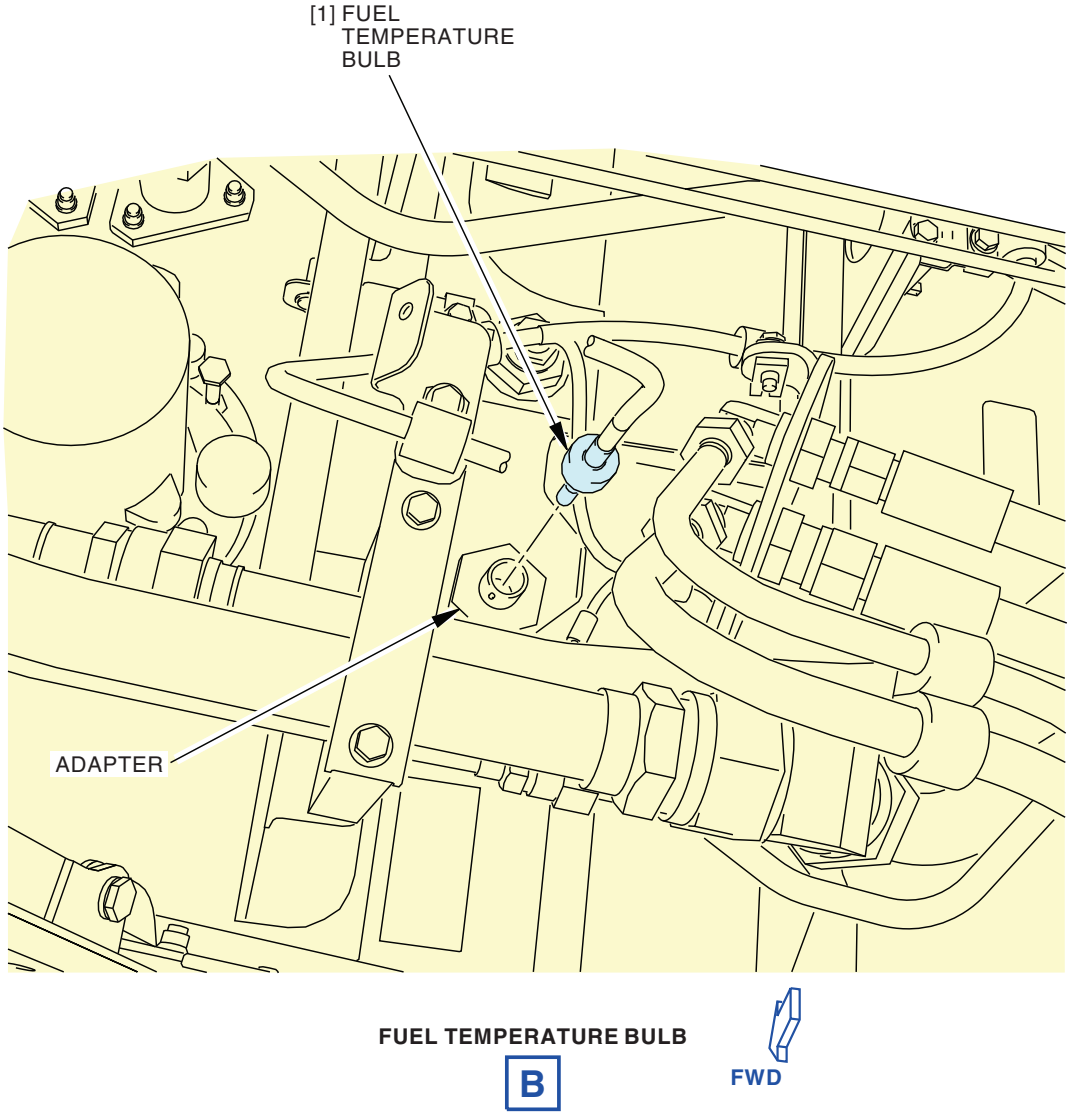
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Temperature Bulb Installation
Figure 401/28-43-11-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fuel Temperature Bulb Installation
Figure 401/28-43-11-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-43-11-400-801

3. Fuel Temperature Bulb Installation

(Figure 401)

A. General

(1) This task gives instructions to install the fuel temperature bulb [1].

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fuel temperature bulb	28-43-11-01-035	PTF ALL

D. Access Panels

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

E. Fuel Temperature Bulb Installation

SUBTASK 28-43-11-420-001

(1) Hold the fuel temperature bulb [1] by the locking cap.

SUBTASK 28-43-11-420-002

(2) Put the fuel temperature bulb [1] into the adapter.

SUBTASK 28-43-11-420-003

(3) Push the fuel temperature bulb [1] into the adapter until it touches the pin.

SUBTASK 28-43-11-420-004

(4) Turn the locking cap to lock it around the pin.

SUBTASK 28-43-11-420-005

(5) Make a splice from the wires on the fuel temperature bulb [1] to the airplane wire bundle.

F. Put the Airplane Back to its Usual Condition

SUBTASK 28-43-11-860-002

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

SUBTASK 28-43-11-860-003

(2) Do this task if necessary: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 28-43-11-710-001

(3) Monitor the fuel temperature on the indicator.

(a) Make sure the indicator shows the correct temperature.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-43-11-410-001

(4) Close this access panel:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL TEMPERATURE INDICATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Fuel temperature indicator removal

(2) Fuel temperature indicator installation

TASK 28-43-21-020-801

2. Fuel Temperature Indicator Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the Fuel Temperature Indicator.

B. Location Zones

Zone	Area
211	Flight Compartment - Left

C. Prepare for the Removal

SUBTASK 28-43-21-860-001

(1) Do these steps to remove the fuel temperature indicator [1] from the P5-2 overhead panel:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

D. Fuel Temperature Indicator Removal

SUBTASK 28-43-21-020-001

(1) Loosen the 6 quarter-turn fasteners that hold the P5-2 overhead panel.

(2) Carefully pull the P5-2 overhead panel away from the rest of the P5 panel until there is sufficient access to the fuel temperature indicator [1] behind the face of the P5-2 panel.

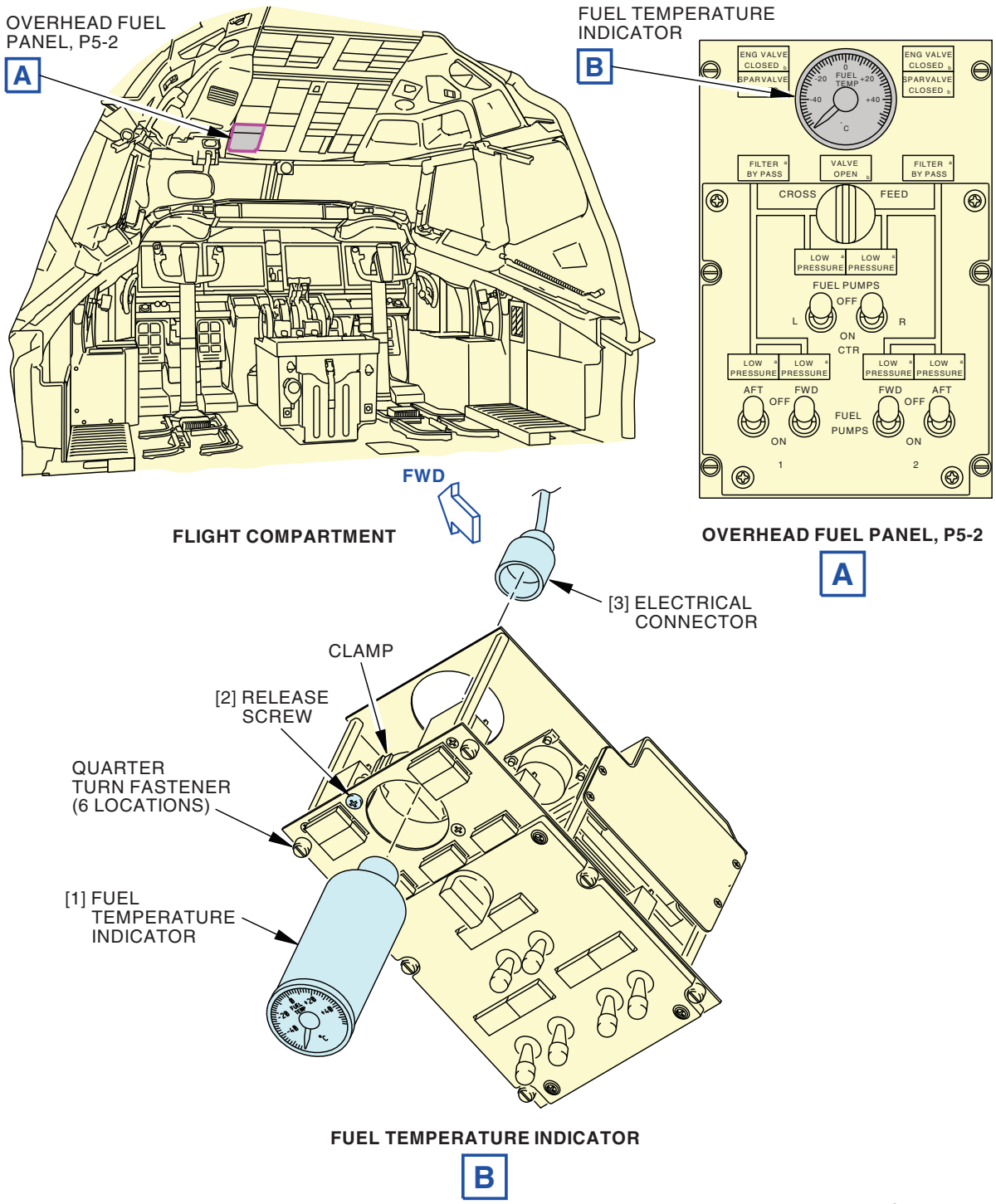
(3) Disconnect the electrical connector [3] from the fuel temperature indicator [1].

(4) Loosen the release screw [2] to loosen the clamp that holds the fuel temperature indicator [1] in its position.

(5) Remove the fuel temperature indicator from the P5-2 panel.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fuel Temperature Indicator Installation
Figure 401/28-43-21-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-43-21-020-802

3. Fuel Temperature Indicator Installation

(Figure 401)

A. General

(1) This task gives instructions to install the Fuel Temperature Indicator.

B. References

Reference	Title
28-43-00-710-801	Fuel Temperature Indicating System Test (Master Thermometer) (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fuel temperature indicator	28-43-21-01-050	PTF ALL

D. Location Zones

Zone	Area
211	Flight Compartment - Left

E. Fuel Temperature Indicator Installation

SUBTASK 28-43-21-020-002

(1) Do these steps to install the fuel temperature indicator on the P5-2 overhead panel:

(a) Put the fuel temperature indicator [1] in its position in the P5-2 overhead panel.

(b) Tighten the release screw [2] to tighten the clamp that holds the fuel temperature indicator [1] in its position.

(c) Connect the electrical connector [3] to the fuel temperature indicator [1].

(d) Carefully put the P5-2 overhead panel into its position in the P5 panel until there is sufficient access to the fuel temperature indicator [1] behind the face of the P5-2 panel.

(e) Tighten the 6 quarter-turn fasteners that hold the P5-2 overhead panel.

F. Fuel Temperature Indicator Installation Test

SUBTASK 28-43-21-860-002

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	7	C00355	FUEL TEMP IND

SUBTASK 28-43-21-720-001

(2) Do this test of the Fuel Temperature Indicating System: Fuel Temperature Indicating System Test (Master Thermometer), TASK 28-43-00-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MEASURING STICK - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the fuel measuring stick

(2) An installation of the fuel measuring stick

(3) A removal of the fuel measuring stick mounting base

(4) An installation of the fuel measuring stick mounting base.

B. To remove the fuel measuring stick, it is not necessary to defuel the fuel tanks.

C. To remove the fuel measuring stick mounting base:

(1) Defuel the applicable fuel tank.

(2) Remove the applicable access panel when necessary.

TASK 28-44-11-000-801

2. Fuel Measuring Stick Removal

(Figure 401)

A. General

(1) This task gives instruction to remove the fuel measuring stick.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-5738	Spanner Wrench - Fuel Stick Retainer Part #: A28016-1 Supplier: 81205 Part #: T5-0497K Supplier: 86831

C. Location Zones

Zone	Area
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Fuel Measuring Stick Removal

SUBTASK 28-44-11-860-001

(1) Turn the screwdriver slot counterclockwise to release the applicable fuel measuring stick (Figure 401).

SUBTASK 28-44-11-020-001

(2) Use snap ring pliers or a spanner wrench, SPL-5738, to remove the bayonet socket [10] or retainer [8] from the stick fitting or mounting base [36] (Figure 401).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-44-11-020-002

(3) Pull the stick subassembly [35] from the housing [26] with the retainer [8], spring [30], and the guides [29].

NOTE: You can feel a small force when the steel armature on the stick subassembly goes through the magnet in the float.

SUBTASK 28-44-11-020-003

(4) Remove the latch [9] from the stick subassembly [35].

SUBTASK 28-44-11-860-002

(5) Keep the latch [9] for installation of the fuel measuring stick.

SUBTASK 28-44-11-790-001

(6) Do these steps to do a check for fuel leakage in the housing of the fuel measuring stick:

(a) Make sure that the applicable tank contains sufficient fuel to cover the base of the fuel measuring stick housing.

Table 401/28-44-11-993-801 FUEL NECESSARY TO COVER THE CENTER TANK MEASURING STICKS

Fuel Measuring Stick Number	Fuel (LBS)	Fuel (KGS)
Stick No. 1	9197	4172
Stick No. 2	13722	6224

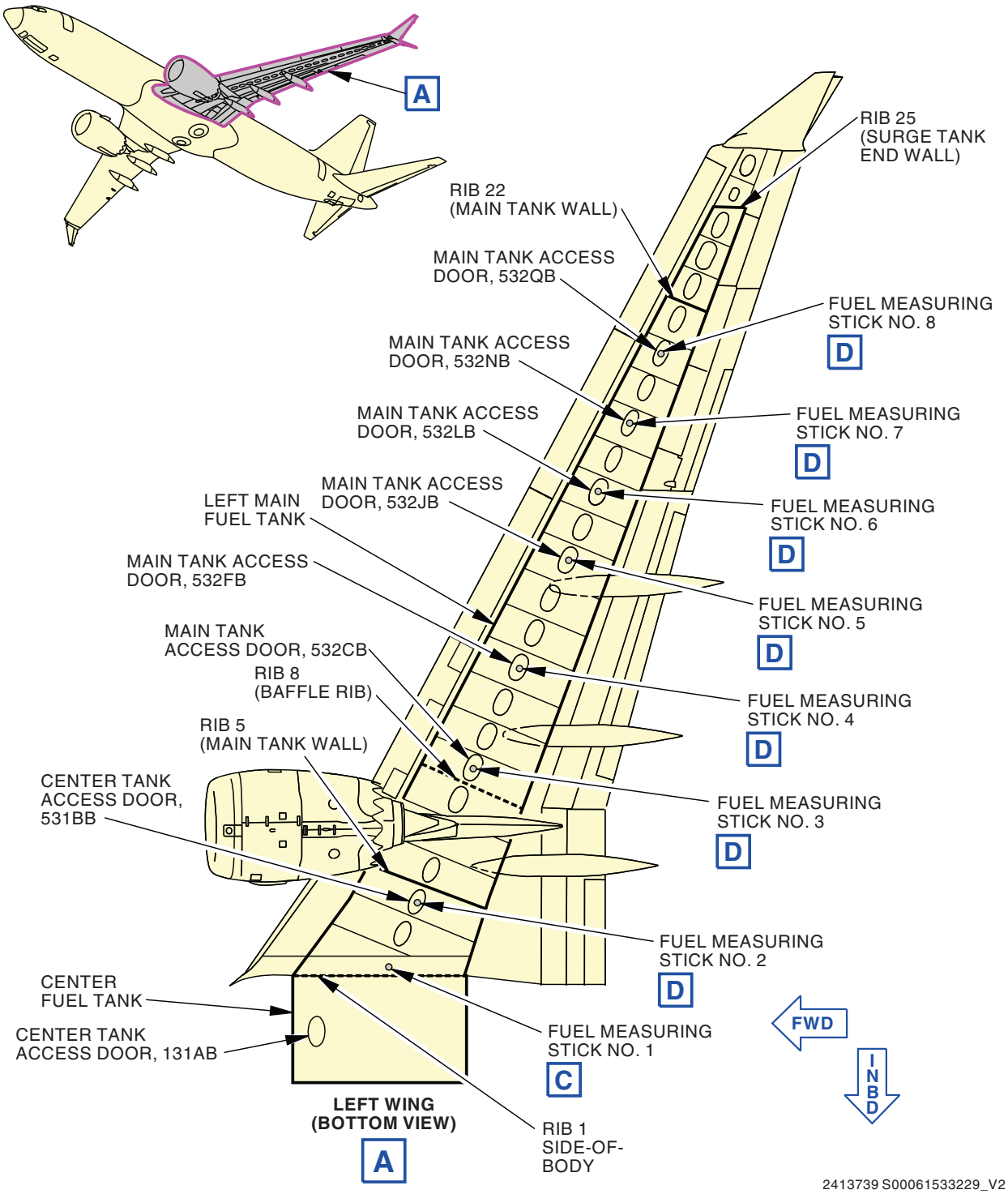
Table 402/28-44-11-993-802 FUEL NECESSARY TO COVER THE NO. 1 OR NO. 2 TANK MEASURING STICKS

Fuel Measuring Stick Number	Fuel (LBS)	Fuel (KGS)
Stick No. 3	1167	529
Stick No. 4	3402	1543
Stick No. 5	5665	2570
Stick No. 6	6710	3044
Stick No. 7	7570	3434
Stick No. 8	8262	3747

(b) Monitor the opening for the fuel measuring stick for five minutes after you remove the fuel measuring stick to make sure that there is no fuel leakage.

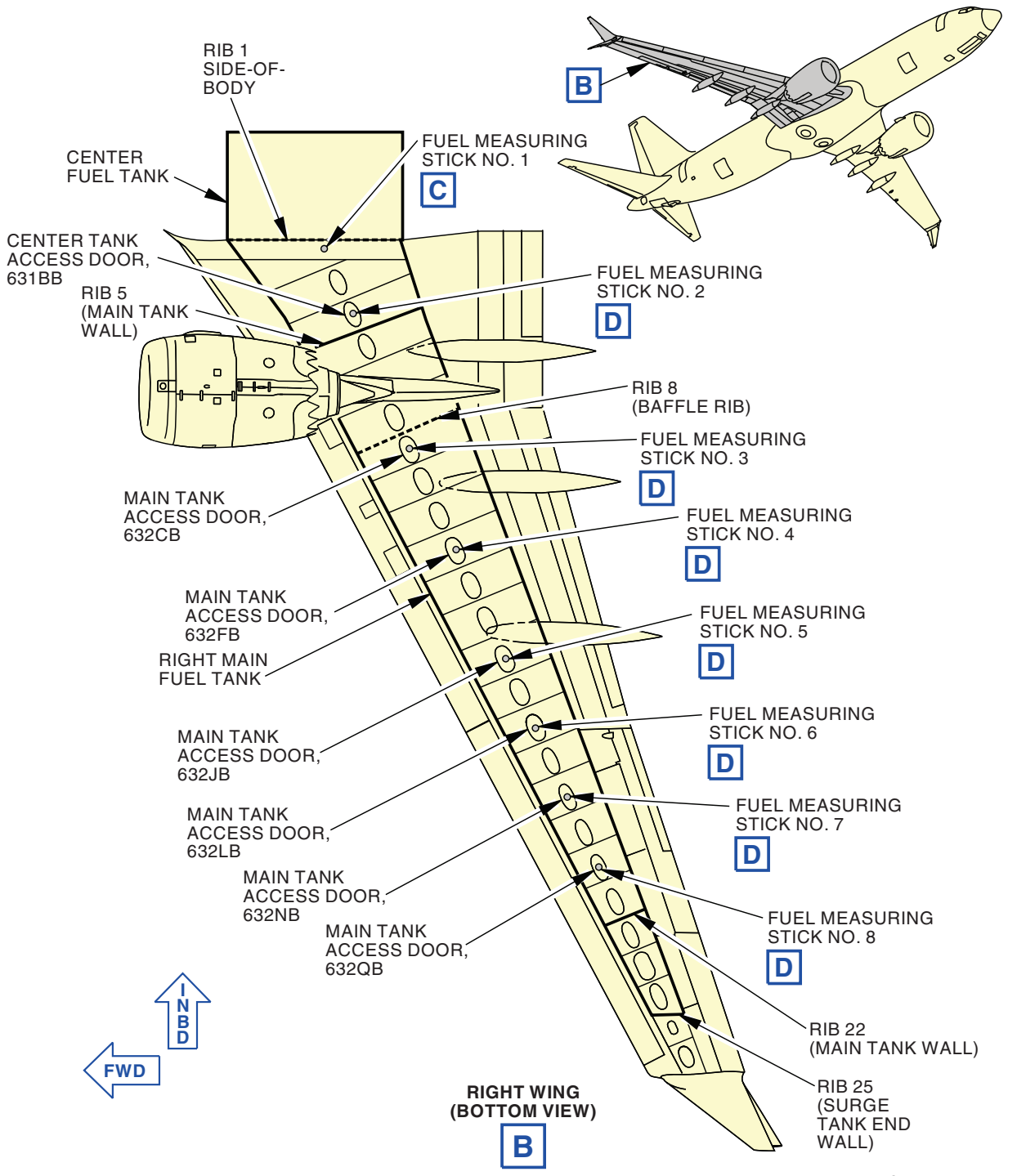
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



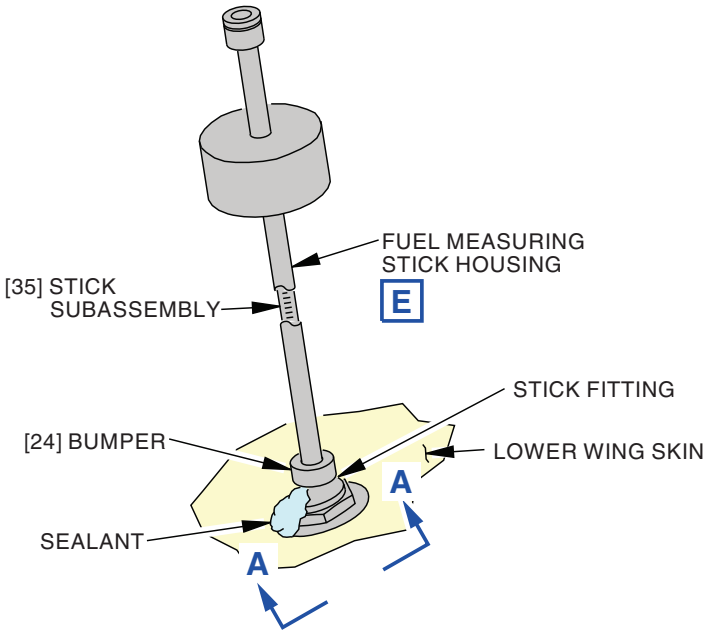
Fuel Measuring Stick Installation
Figure 401/28-44-11-990-801 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



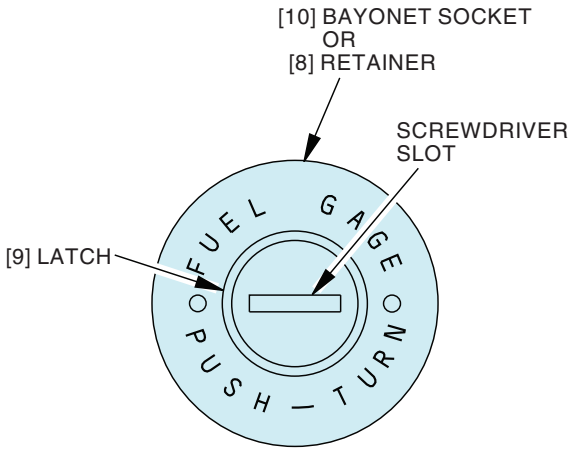
Fuel Measuring Stick Installation
Figure 401/28-44-11-990-801 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**FUEL MEASURING STICK INSTALLATION
(STICK NO. 1)**

C



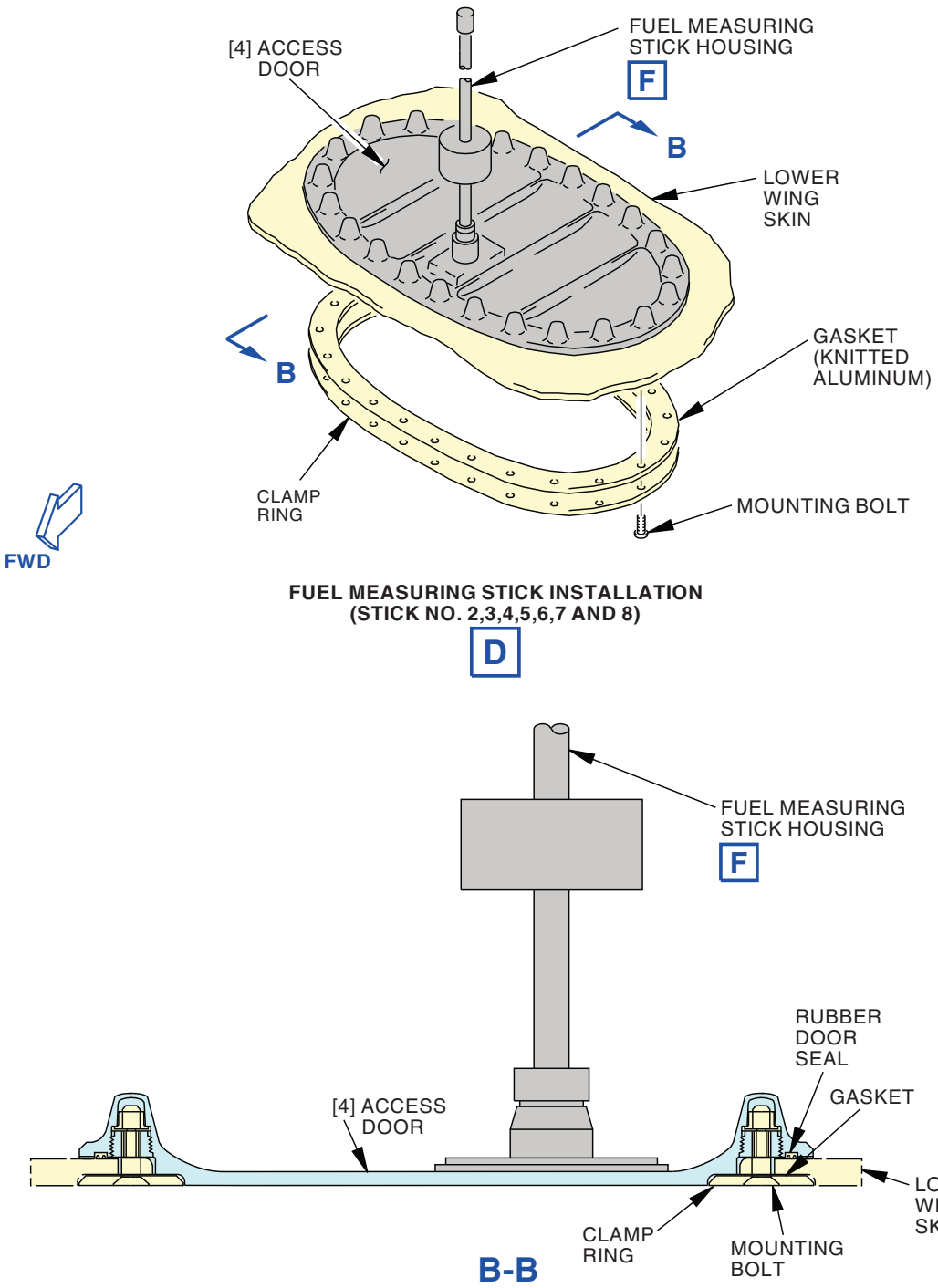
**BAYONET SOCKET OR RETAINER
(VIEW FROM BELOW THE WING,
STICKS NO. 1 THROUGH 8)**

A-A

2413741 S00061533231_V2

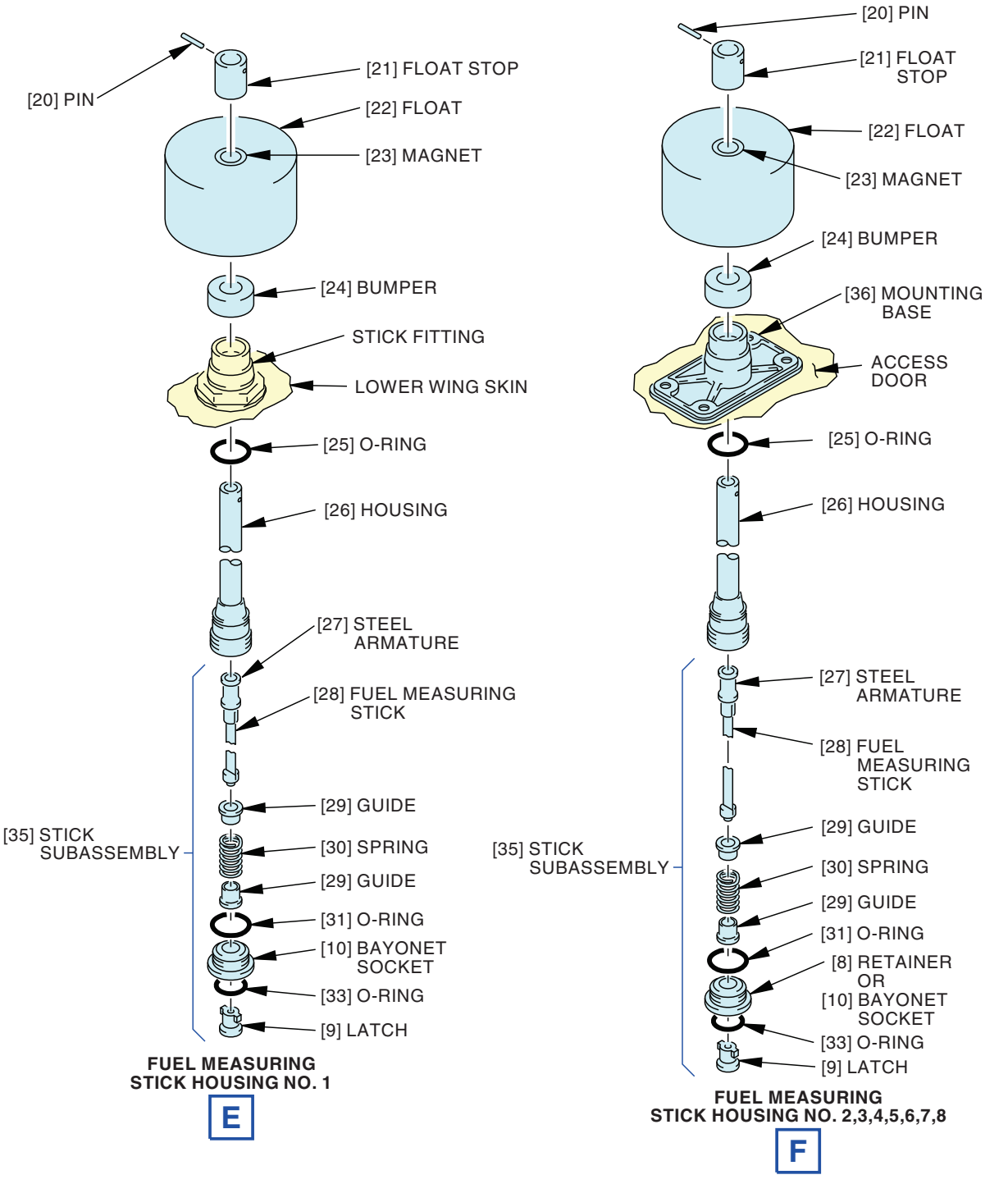
**Fuel Measuring Stick Installation
Figure 401/28-44-11-990-801 (Sheet 3 of 5)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Measuring Stick Installation
Figure 401/28-44-11-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Measuring Stick Installation
Figure 401/28-44-11-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-44-11-400-801

3. Fuel Measuring Stick Installation

(Figure 401)

A. General

(1) This task gives instructions to install the fuel measuring stick.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-5738	Spanner Wrench - Fuel Stick Retainer Part #: A28016-1 Supplier: 81205 Part #: T5-0497K Supplier: 86831

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
31	O-ring	28-44-11-01A-025	PTF ALL
		28-44-11-12-060	PTF ALL
		28-44-11-13-060	PTF ALL
		28-44-11-16-060	PTF ALL
		28-44-11-19-060	PTF ALL
33	O-ring	28-44-11-01A-020	PTF ALL
		28-44-11-12-055	PTF ALL
		28-44-11-13-055	PTF ALL
		28-44-11-16-055	PTF ALL
		28-44-11-19-055	PTF ALL
35	Stick subassembly	28-44-11-01A-010	PTF ALL
		28-44-11-01A-015	PTF ALL
		28-44-11-12-045	PTF ALL
		28-44-11-12-050	PTF ALL
		28-44-11-13-045	PTF ALL
		28-44-11-13-050	PTF ALL
		28-44-11-16-045	PTF ALL
		28-44-11-16-050	PTF ALL
		28-44-11-19-045	PTF ALL

D. Location Zones

Zone	Area
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Fuel Measuring Stick Installation

SUBTASK 28-44-11-420-001

(1) Install the new O-ring [31] and new O-ring [33] on the bayonet socket [10] or retainer [8].

SUBTASK 28-44-11-790-002

(2) Do these steps to do a check for fuel leakage in the housing of the fuel measuring stick:

(a) Make sure that the applicable tank contains sufficient fuel to cover the base of the fuel measuring stick housing.

Table 403/28-44-11-993-803

TABLE 601. Fuel Necessary to Cover the Center Tank Measuring Sticks		
Fuel Measuring Stick Number	**FUEL (LBS)**	**FUEL (KGS)**
Stick No. 1	9197	4172
Stick No. 2	13722	6224

Table 404/28-44-11-993-804

TABLE 602. Fuel Necessary to Cover the No. 1 or No. 2 Tank Measuring Sticks		
Fuel Measuring Stick Number	**FUEL (LBS)**	**FUEL (KGS)**
Stick No. 3	1167	529
Stick No. 4	3402	1543
Stick No. 5	5665	2570
Stick No. 6	6710	3044
Stick No. 7	7570	3434
Stick No. 8	8262	3747

(b) Make sure that there are no indications of fuel leakage from the housing before installation of the fuel measuring stick in the opening.

SUBTASK 28-44-11-420-002

(3) With the latch [9] of the stick subassembly removed, install the stick subassembly in the housing [26].

SUBTASK 28-44-11-420-003

(4) Install the guide [29], spring [30], second guide [29], and the retainer [8] in the housing [26].

SUBTASK 28-44-11-420-004

(5) Use a spanner wrench, SPL-5738, to tighten the retainer [8] in the housing to a torque of 80 ±5 in-lb (9 ±1 N·m).

SUBTASK 28-44-11-420-005

(6) Lower the stick subassembly [35], and install the latch [9] on the stick subassembly [35].

SUBTASK 28-44-11-860-003

(7) Push the stick subassembly [35] into the housing [26].

SUBTASK 28-44-11-860-004

(8) Turn the screwdriver slot on the latch [9] clockwise to lock the fuel measuring stick into the correct position (Figure 401).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 28-44-11-000-802

4. Fuel Measuring Stick Mounting Base Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the Fuel Measuring Stick Mounting Base.

B. References

Reference	Title
28-11-00-910-801	Purging and Fuel Tank Entry Precautions (P/B 201)
28-11-00-910-802	Purging and Fuel Tank Entry (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-26-00-650-801	Fuel Tank Defueling (P/B 201)
28-26-00-650-802	Tank to Tank Fuel Transfer (P/B 201)

C. Tools/Equipment

Reference	Description
STD-625	Pen

D. Location Zones

Zone	Area
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

E. Access Panels

Number	Name/Location
531BB	Center Tank Access Door - Wing Station 192
532CB	Main Tank Access Door - Wing Station 290
532FB	Main Tank Access Door - Wing Station 367
532JB	Main Tank Access Door - Wing Station 443
532LB	Main Tank Access Door - Wing Station 496
532NB	Main Tank Access Door - Wing Station 549
532QB	Main Tank Access Door - Wing Station 602
631BB	Center Tank Access Door - Wing Station 192
632CB	Main Tank Access Door - Wing Station 290
632FB	Main Tank Access Door - Wing Station 367
632JB	Main Tank Access Door - Wing Station 443
632LB	Main Tank Access Door - Wing Station 496
632NB	Main Tank Access Door - Wing Station 549
632QB	Main Tank Access Door - Wing Station 602

F. Prepare for the Removal

SUBTASK 28-44-11-650-001

(1) Defuel the applicable main tank (TASK 28-26-00-650-801) or transfer fuel out of the applicable tank (TASK 28-26-00-650-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 28-44-11-860-005

WARNING OBEY THE PRECAUTIONS FOR PURGING AND FUEL TANK ENTRY. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Obey the fuel tank entry precautions (TASK 28-11-00-910-801).

SUBTASK 28-44-11-650-002

(3) Drain and purge the applicable main fuel tank (TASK 28-11-00-910-802).

SUBTASK 28-44-11-020-004

(4) Do this task: Main Tank Access Door Removal, TASK 28-11-11-000-801.

(a) Remove the applicable fuel tank access door(s):

Number	Name/Location
531BB	Center Tank Access Door - Wing Station 192
532CB	Main Tank Access Door - Wing Station 290
532FB	Main Tank Access Door - Wing Station 367
532JB	Main Tank Access Door - Wing Station 443
532LB	Main Tank Access Door - Wing Station 496
532NB	Main Tank Access Door - Wing Station 549
532QB	Main Tank Access Door - Wing Station 602
631BB	Center Tank Access Door - Wing Station 192
632CB	Main Tank Access Door - Wing Station 290
632FB	Main Tank Access Door - Wing Station 367
632JB	Main Tank Access Door - Wing Station 443
632LB	Main Tank Access Door - Wing Station 496
632NB	Main Tank Access Door - Wing Station 549
632QB	Main Tank Access Door - Wing Station 602

G. Fuel Measuring Stick Mounting Base Removal

SUBTASK 28-44-11-020-005

(1) Remove the fuel measuring stick (TASK 28-44-11-000-801).

SUBTASK 28-44-11-020-006

(2) Remove the old sealant from the mounting base [36] and fasteners.

SUBTASK 28-44-11-860-006

(3) Make a mark on the mounting base with a pen, STD-625, of an arrow that points in the forward direction.

NOTE: This arrow will make sure the mounting base and fuel measuring stick are installed with the correct orientation.

SUBTASK 28-44-11-020-007

(4) Remove the bolts, collars, and washers from the mounting base (four locations).

(a) Keep the fasteners for the installation.

SUBTASK 28-44-11-020-008

(5) Carefully remove the mounting base and fuel measuring stick housing from the fuel tank access door.

NOTE: The mounting base and fuel tank access door are bonded with faying surface sealant.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Keep the mounting base and fuel measuring stick housing for the installation.

——— END OF TASK ———

TASK 28-44-11-400-802

5. Fuel Measuring Stick Mounting Base Installation

(Figure 401)

A. General

(1) This task gives instructions to install the Fuel Measuring Stick Mounting Base.

(2) For faying surface seal applications:

(a) Inside the fuel tank, use BMS 5-45 Class B-2 sealant, A50110, or BMS 5-45 Class C sealant, A50155.

(b) Outside the fuel tank, use BMS 5-45 Class B-2 sealant, A50110, BMS 5-45 Class C sealant, A50155, BMS 5-95 Class B sealant, A50231, or BMS 5-95 Class C sealant, A50296.

(c) Do not apply a faying surface seal with a temperature higher than 100°F (38°C).

(d) Do not use alternatives for faying surface sealants.

B. References

Reference	Title
28-11-00-300-803	Repair of Sealant Leaks in the Fuel Tank Structure (P/B 801)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)
51-22-00-390-806	Fillet Seal Application (P/B 201)

C. Tools/Equipment

Reference	Description
STD-123	Brush - Soft Bristle

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50021	Pad - General Purpose Scrubbing - Scotch-Brite 96W	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
36	Mounting base	28-44-11-12-130	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
36 (cont.)		28-44-11-13-130	PTF ALL
		28-44-11-16-130	PTF ALL
		28-44-11-16-135	PTF ALL
		28-44-11-19-120	PTF ALL

F. Location Zones

Zone	Area
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

G. Access Panels

Number	Name/Location
531BB	Center Tank Access Door - Wing Station 192
532CB	Main Tank Access Door - Wing Station 290
532FB	Main Tank Access Door - Wing Station 367
532JB	Main Tank Access Door - Wing Station 443
532LB	Main Tank Access Door - Wing Station 496
532NB	Main Tank Access Door - Wing Station 549
532QB	Main Tank Access Door - Wing Station 602
631BB	Center Tank Access Door - Wing Station 192
632CB	Main Tank Access Door - Wing Station 290
632FB	Main Tank Access Door - Wing Station 367
632JB	Main Tank Access Door - Wing Station 443
632LB	Main Tank Access Door - Wing Station 496
632NB	Main Tank Access Door - Wing Station 549
632QB	Main Tank Access Door - Wing Station 602

H. Fuel Measuring Stick Mounting Base Installation

SUBTASK 28-44-11-110-001

(1) Remove the old sealant from these components:

(a) mounting base [36] (if re-used)

(b) bolts, collars, and washers.

SUBTASK 28-44-11-110-002

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE. OBEY THE MATERIAL SAFETY DATA SHEETS (MSDS) FOR SOLVENTS. OBEY LOCAL REGULATIONS FOR THE CORRECT PROCEDURES TO USE OR DISCARD SOLVENTS. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) To remove the remaining sealant, grease, oil, dirt, and other contamination from the components, use one of these brushed/pads soaked with solvent, B00184, or solvent, B00666:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) soft bristle brush, STD-123

(b) cotton wiper, G00034

(c) Scotch-Brite 96W pad, G50021.

SUBTASK 28-44-11-760-001

(3) Do these steps to prepare the mounting base and the fuel tank access door for an electrical faying surface bond (SWPM 20-20-00):

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE. OBEY THE MATERIAL SAFETY DATA SHEETS (MSDS) FOR SOLVENTS. OBEY LOCAL REGULATIONS FOR THE CORRECT PROCEDURES TO USE OR DISCARD SOLVENTS. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Final clean the contact surfaces of the mounting base and the fuel tank access door with a cotton wiper, G00034, soaked with alcohol, B00130, or solvent, B00184.

(b) Rub dry with a clean, dry cotton wiper, G00034.

(c) Continue to clean and dry the surface until the dry cotton wiper, G00034, stays clean.

(d) Apply a thin continuous layer of BMS 5-45 Class B-2 sealant, A50110, or BMS 5-45 Class C sealant, A50155, to the contact area of the mounting base.

1) For the use of faster cure time sealants, refer to the figure "Sealant Usable Time" in TASK 28-11-00-300-803 for BMS 5-45 Class B or Class C sealants.

SUBTASK 28-44-11-420-006

(4) Do these steps to install the mounting base to the fuel tank access door:

(a) Put the mounting base onto the fuel tank access door.

1) Make sure that the arrow on the mounting base is pointing in the forward direction.

(b) Install the bolts, washers, and collars (four locations).

SUBTASK 28-44-11-390-001

(5) On the in-tank side of the fuel tank access door, apply sealant, A50110, to these components:

(a) A cap seal on the four bolts

(b) A fillet seal around the periphery of the mounting base and the fuel tank access door (TASK 51-22-00-390-806).

SUBTASK 28-44-11-760-002

(6) Do these steps to do a check of the electrical bonding resistance for all fuel tank access doors except 531BB and 631BB (SWPM 20-20-00):

(a) Measure the electrical bonding resistance between the mounting base and the fuel tank access door.

1) Make sure that the resistance is less than 0.001 ohm (1 milliohm).

SUBTASK 28-44-11-760-003

(7) Do these steps to do a check of the electrical bonding resistance for fuel tank access doors 531BB and 631BB only (SWPM 20-20-00):

(a) Measure the electrical bonding resistance between the mounting base and the unpainted periphery of the aluminum face sheet.

1) Make sure that the resistance is less than 0.0025 ohm (2.5 milliohms).

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I. Put the Airplane Back to Its Usual Condition

SUBTASK 28-44-11-420-007

(1) Do this task: Main Tank Access Door Installation, TASK 28-11-11-400-801.

(a) Install the applicable fuel tank access door(s):

Number	Name/Location
531BB	Center Tank Access Door - Wing Station 192
532CB	Main Tank Access Door - Wing Station 290
532FB	Main Tank Access Door - Wing Station 367
532JB	Main Tank Access Door - Wing Station 443
532LB	Main Tank Access Door - Wing Station 496
532NB	Main Tank Access Door - Wing Station 549
532QB	Main Tank Access Door - Wing Station 602
631BB	Center Tank Access Door - Wing Station 192
632CB	Main Tank Access Door - Wing Station 290
632FB	Main Tank Access Door - Wing Station 367
632JB	Main Tank Access Door - Wing Station 443
632LB	Main Tank Access Door - Wing Station 496
632NB	Main Tank Access Door - Wing Station 549
632QB	Main Tank Access Door - Wing Station 602

SUBTASK 28-44-11-420-011

(2) Install the fuel measuring stick (TASK 28-44-11-400-801).

——— END OF TASK ———

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FUEL MEASURING STICK - REPAIRS

1. General

A. This procedure contains a task to replace the O-ring(s) that make a seal between the tube assembly and the housing for the magnetic fuel measuring sticks. If this location is not sealed correctly, it can be a cause of fuel leakage.

TASK 28-44-11-360-801

2. Replacement of the O-Ring Seal Between the Housing and the Base Assembly

(Figure 801)

A. References

Reference	Title
28-44-11-000-801	Fuel Measuring Stick Removal (P/B 401)
28-44-11-400-801	Fuel Measuring Stick Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	O-Ring	28-44-11-12-115	PTF ALL
		28-44-11-13-115	PTF ALL
		28-44-11-16-115	PTF ALL
		28-44-11-19-110	PTF ALL

C. Location Zones

Zone	Area
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Replacement of the O-Ring Seal Between the Housing and the Base Assembly

SUBTASK 28-44-11-860-007

(1) Make sure that the airplane has a roll between -1 degree and +1 degree (less than one degree left or right roll).

SUBTASK 28-44-11-860-008

(2) Make sure that the airplane has a pitch between -1 and +1 degrees (less than one degree pitch up or down).

SUBTASK 28-44-11-860-009

WARNING OBEY THESE PRECAUTIONS. IF YOU SPILL FUEL, IT CAN CAUSE FIRES AND EXPLOSIONS. THESE CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(3) Make sure that the fuel tank for the applicable fuel measuring stick has less than the maximum quantity of fuel given in the tables below (Table 801, Table 802).

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Table 801/28-44-11-993-805 Maximum Fuel Quantity in Center Tank When the Fuel Measuring Stick Housing is Loosened

Fuel Measuring Stick	FUEL (LBS) *[1]	FUEL (KGS) *[2]
1	0	0
2	7774	3533

*[1] Based on a nominal density of 6.76 pounds per U.S. gallon

*[2] Based on a nominal density of 0.810 kilograms per liter

Table 802/28-44-11-993-806 Maximum Fuel Quantity in No. 1 or No. 2 Tank When the Fuel Measuring Stick Housing is Loosened

Fuel Measuring Stick	FUEL (LBS) *[1]	FUEL (KGS) *[2]
3	608	276
4	1960	891
5	4394	1997
6	5746	2611
7	6760	3072
8	7436	3380

*[1] Based on a nominal density of 6.76 pounds per U.S. gallon

*[2] Based on a nominal density of 0.810 kilograms per liter

SUBTASK 28-44-11-010-001

(4) Remove the fuel measuring stick subassembly [1] from its housing (TASK 28-44-11-000-801).

SUBTASK 28-44-11-020-009

(5) Put a nine sixteenth inch (9/16 inch) hexagonal wrench through the bottom of the base and into the bottom of the housing until it has a good fit.

SUBTASK 28-44-11-020-010

(6) Loosen the housing [3] from the base with the wrench.

SUBTASK 28-44-11-020-011

CAUTION

DO NOT MOVE THE HOUSING MORE THAN NECESSARY. YOU CAN CAUSE DAMAGE TO THE FLOAT AND FLOAT STOP.

(7) Carefully move the housing down until it is possible to remove the O-Ring [2] from the end of the housing.

SUBTASK 28-44-11-020-012

(8) Use a plastic pick to remove the O-Ring [2] from the housing.

SUBTASK 28-44-11-360-001

(9) Install new O-Ring [2] on the housing.

SUBTASK 28-44-11-420-008

(10) Lift the housing into the base assembly.

SUBTASK 28-44-11-420-009

(11) Tighten the housing into the base assembly with the hexagonal wrench to 40 in-lb (4.5 N·m) - 50 in-lb (5.6 N·m).

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SUBTASK 28-44-11-650-003

(12) Refuel the applicable tank until the fuel quantity is more than the minimum quantity shown in the tables below (Table 803, Table 804).

Table 803/28-44-11-993-807 Minimum Fuel Quantity in Center Tank To Do a Leakage Check After the Repair

Fuel Measuring Stick	FUEL (LBS) *[1]	FUEL (KGS) *[2]
1	2840	1291
2	16224	7375

*[1] Based on a nominal density of 6.76 pounds per U.S. gallon

*[2] Based on a nominal density of 0.810 kilograms per liter

Table 804/28-44-11-993-808 Minimum Fuel Quantity in No. 1 or No. 2 Tank To Do a Leakage Check After the Repair

Fuel Measuring Stick	FUEL (LBS) *[1]	FUEL (KGS) *[2]
3	1555	707
4	4124	1875
5	6152	2797
6	7098	3227
7	7842	3565
8	8450	3841

*[1] Based on a nominal density of 6.76 pounds per U.S. gallon

*[2] Based on a nominal density of 0.810 kilograms per liter

SUBTASK 28-44-11-210-001

(13) Do a check for fuel leakage in the area of the fuel measuring stick.

SUBTASK 28-44-11-410-001

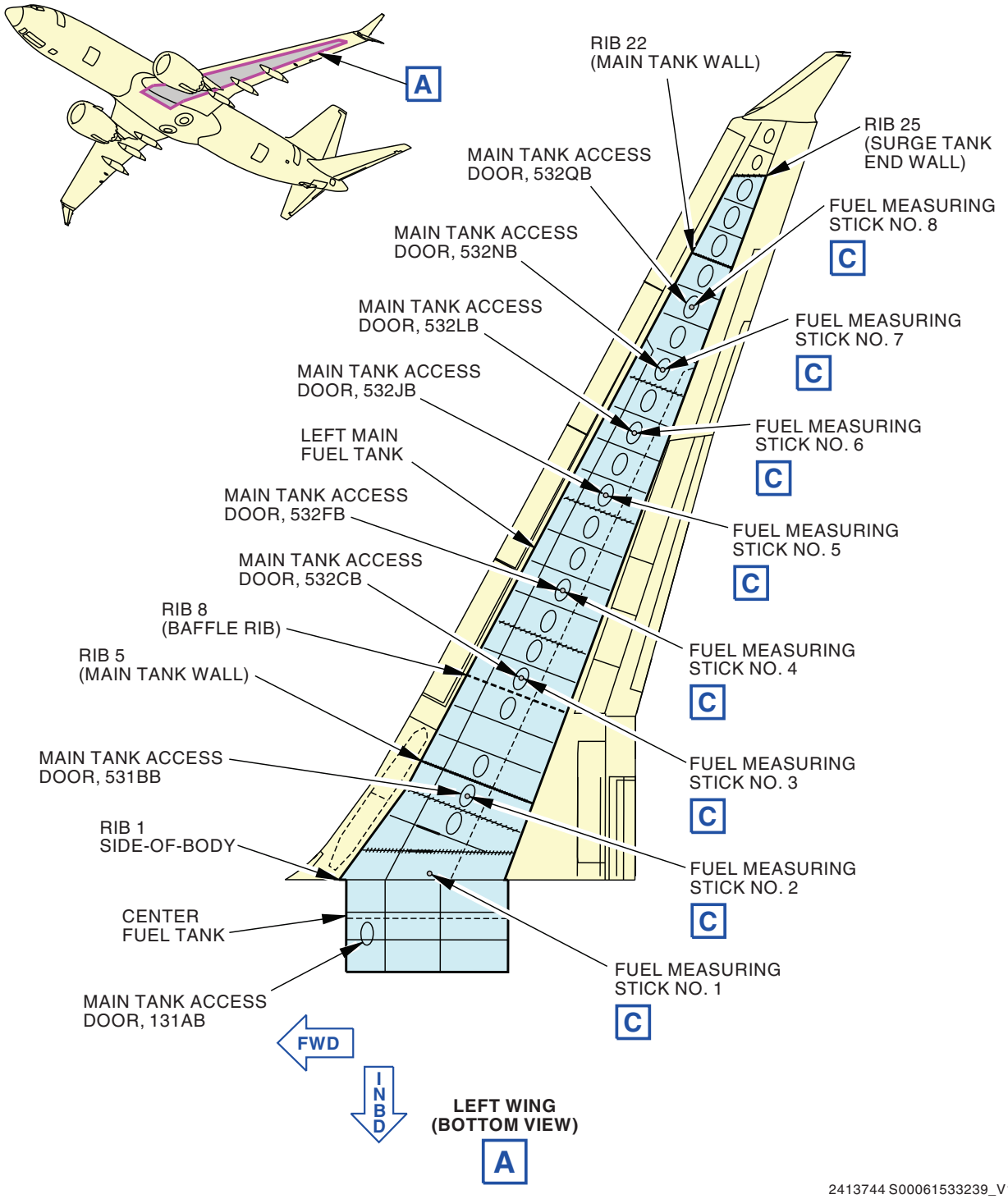
(14) If there is no fuel leakage, install the fuel measuring stick again (TASK 28-44-11-400-801).

SUBTASK 28-44-11-420-010

(15) If the fuel leakage continues, replace the housing for the fuel measuring stick.

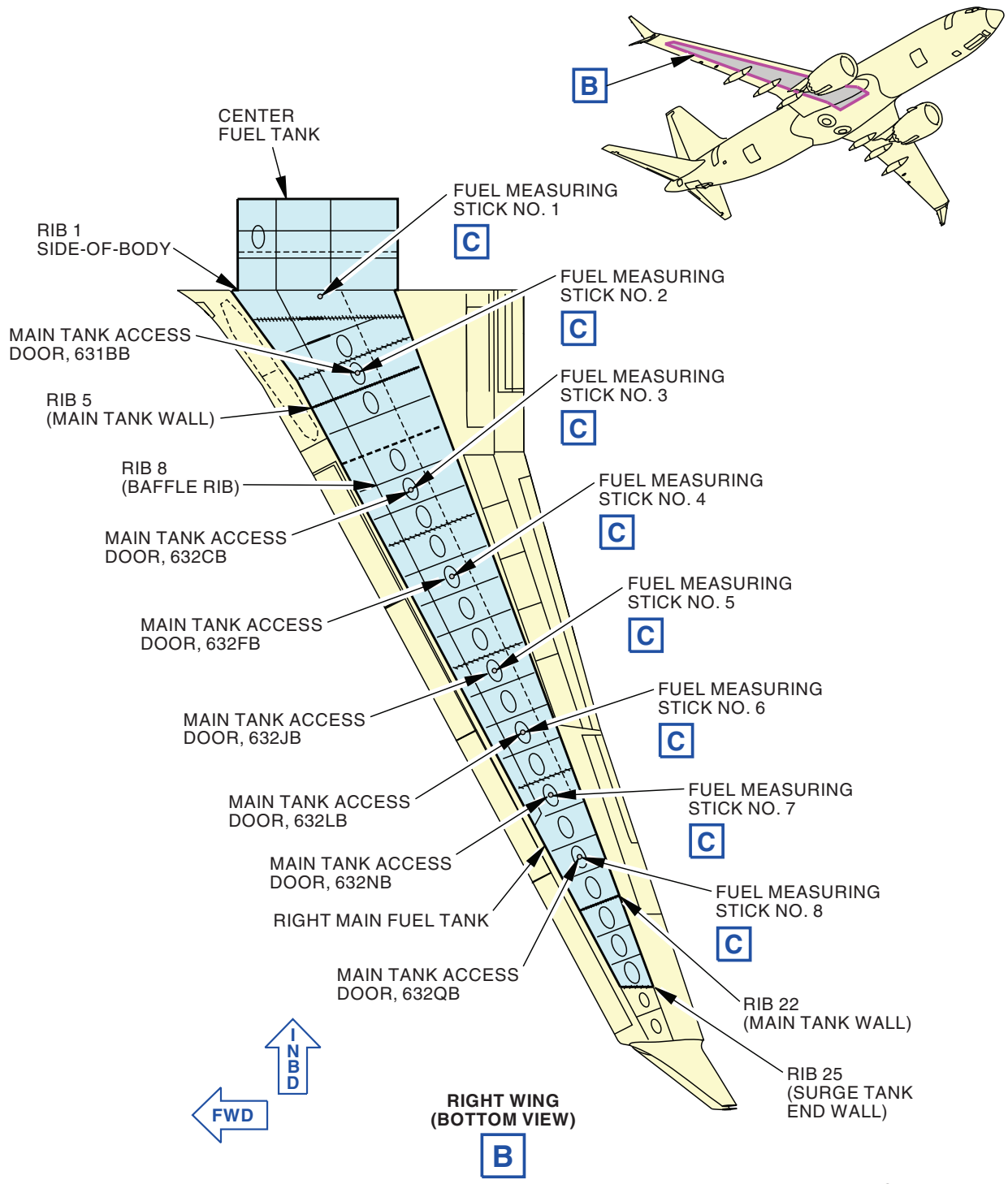
——— END OF TASK ———

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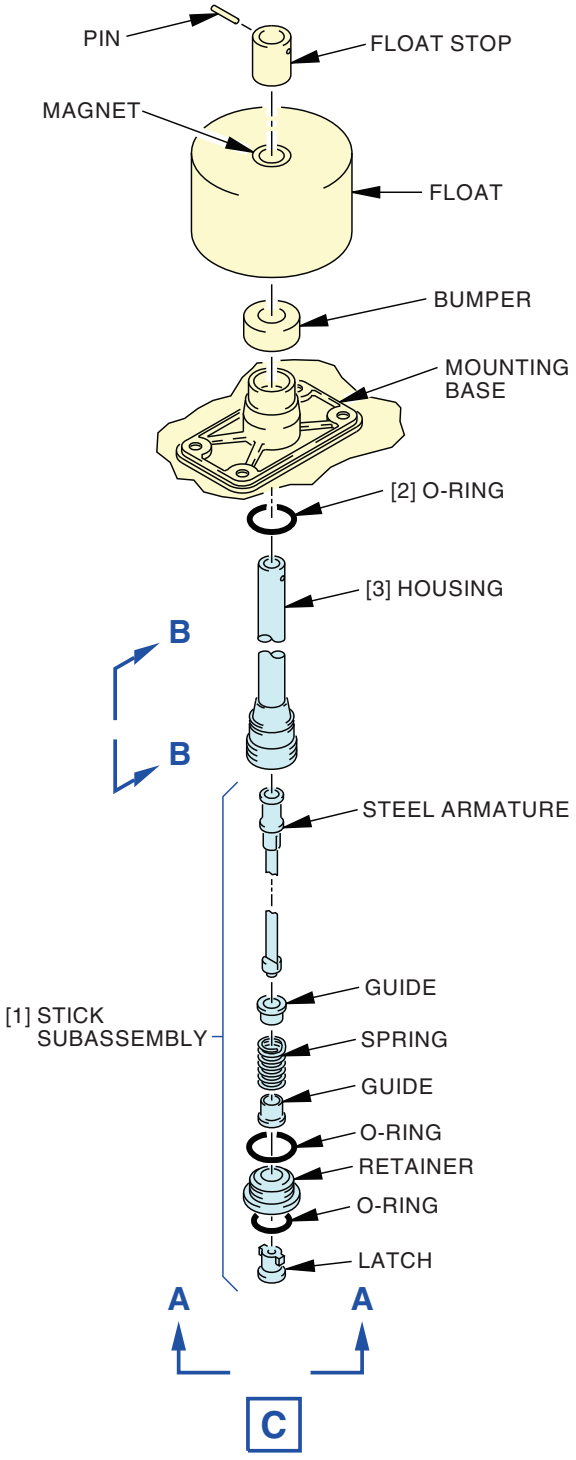
Fuel Measuring Stick Installation
Figure 801/28-44-11-990-802 (Sheet 1 of 3)

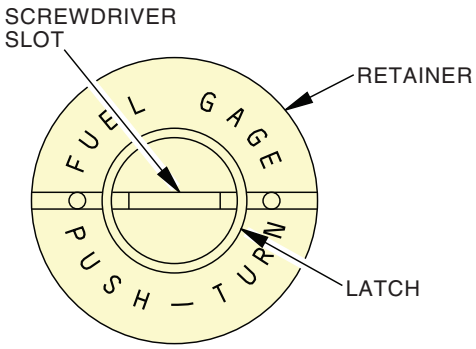
ECCN 9E991 BOEING PROPRIETARY - See title page for details



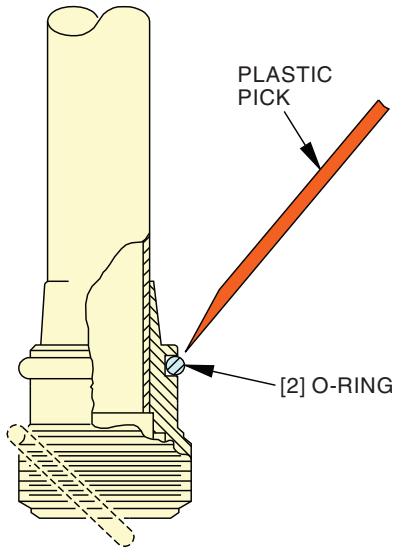
Fuel Measuring Stick Installation
Figure 801/28-44-11-990-802 (Sheet 2 of 3)

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RETAINER
(VIEWED FROM BELOW THE WING)
A-A



REMOVAL OF O-RING
B-B

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Fuel Measuring Stick Installation
Figure 801/28-44-11-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details